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TREATMENT OF DISEASES RELATED TO ATP-BINDING CASSETTE TRANSPORTER 1 DYSFUNCTION USING TREM2 AGONISTS

Abstract

The present invention provides a method of treating a disease or disorder caused by and/or associated with ABCD1 dysfunction in a human patient, the method comprising administering to the patient in need thereof an effective amount of a compound that increases the activity of triggering receptor expressed on myeloid cells 2 (TREM2). In some embodiments, compound that increases the activity of TREM2 is an agonist of TREM2. In some embodiments, the agonist of TREM2 is a small molecule agonist of TREM2 or an antibody agonist of TREM2.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. application Ser. No. 17/457,727, filed Dec. 6, 2021, which claims priority benefit of U.S. Provisional Application No. 63/121,404, filed Dec. 4, 2020, each of which is hereby incorporated by reference in its entirety.

REFERENCE TO SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Dec. 17, 2024, is named 2024-12-17-Sequence-Listing-5258.0070003.txt and is 2,755,362 bytes in size.

TECHNICAL FIELD OF THE INVENTION

[0003] The present invention relates to compounds and methods of use thereof for treating diseases and disorders caused by ATP-binding cassette transporter 1 (ABCD1) dysfunction.

BACKGROUND OF THE INVENTION

[0004] Microglia are brain-resident macrophages with many homeostatic and injury responsive roles, including trophic and phagocytic functions. Microglia are highly dependent on peroxidation for maintaining normal function. The ATP-binding cassette transporter 1 (ABCD1) gene encodes a key peroxisomal protein responsible for transport of activated very long chain fatty acids (VLCFA) into the peroxisome for further degradation and beta-oxidation for energy production. Therefore, mutations in the ABCD1 gene can lead to microglial dysfunction and damage due to accumulation of VLCFA, resulting in neurological and adrenal gland diseases and disorders. X-linked adrenoleukodystrophy (X-ALD) is one such condition associated with ABCD1 mutations, characterized by cerebral and spinal cord white matter degeneration with demyelination and adrenal insufficiency, which lead to progressive cognitive and motor dysfunction and ultimately death. To date, there are no known treatments for diseases and disorders caused by ABCD1 dysfunction, and patients are usually treated by managing the symptoms of the disease. Therefore, there remains a need in the art for methods of treating diseases and disorders caused by ABCD1 loss of function mutations.

SUMMARY OF THE INVENTION

[0005] In one aspect, the present invention provides a method of treating a disease or disorder caused by and/or associated with a dysfunction in ABCD1 in a human patient, the method comprising administering to the patient an effective amount of a compound that increases the activity of triggering receptor expressed on myeloid cells 2 (TREM2). In some embodiments, the compound that increases the activity of TREM2 is an agonist of TREM2. In some embodiments, the agonist of TREM2 is a small molecule agonist of TREM2 or an antibody agonist of TREM2. In some embodiments, the disease or disorder caused by and/or associated with a dysfunction in ABCD1 is x-ALD.

Description

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

TREM2, ABCD1 and X-ALD

[0006] TREM2 is a member of the Ig superfamily of receptors that is expressed on cells of myeloid

lineage, including macrophages, dendritic cells, and microglia (Schmid et al., *Journal of Neurochemistry*, Vol. 83: 1309-1320, 2002; Colonna, *Nature Reviews Immunology*, Vol. 3: 445-453, 2003; Kialainen et al., *Neurobiology of Disease*, 2005, 18: 314-322). TREM2 is an innate immune receptor that binds many endogenous substrates, and signals through a short intracellular domain that complexes with the adaptor protein DAP12, the cytoplasmic domain of which comprises an ITAM motif (Bouchon et al., *The Journal of Experimental Medicine*, 2001, 194: 1111-1122). Upon activation of TREM2, tyrosine residues within the ITAM motif in DAP12 are phosphorylated by the Src family of kinases, providing docking sites for the tyrosine kinase ξ -chain-associated protein 70 (ZAP70) and spleen tyrosine kinase (Syk) via their SH2 domains (Colonna, *Nature Reviews Immunology*, 2003, 3:445-453; Ulrich and Holtzman, *ACS Chem. Neurosci.*, 2016, 7:420-427). The ZAP70 and Syk kinases induce activation of several downstream signaling cascades, including phosphatidylinositol 3-kinase (PI3K), protein kinase C (PKC), extracellular regulated kinase (ERK), and elevation of intracellular calcium (Colonna, *Nature Reviews Immunology*, 2003, 3:445-453; Ulrich and Holtzman, *ACS Chem. Neurosci.*, 2016, 7:420-427). The wild-type human TREM2 amino acid sequence is provided as SEQ ID NO: 1.

[0007] TREM2 has been implicated in several myeloid cell processes, including phagocytosis, proliferation, survival, and regulation of inflammatory cytokine production (Ulrich and Holtzman, *ACS Chem. Neurosci.*, 2016, 7: 420-427). One of the key TREM2 functions is regulating myeloid cell number. Knocking down expression of TREM2 in primary microglia using translation blockers leads to reduced cell number (Zheng, et al., *Neurobiol. Aging*, 2016; 42: 132-141). Evidence suggests that in various contexts, TREM2 is important for myeloid cell survival, proliferation and chemotaxis, all of which could lead to disease-associated increases in myeloid cell number including microglia (Jay, et al., *Mol Neurodegener.* 2017; 12(1):56).

[0008] A well-characterized function of TREM2 is to enhance phagocytosis. TREM2 is expressed in a subset of myeloid cells within the CNS that have high phagocytic capacity (Bisht et al., *Glia*. 2016; 64: 826-839). Across numerous in vitro studies, loss of TREM2 results in reduced phagocytosis of a variety of substrates, including apoptotic neurons or neuronal cell lines (Takahashi et al., *Exp Med.* 2005; 201(4):647-657.; Hsieh et al., *J Neurochem.* 2009; 109(4): 1144-1156). Conversely, TREM2 activation or overexpression enhanced uptake of these substrates (Takahashi et al., *J Exp Med.* 2005; 201(4):647-657; Takahashi et al., *PLoS Med.* 2007; 4(4):e124; Jiang et al., *Neuropsychopharmacology.* 2014; 39(13): 2949-2962.). TREM2 is important for clearance of myelin debris in experimental autoimmune encephalomyelitis (EAE) (Takahashi et al., *PLoS Med.* 2007; 4(4):e124) and peri-infarct tissue in mice following middle coronary artery occlusion (MCAO) (Kawabori et al., *J Neurosci.* 2015; 35(8): 3384-3396).

[0009] TREM2 has been classically described as being anti-inflammatory and several in vitro and in vivo studies are supportive of an anti-inflammatory role for TREM2 in certain contexts (Yin et al., *Traffic.* 2016; 17(12): 1286-1296). Knocking down TREM2 in cell lines increases levels of proinflammatory mediators such as iNOS, TNF α , IL1 β and IL δ (Yin et al., *Traffic.* 2016; 17(12): 1286-1296) in response to apoptotic neuronal membrane debris (Takahashi et al., *J Exp Med.* 2005; 201(4):647-657.), TLR ligands (Turnbull et al., *J Immunol.* 2006; 177(6):3520-3524.), including LPS (Gawish et al., *FASEB J.* 2015 Apr; 29(4):1247-1257.; Gao et al., *Mol Med Rep.* 2013 March; 7(3):921-926.; Takahashi et al., *PLoS Med.* 2007; 4(4):e124.) and A β 42 (Jiang et al., *Neuropsychopharmacology.* 2014; 39(13): 2949-2962.). Moreover, overexpressing TREM2 in cell lines or amyloid (Jiang et al., *Neuropsychopharmacology.* 2014; 39(13): 2949-2962.) and tau mouse models of AD (Jiang et al., *Neuropharmacology.* 2016; 105:196-206.) reduced levels of these pro-inflammatory transcripts. Together, these studies suggest that in some contexts, TREM2 can attenuate inflammatory responses.

[0010] TREM2 has been linked to several serious diseases. For instance, mutations in both TREM2 and DAP12 have been linked to the autosomal recessive disorder Nasu-Hakola Disease, which is characterized by bone cysts, muscle wasting and demyelination phenotypes (Guerreiro et al., *New*

England Journal of Medicine, 2013, 368: 117-127). Variants in the TREM12 gene have been linked to increased risk for Alzheimer's disease (AD) and other forms of dementia including frontotemporal dementia and amyotrophic lateral sclerosis (Jonsson et al., New England Journal of Medicine, 2013, 368:107-116; Guerreiro et al., JAMA Neurology, 2013, 70:78-84; Jay et al., Journal of Experimental Medicine, 2015, 212: 287-295; Cady et al., JAMA Neurol. 2014 April; 71(4):449-53). Impairment in microgliosis has been reported in animal models of prion disease, multiple sclerosis, and stroke, suggesting that TREM2 may play an important role in supporting microgliosis in response to pathology or damage in the central nervous system (Ulrich and Holtzman, ACS Chem. Neurosci., 2016, 7: 420-427).

[0011] The ABCD1 gene provides instructions for producing the adrenoleukodystrophy protein (ALDP). ABCD1 (ALDP) maps to Xq28. ABCD1 is a member of the ATP-binding cassette (ABC) transporter superfamily. The superfamily contains membrane proteins that translocate a wide variety of substrates across extra- and intracellular membranes, including metabolic products, lipids and sterols, and drugs. ALDP is located in the membranes of cell structures called peroxisomes. Peroxisomes are small sacs within cells that process many types of molecules. ALDP brings a group of fats called very long-chain fatty acids (VLCFAs) into peroxisomes, where they are broken down. As ABCD1 is highly expressed in microglia, it is possible that microglial dysfunction and their close interaction with other cell types actively participates in neurodegenerative processes (Gong et al., Annals of Neurology. 2017; 82(5):813-827.). It has been shown that severe microglia loss and damage is an early feature in patients with cerebral form of X-linked ALD (cALD) carrying ABCD1 mutations (Bergner et al., Glia. 2019; 67: 1196-1209). It has also been shown that ABCD1-deficiency leads to an impaired plasticity of myeloid lineage cells that is reflected in incomplete establishment of anti-inflammatory responses, thus possibly contributing to the devastating rapidly progressive demyelination in cerebral adrenoleukodystrophy (Weinhor et al., BRAIN 2018: 141; 2329-2342). It has also been shown in a recent report of 83 young males with cALD that patients harboring the APOE4 allele, a known ligand of TREM2, have an increased burden of cerebral disease involvement as determined by Loes score, gadolinium intensity score (GIS), and neurologic function score (NFS) (Orchard et al., Nature Scientific Reports 2019 9:7858). These findings emphasize microglia/monocytes/macrophages as crucial therapeutic targets for preventing or stopping myelin destruction in patients with X-linked adrenoleukodystrophy.

[0012] The present invention relates to the unexpected discovery that administration of a TREM2 agonist can rescue the loss of microglia in cells having mutations in the ABCD1 gene. It has been previously shown that TREM2 agonist antibody 4D9 increases ATP luminescence (a measure of cell number and activity) in a dose dependent manner when the levels of M-CSF in media are reduced to 5 ng/mL (Schlepckow et al, EMBO Mol Med., 2020) and that TREM2 agonist AL002c increases ATP luminescence when M-CSF is completely removed from the media (Wang et al, J. Exp. Med.; 2020, 217(9): e20200785). This finding suggests that TREM2 agonism can compensate for deficiency in ABCD1 function leading to sustained activation, proliferation, chemotaxis of microglia, maintenance of anti-inflammatory environment and reduced astrogliosis caused by a decrease in ABCD1 and accumulation of VLCFAs. The present invention relates to the unexpected discovery that activation of TREM2 can rescue microglia harboring the ABCD1 mutation and challenged with an increase in VLCFA, and that this effect may be also observed in patients suffering from loss of functional microglia due to ABCD1 mutation. This discovery has not been previously taught or suggested in the available art.

[0013] To date, no prior study has shown that TREM2 agonism can rescue the loss of microglia in cells where mutations in the ABCD1 and a VLCFA increase is present. No prior study has taught or suggested that reversal of the loss of microglia due to an ABCD1 mutation through TREM2 agonism can be used to treat a disease or disorder caused by and/or associated with an ABCD1 mutation.

[0014] X-linked adrenoleukodystrophy (x-ALD) is an X chromosome-linked central nervous

system disease caused by an ABCD1 mutation that manifests in the form of variable developmental behavioral, cognitive, motor and sensory function changes in patients suffering from the disease. Three main phenotypes are seen in affected males: (1) The childhood cerebral form manifests most commonly between ages four and eight years. It initially resembles attention-deficit disorder or hyperactivity; progressive impairment of cognition, behavior, vision, hearing, and motor function follow the initial symptoms and often lead to total disability within six months to two years and death within 5 years. (2) Adrenomyeloneuropathy (AMN) manifests most commonly between the second and fourth decades as progressive stiffness and weakness of the legs, sphincter disturbances, sexual dysfunction, and often, impaired adrenocortical function; all symptoms are progressive over decades. (3) “Addison disease only” presents with primary adrenocortical insufficiency between age two years and adulthood and most commonly by age 7.5 years, without evidence of neurologic abnormality. The cerebral childhood form of x-ALD also known as cerebral ALD (cALD) is characterized by patchy cerebral white matter abnormalities visible by magnetic resonance imaging. However, the clinical symptoms and MRI changes are not specific to cALD and are common for other neurological conditions, including Adult-onset leukoencephalopathy with axonal spheroids and pigmented glia (ALSP), Nasu-Hakola disease (NHD) and other leukodystrophies, making diagnosis and treatment of cALD very difficult.

[0015] Studies have discovered that x-ALD is a genetic disorder in which male patients that carry a loss of function mutation in the peroxisomal transporter gene ABCD1, leading to VLCFA increase and activation of inflammatory processes leading to demyelination and axonal degeneration. In one aspect, the present invention relates to the surprising discovery that activation of the TREM2 pathway can rescue the loss of microglia in patients carrying ABCD1 mutations, preventing microglia apoptosis, thereby treating ABCD1-related conditions, such as, but not limited to, x-ALD.

[0016] The present invention also relates to the surprising discovery that neurofilament light chain and neurofilament heavy chain proteins can serve as a therapeutic biomarker to determine treatment efficacy in patients suffering from a disease or disorder caused by and/or associated with a ABCD1 dysfunction, such as x-ALD. Neurofilament light chain (NfL) is highly elevated in the plasma, serum and CSF of patients with x-ALD, (van Ballegoij, et al., Ann Clin Transl Neurol, 7: 2127-2136.). cALD is characterized by severe and rapid myelin breakdown followed by neurodegeneration. Mice exposed to cuprizone, a model of acute demyelination, show elevations in plasma NfL (Taylor Meadows et al, European Charcot Foundation 25th Annual Meeting; November 30-Dec. 2, 2017; Baveno, Italy). Additionally, TREM2 knockout mice exposed to cuprizone show increased neurotoxicity and further increases in plasma and CSF NfL (Nugent et al, Neuron; 2020, 105(5): 837-854; O’Loughlin et al, Poster #694 ADPD Symposium, Lisbon Portugal, April 2019.). Patients with cALD have quantitatively fewer microglia than healthy individuals in multiple regions of the brain (Bergner et al., Glia. 2019; 67(6):1196-1209). The present invention relates to the unexpected discovery that neurofilament is broken down in the neurons of animals suffering from a disease or disorder caused by and/or associated with a ABCD1 dysfunction, such as x-ALD, resulting in an increase in neurofilament breakdown products in the plasma, serum and cerebral spinal fluid (CSF), and that efficacy of treatment of the disease or disorder with a TREM2 agonist can be determined by measuring central levels of neurofilament and central nervous system (CNS), plasma and serum levels of its degradation products, namely neurofilament light chain and neurofilament heavy chain proteins. In one aspect, the present invention provides methods for selecting x-ALD patients that are likely to experience progression of their neurodegenerative or other disease phenotypes based on neurofilament light chain or neurofilament heavy chain levels, thereby informing the timing of treatment with a TREM2 agonist.

Definitions

[0017] Unless defined otherwise, all technical and scientific terms used herein have the same

meaning as commonly understood by one of ordinary skill in the art. Accordingly, the following terms are intended to have the following meanings.

[0018] “Agonist” or an “activating” agent, such as a compound or antibody, is an agent that induces (e.g., increases) one or more activities or functions of the target (e.g., TREM2) of the agent after the agent binds the target.

[0019] “Antagonist” or a “blocking” agent, such as a compound or antibody, is an agent that reduces or eliminates (e.g., decreases) binding of the target to one or more ligands after the agent binds the target, and/or that reduces or eliminates (e.g., decreases) one or more activities or functions of the target after the agent binds the target. In some embodiments, antagonist agent, or blocking agent substantially or completely inhibits target binding to one or more of its ligand and/or one or more activities or functions of the target.

[0020] “Antibody” is used in the broadest sense and refers to an immunoglobulin or fragment thereof, and encompasses any such polypeptide comprising an antigen-binding fragment or region of an antibody. The recognized immunoglobulin genes include the kappa, lambda, alpha, gamma, delta, epsilon and mu constant region genes, as well as myriad immunoglobulin variable region genes. Light chains are generally classified as either kappa or lambda. Heavy chains are classified as gamma, mu, alpha, delta, or epsilon, which in turn define the immunoglobulin classes, IgG, IgM, IgA, IgD and IgE, respectively. Immunoglobulin classes may also be further classified into subclasses, including IgG subclasses IgG.sub.1, IgG.sub.2, IgG.sub.3, and IgG.sub.4; and IgA subclasses IgA.sub.1 and IgA.sub.2. The term includes, but is not limited to, polyclonal, monoclonal, monospecific, multispecific (e.g., bispecific antibodies), natural, humanized, human, chimeric, synthetic, recombinant, hybrid, mutated, grafted, antibody fragments (e.g., a portion of a full-length antibody, generally the antigen binding or variable region thereof, e.g., Fab, Fab', F(ab')₂, and Fv fragments), and in vitro generated antibodies so long as they exhibit the desired biological activity. The term also includes single chain antibodies, e.g., single chain Fv (scFv or sFv) antibodies, in which a variable heavy and a variable light chain are joined together (directly or through a peptide linker) to form a continuous polypeptide.

[0021] “Isolated” refers to a change from a natural state, that is, changed and/or removed from its original environment. For example, a polynucleotide or polypeptide (e.g., an antibody) is isolated when it is separated from material with which it is naturally associated in the natural environment. Thus, an “isolated antibody” is one which has been separated and/or recovered from a component of its natural environment.

[0022] “Purified antibody” refers to an antibody preparation in which the antibody is at least 80% or greater, at least 85% or greater, at least 90% or greater, at least 95% or greater by weight as compared to other contaminants (e.g., other proteins) in the preparation, such as by determination using SDS-polyacrylamide gel electrophoresis (PAGE) or capillary electrophoresis- (CE) SDS under reducing or non-reducing conditions.

[0023] “Extracellular domain” and “ectodomain” are used interchangeably when used in reference to a membrane bound protein and refer to the portion of the protein that is exposed on the extracellular side of a lipid membrane of a cell.

[0024] “Binds specifically” in the context of any binding agent, e.g., an antibody, refers to a binding agent that binds specifically to an antigen or epitope, such as with a high affinity, and does not significantly bind other unrelated antigens or epitopes.

[0025] “Functional” refers to a form of a molecule which possesses either the native biological activity of the naturally existing molecule of its type, or any specific desired activity, for example as judged by its ability to bind to ligand molecules. Examples of “functional” polypeptides include an antibody binding specifically to an antigen through its antigen-binding region.

[0026] “Antigen” refers to a substance, such as, without limitation, a particular peptide, protein, nucleic acid, or carbohydrate which can bind to a specific antibody.

[0027] “Epitope” or “antigenic determinant” refers to that portion of an antigen capable of being

recognized and specifically bound by a particular antibody. When the antigen is a polypeptide, epitopes can be formed from contiguous amino acids and/or noncontiguous amino acids juxtaposed by tertiary folding of a protein. Linear epitope is an epitope formed from contiguous amino acids on the linear sequence of amino acids. A linear epitope may be retained upon protein denaturing. Conformational or structural epitope is an epitope composed of amino acid residues that are not contiguous and thus comprised of separated parts of the linear sequence of amino acids that are brought into proximity to one another by folding of the molecule, such as through secondary, tertiary, and/or quaternary structures. A conformational or structural epitope may be lost upon protein denaturation. In some embodiments, an epitope can comprise at least 3, and more usually, at least 5 or 8-10 amino acids in a unique spatial conformation. Thus, an epitope as used herein encompasses a defined epitope in which an antibody binds only portions of the defined epitope. There are many methods known in the art for mapping and characterizing the location of epitopes on proteins, including solving the crystal structure of an antibody-antigen complex, competition assays, gene fragment expression assays, mutation assays, and synthetic peptide-based assays, as described, for example, in *Using Antibodies: A Laboratory Manual*, Chapter 11, Harlow and Lane, eds., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, New York (1999).

[0028] “Protein,” “polypeptide,” or “peptide” denotes a polymer of at least two amino acids covalently linked by an amide bond, regardless of length or post-translational modification (e.g., glycosylation, phosphorylation, lipidation, myristoylation, ubiquitination, etc.). Included within this definition are D- and L-amino acids, and mixtures of D- and L-amino acids. Unless specified otherwise, the amino acid sequences of a protein, polypeptide, or peptide are displayed herein in the conventional N-terminal to C-terminal orientation.

[0029] “Polynucleotide” and “nucleic acid” are used interchangeably herein and refer to two or more nucleosides that are covalently linked together. The polynucleotide may be wholly comprised of ribonucleosides (i.e., an RNA), wholly comprised of 2' deoxyribonucleotides (i.e., a DNA) or mixtures of ribo- and 2' deoxyribonucleosides. The nucleosides will typically be linked together by sugar-phosphate linkages (sugar-phosphate backbone), but the polynucleotides may include one or more non-standard linkages. Non-limiting example of such non-standard linkages include phosphoramidates, phosphorothioates, and amides (see, e.g., Eckstein, F., *Oligonucleotides and Analogues: A Practical Approach*, Oxford University Press (1992)).

[0030] “Operably linked” or “operably associated” refers to a situation in which two or more polynucleotide sequences are positioned to permit their ordinary functionality. For example, a promoter is operably linked to a coding sequence if it is capable of controlling the expression of the sequence. Other control sequences, such as enhancers, ribosome binding or entry sites, termination signals, polyadenylation sequences, and signal sequences are also operably linked to permit their proper function in transcription or translation.

[0031] “Amino acid position” and “amino acid residue” are used interchangeably to refer to the position of an amino acid in a polypeptide chain. In some embodiments, the amino acid residue can be represented as “XN”, where X represents the amino acid and the N represents its position in the polypeptide chain. Where two or more variations, e.g., polymorphisms, occur at the same amino acid position, the variations can be represented with a “/” separating the variations. A substitution of one amino acid residue with another amino acid residue at a specified residue position can be represented by XNY, where X represents the original amino acid, N represents the position in the polypeptide chain, and Y represents the replacement or substitute amino acid. When the terms are used to describe a polypeptide or peptide portion in reference to a larger polypeptide or protein, the first number referenced describes the position where the polypeptide or peptide begins (i.e., amino end) and the second referenced number describes where the polypeptide or peptide ends (i.e., carboxy end).

[0032] “Polyclonal” antibody refers to a composition of different antibody molecules which is capable of binding to or reacting with several different specific antigenic determinants on the same

or on different antigens. A polyclonal antibody can also be considered to be a “cocktail of monoclonal antibodies.” The polyclonal antibodies may be of any origin, e.g., chimeric, humanized, or fully human.

[0033] “Monoclonal antibody” refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally occurring mutations that may be present in minor amounts. Each monoclonal antibody is directed against a single determinant on the antigen. In some embodiments, monoclonal antibodies to be used in accordance with the present disclosure can be made by the hybridoma method described by Kohler et al., 1975, *Nature* 256:495-7, or by recombinant DNA methods. The monoclonal antibodies can also be isolated, e.g., from phage antibody libraries.

[0034] “Chimeric antibody” refers to an antibody made up of components from at least two different sources. A chimeric antibody can comprise a portion of an antibody derived from a first species fused to another molecule, e.g., a portion of an antibody derived from a second species. In some embodiments, a chimeric antibody comprises a portion of an antibody derived from a non-human animal, e.g., mouse or rat, fused to a portion of an antibody derived from a human. In some embodiments, a chimeric antibody comprises all or a portion of a variable region of an antibody derived from a non-human animal fused to a constant region of an antibody derived from a human.

[0035] “Humanized antibody” refers to an antibody that comprises a donor antibody binding specificity, e.g., the CDR regions of a donor antibody, such as a mouse monoclonal antibody, grafted onto human framework sequences. A “humanized antibody” typically binds to the same epitope as the donor antibody.

[0036] “Fully human antibody” or “human antibody” refers to an antibody that comprises human immunoglobulin protein sequences only. A fully human antibody may contain murine carbohydrate chains if produced in a non-human cell, e.g., a mouse, in a mouse cell, or in a hybridoma derived from a mouse cell.

[0037] “Full-length antibody,” “intact antibody” or “whole antibody” are used interchangeably to refer to an antibody, such as an anti-TREM2 antibody of the present disclosure, in its substantially intact form, as opposed to an antibody fragment. Specifically whole antibodies include those with heavy and light chains including an Fc region. The constant domains may be native sequence constant domains {e.g., human native sequence constant domains) or amino acid sequence variants thereof. In some cases, the intact antibody may have one or more effector functions.

[0038] “Antibody fragment” or “antigen-binding moiety” refers to a portion of a full length antibody, generally the antigen binding or variable domain thereof. Examples of antibody fragments include Fab, Fab', F(ab')₂, and Fv fragments; diabodies; linear antibodies; single-chain antibodies; and multispecific antibodies formed from antibody fragments that bind two or more different antigens. Several examples of antibody fragments containing increased binding stoichiometries or variable valencies (2, 3 or 4) include triabodies, trivalent antibodies and trimerbodies, tetrabodies, tandAbs®, di-diabodies and (sc(Fv)₂).sub.2 molecules, and all can be used as binding agents to bind with high affinity and avidity to soluble antigens (see, e.g., Cuesta et al., 2010, *Trends Biotech.* 28:355-62).

[0039] “Single-chain Fv” or “sFv” antibody fragment comprises the VH and VL domains of an antibody, where these domains are present in a single polypeptide chain. Generally, the Fv polypeptide further comprises a polypeptide linker between the VH and VL domains which enables the sFv to form the desired structure for antigen binding. For a review of sFv, see Pluckthun in *The Pharmacology of Monoclonal Antibodies*, Vol. 113, pp. 269-315, Rosenberg and Moore, eds., Springer-Verlag, New York (1994).

[0040] “Diabodies” refers to small antibody fragments with two antigen-binding sites, which comprise a heavy chain variable domain (VH) connected to a light chain variable domain (VL) in the same polypeptide chain (VH-VL). By using a linker that is short to allow pairing between the two domains on the same chain, the domains are forced to pair with the complementary domains of

another chain and create two antigen-binding sites.

[0041] “Antigen binding domain” or “antigen binding portion” refers to the region or part of the antigen binding molecule that specifically binds to and complementary to part or all of an antigen. In some embodiments, an antigen binding domain may only bind to a particular part of the antigen (e.g., an epitope), particularly where the antigen is large. An antigen binding domain may comprise one or more antibody variable regions, particularly an antibody light chain variable region (VL) and an antibody heavy chain variable region (VH), and particularly the complementarity determining regions (CDRs) on each of the VH and VL chains.

[0042] “Variable region” and “variable domain” are used interchangeably to refer to the polypeptide region that confers the binding and specificity characteristics of each particular antibody. The variable region in the heavy chain of an antibody is referred to as “VH” while the variable region in the light chain of an antibody is referred to as “VL”. The major variability in sequence is generally localized in three regions of the variable domain, denoted as “hypervariable regions” or “CDRs” in each of the VL region and VH region, and forms the antigen binding site. The more conserved portions of the variable domains are referred to as the framework region FR.

[0043] “Complementarity-determining region” and “CDR” are used interchangeably to refer to non-contiguous antigen binding regions found within the variable region of the heavy and light chain polypeptides of an antibody molecule. In some embodiments, the CDRs are also described as “hypervariable regions” or “HVR”. Generally, naturally occurring antibodies comprise six CDRs, three in the VH (referred to as: CDR H1 or H1; CDR H2 or H2; and CDR H3 or H3) and three in the VL (referred to as: CDR L1 or L1; CDR L2 or L2; and CDR L3 or L3). The CDR domains have been delineated using various approaches, and it is to be understood that CDRs defined by the different approaches are to be encompassed herein. The “Kabat” approach for defining CDRs uses sequence variability and is the most commonly used (Kabat et al., 1991, “Sequences of Proteins of Immunological Interest, 5th Ed.” NIH 1:688-96). “Chothia” uses the location of structural loops (Chothia and Lesk, 1987, J Mol Biol. 196:901-17). CDRs defined by “AbM” are a compromise between the Kabat and Chothia approach, and can be delineated using Oxford Molecular AbM antibody modeling software (see, Martin et al., 1989, Proc. Natl Acad Sci USA. 86:9268; see also, world wide web www.bioinf-uk.org/abs). The “Contact” CDR delineations are based on analysis of known antibody-antigen crystal structures (see, e.g., MacCallum et al., 1996, J. Mol. Biol. 262, 732-45). The CDRs delineated by these methods typically include overlapping or subsets of amino acid residues when compared to each other.

[0044] It is to be understood that the exact residue numbers which encompass a particular CDR will vary depending on the sequence and size of the CDR, and those skilled in the art can routinely determine which residues comprise a particular CDR given the amino acid sequence of the variable region of an antibody.

[0045] Kabat, supra, also defined a numbering system for variable domain sequences that is applicable to any antibody. The Kabat numbering system is generally used when referring to a residue in the variable domain (approximately residues 1-107 of the light chain and residues 1-113 of the heavy chain) (e.g., Kabat et al., Sequences of Immunological Interest. 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)). The “EU or, Kabat numbering system” or “EU index” is generally used when referring to a residue in an immunoglobulin heavy chain constant region (e.g., the EU index reported in Kabat et al., supra). The “EU index as in Kabat” refers to the residue numbering of the human IgG1 EU antibody. References to residue numbers in the variable domain of antibodies means residue numbering by the Kabat numbering system. References to residue numbers in the constant domain of antibodies means residue numbering by the EU or, Kabat numbering system {e.g., see United States Patent Publication No. 2010-280227}. One of skill in the art can assign this system of “Kabat numbering” to any variable domain sequence. Accordingly, unless otherwise specified, references to the number of specific amino acid residues in an antibody or antigen binding fragment are according to the Kabat

numbering system.

[0046] “Framework region” or “FR region” refers to amino acid residues that are part of the variable region but are not part of the CDRs (e.g., using the Kabat, Chothia or AbM definition). The variable region of an antibody generally contains four FR regions: FR1, FR2, FR3 and FR4. Accordingly, the FR regions in a VL region appear in the following sequence: FR.sub.L1-CDR L1-FR.sub.L2-CDR L2-FR.sub.L3-CDR L3-FR.sub.L4, while the FR regions in a VH region appear in the following sequence: FR1.sub.H-CDR H1-FR.sub.H2-CDR H2-FR.sub.H3-CDR H3-FR.sub.H4.

[0047] “Constant region” or “constant domain” refers to a region of an immunoglobulin light chain or heavy chain that is distinct from the variable region. The constant domain of the heavy chain generally comprises at least one of: a CH1 domain, a Hinge (e.g., upper, middle, and/or lower hinge region), a CH2 domain, and a CH3 domain. In some embodiments, the antibody can have additional constant domains CH4 and/or CH5. In some embodiments, an antibody described herein comprises a polypeptide containing a CH1 domain; a polypeptide comprising a CH1 domain, at least a portion of a Hinge domain, and a CH2 domain; a polypeptide comprising a CH1 domain and a CH3 domain; a polypeptide comprising a CH1 domain, at least a portion of a Hinge domain, and a CH3 domain, or a polypeptide comprising a CH1 domain, at least a portion of a Hinge domain, a CH2 domain, and a CH3 domain. In some embodiments, the antibody comprises a polypeptide which includes a CH3 domain. The constant domain of a light chain is referred to a CL, and in some embodiments, can be a kappa or lambda constant region. However, it will be understood by one of ordinary skill in the art that these constant domains (e.g., the heavy chain or light chain) may be modified such that they vary in amino acid sequence from the naturally occurring immunoglobulin molecule.

[0048] “Fc region” or “Fc portion” refers to the C terminal region of an immunoglobulin heavy chain. The Fc region can be a native-sequence Fc region or a non-naturally occurring variant Fc region. Generally, the Fc region of an immunoglobulin comprises constant domains CH2 and CH3. Although the boundaries of the Fc region can vary, in some embodiments, the human IgG heavy chain Fc region can be defined to extend from an amino acid residue at position C226 or from P230 to the carboxy terminus thereof. In some embodiments, the “CH2 domain” of a human IgG Fc region, also denoted as “Cy2”, generally extends from about amino acid residue 231 to about amino acid residue 340. In some embodiments, N-linked carbohydrate chains can be interposed between the two CH2 domains of an intact native IgG molecule. In some embodiments, the CH3 domain” of a human IgG Fc region comprises residues C-terminal to the CH2 domain, e.g., from about amino acid residue 341 to about amino acid residue 447 of the Fc region. A “functional Fc region” possesses an “effector function” of a native sequence Fc region. Exemplary Fc “effector functions” include, among others, C1q binding; complement dependent cytotoxicity (CDC); Fc receptor binding; antibody dependent cell-mediated cytotoxicity (ADCC); phagocytosis; down regulation of cell-surface receptors (e.g., LT receptor); etc. Such effector functions generally require the Fc region to be combined with a binding domain (e.g., an antibody variable domain) and can be assessed using various assays known in the art.

[0049] “Native sequence Fc region” comprises an amino acid sequence identical to the amino acid sequence of an Fc region found in nature. Native sequence human Fc regions include a native sequence human IgG1 Fc region (non-A and A allotypes); native sequence human IgG2 Fc region; native sequence human IgG3 Fc region; and native sequence human IgG4 Fc region as well as naturally occurring variants thereof.

[0050] “Variant Fc region” comprises an amino acid sequence which differs from that of a native sequence Fc region by virtue of at least one amino acid modification, preferably one or more amino acid substitution(s). Preferably, the variant Fc region has at least one amino acid substitution compared to a native sequence Fc region or to the Fc region of a parent polypeptide, e.g. from about one to about ten amino acid substitutions, and preferably from about one to about five amino

acid substitutions in a native sequence Fc region or in the Fc region of the parent polypeptide. The variant Fc region herein will preferably possess at least about 80% homology with a native sequence Fc region and/or with an Fc region of a parent polypeptide, and most preferably at least about 90% homology therewith, more preferably at least about 95% homology therewith.

[0051] “Affinity-matured” antibody, such as an affinity matured anti-TREM2 antibody of the present disclosure, is one with one or more alterations in one or more HVRs thereof that result in an improvement in the affinity of the antibody for antigen, compared to a parent antibody that does not possess those alteration(s). In one embodiment, an affinity-matured antibody has nanomolar or even picomolar affinities for the target antigen. Affinity-matured antibodies are produced by procedures known in the art. For example, Marks et al., *Bio/Technology*, 1992, 10:779-783 describes affinity maturation by VH- and VL-domain shuffling. Random mutagenesis of HVR and/or framework residues is described by, for example: Barbas et al., *Proc Nat. Acad. Sci. USA.*, 1994, 91:3809-3813; Schier et al. *Gene*, 1995, 169: 147-155; Yelton et al., *Immunol.*, 1995, 155: 1994-2004; Jackson et al., *Immunol.*, 1995, 154(7):3310-9; and Hawkins et al, *J. Mol. Biol.*, 1992, 226:889-896.

[0052] “Binding affinity” refers to strength of the sum total of noncovalent interactions between a ligand and its binding partner. In some embodiments, binding affinity is the intrinsic affinity reflecting a one-to-one interaction between the ligand and binding partner. The affinity is generally expressed in terms of equilibrium association ($K_{sub.A}$) or dissociation constant ($K_{sub.D}$), which are in turn reciprocal ratios of dissociation ($k_{sub.off}$) and association rate constants ($k_{sub.on}$).

[0053] “Percent (%) sequence identity” and “percentage sequence homology” are used interchangeably herein to refer to comparisons among polynucleotides or polypeptides, and are determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide or polypeptide sequence in the comparison window may comprise gaps as compared to the reference sequence for optimal alignment of the two sequences. The percentage may be calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. Alternatively, the percentage may be calculated by determining the number of positions at which either the identical nucleic acid base or amino acid residue occurs in both sequences or a nucleic acid base or amino acid residue is aligned with a gap to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. Those of skill in the art appreciate that there are many established algorithms available to align two sequences. Optimal alignment of sequences for comparison can be conducted, e.g., by the local homology algorithm of Smith and Waterman, 1981, *Adv Appl Math.* 2:482, by the homology alignment algorithm of Needleman and Wunsch, 1970, *J Mol Biol.* 48:443, by the search for similarity method of Pearson and Lipman, 1988, *Proc Natl Acad Sci USA.* 85:2444-8, and particularly by computerized implementations of these algorithms (e.g., BLAST, ALIGN, GAP, BESTFIT, FASTA, and TFASTA; see, e.g., Mount, D.W., *Bioinformatics: Sequence and Genome Analysis*, 2nd Ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, New York (2013))

[0054] Examples of algorithms that are suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0, FASTDB, or ALIGN algorithms, which are publically available (e.g., NCBI: National Center for Biotechnology Information). Those skilled in the art can determine appropriate parameters for aligning sequences. For example, the BLASTN program (for nucleotide sequences) can use as defaults a wordlength (W) of 11, an expectation (E) of 10, M=5, N=-4, and a comparison of both strands. Comparison of amino acid sequences using BLASTP can use as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62

scoring matrix (see Henikoff and Henikoff, 1989, Proc Natl Acad Sci USA. 89:10915-9).

[0055] “Amino acid substitution” refers to the replacement of one amino acid in a polypeptide with another amino acid. A “conservative amino acid substitution” refers to the interchangeability of residues having similar side chains, and thus typically involves substitution of the amino acid in the polypeptide with amino acids within the same or similar defined class of amino acids. By way of example and not limitation, an amino acid with an aliphatic side chain may be substituted with another aliphatic amino acid, e.g., alanine, valine, leucine, isoleucine, and methionine; an amino acid with hydroxyl side chain is substituted with another amino acid with a hydroxyl side chain, e.g., serine and threonine; an amino acid having aromatic side chains is substituted with another amino acid having an aromatic side chain, e.g., phenylalanine, tyrosine, tryptophan, and histidine; an amino acid with a basic side chain is substituted with another amino acid with a basic side chain, e.g., lysine, arginine, and histidine; an amino acid with an acidic side chain is substituted with another amino acid with an acidic side chain, e.g., aspartic acid or glutamic acid; and a hydrophobic or hydrophilic amino acid is replaced with another hydrophobic or hydrophilic amino acid, respectively.

[0056] “Amino acid insertion” refers to the incorporation of at least one amino acid into a predetermined amino acid sequence. An insertion can be the insertion of one or two amino acid residues; however, larger insertions of about three to about five, or up to about ten or more amino acid residues are contemplated herein.

[0057] “Amino acid deletion” refers to the removal of one or more amino acid residues from a predetermined amino acid sequence. A deletion can be the removal of one or two amino acid residues; however, larger deletions of about three to about five, or up to about ten or more amino acid residues are contemplated herein.

[0058] “Subject” refers to a mammal, including, but not limited to humans, non-human primates, and non-primates, such as goats, horses, and cows. In some embodiments, the terms “subject” and “patient” are used interchangeably herein in reference to a human subject.

[0059] “Therapeutically effective dose” or “therapeutically effective amount” or “effective dose” refers to that quantity of a compound, including a biologic compound, or pharmaceutical composition that is sufficient to result in a desired activity upon administration to a mammal in need thereof. As used herein, with respect to the pharmaceutical compositions comprising an antibody, the term “therapeutically effective amount/dose” refers to the amount/dose of the antibody or pharmaceutical composition thereof that is sufficient to produce an effective response upon administration to a mammal.

[0060] “Pharmaceutically acceptable” refers to compounds or compositions which are generally safe, non-toxic and neither biologically nor otherwise undesirable, and includes a compound or composition that is acceptable for human pharmaceutical and veterinary use. The compound or composition may be approved or approvable by a regulatory agency or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in animals, including humans.

[0061] “Pharmaceutically acceptable excipient, carrier or adjuvant” refers to an excipient, carrier or adjuvant that can be administered to a subject, together with at least one therapeutic agent (e.g., an antibody of the present disclosure), and which does not destroy the pharmacological activity thereof and is generally safe, nontoxic and neither biologically nor otherwise undesirable when administered in doses sufficient to deliver a therapeutic amount of the agent.

[0062] The term “treatment” is used interchangeably herein with the term “therapeutic method” and refers to both 1) therapeutic treatments or measures that cure, slow down, lessen symptoms of, and/or halt progression of a diagnosed pathologic conditions, disease or disorder, and 2) and prophylactic/preventative measures. Those in need of treatment may include individuals already having a particular medical disease or disorder as well as those who may ultimately acquire the disorder (i.e., those at risk or needing preventive measures).

[0063] The term “subject” or “patient” as used herein refers to any individual to which the subject

methods are performed. Generally, the subject is human, although as will be appreciated by those in the art, the subject may be any animal.

[0064] In some embodiments, compounds of the present invention are able to cross the blood-brain barrier (BBB). The term “blood-brain barrier” or “BBB”, as used herein, refers to the BBB proper as well as to the blood-spinal barrier. The blood-brain barrier, which consists of the endothelium of the brain vessels, the basal membrane and neuroglial cells, acts to limit penetration of substances into the brain. In some embodiments, the brain/plasma ratio of total drug is at least approximately 0.01 after administration (e.g. oral or intravenous administration) to a patient. In some embodiments, the brain/plasma ratio of total drug is at least approximately 0.03. In some embodiments, the brain/plasma ratio of total drug is at least approximately 0.06. In some embodiments, the brain/plasma ratio of total drug is at least approximately 0.1. In some embodiments, the brain/plasma ratio of total drug is at least approximately 0.2.

[0065] The term “homologue,” especially “TREM homologue” as used herein refers to any member of a series of peptides or nucleic acid molecules having a common biological activity, including antigenicity/immunogenicity and inflammation regulatory activity, and/or structural domain and having sufficient amino acid or nucleotide sequence identity as defined herein. TREM homologues can be from either the same or different species of animals.

[0066] The term “variant” as used herein refers either to a naturally occurring allelic variation of a given peptide or a recombinantly prepared variation of a given peptide or protein in which one or more amino acid residues have been modified by amino acid substitution, addition, or deletion.

[0067] The term “derivative” as used herein refers to a variation of given peptide or protein that are otherwise modified, i.e., by covalent attachment of any type of molecule, preferably having bioactivity, to the peptide or protein, including non-naturally occurring amino acids.

Description of Treatment Methods of the Present Invention

[0068] In one aspect, the present invention provides a method of treating a disease or disorder caused by and/or associated with an ABCD1 dysfunction in a human patient, the method comprising administering to the patient a compound that increases activity of TREM2. In some embodiments, the compound that increases activity of TREM2 is an agonist of TREM2. In some embodiments, the compound that increases activity of TREM2 is a compound that prevents the degradation of TREM2.

[0069] In one aspect, the present invention provides a method of treating a disease or disorder caused by and/or associated with an ABCD1 dysfunction in a human patient, the method comprising administering to the patient an effective amount of an agonist of TREM2. In some embodiments, administration of the agonist of TREM2 activates DAP12 signaling pathways in the patient, resulting in an increase in microglia proliferation, microglia survival and microglia phagocytosis, which in turn results in a slowing of disease progression. In some embodiments, the agonist of TREM2 is an antibody or a small molecule.

[0070] In some embodiments, the agonist of TREM2 activates TREM2/DAP12 signaling in myeloid cells, including monocytes, dendritic cells, microglial cells and macrophages. In some embodiments, an agonist of TREM2 activates, induces, promotes, stimulates, or otherwise increases one or more TREM2 activities. TREM2 activities that are activated or increased by the agonist, include but are not limited to: TREM2 binding to DAP12; DAP12 binding to TREM2; TREM2 phosphorylation, DAP12 phosphorylation; PI3K activation; increased levels of soluble TREM2 (sTREM2); increased levels of soluble CSF1R (sCSF1R); increased expression of one or more anti-inflammatory mediators (e.g., cytokines) selected from the group consisting of IL-12p70, IL-4, IL-6, and IL-1 β ; reduced expression of one or more pro-inflammatory mediators selected from the group consisting of IFN- α 4, IFN-b, IL-6, IL-12 p70, IL-12 p40, IL-1 β , TNF, TNF- α , IL-1 β , IL-8, CRP, TGF-beta members of the chemokine protein families, IL-20 family members, IL-33, LIF, IFN-gamma, OSM, CNTF, TGF-beta, GM-CSF, IL-11, IL-12, IL-17, IL-18, and CRP; increased expression of one or more chemokines selected from the group consisting of CCL2,

CCL4, CXCL10, CCL3 and CST7; reduced expression of TNF- α , IL-6, or both; extracellular signal-regulated kinase (ERK) phosphorylation; increased expression of C-C chemokine receptor 7 (CCR7); induction of microglial cell chemotaxis toward CCL19 and CCL21 expressing cells; an increase, normalization, or both of the ability of bone marrow-derived dendritic cells to induce antigen-specific T-cell proliferation; induction of osteoclast production, increased rate of osteoclastogenesis, or both; increasing the survival and/or function of one or more of dendritic cells, macrophages, microglial cells, M1 macrophages and/or microglial cells, activated M1 macrophages and/or microglial cells, M2 macrophages and/or microglial cells, monocytes, osteoclasts, Langerhans cells of skin, and Kupffer cells; induction of one or more types of clearance selected from the group consisting of apoptotic neuron clearance, nerve tissue debris clearance, non-nerve tissue debris clearance, bacteria or other foreign body clearance, disease-causing protein clearance, disease-causing peptide clearance, and disease-causing nucleic acid clearance; induction of phagocytosis of one or more of apoptotic neurons, nerve tissue debris, non-nerve tissue debris, bacteria, other foreign bodies, disease-causing proteins, disease-causing peptides, or disease-causing nucleic acids; normalization of disrupted TREM2/DAP12-dependent gene expression; recruitment of Syk, ZAP70, or both to the TREM2/DAP12 complex; Syk phosphorylation; increased expression of CD83 and/or CD86 on dendritic cells, macrophages, monocytes, and/or microglia; reduced secretion of one or more inflammatory cytokines selected from the group consisting of TNF- α , IL-1 β , IL-6, MCP-1, IFN- α 4, IFN-b, IL-1 β , IL-8, CRP, TGF-beta members of the chemokine protein families, IL-20 family members, IL-33, LIF, IFN-gamma, OSM, CNTF, TGF-beta, GM-CSF, IL-11, IL-12, IL-17, IL-18, and CRP; reduced expression of one or more inflammatory receptors; increasing phagocytosis by macrophages, dendritic cells, monocytes, and/or microglia under conditions of reduced levels of MCSF; decreasing phagocytosis by macrophages, dendritic cells, monocytes, and/or microglia in the presence of normal levels of MCSF; increasing activity of one or more TREM2-dependent genes; increased levels of one or more of CSF1, CSF2 and IL-34; or any combination thereof. In another aspect, the invention provides a TREM2 agonist for the manufacture of a medicament for the treatment of a disease or disorder caused by and/or associated with an ABCD1 dysfunction.

[0071] In another aspect, the invention provides a TREM2 agonist for use in treating a disease or disorder caused by and/or associated with an ABCD1 dysfunction in a human patient.

I. Diseases and Disorders

[0072] The methods of the present invention can be used to treat any disease or disorder related to a dysfunction in ABCD1. In some embodiments, the patient is selected for treatment based on a diagnosis that includes the presence of a mutation in an ABCD1 gene affecting the function of ABCD1. In some embodiments, the mutation in the ABCD1 gene is a mutation that causes a decrease in ABCD1 activity or a cessation of ABCD1 activity. In some embodiments, the disease or disorder is caused by a heterozygous ABCD1 mutation. In some embodiments, the disease or disorder is caused by a homozygous ABCD1 mutation. In some embodiments, [0073] the disease or disorder is caused by a splice mutation in the ABCD1 gene. In some embodiments, the disease or disorder is caused by a missense mutation in the ABCD1 gene.

[0074] In some embodiments, the disease or disorder is a disease or disorder resulting from a change (e.g. increase, decrease or cessation) in the activity of ABCD1. In some embodiments, the disease or disorder is a disease or disorder resulting from a decrease or cessation in the activity of ABCD1. ABCD1 related activities that are changed in the disease or disorder include, but are not limited to peroxisomal import of fatty acids and/or fatty acyl-CoAs and production of adrenoleukodystrophy protein (ALDP).

[0075] In some embodiments, the disease or disorder is caused by a loss-of-function mutation in ABCD1. In some embodiments, the loss-of-function mutation results in a complete cessation of ABCD1 function. In some embodiments, the loss-of-function mutation results in a partial loss of ABCD1 function, or a decrease in ABCD1 activity. In some embodiments, the disease or disorder

is caused by a homozygous mutation in ABCD1.

[0076] In some embodiments, the disease or disorder is a neurodegenerative disorder. In some embodiments, the disease or disorder is a neurodegenerative disorder caused by and/or associated with an ABCD1 dysfunction.

[0077] In some embodiments, the disease or disorder is an immunological disorder. In some embodiments, the disease or disorder is an immunological disorder caused by and/or associated with an ABCD1 dysfunction.

[0078] In some embodiments, the disease or disorder is selected from X-linked adrenoleukodystrophy (x-ALD), Globoid cell leukodystrophy (also known as Krabbe disease), Metachromatic leukodystrophy (MLD), Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL), Alexander disease, fragile X-associated tremor ataxia syndrome (FXTAS), adult-onset autosomal dominant leukodystrophy (ADLD), and X-linked Charcot-Marie-Tooth disease (CMTX).

[0079] In some embodiments, the disease or disorder is selected from X-linked adrenoleukodystrophy (x-ALD), Globoid cell leukodystrophy (also known as Krabbe disease), Metachromatic leukodystrophy (MLD), Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL), Vanishing white matter disease (VWM), Alexander disease, fragile X-associated tremor ataxia syndrome (FXTAS), adult-onset autosomal dominant leukodystrophy (ADLD), and X-linked Charcot-Marie-Tooth disease (CMTX), wherein any of the aforementioned diseases or disorders are present in a patient exhibiting ABCD1 dysfunction, or having a mutation in a gene affecting the function of ABCD1.

[0080] In some embodiments, the disease or disorder is X-linked adrenoleukodystrophy (x-ALD). In some embodiments, the x-ALD is a cerebral form of X-linked ALD (cALD).

[0081] In some embodiments, the disease or disorder is Addison disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is Addison disease, wherein the patient has a loss-of-function mutation in ABCD1.

[0082] In some embodiments, the disease or disorder is a white matter disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is a white matter disease, wherein the patient has a loss-of-function mutation in ABCD1.

[0083] In some embodiments, the disease or disorder is vanishing white matter disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is vanishing white matter disease, wherein the patient has a loss-of-function mutation in ABCD1.

[0084] In some embodiments, the disease or disorder is selected from Nasu-Hakola disease, Alzheimer's disease, frontotemporal dementia, multiple sclerosis, Guillain-Barre syndrome, amyotrophic lateral sclerosis (ALS), or Parkinson's disease, wherein any of the aforementioned diseases or disorders are present in a patient exhibiting ABCD1 dysfunction, or having a mutation in a gene affecting the function of ABCD1.

[0085] In some embodiments, the disease or disorder is Alzheimer's disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the patient has been diagnosed with Alzheimer's disease based on neuropathology, and also has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is Alzheimer's disease, wherein the patient has a loss-of-function mutation in ABCD1.

[0086] In some embodiments, the disease or disorder is Nasu-Hakola disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the patient has been diagnosed with Nasu-Hakola disease based on neuropathology, and also has been found to have a mutation in one or more ABCD1 genes affecting

ABCD1 function. In some embodiments, the disease or disorder is Nasu-Hakola disease, wherein the patient has a loss-of-function mutation in ABCD1.

[0087] In some embodiments, the disease or disorder is Parkinson's disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the patient has been diagnosed with Parkinson's disease based on neuropathology, and also has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is Parkinson's disease, wherein the patient has a loss-of-function mutation in ABCD1.

[0088] In some embodiments, the disease or disorder is multiple sclerosis wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the patient has been diagnosed with multiple sclerosis based on neuropathology, and also has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is multiple sclerosis, wherein the patient has a loss-of-function mutation in ABCD1.

[0089] In some embodiments, the disease or disorder is ALS wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the patient has been diagnosed with ALS based on neuropathology, and also has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is ALS, wherein the patient has a loss-of-function mutation in ABCD1.

[0090] In some embodiments, the disease or disorder is Guillain-Barre syndrome wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the patient has been diagnosed with Guillain-Barre syndrome based on neuropathology, and also has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function. In some embodiments, the disease or disorder is Guillain-Barre syndrome, wherein the patient has a loss-of-function mutation in ABCD1.

[0091] In some embodiments, the patient also possesses a mutation in one or more of NOTCH3, HTRA1, TREX1, ARSA, EIF2B1, EIF2B2, EIF2B3, EIF2B4, and EIF2B5.

[0092] In some embodiments, the disease or disorder presents one or more symptoms selected from abnormal motor control, parkinsonism, slow movement (bradykinesia), involuntary trembling (tremor), muscle stiffness (rigidity), cognitive decline, dementia, inability to speak, inability to walk, memory loss, personality changes, seizures, depression, loss of executive function, loss of impulse control, loss of attention span, adrenal insufficiency, vision impairment, hearing impairment, sexual dysfunction, impaired adrenocortical function, attention-deficit, hyperactivity, and incontinence.

[0093] In one aspect, the present invention provides a method of treating x-ALD in a human patient, the method comprising administering to the patient a compound that increases activity of TREM2. In some embodiments, the compound that increases activity of TREM2 is an agonist of TREM2. In some embodiments, the compound that increases activity of TREM2 is a compound that prevents the degradation of TREM2.

[0094] In one aspect, the present invention provides a method of treating x-ALD in a human patient, the method comprising administering to the patient an effective amount of an agonist of TREM2. In some embodiments, administration of the agonist of TREM2 activates DAP12 signaling pathways in the patient, resulting in an increase in microglia proliferation, microglia survival and microglia phagocytosis, which in turn results in a slowing of disease progression in x-ALD. In some embodiments, the agonist of TREM2 is an antibody or a small molecule.

[0095] In another aspect, the invention provides a TREM2 agonist for the manufacture of a medicament for the treatment of a disease or disorder related to an ABCD1 dysfunction. In another aspect, the invention provides a TREM2 agonist for the manufacture of a medicament for the treatment of x-ALD.

[0096] In another aspect, the invention provides a TREM2 agonist for use in treating a disease or

disorder related to an ABCD1 dysfunction in a human patient. In another aspect, the invention provides a TREM2 agonist for use in treating x-ALD in a human patient.

Huntington's Disease

[0097] In one aspect, the present invention provides a method of treating Huntington's disease in a human patient, the method comprising administering to the patient an effective amount of an agonist of TREM2. In some embodiments, administration of the agonist of TREM2 activates DAP12 signaling pathways in the patient, resulting in an increase in microglia proliferation, microglia survival and microglia phagocytosis, which in turn results in a slowing of disease progression in Huntington's disease. In some embodiments, the agonist of TREM2 is an antibody or a small molecule. In some embodiments, the agonist of TREM2 is an antibody or a small molecule disclosed elsewhere herein. In some embodiments, the agonist of TREM2 is an antibody disclosed elsewhere herein. In some embodiments, the agonist of TREM2 is a small molecule disclosed elsewhere herein. In another aspect, the invention provides a TREM2 agonist for the manufacture of a medicament for the treatment of Huntington's disease. In another aspect, the invention provides a TREM2 agonist for use in treating Huntington's disease in a human patient.

II. Antibodies

[0098] In one aspect, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of an antigen binding protein or an antibody, or an antigen-binding fragment thereof, which increases the activity of TREM2. In some embodiments, the antibody is an agonist of TREM2. In some embodiments, the antibody is an agonist of TREM2 that specifically binds to and activates human TREM2.

[0099] The TREM2 agonist antibodies specifically bind to human TREM2 (SEQ ID NO: 1) or an extra cellular domain (ECD) of human TREM2 (e.g. ECD set forth in SEQ ID NO: 2), for example with an equilibrium dissociation constant ($K_{sub.D}$) less than 50 nM, less than 25 nM, less than 10 nM, or less than 5 nM. In some embodiments, the TREM2 agonist antibodies do not cross-react with other TREM proteins, such as human TREM1. In some embodiments, the TREM2 agonist antibodies do not bind to human TREM1 (SEQ ID NO: 4).

[0100] In some embodiments, the TREM2 antibody specifically bind to human TREM2 human TREM2 residues 19-174. In some embodiments, the TREM2 antibody specifically bind to IgV region of human TREM2, for example human TREM2 residues 19-140.

[0101] In certain embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 29-112 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 29-112 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 29-41 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 29-41 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 47-69 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 47-69 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 76-86 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 76-86 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 91-100 of human TREM2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 91-100 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 99-115 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 99-115 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 104-112 of human TREM 2 (SEQ ID NO: 1),

or within amino acid residues on a TREM2 protein corresponding to amino acid residues 104-112 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 114-118 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 114-118 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 130-171 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 130-171 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 139-153 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 139-153 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 139-146 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 139-146 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 130-144 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 130-144 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 158-171 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 158-171 of SEQ ID NO: 1.

[0102] In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 43-50 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 43-50 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 49-57 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 49-57 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 139-146 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 139-146 of SEQ ID NO: 1. In some embodiments, anti-TREM2 antibodies of the present disclosure bind to one or more amino acids within amino acid residues 140-153 of human TREM 2 (SEQ ID NO: 1), or within amino acid residues on a TREM2 protein corresponding to amino acid residues 140-153 of SEQ ID NO: 1. In some embodiments, the TREM2 antibody specifically binds to the stalk region of human TREM2, for example amino acid residues 145-174 of human TREM2.

[0103] In some embodiments, the antibody, or an antigen-binding fragment thereof, specifically binds TREM2 and prevents the degradation or cleavage of TREM2.

[0104] In some embodiments, the antibody is a polyclonal antibody. In some embodiments, the antibody is a monoclonal antibody. In some embodiments, the antibody is a chimeric antibody. In some embodiments, the antibody is a humanized antibody. In some embodiments, the antibody is a human antibody, particularly a fully human antibody. In some embodiments, the antibody is a bispecific or other multivalent antibody. In some embodiments, the antibody is a single chain antibody.

[0105] In some embodiments, a TREM2 activating antibody comprise a light chain variable region comprising complementarity determining regions CDRL1, CDRL2, and CDRL3 and a heavy chain variable region comprising complementarity determining regions CDRH1, CDRH2, and CDRH3 described herein.

[0106] In certain embodiments, the TREM2 agonist antigen binding proteins of the invention comprise at least one light chain variable region comprising a CDRL1, CDRL2, and CDRL3, and at least one heavy chain variable region comprising a CDRH1, CDRH2, and CDRH3 from an anti-TREM2 agonist antibody described herein.

[0107] In some embodiments, a TREM2 activating antibody comprise a light chain variable region and a heavy chain variable region described herein. The light chain and heavy chain variable regions or CDRs may be from any of the anti-TREM2 antibodies or a variant thereof described herein.

Sequence Information

A. PCT Patent Application Publication No. WO2018/195506A1

[0108] In some embodiments, the TREM2 agonist is an antigen binding protein or an antibody, or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2018/195506A1, which is incorporated by reference herein, in its entirety.

[0109] In some embodiments, the TREM2 agonist antigen binding protein comprises a CDRL1 or a variant thereof having one, two, three or four amino acid substitutions; a CDRL2, or a variant thereof having one, two, three or four amino acid substitutions; a CDRL3, or a variant thereof having one, two, three or four amino acid substitutions; a CDRH1, or a variant thereof having one, two, three or four amino acid substitutions; a CDRH2, or a variant thereof having one, two, three or four amino acid substitutions; and a CDRH3, or a variant thereof having one, two, three or four amino acid substitutions, where the amino acid sequences of the CDRL1, CDRL2, CDRL3, CDRH1, CDRH2, and CDRH3 are provided in TABLES A1 and A2 below, along with exemplary light chain and variable regions.

TABLE-US-00001	TABLE A1	Exemplary Anti-Human TREM2 Antibody Light Chain Variable Region Amino Acid Sequence	Amino Acid Sequences Ab ID. VL Group VL Amino Acid Sequence
CDRL1	CDRL2	CDRL3	12G10 LV-01
QAVPTQPSSLSASPGVLA	SLTCTLRSGINVG	TLRSGI YKSDSD MIWYSS	
TYRIYWYQQKPGSPQYLL	RYKSDSDKQQ NVGTY	KQQGS AVV	
GSGVPSRFS	GSKDASANAGILLISGLQSEDE	RIY (SEQ ID (SEQ ID	
ADYYCMIWYSSAVVFGGGTKLTVL	(SEQ ID NO: 19) NO: 31)	(SEQ ID NO: 46) NO: 5)	26A10 LV-02
SYELTQPPSVSVSPGQTASITCSGDKLGD	KY SGDKL QDSKRP		
QAWDS VCWYQQKPGQSPVLVIYQDSKRPSGIPERF	GDKYVC S NTVV		
SGSNSGNTATLTISGTQAMDEADYYCQAW	(SEQ ID (SEQ ID (SEQ ID		
DSNTVVFGGGTKLTVL	(SEQ ID NO: 47) NO: 6) NO: 20) NO: 32)		26C10 LV-03
SFELTQPPSVSVSPGQTASITCSGDKLGD	KY SGDKL QDTKRP QAWDSS		
VCWYQQKPGQSPMLVIYQDTKRPSGIPERF	GDKYVC S TVV		
SGSNSGNTATLTISGTQAMDEADYYCQAW	(SEQ ID (SEQ ID (SEQ ID		
DSSTVVFGGGTKLTVL	(SEQ ID NO: 48) NO: 6) NO: 21) NO: 33)		26F2 LV-04
SYELTQPPSVSVSPGQTASITCSGDKLGD	KY SGDKL QDSKRP QAWDSS		
VCWYQQKPGQSPVLVIFQDSKRPSGIPERFS	GDKYVC S TVV		
GSNSGNTATLTISGTQAMDEADYYCQAWD	(SEQ ID (SEQ ID (SEQ ID		
SSTVVFGGGTKLTVL	(SEQ ID NO: 49) NO: 6) NO: 20) NO: 33)		33B12 LV-05
SYELTQPPSVSVSPGQTASITCSGDKLGD	KY SGDKL QDSKRP QAWDSS		
VCWYQQKPGQSPVLVIYQDSKRPSGIPERF	GDKYVC S TVV		
SGSNSGNTATLTISGTQAMDEADYYCQAW	(SEQ ID (SEQ ID (SEQ ID		
DSSTVVFGGGTKLTVL	(SEQ ID NO: 50) NO: 6) NO: 20) NO: 33)		24C12 LV-06
GIVMTQSPDSLAVSLGERATINCKSSRSVLY	KSSRSV WASTRE QQYYIT		
SSNNKNYLAWYQQKPGQPPKVLIIYWASTR	LYSSN S PIT		
ESGVPDRFSGSGSGTDFTLTISLQAEDVAV	KNYLA (SEQ ID (SEQ ID		
YNCQQYYITPITFGQGTRLEIK	(SEQ ID NO: 22) NO: 34) (SEQ ID NO: 51) NO: 7)		24G6 LV-07
DIVMTQSPDSLAVSLGERATINCKSSQSVLY	KSSQSV WASTRE QQYYST		
SSNNKHFLAWYQQKPGQPPKLLIYWASTR	LYSSN S PLT		
ESGVPDRFSGSGSGTDFTLTISLQAEDVAF	KFIFLA (SEQ ID (SEQ ID		
YYCQQYYSTPLTFGGGTKVEIK	(SEQ ID NO: 22) NO: 35) (SEQ ID NO: 52) NO: 8)		24A10 LV-08
DIVMTQSPDSLAVSLGERATITCKSSHNVL	KSSHN WASTRE		

HQYYST YSSNNKYNLA TWPPT
RESGVPDRFSGSGSGTDFTLTISLQAEDVA NKNYLA (SEQ ID NO: 22) NO: 36) (SEQ ID NO: 53) NO: 9) 10E3 LV-09 EIVMTQSPATLSVSPGERATLSCRASQSVSS RASQS GASTRA LQDNN
NLAWFQQKPGQAPRLLIYGASTRATGIPAR VSSNLA TWPPT
FSVSGSGTEFTLTISLQSEDFAFYYCLQDN (SEQ ID NO: 54) NO: 10) NO: 23) NO: 37) 13E7 LV-10
EIVMTQSPATLSVSPGERATLSCRASQSVSS RASQS GASTRA LQDNN 14C12
NLAWFQQKPGQAPRLLIYGASTRATGIPAR VSSNLA TWPPT
FSVSGSGTEFTLTISLQSEDFAVYYCLQDN (SEQ ID NO: 55) NO: 10) NO: 23) NO: 37) 25F12 LV-11
EKVMTQSPATLSVSPGERATLSCRASQSVN RASQS GASTRA QQYNN
NNLAWYQQKPGQAPRLLIYGASTRATGIPA VNNNL TWPPT
RFSGSGSGTEFTLTISLQSEDFAVYYCQQY A (SEQ ID NO: 56) (SEQ ID NO: 23) NO: 38) NO: 11) 32E3 LV-12 EFVLTQSPGTLSPGERATLSCRASQIISN RASQIIS SASSRA QQFDSS
YLAWYQQKPGQAPRLLIYSASSRATGIPDR SNYLA T PIT
FSGSGSGTDFTLTISRLEPEDFAVYYCQQFD (SEQ ID NO: 57) NO: 12) NO: 24) NO: 39) 24F4 LV-13
EIVLTQSPGTLSPGERATLSCRASQSVSSS RASQS GASSRA QQYDTS
YLAWYQQKPGQAPRLLIYGASSRATGIPDR VSSSYLA T PFT
FSGSGSGTDFTLTISRLEPEDFALYYCQQYD (SEQ ID NO: 58) NO: 13) NO: 25) NO: 40) 16B8 LV 14
DIQMTQSPSSVSASVGDRVTVTCRASQDIN RASQDI AASSLQ QQSNSF
SWLAWYQQKPGKAPKLLIYAASSLQTGVPL NSWLA T PIT
SRFSGSGSGTDFTLTISLQPEDFATYSCQQS (SEQ ID NO: 59) NO: 14) NO: 26) NO: 41) 4C5 LV-15
DIQMTQSPSSVSASVGDRVTTTCRASQGISN RASQGI AASSLQ QQADSF
WLAWYQQKPGKAPKLLIYAASSLQGVPL SNWLA V PRN
RFSGSGSGTDFTLTISLQPEDFATYYCQQA (SEQ ID NO: 60) NO: 15) NO: 27) NO: 42) 6E7 LV-16
DSFPRNFGQGTKLEIK (SEQ ID NO: 61) NO: 16) NO: 28) NO: 43) 5E3 LV-17
DIQMTQSPSSLSASVGDRVTTTCRASQGISN RASQGI AASSLQ QQYSTY
YLAWFQQKPGKAPKSLIYAASSLQSGVPSK SNYLA S PFT
FSGSGSGTDFTLTISLQPEDFATYYCQQYS (SEQ ID NO: 62) NO: 17) NO: 29) NO: 44) 4G10 LV-18
DIQMTQSPSSLSASVGDRVTTTCRASQGIRN RASQGI AASSLPS LQHNYSY
DLGWYQQKPGNAPKRLIYAASSLPSGVPSR RNDLG (SEQ ID NO: 30) (SEQ ID NO: 63) NO: 18) NO: 45)
SYPWTFGQGTKVEIT (SEQ ID NO: 63) NO: 18) NO: 45)
TABLE-US-00002 TABLE A2 Exemplary Anti-Human TREM2 Antibody Heavy Chain Variable Region Amino Acid Sequences Ab ID. VH Group VH Amino Acid Sequence CDRH1 CDRH2 CDRH3 12G10 HV-01
EVQLLES GGGGLVQP GGSRLRLSCAASGFTFS SYAMS AIGGGG FYIAVA 24C12
SYAMSWVRQAPGKGLEWVSAIGGGGVST (SEQ ID NO: 77) DSVKG (SEQ ID NO: 95) TVSS (SEQ ID NO: 95)
EDTAVYYCAKFYIAVAGSHFDYWGQGLTLV (SEQ ID NO: 95) TVSS (SEQ ID NO: 95)

110) NO: 87) 26A10 HV-02 EVQLVESGGALVQRGGSRLSCAASGFTFS SFGMS YISSSSF
EGGLTM SFGMSWVRQAPGKGLEWVSYISSSSF TIYY (SEQ ID TIYYAD VRGVSS
ADSVKGRFTISRDN AKNSFY LQMNSLRDED NO: 78) SVKG YGLDV
TAVYYCAREGGLTMVRGVSSYGLDVWGQ (SEQ ID (SEQ ID GTTVTVSS (SEQ
ID NO: 111) NO: 88) NO: 96) 26C10 HV-03
EVQLVESGGALVQPGGSLRLSCAASGFTFS SFGMS YISSSSF EGGITM
SFGMSWVRQAPGKGLEWVSYISSSSF TIYY (SEQ ID TIYYAD VRGVSS
ADSVKGRFTISRDN AKNSFY LQMNSLRDED NO: 78) SVKG YGMDV
TAVYFCVREGGITMVRGVSSYGM DVWGQ (SEQ ID (SEQ ID GTTVTVSS (SEQ ID
NO: 112) NO: 88) NO: 97) 26F2 HV-04 EVQLVESGGALVQPGGSLRLSCAASGFTFS
SFGMS YISSSSF EGGITM SFGMSWVRQAPGKGLEWISYISSSSF TIYYA (SEQ ID TIYYAD
VRGVSS DSVKGRFTISRDN AKNSFY LQMNSLRDED NO: 78) SVKG YGMDV
AVYFCAREGGITMVRGVSSYGM DVWGQ (SEQ ID (SEQ ID TTVTVSS (SEQ ID
NO: 113) NO: 88) NO: 97) 33B12 HV-05 EVQLVESGGALVQPGGSLRLSCAASGFTFS
SFGMS YISKSSF EGGLTM SFGMSWVRQAPGKGLEWVSYISKSSF TIYY (SEQ ID
TIYYAD VRGVSS ADSVKGRFTISRDN AKNSFY LQMNSLRDED NO: 78) SVKG YGLDV
TAVYYCAREGGLTMVRGVSSYGLDVWGQ (SEQ ID (SEQ ID GTTVTVSS (SEQ
ID NO: 114) NO: 89) NO: 96) 24G6 HV-06
EVQLLESGGGLVQPGGSLRLSCAASGFTFS SYAMS AISGSGG AYTPMA
SYAMSWVRQAPGKGLEWVSAISGSGGSTY (SEQ ID STYYAD FFDY
YADSVKGRFTISRDN SKNTLY LQMNSLRAE NO: 77) SVKG (SEQ ID
DTAVYYCAKAYTPMAFFDYWGQGT LVT (SEQ ID NO: 98) SS (SEQ ID NO:
115) NO: 90) 24A10 HV-07 EVQVLES GGGLVQPGGSLRLSCAASGFTFS NYAMS AISGSGG
GGWELF NYAMSWVRQAPGKGLEWVSAISGSGGSTY (SEQ ID STYYAD Y
YADSVKGRFTISRDN SKNTLY LQMNSLRAE NO: 79) SVKG (SEQ ID
DTAVYYCAKGGWELFYWGQGT LVT VSS (SEQ ID NO: 99) (SEQ ID NO: 116)
NO: 90) 10E3 HV-08 EVQLVQSGAEVKKPGESLMISCKGSGYSFT NYWIG ITYPGDS
RRQGIW NYWIGWVRQMPGKGLEWMGITYPGDS DTR (SEQ ID DTRYSP GDALDI
YSPSFQGGQVTISADKSISTAYLQWSSLKASD NO: 80) SFQG (SEQ ID
TAMYFCARRRQGIWGDALDIWGQGT LVT (SEQ ID NO: 100) SS (SEQ ID NO:
117) NO: 91) 13E7 HV-09 EVQLVQSGAEVKKPGESLMISCKGSGYSFT SWIG ITYPGDS
RRQGIW 14C12 SYWIGWVRQMPGKGLEWMGITYPGDS DTR (SEQ ID DTRYSP
GDALDF YSPSFQGGQVTISADKSISTAYLQWSSLKASD NO: 81) SFQG (SEQ ID
TAMYFCARRRQGIWGDALDFWGQGT LVT (SEQ ID NO: 101) VSS (SEQ ID NO:
118) NO: 91) 25F12 HV-10 QVQLQQWGAGLLKPSETLSLTC AVYGGSF SYYWS EINHSG
EGYYDI SSYYWSWIRQPPGKGLEWIGEINHSGNTNY (SEQ ID NTNYP LTGYHD
NPSLKSRVTISVDTSKNQFSLKLSSVTAADT NO: 82) SLKS AFDI
AVYYCAREGYDILTGYHDAFDIWDQGT (SEQ ID (SEQ ID VTVFS (SEQ ID
NO: 119) NO: 92) NO: 102) 32E3 HV-11 EVQLVQSGAEVKKPGESLKISCKGSGYSFT
SWIG IYPGDS HDIIPAA SYWIGWVRQMPGKGLEWMGITYPGDS DTR (SEQ ID DTRYSP
PGA FDI YSPSFQGGQVTISADKSISTAYLQWSTLKASD NO: 81) SFQG (SEQ ID
TAIYYCAR HDIIPAA PGA FDIWGQGT MVTV (SEQ ID NO: 103) SS (SEQ ID NO:
120) NO: 91) 24F4 HV-12 EVQLVQSGAEVKKPGESLKISCKGSGYTFT SWIG IYPGDS
QAI AVT SYWIGWVRQMPGKGLEWMGITYPGDS DTR (SEQ ID DTRYSP GLGGFD
YSPSFQGGQVTISVDKSSSTAYLQWSSLKAS NO: 81) SFQG P
DTAIYYCTRQAI AVTGLGGFDPWGQGT LVT (SEQ ID (SEQ ID TVSS (SEQ ID NO:
121) NO: 91) NO: 104) 16B8 HV-13 QVQLVQSGAEVKKPGASVKVSC KASGYTF NYGIS
WISAYN RGYSYG TNYGISWVRQAPGQGLEWMGWISAYNGN (SEQ ID GNTNYA SFDY
TNYAQKLQGRVTMTTDTSTSTVYME LRSL NO: 83) QKLQG (SEQ ID
RSDDTAVYYCARRGYSGSFDYWGQGT LVT (SEQ ID NO: 105) VTVSS (SEQ ID

NO: 122) NO: 93) 4C5 HV-14 EVQLVQSGAEVKKPGESLKISCKGSGHSFT NYWIA
 IIYPGDS QRTFYY NYWIAWVRQMPGKGLEWMGITYPGDSDTR (SEQ ID DTRYSP
 DSSGYF YSPSFQGGQVTISADKSISTAYLQWSSLKASD NO: 84) SFQG DY
 TAVYFCARQRTFYYDSSGYFDYWGQGTLV (SEQ ID (SEQ ID TVSS (SEQ ID
 NO: 123) NO: 91) NO: 106) 6E7 HV-15 EVQLVQSGAEVKKPGESLKISCKGSGYSFT
 SYWIA ITYPGDS QRTFYY SYWIAWVRQMPGKGLEWMGITYPGDSDTR (SEQ ID
 DTRYSP DSSDYF YSPSFQGGQVTISADKSISTAYLQWSSLKASD NO: 85) SFQG DY
 TAMYFCARQRTFYYDSSDYFDYWGQGT (SEQ ID (SEQ ID VTVSS (SEQ ID
 NO: 124) NO: 91) NO: 107) 5E3 HV-16 QVQLVQSGAEVKKPGASVKVSKASGYTF
 GYYIH WINPYS DGGYLA TGYIHWVRQAPGLGLEWMGWINPYSGG (SEQ ID GGTTSA
 LYGTDV TTSQKFKQGRVTMTRDTSISSAYMELSRLR NO: 86) QKFQG (SEQ ID
 SDDTAVYYCARDGGYLALYGTDVWGQGT (SEQ ID NO: 108) TVTVSS (SEQ ID
 NO: 125) NO: 94) 4G10 HV-17 EVQLVQSGAEVKKPGESLKISCKGSGYSFP SYWIA
 ITYPGDS QGIEVT SYWIAWVRQMPGKGLEWMGITYPGDSDTR (SEQ ID DTRYSP
 GTGGLD YSPSFQGGQVTISADKSISTAFLLKQWSSLKASD NO: 85) SFQG V
 TAMYFCARQGIEVTGTGGLDVWGQGT (SEQ ID (SEQ ID VSS (SEQ ID NO:
 126) NO: 91) NO: 109)

[0110] As noted above, a TREM2 agonist antigen binding protein may comprise one or more of the CDRs presented in TABLE A1 (light chain CDRs; i.e. CDRLs) and TABLE A2 (heavy chain CDRs, i.e. CDRHs).

[0111] In some embodiments, the TREM2 agonist antigen binding protein comprises one or more light chain CDRs selected from (i) a CDRL1 selected from SEQ ID NOS:5 to 18, (ii) a CDRL2 selected from SEQ ID NOS:19 to 30, and (iii) a CDRL3 selected from SEQ ID NOS:31 to 45, and (iv) a CDRL of (i), (ii) and (iii) that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids. In these and other embodiments, the TREM2 agonist antigen binding proteins comprise one or more heavy chain CDRs selected from (i) a CDRH1 selected from SEQ ID NOS:77 to 86, (ii) a CDRH2 selected from SEQ ID NOS:87 to 94, and (iii) a CDRH3 selected from SEQ ID NOS:95 to 109, and (iv) a CDRH of (i), (ii) and (iii) that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids.

[0112] In some embodiments, the TREM2 agonist antigen binding protein may comprise 1, 2, 3, 4, 5, or 6 variant forms of the CDRs listed in TABLES A1 and A2, each having at least 80%, 85%, 90% or 95% sequence identity to a CDR sequence listed in TABLES A1 and A2. In some embodiments, the TREM2 agonist antigen binding protein includes 1, 2, 3, 4, 5, or 6 of the CDRs listed in TABLES A1 and A2, each differing by no more than 1, 2, 3, 4 or 5 amino acids from the CDRs listed in these tables.

[0113] In some embodiments, the TREM2 agonist antigen binding protein comprises a CDRL1 comprising a sequence selected from SEQ ID NOS:5-18 or a variant thereof having one, two, three or four amino acid substitutions; a CDRL2 comprising a sequence selected from SEQ ID NOS: 19-30 or a variant thereof having one, two, three or four amino acid substitutions; a CDRL3 comprising a sequence selected from SEQ ID NOS:31-45 or a variant thereof having one, two, three or four amino acid substitutions; a CDRH1 comprising a sequence selected from SEQ ID NOS:77-86 or a variant thereof having one, two, three or four amino acid substitutions; a CDRH2 comprising a sequence selected from SEQ ID NOS:87-94 or a variant thereof having one, two, three or four amino acid substitutions; and a CDRH3 comprising a sequence selected from SEQ ID NOS:95-109 or a variant thereof having one, two, three or four amino acid substitutions.

[0114] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a CDRL1 comprising a sequence selected from SEQ ID NOS:5-18; a CDRL2 comprising

a sequence selected from SEQ ID NOS:19-30; a CDR3 comprising a sequence selected from SEQ ID NOS:31-45; a CDRH1 comprising a sequence selected from SEQ ID NOS:77-86; a CDRH2 comprising a sequence selected from SEQ ID NOS:87-94; and a CDRH3 comprising a sequence selected from SEQ ID NOS:95-109.

[0115] In some embodiments, the TREM2 agonist antigen binding protein comprise a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3, wherein: [0116] (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 5, 19, and 31, respectively; [0117] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 20, and 32, respectively; [0118] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 21, and 33, respectively; [0119] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 20, and 33, respectively; [0120] (e) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:7, 22, and 34, respectively; [0121] (f) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:8, 22, and 35, respectively; [0122] (g) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:9, 22, and 36, respectively; [0123] (h) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 10, 23, and 37, respectively; [0124] (i) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 11, 23, and 38, respectively; [0125] (j) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 12, 24, and 39, respectively; [0126] (k) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 13, 25, and 40, respectively; [0127] (l) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 14, 26, and 41, respectively; [0128] (m) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 15, 27, and 42, respectively; [0129] (n) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively; [0130] (o) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 17, 29, and 44, respectively, or [0131] (p) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 18, 30, and 45, respectively.

[0132] In some embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: [0133] (a) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 87, and 95, respectively; [0134] (b) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 88, and 96, respectively; [0135] (c) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 88, and 97, respectively; [0136] (d) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 89, and 96, respectively; [0137] (e) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 90, and 98, respectively; [0138] (f) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:79, 90, and 99, respectively; [0139] (g) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:80, 91, and 100, respectively; [0140] (h) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 101, respectively; [0141] (i) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:82, 92, and 102, respectively; [0142] (j) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 103, respectively; [0143] (k) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 104, respectively; [0144] (l) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:83, 93, and 105, respectively; [0145] (m) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:84, 91, and 106, respectively; [0146] (n) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0147] (o) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:86, 94, and 108, respectively; or [0148] (p) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 109, respectively.

[0149] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: [0150] (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:5, 19, and 31, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 87, and 95, respectively; [0151] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 20, and 32, respectively, and CDRH1,

CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 88, and 96, respectively; [0152] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 21, and 33, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 88, and 97, respectively; [0153] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 20, and 33, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 88, and 97, respectively; [0154] (e) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:6, 20, and 33, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:78, 89, and 96, respectively; [0155] (f) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:7, 22, and 34, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 87, and 95, respectively; [0156] (g) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:8, 22, and 35, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 90, and 98, respectively; [0157] (h) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:9, 22, and 36, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:79, 90, and 99, respectively; [0158] (i) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 10, 23, and 37, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:80, 91, and 100, respectively; [0159] (j) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 10, 23, and 37, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 101, respectively; [0160] (k) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:11, 23, and 38, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:82, 92, and 102, respectively; [0161] (l) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 12, 24, and 39, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 103, respectively; [0162] (m) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 13, 25, and 40, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 104, respectively; [0163] (n) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 14, 26, and 41, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:83, 93, and 105, respectively; [0164] (o) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 15, 27, and 42, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:84, 91, and 106, respectively; [0165] (p) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0166] (q) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 17, 29, and 44, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:86, 94, and 108, respectively; or [0167] (r) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:18, 30, and 45, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 109, respectively.

[0168] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:10, 23, and 37, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:80, 91, and 100, respectively. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:10, 23, and 37, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 91, and 101, respectively. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 15, 27, and 42, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:84, 91, and 106, respectively. In some

embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:17, 29, and 44, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:86, 94, and 108, respectively. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:8, 22, and 35, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 90, and 98, respectively.

[0169] In some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region comprising a sequence selected from SEQ ID NOS:46-63 and a heavy chain variable region comprising a sequence selected from SEQ ID NOS: 110-126. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:46 and a heavy chain variable region comprising the sequence of SEQ ID NO:110. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:47 and a heavy chain variable region comprising the sequence of SEQ ID NO:111. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:48 and a heavy chain variable region comprising the sequence of SEQ ID NO: 112. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:49 and a heavy chain variable region comprising the sequence of SEQ ID NO: 113. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:50 and a heavy chain variable region comprising the sequence of SEQ ID NO:114. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:51 and a heavy chain variable region comprising the sequence of SEQ ID NO: 110. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:53 and a heavy chain variable region comprising the sequence of SEQ ID NO: 116. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:54 and a heavy chain variable region comprising the sequence of SEQ ID NO: 117. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:55 and a heavy chain variable region comprising the sequence of SEQ ID NO:118. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:56 and a heavy chain variable region comprising the sequence of SEQ ID NO: 119. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:57 and a heavy chain variable region comprising the sequence of SEQ ID NO: 120. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:58 and a heavy chain variable region comprising the sequence of SEQ ID NO:121. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:59 and a heavy chain variable region comprising the sequence of SEQ ID NO:122. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:60 and a heavy chain variable region comprising the

sequence of SEQ ID NO: 123. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:61 and a heavy chain variable region comprising the sequence of SEQ ID NO: 124. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:62 and a heavy chain variable region comprising the sequence of SEQ ID NO: 125. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:63 and a heavy chain variable region comprising the sequence of SEQ ID NO:126. In yet another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:52 and a heavy chain variable region comprising the sequence of SEQ ID NO:115.

[0170] In some embodiments, the TREM2 agonist antigen binding protein may comprise a light chain variable region selected from LV-01, LV-02, LV-03, LV-04, LV-05, LV-06, LV-07, LV-08, LV-09, LV-10, LV-11, LV-12, LV-13, LV-14, LV-15, LV-16, LV-17, and LV-18, as shown in TABLE A1, and/or a heavy chain variable region selected from HV-01, HV-02, HV-03, HV-04, HV-05, HV-06, HV-07, HV-08, HV-09, HV-10, HV-11, HV-12, HV-13, HV-14, HV-15, HV-16, and HV-17, as shown in TABLE A2, and functional fragments, derivatives, muteins and variants of these light chain and heavy chain variable regions.

[0171] In some embodiments, each of the light chain variable regions listed in TABLE A1 may be combined with any of the heavy chain variable regions listed in TABLE A2 to form an anti-TREM2 binding domain of the antigen binding proteins of the invention. Examples of such combinations include, but are not limited to: LV-01 (SEQ ID NO:46) and HV-01 (SEQ ID NO: 110); LV-02 (SEQ ID NO:47) and HV-02 (SEQ ID NO: 111); LV-03 (SEQ ID NO:48) and HV-03 (SEQ ID NO: 112); LV-04 (SEQ ID NO:49) and HV-04 (SEQ ID NO: 113); LV-05 (SEQ ID NO:50) and HV-05 (SEQ ID NO: 114); LV-06 (SEQ ID NO:51) and HV-0 (SEQ ID NO:110); LV-07 (SEQ ID NO:52) and HV-06 (SEQ ID NO: 115); LV-08 (SEQ ID NO:53) and HV-07 (SEQ ID NO: 116); LV-09 (SEQ ID NO:54) and HV-08 (SEQ ID NO: 117); LV-10 (SEQ ID NO:55) and HV-09 (SEQ ID NO: 118); LV-11 (SEQ ID NO:56) and HV-10 (SEQ ID NO: 119); LV-12 (SEQ ID NO:57) and HV-11 (SEQ ID NO: 120); LV-13 (SEQ ID NO:58) and HV-12 (SEQ ID NO:121); LV-14 (SEQ ID NO:59) and HV-13 (SEQ ID NO: 122); LV-15 (SEQ ID NO:60) and HV-14 (SEQ ID NO: 123); LV-16 (SEQ ID NO:61) and HV-15 (SEQ ID NO: 124); LV-17 (SEQ ID NO:62) and HV-16 (SEQ ID NO: 125); and LV-18 (SEQ ID NO:63) and HV-17 (SEQ ID NO: 126).

[0172] In certain embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising the sequence of LV-09 (SEQ ID NO:54) and a heavy chain variable region comprising the sequence of HV-08 (SEQ ID NO: 117). In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising the sequence of LV-10 (SEQ ID NO:55) and a heavy chain variable region comprising the sequence of HV-09 (SEQ ID NO: 118). In other embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising the sequence of LV-15 (SEQ ID NO:60) and a heavy chain variable region comprising the sequence of HV-14 (SEQ ID NO: 123). In still other embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising the sequence of LV-16 (SEQ ID NO:61) and a heavy chain variable region comprising the sequence of HV-15 (SEQ ID NO: 124). In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising the sequence of LV-17 (SEQ ID NO:62) and a heavy chain variable region comprising the sequence of HV-16 (SEQ ID NO: 125). In certain embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising the sequence of LV-07 (SEQ ID NO:52) and a heavy chain variable region comprising the sequence of HV-06 (SEQ ID NO: 115).

[0173] In some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region comprising a sequence of contiguous amino acids that differs from the sequence of

a light chain variable region in TABLE A1, i.e. a VL selected from LV-01, LV-02, LV-03, LV-04, LV-05, LV-06, LV-07, LV-08, LV-09, LV-10, LV-11, LV-12, LV-13, LV-14, LV-15, LV-16, LV-17, or LV-18, at only 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or 15 amino acid residues, wherein each such sequence difference is independently either a deletion, insertion or substitution of one amino acid, with the deletions, insertions and/or substitutions resulting in no more than 15 amino acid changes relative to the foregoing variable domain sequences. The light chain variable region in some TREM2 agonist antigen binding proteins comprises a sequence of amino acids that has at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97% or at least 99% sequence identity to the amino acid sequences of SEQ ID NOS:46-63 (i.e. the light chain variable regions in TABLE A1). In one embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence that is at least 90% identical to a sequence selected from SEQ ID NOS:46-63. In another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS:46-63. In yet another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence selected from SEQ ID NOS:46-63. In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence of SEQ ID NO:54. In other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence of SEQ ID NO:55. In yet other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence of SEQ ID NO:60. In still other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence of SEQ ID NO:61. In certain embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence of SEQ ID NO:62. In other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a sequence of SEQ ID NO:52.

[0174] In these and other embodiments, the TREM2 agonist antigen binding proteins comprise a heavy chain variable region comprising a sequence of contiguous amino acids that differs from the sequence of a heavy chain variable region in TABLE A2, i.e., a VH selected from HV-01, HV-02, HV-03, HV-04, HV-05, HV-06, HV-07, HV-08, HV-09, HV-10, HV-11, HV-12, HV-13, HV-14, HV-15, HV-16, or HV-17, at only 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or 15 amino acid residues, wherein each such sequence difference is independently either a deletion, insertion or substitution of one amino acid, with the deletions, insertions and/or substitutions resulting in no more than 15 amino acid changes relative to the foregoing variable domain sequences. The heavy chain variable region in some TREM2 agonist antigen binding proteins comprises a sequence of amino acids that has at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97% or at least 99% sequence identity to the amino acid sequences of SEQ ID NOS:110-126 (i.e. the heavy chain variable regions in TABLE A2). In one embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence that is at least 90% identical to a sequence selected from SEQ ID NOS: 110-126. In another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 110-126. In yet another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence selected from SEQ ID NOS: 110-126. In some embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence of SEQ ID NO: 117. In other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence of SEQ ID NO: 118. In yet other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence of SEQ ID NO:123. In still other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence of SEQ ID NO: 124. In certain embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable

region comprising a sequence of SEQ ID NO:125. In other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a sequence of SEQ ID NO: 115.

[0175] In some embodiments, variants of the anti-TREM2 antibodies can be generated by substituting one or more amino acids in the light chain or heavy chain variable regions to address chemical liabilities (e.g., aspartate isomerization, asparagine deamidation, tryptophan and methionine oxidation) or correct covariance violations (see e.g., WO 2012/125495, which is hereby incorporated by reference in its entirety). Such variants can have improved biophysical, expression, and/or stability properties as compared with the parental antibody. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region and/or heavy chain variable region having one or more of the amino acid substitutions set forth in any of TABLES A3-A4 below.

[0176] In some embodiments, additional variants of the anti-TREM2 antibodies described herein can be generated by affinity modulating any of the anti-TREM2 antibodies described herein. An “affinity-modulated antibody” is an antibody that comprises one or more amino acid substitutions in its light chain variable region sequence and/or heavy chain variable region sequence that increases or decreases the affinity of the antibody for the target antigen as compared to the parental antibody that does not contain the amino acid substitutions. Antibody affinity modulation methods are known to those of skill in the art and can include CDR walking mutagenesis (Yang et al., J. Mol. Biol., 254, 392-403, 1995), chain shuffling (Marks et al., Bio/Technology, 10, 779-783, 1992), use of mutation strains of *E. coli* (Low et al., J. Mol. Biol., 250, 350-368, 1996), DNA shuffling (Patten et al., Curr. Opin. Biotechnol., 1997, 8:724-733), phage display (Thompson et al., J. Mol. Biol., 1996, 256:7-88), PCR techniques (Cramer et al., Nature, 1998, 391:288-291), and other mutagenesis strategies (Barbas et al., Proc Nat. Acad. Sci. USA 91:3809-3813, 1994; Schier et al., Gene 169:147-155, 1995; Yelton et al., J. Immunol. 155:1994-2004, 1995; Jackson et al., J. Immunol. 154(7):3310-9, 1995; and Hawkins et al., J. Mol. Biol., 1992, 226:889-896). Methods of affinity modulation are discussed in Hoogenboom, Trends in Biotechnology, 1995, 15:62-70, and Vaughan et al., Nature Biotechnology, 1998, 16:535-539. One specific method for generating affinity-modulated variants of the anti-TREM2 antibodies described herein is the use of a yeast-display Fab mutagenesis library.

[0177] In some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region that is a variant of a light chain variable region of any of the anti-TREM2 antibodies described herein. Thus, in some embodiments, the light chain variable region of the TREM2 agonist antigen binding proteins comprises a sequence that is at least 90% identical, at least 91% identical, at least 92% identical, at least 93% identical, at least 94% identical, or at least 95% identical to a sequence selected from SEQ ID NOS:46-63. In some embodiments, the TREM2 agonist antigen binding proteins can comprise a light chain variable region from any of the engineered anti-TREM2 antibody variants set forth in TABLES A3-A4 below.

[0178] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:54 with a mutation at one or more amino acid positions 64, 79, 80, 85, 94, and/or 100. In some such embodiments, the mutation is V64G, V64A, Q79E, Q79D, S80P, S80A, F85V, F85L, F85A, F85D, F85I, F85L, F85M, F85T, W94F, W94Y, W94S, W94T, W94A, W94H, W94I, W94Q, P100R, P100Q, P100G, or combinations thereof. In another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:55 with a mutation at one or more amino acid positions 64, 79, 80, 94, and/or 100. Such mutations can include V64G, V64A, Q79E, Q79D, S80P, S80A, W94F, W94Y, W94S, W94T, W94A, W94H, W94I, W94Q, P100R, P100Q, P100G, or combinations thereof. In certain embodiments, the mutation is V64G, V64A, Q79E, S80P, S80A, W94Y, W94S, P100R, P100Q, or combinations thereof. In another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the

sequence of SEQ ID NO:60 with a mutation at one or more amino acid positions 60, 92, and/or 93. The mutation in such embodiments can be selected from L60S, L60P, L60D, L60A, D92E, D92Q, D92T, D92N, S93A, S93N, S93Q, S93V, or combinations thereof. In yet another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:61 with a mutation at one or more amino acid positions 56, 57, 92, and/or 93. In such embodiments, the mutation can be N56S, N56T, N56Q, N56E, G57A, G57V, D92E, D92Q, D92T, D92N, S93A, S93N, S93Q, S93V, or combinations thereof. In certain embodiments, the mutation is N56S, N56Q, G57A, D92E, D92Q, S93A, or combinations thereof. In still another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:62 with a mutation at amino acid position 36, 46, 61 and/or 100. Such mutations can include F36Y, S46L, S46R, S46V, S46F, K61R, P100Q, P100G, P100R or combinations thereof. In particular embodiments, the mutation is F36Y, K61R, P100Q, or combinations thereof. In another embodiment, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:52 with a mutation at amino acid position 91, which can be selected from F91V, F91I, F91T, F91L, or F91D. In one embodiment, the mutation is F91V.

[0179] In some embodiments, the TREM2 agonist antigen binding proteins comprise a heavy chain variable region that is a variant of a heavy chain variable region from any of the anti-TREM2 antibodies described herein. Thus, in some embodiments, the heavy chain variable region of the TREM2 agonist antigen binding proteins comprises a sequence that is at least 90% identical, at least 91% identical, at least 92% identical, at least 93% identical, at least 94% identical, or at least 95% identical to a sequence selected from SEQ ID NOS:110-126. For instance, the TREM2 agonist antigen binding proteins can comprise a heavy chain variable region from any of the engineered anti-TREM2 antibody variants set forth in TABLES A3-A4 below. In one embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 117 with a mutation at one or more amino acid positions 19, 55, 56, 57, 58, and/or 104. In some such embodiments, the mutation is M19K, M19R, M19T, M19E, M19N, M19Q, D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, W104F, W104Y, W104T, W104S, W104A, W104H, W104I, W104Q, or combinations thereof. In another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO:118 with a mutation at one or more amino acid positions 19, 55, 56, 57, 58, and/or 104. Such mutations can include M19K, M19R, M19T, M19E, M19N, M19Q, D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, W104F, W104Y, W104T, W104S, W104A, W104H, W104I, W104Q, or combinations thereof. In certain embodiments, the mutation is M19K, D55E, S56A, D57E, T58A, W104Y, W104T, or combinations thereof. In another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO:123 with a mutation at one or more amino acid positions 27, 55, 56, 57, 58, 105, and/or 106. In some embodiments, the mutation is selected from H27Y, H27D, H27F, H27N, D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, D105E, D105Q, D105T, D105N, D105G, S106A, S106Q, S106V, S106T, or combinations thereof. In yet another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO:124 with a mutation at one or more amino acid positions 55, 56, 57, 58, 105, and/or 106. The mutation in such embodiments can be selected from D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, D105E, D105Q, D105T, D105N, D105G, S106A, S106Q, S106V, S106T, or combinations thereof. In certain embodiments, the mutation is D55E, D55Q, S56A, D57E, T58A, D105E, D105N, S106A, or combinations thereof. In still another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 125 with a mutation at one or more amino acid positions 43, 76, 85, 99, 100, and/or 116. Such mutations can include L43Q, L43K, L43H,

I76T, R85S, R85G, R85N, R85D, D99E, D99Q, D99S, D99T, G100A, G100Y, G100V, T116L, T116M, T116P, T116R, or combinations thereof. In certain embodiments, the mutation is L43Q, R85S, D99E, G100A, G100Y, T116L, or combinations thereof. In another embodiment, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 115 with a mutation at amino acid position 62 and/or 63. In such embodiments, the mutation can be selected from D62E, D62Q, D62T, D62N, S63A, S63Q, S63V, or combinations thereof. In some embodiments, the mutation is D62E, D62Q, S63A, or combinations thereof. In some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region and/or heavy chain variable region from any of the anti-TREM2 variant antibodies set forth in TABLES A3, A4, A4, A5, A6, A7, and A8. Accordingly, in some embodiments, the light chain variable region of the TREM2 agonist antigen binding proteins comprises a sequence that is at least 90% identical, at least 91% identical, at least 92% identical, at least 93% identical, at least 94% identical, or at least 95% identical to a sequence selected from SEQ ID NOS:61, 153-162, and 295-300. In these and other embodiments, the heavy chain variable region of the TREM2 agonist antigen binding proteins comprises a sequence that is at least 90% identical, at least 91% identical, at least 92% identical, at least 93% identical, at least 94% identical, or at least 95% identical to a sequence selected from SEQ ID NOS: 124, 180-190, and 307-312.

[0180] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:54 with a mutation at one or more amino acid positions 64, 79, 80, 85, 94, and/or 100. Such mutations can include V64G, V64A, Q79E, Q79D, S80P, S80A, F85V, F85L, F85A, F85D, F85I, F85L, F85M, F85T, W94F, W94Y, W94S, W94T, W94A, W94H, W94I, W94Q, P100R, P100Q, P100G, or combinations thereof. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 117 with a mutation at one or more amino acid positions 19, 55, 56, 57, 58, and/or 104. In certain embodiments, the mutation is selected from M19K, M19R, M19T, M19E, M19N, M19Q, D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, W104F, W104Y, W104T, W104S, W104A, W104H, W104I, W104Q, or combinations thereof.

[0181] In other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:55 with a mutation at one or more amino acid positions 64, 79, 80, 94, and/or 100. In some embodiments, the mutation is selected from V64G, V64A, Q79E, Q79D, S80P, S80A, W94F, W94Y, W94S, W94T, W94A, W94H, W94I, W94Q, P100R, P100Q, P100G, or combinations thereof. In certain embodiments, the mutation is selected from V64G, V64A, Q79E, S80P, S80A, W94Y, W94S, P100R, P100Q, or combinations thereof. For instance, in some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:55 with one or more mutations selected from V64G, Q79E, S80P, W94Y, and P100Q. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 118 with a mutation at one or more amino acid positions 19, 55, 56, 57, 58, and/or 104. Such mutations can include M19K, M19R, M19T, M19E, M19N, M19Q, D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, W104F, W104Y, W104T, W104S, W104A, W104H, W104I, W104Q, or combinations thereof. In certain embodiments, the mutation is selected from M19K, D55E, S56A, D57E, T58A, W104Y, W104T, or combinations thereof.

[0182] In certain other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:60 with a mutation at one or more amino acid positions 60, 92, and/or 93. The mutation can be selected from L60S, L60P, L60D, L60A, D92E, D92Q, D92T, D92N, S93A, S93N, S93Q, S93V, or combinations thereof. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain

variable region comprising the sequence of SEQ ID NO: 123 with a mutation at one or more amino acid positions 27, 55, 56, 57, 58, 105, and/or 106. In some embodiments, the mutation is selected from H27Y, H27D, H27F, H27N, D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, D105E, D105Q, D105T, D105N, D105G, S106A, S106Q, S106V, S106T, or combinations thereof.

[0183] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:61 with a mutation at one or more amino acid positions 56, 57, 92, and/or 93. In certain embodiments, the mutation is selected from N56S, N56T, N56Q, N56E, G57A, G57V, D92E, D92Q, D92T, D92N, S93A, S93N, S93Q, S93V, or combinations thereof. In some embodiments, the mutation is selected from N56S, N56Q, G57A, D92E, D92Q, S93A, or combinations thereof. In particular embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:61 with one or more mutations selected from N56S, D92E, and S93A. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 124 with a mutation at one or more amino acid positions 55, 56, 57, 58, 105, and/or 106. The mutation can be selected from D55E, D55Q, D55N, D55T, S56A, S56Q, S56V, D57S, D57E, D57Q, T58A, T58V, D105E, D105Q, D105T, D105N, D105G, S106A, S106Q, S106V, S106T, or combinations thereof. In certain embodiments, the mutation is D55E, D55Q, S56A, D57E, T58A, D105E, D105N, S106A, or combinations thereof. In some embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 124 with one or more mutations selected from D55E, S56A, D57E, D105E, and S106A.

[0184] In other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:62 with a mutation at amino acid position 36, 46, 61 and/or 100. In particular embodiments, the mutation is selected from F36Y, S46L, S46R, S46V, S46F, K61R, P100Q, P100G, P100R or combinations thereof. In some embodiments, the mutation is F36Y, K61R, P100Q, or combinations thereof. In some embodiments, the mutation is S46L, P100Q, or combinations thereof. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 125 with a mutation at one or more amino acid positions 43, 76, 85, 99, 100, and/or 116. The mutation can be selected from L43Q, L43K, L43H, I76T, R85S, R85G, R85N, R85D, D99E, D99Q, D99S, D99T, G100A, G100Y, G100V, T116L, T116M, T116P, T116R, or combinations thereof. In certain embodiments, the mutation is L43Q, I76T, R85S, D99E, G100A, G100Y, T116L, or combinations thereof.

[0185] In still other embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO:52 with a mutation at amino acid position 91. The mutation can be selected from F91V, F91I, F91T, F91L, or F91D. In one embodiment, the mutation is F91V. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO:115 with a mutation at amino acid position 62 and/or 63. In particular embodiments, the mutation is selected from D62E, D62Q, D62T, D62N, S63A, S63Q, S63V, or combinations thereof. In some embodiments, the mutation is selected from D62E, D62Q, S63A, or combinations thereof.

TABLE-US-00003 TABLE A3 Engineered Variants of 10E3 Antibody Position in 10E3 VL
Sequence or Parent Amino Acid VH sequence Region Hot Spot Acid Substitutions Light chain variable sequence (SEQ ID NO: 54) 64 FR3 Covariance violator V G, A 79 FR3 Covariance violator Q E, D 80 FR3 Covariance violator S P, A 85 FR3 Covariance violator F V, L, A, D, I, L, M, T 94 CDR3 Potential Tryptophan W F, Y, S, T, A, H, I, Q Oxidation Site 100 FR4 Covariance violator P R, Q, G Heavy chain variable sequence (SEQ ID NO: 117) 19 FR1 Covariance violator M K, R, T, E, N, Q 55-56 CDR2 Potential Isomerization Site DS ES, QS, DA, NS, DQ, TS, DV 57-58 CDR2 Potential Isomerization Site DT ST, ET, DA, DV, QT 104 CDR3 Potential Tryptophan W

F, Y, T, S, A, H, I, Q Oxidation Site

TABLE-US-00004 TABLE A4 Engineered Variants of 13E7 Antibody Position in 13E7 VL Sequence or Parent Amino Acid VH sequence Region Hot Spot Acid Substitutions Light chain variable sequence (SEQ ID NO: 55) 64 FR3 Covariance violator V G, A 79 FR3 Covariance violator Q E, D 80 FR3 Covariance violator S P, A 94 CDR3 Potential Tryptophan W F, Y, S, T, A, H, I, Q Oxidation Site 100 FR4 Covariance violator P R, Q, G Heavy chain variable sequence (SEQ ID NO: 118) 19 FR1 Covariance violator M K, R, T, E, N, Q 55-56 CDR2 Potential Isomerization Site DS ES, QS, DA, DQ, NS, TS, DV 57-58 CDR2 Potential Isomerization Site DT ST, ET, DA, DV, QT 104 CDR3 Potential Tryptophan W F, Y, T, S, A, H, I, Q Oxidation Site

TABLE-US-00005 TABLE A5 Engineered Variants of 4C5 Antibody Position in 4C5 VL Sequence or Parent Amino Acid VH sequence Region Hot Spot Acid Substitutions Light chain variable sequence (SEQ ID NO: 60) 60 FR3 Covariance violator L S, P, D, A 92-93 CDR3 Potential Isomerization Site DS ES, QS, DA, DN, DQ, TS, NS, DV Heavy chain variable sequence (SEQ ID NO: 123) 27 FR1 Covariance violator H Y, D, F, N 55-56 CDR2 Potential Isomerization Site DS ES, QS, DA, DQ, DV, TS, NS 57-58 CDR2 Potential Isomerization Site DT ST, ET, DA, DV, QT 105-106 CDR3 Potential Isomerization Site DS ES, QS, DA, DQ, DV, TS, NS, GT

TABLE-US-00006 TABLE A6 Engineered Variants of 6E7 Antibody Position in 6E7 VL Sequence or Parent Amino Acid VH sequence Region Hot Spot Amino Acid Substitutions Light chain variable sequence (SEQ ID NO: 61) 56-57 CDR2/FR3 Potential Deamidation Site NG SG, TG, QG, NA, EG, boundary NV 92-93 CDR3 Potential Isomerization Site DS ES, QS, DA, DN, DQ, DV, TS, NS Heavy chain variable sequence (SEQ ID NO: 124) 55-56 CDR2 Potential Isomerization Site DS ES, QS, DA, DQ, DV, TS, NS 57-58 CDR2 Potential Isomerization Site DT ST, ET, DA, DV, QT 105-106 CDR3 Potential Isomerization Site DS ES, QS, DA, DQ, DV, TS, NS, GT

TABLE-US-00007 TABLE A7 Engineered Variants of 5E3 Antibody Position in 5E3 VL Sequence or Parent Amino Acid VH sequence Region Hot Spot Amino Acid Substitutions Light chain variable sequence (SEQ ID NO: 62) 36 FR2 Consensus violator F Y 46 FR2 Covariance violator S L, R, V, F 61 FR3 Consensus violator K R 100 FR4 Covariance violator P Q, G, R Heavy chain variable sequence (SEQ ID NO: 125) 43 FR2 Covariance violator L Q, K, H 76 FR3 Covariance violator I T 85 FR3 Covariance violator R S, G, N, D 99-100 CDR3 Potential Isomerization Site DG EG, DA, DY, DV, QG, SG, TG 116 FR4 Covariance violator T L, M, P, R

TABLE-US-00008 TABLE A8 Engineered Variants of 24G6 Antibody Position in 24G6 VL Sequence or Parent Amino Acid VH sequence Region Hot Spot Amino Acid Substitutions Light chain variable sequence (SEQ ID NO: 52) 91 FR3 Covariance violator F V, I, T, L, D Heavy chain variable sequence (SEQ ID NO: 115) 62-63 CDR2 Potential Isomerization Site DS ES, QS, DA, DQ, TS, DV, NS

[0186] In some embodiments, the TREM2 agonist antigen binding proteins comprise one or more CDRs of a variant of the anti-TREM2 antibodies described herein. In some embodiments, the TREM2 agonist antigen binding proteins may comprise one or more CDRs of the anti-TREM2 antibody variants set forth in TABLES A10, A11, A12, A13, and A14, below.

[0187] In certain embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region and/or heavy chain variable region from an affinity-modulated variant of the 6E7 antibody. For instance, in some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region and/or a heavy chain variable region having one or more of the amino acid substitutions set forth in TABLE A9.

TABLE-US-00009 TABLE A9 6E7 Antibody Affinity Modulation Variants Binding Signal (fold over 6E7 Substitutions with respect Parental antibody) to 6E7 VH sequence Substitutions with 1.sup.st (SEQ ID NO: 124) respect to 6E7 VL screen HC (SEQ ID NO: 61) 110 2.sup.nd 2.sup.nd 2.sup.nd Variant FR1- HC HC LC LC LC nM or screen screen screen Ab ID CDR1 CDR2 CDR3 CDR1 CDR2 CDR3 10 nM.sup.a 2 nM 10 nM 100 nM V1 Y32S Q99S Q55T F94Y 1.68 1.29 1.92

V2 Y27S S56G Q99S L54R S93R 2.55 2.23 2.90 V3 T30A G66D Q99G L54R S93R 1.97 1.95
 2.24 V4 T30G Y60V Q99S S53R F94Y 6.00 5.88 5.51 V5 I50T F94H 2.73 1.25 2.84 V6 Y32M
 0.20* 0.56 V7 Y32E 0.11* 0.32 V8 R59K 0.28* 0.77 V9 T101G 0.67* 0.54 V10 A50S 0.76* 0.70
 V11 D92A 0.79* 0.42 V12 S28E T58V Q99G N56R 2.29 1.04 2.58 V13 T30G P62A Q99G N56G
 F94M 1.31 1.15 1.35 V14 T30G S56Q Q99G S53R 4.71 2.57 4.64 V15 T30A I50T Q99S S53W
 F94Y 5.23 4.72 4.78 V16 F29M S56G Q99S S53N 4.01 3.57 4.04 V17 T30G Q99S L54R F94S
 5.37 4.22 5.51 V18 W33H 0.17* 0.42 V19 Y32S 0.59* 0.48 V20 I50R 0.18* 0.52 V21 Y109F
 0.76* 0.68 V22 A50R 0.30* 0.71 V23 R96L 0.40* 0.40 V24 T58V Q99S N56K R96H 2.64 1.42
 2.90 V25 T30G I50L Q99S Q55A F94M 4.23 3.15 4.70 V26 A35G I50T F102M, N56R F94Y 3.57
 2.83 3.47 Y112A V27 S61A Q99S N56R 5.50 5.67 5.69 V28 T30Q I50T Y103F N56S F94L 3.08
 2.63 3.61 V29 T30K 1.53 0.84 1.67 V30 Y27S 0.79* 0.72 V31 D57E 0.61* 0.73 V32 P62N 0.82*
 0.89 V33 Y104G 0.23* 0.34 V34 N56D 0.34* 1.02 V35 D92Y 0.21* 0.29 V36 I34L Q99S L54R
 F94Y 3.38 4.00 3.44 V37 F29H Q65A Q99S N56W F94Y 3.46 3.69 3.49 V38 T30G T58V L54R
 F94H 4.34 3.44 4.36 V39 T30G S61N Q99G Q55V F94S 6.15 5.11 5.81 V40 T30G T58V F110S
 N56L S93R 4.48 3.41 4.16 V41 I50T 1.74 0.58 1.72 V42 Y32A 0.45* 0.41 V43 D57G 0.20* 0.33
 V44 G54S 0.65* 0.52 V45 W32F 0.43* 0.53 V46 S53T 0.83* 0.96 V47 R96M 0.42* 0.47 V48
 T30G T58V Q99M N56T F94L 2.42 2.30 2.54 V49 T30N I50T, Q99S L54R F94Y 6.51 5.02 6.58
 Y60L V50 T30G I50V F110L L54R F94L 4.10 3.39 4.16 V51 T58V Q99G, L54R 2.81 1.83 3.18
 Y112N V52 T30E Q99G N56R S93R 3.00 1.78 3.09 V53 S63H 1.25 0.66 1.17 V54 Y32Q 0.55*
 0.54 V55 R59I, 0.24* 0.66 F64H V56 S61Q 0.23* 0.59 V57 R24A 0.84* 0.85 V58 A50K 0.28*
 0.68 V59 Q89M 0.19* 0.60 V60 S28H T58V F110S N56R Q89G 3.26 3.35 3.63 V61 T30S S61N
 Q99G Q55V F94L 5.08 3.63 5.22 V62 T30G S61A D108G N56R Q89G 2.49 1.87 2.89 V63 T30R
 Q99S N56R S93R 3.76 4.91 3.71 V64 T30Q Q99G Q55A F94Y 5.41 4.88 5.48 V65 Q99S 2.05
 1.29 2.75 V66 Y27T 0.25* 0.74 V67 I50M 0.80* 0.84 V68 Y103R 0.44* 0.43 V69 W32Y 0.41*
 0.40 V70 S52G 0.79* 0.84 V71 F94E 0.37* 0.48 V72 A35G Q99G Q55V F94Y 3.64 2.50 4.01
 V73 T30G S63G Q99G L54R F94Y 5.12 4.17 5.44 V74 T30A T58V Q99G N56L 3.94 2.54 4.01
 V75 Q99G N56A F94Y 4.64 3.74 4.52 V76 T30G S63E F110S N56K 4.57 4.34 4.93 V77 L54R
 1.43 0.83 1.38 V78 S28R 0.86* 1.11 V79 R59N 0.70* 0.52 V80 T101N 0.59* 0.50 V81 W32L
 0.17* 0.23 V82 A51G 0.30* 0.79 V83 D92V 0.20* 0.29 V84 S28G F110S A50G 1.44 1.45 1.62
 V85 T30R I50T Q99S L54R 5.41 5.41 5.37 V86 T30G, Q65E Q99S L54R 4.80 5.17 5.02 I34L
 V87 T30R T58V, Q99S N56W 3.84 4.86 3.93 S63D V88 T30G S53R, F94S 4.92 5.57 5.30 N56R
 V89 F94H 1.33 0.94 1.46 V90 Y32E S31R 0.33* 0.36 V91 G54D 0.25* 0.61 V92 Y103H 0.22*
 0.65 V93 S31G 0.35* 1.05 V94 S52A 0.31* 0.87

[0188] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the sequence of SEQ ID NO: 61 with a mutation at one or more amino acid positions 24, 31, 50, 52, 54, 56, 89, 92, 93, 94 and/or 96. In certain embodiments, the mutation is selected from R24A, S31R, A50S, A50G, S52G, L54R, N56K, N56R, N56L, N56T, Q89G, D92V, S93R, F94Y, F94L, R96H, R96L, or combinations thereof. In these and other embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising the sequence of SEQ ID NO: 124 with a mutation at one or more amino acid positions 27, 28, 30, 32, 50, 54, 58, 60, 61, 63, 66, 99, 101, 103, 104, and/or 110. In some embodiments, the mutation is selected from Y27S, S28G, S28H, T30N, T30G, T30E, T30A, Y32E, I50T, G54S, T58V, Y60L, S61A, S63G, S63E, G66D, Q99G, Q99S, Q99M, T101G, Y103R, Y104G, F110S, or combinations thereof. Amino acid sequences for light chain and heavy chain variable regions and associated CDRs of exemplary variants of the 6E7 antibody with improved affinity are set forth below in TABLES A7 and A8, respectively. Amino acid sequences for light chain and heavy chain variable regions and associated CDRs of exemplary variants of the 6E7 antibody with reduced affinity are set forth below in TABLES A10 and A11, respectively. The corresponding sequences for the 6E7 antibody are listed for comparison.

TABLE-US-00010 TABLE A10 Light Chain Variable Region Amino Acid

Sequences for Improved Affinity TREM2 Antibodies Variant VL Ab ID. Group VL

Amino Acid Sequence CDRL1 CDRL2 CDRL3 6E7 LV-16
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADS
 SWLAWYQQKPGKAPKLLIYAASSLQNGV SSWLA N FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSFPRFTFGQGTKLEIK NO: 16) NO: 28) NO: 43) (SEQ ID NO: 61) V3 LV-101
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSRQ QQADR
 SWLAWYQQKPGKAPKLLIYAASSRQNGV SSWLA N FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADRFPRFTFGQGTKLEIK NO: 16) NO: 143) NO: 148) (SEQ ID NO: 153) V24 LV-102
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADS
 SWLAWYQQKPGKAPKLLIYAASSLQKGV SSWLA K FPHT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSFPHFTFGQGTKLEIK NO: 16) NO: 144) NO: 149) (SEQ ID NO: 154) V27 LV-103
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADS
 SWLAWYQQKPGKAPKLLIYAASSLQRGV SSWLA R FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSFPRFTFGQGTKLEIK NO: 16) NO: 145) NO: 43) (SEQ ID NO: 155) V40 LV-104
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADR
 SWLAWYQQKPGKAPKLLIYAASSLQLGV SSWLA L FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADRFPRFTFGQGTKLEIK NO: 16) NO: 146) NO: 148) (SEQ ID NO: 156) V48 LV-105
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADS
 SWLAWYQQKPGKAPKLLIYAASSLQTVGV SSWLA T LPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSLPRFTFGQGTKLEIK NO: 16) NO: 26) NO: 150) (SEQ ID NO: 157) V49 LV-106
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSRQ QQADS V73
 SWLAWYQQKPGKAPKLLIYAASSRQNGV SSWLA N YPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSYPRFTFGQGTKLEIK NO: 16) NO: 143) NO: 151) (SEQ ID NO: 158) V52
 LV-107 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADR
 SWLAWYQQKPGKAPKLLIYAASSLQRGV SSWLA R FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADRFPRFTFGQGTKLEIK NO: 16) NO: 145) NO: 148) (SEQ ID NO: 159) V60 LV-108
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADS
 SWLAWYQQKPGKAPKLLIYAASSLQRGV SSWLA R FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSFPRFTFGQGTKLEIK NO: 16) NO: 145) NO: 152) (SEQ ID NO: 160) V76 LV-109
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI AASSLQ QQADS
 SWLAWYQQKPGKAPKLLIYAASSLQKGV SSWLA K FPRT
 PSRFSGSGSGRDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSFPRFTFGQGTKLEIK NO: 16) NO: 144) NO: 43) (SEQ ID NO: 161) V84 LV-110
 DIQMTQSPSSVSASVGDRVITITCRASQGIS RASQGI GAS SLQ QQADS
 SWLAWYQQKPGKAPKLLIYGASSLQNGV SSWLA N FPRT
 PSRFSGSGSGTDFTLTISLQPEDFATYFC (SEQ ID (SEQ ID (SEQ ID
 QQADSFPRFTFGQGTKLEIK NO: 16) NO: 147) NO: 43) (SEQ ID NO: 162)

TABLE-US-00011 TABLE A11 Heavy Chain Variable Region Amino Acid
 Sequences for Improved Affinity TREM2 Antibodies FR1/Variant VH CDRH1 Ab ID.
 Group VH Amino Acid Sequence Border CDRH1 CDRH2 CDRH3 6E7 HV-15
 EVQLVQSGAEVKKPGESLKIS YSFT SYWIA IYPGDS QRTFYY

CKGSGYSTSYWIAWVRQMP (SEQ ID (SEQ ID DTRYSP DSSDYF
GKGLEWMGITYPGDSDTRYSP NO: 163) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARQRTFY NO:
91) NO: 107) DSSDYFDYWGGQGLTVTVSS (SEQ ID NO: 124) V3 HV-101
EVQLVQSGAEVKKPGESLKIS YSFA SYWIA IIYPGDS GRTFY
CKGSGYSFASYWIAWVRQMP (SEQ ID (SEQ ID DTRYSP DSSDYF
GKGLEWMGITYPGDSDTRYSP NO: 164) NO: 85) SFQD DY
SFQDQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARGRTFY NO:
170) NO: 176) DSSDYFDYWGGQGLTVTVSS (SEQ ID NO: 180) V24 HV-102
EVQLVQSGAEVKKPGESLKIS YSFT SYWIA IIYPGDS SRTFY
CKGSGYSFTSYWIAWVRQMP (SEQ ID (SEQ ID DVRYS DSSDYF
GKGLEWMGIIYPGDS DTRYSP NO: 163) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARSRTFYD
NO: 171) NO: 177) SSDYFDYWGGQGLTVTVSS (SEQ ID NO: 181) V27 HV-103
EVQLVQSGAEVKKPGESLKIS YSFT SYWIA IIYPGDS SRTFY
CKGSGYSFTSYWIAWVRQMP (SEQ ID (SEQ ID DTRYAP DSSDYF
GKGLEWMGIIYPGDS DTRYAP NO: 163) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCVRSRTFYD
NO: 172) NO: 177) SSDYFDYWGGQGLTVTVSS (SEQ ID NO: 182) V40 HV-104
EVQLVQSGAEVKKPGESLKIS YSFG SYWIA IIYPGDS QRTFY
CKGSGYSFGSYWIAWVRQMP (SEQ ID (SEQ ID DVRYS DSSDYS
GKGLEWMGIIYPGDS DTRYSP NO: 165) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARQRTFY NO:
171) NO: 178) DSSDYSDYWGGQGLTVTVSS (SEQ ID NO: 183) V48 HV-105
EVQLVQSGAEVKKPGESLKIS YSFG SYWIA IIYPGDS MRTFY
CKGSGYSFGSYWIAWVRQMP (SEQ ID (SEQ ID DVRYS DSSDYF
GKGLEWMGIIYPGDS DTRYSP NO: 165) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARMRTFY NO:
171) NO: 179) DSSDYFDYWGGQGLTVTVSS (SEQ ID NO: 184) V49 HV-106
EVQLVQSGAEVKKPGESLKIS YSFN SYWIA TIYPGDS SRTFY
CKGSGYSFNSYWIAWVRQMP (SEQ ID (SEQ ID DTRLSPS DSSDYF
GKGLEWMGTIYPGDS DTRLSP NO: 166) NO: 85) FQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARSRTFYD
NO: 173) NO: 177) SSDYFDYWGGQGLTVTVSS (SEQ ID NO: 185) V52 HV-107
EVQLVQSGAEVKKPGESLKIS YSFE SYWIA IIYPGDS GRTFY
CKGSGYSFESYWIAWVRQMP (SEQ ID (SEQ ID DTRYSP DSSDYF
GKGLEWMGIIYPGDS DTRYSP NO: 167) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARGRTFY NO:
91) NO: 176) DSSDYFDYWGGQGLTVTVSS (SEQ ID NO: 186) V60 HV-108
EVQLVQSGAEVKKPGESLKIS YFIFT SYWIA IIYPGDS QRTFY
CKGSGYHFTSYWIAWVRQMP (SEQ ID (SEQ ID DVRYS DSSDYS
GKGLEWMGIIYPGDS DTRYSP NO: 168) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARQRTFY NO:
171) NO: 178) DSSDYSDYWGGQGLTVTVSS (SEQ ID NO: 187) V73 HV-109
EVQLVQSGAEVKKPGESLKIS YSFG SYWIA IIYPGDS GRTFY
CKGSGYSFGSYWIAWVRQMP (SEQ ID (SEQ ID DTRYSP DSSDYF
GKGLEWMGIIYPGDS DTRYSP NO: 165) NO: 85) GFQG DY
GFQGGQVTISADKSISTAYLQW (SEQ ID (SEQ ID SSLKASDTAMYFCARGRTFY NO:
174) NO: 176) DSSDYFDYWGGQGLTVTVSS (SEQ ID NO: 188) V76 HV-110
EVQLVQSGAEVKKPGESLKIS YSFG SYWIA IIYPGDS QRTFY

CKGSGSYFGSYWIAWVRQMP (SEQ ID (SEQ ID DTRYSP DSSDYS
GKGLEWMGITYPGDSDTRYSP NO: 165) NO: 85) EFQG DY
EFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARQRTFYY NO:
175) NO: 178) DSSDYS DYWGQGTLVTVSS (SEQ ID NO: 189) V84 HV-111
EVQLVQSGAEVKKPGESLKIS YGFT SYWIA IIYPGDS QRTFYY
CKGSGYGFYSYWIAWVRQMP (SEQ ID (SEQ ID DTRYSP DSSDYS
GKGLEWMGITYPGDSDTRYSP NO: 169) NO: 85) SFQG DY
SFQGGQVTISADKSISTAYLQWS (SEQ ID (SEQ ID SLKASDTAMYFCARQRTFYY NO:
91) NO: 178) DSSDYS DYWGQGTLVTVSS (SEQ ID NO: 190)

[0189] In some embodiments, the TREM2 agonist antigen binding proteins of the invention may comprise one or more of the CDRs from the improved affinity variants presented in TABLE A10 (light chain CDRs; i.e. CDRLs) and TABLE A11 (heavy chain CDRs, i.e. CDRHs). In some embodiments, the TREM2 agonist antigen binding proteins comprise a consensus CDR sequence derived from the improved affinity variants. For instance, in some embodiments, the TREM2 agonist antigen binding proteins comprise a CDRL2 consensus sequence of X.sub.1ASSX.sub.2QX.sub.3 (SEQ ID NO: 139), where X.sub.1 is A or G; X.sub.2 is L or R; and X.sub.3 is N, K, R, L, or T. In another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRL3 consensus sequence of X.sub.1QADX.sub.2ZX.sub.3PX.sub.4T (SEQ ID NO: 140), where X.sub.1 is Q or G; X.sub.2 is S or R; X.sub.3 is F, L, or Y; and X.sub.4 is R or H. In yet another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRH2 consensus sequence of X.sub.1IYPGDSDX.sub.2ZRX.sub.3X.sub.4PX.sub.5FQX.sub.6 (SEQ ID NO: 141), where X.sub.1 is I or T; X.sub.2 is T or V; X.sub.3 is Y or L; X.sub.4 is S or A; X.sub.5 is S, G, or E; and X.sub.6 is G or D. In some embodiments, the TREM2 agonist antigen binding proteins comprise a CDRH3 consensus sequence of X.sub.1RTFYDYSSDYXzDY (SEQ ID NO: 142), where X.sub.1 is Q, G, S, or M; and X.sub.2 is F or S.

[0190] In some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region comprising complementarity determining regions CDRL1, CDRL2, and CDRL3 and a heavy chain variable region comprising complementarity determining regions CDRH1, CDRH2, and CDRH3, wherein CDRL1 comprises the sequence of SEQ ID NO: 16, CDRL2 comprises the consensus sequence of SEQ ID NO: 139, CDRL3 comprises the consensus sequence of SEQ ID NO: 140, CDRH1 comprises the sequence of SEQ ID NO:85, CDRH2 comprises the consensus sequence of SEQ ID NO:141, and CDRH3 comprises the consensus sequence of SEQ ID NO: 142.

[0191] In some embodiments, the TREM2 agonist antigen binding protein comprises a CDRL1 comprising the sequence of SEQ ID NO: 16; a CDRL2 comprising a sequence selected from SEQ ID NOS:26 and 143-147; a CDRL3 comprising a sequence selected from SEQ ID NOS:43 and 148-152; a CDRH1 comprising the sequence of SEQ ID NO:85; a CDRH2 comprising a sequence selected from SEQ ID NOS:91 and 170-175; and a CDRH3 comprising a sequence selected from SEQ ID NOS:176-179.

[0192] In particular embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3, wherein:

[0193] (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 143, and 148, respectively; [0194] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 144, and 149, respectively; [0195] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 145, and 43, respectively; [0196] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 146, and 148, respectively; [0197] (e) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 26, and 150, respectively; [0198] (f) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 143, and 151, respectively; [0199] (g) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 145, and 148, respectively; [0200] (h) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 145, and 152, respectively; [0201] (i)

CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 144, and 43, respectively; or [0202] (j) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 147, and 43, respectively.

[0203] In related embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: (a) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 170, and 176, respectively; [0204] (b) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 177, respectively; [0205] (c) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 172, and 177, respectively; [0206] (d) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 178, respectively; [0207] (e) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 179, respectively; [0208] (f) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 173, and 177, respectively; [0209] (g) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 176, respectively; [0210] (h) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 174, and 176, respectively; [0211] (i) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 175, and 178, respectively; or [0212] (j) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 178, respectively.

[0213] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: [0214] (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 143, and 148, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 170, and 176, respectively; [0215] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 144, and 149, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 177, respectively; [0216] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 145, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 172, and 177, respectively; [0217] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 146, and 148, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 178, respectively; [0218] (e) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 26, and 150, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 179, respectively; [0219] (f) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 143, and 151, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 173, and 177, respectively; [0220] (g) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 145, and 148, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 176, respectively; [0221] (h) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 145, and 152, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 171, and 178, respectively; [0222] (i) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 143, and 151, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 174, and 176, respectively; [0223] (j) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 144, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 175, and 178, respectively; or [0224] (k) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 147, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 178, respectively.

[0225] In some embodiments, the TREM2 agonist antigen binding proteins of the invention may comprise a light chain variable region selected from LV-101, LV-102, LV-103, LV-104, LV-105, LV-106, LV-107, LV-108, LV-109, and LV-110, as shown in TABLE A10, and/or a heavy chain variable region selected from HV-101, HV-102, HV-103, HV-104, HV-105, HV-106, HV-107, HV-108, HV-109, HV-110, and HV-111, as shown in TABLE A11, or sequences that are at least 80% identical, at least 85% identical, at least 90% identical, or at least 95% identical to any of the

sequences in TABLE A10 and A11. For instance, in some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region comprising (i) a sequence that is at least 90% identical to a sequence selected from SEQ ID NOS: 153-162, (ii) a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 153-162, or (iii) a sequence selected from SEQ ID NOS:153-162. In related embodiments, the TREM2 agonist antigen binding proteins comprise a heavy chain variable region comprising (i) a sequence that is at least 90% identical to a sequence selected from SEQ ID NOS: 180-190, (ii) a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 180-190, or (iii) a sequence selected from SEQ ID NOS:180-190.

[0226] Each of the light chain variable regions listed in TABLE A10 may be combined with any of the heavy chain variable regions listed in TABLE A11 to form an anti-TREM2 binding domain of the antigen binding proteins of the invention. Examples of such combinations include, but are not limited to: LV-101 (SEQ ID NO:153) and HV-101 (SEQ ID NO:180); LV-102 (SEQ ID NO:154) and HV-102 (SEQ ID NO:181); LV-103 (SEQ ID NO: 155) and HV-103 (SEQ ID NO:182); LV-104 (SEQ ID NO: 156) and HV-104 (SEQ ID NO: 183); LV-105 (SEQ ID NO:157) and HV-105 (SEQ ID NO: 184); LV-106 (SEQ ID NO: 158) and HV-106 (SEQ ID NO:185); LV-107 (SEQ ID NO: 159) and HV-107 (SEQ ID NO: 186); LV-108 (SEQ ID NO:160) and HV-108 (SEQ ID NO: 187); LV-106 (SEQ ID NO: 158) and HV-109 (SEQ ID NO:188); LV-109 (SEQ ID NO: 161) and HV-110 (SEQ ID NO: 189); and LV-110 (SEQ ID NO: 162) and HV-111 (SEQ ID NO: 190).

TABLE-US-00012	TABLE A12	Light Chain Variable Region	Amino Acid Sequences for Reduced Affinity TREM2 Antibodies	Variant VL Ab ID. Group VL
Amino Acid Sequence	CDRL1	CDRL2	CDRL3	6E7 LV-16
DIQMTQSPSSVSASVGDRVITTCRASQGI RASQGI AASSLQ QQADS				
SSWLAWYQQKPGKAPKLLIYAASSLQN SSWLA N FPRT				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 16)				
FCQQADSFPRTFGQGGTKLEIK NO: 28) NO: 43) (SEQ ID NO: 61) V9 LV-16				
DIQMTQSPSSVSASVGDRVITTCRASQGI RASQGI AASSLQ QQADS V30				
SSWLAWYQQKPGKAPKLLIYAASSLQN SSWLA N FPRT V33				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 16)				
FCQQADSFPRTFGQGGTKLEIK NO: 28) NO: 43) V68 (SEQ ID NO: 61) V10				
LV-201 DIQMTQSPSSVSASVGDRVITTCRASQGI RASQGI SASSLQ QQADS				
SSWLAWYQQKPGKAPKLLIYSASSLQN SSWLA N FPRT				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 16)				
FCQQADSFPRTFGQGGTKLEIK NO: 292) NO: 43) (SEQ ID NO: 295) V23				
LV-202 DIQMTQSPSSVSASVGDRVITTCRASQGI RASQGI AASSLQ QQADS				
SSWLAWYQQKPGKAPKLLIYAASSLQN SSWLA N FPLT				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 16)				
FCQQADSFPLTFGQGGTKLEIK NO: 28) NO: 294) (SEQ ID NO: 296) V57				
LV-203 DIQMTQSPSSVSASVGDRVITTCRASQGI AASQGI AASSLQ QQADS				
SSWLAWYQQKPGKAPKLLIYAASSLQN SSWLA N FPRT				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 290)				
FCQQADSFPRTFGQGGTKLEIK NO: 28) NO: 43) (SEQ ID NO: 297) V70				
LV-204 DIQMTQSPSSVSASVGDRVITTCRASQGI RASQGI AAGSL QQADS				
SSWLAWYQQKPGKAPKLLIYAAGSLQN SSWLA QN FPRT				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 16)				
FCQQADSFPRTFGQGGTKLEIK NO: 293) NO: 43) (SEQ ID NO: 298) V83				
LV-205 DIQMTQSPSSVSASVGDRVITTCRASQGI RASQGI AASSLQ QQAVS				
SSWLAWYQQKPGKAPKLLIYAASSLQN SSWLA N FPRT				
GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID NO: 16)				
FCQQAVSFPRTFGQGGTKLEIK NO: 28) NO: 271) (SEQ ID NO: 299) V90				

LV-206 DIQMTQSPSSVSASQTISRASQGI RASQGI AASSLQ QQADS
 SRWLAWYQQKPGKAPKLLIYAASSLQN SRWLA N FPRT
 GVPSRFGSGSGTDFTLTISSLQPEDFATY (SEQ ID (SEQ ID (SEQ ID
 FCQQADSFPRTFGQGTKLEIK NO: 291) NO: 28) NO: 43) (SEQ ID NO: 300)
 TABLE-US-00013 TABLE A13 Heavy Chain Variable Region Amino Acid
 Sequences for Reduced Affinity TREM2 Antibodies Variant VH FR1/CDRH1 Ab ID.
 Group VH Amino Acid Sequence border CDRH1 CDRH2 CDRH3 6E7 HV-15
 EVQLVQSGAEVKKPGESLKISCK YSFT SYWIA IIYPGD QRTFYY
 GSGYSFTSYWIAWVRQMPGKG (SEQ ID (SEQ ID SDTRYSDSSDYF
 LEWMGIIYPGDS DTRYSPSFQGG NO: 163) NO: 85) PSFQG DY
 VTISADKSISTAYLQWSSLKASD (SEQ ID (SEQ ID TAMYFCARQRTFYYDSSDYFDY
 NO: 91) NO: 107) WGQGT LVT VSS (SEQ ID NO: 124) V9 HV-
 EVQLVQSGAEVKKPGESLKISCK YSFT SYWIA IIYPGD QRGFYY 201
 GSGYSFTSYWIAWVRQMPGKG (SEQ ID (SEQ ID SDTRYSDSSDYF
 LEWMGIIYPGDS DTRYSPSFQGG NO: 163) NO: 85) PSFQG DY
 VTISADKSISTAYLQWSSLKASD (SEQ ID (SEQ ID TAMYFCARQRGFYYDSSDYFDY
 NO: 91) NO: 304) WGQGT LVT VSS (SEQ ID NO: 307) V10 HV-15
 EVQLVQSGAEVKKPGESLKISCK YSFT SYWIA IIYPGD QRTFYY V23
 GSGYSFTSYWIAWVRQMPGKG (SEQ ID (SEQ ID SDTRYSDSSDYF V57
 LEWMGIIYPGDS DTRYSPSFQGG NO: 163) NO: 85) PSFQG DY V70
 VTISADKSISTAYLQWSSLKASD (SEQ ID (SEQ ID V83
 TAMYFCARQRTFYYDSSDYFDY NO: 91) NO: 107) WGQGT LVT VSS (SEQ ID NO:
 124) V30 HV- EVQLVQSGAEVKKPGESLKISCK SSFT SYWIA IIYPGD QRTFYY 202
 GSGSSFTSYWIAWVRQMPGKGL (SEQ ID (SEQ ID SDTRYSDSSDYF
 EWMGIIYPGDS DTRYSPSFQGGV NO: 301) NO: 85) PSFQG DY
 TISADKSISTAYLQWSSLKASDT (SEQ ID (SEQ ID AMYFCARQRTFYYDSSDYFDY
 NO: 91) NO: 107) WGQGT LVT VSS (SEQ ID NO: 308) V33 HV-
 EVQLVQSGAEVKKPGESLKISCK YSFT SYWIA IIYPGD QRTFYG 203
 GSGYSFTSYWIAWVRQMPGKG (SEQ ID (SEQ ID SDTRYSDSSDYF
 LEWMGIIYPGDS DTRYSPSFQGG NO: 163) NO: 85) PSFQG DY
 VTISADKSISTAYLQWSSLKASD (SEQ ID (SEQ ID TAMYFCARQRTFYGDSSDYFDY
 NO: 91) NO: 305) WGQGT LVT VSS (SEQ ID NO: 309) V44 HV-
 EVQLVQSGAEVKKPGESLKISCK YSFT SYWIA IIYPSDS QRTFYY 204
 GSGYSFTSYWIAWVRQMPGKG (SEQ ID (SEQ ID DTRYSPDSSDYF
 LEWMGIIYPSDSDTRYSPSFQGG NO: 163) NO: 85) SFQG DY
 VTISADKSISTAYLQWSSLKASD (SEQ ID (SEQ ID TAMYFCARQRTFYYDSSDYFDY
 NO: 303) NO: 107) WGQGT LVT VSS (SEQ ID NO: 310) V68 HV-
 EVQLVQSGAEVKKPGESLKISCK YSFT SYWIA IIYPGD QRTFRY 205
 GSGYSFTSYWIAWVRQMPGKG (SEQ ID (SEQ ID SDTRYSDSSDYF
 LEWMGIIYPGDS DTRYSPSFQGG NO: 163) NO: 85) PSFQG DY
 VTISADKSISTAYLQWSSLKASD (SEQ ID (SEQ ID TAMYFCARQRTFRYDSSDYFDY
 NO: 91) NO: 306) WGQGT LVT VSS (SEQ ID NO: 311) V90 HV-
 EVQLVQSGAEVKKPGESLKISCK YSFT SEWIA IIYPGD QRTFYY 206
 GSGYSFTSEWIAWVRQMPGKGL (SEQ ID (SEQ ID SDTRYSDSSDYF
 EWMGIIYPGDS DTRYSPSFQGGV NO: 163) NO: 302) PSFQG DY
 TISADKSISTAYLQWSSLKASDT (SEQ ID (SEQ ID AMYFCARQRTFYYDSSDYFDY
 NO: 91) NO: 107) WGQGT LVT VSS (SEQ ID NO: 312)

[0227] In some embodiments, the TREM2 agonist antigen binding proteins of the invention may comprise one or more of the CDRs from the reduced affinity variants presented in TABLE A12 (light chain CDRs; i.e. CDRLs) and TABLE A13 (heavy chain CDRs, i.e. CDRHs). In some

embodiments, the TREM2 agonist antigen binding proteins comprise a consensus CDR sequence derived from the reduced affinity variants. For instance, in one embodiment, the TREM2 agonist antigen binding proteins comprise a CDRL1 consensus sequence of X.sub.1ASQGISX.sub.2WLA (SEQ ID NO:284), where X.sub.1 is R or A; and X.sub.2 is S or R. In another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRL2 consensus sequence of X.sub.1AX.sub.2SLQN (SEQ ID NO:285), where X.sub.1 is A or S; and X.sub.2 is S or G. In another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRL3 consensus sequence of QQAX.sub.1SFPX.sub.2T (SEQ ID NO:286), where X.sub.1 is D or V; and X.sub.2 is R or L. In another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRH1 consensus sequence of SX.sub.1WIA (SEQ ID NO:287), where X.sub.1 is Y or E. In yet another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRH2 consensus sequence of IIYPX.sub.1DSDTRYSPSFQG (SEQ ID NO:288), where X.sub.1 is G or S. In still another embodiment, the TREM2 agonist antigen binding proteins comprise a CDRH3 consensus sequence of QRX.sub.1FX.sub.2X.sub.3DSSDYFDY (SEQ ID NO:289), where X.sub.1 is T or G; X.sub.2 is Y or R; and X.sub.3 is Y or G. In some embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region comprising complementarity determining regions CDRL1, CDRL2, and CDRL3 and a heavy chain variable region comprising complementarity determining regions CDRH1, CDRH2, and CDRH3, wherein CDRL1 comprises the sequence of SEQ ID NO:284, CDRL2 comprises the consensus sequence of SEQ ID NO:285, CDRL3 comprises the consensus sequence of SEQ ID NO:286, CDRH1 comprises the sequence of SEQ ID NO:287, CDRH2 comprises the consensus sequence of SEQ ID NO:288, and CDRH3 comprises the consensus sequence of SEQ ID NO:289.

[0228] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a CDRL1 comprising a sequence selected from SEQ ID NOS:16, 290, and 291; a CDRL2 comprising a sequence selected from SEQ ID NOS:28, 292, and 293; a CDRL3 comprising a sequence selected from SEQ ID NOS:43, 294, and 271; a CDRH1 comprising the sequence of SEQ ID NO:85 or SEQ ID NO:302; a CDRH2 comprising the sequence of SEQ ID NO:91 or SEQ ID NO:303; and a CDRH3 comprising a sequence selected from SEQ ID NOS:107 and 304-306.

[0229] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3, wherein: (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 28, and 43, respectively;

[0230] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 292, and 43, respectively; [0231] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 294, respectively; [0232] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:290, 28, and 43, respectively; [0233] (e) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 293, and 43, respectively; [0234] (f) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 28, and 271, respectively; or [0235] (g) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:291, 28, and 43, respectively.

[0236] In related embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein:

[0237] (a) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 304, respectively; [0238] (b) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0239] (c) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 305, respectively; [0240] (d) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 303, and 107, respectively; [0241] (e) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 306, respectively; or [0242] (f) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:302, 91, and 107, respectively.

[0243] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: [0244] (a)

CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 304, respectively; [0245] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 292, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0246] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 294, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0247] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0248] (e) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 305, respectively; [0249] (f) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 303, and 107, respectively; [0250] (g) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:290, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0251] (h) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 306, respectively; [0252] (i) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 293, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; [0253] (j) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 28, and 271, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 91, and 107, respectively; or [0254] (k) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:291, 28, and 43, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:302, 91, and 107, respectively.

[0255] In some embodiments, the TREM2 agonist antigen binding proteins of the invention may comprise a light chain variable region selected from LV-16, LV-201, LV-202, LV-203, LV-204, LV-205, and LV-206, as shown in TABLE A12, and/or a heavy chain variable region selected from HV-15, HV-201, HV-202, HV-203, HV-204, HV-205, and HV-206, as shown in TABLE A13, or sequences that are at least 80% identical, at least 85% identical, at least 90% identical, or at least 95% identical to any of the sequences in TABLES A12 and A13. For instance, in certain embodiments, the TREM2 agonist antigen binding proteins comprise a light chain variable region comprising (i) a sequence that is at least 90% identical to a sequence selected from SEQ ID NOS:61 and 295-300, (ii) a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS:61 and 295-300, or (iii) a sequence selected from SEQ ID NOS:61 and 295-300. In related embodiments, the TREM2 agonist antigen binding proteins comprise a heavy chain variable region comprising (i) a sequence that is at least 90% identical to a sequence selected from SEQ ID NOS:124 and 307-312, (ii) a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS:124 and 307-312, or (iii) a sequence selected from SEQ ID NOS: 124 and 307-312.

[0256] In some embodiments, each of the light chain variable regions listed in TABLE A12 may be combined with any of the heavy chain variable regions listed in TABLE A13 to form an anti-TREM2 binding domain of the antigen binding proteins of the invention. Examples of such combinations include, but are not limited to: LV-16 (SEQ ID NO:61) and HV-201 (SEQ ID NO:307); LV-201 (SEQ ID NO:295) and HV-15 (SEQ ID NO: 124); LV-202 (SEQ ID NO:296) and HV-15 (SEQ ID NO: 124); LV-16 (SEQ ID NO:61) and HV-202 (SEQ ID NO:308); LV-16 (SEQ ID NO:61) and HV-203 (SEQ ID NO:309); LV-16 (SEQ ID NO:61) and HV-204 (SEQ ID NO:310); LV-203 (SEQ ID NO:297) and HV-15 (SEQ ID NO: 124); LV-16 (SEQ ID NO:61) and HV-205 (SEQ ID NO:311); LV-204 (SEQ ID NO:298) and HV-15 (SEQ ID NO: 124); LV-205 (SEQ ID NO:299) and HV-15 (SEQ ID NO: 124); and LV-206 (SEQ ID NO:300) and HV-206 (SEQ ID NO:312).

[0257] In some embodiments, the TREM2 agonist antigen binding proteins comprise one or more

CDRs of the anti-TREM2 antibody variants set forth in TABLE A14. In some embodiments, the TREM2 agonist antigen binding proteins comprise the light chain variable region and heavy chain variable region of the anti-TREM2 antibody variants set forth in TABLE A14.

TABLE-US-00014 TABLE A14 Exemplary Variable Region Amino Acid Sequences of Engineered Antibodies

Ab ID	LC variable region	CDRL1	CDRL2	CDRL3	24G6
DIVMTQSPDSLAVSLGERATINCKSSQSVLY KSSQSV WASTR QQYYYS (SST28347	SSNNKHFLAWYQQKPGQPPKLLIYWASTRLYSSN LYSSNN ES TPLT and	ESGVPDRFSGSGSGTDFTLTISLQAEDVAV KHFLA (SEQ ID (SEQ ID SST204812)	YYCQQYYSTPLTFGGG	TKVEIK NO: 8) NO: 22) NO: 35) (SEQ ID NO: 326) 6E7	DIQMTQSPSSVSASVGDRVITTCRASQGISS RASQGI AASSLQ QQADA (SST29857)
WLAWYQQKPGKAPKLLIYAASSLQSGVPS SSWLA S FPRT	RFSGSGSGTDFTLTISLQPEDFATYFCQQA (SEQ ID (SEQ ID (SEQ ID	DAFPRTFGQG	TKLEIK (SEQ ID NO: 328) NO: 16) NO: 369) NO: 370) 13E7	EIVMTQSPATLSVSPGERATLSCRASQSVSS RASQS GASTR LQDNN (SST202443)	NLAWFQQKPGQAPRLLIYGASTRATGIPAR VSSNLA AT FPPT
FSGSGSGTEFTLTISLQPEDFAVYYCLQDN (SEQ ID (SEQ ID (SEQ ID	NFPPTFGQG	TKVDIK (SEQ ID NO: 330) NO: 10) NO: 23) NO: 372) 5E3	DIQMTQSPSSLSASVGDRVITTCRASQGISN RASQGI AASSLQ QQYST (SST29825)	YLAWYQQKPGKAPKSLIYAASSLQSGVPSR SNYLA S YPFT	FSGSGSGTDFTLTISLQPEDFATYFCQQYS (SEQ ID (SEQ ID (SEQ ID
TYPPTFGQG	TKVDIK (SEQ ID NO: 332) NO: 17) NO: 29) NO: 44) Ab ID. HC	variable region	CDRH1 CDRH2 CDRH3 24G6 EVQLLES	GGGLVQPGGSLRLSCAASGFTFS	SYAMS AISGSG AYTPM (SST28347 SYAMSWVRQAPGKGLEWVSAISGSGGSTY (SEQ ID
GSTYY AFFDY and YAESVKGRFTISRDN	SKNTLYLQMNSLRAE NO: 77) AESVK (SEQ ID SST204812)	DTAVYYCAKAYTPMAFFDYWGQGT	LVTV G NO: 98) SS (SEQ ID NO: 327) (SEQ ID NO: 368) 6E7	EVQLVQSGAEVKKPGESLKISCKGSGYSFT SYWIA	IIYPGD QRIFY (SST29857) SYWIAWVRQMPGKGLEWMGITYPGDADA (SEQ ID ADARY
YDSSD RYSPSFQGGQVTISADKSISTAYLQWSSLKAS NO: 85) SPSFQG YFDY	DTAMYFCARQRIFYDSSDYFDYWGQGT (SEQ ID (SEQ ID LTVVSS (SEQ ID	NO: 329) NO: 371) NO: 107) 13E7	EVQLVQSGAEVKKPGESLKISCKGSGYSFT SWIG	IIYPGD RRQGI (SST202443) SYWIGWVRQMPGKGLEWMGITYPGDADA (SEQ ID	ADARY GDALD RYSPSFQGGQVTISADKSISTAYLQWSSLKAS NO: 81) SPSFQG F
DTAMYFCARRRQGI	FGDALDFWGQGT	LVTV (SEQ ID NO: 331) NO: 373) NO: 374) QVQLVQSGAEVKKPGASVKV	SCKASGYTF GYYIH WINPYS	DAGYL TGYIHWVRQAPGQGLEWMGWINPYS	GG (SEQ ID GGTT
ALYGT TTS	AQKFQ DV	SDDTAVYYCARDAGYLALYGTDVWGQGT G (SEQ ID LTVVSS (SEQ ID NO: 333)	(SEQ ID NO: 375 NO: 94		

[0258] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3, wherein: [0259] (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 16, 369, and 370, respectively; [0260] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 10, 23, and 372, respectively; or [0261] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 6, 21, and 33, respectively; [0262] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 6, 20, and 33, respectively.

[0263] In some embodiments, the TREM2 agonist antigen binding protein comprises a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: [0264] (a) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS: 77, 368, and 98, respectively; [0265] (b) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS: 85, 371, and 107, respectively; [0266] (c) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS: 81, 373, and 374,

respectively; or [0267] (d) CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:86, 94, and 375, respectively.

[0268] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3 and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein: [0269] (a) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:8, 22, and 35, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:77, 368, and 98, respectively; [0270] (b) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS:16, 369, and 370, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:85, 371, and 107, respectively; [0271] (c) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 10, 23, and 372, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 373, and 374, respectively; or [0272] (d) CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 17, 29, and 44, respectively, and CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:86, 94, and 375, respectively.

[0273] Accordingly, in some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising a CDRL1, a CDRL2, and a CDRL3, and a heavy chain variable region comprising a CDRH1, a CDRH2, and a CDRH3, wherein the CDRL1, CDRL2, and CDRL3 have the sequence of SEQ ID NOS: 10, 23, and 372, respectively, and the CDRH1, CDRH2, and CDRH3 have the sequence of SEQ ID NOS:81, 373, and 374, respectively.

[0274] In some embodiments therefore, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a CDRL1, CDRL2, and CDRL3 having the sequence of SEQ ID NOS:10, 23, and 372, respectively, and a CDRH1, CDRH2, and CDRH3 having the sequence of SEQ ID NOS:81, 373, and 374, respectively. In certain embodiments, the antibody is human. In some embodiments, the TREM2 agonist antigen binding protein comprises [0275] (a) a light chain variable region comprising the amino acid sequence of SEQ ID NO:326 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:327; [0276] (b) a light chain variable region comprising the amino acid sequence of SEQ ID NO:328 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:329; [0277] (c) a light chain variable region comprising the amino acid sequence of SEQ ID NO:330 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:331; or [0278] (d) a light chain variable region comprising the amino acid sequence of SEQ ID NO:332 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:333.

[0279] In some embodiments, the TREM2 agonist antigen binding protein comprises a light chain variable region comprising the amino acid sequence of SEQ ID NO:330 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:331.

[0280] In some embodiments therefore, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain variable region comprising the amino acid sequence of SEQ ID NO:330 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:331. In certain embodiments, the antibody is human.

[0281] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:326, 328, 330 or 332. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:327, 329, 331 or 333. In a specific embodiment, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region and a heavy chain variable region, wherein the light chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:326 and the heavy chain variable region consisting of or consisting essentially of the amino acid sequence of

SEQ ID NO:327. In a specific embodiment, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region and a heavy chain variable region, wherein the light chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:328 and the heavy chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:329. In a specific embodiment, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region and a heavy chain variable region, wherein the light chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:330 and the heavy chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:331. In a specific embodiment, the TREM2 agonist antigen binding proteins of the invention comprise a light chain variable region and a heavy chain variable region, wherein the light chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:332 and the heavy chain variable region consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:333.

[0282] In some embodiments, each of the light chain variable regions disclosed in TABLES A1, A10, A12, and A14 and each of the heavy chain variable regions disclosed in TABLES A2, A11, A13, and 3E may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

[0283] In some embodiments, exemplary TREM2 agonist antibody having a light chain variable region with a light chain constant domain and a heavy chain variable region with a heavy chain constant region are disclosed in TABLE A15.

TABLE-US-00015 TABLE A15 Light Chain and Heavy Chain Amino Acid Sequences of Exemplary Antibodies Ab ID. Sequence 24G6 LC

MDMRVPAQLLGLLLLWLRGARCDIVMTQSPDSLAVSLGERATIN (SST28347)

CKSSQSVLYSSNNKHFLAWYQQKPGQPPKLLIYWASTRESGVPD

RFSGSGSGTDFTLTISLQAEDVAVYYCQQYYSTPLTFGGGGTKVE

IKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKV

DNALQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYAC

EVTHQGLSPVTKSFNRGEC (SEQ ID NO: 334) HC

MDMRVPAQLLGLLLLWLRGARCEVQLLESGGGLVQPGGSLRLS

CAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTYYAESVK

GRFTISRDNSTNTLYLQMNSLRAEDTAVYYCAKAYTPMAFFDY

WGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP

EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQ

TYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVF

LFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVH

NAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKAL

PAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFY

PSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRW

QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 335) 24G6 LC

MDMRVPAQLLGLLLLWLRGARCDIVMTQSPDSLAVSLGERATIN (SST204812)

CKSSQSVLYSSNNKHFLAWYQQKPGQPPKLLIYWASTRESGVPD

RFSGSGSGTDFTLTISLQAEDVAVYYCQQYYSTPLTFGGGGTKVE

IKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKV

DNALQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKHKVYAC

EVTHQGLSPVTKSFNRGEC (SEQ ID NO: 334) HC

MDMRVPAQLLGLLLLWLRGARCEVQLLESGGGLVQPGGSLRLS

CAASGFTFSSYMSVSTVRQAPKGLEWVSASISGSGGSTYYAESVK
GRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKAYTPMAFFDY
WGQGT LVT VSSASTKGPSVFPLAPSSRSTSESTAALGCLVKDYFP
EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSNFGTQ
TYTCNV DHKPSNTKVDKTV ERKCCVECPPCPAPELLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPS
DIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 336) 6E7 LC
MDMRVPAQLLGLLLLWLRGARCDIQMTQSPSSVSASVGDRVIT (SST29857)
CRASQGISSWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGS
GTDFTLTISLQPEDFATYFCQQADAFPRTFGQGTKLEIKRTVAAP
SVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSG
NSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQGL SSPVTKSFNRGEC
(SEQ ID NO: 337) HC
MDMRVPAQLLGLLLLWLRGARCEVQLVQSGAEVKKPGESLKIS
CKGSGYSFTSYWIAWVRQMPGKGLEWMGITYPGDADARYSPSF
QGQVTISADKSISTAYLQWSSLKASDTAMYFCARQRTFYDSSD
YFDYWGGT LVT VSSASTKGPSVFPLAPSSRSTSESTAALGCLVK
DYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSN
FGTQTYTCNV DHKPSNTKVDKTV ERKCCVECPPCPAPELLGGPS
VFLFPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE
VHNAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNK
ALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 338) 13E7 LC
MDMRVPAQLLGLLLLWLRGARCEIVMTQSPATLSVSPGERATLS (SST202443)
CRASQSVSSNLAWFQQKPGQAPRLLIYGASTRATGIPARFSGSGS
GTEFTLTISLQPEDFAVYYCLQDNNFPPTFGQGTKVDIKRTVAA
PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQS
GNSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQG LSSPVTKSFNRGEC
(SEQ ID NO: 339) HC
MDMRVPAQLLGLLLLWLRGARCEVQLVQSGAEVKKPGESLKIS
CKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDADARYSPSF
QGQVTISADKSISTAYLQWSSLKASDTAMYFCARRRQGIFGDAL
DFWGQGTLVT VSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKD
YFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSL
GTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGG
PSVFLFPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVS
NKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV
KGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 340) 5E3
(SST29825) LC MDMRVPAQLLGLLLLWLRGARCDIQMTQSPSSLSASVGDRVIT
CRASQGISNYLAWYQQKPGKAPKSLIYAASSLQSGVPSRFSGSGS
GTDFTLTISLQPEDFATYYCQYSTYPFTFGQGTKVDIKRTVAA
PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQS
GNSQESVTEQDSKDSSTYLSSTLTLSKADYEKHKVYACEVTHQG LSSPVTKSFNRGEC
(SEQ ID NO: 341) HC

MDMRVPAQLLLGLLQVQLVQSGAEVKKPGASVKV
SCKASGYTFTGYYIHWVRQAPGQGLEWMGWINPYSGGT TSAQK
FQGRVTMTRDTSTSSAYMELSRLRSDDTAVYYCARDAGYLALY
GTDVWGQGT LVT VSSASTKGPSVFPLAPSSRSTSESTAALGCLVK
DYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSN
FGTQTYTCNVDHKPSNTKVDKTVERKCCVECPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVE
VHNAKTKPCEEQYGSTYRCVSVLT VLHQDWLNGKEYKCKVSNK
ALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKG
FYPSDIAVEWESNGQPENNYKTTTPVLDS DGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 342) 24G6-1 LC
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKHFLAWYQQ (SST28347-1)
KPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTIS SLQAEDVA
VYYCQQYYSTPLTFGGG TKVEIKRTVAAPSVFIFPPSDEQLKSGT
ASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDST
YLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC (SEQ ID NO: 2768) HC
EVQLLES GGGVLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKG
LEWVSAISGSGGSTYYAESVKGRFTISRDN SKNTLYLQMNSLRAE
DTAVYYCAKAYTPMAFFDYWGQGT LVT VSSASTKGPSVFPLAPS
SKSTSGGTAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQ
SSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC
DKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVD
VSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVSVLT V
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL P
PSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTP
VLDS DGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQK SLSLSPGK (SEQ
ID NO: 2769) 24G6-1 LC
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKHFLAWYQQ (SST28347-1)
KPGQPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTIS SLQAEDVA
VYYCQQYYSTPLTFGGG TKVEIKRTVAAPSVFIFPPSDEQLKSGT
ASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDST
YLSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC (SEQ ID NO: 2768) HC
EVQLLES GGGVLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKG
LEWVSAISGSGGSTYYAESVKGRFTISRDN SKNTLYLQMNSLRAE
DTAVYYCAKAYTPMAFFDYWGQGT LVT VSSASTKGPSVFPLAPS
SRSTSESTAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQS
SGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTVERKCC
VECPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSH E
DPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVSVLT VLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPS REE
MTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDS D
GSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSP G (SEQ ID NO:
2770) 6E7-1 LC DIQMTQSPSSVSASVGDRV TITCRASQGISSWLAWYQQKPGKAP
(SST29857-1) KLLIYAASSLQSGVPSRFSGSGSGTDFTLTIS SLQPEDFATYFCQQ
ADAFPRTFGQGTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLL
NNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYLSSTLT
LSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC (SEQ ID NO: 2771) HC
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIAWVRQMPGKG
LEWMGIIYPGDADARYSPSFQGQVTISADKSISTAYLQWSSLKAS
DTAMYFCARQRTFYDSSDYFDYWGQGT LVT VSSASTKGPSVFP

LAPSSRSTSESTALGCLVDFPEPVTVSWNSGALTSGVHTFPA
VLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTVE
RKCCVECPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVD
VSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL
PSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPP
VLDSGDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQK SLSLSPG (SEQ ID
NO: 2772) 13E7-1 LC EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWFQQKPGQAPR
(SST202443-1) LLIYGASTRATGIPARFSGSGSGTEFTLTISLQPEDFAVYYCLQD
NNFPPTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLN
NFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLT
SKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 2773) HC
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKG
LEWMGIIYPGDADARYSPSFQGGQVTISADKSISTAYLQWSSLKAS
DTAMYFCARRRQGIFGDALDFWGQGTLLVTVSSASTKGPSVFPLA
PSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV
LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPK
SCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVV
DVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVSVLT
VLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL
PPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPP
VLDSGDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQK SLSLSPG (SEQ ID
NO: 2774) 5E3-1 LC DIQMTQSPSSLSASVGDRVTITCRASQGISNYLAWYQQKPGKAPK
(SST29825-1) SLIYAASSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQY
STYPFTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLN
NFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLT
SKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 2775) HC
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYYIHWVRQAPGQ
GLEWMGWINPYSGGTSAQKFQGRVTMTRDTSTSSAYMELSRL
RSDDTAVYYCARDAGYLALYGTDVWGQGTLLVTVSSASTKGPSV
FPLAPSSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF
PAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKT
VERKCCVECPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV
VDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVSVL
TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYT
LPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
PVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNHYTQ KLSLSPG (SEQ
ID NO: 2776) 13E7 Variant LC
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWFQQKPGQAPR
LLIYGASTRATGIPARFSGSGSGTEFTLTISLQPEDFAVYYCLQD
NNFPPTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLN
NFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLT
SKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 2777) HC
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKG
LEWMGIIYPGDADARYSPSFQGGQVTISADKSISTAYLQWSSLKAS
DTAMYFCARRRQGIFGDALDFWGQGTLLVTVSSASTKGPSVFPLA
PSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV
LQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPK
SCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVV
DVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCVSVLT

VLHQDWNKLYKCKVSNKALPSEKTSKAKGQPREPQVYITL
PPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPP
VLDSGDSFFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQK SLSLSPGK (SEQ
ID NO: 2778)

[0284] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:334 and a heavy chain comprising the sequence of SEQ ID NO:335. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:334 and a heavy chain comprising the sequence of SEQ ID NO:336. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:337 and a heavy chain comprising the sequence of SEQ ID NO:338. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:339 and a heavy chain comprising the sequence of SEQ ID NO:340. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:341 and a heavy chain comprising the sequence of SEQ ID NO:342. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:2768 and a heavy chain comprising the sequence of SEQ ID NO:2769. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:2768 and a heavy chain comprising the sequence of SEQ ID NO:2770. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:2771 and a heavy chain comprising the sequence of SEQ ID NO:2772. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:2773 and a heavy chain comprising the sequence of SEQ ID NO:2774. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain comprising the sequence of SEQ ID NO:2775 and a heavy chain comprising the sequence of SEQ ID NO:2776.

[0285] In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:334 and a heavy chain comprising the sequence of SEQ ID NO:335. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:334 and a heavy chain comprising the sequence of SEQ ID NO:336. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:337 and a heavy chain comprising the sequence of SEQ ID NO:338. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:339 and a heavy chain comprising the sequence of SEQ ID NO:340. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:341 and a heavy chain comprising the sequence of SEQ ID NO:342. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:2768 and a heavy chain comprising the sequence of SEQ ID NO:2769. In some embodiments, the present invention provides a method of treating ALSP in a

human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:2768 and a heavy chain comprising the sequence of SEQ ID NO:2770. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:2771 and a heavy chain comprising the sequence of SEQ ID NO:2772. In some embodiments therefore, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:2773 and a heavy chain comprising the sequence of SEQ ID NO:2774. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:2775 and a heavy chain comprising the sequence of SEQ ID NO:2776. In some embodiments, the present invention provides a method of treating ALSP in a human patient, the method comprising administering to the patient an effective amount of a TREM2 agonist antigen binding protein comprising a light chain comprising the sequence of SEQ ID NO:2777 and a heavy chain comprising the sequence of SEQ ID NO:2778.

[0286] In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:334, 337, 339 or 341. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2768, 2771, 2773, or 2775. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:335, 336, 338, 340, or 342. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2769, 2770, 2772, 2774, or 2776. In a specific embodiment, the TREM2 agonist antigen binding proteins of the invention comprise a light chain and a heavy chain, wherein: [0287] (a) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:334 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:335; [0288] (b) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:334 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:336; [0289] (c) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:337 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:338; [0290] (d) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:339 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:340; or [0291] (e) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:341 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:342. [0292] In a specific embodiment, the TREM2 agonist antigen binding proteins of the invention comprise a light chain and a heavy chain, wherein: [0293] (a) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2768 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2769; [0294] (b) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2768 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2770; [0295] (c) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2771 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2772; [0296] (d) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2773 and the heavy chain consisting of or

consisting essentially of the amino acid sequence of SEQ ID NO:2774; [0297] (e) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2775 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2776; or [0298] (f) the light chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2777 and the heavy chain consisting of or consisting essentially of the amino acid sequence of SEQ ID NO:2778.

[0299] Unless indicated otherwise by reference to a specific sequence and in related discussions, the numbering of the amino acid residues in an immunoglobulin heavy chain or light chain is according to Kabat-EU numbering as described in Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed., US Department of Health and Human Services, NIH publication No. 91-3242, pp 662,680,689 (1991) and Edelman et al., Proc. Natl. Acad. USA, Vol. 63: 78-85 (1969). The Kabat numbering scheme is typically used when referring to the position of an amino acid within the variable regions, whereas the EU numbering scheme is generally used when referring to the position of an amino acid with an immunoglobulin constant region.

[0300] In some embodiments, the TREM2 antigen binding protein comprise an antibody that competes with an antibody comprising CDRL1, CDRL2, CDRL3 or light chain variable region disclosed in TABLES A1, A10, A12, and A14, and a heavy chain variable region disclosed in TABLES A2, A11, A13, and A14. In some embodiments, a suitable assay for detecting competitive binding employs kinetic sensors used with Octet® systems (Pall ForteBio), which measures binding interactions using bio-layer interferometry methodology. One group of antibodies, antibodies 10E3, 13E7, 24F4, 4C5, 4G10, 32E3, and 6E7, competed with each other for binding to human TREM2, indicating that they share the same or similar epitope on human TREM2. Antibodies 16B8, 26A10, 26C10, 26F2, 33B12, and 5E3 compete with each other for TREM2 binding, but does not compete with antibodies in the first group or antibodies 24A10, 24G6, or 25F12, indicating that this second group of antibodies bind to a distinct epitope on human TREM2. Antibodies 24A10 and 24G6 share a similar epitope on human TREM2 as these two antibodies compete with each other for human TREM2 binding, but did not compete with any other antibody. Antibody 25F12 did not compete with any of the other tested antibodies for human TREM2 binding, indicating that this antibody binds to yet another epitope.

[0301] In some embodiments, a TREM2 agonist antigen binding protein competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising a sequence selected from SEQ ID NOS:46-63 and a heavy chain variable region comprising a sequence selected from SEQ ID NOS: 110-126. In other embodiments, a TREM2 agonist antigen binding protein of the invention competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising a sequence selected from SEQ ID NOS: 153-162 and a heavy chain variable region comprising a sequence selected from SEQ ID NOS: 180-190. In still other embodiments, a TREM2 agonist antigen binding protein of the invention competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising a sequence selected from SEQ ID NOS:61 and 295-300 and a heavy chain variable region comprising a sequence selected from SEQ ID NOS: 124 and 307-312. In certain embodiments, a TREM2 agonist antigen binding protein of the invention competes for binding to human TREM2 with one or more of the anti-TREM2 antibodies described herein, including 12G10, 26A10, 26C10, 26F2, 33B12, 24C12, 24G6, 24A10, 10E3, 13E7, 14C12, 25F12, 32E3, 24F4, 16B8, 4C5, 6E7, 5E3, 4G10, V3, V9, V10, V23, V24, V27, V30, V33, V40, V44, V48, V49, V52, V57, V60, V68, V70, V73, V76, V83, V84, and V90.

[0302] In some embodiments, the TREM2 agonist antigen binding protein competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising the sequence of SEQ ID NO:61 and a heavy chain variable region comprising the sequence of SEQ ID NO: 124. In such embodiments, antigen binding proteins that

compete with this reference antibody for binding to human TREM2 would bind the same or similar epitope as antibody 6E7 or any of the other antibodies 10E3, 13E7, 24F4, 4C5, 4G10, and 32E3.

[0303] In some embodiments, the TREM2 agonist antigen binding protein competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising the sequence of SEQ ID NO:62 and a heavy chain variable region comprising the sequence of SEQ ID NO: 125. In such embodiments, antigen binding proteins that compete with this reference antibody for binding to human TREM2 would bind the same or similar epitope as antibody 5E3 or any of the other antibodies 16B8, 26A10, 26C10, 26F2, and 33B12.

[0304] In some embodiments, the TREM2 agonist antigen binding protein competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising the sequence of SEQ ID NO:52 and a heavy chain variable region comprising the sequence of SEQ ID NO: 115. In such embodiments, antigen binding proteins that compete with this reference antibody for binding to human TREM2 would bind the same or similar epitope as antibody 24G6 or antibody 24A10.

[0305] In some embodiments, the TREM2 agonist antigen binding protein competes with a reference antibody for binding to human TREM2, wherein the reference antibody comprises a light chain variable region comprising the sequence of SEQ ID NO:56 and a heavy chain variable region comprising the sequence of SEQ ID NO: 119. In such embodiments, antigen binding proteins that compete with this reference antibody for binding to human TREM2 would bind the same or similar epitope as antibody 25F12.

[0306] In some embodiments, isolated nucleic acids encoding the anti-TREM2 binding domain of the antigen binding proteins of the invention can be used to synthesize the antigen binding protein or used to generate variants. In some embodiments, the polynucleotide may comprise a nucleotide sequence that is at least 80% identical, at least 90% identical, at least 95% identical, or at least 98% identical to any of the nucleotide sequences listed in TABLE A15..

TABLE-US-00016 TABLE A16 Exemplary Anti-TREM2 Antibody Variable Region
Nucleic Acid Sequences VL or VH Group Ab ID. Designation Nucleic Acid Sequence
Light chain variable regions 12G10 LV-01

CAGGCTGTGCCGACTCAGCCGTCTTCCCTCTCTGCATCTCCTGGAGT
ATTAGCCAGTCTCACCTGCACCTTACGCAGTGGCATCAATGTTGGTA
CCTACAGGATATACTGGTACCAGCAGAAGCCAGGGAGTCCTCCCCA
GTATCTCCTGAGGTACAAATCAGACTCAGATAAGCAGCAGGGCTCT
GGAGTCCCCAGCCGCTTCTCTGGATCCAAGGATGCTTCGGCCAATGC
AGGGATTTTACTCATCTCTGGGCTCCAGTCTGAGGATGAGGCTGACT
ATTACTGTATGATTTGGTACAGCAGTGCTGTGGTATTCGGCGGAGGG
ACCAAAGTGACCGTCCTA (SEQ ID NO: 208) 26A10 LV-02

TCCTATGAGCTGACTCAGCCACCCTCAGTGTCCGTGTCCCCAGGACA
GACAGCCAGCATCACCTGCTCTGGAGATAAATTGGGAGATAAGTAT
GTTTGCTGGTATCAGCAGAAGCCAGGCCAGTCCCCTGTGCTGGTCAT
CTATCAAGATAGCAAGCGGCCCTCAGGGATCCCTGAGCGATTCTCTG
GCTCCAAGTCTGGGAACACAGCCACTCTGACCATCAGCGGGACCCA
GGCTATGGATGAGGCTGACTATTACTGTCAGGCGTGGGACAGTAAC
ACTGTGGTATTCGGCGGAGGGACCAAGCTGACCGTCCTA (SEQ ID NO: 209) 26C10

LV-03 TCCTTTGAGCTGACTCAGCCACCCTCAGTGTCCGTGTCCCCAGGACA
GACAGCCAGCATCACCTGCTCTGGAGATAAATTGGGGGATAAGTAT
GTTTGCTGGTATCAGCAGAAGCCAGGCCAGTCCCCTATGTTGGTCAT
CTATCAAGATACCAAGCGGCCCTCAGGGATCCCTGAACGATTCTCTG
GCTCCAAGTCTGGGAACACAGCCACTCTGACCATCAGCGGGACCCA
GGCTATGGATGAGGCTGACTATTACTGTCAGGCGTGGGACAGCAGC
ACTGTGGTCTTCGGCGGAGGGACCAAGCTGACCGTCCTA (SEQ ID NO: 210) 26F2

LV-04 TCCTATGAGCTGACTCAGCCACCCTCAGTGTCCGTGTCCCCAGGACA
GACAGCCAGCATCACCTGCTCTGGAGATAAATTGGGGGATAAGTAT
GTTTGCTGGTATCAGCAGAAGCCAGGCCAGTCCCCTGTGTTGGTTCAT
CTTTCAAGATAGCAAGCGGCCCTCAGGGATCCCTGAGCGATTCTCTG
GCTCCAACCTCTGGGAACACAGCCACTCTGACCATCAGCGGGACCCA
GGCTATGGATGAGGCTGACTATTACTGTCAGGCGTGGGACAGCAGC
ACTGTGGTATTCGGCGGAGGGACCAAGCTGACCGTCCTA (SEQ ID NO: 211) 33B12
LV-05 TCCTATGAGCTGACTCAGCCACCCTCAGTGTCCGTGTCCCCAGGACA
GACAGCCAGCATCACCTGCTCTGGAGATAAATTGGGGGATAAGTAT
GTTTGCTGGTATCAGCAGAAGCCAGGCCAGTCCCCTGTGTTGGTTCAT
CTATCAAGATAGCAAGCGGCCCTCAGGGATCCCTGAGCGATTCTCTG
GCTCCAACCTCTGGGAACACAGCCACTCTGACCATCAGCGGGACCCA
GGCTATGGATGAGGCTGACTATTACTGTCAGGCGTGGGACAGTAGC
ACTGTGGTATTCGGCGGAGGGACCAAGCTGACCGTCCTA (SEQ ID NO: 212) 24C12
LV-06 GGCATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGG
CGAGAGGGGCCACCATCAACTGCAAGTCCAGCCGAGTGTTTTGTAC
AGCTCCAACAATAAGAACTACTTAGCTTGGTACCAGCAGAAACCAG
GACAGCCTCCTAAGGTGCTCATTTACTGGGCATCTACCCGGGAATCC
GGGGTCCCTGACCGATTTCAGTGGCAGCGGGTCTGGGACAGATTTCA
CTCTCACCATCAGCAGCCTGCAGGCTGAAGATGTGGCAGTTTATAAC
TGTCAGCAATATTATATACTCCGATCACCTTCGGCCAAGGGACACG
ACTGGAGATTAAA (SEQ ID NO: 213) 24G6 LV-07
GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGG
CGAGAGGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGTTTTATAC
AGCTCCAACAATAAGCACTTCTTAGCTTGGTACCAGCAGAAACCAG
GACAGCCTCCTAAGCTGCTCATTTACTGGGCATCTACCCGGGAGTCC
GGGGTCCCTGACCGATTTCAGTGGCAGCGGGTCTGGGACAGATTTCA
CTCTCACCATCAGCAGCCTGCAGGCTGAAGATGTGGCAGTTTATTAC
TGTCAGCAATATTATAGTACTCCGCTCACTTTTCGGCGGAGGGACCAA
GGTGGAGATCAAA (SEQ ID NO: 214) 24A10 LV-08
GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGG
CGAGAGGGGCCACCATCACCTGCAAGTCCAGCCACAATGTTTTATACA
GCTCCAACAATAAGAACTACTTAGCTTGGTATCAGCAGAAACCAGG
ACAGCCTCCTAAACTGCTCATTTACTGGGCATCTACCCGGGAATCCG
GGGTCCCTGACCGATTTCAGTGGCAGCGGGTCTGGGACAGATTTCACT
CTCACCATCAGCAGCCTGCAGGCTGAAGATGTGGCAGTTTATTACTG
TCACCAATATTATAGTACTCCGTGCAGTTTTTGGCCAGGGGACCAAGC
TGGAGATCAAA (SEQ ID NO: 215) 10E3 LV-09
GAAATAGTGATGACGCAGTCTCCAGCCACCCTGTCTGTGTCTCCAGG
GGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCAGAGTGTTAGCAGC
AACTTAGCCTGGTTCCAGCAGAAACCTGGCCAGGCTCCCAGGCTCCT
CATCTATGGTGCTTCCACCAGGGCCACTGGTATTCCAGCCAGGTTCA
GTGTCAGTGGGTCTGGGACAGAGTTCACTCTCACCATCAGCAGCCTG
CAGTCTGAAGATTTTGCATTTTATTACTGTCTGCAGGATAATAATTG
GCCTCCCACCTTTCGGGCCCTGGGACCAAAGTGGATATCAAA (SEQ ID NO: 216) 13E7
LV-10 GAAATAGTGATGACGCAGTCTCCAGCCACCCTGTCTGTGTCTCCAGG
GGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCAGAGTGTTAGCAGC
AACTTAGCCTGGTTCCAGCAGAAACCTGGCCAGGCTCCCAGGCTCCT
CATCTATGGTGCTTCCACCAGGGCCACTGGTATTCCAGCCAGGTTCA
GTGTCAGTGGGTCTGGGACAGAGTTCACTCTCACCATCAGCAGCCTG

CAGTCTGATGATTATTAGTCTGTCTGCTGCAGGATAATAATTG
GCCTCCCACCTTTTCGGCCCTGGGACCAAAGTGGATATCAAA (SEQ ID NO: 217) 25F12
LV-11 GAAAAAGTGATGACGCAGTCTCCAGCCACCCTGTCTGTGTCTCCAGG
GGAAAGAGCCACCCTCTCCTGCAGGGGCCAGTCAGAGTGTTAACAAC
AACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCCAGGCTCCT
CATCTATGGTGCATCCACCAGGGGCCACTGGTATCCCAGCCAGGTTCA
GTGGCAGTGGGTCTGGGACAGAGTTCACTCTCACCATCAGCAGCCTG
CAGTCTGAAGATTTTGCAGTTTATTACTGTCAGCAGTATAATAACTG
GCCTCGGACGTTTCGGCCAAGGGACCAAGGTGGAAATCAAA (SEQ ID NO: 218)
32E3 LV-12 GAATTTGTGTTGACGCAGTCTCCAGGCACCCTGTCTTTGTCTCCGGG
GGAAAGAGCCACCCTCTCCTGCAGGGGCCAGTCAGATTATTAGCAGC
AACTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCCAGGC
TCCTCATCTATAGTGCATCCAGCAGGGGCCACTGGCATCCCAGACAGG
TTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAG
ACTGGAGCCTGAAGATTTTGCAGTGTATTACTGTCAGCAGTTTGATA
GCTCACCGATCACCTTCGGCCGAGGGACACGACTGGACATTAAA (SEQ ID NO:
219) 24F4 LV-13 GAAATTGTGTTGACGCAGTCTCCAGGCACCCTGTCTTTGTCTCCAGG
GGAAAGAGCCACCCTCTCCTGCAGGGGCCAGTCAGAGTGTTAGCAGC
AGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCCAGGC
TCCTCATCTATGGTGCATCCAGCAGGGGCCACTGGCATCCCAGACAGG
TTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAG
ACTGGAGCCTGAAGATTTTGCAGTGTATTACTGTCAGCAGTATGATA
CCTCACCATTCACCTTCGGCCCTGGGACCAAAGTGGATATCAAA (SEQ ID NO: 220)
16B8 LV-14 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCGTCACTTGTCTGGGCGAGTCAGGATATTAACAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCTCTTTGCAAACCTGGGGTCCCTTCAAGGTTT
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTCTTGTCAACAGTCTAACAGTT
TCCCGATCACCTTCGGCCAAGGGACACGACTGGAGATTAAA (SEQ ID NO: 221) 4C5
LV-15 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAAC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAAAGTTGGGGTCCCATTAAGGTTT
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGCTGACAGTT
TCCCTCGCAATTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 222) 6E7
LV-16 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG V9
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC V30
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC V33
TGATCTATGCTGCATCCAGTTTGCAAAATGGGGTCCCATCAAGGTTT V44
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT V68
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 223) 5E3
LV-17 GACATCCAGATGACCCAGTCTCCATCCTCACTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGCATTAGCAAT
TATTAGCCTGGTTTCAGCAGAAACCAGGGAAAGCCCCTAAATCCCT
GATCTATGCTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAAGTTCA
GCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCTG
CAGCCTGAAGATTTTGCAACTTATTACTGCCAACAGTATAGTACTTA

CCATTCTTCGGAGCCCTGGAAGCTGGATATGCAA (SEQ ID NO: 224) 4G10
LV-18 GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGCCGGGCAAGTCAGGGCATAAGAAAT
GATTTAGGCTGGTATCAGCAGAAACCAGGGAATGCCCCTAAGCGCC
TGATCTATGCTGCATCCAGTTTGCCAAGTGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGGCAGAATTCCTCTCACAATCAGCAGTCT
GCAGCCTGAAGATTTTGCAACTTATTACTGTCTACAGCATAATAGTT
ACCCGTGGACGTTTCGGCCAAGGGACCAAGGTGGAAATCACA (SEQ ID NO: 225) V3
LV-101 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTAGGCCAAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGGT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 226) V24
LV-102 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAAAAGGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCATACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 227) V27
LV-103 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAACGTGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 228) V40
LV-104 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAACTTGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACCGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 229) V48
LV-105 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAACGGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TGCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 230)
V49 LV-106 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTCGGCAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
ATCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 231) V52

LV-107 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAAAGGGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACCGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 232) V60
LV-108 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAAAGGGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTGGGCAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 233) V73
LV-106 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTCGTCAAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
ATCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 234) V76
LV-109 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGAGAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 235) V84
LV-110 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGGTGCATCCAGTTTGCAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCGCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 236)
V10 LV-201 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATTCTGCATCCAGTTTGCAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 313) V23
LV-202 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGCAAATGGGGTCCCATCAAGGTTC
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGCAACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCTTACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 314) V57
LV-203 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG

AGACAGTACGATCCTCACTTGTGCGGAGTCCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGC AAAATGGGGTCCCATCAAGGTTT
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGC AACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 315) V70
LV-204 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCAGGGAGTTTGC AAAATGGGGTCCCATCAAGGTTT
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGC AACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 316) V83
LV-205 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGC
TGGTTAGCCTGGTATCAGCAGAAACCAGGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGC AAAATGGGGTCCCATCAAGGTTT
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGC AACTTACTTTTGTCAACAGGCTGTGAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 317) V90
LV-206 GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGG
AGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGA
TGGTTAGCCTGGTATCAGCAGAAACCAGGGGAAAGCCCCTAAGCTCC
TGATCTATGCTGCATCCAGTTTGC AAAATGGGGTCCCATCAAGGTTT
AGCGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCT
GCAGCCTGAAGATTTTGC AACTTACTTTTGTCAACAGGCTGACAGTT
TCCCTCGCACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA (SEQ ID NO: 318)

Heavy chain variable regions 12G10 HV-01

GAGGTGCAGCTGTTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGG 24C12
GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTTAGCAGC
TATGCCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGT
GGGTCTCAGCTATTGGTGGTGGTGGTGTAGCACATACTGCGCAGAC
TCCGTGAAGGGCCGTTTACCATCTCCAGAGACAATTCCAAGAATA
CGCTGTATCTGCAAATGAACAGCCTGAGAGCCGAGGACACGGCCGT
ATATTACTGTGCGAAATTTTATATAGCAGTGGCTGGTTCTCACTTTG
ACTACTGGGGGCCAGGGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 237) 26A10
HV-02 GAGGTGCAACTGGTGGAGTCTGGGGGAGCCTTGGTACAGCGGGGGG
GGTCCCTGAGACTCTCCTGTGCAGCCTCTAGATTACCTTCAGTAGC
TTTGGCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGT
GGGTTTCATACATTAGTAGTAGTAGTTTTACCATATATTACGCAGAC
TCTGTGAAGGGCCGATTACCATCTCCAGAGACAATGCCAAGAATTC
ATTCTATCTGCAAATGAACAGCCTGAGAGACGAGGACACGGCTGTG
TATTACTGTGCGAGAGAGGGGGGTCTTACTATGGTTCGGGGAGTCTC
TTCCTACGGTTTGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCT CCTCA (SEQ
ID NO: 238) 26C10 HV-03

GAGGTGCAACTGGTGGAGTCTGGGGGAGCCTTGGTACAGCCTGGGG
GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTAGC
TTTGGCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGT
GGGTTTCATACATTAGTAGTAGTAGTTTTACCATATACTACGCAGAC
TCTGTGAAGGGCCGATTACCATCTCCAGAGACAATGCCAAGAATTC

GTTCCTATCTGCAAATGAGACGAGGACACGGCTGTG
TATTTCTGTGTGAGAGAGGGGGGTATAACTATGGTTCGGGGAGTCTC
TTCCTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCT CCTCA (SEQ
ID NO: 239) 26F2 HV-04

GAGGTGCAACTGGTGGAGTCTGGGGGAGCCTTGGTACAGCCTGGGG
GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTAGC
TTTGGCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGT
GGATTTCATACATTAGTAGTAGTAGTTTTACCATATACTACGCAGAC
TCTGTGAAGGGCCGATTACCATCTCCAGAGACAATGCCAAGAATTC
ATTCTATCTGCAAATGAACAGCCTGAGAGACGAGGACACGGCTGTG
TATTTCTGTGCGAGAGAGGGGGGTATTACTATGGTTCGGGGAGTCTC
TTCCTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCT CCTCA (SEQ
ID NO: 240) 33B12 HV-05

GAGGTGCAACTGGTGGAGTCTGGGGGAGCCTTGGTACAGCCTGGGG
GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTAGC
TTTGGCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGCCTGGAGT
GGGTTCATACATTAGTAAAAGTAGTTTTACCATATACTACGCAGAC
TCTGTGAAGGGCCGATTACCATCTCCAGAGACAATGCCAAGAATTC
ATTCTATCTGCAAATGAACAGCCTGAGAGACGAGGACACGGCTGTG
TATTACTGTGCGAGAGAGGGGGGTCTTACTATGGTTCGGGGAGTCTC
TTCCTACGGTTTGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCT CCTCA (SEQ
ID NO: 241) 24G6 HV-06

GAGGTGCAGCTGTTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGG
GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTTAGCAGC
TATGCCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGACTGGAGT
GGGTCTCAGCTATTAGTGGTAGTGGTGGTAGCACATACTACGCAGAC
TCCGTGAAGGGCCGGTTCACCATCTCCAGAGACAATTCCAAGAACA
CGCTGTATCTGCAAATGAACAGCCTGAGAGCCGAGGACACGGCCGT
ATATTACTGTGCGAAGGCGTATACACCTATGGCATTCTTTGACTACT
GGGGCCAGGGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 242) 24A10 HV-07

GAGGTGCAGGTGTTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGG
GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTTAGCAAC
TATGCCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGT
GGGTCTCAGCTATTAGTGGTAGTGGTGGTAGCACATACTACGCAGAC
TCCGTGAAGGGCCGGTTCACCATCTCCAGAGACAATTCCAAGAACA
CGCTGTATCTGCAAATGAACAGCCTGAGAGCCGAGGACACGGCCGT
ATATTACTGTGCGAAAGGAGGGTGGGAGCTATTTTACTGGGGCCAG
GGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 243) 10E3 HV-08

GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGATGATCTCCTGTAAGGGTTCTGGATACAGCTTTACCAAC
TACTGGATCGGCTGGGTGCGCCAGATGCCCAGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGAGACTCTGATACCAGATACAGCCC
GTCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCA
CCGCCTACCTGCAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT
GTATTTCTGTGCGAGACGGAGACAGGGGATCTGGGGTGATGCTCTTG
ATATCTGGGGCCAAGGGACATTGGTCACCGTCTCTTCA (SEQ ID NO: 244) 13E7 HV-
09 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGATGATCTCCTGTAAGGGTTCTGGATACAGCTTTACCAAC
TACTGGATCGGCTGGGTGCGCCAGATGCCCAGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGAGACTCTGATACCAGATACAGCCC

GTCTTCCAAGTCCATCAGTCCGACCAAGTCCATCAGTCA
CCGCCTACCTGCAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT
GTATTTCTGTGCGAGACGGAGACAGGGGATCTGGGGTGATGCTCTTG
ATTTCTGGGGCCAAGGGACATTGGTCACCGTCTCTTCA (SEQ ID NO: 245) 25F12
HV-10 CAGGTGCAGCTACAGCAGTGGGGCGCAGGACTGTTGAAGCCTTCGG
AGACCCTGTCCCTCACCTGCGCTGTCTATGGTGGGTCCTTCAGTAGT
TACTACTGGAGCTGGATCCGCCAGCCCCCAGGGAAGGGGCTGGAGT
GGATTGGGGAAATCAATCATAGTGGAAACACCAACTACAACCCGTC
CCTCAAGAGTCGAGTCACCATATCAGTAGACACGTCCAAGAACCAG
TTCTCCCTGAAGCTGAGCTCTGTGACCGCCGCGGACACGGCTGTGTA
TTACTGTGCGAGAGAGGGGTATTACGATATCTTGACTGGTTATCATG
ATGCTTTTGATATTTGGGACCAAGGGACAATGGTCACCGTNTTTTCA (SEQ ID NO:
246) 32E3 HV-11 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGCTTTACCAGC
TACTGGATCGGCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGA CTCTGATACCAGATACAGCCCG
TCCTTCCAAGGCCAGGTCAACATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTGCAGTGGAGCACCTGAAGGCCTCGGACACCGCCATA
TATTACTGTGCGCGACATGACATTATACCAGCAGCCCCTGGTGCTTT
TGATATCTGGGGCCAAGGGACAATGGTCACCGTCTCTTCA (SEQ ID NO: 247) 24F4
HV-12 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACACCTTTACCAGC
TACTGGATCGGCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGA CTCTGATACCAGATACAGCCCG
TCCTTCCAAGGCCAGGTCAACATCTCAGTCGACAAGTCCAGCAGCAC
CGCCTACCTGCAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATA
TATTACTGTACGAGACAGGCCATAGCAGTGA CTGGTTTGGGGGGTTT
CGACCCCTGGGGCCAGGGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 248) 16B8
HV-13 CAGGTT CAGCTGGTGCAGTCTGGAGCTGAGGTGAAGAAGCCTGGGG
CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACCAAC
TATGGTATCAGCTGGGTGCGACAGGCCCTGGACAAGGGCTTGAGT
GGATGGGATGGATCAGCGCTTACAATGGTAACACAAACTATGCACA
GAAGCTCCAGGGCAGAGTCACCATGACCACAGACACATCCACGAGT
ACAGTCTACATGGAGCTGAGGAGCCTGAGATCTGACGACACGGCCG
TGTATTACTGTGCGAGACGGGGATACAGCTATGGTTCCTTTGACTAC
TGGGGCCAGGGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 249) 4C5 HV-14
GAGGTGCAGCTGGTGCAGTCTGGAGCAGAAGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGACACAGTTTTTACCAAC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGA CTCTGATACCAGATACAGCCCG
TCCTTCCAAGGCCAGGTCAACATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTGCAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCGTG
TATTTCTGTGCGAGACAAAGGACGTTTTTACTATGATAGTAGTGGTTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 250)
6E7 HV-15 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGA CTCTGATACCAGATACAGCCCG
TCCTTCCAAGGCCAGGTCAACATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG

TATTCTGTGCGAGAGCAAAAGGACGTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTCTCCTCA (SEQ ID NO: 251)
5E3 HV-16 CAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAGCCTGGGG
CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGATACACCTTCACCGGC
TACTATATACTGGGTGCGACAGGCCCCTGGACTAGGGCTTGAGTG
GATGGGATGGATCAACCCTTACAGTGGTGGCACAACCTCTGCACAG
AAGTTTCAGGGCAGGGTCACCATGACCAGGGACACGTCCATCAGCT
CAGCCTACATGGAAGTGAAGCAGGCTGAGATCTGACGACACGGCCGT
GTATTACTGTGCGAGAGATGGAGGCTACCTGGCCCTCTACGGTACGG
ACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA (SEQ ID NO: 252) 4G10
HV-17 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGCTTTCCCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGAAGTCTGATACCAGATACAGCCCCG
TCCTTCCAAGGCCAGGTACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTTTTTGAAGTGGAGTAGCCTGAAGGCCTCGGACACCGCCATGT
ATTTCTGTGCGGACAGGGTATAGAAGTGAAGTGGTACGGGAGGTTT
GGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA (SEQ ID NO: 253) V3
HV-101 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTGCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGAAGTCTGATACCAGATACAGCCCCG
TCCTTCCAAGATCAGGTACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGAGGGAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCCTCA (SEQ ID NO: 254)
V24 HV-102 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTACCAGC
TACTGGATTGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGAAGTCTGATGTGAGATACAGCCCCG
TCCTTCCAAGGCCAGGTACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGATCTAGGACGTTTTATTATGATAGTAGTGATTAT
TTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCCTCA (SEQ ID NO: 255)
V27 HV-103 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGAAGTCTGATACCAGATACGCTCCG
TCCTTCCAAGGCCAGGTACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGTGAGAAGTAGGACGTTTTATTATGATAGTAGTGATTAT
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCCTCA (SEQ ID NO: 256)
V40 HV-104 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTGGGAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGAAGTCTGATGTTAGATACAGCCCCG
TCCTTCCAAGGCCAGGTACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGTGAGACAAAGGACGTTTTATTATGATAGTAGTGATTAT
TTCGGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCCTCA (SEQ ID NO: 257)

V48 HV-105 GAGGTGCAGCTGGTGCAGAGCTGCCGAGGTGAGAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTGGTAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATGTGAGATACAGCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGAATGAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCA (SEQ ID NO: 258)
V49 HV-106 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTTAATAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGACGATCTATCCTGGTGACTCTGATACCAGACTGAGCCC
GTCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCA
CCGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT
GTATTTCTGTGCGAGAAGTAGGACGTTTTATTATGATAGTAGTGATT
ATTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCA (SEQ ID NO: 259)
V52 HV-107 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTGAGAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGAGGGAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCA (SEQ ID NO: 260)
V60 HV-108 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACCATTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATGTGAGATACAGCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATTA
TAGTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCA (SEQ ID NO: 261)
V73 HV-109 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTGGTAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCG
GGGTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCA
CCGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT
GTATTTCTGTGCGAGAGGGAGGACGTTTTATTATGATAGTAGTGATT
ATTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCA (SEQ ID NO: 262)
V76 HV-110 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTGGGAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCG
GAGTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCA
CCGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT
GTATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATT
ATAGTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCA (SEQ ID NO: 263)
V84 HV-111 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACGGGTTTACCAGC

TACTGGATCGGCTGCGGATGCGCCAGATGCCCCGGGAAAGGCTGGAGT
GGATGGGGATCATCTATCCTGGTGACAGTGATACCAGATACAGCCC
GTCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCA
CCGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCAT
GTATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATT
ATTCGGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 264)
V9 HV-201 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGGGGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 319)
V10 HV-15 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG V23
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTACCAGC V57
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT V70
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCCG V83
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 320)
V30 HV-202 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATCGAGTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 321)
V33 HV-203 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTTATGGGGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 322)
V44 HV-204 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTAGTGACTCTGATACCAGATACAGCCCCG
TCCTTCCAAGGCCAGGTCACCATCTCAGCCGACAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 323)
V68 HV-205 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTACCAGC
TACTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGACTCTGATACCAGATACAGCCCCG

TCCTTCCAAGGCCAGGTCAACCTACAGCCGACCAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTAGGTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 324)
V90 HV-206 GAGGTGCAGCTGGTGCAGTCTGGAGCAGAGGTGAAAAAGCCCCGGGG
AGTCTCTGAAGATCTCCTGTAAGGGTTCTGGATACAGTTTTACCAGC
GAGTGGATCGCCTGGGTGCGCCAGATGCCCCGGGAAAGGCCTGGAGT
GGATGGGGATCATCTATCCTGGTGA CTCTGATACCAGATAACAGCCCG
TCCTTCCAAGGCCAGGTCAACCTCTCAGCCGACAAAGTCCATCAGCAC
CGCCTACCTACAGTGGAGCAGCCTGAAGGCCTCGGACACCGCCATG
TATTTCTGTGCGAGACAAAGGACGTTTTATTATGATAGTAGTGATTA
TTTTGACTACTGGGGCCAGGGAACCCCTGGTCACCGTGTCTCTCA (SEQ ID NO: 325)

[0307] In some embodiments, an isolated nucleic acid encoding an anti-TREM2 antibody light chain variable region comprises a sequence that is at least 80% identical, at least 90% identical, at least 95% identical, or at least 98% identical to a sequence selected from SEQ ID NOS:208-236 and 313-318. In certain embodiments, an isolated nucleic acid encoding an anti-TREM2 antibody light chain variable region comprises a sequence selected from SEQ ID NOS:208-236 and 313-318. In related embodiments, an isolated nucleic acid encoding an anti-TREM2 antibody heavy chain variable region comprises a sequence that is at least 80% identical, at least 90% identical, at least 95% identical, or at least 98% identical to a sequence selected from SEQ ID NOS:237-264 and 319-325. In other related embodiments, an isolated nucleic acid encoding an anti-TREM2 antibody heavy chain variable region comprises a sequence selected from SEQ ID NOS:237-264 and 319-325.

[0308] In some embodiments, the polynucleotide encodes the full length light chain and full length heavy chain. Exemplary polynucleotide sequences are provided in TABLE A15.

B. U.S. Pat. No. 8,231,878

[0309] In some embodiments, the TREM2 agonist is antibody, or an antigen-binding fragment thereof, as described in U.S. Pat. No. 8,231,878, which is incorporated by reference herein, in its entirety. In some embodiments, the TREM2 antibody is monoclonal antibody 29E3, or a fragment, homologue, derivative or variant thereof.

[0310] In some embodiments, the TREM2 antigen bind protein comprises a CDRL1, CDRL2, and CDRL3 of the light chain variable region, and a CDRH1, CDRH2, and CDRH3 of the heavy chain variable region of monoclonal antibody 29E3. Monoclonal antibody 29E3 is further described in Bouchon et al., J Exp Med., 2001, 194(8):1111-1122.

[0311] In some embodiments, the TREM2 antigen bind protein comprises a light chain variable region and a heavy chain variable region of monoclonal antibody 29E3.

[0312] In some embodiments, the TREM2 antigen bind protein is a chimeric antibody containing the light chain variable region and the heavy chain variable region of monoclonal antibody 29E3, and a human heavy chain constant region, such as a human Fc region, or an engineered variant thereof.

[0313] In some embodiments, the TREM2 antigen bind protein, e.g., a TREM2 antibody, competes with binding of monoclonal antibody 29E3 to TREM2.

C. U.S. Patent Application Publication No. US2019/0010230A1

[0314] In some embodiments, the TREM2 agonist is an antibody, or an antigen-binding fragment thereof, as described in U.S. Patent Application Publication No. US2019/0010230A1 ("the '230 application"), which is incorporated by reference herein, in its entirety.

[0315] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3 (also referred to as HVR-L1, HVR-L2, and HVR-L3, respectively), and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 (also referred to as HVR-H1, HVR-H2, and HVR-H3, respectively).

disclosed in the '230 application specification. In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '230 application specification.

[0316] In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises the HVR-H1, HVR-H2, and/or HVR-H3 of the monoclonal antibody Ab52; and/or wherein the light chain variable domain comprises the HVR-L1, HVR-L2, and/or HVR-L3 of the monoclonal antibody Ab52. In some embodiments, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:772. In some embodiments, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:773. In some embodiments, the HVR-H3 comprises the amino acid sequence of SEQ ID NO:774. In some embodiments, the HVR-L1 comprises the amino acid sequence of SEQ ID NO:775. In some embodiments, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:776. In some embodiments, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:777. In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises: (a) an HVR-H1 comprising the amino acid sequence of SEQ ID NO:772, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:772; (b) an HVR-H2 comprising the amino acid sequence of SEQ ID NO:773, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:773; and; and/or (c) an HVR-H3 comprising the amino acid sequence of SEQ ID NO:774, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:774; and/or wherein the light chain variable domain comprises: (a) an HVR-L1 comprising the amino acid sequence of SEQ ID NO:775, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:775; (b) an HVR-L2 comprising the amino acid sequence of SEQ ID NO:776, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:776; and/or (c) an HVR-L3 comprising the amino acid sequence of SEQ ID NO:777, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:777. In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises the HVR-H1, HVR-H2, and/or HVR-H3 of the monoclonal antibody Ab21; and/or wherein the light chain variable domain comprises the HVR-L1, HVR-L2, and/or HVR-L3 of the monoclonal antibody Ab21. In some embodiments, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:778. In some embodiments, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:779. In some embodiments, the HVR-H3 comprises the amino acid sequence of SEQ ID NO:780. In some embodiments, the HVR-L1 comprises the amino acid sequence of SEQ ID NO:781. In some embodiments, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:782. In some embodiments, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:783. In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises: (a) an HVR-H1 comprising the amino acid sequence of SEQ ID NO:778, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:778; (b) an HVR-H2 comprising the amino acid sequence of SEQ ID NO:779, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:779; and/or (c) an HVR-H3 comprising the amino acid sequence of SEQ ID NO:780, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:780, and/or wherein the light chain variable domain comprises: (a) an HVR-L1 comprising the amino acid sequence of SEQ ID NO:781, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:781; (b) an HVR-L2 comprising the amino acid sequence of SEQ ID NO:782, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:782; and/or (c) an HVR-L3 comprising the amino acid sequence of SEQ ID NO:783, or an amino acid

sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:783.

[0317] In some embodiments, the heavy chain variable domain comprises the HVR-H1, HVR-H2, and/or HVR-H3 of the monoclonal antibody Ab52; and/or wherein the light chain variable domain comprises the HVR-L1, HVR-L2, and/or HVR-L3 of the monoclonal antibody Ab52. In some embodiments, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:772. In some embodiments, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:773. In some embodiments, the HVR-H3 comprises the amino acid sequence of SEQ ID NO:774. In some embodiments, the HVR-L1 comprises the amino acid sequence of SEQ ID NO:775. In some embodiments, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:776. In some embodiments, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:777. In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises an HVR-H1 comprising the amino acid sequence of SEQ ID NO:772, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:773, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:774, and/or wherein the light chain variable domain comprises an HVR-L1 comprising the amino acid sequence of SEQ ID NO:775, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:776, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:777.

[0318] In some embodiments, the heavy chain variable domain comprises: (a) an HVR-H1 comprising the amino acid sequence of SEQ ID NO:772, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:772; (b) an HVR-H2 comprising the amino acid sequence of SEQ ID NO:773, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:773; and; and/or (c) an HVR-H3 comprising the amino acid sequence of SEQ ID NO:774, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:774; and/or wherein the light chain variable domain comprises: (a) an HVR-L1 comprising the amino acid sequence of SEQ ID NO:775, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:775; (b) an HVR-L2 comprising the amino acid sequence of SEQ ID NO:776, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:776; and/or (c) an HVR-L3 comprising the amino acid sequence of SEQ ID NO:777, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:777.

[0319] In some embodiments, the heavy chain variable domain comprises the HVR-H1, HVR-H2, and/or HVR-H3 of the monoclonal antibody Ab21; and/or wherein the light chain variable domain comprises the HVR-L1, HVR-L2, and/or HVR-L3 of the monoclonal antibody Ab21. In some embodiments, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:778. In some embodiments, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:779. In some embodiments, the HVR-H3 comprises the amino acid sequence of SEQ ID NO:780. In some embodiments, the HVR-L1 comprises the amino acid sequence of SEQ ID NO:781. In some embodiments, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:782. In some embodiments, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:783. In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises an HVR-H1 comprising the amino acid sequence of SEQ ID NO:778, an HVR-H2 comprising the amino acid sequence of SEQ ID NO:779, and an HVR-H3 comprising the amino acid sequence of SEQ ID NO:780, and/or wherein the light chain variable domain comprises an HVR-L1 comprising the amino acid sequence of SEQ ID NO:781, an HVR-L2 comprising the amino acid sequence of SEQ ID NO:782, and an HVR-L3 comprising the amino acid sequence of SEQ ID NO:783.

[0320] In some embodiments, the heavy chain variable domain comprises: (a) an HVR-H1 comprising the amino acid sequence of SEQ ID NO:778, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:778; (b) an HVR-H2 comprising the amino acid sequence of SEQ ID NO:779, or an amino acid sequence with at least about 95%

homology to the amino acid sequence of SEQ ID NO:779; and/or (c) an HVR-H3 comprising the amino acid sequence of SEQ ID NO:780, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:780, and/or wherein the light chain variable domain comprises: (a) an HVR-L1 comprising the amino acid sequence of SEQ ID NO:781, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:781; (b) an HVR-L2 comprising the amino acid sequence of SEQ ID NO:782, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:782; and/or (c) an HVR-L3 comprising the amino acid sequence of SEQ ID NO:783, or an amino acid sequence with at least about 95% homology to the amino acid sequence of SEQ ID NO:783.

[0321] In some embodiments, the antibody comprises a heavy chain variable domain and a light chain variable domain, wherein the heavy chain variable domain comprises: (a) an HVR-H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:3-24, 772, and 778; an HVR-H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:25-49, 773, and 779; and (c) an HVR-H3 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:50-119, 774, and 780; and/or wherein the light chain variable domain comprises: (a) an HVR-L1 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 120-137, 775, and 781; (b) an HVR-L2 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 138-152, 776, and 782; and (c) an HVR-L3 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 153-236, 777, and 783. In any of the above embodiments, the light chain variable domain and/or heavy chain variable domain comprises an amino acid sequence with at least about 90% homology to the amino acid sequence indicated.

[0322] In some embodiments, the antibody is an antibody disclosed in Tables 1A, 1B and 8 and FIGS. 20A and 20B of U.S. Patent Application Publication No. US2019/0010230A1, reproduced below as TABLES C1-C2.

TABLE-US-00017 TABLE C1 Kabat heavy chain CDR sequences Antibody Name CDR L1 CDR L2 CDR L3 Ab21 YSFTTYWIG IIYPGDS DTRYSPSFQG ARAGHYDGGHLGMDV (SEQ ID NO: 778) (SEQ ID NO: 779) (SEQ ID NO: 780) Ab52 YTFTSY YIH IINPSGGSTSYAQKFQG AREADDSSGYPLGLDV (SEQ ID NO: 772) (SEQ ID NO: 773) (SEQ ID NO: 774)

TABLE-US-00018 TABLE C2 Kabat light chain CDR sequences Antibody Name CDR L1 CDR L2 CDR L3 Ab21 RASQSVSSSYLA GASNRAT QQDDSA PYT (SEQ ID NO: 781) (SEQ ID NO: 782) (SEQ ID NO: 783) Ab52 RASQSVSSNLA GASTRAT QQVNSLPPT (SEQ ID NO: 775) (SEQ ID NO: 776) (SEQ ID NO: 777)

TABLE-US-00019 TABLE C3 Kabat CDR sequences Antibody Name CDR H1 CDR H2 CDR H3 CDR L1 CDR L2 CDR L3 Ab1 FTFSSYAMS VISGSGGSTYYADSVKG AKGTP TLLFQH RASQSVSSNLA GASTRAT QQLPYWPPT (SEQ ID NO: 377) (SEQ ID NO: 399) (SEQ ID NO: 424) (SEQ ID NO: 494) (SEQ ID NO: 512) (SEQ ID NO: 527) Ab2 FTFSSSAMS AISGSGGSTYYAD AKVPSYDYWSGYSNYYYY RASQSVGSNLA GASTRAT QQYFFYPPT (SEQ ID NO: 378) SVKG (SEQ ID NO: 400) MDV (SEQ ID NO: 425) (SEQ ID NO: 495) (SEQ ID NO: 512) (SEQ ID NO: 528) Ab3 GTFSSYAIS GIPIFGTANYAQK AREQYHVGMDV QASQDISNYLN DASNLAT QQP FNFPYT (SEQ ID NO: 379) FQG (SEQ ID NO: 401) (SEQ ID NO: 426) (SEQ ID NO: 496) (SEQ ID NO: 513) (SEQ ID NO: 529) Ab4 GTFSSYAIS GIPIFGTASYAQK ARGVDSIMDY RASQSVSSNLA SASTRAT QQDHDYPPT (SEQ ID NO: 379) FQG (SEQ ID NO: 402) (SEQ ID NO: 427) (SEQ ID NO: 494) (SEQ ID NO: 514) (SEQ ID NO: 530) Ab5 YTFTSY YIH IINPSGGSTSYAQK ARAPQESP YVFDI RASQSVSSSYLA GASSRAT QQYFSSPPT (SEQ ID NO: 380) FQG (SEQ ID NO: 403) (SEQ ID NO: 428) (SEQ ID NO: 497) (SEQ ID NO: 515) (SEQ ID NO: 531) Ab6 YTFTSY YMH IINPGGGSTSYAQKFQG

ARGSPYGYLYDP RASQSVSSYLA DASKRAT QQRVNLPPT (SEQ ID NO: 381)
(SEQ ID NO: 404) (SEQ ID NO: 429) (SEQ ID NO: 498) (SEQ ID NO: 516) (SEQ ID NO: 532) Ab7 YTFTSYMH IINPSGGSTTYAQKFQG ARTSSKERDY
RASQSVSSYLA DASKRAT QQRISYPIT (SEQ ID NO: 381) (SEQ ID NO: 405)
(SEQ ID NO: 430) (SEQ ID NO: 498) (SEQ ID NO: 516) (SEQ ID NO: 533) Ab8 GSISSSSYYWG SISYSGSTYYNPSL ARGPYRLLLGMDV RASQSISSYLN
GASSLQS QQIDDPIT (SEQ ID NO: 382) KS (SEQ ID NO: 406) (SEQ ID NO: 431) (SEQ ID NO: 499) (SEQ ID NO: 517) (SEQ ID NO: 534) Ab9
YSFTSYWIG IIPGDSDDTYSPS ARLHISGEVNWFD RASQSVSSYLA DASN RAT
QQFSYWPWT (SEQ ID NO: 383) FQG (SEQ ID NO: 407) P (SEQ ID NO: 432) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 535) Ab10
YSFTSNWIG IIPGDSDDTRYSPS AREAGYDYGELAF RASQSVSSSYLA GASSRAT
QQHDSPPPT (SEQ ID NO: 384) FQG (SEQ ID NO: 408) DI (SEQ ID NO: 433) (SEQ ID NO: 497) (SEQ ID NO: 515) (SEQ ID NO: 536) Ab11
YSFTTYWIG IIPGDSDDTRYSPS ARAGHYDGGH LGMDV RASQSVSSDYLA GASSRAT
QQDYSYPWT (SEQ ID NO: 385) FQG (SEQ ID NO: 408) (SEQ ID NO: 434)
(SEQ ID NO: 500) (SEQ ID NO: 515) (SEQ ID NO: 537) Ab12 YSFTSYWIG
IIPGDSDDTRYSPSFQG ARLGHYSGTVSSYGMDV RASQSISSYLN AASSLQS QQEYAVPYT
(SEQ ID NO: 383) (SEQ ID NO: 408) (SEQ ID NO: 435) (SEQ ID NO: 499) (SEQ ID NO: 519) (SEQ ID NO: 538) Ab13 YTFTSYGIS WISAYNGNTNYA
ARGPSHYDLA RASQSVSSYLA DASN RAT QQVSNYPIT (SEQ ID NO: 386)
QKLQG (SEQ ID NO: 409) (SEQ ID NO: 436) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 539) Ab14 GSISSGGYYWS NIYYSGSTVYNPS
ARGLYGYGVLDV QASQDISNYLN DASNLET QQVDNIPPT (SEQ ID NO: 387) LKS
(SEQ ID NO: 410) (SEQ ID NO: 437) (SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 540) Ab15 GSISSGGYYWS NIYYSGSTVYNPS ARGLYGYGVLDV
QASQDISNYLN DASNLET QQFD TYPT (SEQ ID NO: 387) LKS (SEQ ID NO: 410) (SEQ ID NO: 437) (SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 541) Ab16 GSISSNSYYWG SIYYSGSTYYNPS ARGVLGYGVFDY QASQDISNYLN
DASNLET QQFLNFPT (SEQ ID NO: 388) LKS (SEQ ID NO: 411) (SEQ ID NO: 438) (SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 542) Ab17
GSISSNSYYWG SIYYSGSTYYNPSL ARGVLGYGVFDY QASQDISNYLN DASNLET
QQFFNFPT (SEQ ID NO: 388) KS (SEQ ID NO: 411) (SEQ ID NO: 438)
(SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 543) Ab18 GSISSYYWS
SIYYSGSTNYNPSL ARDGGGEYPSGTP QASQDISNYLN DASNLET QQFIDL PFT (SEQ ID NO: 389) KS (SEQ ID NO: 412) FDI (SEQ ID NO: 439) (SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 544) Ab19 GSISSYYWS SIYYSGSTNYNPSL
ARDGGGEYPSGTP QASQDISNYLN DASNLET QQYYDL PFT (SEQ ID NO: 389) KS
(SEQ ID NO: 412) FDI (SEQ ID NO: 439) (SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 545) Ab20 GSISSYYWS SIYYSGSTNYNPSL ARSGMASFFDY
RASQSVSSDYLA GASSRAT QQFSSHPFT (SEQ ID NO: 389) KS (SEQ ID NO: 412) (SEQ ID NO: 440) (SEQ ID NO: 500) (SEQ ID NO: 515) (SEQ ID NO: 546) Ab22 YSFTTYWIG IIPGDSDDTRYSPS ARAGHYDGGH LG RASQSVSSSYLA
GASSRAT QQDDRSPYT (SEQ ID NO: 385) FQG (SEQ ID NO: 408) MDV
(SEQ ID NO: 434) (SEQ ID NO: 497) (SEQ ID NO: 515) (SEQ ID NO: 547) Ab23 FTFSSYAMS AISGSGGSTYYAD AKLGGHSM DV KSSQSVLYSSNNKNYLA
WASTRES QQAYLPIT (SEQ ID NO: 377) SVKG (SEQ ID NO: 400) (SEQ ID NO: 441) (SEQ ID NO: 501) (SEQ ID NO: 521) (SEQ ID NO: 548) Ab24
FTFSSYAMS AISGSGGSTYYAD AKPLKRGRGFY RASQSISSYLN AASSLQS
QQA FSPPPWT (SEQ ID NO: 377) SVKG (SEQ ID NO: 400) (SEQ ID NO:

442)(SEQ ID NO: 499)(SEQ ID NO: 519)(SEQ ID NO: 549) Ab25
FTFSSYAMS VISGSGGSTYYAD AKEGRTITMD RASQSVSSSYLA GASSRAT QQDDRSPT
(SEQ ID NO: 377)SVKG (SEQ ID NO: 399)(SEQ ID NO: 443)(SEQ ID
NO: 497)(SEQ ID NO: 515)(SEQ ID NO: 550) Ab26 FTFSSYAMS
VISGSGGSTYYAD AKDQYSVLDY RASQSVSSSYLA DASNRAT QQEFDLPFT (SEQ ID
NO: 377)SVKG (SEQ ID NO: 399)(SEQ ID NO: 444)(SEQ ID NO: 498)
(SEQ ID NO: 518)(SEQ ID NO: 551) Ab27 FTFSSYAMS AISGSGGSTYYAD
AKKYSSRGVYFDY RASQSVSSSYLA DASNRAT QQYNNFPPT (SEQ ID NO: 377)
SVKG (SEQ ID NO: 400)(SEQ ID NO: 445)(SEQ ID NO: 498)(SEQ ID
NO: 518)(SEQ ID NO: 552) Ab28 FTFSSYAMS AISGSGGSTYYAD
ARLGGAVGARHVT RASQSVSSSYLA DASKRAT QQRYLRPIT (SEQ ID NO: 377)
SVKG (SEQ ID NO: 400)YFDY (SEQ ID NO: 446)(SEQ ID NO: 498)
(SEQ ID NO: 516)(SEQ ID NO: 553) Ab29 FTFSSYGMH VISYDGSNKYYAD
ARGQYYGGSGWF RASQSVSSSYLA GASSRAT QQPGAVPT (SEQ ID NO: 390)
SVKG (SEQ ID NO: 413)DP (SEQ ID NO: 447)(SEQ ID NO: 497)(SEQ
ID NO: 515)(SEQ ID NO: 554) Ab30 FTFSSYAMS AISGSGGSTYYAD
ARLGQEYAYFQH RASQSISSYLN GASSLQS QQVYITPIT (SEQ ID NO: 377)SVKG
(SEQ ID NO: 400)(SEQ ID NO: 448)(SEQ ID NO: 499)(SEQ ID NO:
517)(SEQ ID NO: 555) Ab31 FTFSSYGMH LIWYDGSNKYYA ARRRDGYDEVFD
QASQDISNFLN DASNLET QQPVDLPFT (SEQ ID NO: 390)DSVKG (SEQ ID
NO: 414)I (SEQ ID NO: 449)(SEQ ID NO: 502)(SEQ ID NO: 520)(SEQ
ID NO: 556) Ab32 FTFSSYAMS AISGSGGSTYYAD ARVPKHYVVDY RASQSVSSSYLA
DASNRAT QQYSFFPPT (SEQ ID NO: 377)SVKG (SEQ ID NO: 400)(SEQ ID
NO: 450)(SEQ ID NO: 498)(SEQ ID NO: 518)(SEQ ID NO: 557) Ab33
FTFSSYGMH VISYDGSNKYYAD ARAGGHLFDY RASQSVSSSYLA DASNRAT QQDSSFPPT
(SEQ ID NO: 390)SVKG (SEQ ID NO: 413)(SEQ ID NO: 451)(SEQ ID
NO: 498)(SEQ ID NO: 518)(SEQ ID NO: 558) Ab34 FTFSSYGMH
VISYDGSNKYYAD ARDRGGEYVDFAF RASQSISSYLN AASSLQS QQSDFPWPWT (SEQ
ID NO: 390)SVKG (SEQ ID NO: 413)DI (SEQ ID NO: 452)(SEQ ID
NO: 499)(SEQ ID NO: 519)(SEQ ID NO: 559) Ab35 FTFSSYAMS
AISGSGGSTYYAD ARTRSGYGASNYF RASQSISSYLN AASSLQS QQGYSAIPIT (SEQ
ID NO: 377)SVKG (SEQ ID NO: 400)DY (SEQ ID NO: 453)(SEQ ID
NO: 499)(SEQ ID NO: 519)(SEQ ID NO: 560) Ab36 FTFSTYGMH
VIWYDGSNKYYA ARGTGAAAASPAF RASQSVSSSYLA DASNRAT QQLFDWPT (SEQ
ID NO: 391)DSVKG (SEQ ID NO: 415)DI (SEQ ID NO: 454)(SEQ ID
NO: 498)(SEQ ID NO: 518)(SEQ ID NO: 561) Ab37 FTFSSYAMS
AISGSGGSTYYAD ARVGQYMLGMDV RASQSVSSSYLA DASNRAT QQRAFLFT (SEQ
ID NO: 377)SVKG (SEQ ID NO: 400)(SEQ ID NO: 455)(SEQ ID NO:
498)(SEQ ID NO: 518)(SEQ ID NO: 562) Ab38 FTFSTYGMH VIWYDGSNKYYAD
ARGAPVDYGGIEP RASQSVSSSYLA DASNRAT QQIDFLPYT (SEQ ID NO: 391)
SVKG (SEQ ID NO: 415)EYFQH (SEQ ID NO: 456)(SEQ ID NO: 498)
(SEQ ID NO: 518)(SEQ ID NO: 563) Ab39 FTFSSYAMS AISGSGGSTYYAD
AKHYHVGIAFDI RASQSISSYLN AASSLQS QQVYSPPT (SEQ ID NO: 377)SVKG
(SEQ ID NO: 400)(SEQ ID NO: 457)(SEQ ID NO: 499)(SEQ ID NO:
519)(SEQ ID NO: 564) Ab40 FTFSSYAMS AISGSGGSTYYAD ARTRSGYGASNYF
RASQSISSYLN AASSLQS QQGYAAPIT (SEQ ID NO: 377)SVKG (SEQ ID NO:
400)DY (SEQ ID NO: 453)(SEQ ID NO: 499)(SEQ ID NO: 519)(SEQ
ID NO: 565) Ab41 FTFSTYAMS AISGSGGSTYYAD ARAMARKSVAFDI RASQSVSSSYLA
DASNRAT QQRYPALPIT (SEQ ID NO: 392)SVKG (SEQ ID NO: 400)(SEQ ID
NO: 458)(SEQ ID NO: 498)(SEQ ID NO: 518)(SEQ ID NO: 566) Ab42

FTFSSMS AISGSGSTYYAD AKSPAVAGIYRAD RASQSVSSSYLA GASSRAT
QQYASPPIT (SEQ ID NO: 378) SVKG (SEQ ID NO: 400) P (SEQ ID NO:
459) (SEQ ID NO: 497) (SEQ ID NO: 515) (SEQ ID NO: 567) Ab43
FTFSSSAMS AISGSGGSTYYAD AKSPAVAGIYRAD RASQSVSSSYLA AASSLQS
QQVYSTPIT (SEQ ID NO: 378) SVKG (SEQ ID NO: 400) Y (SEQ ID NO:
460) (SEQ ID NO: 503) (SEQ ID NO: 519) (SEQ ID NO: 568) Ab44
FTFSTYGMH VIWYDGSNKYYAD ARGTGAAAASPAF RASQSVSSSYLA DSSNRAT
QQLVHWPT (SEQ ID NO: 391) SVKG (SEQ ID NO: 415) DI (SEQ ID NO:
454) (SEQ ID NO: 498) (SEQ ID NO: 522) (SEQ ID NO: 569) Ab45
YTFTSYMH IINPSGGSTSYAQK ARGPYTTALDYY RASQSVSSNLA GASTRAT
QQLDDWFT (SEQ ID NO: 381) FQG (SEQ ID NO: 403) YMDV (SEQ ID
NO: 461) (SEQ ID NO: 494) (SEQ ID NO: 512) (SEQ ID NO: 570) Ab46
YTFTSYMH IINPSGGSTSYAQK ARPAKTADY RASQSVSSSYLA DSSNRAT QQRSNYPIT
(SEQ ID NO: 381) FQG (SEQ ID NO: 403) (SEQ ID NO: 462) (SEQ ID
NO: 498) (SEQ ID NO: 522) (SEQ ID NO: 571) Ab47 YTFTSYMH
IINPSGGSTTYAQK ARPGKSMDV RASQSVSSSYLA DASNRAT QQRILYPIT (SEQ ID
NO: 381) FQG (SEQ ID NO: 405) (SEQ ID NO: 463) (SEQ ID NO: 498)
(SEQ ID NO: 518) (SEQ ID NO: 572) Ab48 YTFTSYMH IINPSGGSTTYAQK
ARPGKSMDV RASQSVSSSYLA DASNRAT QQRAAYPIT (SEQ ID NO: 381) FQG
(SEQ ID NO: 405) (SEQ ID NO: 463) (SEQ ID NO: 498) (SEQ ID NO:
518) (SEQ ID NO: 573) Ab49 YTFTSYMH IINPSGGSTSYAQK ARPAKTADY
RASQSVSSSYLA DASKRAT QQRTSHPIT (SEQ ID NO: 381) FQG (SEQ ID NO:
403) (SEQ ID NO: 462) (SEQ ID NO: 498) (SEQ ID NO: 516) (SEQ ID
NO: 574) Ab50 YTFTSYIH IINPSGGSTSYAQK ARAPQESPYVFDI RASQSVSSSYLA
GASSRAT QQYAGSPFT (SEQ ID NO: 380) FQG (SEQ ID NO: 403) (SEQ ID
NO: 428) (SEQ ID NO: 497) (SEQ ID NO: 515) (SEQ ID NO: 575) Ab51
YTFTSYMH IINPSGGSTSYAQK ARGVGGQDYHYM RASQSISSYLN AASSLQS
QQFDDVFT (SEQ ID NO: 381) FQG (SEQ ID NO: 403) DV (SEQ ID NO:
464) (SEQ ID NO: 499) (SEQ ID NO: 519) (SEQ ID NO: 576) Ab53
YTFTSYIH IINPSGGSTSYAQK ARAPQESPYVFDI RASQSVSSSYLA GASSRAT
QQYVNSPFT (SEQ ID NO: 380) FQG (SEQ ID NO: 403) (SEQ ID NO: 428)
(SEQ ID NO: 497) (SEQ ID NO: 515) (SEQ ID NO: 577) Ab54 YTFTSYMH
IINPSGGSTSYAQK ARGPYTTALDYY RASQSINSYLN AASSLQS QQSDDDPFT (SEQ
ID NO: 381) FQG (SEQ ID NO: 403) YMDV (SEQ ID NO: 461) (SEQ ID
NO: 504) (SEQ ID NO: 519) (SEQ ID NO: 578) Ab55 YTFTGSYMH
WINPNSGGTNYAQ ARGPLYHPMIFDY RASQSVSSSYLA DASNRAT QQLSTYPLT (SEQ
ID NO: 393) KFQG (SEQ ID NO: 416) (SEQ ID NO: 465) (SEQ ID NO:
498) (SEQ ID NO: 518) (SEQ ID NO: 579) Ab56 YTFTGYMH SINPNSGGTNYAQ
ARASSVDN RASQSVSSSYLA DASNRAT QQRSVYPIT (SEQ ID NO: 394) KFQG
(SEQ ID NO: 417) (SEQ ID NO: 466) (SEQ ID NO: 498) (SEQ ID NO:
518) (SEQ ID NO: 580) Ab57 YTFTNYGIS WISAYNGNTNYA ARGPTKAYYGS GS
RASQSVSSSYLA DASKRAT QQVSLFPLT (SEQ ID NO: 395) QKLQG (SEQ ID
NO: 409) YVVFDP (SEQ ID NO: 467) (SEQ ID NO: 498) (SEQ ID NO: 516)
(SEQ ID NO: 581) Ab58 YSFTSYWIG IYPGDS DTRYSPS ARLGIYSTGATAF
RASQSISSWLA DASSLES LDYNSYSPIT (SEQ ID NO: 383) FQG (SEQ ID NO:
408) DI (SEQ ID NO: 468) (SEQ ID NO: 505) (SEQ ID NO: 523) (SEQ ID
NO: 582) Ab59 YTFTGSYMH WINPNSGGTNYAQ ARGGVWYSLFDI QASQDISNYLN
DASNLET QQHIALPFT (SEQ ID NO: 393) KFQG (SEQ ID NO: 416) (SEQ ID
NO: 469) (SEQ ID NO: 496) (SEQ ID NO: 520) (SEQ ID NO: 583) Ab60
YTFTGYMH WINPNSGGTSYAQ ARASKMGDD RASQSVSSSYLA DASKRAT QQRASMPIT

(SEQ ID NO: 394) KFQG (SEQ ID NO: 418) (SEQ ID NO: 470) (SEQ ID NO: 498) (SEQ ID NO: 516) (SEQ ID NO: 584) Ab61 YTFTSYGIH WISAYNGNTNYA ARGGVPRVS YFQH RASQSVSSYLA DSSNRAT QQAFNR PPT (SEQ ID NO: 396) QKLQG (SEQ ID NO: 409) (SEQ ID NO: 471) (SEQ ID NO: 498) (SEQ ID NO: 522) (SEQ ID NO: 585) Ab62 YSFTSYWIG IYPGDS DTRYSPS ARAGHYDDWSGL RASQSVSSYLA DASKRAT QQSSVHPYT (SEQ ID NO: 383) FQG (SEQ ID NO: 408) GLDV (SEQ ID NO: 472) (SEQ ID NO: 498) (SEQ ID NO: 516) (SEQ ID NO: 586) Ab63 YTFTSYGIS WISTYNGNTNYAQ ARGSGSGYDSWY RASQGIDSWLA AASSLQS QQAYSLPPT (SEQ ID NO: 386) KLQG (SEQ ID NO: 419) D (SEQ ID NO: 473) (SEQ ID NO: 506) (SEQ ID NO: 519) (SEQ ID NO: 587) Ab64 YSFTSYWIG IYPGDS DTRYSPS ARLGRWSSGSTAF RASQSVSSNLA GASTRAT QQDDDGYT (SEQ ID NO: 383) FQG (SEQ ID NO: 408) DI (SEQ ID NO: 474) (SEQ ID NO: 494) (SEQ ID NO: 512) (SEQ ID NO: 588) Ab65 YSFTSYWIG IYPGDS DTRYSPS ARLGRKPSGSVAF RASQSVSSYLA DASNRAT QQDYSWPYT (SEQ ID NO: 383) FQG (SEQ ID NO: 408) DI (SEQ ID NO: 475) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 589) Ab66 YTFTGSYMH WINPNSGGTNYAQ ARAGHKTHDY RASQSVSSYLA DASNRAT QQRSAYPIT (SEQ ID NO: 393) KFQG (SEQ ID NO: 416) (SEQ ID NO: 476) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 590) Ab67 YTFTSYMH IINPSGGSTTYAQK ARPGKSMDV RASQSVSSYLA DASNRAT QQRSHFPIT (SEQ ID NO: 381) FQG (SEQ ID NO: 405) (SEQ ID NO: 463) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 591) Ab68 FTFSSYGMH LIWYDGSNKYYA AKPGSMTDY RASQSVSSYLA DASNRAT QQRANYPIT (SEQ ID NO: 390) DSVKG (SEQ ID NO: 414) (SEQ ID NO: 477) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 592) Ab69 YTFTGSYMH WINPNSGGTNYAQ ARAKSVDHDY RASQSVSSYLA DASNRAT QQRADYPIT (SEQ ID NO: 393) KFQG (SEQ ID NO: 416) (SEQ ID NO: 478) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 593) Ab70 YTFTGYMH WINPNSGGTSYQA ARASKMGDD RASQSVSSYLA DASNRAT QQRSVYPIT (SEQ ID NO: 394) KFQG (SEQ ID NO: 418) (SEQ ID NO: 470) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 580) Ab71 YTFTSYMH IINPSGGSTSYAQK ARDISTHDYDLAF RASQSVSSSYLA GASNRAT QQAGSHPFT (SEQ ID NO: 381) FQG (SEQ ID NO: 403) DI (SEQ ID NO: 479) (SEQ ID NO: 497) (SEQ ID NO: 524) (SEQ ID NO: 594) Ab72 GSISSYYWS SIYYSGSTNYNPSL ARSGTETLFDY QASQDITNYLN DASNLET QQDVNYPPT (SEQ ID NO: 389) KS (SEQ ID NO: 412) (SEQ ID NO: 480) (SEQ ID NO: 507) (SEQ ID NO: 520) (SEQ ID NO: 595) Ab73 YSFTSYWIG IYPGDS DTTYSPS ARAKMLDDGYAF RASQSVSSNLA GASTRAT QQDDNYPYT (SEQ ID NO: 383) FQG (SEQ ID NO: 407) DI (SEQ ID NO: 481) (SEQ ID NO: 494) (SEQ ID NO: 512) (SEQ ID NO: 596) Ab74 YTFTGSYMH WINPNSGGTNYAQ ARAGHKTHDY RASQSVSSYLA DASNRAT QQRSTFPIT (SEQ ID NO: 393) KFQG (SEQ ID NO: 416) (SEQ ID NO: 476) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 597) Ab75 YTFTGYMH WINPNSGGTNYAQ ARDLGYSSLLALDI RASQSVSSYLA DASNRAT QQVSNYPFT (SEQ ID NO: 394) KFQG (SEQ ID NO: 416) (SEQ ID NO: 482) (SEQ ID NO: 498) (SEQ ID NO: 518) (SEQ ID NO: 598) Ab76 FTFSSYSMN SSSSSSYIYYADS ARGGGRRGDNNW KSSQSVLYSSNNKNYLA WASTRES QQYHDAPIT (SEQ ID NO: 397) VKG (SEQ ID NO: 420) FDP (SEQ ID NO: 483) (SEQ ID NO: 501) (SEQ ID NO: 521) (SEQ ID NO: 599) Ab77 FTFSSYGMH VISYDGSNKYYAD ARGPPHEMDY KSSQSVLYSSNNKNYLA WASTRES QQAYVVPPT (SEQ ID NO: 390) SVKG (SEQ ID NO: 413) (SEQ ID NO: 484) (SEQ ID

NO: 501)(SEQ ID NO: 521)(SEQ ID NO: 600)Ab78 FTFSSYGMH
VIWYDGSNKYYAARTPYWYFDL RASQSVSSYLA DASN RAT QQADNWPFT (SEQ
ID NO: 390)DSVKG (SEQ ID NO: 415)(SEQ ID NO: 485)(SEQ ID NO:
498)(SEQ ID NO: 518)(SEQ ID NO: 601)Ab79 FTFSSYSMN YISGSSSTIYYADS
ARGGRRHYGGMD RSSQSLLSNGY LGSHRAS MQALESPT (SEQ ID NO: 397)
VKG (SEQ ID NO: 421)V (SEQ ID NO: 486)NYLD (SEQ ID NO: 508)
(SEQ ID NO: 525)(SEQ ID NO: 602)Ab80 GTFSSY AIS GIPIFGTANYAQK
ARGGGTFWGSW RASQSVSSYLA DASN RAT QQYVNW PFT (SEQ ID NO: 379)
FQG (SEQ ID NO: 401)ALY (SEQ ID NO: 487)(SEQ ID NO: 498)(SEQ
ID NO: 518)(SEQ ID NO: 603)Ab81 GTFSSY AIS GIPIFGTANYAQK
ARDSGNYDYWSG RASQSVSSYLA DASN RAT QQSSNWPWT (SEQ ID NO: 379)
FQG (SEQ ID NO: 401)ALRY (SEQ ID NO: 488)(SEQ ID NO: 498)
(SEQ ID NO: 518)(SEQ ID NO: 604)Ab82 GSISGGYYWS YIYSGSTVYNPS
ARVSSSWYKA RASQGISSWLA AASSLQS QQASTFPIT (SEQ ID NO: 387)LKS
(SEQ ID NO: 422)(SEQ ID NO: 489)(SEQ ID NO: 509)(SEQ ID NO:
519)(SEQ ID NO: 605)Ab83 GSFSGGYYWS EIDHSGSTKYNPSL ARVGVVVGRPGYS
RASQGISSWLA AASSLQS QQRNSLPLT (SEQ ID NO: 398)KS (SEQ ID NO:
423)AFDI (SEQ ID NO: 490)(SEQ ID NO: 509)(SEQ ID NO: 519)(SEQ
ID NO: 606)Ab84 YTFTSYGIS WISTYNGNTNYAQ ARGSGSGYDSWY RASQSISSYLN
AASSLQS QQSDFPIT (SEQ ID NO: 386)KLQG (SEQ ID NO: 419)D (SEQ
ID NO: 473)(SEQ ID NO: 499)(SEQ ID NO: 519)(SEQ ID NO: 607)Ab85
FTFSSYGMH VIWYDGSNKYYAD AKDLGGYYGGAA RASQDISSWLA AASSLQS
QQEVDYPPLT (SEQ ID NO: 390)SVKG (SEQ ID NO: 415)YGMDV (SEQ
ID NO: 491)(SEQ ID NO: 510)(SEQ ID NO: 519)(SEQ ID NO: 608)Ab86
FTFSSYGMH VISYDGSNKYYAD AKDGVYYGLGNW RASQSISSWLA KASSLES
QQLNSYSPT (SEQ ID NO: 390)SVKG (SEQ ID NO: 413)FDP (SEQ ID
NO: 492)(SEQ ID NO: 505)(SEQ ID NO: 526)(SEQ ID NO: 609)Ab87
GSISYYWS SIYSGSTNYPNPSL ARHGWD RVGWFD RASQSVSRYLA DASN RAT
QQYIFWPPT (SEQ ID NO: 389)KS (SEQ ID NO: 412)P (SEQ ID NO: 493)
(SEQ ID NO: 511)(SEQ ID NO: 518)(SEQ ID NO: 610)

TABLE-US-00020 TABLE C4 Heavy chain variable regions Ab ID HC Variable
Region Sequences Ab 1

EVQLLES GGGGLVQP GGSRLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSVISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKGTPTLLFQHWGQGTLVTVSS
(SEQ ID NO: 616)Ab 2

EVQLLES GGGGLVQP GGSRLRLSCAASGFTFSSSAMSWSVRQAPGKGLEWVSAISGSGGSTYY
ADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKVPSYDYWSGYSNYYYYMDV
WGKGTTVTVSS (SEQ ID NO: 618)Ab 3

QVQLVQSGAEVKKPGSSVKV SCKASGGTFSSY AISWVRQAPGQGLEWMGGIPIFGTANY
AQKFQGRVTITADESTSTAYMELSSLRSED TAVYYCAREQYHVGMDVWGKGTTVTVSS
(SEQ ID NO: 620)Ab 4

QVQLVQSGAEVKKPGSSVKV SCKASGGTFSSY AISWVRQAPGQGLEWMGGIPIFGTASY
AQKFQGRVTITADESTSTAYMELSSLRSED TAVYYCARGVDSIMDYWGQGTLVTVSS
(SEQ ID NO: 622)Ab 5

QVQLVQSGAEVKKPGASVKV SCKASGYTFTSY YTHWVRQAPGQGLEWMGIINPSGGSTS
YAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYCARAPQESPYVFDI WGQGTMTVTV
SS (SEQ ID NO: 624)Ab 6

QVQLVQSGAEVKKPGASVKV SCKASGYTFTSY YMHWVRQAPGQGLEWMGIINPGGGST
SYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYCARGSPTYGYLYDPWGQGTLT
VSS (SEQ ID NO: 626)Ab 7

QVQLVQSGAEVKKPGESLKISCKGSGYSFTTSYMHVWRQAPGQGLEWMGIINPSGGST
TYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYCARTSSKERDYWGQGGLTLTVSS
(SEQ ID NO: 628) Ab 8

QLQLQESGPGLVKPSETLSLTCTVSGGSISSSSYYWGWIRQPPGKGLEWIGSISYSGSTYYN
PSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARGPYRLLLGMDVWGQGTTVTVSS
(SEQ ID NO: 630) Ab 9

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDSDDT
YSPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYYCARLHISGEVNWFDPWGQGGLTLTV
SS (SEQ ID NO: 632) Ab 10

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSNWIGWVRQMPGKGLEWMGITYPGDSDDTR
YSPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYYCAREAGYDYGELAFDIWGQGTMTV
TVSS (SEQ ID NO: 634) Ab 11

EVQLVQSGAEVKKPGESLKISCKGSGYSFTTYWIGWVRQMPGKGLEWMGITYPGDSDDTR
YSPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYYCARAGHYDGGHLGMDVWGQGTT
VTVSS (SEQ ID NO: 636) Ab 12

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDSDDTR
YSPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYYCARLGHYSGTVSSYGMDVWGQGT
TVTVSS (SEQ ID NO: 638) Ab 13

QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYGISWVRQAPGQGLEWMGWISAYNGNT
NYAQKLQGRVTMTTDTSTSTAYMELRSLRSDDTAVYYCARGPSHYDLAWGQGGLTLTV
SS (SEQ ID NO: 640) Ab 14

QVQLQESGPGLVKPSQTLSTCTVSGGSISSGGYYWSWIRQHPGKGLEWIGNIYYSGSTVY
NPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARGLYGYGVLDVWGQGTMTVTVSS
(SEQ ID NO: 642) Ab 15

QVQLQESGPGLVKPSQTLSTCTVSGGSISSGGYYWSWIRQHPGKGLEWIGNIYYSGSTVY
NPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARGLYGYGVLDVWGQGTMTVTVSS
(SEQ ID NO: 642) Ab 16

QLQLQESGPGLVKPSETLSLTCTVSGGSISSNSYYWGWIRQPPGKGLEWIGSIYYSGSTYY
NPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARGVLGYGVFDYWGQGGLTLTVSS
(SEQ ID NO: 645) Ab 17

QLQLQESGPGLVKPSETLSLTCTVSGGSISSNSYYWGWIRQPPGKGLEWIGSIYYSGSTYY
NPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARGVLGYGVFDYWGQGGLTLTVSS
(SEQ ID NO: 645) Ab 18

QVQLQESGPGLVKPSETLSLTCTVSGGSISSYYWSWIRQPPGKGLEWIGSIYYSGSTNYNPS
LKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARDGGGEYPSGTPFDIWGQGTMTVTVSS
(SEQ ID NO: 648) Ab 19

QVQLQESGPGLVKPSETLSLTCTVSGGSISSYYWSWIRQPPGKGLEWIGSIYYSGSTNYNPS
LKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARDGGGEYPSGTPFDIWGQGTMTVTVSS
(SEQ ID NO: 648) Ab 20

QVQLQESGPGLVKPSETLSLTCTVSGGSISSYYWSWIRQPPGKGLEWIGSIYYSGSTNYNPS
LKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARSGMASFFDYWGQGGLTLTVSS (SEQ
ID NO: 651) Ab 22

EVQLVQSGAEVKKPGESLKISCKGSGYSFTTYWIGWVRQMPGKGLEWMGITYPGDSDDTR
YSPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYYCARAGHYDGGHLGMDVWGQGTT
VTVSS (SEQ ID NO: 636) Ab 23

EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKLGGHSM DVWGQGTTVTVSS
(SEQ ID NO: 654) Ab 24

EVQLLESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKPLKRGRGFYWGQGGLTLTVSS

(SEQ ID NO: 656) Ab 25
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSVISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKEGRTITMDWGQGT LVT VSS
(SEQ ID NO: 658) Ab 26
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSVISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKDQYSVLDYWGQGT LVT VSS
(SEQ ID NO: 660) Ab 27
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKKYSSRGVYFDYWGQGT LVT
VSS (SEQ ID NO: 662) Ab 28
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARLGGA VGARHVTYFDYWGQGT
LVT VSS (SEQ ID NO: 664) Ab 29
QVQLVES GGGGVVQP GRS LRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVISYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARGQYYGGSGWFD P WGQGT L
VT VSS (SEQ ID NO: 666) Ab 30
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARLGQEYAYFQHWGQGT LVT V
SS (SEQ ID NO: 668) Ab 31
QVQLVES GGGGVVQP GRS LRLSCAASGFTFSSYGMHWVRQAPGKGLEWVALIWYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARRRDGY YDEVFDI WGQGT M
VT VSS (SEQ ID NO: 670) Ab 32
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARVPKH YV VLDYWGQGT LVT V
SS (SEQ ID NO: 672) Ab 33
QVQLVES GGGGVVQP GRS LRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVISYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARAGGH LFDYWGQGT LVT V
S (SEQ ID NO: 674) Ab 34
QVQLVES GGGGVVQP GRS LRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVISYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARDRGGEY VDFAFDI WGQGT
MVT VSS (SEQ ID NO: 676) Ab 35
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARTRSGY GASNYFDYWGQGT L
VT VSS (SEQ ID NO: 678) Ab 36
QVQLVES GGGGVVQP GRS LRLSCAASGFTFSTYGMHWVRQAPGKGLEWVAVIWYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARGTGAAA SPAFDI WGQGT
MVT VSS (SEQ ID NO: 680) Ab 37
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARVGQYMLGMDVWGQGT T VT
VSS (SEQ ID NO: 682) Ab 38
QVQLVES GGGGVVQP GRS LRLSCAASGFTFSTYGMHWVRQAPGKGLEWVAVIWYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARGAPVDYGGIEPEYFQHWGQ
GT LVT VSS (SEQ ID NO: 684) Ab 39
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKH YHV GIAFDI WGQGT MVT V
S (SEQ ID NO: 686) Ab 40
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSSYAMSWVRQAPGKGLEWVSAISGSGGSTY
YADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARTRSGY GASNYFDYWGQGT L
VT VSS (SEQ ID NO: 678) Ab 41
EVQLLES GGGGLVQP GGS LRLSCAASGFTFSTYAMSWVRQAPGKGLEWVSAISGSGGSTY

YADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCARAMARKSVAFDIWGQGTMTV
VSS (SEQ ID NO: 689) Ab 42
EVQLLES GGGGLVQP GGSRLSCAASGFTFSSSAMS WVRQAPGKGLEWVSAISGSGGSTYY
ADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAKVPSYQRGTAFDPWGQGT LTV
SS (SEQ ID NO: 691) Ab 43
EVQLLES GGGGLVQP GGSRLSCAASGFTFSSSAMS WVRQAPGKGLEWVSAISGSGGSTYY
ADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCAKSPAVAGIYRADYWGQGT LTV
SS (SEQ ID NO: 693) Ab 44
QVQLVES GGGGVVQP GRSLRLSCAASGFTFSTYGMHWVRQAPGKGLEWVAVIWDGSNK
YYADSVKGRFTISRDN SKNTLYLQMN SLRAEDTAVYYCARGTGAAAASPAFDIWGQGT
MTVTVSS (SEQ ID NO: 680) Ab 45
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
SYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARGPGYTTALDYYYMDVWGK
GTTVTVSS (SEQ ID NO: 696) Ab 46
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
SYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARPAKTADYWGQGT LTVTVSS
(SEQ ID NO: 698) Ab 47
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
TYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARPGKSMDVWGQGT TTVTVSS
(SEQ ID NO: 700) Ab 48
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
TYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARPGKSMDVWGQGT TTVTVSS
(SEQ ID NO: 700) Ab 49
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
SYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARPAKTADYWGQGT LTVTVSS
(SEQ ID NO: 698) Ab 50
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYIHWVRQAPGQGLEWMGIINPSGGSTS
Y AQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARAPQESPYVFDIWGQGTMTV
SS (SEQ ID NO: 624) Ab 51
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
SYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARGVGGQDYYYMDVWGKGT
TVTVTVSS (SEQ ID NO: 705) Ab 53
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYIHWVRQAPGQGLEWMGIINPSGGSTS
Y AQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARAPQESPYVFDIWGQGTMTV
SS (SEQ ID NO: 624) Ab 54
QVQLVQSGAEVKKPGASVKV SCKASGYTFTSYMH WVRQAPGQGLEWMGIINPSGGST
SYAQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCARGPGYTTALDYYYMDVWGK
GTTVTVTVSS (SEQ ID NO: 696) Ab 55
QVQLVQSGAEVKKPGASVKV SCKASGYTFTGSYMH WVRQAPGQGLEWMGWINPNSGG
TNYAQKFQGRVTMTRDTSISTAYME LSR LRSDDTAVYYCARGPLYHPMIFDYWGQGT LTV
TVSS (SEQ ID NO: 709) Ab 56
QVQLVQSGAEVKKPGASVKV SCKASGYTFTGYMH WVRQAPGQGLEWMGSINPNSGG
TNYAQKFQGRVTMTRDTSISTAYME LSR LRSDDTAVYYCARASSVDNWGQGT LTVTVSS
(SEQ ID NO: 711) Ab 57
QVQLVQSGAEVKKPGASVKV SCKASGYTFTNYGISWVRQAPGQGLEWMGWISAYNGNT
NYAQKLQGRVTMTTDTSTSTAYME LSR LRSDDTAVYYCARGPTKAYYGSGSYVFDPW
GQGT LTVTVSS (SEQ ID NO: 713) Ab 58
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDS DTR
YSPSFQGQVTISADKSISTAYLQWSSLKASDTAMYYCARLGIYSTGATAFDIWGQGTMTV
VSS (SEQ ID NO: 715) Ab 59

QVQLVQSGAEVKKPGASVKVSCASGYTFTGSYMHWVRQAPGQGLEWMGWINPNSGG
TNYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARGGVWYSLFDIWGQGTMTV
TVSS (SEQ ID NO: 717)Ab 60

QVQLVQSGAEVKKPGASVKVSCASGYTFTGYMHVVRQAPGQGLEWMGWINPNSGG
TSYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARASKMGDDWGQGTTLVTVS
S (SEQ ID NO: 719)Ab 61

QVQLVQSGAEVKKPGASVKVSCASGYTFTSYGIHWVRQAPGQGLEWMGWISAYNGNT
NYAQKLQGRVTMTTDTSTSTAYMELSLRSDDTAVYYCARGGVPRVSFQHWGQGTTLV
TVSS (SEQ ID NO: 721)Ab 62

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDS DTR
YSPSFQGGVTISADKSISTAYLQWSSLKASDTAMYYCARAGHYDDWSGLGLDVWGQGT
MTVTVSS (SEQ ID NO: 723)Ab 63

QVQLVQSGAEVKKPGASVKVSCASGYTFTSYGISWVRQAPGQGLEWMGWISTYNGNT
NYAQKLQGRVTMTTDTSTSTAYMELSLRSDDTAVYYCARGSGSGYDSWYDWGQGT
LTVTVSS (SEQ ID NO: 725)Ab 64

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDS DTR
YSPSFQGGVTISADKSISTAYLQWSSLKASDTAMYYCARLGRWSSGSTAFDIWGQGTMTV
TVSS (SEQ ID NO: 727)Ab 65

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDS DTR
YSPSFQGGVTISADKSISTAYLQWSSLKASDTAMYYCARLGRKPSGSVAFDIWGQGTMTV
TVSS (SEQ ID NO: 729)Ab 66

QVQLVQSGAEVKKPGASVKVSCASGYTFTGSYMHVVRQAPGQGLEWMGWINPNSGG
TNYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARAGHKTHDYWGQGTTLVTV
SS (SEQ ID NO: 731)Ab 67

QVQLVQSGAEVKKPGASVKVSCASGYTFTSYMHVVRQAPGQGLEWMGIINPSGGST
TYAQKFQGRVTMTRDTSTSTVYMESSLRSED TAVYYCARPGKSMDVWGQGTTLVTVSS
(SEQ ID NO: 700)Ab 68

QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWVALIWYDGSNK
YYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAKPGSMTDYWGQGTTLVTVSS
(SEQ ID NO: 734)Ab 69

QVQLVQSGAEVKKPGASVKVSCASGYTFTGSYMHVVRQAPGQGLEWMGWINPNSGG
TNYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARAKSVDHDYWGQGTTLVTV
SS (SEQ ID NO: 736)Ab 70

QVQLVQSGAEVKKPGASVKVSCASGYTFTGYMHVVRQAPGQGLEWMGWINPNSGG
TSYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARASKMGDDWGQGTTLVTVS
S (SEQ ID NO: 719)Ab 71

QVQLVQSGAEVKKPGASVKVSCASGYTFTSYMHVVRQAPGQGLEWMGIINPSGGST
SYAQKFQGRVTMTRDTSTSTVYMESSLRSED TAVYYCARDISTHDYDLAFDIWGQGT
MTVTVSS (SEQ ID NO: 739)Ab 72

QVQLQESGPGLVKPSETLSLTCTVSGGSISSYYWSWIRQPPGKGLEWIGSIYYSGSTNYPN
LKSRTISVDTSKNQFSLKLSSVTAADTAVYYCARSGTETLFDYWGQGTTLVTVSS (SEQ
ID NO: 741)Ab 73

EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDS DTT
YSPSFQGGVTISADKSISTAYLQWSSLKASDTAMYYCARAKMLDDGYAFDIWGQGTMTV
TVSS (SEQ ID NO: 743)Ab 74

QVQLVQSGAEVKKPGASVKVSCASGYTFTGSYMHVVRQAPGQGLEWMGWINPNSGG
TNYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARAGHKTHDYWGQGTTLVTV
SS (SEQ ID NO: 731)Ab 75

QVQLVQSGAEVKKPGASVKVSCASGYTFTGYMHVVRQAPGQGLEWMGWINPNSGG
TNYAQKFQGRVTMTRDTSISTAYMELSLRSDDTAVYYCARDLGYSSLLALDIWGQGTMT

VTVSS (SEQ ID NO: 746) Ab 76
EVQLVESGGGLVKPGGSLRLSCAASGFTFSSYSMNWVRQAPGKGLEWVSSISSSSSSYIYY
ADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARGGGRRGDNNWFDPPWGQGT
TVSS (SEQ ID NO: 748) Ab 77
QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVISYDGSNK
YYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCARGPPHEMDYWGQGT
S (SEQ ID NO: 750) Ab 78
QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVIWDGSNK
YYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCARTPYPIWYFDLWGRGTLTV
SS (SEQ ID NO: 752) Ab 79
EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYSMNWVRQAPGKGLEWVSYISGSSTIYY
ADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCARGGRRHYGGMDVWGQGT
VSS (SEQ ID NO: 754) Ab 80
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGGIPIFGTANY
AQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARGGGTFWGSWALYWGQGT
VSS (SEQ ID NO: 756) Ab 81
QVQLVQSGAEVKKPGSSVKVSCKASGGTFSSYAISWVRQAPGQGLEWMGGIPIFGTANY
AQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARD
SGNYDYWSGALRYWGQGT
LTVSS (SEQ ID NO: 758) Ab 82
QVQLQESGPGLVKPSQTLSTCTVSGGSISSGGYYWSWIRQHPGKGLEWIGYIYYSGSTVY
NPSLKS
RVTISVDTSKNQFSLKLSSVTAADTAVYYCARVSSSWYKAWGQGT
MTVSS (SEQ ID NO: 760) Ab 83
QVQLQQWGAGLLKPSETLSLTCAVYGGSFSGYYWSWIRQPPGKGLEWIGEIDHSGSTKY
NPSLKS
RVTISVDTSKNQFSLKLSSVTAADTAVYYCARVGVVVG
RPGYSAFDIWGQGT
MTVSS (SEQ ID NO: 762) Ab 84
QVQLVQSGAEVKKPGASVKVSCKASGYTFTSYGISWVRQAPGQGLEWMGWISTYNGNT
NYAQKLQGRVTMTTDTSTSTAYMELRSLRSDDTAVYYCARGSGSGYDSWYDWGQGT
LTVSS (SEQ ID NO: 725) Ab 85
QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVIWDGSNK
YYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCAKDLGGYYGGAAYGMDVWG
QGT
TVSS (SEQ ID NO: 765) Ab 86
QVQLVESGGGVVQPGRSLRLSCAASGFTFSSYGMHWVRQAPGKGLEWVAVISYDGSNK
YYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCAKDG
VYYGLGNWFDPPWGQGT
LTVSS (SEQ ID NO: 767) Ab 87
QVQLQESGPGLVKPSETLSLTCTVSGGSISSYYWSWIRQPPGKGLEWIGSIYYSGSTNYNPS
LKS
RVTISVDTSKNQFSLKLSSVTAADTAVYYCARHGWDRVGVWFDPPWGQGT
LTVSS (SEQ ID NO: 769)

TABLE-US-00021 TABLE C5 Light chain variable regions Ab ID LC Variable
Region Sequences Ab 1
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWYQQKPGQAPRLLIYGA
STRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQLPYWPPTFGGGTKVEIK (SEQ ID NO: 617)
Ab 2
EIVLTQSPATLSVSPGERATLSCRASQSVGSNLAWYQQKPGQAPRLLIYGA
STRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQYFFYPPTFGGGTKVEIK (SEQ ID NO: 619)
Ab 3
DIQMTQSPSSLSASVGDRTITCQASQDISNYLNWYQQKPGKAPKWDASNLATGVPSR
FSGSGSGTDFTFTISSLQPEDATYYCQQPFNFPYTFGGGTKVEIK (SEQ ID NO: 621)
Ab 4
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWYQQKPGQAPRLLIYSA
STRATGIPARFSGSGSGTEFTLTISSLQSEDFAVYYCQQDHDYPPTFGGGTKVEIK (SEQ ID NO: 623)

Ab 5
EIVMTQSPGTLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQYFSSPFTFGGGTKVEIK (SEQ ID NO:
625) Ab 6
EIVLTQSPATLSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISLEPEDFAVYYCQQRVNLPPTFGGGTKVEIK (SEQ ID NO: 627)
Ab 7
EIVLTQSPATLSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISLEPEDFAVYYCQQRISYPITFGGGTKVEIK (SEQ ID NO: 629)
Ab 8
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKWYGASSLQSGVPSRF
SGSGSGTDFTLTISLQPEDFATYYCQQIDDPITFGGGTKVEIK (SEQ ID NO: 631)
Ab 9
EIVLTQSPATLSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISLEPEDFAVYYCQQFSYWPWTFGGGKVEIK (SEQ ID NO: 633)
Ab 10
EIVLTQSPGTLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQHDSSPPTFGGGTKVEIK (SEQ ID NO: 635)
Ab 11
EIVLTQSPGTLSPGERATLSCRASQSVSSDYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQDYSYPWTFGGGKVEIK (SEQ ID NO:
637) Ab 12
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISLQPEDFATYYCQQEYAVPYTFGGGKVEIK (SEQ ID NO: 639)
Ab 13
EIVLTQSPATLSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISLEPEDFAVYYCQQVSNYPITFGGGTKVEIK (SEQ ID NO: 641)
Ab 14
DIQMTQSPSSLSASVGDRVITTCQASQDISNYLNWYQQKPGKAPKWDASNLETGVPSR
FSGSGSGTDFTFTISLQPEDATYYCQQVDNIPPTFGGGTKVEIK (SEQ ID NO: 643)
Ab 15
DIQMTQSPSSLSASVGDRVITTCQASQDISNYLNWYQQKPGKAPKWDASNLETGVPSR
FSGSGSGTDFTFTISLQPEDATYYCQQFDYPTFGGGTKVEIK (SEQ ID NO: 644)
Ab 16
DIQMTQSPSSLSASVGDRVITTCQASQDISNYLNWYQQKPGKAPKWDASNLETGVPSR
FSGSGSGTDFTFTISLQPEDATYYCQQFLNFPPTFGGGTKVEIK (SEQ ID NO: 646)
Ab 17
DIQMTQSPSSLSASVGDRVITTCQASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSR
FSGSGSGTDFTFTISLQPEDATYYCQQFFNFPPTFGGGTKVEIK (SEQ ID NO: 647)
Ab 18
DIQMTQSPSSLSASVGDRVITTCQASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSR
FSGSGSGTDFTFTISLQPEDATYYCQQFIDLFPPTFGGGTKVEIK (SEQ ID NO: 649)
Ab 19
DIQMTQSPSSLSASVGDRVITTCQASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSR
FSGSGSGTDFTFTISLQPEDATYYCQQYYDLFPPTFGGGTKVEIK (SEQ ID NO: 650)
Ab 20
EIVLTQSPGTLSPGERATLSCRASQSVSSDYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQFSSHPPTFGGGTKVEIK (SEQ ID NO: 652)
Ab 22
EIVMTQSPGTLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR

RFSGSGSGTDFTLTISRLEPEDFAVYYCQQDDRSPTYTFGGGGTKVEIK (SEQ ID NO: 653) Ab 23
DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLISWASTR
ESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQAYLPITFGGGGTKVEIK (SEQ ID NO: 655) Ab 24
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQAFSPPPWTFGGGGTKVEIK (SEQ ID NO: 657) Ab 25
EIVLTQSPGTLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQDDRSPTFGGGGTKVEIK (SEQ ID NO: 659) Ab 26
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPA
RFSGSGSGTDFTLTISSLEPEDFAVYYCQQEFDLPFTFGGGGTKVEIK (SEQ ID NO: 661) Ab 27
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQYNNFPPTFGGGGTKVEIK (SEQ ID NO: 663) Ab 28
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQRYLRPITFGGGGTKVEIK (SEQ ID NO: 665) Ab 29
EIVLTQSPGTLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQPGAVPTFGGGGTKVEIK (SEQ ID NO: 667) Ab 30
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYGAS
SLQSGVPSRF SGSGSGTDFTLTISSLQPEDFATYYCQQVYITPITFGGGGTKVEIK (SEQ ID NO: 669) Ab 31
DIQLTQSPSSLSASVGDRVITTCQASQDISNFLNWYQQKPGKAPKLLIYDASNLETGVPSRF
SGSGSGTDFTFITISLQPEDIATYYCQQPVLDLPFTFGGGGTKVEIK (SEQ ID NO: 671) Ab 32
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQYSFFPPTFGGGGTKVEIK (SEQ ID NO: 673) Ab 33
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQDSSFPPPTFGGGGTKVEIK (SEQ ID NO: 675) Ab 34
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQSDFPFWTFGGGGTKVEIK (SEQ ID NO: 677) Ab 35
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQGYSAIPITFGGGGTKVEIK (SEQ ID NO: 679) Ab 36
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQLFDWPTFGGGGTKVEIK (SEQ ID NO: 681) Ab 37
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQRAFLFTFGGGGTKVEIK (SEQ ID NO: 683) Ab 38
EIVLTQSPATLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQIDFLPYTFGGGGTKVEIK (SEQ ID NO: 685) Ab 39
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF

SGSGSGTDFTLTISSLQPEDFATYYCQQVYSPITFGGGGTKVEIK (SEQ ID NO: 687)
Ab 40
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQGYAAPITFGGGGTKVEIK (SEQ ID NO: 688)
Ab 41
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFTVYYCQQRYPITFGGGGTKVEIK (SEQ ID NO: 690)
Ab 42
EIVLTQSPGTLTLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQYASPPITFGGGGTKVEIK (SEQ ID NO: 692)
Ab 43
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQVYSTPITFGGGGTKVEIK (SEQ ID NO: 694)
Ab 44
EIVMTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDSSNRATGIPAR
FSGSGSGTDFTLTISSLEPEDFAVYYCQQQLVHWPTFGGGGTKVEIK (SEQ ID NO: 695)
Ab 45
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWYQQKPGQAPRLLIYGASTRATGIPAR
FSGSGSGTEFTLTISLQSEDFAVYYCQQQLDDWFTFGGGGTKVEIK (SEQ ID NO: 697)
Ab 46
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDSSNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQRSNYPITFGGGGTKVEIK (SEQ ID NO: 699)
Ab 47
EIVLTQSPGTLTLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQRILYPITFGGGGTKVEIK (SEQ ID NO: 701)
Ab 48
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQRAAYPITFGGGGTKVEIK (SEQ ID NO: 702)
Ab 49
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQRTSHYPITFGGGGTKVEIK (SEQ ID NO: 703)
Ab 50
EIVLTQSPGTLTLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQYAGSPFTFGGGGTKVEIK (SEQ ID NO: 704)
Ab 51
DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQFDDVFTFGGGGTKVEIK (SEQ ID NO: 706)
Ab 53
EIVLTQSPGTLTLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDR
FSGSGSGTDFTLTISRLEPEDFAVYYCQQYVNSPFTFGGGGTKVEIK (SEQ ID NO: 707)
Ab 54
DIQMTQSPSSLSASVGDRVITTCRASQSINSYLNWYQQKPGKAPKLLIYAASSLQSGVPSR
FSGSGSGTDFTLTISSLQPEDFATYYCQQSDDDPFTFGGGGTKVEIK (SEQ ID NO: 708)
Ab 55
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSLEPEDFAVYYCQQQLSTYPLTFGGGGTKVEIK (SEQ ID NO: 710)
Ab 56
EIVMTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPAR
FSGSGSGTDFTLTISSLEPEDFAVYYCQQRSVYPITFGGGGTKVEIK (SEQ ID NO: 712)
Ab 57

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQVSLFPLTFGGGGTKVEIK (SEQ ID NO: 714)
Ab 58
DIQMTQSPSTLSASVGDRVTITCRASQSISSWLAWYQQKPGKAPKWDASSLESGVPSR
FSGSGSGTEFTLTISSLQPDDFATYYCLDYNSSPITFGGGGTKVEIK (SEQ ID NO: 716)
Ab 59
DIQMTQSPSSLSASVGDRVTITCQASQDISNYLNWYQQKPGKAPKLLIYDASNLETGVPSR
FSGSGSGTDFTFTISSLQPEDIATYYCQQHIALPFTFGGGGTKVEIK (SEQ ID NO: 718)
Ab 60
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRASMPITFGGGGTKVEIK (SEQ ID NO: 720)
Ab 61
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDSSNRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQAFNRPPTFGGGGTKVEIK (SEQ ID NO: 722)
Ab 62
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASKRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQSSVHPYTFGGGGTKVEIK (SEQ ID NO: 724)
Ab 63
DIQMTQSPSSVSASVGDRVTITCRASQGIDSWLAWYQQKPGKAPKLLIYAASSLQSGVPSR
FSGSGSGTDFTLTISSLQPEDFATYYCQQAYSLPPTFGGGGTKVEIK (SEQ ID NO: 726)
Ab 64
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWYQQKPGQAPRLLIYGASTRATGIPAR
FSGSGSGTEFTLTISSLQSEDFAVYYCQQDDDGYTFGGGGTKVEIK (SEQ ID NO: 728)
Ab 65
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQDYSWPYTFGGGGTKVEIK (SEQ ID NO: 730)
Ab 66
EIVLTQSPGTLTLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRSAYPITFGGGGTKVEIK (SEQ ID NO: 732)
Ab 67
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRSHFPITFGGGGTKVEIK (SEQ ID NO: 733)
Ab 68
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRANYPITFGGGGTKVEIK (SEQ ID NO: 735)
Ab 69
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRADYPITFGGGGTKVEIK (SEQ ID NO: 737)
Ab 70
EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRSVYPITFGGGGTKVEIK (SEQ ID NO: 738)
Ab 71
EIVMTQSPGTLTLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYGASNATGIPD
RFSGSGSGTDFTLTISRLEPEDFAVYYCQQAGSHPTFGGGGTKVEIK (SEQ ID NO:
740) Ab 72
DIQMTQSPSSLSASVGDRVTITCQASQDITNYLNWYQQKPGKAPKWDASNLETGVPSR
FSGSRSGTDFTFTISSLQPEDIATYYCQQDVNYPPTFGGGGTKVEIK (SEQ ID NO: 742)
Ab 73
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWYQQKPGQAPRLLIYGASTRATGIPAR
FSGSGSGTEFTLTISSLQSEDFAVYYCQQDDNYPYTFGGGGTKVEIK (SEQ ID NO:

744) Ab 74

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQRSTFPITFGGGGTKVEIK (SEQ ID NO: 745)

Ab 75

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQVSNYPFTFGGGGTKVEIK (SEQ ID NO: 747)

Ab 76

DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYWASTR
ESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQYHDAPITFGGGGTKVEIK (SEQ ID

NO: 749) Ab 77

DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQPPKLLIYWASTR
ESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQQAYVVPPTFGGGGTKVEIK (SEQ ID

NO: 751) Ab 78

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQADNWPFTFGGGGTKVEIK (SEQ ID NO: 753)

Ab 79

DIVMTQSPSLPVTGPGEPAISCRSSQSLLHSNGYNYLDWYLQKPGQSPQLLIYLGSHRAS
GVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQALESPTFGGGGTKVEIK (SEQ ID

NO: 755) Ab 80

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQYVNWPFTFGGGGTKVEIK (SEQ ID NO: 757)

Ab 81

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRATGIPARF
SGSGSGTDFTLTISSELEPEDFAVYYCQQSSNWPWTFGGGGTKVEIK (SEQ ID NO: 759)

Ab 82

DIQMTQSPSSVSASVGDRVITTCRASQGISSWLAWYQQKPGKAPKLLIYAASSLQSGVPSR
FSGSGSGTDFTLTISSLQPEDFATYYCQQASTFPITFGGGGTKVEIK (SEQ ID NO: 761)

Ab 83

DIQMTQSPSSVSASVGDRVITTCRASQGISSWLAWYQQKPGKAPKLLIYAASSLQSGVPSR
FSGSGSGTDFTLTISSLQPEDFATYYCQQRNSLPLTFGGGGTKVEIK (SEQ ID NO: 763)

Ab 84

DIQMTQSPSSLSASVGDRVITTCRASQSISSYLNWYQQKPGKAPKLLIYAASSLQSGVPSRF
SGSGSGTDFTLTISSLQPEDFATYYCQQSYDFPITFGGGGTKVEIK (SEQ ID NO: 764)

Ab 85

DIQLTQSPSSVSASVGDRVITTCRASQDISSWLAWYQQKPGKAPKLLIYAASSLQSGVPSR
FSGSGSGTDFTLTISSLQPEDFATYYCQQEVDYPPLTFGGGGTKVEIK (SEQ ID NO:

766) Ab 86

DIQMTQSPSTLSASVGDRVITTCRASQSISSWLAWYQQKPGKAPKWKASSLESGVPSR
FSGSGSGTEFTLTISSLQPDDFATYYCQQQLNSYSPTFGGGGTKVEIK (SEQ ID NO: 768)

Ab 87

EIVLTQSPATLSLSPGERATLSCRASQSVSRYLAWYQQKPGQAPRLLIYDASNRATGIPAR
FSGSGSGTDFTLTISSELEPEDFAVYYCQQYIFWPPTFGGGGTKVEIK (SEQ ID NO: 770)

[0323] In some embodiments, anti-TREM2 antibodies of the present disclosure comprise (a) a heavy chain variable region comprising at least one, two, or three HVRs selected from HVR-H1, HVR-H2, and HVR-H3 of any one of the antibodies listed in TABLE C3 or selected from Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab100, Ab111, Ab12, Ab13, Ab14, Ab15, Ab16, Ab17, Ab18, Ab19, Ab20, Ab21, Ab22, Ab23, Ab24, Ab25, Ab26, Ab27, Ab28, Ab29, Ab30, Ab31, Ab32, Ab33, Ab34, Ab35, Ab36, Ab37, Ab38, Ab39, Ab4, Ab41, Ab42, Ab43, Ab44, Ab45, Ab46, Ab47, Ab48, Ab49, Ab50, Ab51, Ab52, Ab53, Ab54, Ab55, Ab56, Ab57, Ab58, Ab59, Ab60, Ab61, Ab62, Ab63, Ab64, Ab65, Ab66, Ab67, Ab68, Ab69, Ab70, Ab71, Ab72, Ab73, Ab74, Ab75, Ab76, Ab77,

Ab78, Ab79, AbM, Ab81, Ab82, Ab83, Ab84, Ab85, Ab86, and Ab87; and/or (b) a light chain variable region comprising at least one, two, or three HVRs selected from HVR-L1, HVR-L2, and HVR-L3 of any one of the antibodies selected from Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab100, Ab11, Ab12, Ab13, Ab14, Ab15, Ab16, Ab17, Ab18, Ab19, Ab20, Ab21, Ab22, Ab23, Ab24, Ab25, Ab26, Ab27, Ab28, Ab29, Ab30, Ab31, Ab32, Ab33, Ab34, Ab35, Ab36, Ab37, Ab38, Ab39, Ab4, Ab41, Ab42, Ab43, Ab44, Ab45, Ab46, Ab47, Ab48, Ab49, Ab50, Ab51, Ab52, Ab53, Ab54, Ab55, Ab56, Ab57, Ab58, Ab59, Ab60, Ab61, Ab62, Ab63, Ab64, Ab65, Ab66, Ab67, Ab68, Ab69, Ab70, Ab71, Ab72, Ab73, Ab74, Ab75, Ab76, Ab77, Ab78, Ab79, Ab80, Ab81, Ab82, Ab83, Ab84, Ab85, Ab86, and Ab87.

[0324] In some embodiments, the anti-TREM2 antibody comprises a light chain variable domain and a heavy chain variable region, wherein the light chain variable region comprises a HVR-L1, HVR-L2, and HVR-L3, and the heavy chain variable domain comprises a HVR-H1, HVR-H2, and HVR-H3 of an antibody listed in TABLE C3 or selected from the group consisting of: Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab10, Ab11, Ab12, Ab13, Ab14, Ab15, Ab16, Ab17, Ab18, Ab19, Ab20, Ab21, Ab22, Ab23, Ab24, Ab25, Ab26, Ab27, Ab28, Ab29, Ab30, Ab31, Ab32, Ab33, Ab34, Ab35, Ab36, Ab37, Ab38, Ab39, Ab40, Ab41, Ab42, Ab43, Ab44, Ab45, Ab46, Ab47, Ab48, Ab49, Ab50, Ab51, Ab52, Ab53, Ab54, Ab55, Ab56, Ab57, Ab58, Ab59, Ab60, Ab61, Ab62, Ab63, Ab64, Ab65, Ab66, Ab67, Ab68, Ab69, Ab70, Ab71, Ab72, Ab73, Ab74, Ab75, Ab76, Ab77, Ab78, Ab79, Ab80, Ab81, Ab82, Ab83, Ab84, Ab85, Ab86, and Ab87.

[0325] In some embodiments, an anti-human TREM2 antibody is an antibody which competes with a monoclonal antibody selected from the group consisting of: Ab1, Ab2, Ab3, Ab4, Ab5, Ab6, Ab7, Ab8, Ab9, Ab10, Ab11, Ab12, Ab13, Ab14, Ab15, Ab16, Ab17, Ab18, Ab19, Ab20, Ab21, Ab22, Ab23, Ab24, Ab25, Ab26, Ab27, Ab28, Ab29, Ab30, Ab31, Ab32, Ab33, Ab34, Ab35, Ab36, Ab37, Ab38, Ab39, Ab40, Ab41, Ab42, Ab43, Ab44, Ab45, Ab46, Ab47, Ab48, Ab49, Ab50, Ab51, Ab52, Ab53, Ab54, Ab55, Ab56, Ab57, Ab58, Ab59, Ab60, Ab61, Ab62, Ab63, Ab64, Ab65, Ab66, Ab67, Ab68, Ab69, Ab70, Ab71, Ab72, Ab73, Ab74, Ab75, Ab76, Ab77, Ab78, Ab79, Ab80, Ab81, Ab82, Ab83, Ab84, Ab85, Ab86, and Ab87 for binding to TREM2.

[0326] In some embodiments, each of the light chain variable regions disclosed in TABLE C5 and each of the heavy chain variable regions disclosed in TABLE C4 may be attached to the light chain constant regions (EN1) and heavy chain constant regions (EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

D. PCT Patent Application Publication No. WO2017/062672A1

[0327] In some embodiments, the TREM2 agonist is an antibody or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2017/062672A1 (“the '672 application”), which is incorporated by reference herein, in its entirety.

[0328] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3 (also referred to as HVR-L1, HVR-L2, and HVR-L3, respectively), and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 (also referred to as HVR-H1, HVR-H2, and HVR-H3, respectively) disclosed in the '672 application specification. In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '672 application specification.

[0329] In some embodiments, the antibody comprises a light chain variable domain and a heavy chain variable domain, wherein the light chain variable domain, or the heavy chain variable domain, or both comprise at least one, two, three, four, five, or six HVRs selected from HVR-L1, HVR-L2, HVR-L3, HVR-H1, HVR-H2, and HVR-H3 such that: (a) the HVR-L1 comprises an

amino acid sequence selected from the group consisting of SEQ ID NOS:829-843, 1401, 1510-1514, 1554-1558, and 1646-1648; (b) the HVR-L2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOS:844-853, 1515-1517, and 1559-1563; (c) the HVR-L3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOS:854-867, 1402, 1403, 1518-1522, and 1564-1566; (d) the HVR-H1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOS:868-885, 1404, 1523-1525, 1567-1574, and 1649-1655; (e) the HVR-H2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOS:886-904, 1405-1407, 1526-1528, 1575-1582, 1656-1662, and 1708; or (f) the HVR-H3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOS:905-992, 1408, 1409, 1529, 1530, and 1583-1590. In some embodiments: (a) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:831, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:846, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:856, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:871, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:889, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:908; (b) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:834, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:848, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:859, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:873, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:891, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:910; (c) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:831, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:846, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:856, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:871, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:889, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:908; (d) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:836, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:849, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:855, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:875, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:893, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:912; (e) the HVR-H1 comprises the amino acid sequence of SEQ ID NO:978, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:896, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:915; (f) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:839, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:848, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:863, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:880, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:898, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:917; (g) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:840, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:848, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:868, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:881, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:899, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:918; (h) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:841, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:852, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:865, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:882, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:900, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:919; (i) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:842, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:849, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:866, the HVR-H1 comprises the amino acid sequence of SEQ ID NO:883, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:902, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:920; or (j) the HVR-L1 comprises the amino acid sequence of SEQ ID NO:936, the HVR-L2 comprises the amino acid sequence of SEQ ID NO:849, the HVR-L3 comprises the amino acid sequence of SEQ ID NO:855, the HVR-H1 comprises the amino acid sequence of SEQ

ID NO:885, the HVR-H2 comprises the amino acid sequence of SEQ ID NO:904, and the HVR-H3 comprises the amino acid sequence of SEQ ID NO:922. In some embodiments, the antibody comprises a light chain variable domain and a heavy chain variable domain, wherein the light chain variable domain comprises: (a) an HVR-L1 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:829-843, 1401, 1510-1514, 1554-1558, and 1646-1648, or an amino acid sequence with at least about 90% homology to an amino acid sequence selected from the group consisting of SEQ ID NOS:829-843, 1401, 1510-1514, 1554-1558, and 1646-1648; (b) an HVR-L2 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:844-853, 1515-1517, and 1559-1563, or an amino acid sequence with at least about 90% homology to an amino acid sequence selected from the group consisting of SEQ ID NOS:844-853, 1515-1517, and 1559-1563; and (c) an HVR-L3 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:854-867, 1402, 1403, 1518-1522, and 1564-1566, or an amino acid sequence with at least about 90% homology to an amino acid sequence selected from the group consisting of SEQ ID NOS:854-867, 1402, 1403, 1518-1522, and 1564-1566; and wherein the heavy chain variable domain comprises: (a) an HVR-H1 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:868-885, 1404, 1523-1525, 1567-1574, and 1649-1655, or an amino acid sequence with at least about 90% homology to an amino acid sequence selected from the group consisting of SEQ ID NOS:868-885, 1404, 1523-1525, 1567-1574, and 1649-1655; (b) an HVR-H2 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:886-904, 1405-1407, 1526-1528, 1575-1582, 1656-1662, and 1708, or an amino acid sequence with at least about 90% homology to an amino acid sequence selected from the group consisting of SEQ ID NOS:886-904, 1405-1407, 1526-1528, 1575-1582, 1656-1662, and 1708; and (c) an HVR-H3 comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:905-992, 1408, 1409, 1529, 1530, and 1583-1590, or an amino acid sequence with at least about 90% homology to an amino acid sequence selected from the group consisting of SEQ ID NOS:905-992, 1408, 1409, 1529, 1530, and 1583-1590. In some embodiments, the antibody comprises a light chain variable domain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1039-1218, 1422-1454, 1499-1509, 1544-1550, 1629-1636, 1641, 1643, 1664, 1669, and 1670; and/or a heavy chain variable domain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1219-1400, 1455-1498, 1551-1553, and 1637-1640, 1642-1645, and 1665-1667.

[0330] In some embodiments, the antibody comprises a light chain variable domain and a heavy chain variable domain, wherein: (a) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1153 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1341; (b) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1670 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1341; (c) the light chain variable domain comprises the amino acid sequence of SEQ ID NO:1154 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1342; (d) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1155 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1343; (e) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1156 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1344; (f) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1157 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1345; (g) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1158 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1346; (h) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1159 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1346; (i) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1160 and the heavy chain variable domain comprises the amino acid

[illegible]

sequence of SEQ ID NO:1361; (ii) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1216 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1371; (jj) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1217 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1381; (kk) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1218 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1391; (ll) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1544 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (mm) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1629 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (nn) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1545 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1552; (oo) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1546 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (pp) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1546 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1637; (qq) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1547 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (rr) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1548 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1553; (ss) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1630 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1638; (tt) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1631 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1553; (uu) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1549 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (vv) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1632 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1639; (ww) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1549 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1640; (xx) the light chain variable domain comprises the amino acid sequence of SEQ ID NO:1550 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (yy) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1633 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1551; (zz) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1634 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1642; (aaa) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1635 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO: 1644; or (bbb) the light chain variable domain comprises the amino acid sequence of SEQ ID NO: 1636 and the heavy chain variable domain comprises the amino acid sequence of SEQ ID NO:1645. In any of the above embodiments, the light chain variable domain and/or heavy chain variable domain comprises an amino acid sequence with at least about 90% homology to the amino acid sequence indicated.

[0331] In some embodiments, the antibody is an antibody disclosed in Tables 2A, 2B, 3A, 3B, 4A, 4B, 7A and 7B of PCT Patent Application Publication No. WO2017/062672A1, reproduced below as TABLES D1-D8.

TABLE-US-00022 TABLE D1 EU or Kabat light chain HVR sequences Ab ID
HVRL1 HVRL2 HVRL3 4D11 RASENIYSFLA NSKTFAE QHHYGTPPWT (SEQ ID NO:
829) (SEQ ID NO: 844) (SEQ ID NO: 854) 78C5 RASENIYSFLA NSKTFAE
QHHYGTPPWT (SEQ ID NO: 829) (SEQ ID NO: 844) (SEQ ID NO: 854) 6G12

KSSQSLLYSNQKNLA WASTRES QQYYSYPLT (SEQ ID NO: 830) (SEQ ID NO: 845) (SEQ ID NO: 855) 8F11 KSSQSLLYSSNQKNCLA LVSKLDS MQGTHFPLT (SEQ ID NO: 830) (SEQ ID NO: 846) (SEQ ID NO: 856) 8E10 KSSQSLLDSDGKTYLN LVSKLDS WQGTHFPYT (SEQ ID NO: 832) (SEQ ID NO: 846) (SEQ ID NO: 857) 7E5 KSSQSLLYSNGKTFLS LVSKLDS MQGTHFPLT (SEQ ID NO: 831) (SEQ ID NO: 846) (SEQ ID NO: 856) 7F8 SASSSVSYMY LTSILAS QQWSFNPYT (SEQ ID NO: 833) (SEQ ID NO: 847) (SEQ ID NO: 858) 8F8 RSSQSLVHSNGNTYLH KVSNRFS SQSTHVPLT (SEQ ID NO: 834) (SEQ ID NO: 848) (SEQ ID NO: 859) H7 SASSSVSYMY LTSILAS QQWSFNPYT (SEQ ID NO: 833) (SEQ ID NO: 847) (SEQ ID NO: 858) 2H8 SASSSVSYMY LTSILAS QQWSFNPYT (SEQ ID NO: 833) (SEQ ID NO: 847) (SEQ ID NO: 858) 3A2 RSSQTIHSNGNTYLE KVSNRFS FQGSHVPYT (SEQ ID NO: 835) (SEQ ID NO: 848) (SEQ ID NO: 860) 3A7 KSSQSLLYSNGKTFLS LVSKLDS MQGTHFPLT (SEQ ID NO: 831) (SEQ ID NO: 846) (SEQ ID NO: 856) 3B10 KSSQSLLYSSDQKNYLA WASTRES QQYYSYPLT (SEQ ID NO: 836) (SEQ ID NO: 849) (SEQ ID NO: 855) 4F11 RSSQTIHSNGNTYLE KVSNRFS FQGSHVPYT (SEQ ID NO: 835) (SEQ ID NO: 848) (SEQ ID NO: 860) 6H6 KSSQSVFYSSNQKNYLA WASTRES HQYLSSLT (SEQ ID NO: 1401) (SEQ ID NO: 849) (SEQ ID NO: 1402) 7A9 RASENIYSYLA KAKTLAE QHHYGTPFT (SEQ ID NO: 837) (SEQ ID NO: 850) (SEQ ID NO: 861) 8A1 RTSENVYSNLA AATNLAD HHFWGTPYT (SEQ ID NO: 838) (SEQ ID NO: 851) (SEQ ID NO: 862) 9F5 RSSQSLVHSNGYTYLH KVSNRFS SQSTRVPYT (SEQ ID NO: 839) (SEQ ID NO: 848) (SEQ ID NO: 863) 9G1 RFSQSLVHSNGNTYLH KVSNRFS SQSTRVPPT (SEQ ID NO: 840) (SEQ ID NO: 848) (SEQ ID NO: 864) 9G3 KASSNVNYMS FTSNLPS SGEVTQFT (SEQ ID NO: 841) (SEQ ID NO: 852) (SEQ ID NO: 865) 10A9 RSSQTIHSNGNTYLE KVSNRFC FQGSHVPYT (SEQ ID NO: 835) (SEQ ID NO: 853) (SEQ ID NO: 860) 11A8 KSSQSLLNSGNQKKYLT WASTRES QNDYGFPLT (SEQ ID NO: 842) (SEQ ID NO: 849) (SEQ ID NO: 866) 12D9 KSSQSLLYSGNQKNFLA WASTRES QQYYSYPFT (SEQ ID NO: 843) (SEQ ID NO: 849) (SEQ ID NO: 867) 12F9 KSSQSLLYSSDQKNYLA WASTRES QQYYSYPLT (SEQ ID NO: 836) (SEQ ID NO: 849) (SEQ ID NO: 855) 10C1 KSSQSVFYSSNQKNYLA WASTRES HQYLSSLT (SEQ ID NO: 1401) (SEQ ID NO: 849) (SEQ ID NO: 1402) 7E9 KSSQSLLYSSNQKNCLA WASTRES QQYYSYPLT (SEQ ID NO: 830) (SEQ ID NO: 849) (SEQ ID NO: 855) 8C3 RS SQSLVHSNGNTYLH KVSNRFS SQSTHVPPPT (SEQ ID NO: 834) (SEQ ID NO: 848) (SEQ ID NO: 1403) IB4v1 SQDVSTTVA SASRYT QQHYSTPPT (SEQ ID NO: 1510) (SEQ ID NO: 1515) (SEQ ID NO: 1518) IB4v2 SQSLVHSNGNTYLH KVSNRVS SQSTHVPLT (SEQ ID NO: 1554) (SEQ ID NO: 1559) (SEQ ID NO: 859) 6H2 QSIVHSNGNTYLE KVSNRFS FQGSHVPFT (SEQ ID NO: 1511) (SEQ ID NO: 848) (SEQ ID NO: 1519) 7B11 SQGVSTAVA WASTRHT QHGYSTYT (SEQ ID NO: 1512) (SEQ ID NO: 1516) (SEQ ID NO: 1520) 18D8 SQDVRTAVA SASRYT QQHYGTPPWT (SEQ ID NO: 1513) (SEQ ID NO: 1515) (SEQ ID NO: 1521) 18E4v1 SENVVITYVS GASNRYT GQGYSYPYT (SEQ ID NO: 1514) (SEQ ID NO: 1517) (SEQ ID NO: 1522) 18E4v2 SQSLVHSNGNTYLH KVSDRFS SQSTHVPLT (SEQ ID NO: 1554) (SEQ ID NO: 1560) (SEQ ID NO: 859) 29F6v1 SQDVRTAVA SASRYT QQHYGTPPWT (SEQ ID NO: 1513) (SEQ ID NO: 1515) (SEQ ID NO: 1521) 29F6v2 SQSLVHSNGDTYLH KVSNRFS SQSTHVPLT (SEQ ID NO: 1555) (SEQ ID NO: 848) (SEQ ID NO: 859) 40D5 SQDVRTAVA SASRYT QQHYGTPPWT (SEQ ID NO: 1513) (SEQ ID NO: 1515) (SEQ ID NO: 1521) 43B9 SQDVRTAVA SASRYT QQHYGTPPWT (SEQ ID NO: 1513) (SEQ ID NO:

1515)(SEQ ID NO: 1521) 44A8v1 SQDVSTVA SASYRYT QQHYSTPPT (SEQ ID NO: 1510)(SEQ ID NO: 1515)(SEQ ID NO: 1518) 44A8v2 SESVDYHGTSLMQ AASNVES QQNRKILWT (SEQ ID NO: 1556)(SEQ ID NO: 1561)(SEQ ID NO: 1564) 44B4v1 SQDVRTAVA SASYRYT QQHYGTPPWT (SEQ ID NO: 1513) (SEQ ID NO: 1515)(SEQ ID NO: 1521) 44B4v2 SENIZYSLA NANSLED KQAYDVPWT (SEQ ID NO: 1557)(SEQ ID NO: 1562)(SEQ ID NO: 1565) 29F7 RASQSIGTSIH FASESIS QQTNTWPIT (SEQ ID NO: 1558)(SEQ ID NO: 1563)(SEQ ID NO: 1566) 32G1 RSSQSLVHSNGNTYLH KVSNRFS SQSTHVPLT (SEQ ID NO: 834)(SEQ ID NO: 848)(SEQ ID NO: 859)

TABLE-US-00023 TABLE D2 EU or Kabat light chain HVR consensus sequences HVR 1,1 Consensus 1 RXSENXYSLA (SEQ ID NO: 1646) Consensus 2 RSSQXXXHSNGXTYLX (SEQ ID NO: 1647) Consensus 3 KSSQSXXXSXXQKXXLX (SEQ ID NO: 1648)

TABLE-US-00024 TABLE D3 EU or Kabat heavy chain HVR sequences Ab ID HVR H1 HVR H2 HVR H3 4D11 FTLSSYAMS VASISRGGSTYYP TRGYGYRTPFAN (SEQ ID NO: 868)(SEQ ID NO: 886)(SEQ ID NO: 905) 78C5 FTLSSYAMS VASISRGGSTYYP TRGYGYRTPFAN (SEQ ID NO: 868)(SEQ ID NO: 886) (SEQ ID NO: 905) 6G12 YTFTEYTMH IGGINPNNGGTSYS ARGGSHYYAMDY (SEQ ID NO: 869)(SEQ ID NO: 887)(SEQ ID NO: 906) 8E10 YTFSTDYEMH IGVIDPETGGTAYN TSPDYYGSSYPLYAMDY (SEQ ID NO: 870)(SEQ ID NO: 888)(SEQ ID NO: 907) 7E5 FTFSDAWMG VAEIRDKVKNHATYYA RLGVFDY (SEQ ID NO: 908)(SEQ ID NO: 871)(SEQ ID NO: 889) 7F8 FSFNTYAMN IARIRSKSNNYATYYA VRHGDGNLWYIDV (SEQ ID NO: 872)(SEQ ID NO: 890) (SEQ ID NO: 909) 8F8 YTVSRYWMH IGRIDPNSGGTKYN VLTGTDFDY (SEQ ID NO: 873)(SEQ ID NO: 891)(SEQ ID NO: 910) 1H7 FSFNTYAMN IARIRSKSNNYATYY VRHGDGNLWYIDV (SEQ ID NO: 872) A (SEQ ID NO: 890)(SEQ ID NO: 909) 2H8 FSFNTYAMN IARIRSKSNNYATYY VRHGDGNLWYIDV (SEQ ID NO: 872) A (SEQ ID NO: 890)(SEQ ID NO: 909) 3A2 YPFSNFWIT IGDIYPGSDNSNYN AREAYYTNP GFAY (SEQ ID NO: 874)(SEQ ID NO: 892) (SEQ ID NO: 911) 3A7 FTFSDAWMG VAEIRDKVKNHATYYA RLGVFDY (SEQ ID NO: 908)(SEQ ID NO: 871)(SEQ ID NO: 889) 3B10 LTSNTYTQT ESVIRSKSNNFSTLYA VRHKSNNRYPGVY (SEQ ID NO: 875)(SEQ ID NO: 893) (SEQ ID NO: 912) 4F11 YPFSNFWIT IGDIYPGSDNSNYN AREAYYTNP GFAY (SEQ ID NO: 874)(SEQ ID NO: 892)(SEQ ID NO: 911) 6H6 FTFSDAWMD VAEIRNKVNNHATYYA TSLYDGYYLRFAY (SEQ ID NO: 876)(SEQ ID NO: 894) (SEQ ID NO: 913) 7A9 FTFNTYSMN VAHIKTKZNNFATFYA VZHZSNNYPFAY (SEQ ID NO: 877)(SEQ ID NO: 895)(SEQ ID NO: 914) 7B3 YTFTTYWIH IGRNDPNSGGSNYN VRTNWDGDF (SEQ ID NO: 878)(SEQ ID NO: 896)(SEQ ID NO: 915) 8A1 YAFSNYWMS IGQIYPGDGDTKYN SREKGADYYGSTYSAWFSY (SEQ ID NO: 879)(SEQ ID NO: 897)(SEQ ID NO: 916) 9F5 YAFSSSWMN IGRIYPGDGDTNYN ARLLRNQPGESYAMDY (SEQ ID NO: 880)(SEQ ID NO: 898)(SEQ ID NO: 917) 9F5a YAFSSSWMN RIYPGDGDTNYNGEFRV ARLLRNQPGESYAMDY (SEQ ID NO: 880)(SEQ ID NO: 1708)(SEQ ID NO: 917) 9G1 YIFTTYWIH IGRIDPNNGDTNYN VMTGTDFDY (SEQ ID NO: 881)(SEQ ID NO: 899)(SEQ ID NO: 918) 9G3 FNFNTYAMK IARIRSNSNDYATNYS VGHKINNYPFAH (SEQ ID NO: 882)(SEQ ID NO: 900)(SEQ ID NO: 919) 10A9 YPFSNFWIT IGDIYPGSDNRNFN AREAYYTNP GFAY (SEQ ID NO: 874)(SEQ ID NO: 901)(SEQ ID NO: 911) 11A8 FNFNTYAMN VARIRSKSNNYATYYA VRHYSNYGWGFAY (SEQ ID NO: 883)(SEQ ID NO: 902)(SEQ ID NO: 920) 12D9 YTFSDYYIH IGYIYPNNGDNGYN ARRGYYGGSYDY (SEQ ID NO: 884)(SEQ

ID NO: 903)(SEQ ID NO: 921)12F9 FRNTYAMT EGVIRKSSNFATLYA
VRHKS NKYPFVY (SEQ ID NO: 885)(SEQ ID NO: 904)(SEQ ID NO: 922)
10C1 FTFSDAWMD VAEIRNKINN HATYYA TSLYDGSYLRFAY (SEQ ID NO: 876)
(SEQ ID NO: 1405)(SEQ ID NO: 1408)7E9 YTFTEYTMH IGGINPNNGGTSYK
ARGGSHYYAMDY (SEQ ID NO: 869)(SEQ ID NO: 1406)(SEQ ID NO: 906)
8C3 YSFTGYMH IGRVNPNNGGTSYN VLTGGYFDY (SEQ ID NO: 1404)(SEQ ID
NO: 1407)(SEQ ID NO: 1409)1B4 SRFTFSSYAMS VAAISGGGRYTYYP
ARHYDGYLDY (SEQ ID NO: 1523)(SEQ ID NO: 1526)(SEQ ID NO: 1529)
6H2 SAFSLTNYAVH LGVIWSSGGSTAFN ATHYYRSTYAFSY (SEQ ID NO: 1524)
(SEQ ID NO: 1527)(SEQ ID NO: 1530)7B11v1 SRFTFSSYAMS
VAAISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1523)(SEQ ID NO: 1526)
(SEQ ID NO: 1529)7B11v2 SGYTFTDFYMN IGDINPNNGHTTYN
AREPYSYGSSPWYFLV (SEQ ID NO: 1567)(SEQ ID NO: 1575)(SEQ ID NO:
1583)18D8 SRFTFSSYAMS VAAISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1523)
(SEQ ID NO: 1526)(SEQ ID NO: 1529)18E4v1 SRFTFSSYAVS
VATISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1525)(SEQ ID NO: 1528)
(SEQ ID NO: 1529)18E4v2 SGYTFTAYWMH IGRTHPSDSDTNYN ATYSNYVTGAMDS
(SEQ ID NO: 1568)(SEQ ID NO: 1576)(SEQ ID NO: 1584)29F6v1
SRFTFSSYAMS VAAISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1523)(SEQ ID
NO: 1526)(SEQ ID NO: 1529)29F6v2 SGFNIKNTYIH IGRIDPAIGNTNYA VSPGMDY
(SEQ ID NO: 1569)(SEQ ID NO: 1577)(SEQ ID NO: 1585)40D5v1
SRFTFSSYAMS VAAISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1523)(SEQ ID
NO: 1526)(SEQ ID NO: 1529)40D5v2 SGYTFTNYWIH IGRIHPSDSDINYN
VKTGTSFAS (SEQ ID NO: 1570)(SEQ ID NO: 1578)(SEQ ID NO: 1586)43B9
SRFTFSSYAMS VAAISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1523)(SEQ ID
NO: 1526)(SEQ ID NO: 1529)44A8 SRFTFSSYAMS VAAISGGGRYTYYP
ARHYDGYLDY (SEQ ID NO: 1523)(SEQ ID NO: 1526)(SEQ ID NO: 1529)
44B4v1 SRFTFSSYAMS VAAISGGGRYTYYP ARHYDGYLDY (SEQ ID NO: 1523)
(SEQ ID NO: 1526)(SEQ ID NO: 1529)44B4v2 SGYTFTSATMH
IGYINPNSGYSKYN ARWGIDGNYGGGFFDV (SEQ ID NO: 1571)(SEQ ID NO:
1579)(SEQ ID NO: 1587)45D6 YSFTDYNH IGYINPNSDNTRYI TRGF SNLGAMDY
(SEQ ID NO: 1572)(SEQ ID NO: 1580)(SEQ ID NO: 1588)29F7
FTLSNYWMN VAQIRLKSDNYATHYA TGAGGNHENY (SEQ ID NO: 1573)(SEQ
ID NO: 1581)(SEQ ID NO: 1589)32G1 YFTTDYNIH IGYINPNNGGTTYN
ATTYVSFSY (SEQ ID NO: 1574)(SEQ ID NO: 1582)(SEQ ID NO: 1590)
TABLE-US-00025 TABLE D4 EU or Kabat heavy chain HVR consensus sequences
HVR H1 HVR H2 Consensus 1 YX.sub.1X.sub.2X.sub.3XYXXH
IGXXXPX.sub.1X.sub.2X.sub.3X.sub.4X.sub.5XYX.sub.6 X.sub.1 is T or S X.sub.1
is N or E X.sub.2 is F or V X.sub.2 is N, S, or T X.sub.3 is T or S
X.sub.3 is G or D (SEQ ID NO: 1649) X.sub.4 is G, D, or N X.sub.5 is
T, S, or N X.sub.6 is N, 5, K, or I (SEQ ID NO: 1656) Consensus 2
YTFTXYXXH IGXXXPNNGGTXYN (SEQ ID NO: 1657)(SEQ ID NO: 1650)
Consensus 3 FTFSDAWMX.sub.1 VAEIRX.sub.1KX.sub.2X.sub.3NHATYYA X.sub.1 is
D or G (SEQ ID NO: 1651) X.sub.1 is N or D X.sub.2 is V or I
X.sub.3 is N or K (SEQ ID NO: 1658) Consensus 4
FXX.sub.1X.sub.2X.sub.3YX.sub.4MX.sub.5
XX.sub.1XIX.sub.2X.sub.3X.sub.4X.sub.5X.sub.6X.sub.7X.sub.8ATXYX.sub.9 x.sub.1 is F
or L X.sub.1 is A or G X.sub.2 is N or S X.sub.2 is R or K X.sub.3 is
T or N X.sub.3 is S, T, R, or L X.sub.4 is A, S, or W X.sub.4 is K
or N X.sub.5 is N, K, or T X.sub.5 is S, E, or Q (SEQ ID NO: 1652)

X.sub.6 is N, S, or D X.sub.7 is N or D X.sub.8 is Y or F X.sub.9 is A or S (SEQ ID NO: 1659) Consensus 5 FXFNTYAMN XAXIRSKSNNYATXYA (SEQ ID NO: 1653)(SEQ ID NO: 1660) Consensus 6 YYFX.sub.1X.sub.2XWX.sub.3X IGXIX.sub.1PX.sub.2XX.sub.3X.sub.4X.sub.5X.sub.6X.sub.7N X.sub.1 is S or T X.sub.1 is Y or D X.sub.2 is N, S, or T X.sub.2 is G or N X.sub.3 is S I or M (SEQ ID NO: 1654) X.sub.3 is G or D X.sub.4 is N or D X.sub.5 is T, R, or S X.sub.6 is N or K X.sub.7 is Y or F (SEQ ID NO: 1661) Consensus 7 YXFSNMVIX IGXIYPGXGDTNYN (SEQ ID NO: 1662)(SEQ ID NO: 1655)

TABLE-US-00026 TABLE D5 EU or Kabat light chain Framework sequences Ab ID VL FR1 VL FR2 VL FR3 VL FR4 4D11 DIZVTQSPASLSAS WYQLKQGKSPQ GVPSRFGSGSGTGQFS FGGGTKLEIK VGETVTITC LLVY LRINSLQPEDFGSYYC (SEQ ID NO: 968)(SEQ ID NO: 923)(SEQ ID NO: 940)(SEQ ID NO: 950) 78C5 DIZVTQSPASLSAS WYQLKQGKSPQ GVPSRFGSGSGTGQFS FGGGTKLEIK VGETVTITC LLVY LRINSLQPEDFGSYYC (SEQ ID NO: 968)(SEQ ID NO: 923)(SEQ ID NO: 940)(SEQ ID NO: 950) 6G12 TMSQSPSSLAVSV WYQQKPGQSPK GVPDRFTGSGSGTDF FGAGTKLELK GEKVTMSC LLIY TLTISVKAEDLAVY (SEQ ID NO: 969)(SEQ ID NO: 924)(SEQ ID NO: 941) YC (SEQ ID NO: 951) 8F11 DVZMTQTPLTSLV WLLQRPGQSPK GVPDRFAGSGSGTDF FGAGTKLELK TIGQPASISC RLIY TLKISRLEADDLGIYY (SEQ ID NO: 969)(SEQ ID NO: 925)(SEQ ID NO: 942) C (SEQ ID NO: 952) 8E10 DVZMTQTPLTSLV WLLQRPGQSPK GVPDRFTGSGSGTDF FGAGTKLELK TIGQPASISC RLIY TLKISRVEAEDLG VY (SEQ ID NO: 968)(SEQ ID NO: 925)(SEQ ID NO: 942) YC (SEQ ID NO: 953) 7E5 DVZMTQTPLTSLV WLLQRPGQSPK GVPDRFAGSGSGTDF FGAGTKLELK TIGQPASISC RLIY TLKISRLEADDLGIYY (SEQ ID NO: 969)(SEQ ID NO: 925)(SEQ ID NO: 942) C (SEQ ID NO: 952) 7E5v2 DVVMTQTPLTSLV WLLQRPGQSPK GVPDRFAGSGSGTDF FGAGTKLELK TIGQPASISC RLIY TLKISRLEADDLGIYY (SEQ ID NO: 969)(SEQ ID NO: 931)(SEQ ID NO: 942) C (SEQ ID NO: 952) 7F8 VLTQSPALMSASP WYQQKPRSSPK GVPARFSGSGSGTSY FGGGTKLVK GEKVTMTC PWIY SLTINNMEAEDAATY (SEQ ID NO: 970)(SEQ ID NO: 926)(SEQ ID NO: 943) YC (SEQ ID NO: 954) 8F8 DVZMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGAGTKLELK SLGDQASISC LLIY TLKISRVEAEDLG VY (SEQ ID NO: 969)(SEQ ID NO: 927)(SEQ ID NO: 944) FC (SEQ ID NO: 955) 1H7 VLTQSPAIMZASP WYQQKPRSSPK GVPARFSGSGSGTSY FGGGTKLVK GEKVTMTC PWIY SLTISSMEAEDAATY (SEQ ID NO: 970)(SEQ ID NO: 928)(SEQ ID NO: 943) YC (SEQ ID NO: 956) 2H8 NVLTQSPALMSAS WYQQKPRSSPK GVPARFSGSGSGTSY FGGGTKLVK PGEKVTMTC PWIY SLTISSMEAEDAATY (SEQ ID NO: 970)(SEQ ID NO: 929)(SEQ ID NO: 943) YC (SEQ ID NO: 956) 3A2 DVVMTQTPLSLPV WYLRKPGQSPK GVPDRFSGSGSGTDF FGGGTELEIK SLGDQASISC LLIY TLKISRVEAEDLG VY (SEQ ID NO: 971)(SEQ ID NO: 930)(SEQ ID NO: 945) YC (SEQ ID NO: 957) 3A7 DVVMTQTPLTSLV WLLQRPGQSPK GVPDRFAGSGSGTDF FGGGTKLEMK TIGQPASISC RLIY TZKISRLEADDLGIYY (SEQ ID NO: 972)(SEQ ID NO: 931)(SEQ ID NO: 942) C (SEQ ID NO: 958) 3B10 ITMSQSPSSLAVSV WYQQKPGQSPK GVPDRFTGSGSGTDF FGAGTKLELK GEKVTMSC LLIY TLTISVKAEDLAVY (SEQ ID NO: 969)(SEQ ID NO: 932)(SEQ ID NO: 941) CC (SEQ ID NO: 959) 4F11 DVZMTQTPLSLPV WYLRKPGQSPK GVPDRFSGSGSGTDF FGGGTELEIK SLGDQASISC LLIY TLKISRVEGEDLG VY (SEQ ID NO: 971)(SEQ ID NO: 927)(SEQ ID NO: 945) YC (SEQ ID NO: 960) 6H6 QTQSPSSLAVSAG WYQQKPGQSPK

GVPDFRFTGSGFGTDF FGAGTKLEL EKVTLSC LLIS TLTISSVQGEDLAVY (SEQ ID NO: 969) (SEQ ID NO: 1410) (SEQ ID NO: 1413) YC (SEQ ID NO: 1414) 7A9 QMSQSPACLZAZV WYQQKQGKSPK GVPSRFSGRGSGTQF FGSGTKLEIK GESVTITC LVVY FLKINSZQREDFGSY (SEQ ID NO: 973) (SEQ ID NO: 933) (SEQ ID NO: 946) YC (SEQ ID NO: 961) 8A1 DIQMTQSPASLSVS WYQQKQGKSPQ GVPSRFSASGSATQFS FGGGTKLEMN VGETVTITC LVVY LKINSLQSADEFGSY (SEQ ID NO: 974) (SEQ ID NO: 934) (SEQ ID NO: 947) C (SEQ ID NO: 962) 9F5 DVZMTQNPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGGGTKLEIK SLGDQASISC LLIY TLKISRVEADDLGVY (SEQ ID NO: 968) (SEQ ID NO: 935) (SEQ ID NO: 944) LC (SEQ ID NO: 963) 9F5v2 DVVMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGGGTKLEIK SLGDQASISC LLIY TLKISRVEADDLGVY (SEQ ID NO: 968) (SEQ ID NO: 930) (SEQ ID NO: 944) FC (SEQ ID NO: 1668) 9G1 DVLMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGGGTKLEIK SLGDQASISC LLIY TLRISGVEAEDDLGVY (SEQ ID NO: 968) (SEQ ID NO: 936) (SEQ ID NO: 944) FC (SEQ ID NO: 964) 9G3 NVLTQSPALIWAZ WXXXKPRSSPK GVPGRFSGSGSGTYX FGGGTKLEMK PGEKVTMTC PGIY SFKISSMEGKMGPLII (SEQ ID NO: 975) (SEQ ID NO: 937) (SEQ ID NO: 948) FC (SEQ ID NO: 965) 10A9 DVVMTQTPLSLPV WYLKPGQSPK GVPDRFSGSGSGTDF FGGGTELEIK SLGDQASISC LLIY TLKISRVEAEDDLGVY (SEQ ID NO: 971) (SEQ ID NO: 930) (SEQ ID NO: 945) YC (SEQ ID NO: 957) 11A8 DIZMTQSPSSLTVT WYQQKPGQPZK GVRDRFTGSGZGTDF FGGGTKLEMK AGEKVTMSC LLIY TLTISSVQGEDLAIYY (SEQ ID NO: 972) (SEQ ID NO: 938) (SEQ ID NO: 949) C (SEQ ID NO: 966) 12D9 TQSPSSLAVSVGE WYQQKPGQSPK GVPDRFTGSGSGTDF FGSGTKLEIK KVTMTC LLIY TLTISTVKAEDLAVY (SEQ ID NO: 973) (SEQ ID NO: 939) (SEQ ID NO: 941) YC (SEQ ID NO: 967) 12F9 TMSQSPSSLAVSV WYQQKPGQSPK GVPDRFTGSGSGTDF FGAGTKLELK GEKVTMSC LLIY TLTISSVKAEDLAVY (SEQ ID NO: 969) (SEQ ID NO: 924) (SEQ ID NO: 941) CC (SEQ ID NO: 959) 10C1 QTQVFLSLLLWVS WYQQKPGQSPK GVPDRFTGSGSGTDF FGAGTKLELK GTCGNIMLTQSPSS LLIS TLTINSVQAEDLAVY (SEQ ID NO: 969) LAVSAGEKVTLS (SEQ ID NO: 1413) YC (SEQ ID NO: 1415) (SEQ ID NO: 1411) 7E9 DIVMSQSPSSLAVS WYQQKPGQSPK GVPDRFTGSGSGTDF FGAGTKLELK VGEKVTMSC LLIY TLTISSVKAEDLAVY (SEQ ID NO: 969) (SEQ ID NO: 1412) (SEQ ID NO: 941) YC (SEQ ID NO: 951) 8C3 DVVMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGSGTKLEIK SLGDQASISC LLIY TLKISRVEAEDDLGVY (SEQ ID NO: 973) (SEQ ID NO: 930) (SEQ ID NO: 944) FC (SEQ ID NO: 955) IB4v1 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGFGTDF FGGGTKLEIK SVGDRVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1531) (SEQ ID NO: 941) YC (SEQ ID NO: 1535) IB4v2 ZVVZTQTPLSLPVS WFLQKPGQSPK GVPDRFSGSGSGTDF FGAGTKLELK LGDQASFSCRS LLIF TLKISRVEAEDDLGVY (SEQ ID NO: 969) (SEQ ID NO: 1591) (SEQ ID NO: 1597) FC (SEQ ID NO: 955) 6H2 DVLMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGSGTKLEIK SLGDQASISCRS LLIY TLKISRVEAEDDLGVY (SEQ ID NO: 973) (SEQ ID NO: 1532) (SEQ ID NO: 944) YC (SEQ ID NO: 957) 7B11 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGSGTDY FGGGTKLEIK SVGDRVSITCKA LLIY TLTISSVQAEDLALY (SEQ ID NO: 968) (SEQ ID NO: 1531) (SEQ ID NO: 941) YC (SEQ ID NO: 1536) 18D8 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGFGTDF FGGGTKLEIK SIGARVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1533) (SEQ ID NO: 941) YC (SEQ ID NO: 1535) 18E4v1 DIVMTQSPKSMMSM WYQQKPGQSPK

GVPDRFTGSGSATDF FGGGTKLEIK SVGERVTLTCKA LLIY TLTISSVQAEDLADY (SEQ ID NO: 968) (SEQ ID NO: 1534) (SEQ ID NO: 941) HC (SEQ ID NO: 1537) 18E4v2 NIVMTQSPKSMMSM WYQQKPGQSPK GVPDRFTGSGSATDF FGGGTKLEIK SVGERVTLTCKA LLIY TLTISSVQAEDLADY (SEQ ID NO: 968) (SEQ ID NO: 1592) (SEQ ID NO: 941) HC (SEQ ID NO: 1537) 18E4v3 DVVMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGAGTKLELK SLGDQASISCRS LLIY TLRISRVEAEDLG VY (SEQ ID NO: 969) (SEQ ID NO: 1593) (SEQ ID NO: 944) FC (SEQ ID NO: 1601) 29F6v1 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGSGTDF FGGGTKLEIK SIGARVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1533) (SEQ ID NO: 941) YC (SEQ ID NO: 1538) 29F6v2 DVVMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGAGTKLELK SLGDQASISCRS LLIY TLKISRVEAEDLG VY (SEQ ID NO: 969) (SEQ ID NO: 1593) (SEQ ID NO: 944) FC (SEQ ID NO: 955) 40D5 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGSGTDF FGGGTKLEIK SIGARVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1533) (SEQ ID NO: 941) YC (SEQ ID NO: 1538) 43B9 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGSGTDF FGGGTKLEIK SIGARVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1533) (SEQ ID NO: 941) YC (SEQ ID NO: 1538) 44A8v1 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGSGTDF FGGGTKLEIK SVGDRVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1531) (SEQ ID NO: 941) YC (SEQ ID NO: 1538) 44A8v2 DIVLTQSPASLAVS WYQQKPGQPPK GVPARFSGSGSGTDF FGGGTKLEIK LGQRATISCRA LLIY SLNIHPVEEDDIAMY (SEQ ID NO: 968) (SEQ ID NO: 1594) (SEQ ID NO: 1598) FC (SEQ ID NO: 1602) 44B4v1 DIVMTQSHKFMST WYQQKPGQSPK GVPDRFTGSGSGTDF FGGGTKLEIK SIGARVSITCKA LLIY TFTISSVQAEDLAVY (SEQ ID NO: 968) (SEQ ID NO: 1533) (SEQ ID NO: 941) YC (SEQ ID NO: 1538) 44B4v2 DIQMTQFPASLAA WYQQKQKGKSPQ GVPSRFSGSGSGTQY FGGGTKLEIK XVGESVTITCRA LLIY SMKINSMQPEDTAIY (SEQ ID NO: 968) (SEQ ID NO: 1595) (SEQ ID NO: 1599) FC (SEQ ID NO: 1603) 29F7 ILLTQSPAILSVSPG WYQQRTNGSPR GIPSRFSGSGSGTDF FGAGTKLELK ERVSFSC LLIK LNINSVESEDIADYYC (SEQ ID NO: 969) (SEQ ID NO: 1596) (SEQ ID NO: 1600) (SEQ ID NO: 1604) 32G1 DVVMTQTPLSLPV WYLQKPGQSPK GVPDRFSGSGSGTDF FGAGTKLELK SLGDQASISC LLIY TLKISRVEAEDLG VY (SEQ ID NO: 969) (SEQ ID NO: 930) (SEQ ID NO: 944) FC (SEQ ID NO: 955) TABLE-US-00027 TABLE D6 EU or Kabat heavy chain Framework sequences Ab ID VH FR1 VH FR2 VH FR3 VH FR4 44D11 EVKLVESGGGLV WVRQTPEKRLEW DSVQGRFTFSRDNA WGQGTLVTVSA KPGGSLKLSCAAS (SEQ ID NO: 995) RNILYLQMSSLRSED (SEQ ID NO: 1029) G (SEQ ID NO: 976) TAMYCYC (SEQ ID NO: 1008) 78C5 EVKLVESGGGLV WVRQTPEKRLEW DSVQGRFTFSRDNA WGQGTLVTVSA KPGGSLKLSCAAS (SEQ ID NO: 995) RNILYLQMSSLRSED (SEQ ID NO: 1029) G (SEQ ID NO: 976) TAMYCYC (SEQ ID NO: 1008) 6G12 EVQLQQSGPELVK WVKQSHGKSLEW QKFKGKASLTVDKS WGQGTSVTVSS PGTSVKISCKTSG (SEQ ID NO: 996) SSTAYMELHSLASD (SEQ ID NO: 1030) (SEQ ID NO: 977) DSAVYYC (SEQ ID NO: 1009) 8E10 QVQLQQSGAELV WVKQTPVHGLEW QKFKGKAILTADKS WGQGTSVTVSS RPGASVTLSCKAS (SEQ ID NO: 997) SSTAYMELRSLTSED (SEQ ID NO: 1030) G (SEQ ID NO: 978) SAVYYC (SEQ ID NO: 1010) 7E5 EVKLEESGGGLVQP WVRQSPEKGLEW ESVKGRFTISRDDSK WGQGTTLTVSS GGSMKLSCAASG (SEQ ID NO: 998) STVYLQMNTLRADD (SEQ ID NO: 1031) (SEQ ID NO: 979) TGIYYC (SEQ ID NO: 1011) 7F8 EVQLVESGGGLV WVRQAPGKGLEW DSVKDRITCSRDDSE

WGTGTTVTST QPKGSLKLSCAA (SEQ ID NO: 999) NMFYLQLSSLKTED (SEQ ID NO: 1032) SG TAMYYC (SEQ ID NO: 980) (SEQ ID NO: 1012) 8F8 QVQLQQSGAELVK WVKQRPGRGLEW EKFKTKATLTVDKP WGQGTTTLTVSS PGASVKLSCKASG (SEQ ID NO: 1000) SSTAYMQVSSLTSE (SEQ ID NO: 1031) (SEQ ID NO: 981) DSAVYYC (SEQ ID NO: 1013) IH7 ZVQLVESGGGLV WVRQAPGKGLEW DSVKDRFTCSRDDS WGTGTTVTTVSS QPKGSLKLSCAA (SEQ ID NO: 999) ENMFYLQLSSLKTE (SEQ ID NO: 1033) SG DTAIYYC (SEQ ID NO: 982) (SEQ ID NO: 1014) 2HS EVQLVESGGGLVQP WVRQAPGKGLEW DSVKDRFTCSRDDS WGTGTTVTTVSS KGSLKLSCAASG (SEQ ID NO: 999) ENMFYLQLSSLKTE (SEQ ID NO: 1033) (SEQ ID NO: 980) DTAMYYC (SEQ ID NO: 1015) 3A2 QVQLQQSGAELVK WVKQRPQGGLV EKFKTKATLTVDTS WGQGTTLTVST PGASVKMSCKTSG W STAYMHLSSLTSEDS (SEQ ID NO: 1034) (SEQ ID NO: 983) (SEQ ID NO: 1001) AVYFC (SEQ ID NO: 1016) 3A7 EVKLEESGGGLVQP WVRQSPEKGLEW ESVKGRFTISRDDSK WGQGTTTLTVSS GGSMKLSCAASG (SEQ ID NO: 998) STVYLQMNTLRADD (SEQ ID NO: 1031) (SEQ ID NO: 979) TGIYYC (SEQ ID NO: 1011) 3B10 EVQLVZZGRGZSQ GVPQGPKGREW DSVKDRFTZSRDDS WGQGTIVTVS GKGSXZZGRAZRC (SEQ ID NO: 1002) ESLFYZQMSZZKZE (SEQ ID NO: 1035) (SEQ ID NO: 984) DTAMYYZ (SEQ ID NO: 1017) 4F11 QVQLQQSGAELVK WVKQRPQGGLV EKFKTKATLTVDTS WGQGTTLTVST PGASVKMSCKTSG W STAYMHL (SEQ ID NO: 1034) (SEQ ID NO: 983) (SEQ ID NO: 1001) SLTSEDSAVYFC (SEQ ID NO: 1016) 6H6 EVKLEESGGGLVQP WVRQSPEKGLEW ESVKGRFTISRDDSK WGQGTTLTVSA GGSMKLSCASG (SEQ ID NO: 998) STVYLQMNSLRTE (SEQ ID NO: 1029) (SEQ ID NO: 985) TGIYYC (SEQ ID NO: 1018) 7A9 LSCAASG WVRQAPGKGLEW DSVKDRFTISRDDSE WGQGTTLTVSA (SEQ ID NO: 986) (SEQ ID NO: 999) SMLYLQMZNLKTED (SEQ ID NO: 1029) TAMYYC (SEQ ID NO: 1019) 7B3 QVQLQQSGAVLVK WVKQRPGRGPEW EKFRNKAILTVDKPS WGQGTTTLTVSS PGASVKLSCKASG (SEQ ID NO: 1003) STAYMQLNSLTSED (SEQ ID NO: 1031) (SEQ ID NO: 987) ZAVYYC (SEQ ID NO: 1020) 8A1 EVQLQQSGAELVK WVKQRPKGKLEW GKFEKGATLTADKS WGQGTTLTVSA PGASVKISCKASG (SEQ ID NO: 1004) SSTAYMQLSSLTSED (SEQ ID NO: 1029) (SEQ ID NO: 988) SAVYFC (SEQ ID NO: 1021) 9F5 QVQLQQSGPELVK WVKQRPKGKLEW GEFVRATLTADTS WGQGASVTVSS GASLKISCKASG (SEQ ID NO: 1004) TTAYMQLSSLTSEDS (SEQ ID NO: 1036) (SEQ ID NO: 989) AVYFC (SEQ ID NO: 1022) 9G1 QVQLQQSGAELVK WVKQRPGRGPEW EKFKTKATLTVDKP WGQGTTTLTVSS PGASVKLSCKASG (SEQ ID NO: 1003) SSTADMQLSSLTSED (SEQ ID NO: 1031) (SEQ ID NO: 981) SAVYYC (SEQ ID NO: 1023) 9G3 EVQLVESGGGLVQP WVRQTPGKGLEW DSVKDRFTISRDDSE WGRGTLV KGSLKLSCAAFG (SEQ ID NO: 1005) SIVYVQMNNLKTED (SEQ ID NO: 1037) (SEQ ID NO: 990) TGMYS (SEQ ID NO: 1024) 10A9 QVQLQQSGAEVVK WVKQRPQGGLV ERFKTKATLTVDTS WGQGTTLTVSA PGASVKMSCKTSG W STAYMHLSSLTSEDS (SEQ ID NO: 1029) (SEQ ID NO: 991) (SEQ ID NO: 1001) AVYFC (SEQ ID NO: 1025) 11A8 EVQLVESGGRLVQP WVRQAPGKGLEW DSVKDRFTISRDDSE WGQGTTLTVSA KGSLKLSCAASG (SEQ ID NO: 999) SMLYLQMNNLKTE (SEQ ID NO: 1029) (SEQ ID NO: 992) DTAMYYC (SEQ ID NO: 1026) 12D9 QVQLQQYGPELVK WMKQSHGKSLE QEFKGKATLTVDKS WGQGT PGASVKMSCKVSG W SS (SEQ ID NO: 1038) (SEQ ID NO: 993) (SEQ ID NO: 1006) TAYMELRSLTFEDS AV YZC (SEQ ID NO: 1027) 12F9 WRIGQGKGSCLKA RVRQGPKGKREW DSVKDRFRASRDDS WGQGTTLTVSA RAARG (SEQ ID NO: 1007) ES (SEQ ID NO: 1029) (SEQ ID NO: 994) MLYVQMSNWKQED T AMYYG (SEQ ID NO: 1028) iOCi

GVQSEVKFEESGG VWRQSPKGLEW ESKVGRFTISRDDSK WGQGTTLTVSA
GLVQPGGSMKLSC (SEQ ID NO: 998) SSVSLQMNSLRTE (SEQ ID NO: 1029)
TASG TGIYYC (SEQ ID NO: 1416) (SEQ ID NO: 1419) 7E9 QVQLQQSGPELVK
WVKQSHGKSLEW QKFKGKATLTVDKS WGQGTSTVTSS PGASVKISCKTSG (SEQ ID
NO: 996) SSTAYMELRSLTSED (SEQ ID NO: 1030) (SEQ ID NO: 1417) SAVYYC
(SEQ ID NO: 1420) SC3 QVQLQQSGPDLVKP WVKQSHGKSLEW QKFKGKATLTVDKS
WGQGTTLTVSS GASVKISCKASG (SEQ ID NO: 996) SSTAYMELRSLTSED (SEQ
ID NO: 1031) (SEQ ID NO: 1418) SAVYYC (SEQ ID NO: 1421) 1B4
EVQLVESGGGLVKP WVRQTPEKRLEW DSMKGRFTISRDN WGQGTTLTVSS
GGSLKLSCEA (SEQ ID NO: 995) KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ
ID NO: 1539) DTAMYYC (SEQ ID NO: 1542) 6H2 QVQLQESGPGLVQP
WIRQSPGKGLEW AAFISRLNISKDNSK WGQGTTLTVSA SQSLSICTV (SEQ ID NO:
1541) SQVFFKMNSLQSD (SEQ ID NO: 1029) (SEQ ID NO: 1540) TAIYYC
(SEQ ID NO: 1543) 7B11v1 EVQLVESGGGLVKP WVRQTPEKRLEW
DSMKGRFTISRDN WGQGTTLTVSS GGSLKLSCEA (SEQ ID NO: 995)
KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ ID NO: 1539) DTAMYYC (SEQ
ID NO: 1542) 7B11v2 EVQZQQSGPELVK WVKQSLGKSLEW QKFKGKATLTVDKS
RGTGTTVTV GASVKISCKA (SEQ ID NO: 1613) SSTAYMELRSLTXEE (SEQ ID
NO: 1626) (SEQ ID NO: 1605) SAVYYC (SEQ ID NO: 1618) 18D8
EVQLVESGGGLVKP WVRQTPEKRLEW DSMKGRFTISRDN WGQGTTLTVSS
GGSLKLSCEA (SEQ ID NO: 995) KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ
ID NO: 1539) DTAMYYC (SEQ ID NO: 1542) 18E4v1 EVQLVESGGGLVKP
WVRQTPEKRLEW DSMKGRFTISRDN WGQGTTLTVSS GGSLKLSCEA (SEQ ID
NO: 995) KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ ID NO: 1539)
DTAMYYC (SEQ ID NO: 1542) 18E4v2 QVQLQQPGAELVK WVKEKPGQGLEW
HNFKGKATLTVDKS WGQGTSTVTSS PGASVKVSCA (SEQ ID NO: 1614)
SSTAYMQLNSLTSE (SEQ ID NO: 1030) (SEQ ID NO: 1606) DSAVYYC (SEQ
ID NO: 1619) 29F6v1 EVQLVESGGGLVKP WVRQTPEKRLEW DSMKGRFTISRDN
WGQGTTLTVSS GGSLKLSCEA (SEQ ID NO: 995) KNFLYLQMSSLRSE (SEQ ID
NO: 1031) (SEQ ID NO: 1539) DTAMYYC (SEQ ID NO: 1542) 29F6v2
QVQLQQSVAELVRP WVKQRPEQGLEW PKFQATATITVATSS WGHGTSVTSS
GASVKLSCTA (SEQ ID NO: 1615) NSAYLQLSSLASEDT (SEQ ID NO: 1627)
(SEQ ID NO: 1607) AIYYC (SEQ ID NO: 1620) 40D5v1 EVQLVESGGGLVKP
WVRQTPEKRLEW DSMKGRFTISRDN WGQGTTLTVSS GGSLKLSCEA (SEQ ID
NO: 995) KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ ID NO: 1539)
DTAMYYC (SEQ ID NO: 1542) 40D5v2 QVQLQQSGAELVK WVKQRPGQGLEW
QKFKGKATLTVDKS WSQGTTLTVS PGASVKVSCA (SEQ ID NO: 1616)
SSTAYMQLSSLTSE (SEQ ID NO: 1628) (SEQ ID NO: 1608) DSAVYYC (SEQ
ID NO: 1621) 43B9 EVQLVESGGGLVKP WVRQTPEKRLEW DSMKGRFTISRDN
WGQGTTLTVSS GGSLKLSCEA (SEQ ID NO: 995) KNFLYLQMSSLRSE (SEQ ID
NO: 1031) (SEQ ID NO: 1539) DTAMYYC (SEQ ID NO: 1542) 44A8
EVQLVESGGGLVKP WVRQTPEKRLEW DSMKGRFTISRDN WGQGTTLTVSS
GGSLKLSCEA (SEQ ID NO: 995) KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ
ID NO: 1539) DTAMYYC (SEQ ID NO: 1542) 44B4v1 EVQLVESGGGLVKP
WVRQTPEKRLEW DSMKGRFTISRDN WGQGTTLTVSS GGSLKLSCEA (SEQ ID
NO: 995) KNFLYLQMSSLRSE (SEQ ID NO: 1031) (SEQ ID NO: 1539)
DTAMYYC (SEQ ID NO: 1542) 44B4v2 XXXXXQSGTELARP WVKQRPGQGLEW
QKFKDKATLTADKS WGTGTTVTSS GASVKMPCKA (SEQ ID NO: 1616)
SSTAYMQLSSLTSEE (SEQ ID NO: 1033) (SEQ ID NO: 1609) SAVYYC (SEQ
ID NO: 1622) 45D6 QVQLQQSGRELVK WVIQSHGESLEW QKFKGKATLTVNKS

WGQGTSTVTVSS GASVKMSCMSSG (SEQ ID NO: 1617) SSTAYMELRSLTSED (SEQ ID NO: 1030) (SEQ ID NO: 1610) SAVYYC (SEQ ID NO: 1623) 29F7 QVKLEESGGGLVQP WVRQSPEKGLEW ESVKGRFTISRDDSK WGQGTTTLTVSS GGSMKLSCVASG (SEQ ID NO: 998) SSVYLQMNNLRAVD (SEQ ID NO: 1031) (SEQ ID NO: 1611) TGIYYC (SEQ ID NO: 1624) 32G1 QVQLQQSGPELVKP WVKQSHGKSLEW QKFKGKATLTVNKS WGQGTTLTVSA GASVQMSCEASG (SEQ ID NO: 996) SSTAYIELRSLTSED (SEQ ID NO: 1029) (SEQ ID NO: 1612) AVYHC (SEQ ID NO: 1625)

TABLE-US-00028 TABLE D7 Humanized light chain variable region sequences Antibody variant Humanized sequences Antibody 4D11 4D11V3-15

EIVMTQSPATLSVSPGERATLSCRASENIYSFLAWYQQKPGQAPRLLIY
NSKTFAEGIPARFSGSGSGTEFTLTISLQSEDFAVYYCQHGHYGTTPWTF GQGTKVEIK
(SEQ ID NO: 1040) 4D11V1-9

DIQLTQSPSFLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1041) 4D11V3-11

EIVLTQSPATLSLSPGERATLSCRASENIYSFLAWYQQKPGQAPRLLIY
NSKTFAEGIPARFSGSGSGTDFTLTISLQPEDFATYYCQHGHYGTTPW TFGQGTKVEIK
(SEQ ID NO: 1042) 4D11V1-5

DIQMTQSPSTLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1043) 4D11V1-39

DIQMTQSPSSLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1044) 4D11V1-33

DIQMTQSPSSLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSGSGSGTDFTFTISLQPEDFATYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1045) 4D11V3-20

EIVLTQSPGTLSPGERATLSCRASENIYSFLAWYQQKPGQAPRLLIY
NSKTFAEGIPDRFSGSGSGTDFTLTISRLEPEDFATYYCQHGHYGTTPW TFGQGTKVEIK
(SEQ ID NO: 1046) 4D11V2-28

DIVMTQSPLSLPVTPGEPASISCRASENIYSFLAWYLQKPGQSPQLLIY
NSKTFAEGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1047) 4D11V2-30

DVVMQTQSPLSLPVTLGQPASISCRASENIYSFLAWFQQRPQGSPRRLI
YNSKTFAEGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQHGHYGTTP
PWTFGQGTKVEIK (SEQ ID NO: 1048) 4D11V4-1

DIVMTQSPDSLAVSLGERATINCRASENIYSFLAWYQQKPGQPPKLLI
YNSKTFAEGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCQHGHYGTTP
PWTFGQGTKVEIK (SEQ ID NO: 1049) Antibody 7C5 7C5V3-15

EIVMTQSPATLSVSPGERATLSCRASENIYSFLAWYQQKPGQAPRLLI
YNSKTFAEGIPARFSGSGSGTEFTLTISLQSEDFAVYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1040) 7C5V1-9

DIQLTQSPSFLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQHGHYGTTP WTFGQGTKVEIK
(SEQ ID NO: 1041) 7C5V3-11

EIVLTQSPATLSLSPGERATLSCRASENIYSFLAWYQQKPGQAPRLLIY
NSKTFAEGIPARFSGSGSGTDFTLTISLQPEDFATYYCQHGHYGTTPW TFGQGTKVEIK
(SEQ ID NO: 1042) 7C5V1-5

DIQMTQSPSTLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI

YNSKTFAEGVPSRFSFGSGSGTEFTLTISLQPDGFATYYCQHGYGTPP WTFGQGGTKVEIK
(SEQ ID NO: 1043) 7C5V1-39
DIQMTQSPSSLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSFGSGSGTDFTLTISLQPEDFATYYCQHGYGTPP WTFGQGGTKVEIK
(SEQ ID NO: 1044) 7C5V1-33
DIQMTQSPSSLSASVGDRVITTCRASENIYSFLAWYQQKPGKAPKLLI
YNSKTFAEGVPSRFSFGSGSGTDFTFTISLQPEDATYYCQHGYGTPP WTFGQGGTKVEIK
(SEQ ID NO: 1045) 7C5V3-20
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NSKTFAEGIPDRFSFGSGSGTDFTLTISRLEPEDFAVYYCQHGYGTPPW TFGQGGTKVEIK
(SEQ ID NO: 1046) 7C5V2-28
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NSKTFAEGVPDRFSFGSGSGTDFTLKISRVEAEDVGVYYCQHGYGTPP WTFGQGGTKVEIK
(SEQ ID NO: 1047) 7C5V2-30
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PWTFGQGGTKVEIK (SEQ ID NO: 1048) 7C5V4-1
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YNSKTFAEGVPDRFSFGSGSGTDFTLTISLQAEDVAVYYCQHGYGTP
PWTFGQGGTKVEIK (SEQ ID NO: 1049) Antibody 6G12 6G12V4-1
DIVMTQSPDSLAVSLGERATTNCKSSQSLLYSSNQKNCLAWYQQKPG
QPPKLLIYWAFTRESGVDRFSFGSGSGTDFTLTISLQAEDVAVYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1051) 6G12V2-30
DVVMTQSPLSLPVTTLGQPASISCKSSQSLLYSSNQKNCLAWFQQRPG
QSPRRLIYWAFTRESGVDRFSFGSGSGTDFTLKISRVEAEDVGVYYC
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1052) 6G12V2-28
DIVMTQSPLSLPVTTPGEPASISCKSSQSLLYSSNQKNCLAWYLQKPG
QSPQLLIYWAFTRESGVDRFSFGSGSGTDFTLKISRVEAEDVGVYYC
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1053) 6G12V1-9
DIQLTQSPSFLSASVGDRVITTCSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWAFTRESGVPSRFSFGSGSGTEFTLTISLQPEDFATYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1054) 6G12V1-5
DIQMTQSPSTLSASVGDRVITTCSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWAFTRESGVPSRFSFGSGSGTEFTLTISLQPDGFATYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1055) 6G12V3-15
EIVMTQSPATLSVSPGERATLSCSSQSLLYSSNQKNCLAWYQQKPG
QAPRLLIYWAFTRESGIPARFSFGSGSGTEFTLTISLQSEDFAVYYCQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1056) 6G12V1-33
DIQMTQSPSSLSASVGDRVITTCSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWAFTRESGVPSRFSFGSGSGTDFTFTISLQPEDATYYCQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1057) 6G12V1-39
DIQMTQSPSSLSASVGDRVITTCSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWAFTRESGVPSRFSFGSGSGTDFTLTISLQPEDFATYYCQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1058) 6G12V3-11
EIVLTQSPATLSLSPGERATLSCSSQSLLYSSNQKNCLAWYQQKPG
QAPRLLIYWAFTRESGIPARFSFGSGSGTDFTLTISLEPEDFAVYYCQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1059) 6G12V3-20
EIVLTQSPGTLSPGERATLSCSSQSLLYSSNQKNCLAWYQQKPG
QAPRLLIYWAFTRESGIPDRFSFGSGSGTDFTLTISRLEPEDFAVYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1060) Antibody 8F11 8F11V2-30

DVVMTQSPSLSLPVTPGEPASISCKSSQSLLYSNGKTFLSWYQQKPGQS
PRRLIYLVSKLDSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1062) 8F11V2-28
DIVMTQSPSLSLPVTPGEPASISCKSSQSLLYSNGKTFLSWYLQKPGQS
PQLLIYLVSKLDSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1063) 8F11V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSNGKTFLSWYQQKPGQ
PPKWYLVSKLDSGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1064) 8F11V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKWYLVSKLDSGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1065) 8F11V1-9
DIQLTQSPSFLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKWYLVSKLDSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1066) 8F11V1-39
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKWYLVSKLDSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1067) 8F11V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKWYLVSKLDSGVPSRFSGSGSGTDFTTISLQPEDFATYYCMQ
THFPLTFGQGGTKVEIK (SEQ ID NO: 1068) 8F11V3-15
EIVMTQSPATLSVSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQ
APRLLIYLVSKLDSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCMQ
THFPLTFGQGGTKVEIK (SEQ ID NO: 1069) 8F11V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQ
APRLLIYLVSKLDSGIPARFSGSGSGTDFTLTISLQPEDFATYYCMQ
GTHFPLTFGQGGTKVEIK (SEQ ID NO: 1070) 8F11V3-20
EIVLTQSPGTLSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQA
PRLLIYLVSKLDSGIPDRFSGSGSGTDFTLTISRLEPEDFATYYCMQ
THFPLTFGQGGTKVEIK (SEQ ID NO: 1071) Antibody 8E10 8E10V2-30
DVVMTQSPSLSLPVTLGQPASISCKSSQSLLDSDGKTYLNWFQQRPGQ
SPRRLIYLVSKLDSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCWQ
GTHFPYTFGQGGTKVEIK (SEQ ID NO: 1073) 8E10V2-28
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PQLLIYLVSKLDSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCWQ
GTHFPYTFGQGGTKVEIK (SEQ ID NO: 1074) 8E10V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLDSDGKTYLNWYQQKPGQ
PPKWYLVSKLDSGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCWQ
GTHFPYTFGQGGTKVEIK (SEQ ID NO: 1075) 8E10V1-9
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APKWYLVSKLDSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCWQ
THFPYTFGQGGTKVEIK (SEQ ID NO: 1076) 8E10V1-5
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APKWYLVSKLDSGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCWQ
GTHFPYTFGQGGTKVEIK (SEQ ID NO: 1077) 8E10V1-39
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GTHFPYTFGQGGTKVEIK (SEQ ID NO: 1078) 8E10V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLDSDGKTYLNWYQQKPGK
APKWYLVSKLDSGVPSRFSGSGSGTDFTTISLQPEDFATYYCWQ

GTHFPYTFGQGTKVEIK (SEQ ID NO: 1079) 8E10V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLDSDGKTYLNWYQQKPGQ
APRLLIYLVSKLDSGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCWQG
THFPYTFGQGTKVEIK (SEQ ID NO: 1080) 8E10V3-15
EIVMTQSPATLSVSPGERATLSCKSSQSLDSDGKTYLNWYQQKPGQ
APRLLIYLVSKLDSGIPARFSGSGSGTEFTLTISSELEPEDFAVYYCWQ
GTHFPYTFGQGTKVEIK (SEQ ID NO: 1081) 8E10V3-20
EIVLTQSPGTLSPGERATLSCKSSQSLDSDGKTYLNWYQQKPGQ
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GTHFPYTFGQGTKVEIK (SEQ ID NO: 1082) Antibody 7E5 7E5V2-30
DVVMTQSPLSLPVTLGQPASISCKSSQSLLYSNGKTFLSWFQQRPGQS
PRRLIYLVSKLDSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1062) 7E5V2-28
DIVMTQSPLSLPVTGPGEASISCKSSQSLLYSNGKTFLSWYLQKPGQS
PQLLIYLVSKLDSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1063) 7E5V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSNGKTFLSWYQQKPGQ
PPKWYLVSKLDSGVPDRFSGSGSGTDFTLTISSELEPEDFAVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1064) 7E5V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKWYLVSKLDSGVPDRFSGSGSGTEFTLTISSELEPEDFAVYYCMQ
THFPLTFGQGTKVEIK (SEQ ID NO: 1065) 7E5V1-9
DIQMTQSPSFLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKLLIYLVSKLDSGVPDRFSGSGSGTEFTLTISSELEPEDFAVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1066) 7E5V1-39
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HFPLTFGQGTKVEIK (SEQ ID NO: 1067) 7E5V1-33
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HFPLTFGQGTKVEIK (SEQ ID NO: 1068) 7E5V3-15
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APRLLIYLVSKLDSGIPARFSGSGSGTEFTLTISSELEPEDFAVYYCMQ
THFPLTFGQGTKVEIK (SEQ ID NO: 1069) 7E5V3-11
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HFPLTFGQGTKVEIK (SEQ ID NO: 1070) 7E5V3-20
EIVLTQSPGTLSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQA
PRLLIYLVSKLDSGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCMQ
THFPLTFGQGTKVEIK (SEQ ID NO: 1071) Antibody 7F8 7F8V3-11
EIVLTQSPATLSLSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLIY
TSILASGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQQWSFNPYTFGQ GTKVEIK (SEQ
ID NO: 1084) 7F8V1-39
DIQMTQSPSSLSASVGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
YLTASILASGVPSRFSGSGSGTDFTLTISSELEPEDFAVYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1085) 7F8V1-5
DIQMTQSPSTLSASVGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
YLTASILASGVPSRFSGSGSGTEFTLTISSELEPEDFAVYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1086) 7F8V3-15
EIVMTQSPATLSVSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLI

YLTSILASGIPARFSGSGTEFTLTISLQSEDFAVYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1087) 7F8V1-9
DIQLTQSPSFLSASVGDRVITITCSASSSVSYMYWYQQKPGKAPKWYL
TSILASGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQQWSFNPYTFGQ GTKVEIK
(SEQ ID NO: 1088) 7F8V1-33
DIQMTQSPSSLSASVGDRVITITCSASSSVSYMYWYQQKPGKAPKLLI
YLTSILASGVPSRFSGSGSGTDFTFTISSLQPEDATYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1089) 7F8V3-20
EIVLTQSPGTLSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLIYL
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(SEQ ID NO: 1090) 7F8V2-28
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LTSILASGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQQWSFNPY TFGQGTKVEIK
(SEQ ID NO: 1091) 7F8V2-30
DVVMTQSPLSLPVTLGQPASISCSASSSVSYMYWFQQRPGQSPRRLIY
LTSILASGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1092) 7F8V4-1
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(SEQ ID NO: 1093) Antibody 8F8 8F8V2-30
DVVMTQSPLSLPVTLGQPASISCRSSQSLVHSNGNTYLHWFQQRPGQS
PRRLIYKVSNNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQS
THVPLTFGQGTKVEIK (SEQ ID NO: 1095) 8F8V2-28
DIVMTQSPLSLPVTPGEPASISCRSSQSLVHSNGNTYLHWYQLKPGQ
SPQLLIYKVSNNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQ
STHVPLTFGQGTKVEIK (SEQ ID NO: 1096) 8F8V4-1
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THVPLTFGQGTKVEIK (SEQ ID NO: 1097) 8F8V3-11
EIVLTQSPATLSLSPGERATLSCRSSQSLVHSNGNTYLHWYQQKPGQ
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HVPLTFGQGTKVEIK (SEQ ID NO: 1098) 8F8V1-39
DIQMTQSPSSLSASVGDRVITITCRSSQSLVHSNGNTYLHWYQQKPGK
APKWYKVSNNRFSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCSQS
THVPLTFGQGTKVEIK (SEQ ID NO: 1099) 8F8V1-33
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APKLLIYKVSNNRFSGVPSRFSGSGSGTDFTFTISSLQPEDATYYCSQST
HVPLTFGQGTKVEIK (SEQ ID NO: 1100) 8F8V3-15
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PRLLIYKVSNNRFSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCSQSTHVP
LTFGQGTKVEIK (SEQ ID NO: 1101) 8F8V1-5
DIQMTQSPSTLSASVGDRVITITCRSSQSLVHSNGNTYLHWYQQKPGK
APKWYKVSNNRFSGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCSQS
THVPLTFGQGTKVEIK (SEQ ID NO: 1102) 8F8V1-9
DIQLTQSPSFLSASVGDRVITITCRSSQSLVHSNGNTYLHWYQQKPGK
APKWYKVSNNRFSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCSQST
HVPLTFGQGTKVEIK (SEQ ID NO: 1103) 8F8V3-20
EIVLTQSPGTLSPGERATLSCRSSQSLVHSNGNTYLHWYQQKPGQA
PRLLIYKVSNNRFSGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCSQSTH
VPLTFGQGTKVEIK (SEQ ID NO: 1104) Antibody1H7 1H7V1-39

DIQMTQSPSSLASVSGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
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(SEQ ID NO: 1085) 1H7V3-11
EIVLTQSPATLSLSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLIYL
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ID NO: 1084) 1H7V1-5
DIQMTQSPSTLSASVGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
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(SEQ ID NO: 1086) 1H7V1-9
DIQMTQSPSFLSASVGDRVTITCSASSSVSYMYWYQQKPGKAPKWYL
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(SEQ ID NO: 1088) 1H7V3-15 EIVMTQ
SPATLSVSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLI
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(SEQ ID NO: 1087) 1H7V1-33
DIQMTQSPSSLASVSGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
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(SEQ ID NO: 1089) 1H7V3-20
EIVLTQSPGTLSLSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLIYL
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(SEQ ID NO: 1090) 1H7V2-28
DIVMTQSPSLPVTGPGEASISCSASSSVSYMYWYQLKPGQSPQLLIY
LTSILASGVPRFSGSGSGTDFTLKISRVEAEDVGVYYCQQWSFNPY TFGQGTKVEIK
(SEQ ID NO: 1091) 1H7V2-30
DVVMTQSPSLPVTGLGPASISCSASSSVSYMYWFQQRPGQSPRRLIY
LTSILASGVPRFSGSGSGTDFTLKISRVEAEDVGVYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1092) 1H7V4-1
DIVMTQSPDSLAVSLGERATTNCSASSSVSYMYWYQQKPGQPPKLLI
YLTSILASGVPRFSGSGSGTDFTLTISLQAEDVAVYYCQQWSFNPY TFGQGTKVEIK
(SEQ ID NO: 1093) Antibody 2H8 2H8V3-11
EIVLTQSPATLSLSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLIYL
TSILASGIPARFSGSGSGTDFTLTISLQPEDFAVYYCQQWSFNPYTFGQ GTKVEIK (SEQ
ID NO: 1084) 2H8V1-39
DIQMTQSPSSLASVSGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
YLTSILASGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQWSFNPY TFGQGTKVEIK
(SEQ ID NO: 1085) 2H8V1-5
DIQMTQSPSTLSASVGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
YLTSILASGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1086) 2H8V3-15
EIVMTQSPATLSVSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLI
YLTSILASGIPARFSGSGSGTEFTLTISLQSEDFAVYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1087) 2H8V1-9
DIQMTQSPSFLSASVGDRVTITCSASSSVSYMYWYQQKPGKAPKWYL
TSILASGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQQWSFNPYTFGQ GTKVEIK
(SEQ ID NO: 1088) 2H8V1-33
DIQMTQSPSSLASVSGDRVTITCSASSSVSYMYWYQQKPGKAPKLLI
YLTSILASGVPSRFSGSGSGTDFTFTISLQPEDFATYYCQQWSFNPYT FGQGTKVEIK
(SEQ ID NO: 1089) 2H8V3-20
EIVLTQSPGTLSLSPGERATLSCSASSSVSYMYWYQQKPGQAPRLLIYL
TSILASGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQWSFNPYTFGQ GTKVEIK

(SEQ ID NO: 1090) 2H8V2-28
DIVMTQSPSLSPVTPGEPASISCSASSSVSYMYWYLQKPGQSPQLLIY
LTSILASGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQQWSFNPHY TFGQGTKVEIK
(SEQ ID NO: 1091) 2H8V2-30
DVVMTQSPSLSPVTLGQPASISCSASSSVSYMYWFQQRPGQSPRRLIY
LTSILASGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQQWSFNPHY TFGQGTKVEIK
(SEQ ID NO: 1092) 2H8V4-1
DIVMTQSPDSLAVSLGERATINCSASSSVSYMYWYQQKPGQPPKLLI
YLTSLASGVDPDRFSGSGSGTDFTLTISLQAEDVAVYYCQQWSFNPHY TFGQGTKVEIK
(SEQ ID NO: 1093) Antibody 3A2 3A2 V2-30
DVVMTQSPSLSPVTLGQPASISCRSSQTIIHSNGNTYLEWFQQRPGQS
PRRLIYKVSNNRFSGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1108) 3A2 V2-28
DIVMTQSPSLSPVTPGEPASISCRSSQTIIHSNGNTYLEWYLQKPGQSP
QLLIYKVSNNRFSGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQGS
HVPYTFGQGTKVEIK (SEQ ID NO: 1109) 3A2 V4-1
DIVMTQSPDSLAVSLGERATINCRSSQTIIHSNGNTYLEWYQQKPGQ
PPKWKVSNNRFSGVDPDRFSGSGSGTDFTLTISLQAEDVAVYYCFQ
GSHVPYTFGQGTKVEIK (SEQ ID NO: 1110) 3A2 V3-11
EIVLTQSPATLSLSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQAP
RLLIYKVSNNRFSGIPARFSGSGSGTDFTLTISLEPEDFAVYYCFQGS
VPYTFGQGTKVEIK (SEQ ID NO: 1111) 3A2 I-9
DIQLTQSPSFLSASVGDRVITICRSSQTIIHSNGNTYLEWYQQKPGKA
PKWKVSNNRFSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCFQGS
HVPYTFGQGTKVEIK (SEQ ID NO: 1112) 3A2 I-33
DIQMTQSPSSLSASVGDRVITICRSSQTIIHSNGNTYLEWYQQKPGK
APKWKVSNNRFSGVPSRFSGSGSGTDFTFTISLQPEDFATYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1113) 3A2 VI-39
DIQMTQSPSSLSASVGDRVITICRSSQTIIHSNGNTYLEWYQQKPGKA
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HVPYTFGQGTKVEIK (SEQ ID NO: 1114) 3A2 V3-15
EIVMTQSPATLSVSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQ
APRLLIYKVSNNRFSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1115) 3A2 VI-5
DIQMTQSPSTLSASVGDRVITICRSSQTIIHSNGNTYLEWYQQKPGK
APKWKVSNNRFSGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1116) 3A2 V3-20
EIVLTQSPGTLSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQA
PRLLIYKVSNNRFSGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCFQGS
HVPYTFGQGTKVEIK (SEQ ID NO: 1117) Antibody 3A7 3A7 V2-30
DVVMTQSPSLSPVTLGQPASISCKSSQSLLYSNGKTFLSWFQQRPGQS
PRRLIYLVSKLDSGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1062) 3A7 V2-28
DIVMTQSPSLSPVTPGEPASISCKSSQSLLYSNGKTFLSWYLQKPGQS
PQLLIYLVSKLDSGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1063) 3A7 V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSNGKTFLSWYQQKPGQ
PPKWLVSCLDSGVDPDRFSGSGSGTDFTLTISLQAEDVAVYYCMQ
GTHFPLTFGQGTKVEIK (SEQ ID NO: 1064) 3A7 VI-39
DIQMTQSPSSLSASVGDRVITICKSSQSLLYSNGKTFLSWYQQKPGK

APKLLIYLVSKLD SGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCMQ
GTHFPLTFGQG GTKVEIK (SEQ ID NO: 1067) 3A7 I-9
DIQLTQSPSFLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKLLIYLVSKLD SGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCMQ
GTHFPLTFGQG GTKVEIK (SEQ ID NO: 1066) 3A7 I-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKLLIYLVSKLD SGVPSRFSGSGSGTDFTLTISSLQPDDEFATYYCMQ
GTHFPLTFGQG GTKVEIK (SEQ ID NO: 1065) 3A7 VI-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSNGKTFLSWYQQKPGK
APKLLIYLVSKLD SGVPSRFSGSGSGTDFTLTISSLQPEDIATYYCMQ
THFPLTFGQG GTKVEIK (SEQ ID NO: 1068) 3A7 V3-15
EIVMTQSPATLSVSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQ
APRLLIYLVSKLD SGIPARFSGSGSGTDFTLTISSLQSEDFAVYYCMQ
THFPLTFGQG GTKVEIK (SEQ ID NO: 1069) 3A7 V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQ
APRLLIYLVSKLD SGIPARFSGSGSGTDFTLTISSLEPEDFAVYYCMQ
GTHFPLTFGQG GTKVEIK (SEQ ID NO: 1070) 3A7 V3-20
EIVLTQSPGTLSLSPGERATLSCKSSQSLLYSNGKTFLSWYQQKPGQ
APRLLIYLVSKLD SGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCMQ
GTHFPLTFGQG GTKVEIK (SEQ ID NO: 1071) Antibody 3B10 3B10V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSSDQKNYLAWYQQKPG
QPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQAEDVAVYYCQ
QYYSYPLTFGQG GTKVEIK (SEQ ID NO: 1120) 3B10V2-28
DIVMTQSPLSLPVTPGEPASISCKSSQSLLYSSDQKNYLAWYLQKPG
QSPQLLIYWASTRESGVPDRFSGSGSGTDFTLKISRVEAEDVGYYC
QYYSYPLTFGQG GTKVEIK (SEQ ID NO: 1121) 3B10V2-30
DVVMTQSPLSLPVTLGQPASISCKSSQSLLYSSDQKNYLAWFQQRPG
QSPRRLIYWASTRESGVPDRFSGSGSGTDFTLKISRVEAEDVGYYC
QYYSYPLTFGQG GTKVEIK (SEQ ID NO: 1122) 3B10V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLYSSDQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISSLQPDDEFATYYCQ
QYYSYPLTFGQG GTKVEIK (SEQ ID NO: 1123) 3B10V1-9
DIQLTQSPSFLSASVGDRVTITCKSSQSLLYSSDQKNYLAWYQQKPG
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YYSYPLTFGQG GTKVEIK (SEQ ID NO: 1124) 3B10V3-15
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YYSYPLTFGQG GTKVEIK (SEQ ID NO: 1125) 3B10V1-39
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSSDQKNYLAWYQQKPG
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YYSYPLTFGQG GTKVEIK (SEQ ID NO: 1126) 3B10V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLLYSSDQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISSLEPEDFAVYYCQ
YYSYPLTFGQG GTKVEIK (SEQ ID NO: 1127) 3B10v1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSSDQKNYLAWYQQKPG
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YYSYPLTFGQG GTKVEIK (SEQ ID NO: 1128) 3B10V3-20
EIVLTQSPGTLSLSPGERATLSCKSSQSLLYSSDQKNYLAWYQQKPG
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QYYSYPLTFGQG GTKVEIK (SEQ ID NO: 1129) Antibody 4F11 4F11V2-30

DVVMTQSPSLPVTLGQPASISCRSSQTIIHSNGNTYLEWQFQQRPQGS
PRRLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQ
GSHVPYTFGQGGTKVEIK (SEQ ID NO: 1108) 4F11V2-28
DIVMTQSPSLPVTGPGEASISCRSSQTIIHSNGNTYLEWYLQKPGQSP
QLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQG
SHVPYTFGQGGTKVEIK (SEQ ID NO: 1109) 4F11V4-1
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KLLIYKVSNRFSGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCFQGSHV
PYTFGQGGTKVEIK (SEQ ID NO: 1110) 4F11V3-11
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RLLIYKVSNRFSGIPARFSGSGSGTDFTLTISLQPEDFAVYYCFQGSHV
PYTFGQGGTKVEIK (SEQ ID NO: 1111) 4F11V3-15
EIVMTQSPATLSVSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQ
APRLLIYKVSNRFSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCFQG
SHVPYTFGQGGTKVEIK (SEQ ID NO: 1115) 4F11V1-33
DIQMTQSPSSLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGK
APKWKVSNRFSGVPSPRFSGSGSGTDFTTISLQPEDATYYCFQG
SHVPYTFGQGGTKVEIK (SEQ ID NO: 1113) 4F11V1-39
DIQMTQSPSSLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGKA
PKWKVSNRFSGVPSPRFSGSGSGTDFTLTISLQPEDFATYYCFQGS
HVPYTFGQGGTKVEIK (SEQ ID NO: 1114) 4F11V1-9
DIQLTQSPSFLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGKA
PKWKVSNRFSGVPSPRFSGSGSGTEFTLTISLQPEDFATYYCFQGS
HVPYTFGQGGTKVEIK (SEQ ID NO: 1112) 4F11V1-5
DIQMTQSPSTLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGKA
PKWKVSNRFSGVPSPRFSGSGSGTEFTLTISLQPDDEFATYYCFQGS
HVPYTFGQGGTKVEIK (SEQ ID NO: 1116) 4F11V3-20
EIVLTQSPGTLSLSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQAP
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VPYTFGQGGTKVEIK (SEQ ID NO: 1117) Antibody 6H6 6H6V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSVFYSSNQKNYLAWYQQKPG
QPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCH
QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1500) 6H6V2-28
DIVMTQSPSLPVTGPGEASISCKSSQSVFYSSNQKNYLAWYLLQKPGQ
SPQLLIYWASTRESGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCHQ
YLSSLTFGQGGTKVEIK (SEQ ID NO: 1501) 6H6V2-30
DVVMTQSPSLPVTLGQPASISCKSSQSVFYSSNQKNYLAWFQQRPG
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QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1502) 6H6V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSVFYSSNQKNYLAWYQQKPG
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QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1503) 6H6V1-9
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YLSSLTFGQGGTKVEIK (SEQ ID NO: 1504) 6H6V3-15
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YLSSLTFGQGGTKVEIK (SEQ ID NO: 1505) 6H6V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSVFYSSNQKNYLAWYQQKPG
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YLSSLTFGQGGTKVEIK (SEQ ID NO: 1506) 6H6V3-11
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YLSSLTFGQGGTKVEIK (SEQ ID NO: 1507) 6H6V1-39
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YLSSLTFGQGGTKVEIK (SEQ ID NO: 1508) 6H6V3-20
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YLSSLTFGQGGTKVEIK (SEQ ID NO: 1509) Antibody 7A9 7A9V1-9
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(SEQ ID NO: 1132) 7A9V3-11
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(SEQ ID NO: 1133) 7A9V1-5
DIQMTQSPSTLSASVGDRVTITCRASENIYSYLAWYQQKPGKAPKLL
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(SEQ ID NO: 1134) 7A9V3-15
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(SEQ ID NO: 1135) 7A9V1-39
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(SEQ ID NO: 1136) 7A9V1-33
DIQMTQSPSSLSASVGDRVTITCRASENIYSYLAWYQQKPGKAPKLLI
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(SEQ ID NO: 1137) 7A9V3-20
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(SEQ ID NO: 1138) 7A9V2-28
DIVMTQSPLSLPVTLPGEPAISCRASENIYSYLAWYLQKPGQSPQLLIY
KAKTLAEGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCQHHYGTPF TFGQGGTKVEIK
(SEQ ID NO: 1139) 7A9V4-1
DIVMTQSPDSLAVSLGERATINCRASENIYSYLAWYQQKPGQPPKLL
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(SEQ ID NO: 1140) 7A9V2-30
DVVMTQSPLSLPVTLGQPASISCRASENIYSYLAWFQQRPQGSPRRLI
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(SEQ ID NO: 1141) Antibody 8A1 8A1V3-15 EIVMTQ
SPATLSVSPGERATLSCRTSENVYSNLAWYQQKPGQAPRLI
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(SEQ ID NO: 1143) 8A1V3-11 EIVLTQ
SPATLSLSPGERATLSCRTSENVYSNLAWYQQKPGQAPRLLI
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(SEQ ID NO: 1144) 8A1V1-9
DIQLTQSPSFLSASVGDRVTITCRTSENVYSNLAWYQQKPGKAPKLLI
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(SEQ ID NO: 1145) 8A1V1-5
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(SEQ ID NO: 1146) 8A1V1-39
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(SEQ ID NO: 1147) 8A1V1-33
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(SEQ ID NO: 1148) 8A1V3-20
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(SEQ ID NO: 1149) 8A1V2-28
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(SEQ ID NO: 1150) 8A1V2-30
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YAATNLADGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCHHFWGTP YTFGQGGTKVEIK
(SEQ ID NO: 1151) 8A1V4-1
DIVMTQSPDSLAVSLGERATINCRTESENVYSNLAWYQQKPGQPPKLL
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(SEQ ID NO: 1152) Antibody 9F5 9F5V2-30
DVVMTQSPLSLPVTLGQPASISCRSSQSLVHSNGYTYLHWYQQRPQG
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STRVPYTFGQGGTKVEIK (SEQ ID NO: 1154) 9F5V2-28
DIVMTQSPLSLPVTPGEPASISCRSSQSLVHSNGYTYLHWYLLQKPGQ
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STRVPYTFGQGGTKVEIK (SEQ ID NO: 1155) 9F5V4-1
DIVMTQSPDSLAVSLGERATINCRSSQSLVHSNGYTYLHWYQQKPGQ
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TRVPYTFGQGGTKVEIK (SEQ ID NO: 1156) 9F5V3-11
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RVPYTFGQGGTKVEIK (SEQ ID NO: 1157) 9F5V1-33
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RVPYTFGQGGTKVEIK (SEQ ID NO: 1158) 9F5V3-15
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RVPYTFGQGGTKVEIK (SEQ ID NO: 1159) 9F5V1-5
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STRVPYTFGQGGTKVEIK (SEQ ID NO: 1160) 9F5V1-39
DIQMTQSPSSLSASVGDRVITTCRSSQSLVHSNGYTYLHWYQQKPGK
APKWKVSNNRFSGVPSRFSGSGSGTDFTLTISLQPDFATYYCSQS
TRVPYTFGQGGTKVEIK (SEQ ID NO: 1161) 9F5V1-9
DIQLTQSPSFLSASVGDRVITTCRSSQSLVHSNGYTYLHWYQQKPGKA
PKWKVSNNRFSGVPSRFSGSGSGTEFTLTISLQPDFATYYCSQSTR
VPYTFGQGGTKVEIK (SEQ ID NO: 1162) 9F5V3-20
EIVLTQSPGTLSLSPGERATLSCRSSQSLVHSNGYTYLHWYQQKPGQ
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RVPYTFGQGGTKVEIK (SEQ ID NO: 1163) 9F5-L1

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SPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQ
STRVPYTFGQGTKLEIK (SEQ ID NO: 1663) 9F5-L2
DVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWYLQKPGQ
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STRVPYTFGQGTKLEIK (SEQ ID NO: 1664) Antibody 9G1 9G1V2-30
DVVMTQSPLSLPVTLGQPASISCRFSQSLVHSNGNTYLHWFQQRPGQ
SPRRLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQ
STRVPPTFGQGTKVEIK (SEQ ID NO: 1165) 9G1V2-28
DIVMTQSPLSLPVTGPGEPAISCRFSQSLVHSNGNTYLHWYLQKPGQSPQ
LLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQSTRVP
PTFGQGTKVEIK (SEQ ID NO: 1166) 9G1V4-1
DIVMTQSPDSLAVSLGERATINCRFSQSLVHSNGNTYLHWYQQKPGQ
PPKWKVSNRFSGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCSQS
TRVPPTFGQGTKVEIK (SEQ ID NO: 1167) 9G1V3 -11
EIVLTQSPATLSLSPGERATLSCRFSQSLVHSNGNTYLHWYQQKPGQ
APRLLIYKVSNRFSGIPARFSGSGSGTDFTLTISLEPEDFAVYYCSQST
RVPPTFGQGTKVEIK (SEQ ID NO: 1168) 9G1V3-15
EIVMTQSPATLSVSPGERATLSCRFSQSLVHSNGNTYLHWYQQKPGQ
APRLLIYKVSNRFSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCSQS
TRVPPTFGQGTKVEIK (SEQ ID NO: 1169) 9G1V1-9
DIQLTQSPSFLSASVGDRVITICRFSQSLVHSNGNTYLHWYQQKPGK
APKWKVSNRFSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCSQST
RVPPTFGQGTKVEIK (SEQ ID NO: 1170) 9G1V1-5
DIQMTQSPSTLSASVGDRVITICRFSQSLVHSNGNTYLHWYQQKPGK
APKWKVSNRFSGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCSQST
RVPPTFGQGTKVEIK (SEQ ID NO: 1171) 9G1V1-39
DIQMTQSPSSLSASVGDRVITICRFSQSLVHSNGNTYLHWYQQKPGKAP
KLLIYKVSNRFSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCSQSTRVPP
TFGQGTKVEIK (SEQ ID NO: 1172) 9G1V1-33
DIQMTQSPSSLSASVGDRVITICRFSQSLVHSNGNTYLHWYQQKPGK
APKWKVSNRFSGVPSRFSGSGSGTDFTFTISLQPEDFATYYCSQST
RVPPTFGQGTKVEIK (SEQ ID NO: 1173) 9G1V3 -20
EIVLTQSPGTLSPGERATLSCRFSQSLVHSNGNTYLHWYQQKPGQ
APRLLIYKVSNRFSGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCSQS
TRVPPTFGQGTKVEIK (SEQ ID NO: 1174) Antibody 9G3 9G3V1-33
DIQMTQSPSSLSASVGDRVITICKASSNVNYMSWYQQKPGKAPKLLIY
FTSNLPSGVPSRFSGSGSGTDFTFTISLQPEDFATYYCSGEVTQFTFGQG TKVEIK (SEQ
ID NO: 1176) 9G3V1-9
DIQLTQSPSFLSASVGDRVITICKASSNVNYMSWYQQKPGKAPKLLIY
FTSNLPSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCSGEVTQFTFGQ GTKVEIK (SEQ
ID NO: 1177) 9G3V1-39
DIQMTQSPSSLSASVGDRVITICKASSNVNYMSWYQQKPGKAPKLLIY
FTSNLPSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCSGEVTQFTFGQ GTKVEIK (SEQ
ID NO: 1178) 9G3V3-11
EIVLTQSPATLSLSPGERATLSCKASSNVNYMSWYQQKPGQAPRLLIYF
TSNLPSGIPARFSGSGSGTDFTLTISLEPEDFAVYYCSGEVTQFTFGQG TKVEIK (SEQ
ID NO: 1641) 9G3V1-5
DIQMTQSPSTLSASVGDRVITICKASSNVNYMSWYQQKPGKAPKLLIYF
TSNLPSGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCSGEVTQFTFGQGT KVEIK (SEQ

ID NO: 1179) 9G3V3-15
EIVMTQSPATLSVSPGERATLSCKASSNVNYMSWYQQKPGQAPRLLIY
FTSNLPSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCSGEVTQFTFGQG TKVEIK (SEQ
ID NO: 1180) 9G3V3-20
EIVLTQSPGTLSPGERATLSCKASSNVNYMSWYQQKPGQAPRLLIYF
TSNLPSGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCSGEVTQFTFGQG TKVEIK (SEQ
ID NO: 1181) 9G3V2-28
DIVMTQSPLSLPVTGPGEPAISCKASSNVNYMSWYLQKPGQSPQLLIY
FTSNLPSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSGEVTQFTF GQGTKVEIK
(SEQ ID NO: 1182) 9G3V2-30
DVVMTQSPLSLPVTLGQPASISCKASSNVNYMSWFQQRPGQSPRRLIYF
TSNLPSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSGEVTQFTFGQ GTKVEIK
(SEQ ID NO: 1183) 9G3V4-1
DIVMTQSPDSLAVSLGERATTNCKASSNVNYMSWYQQKPGQPPKLLI
YFTSNLPSGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCSGEVTQFTF GQGTKVEIK
(SEQ ID NO: 1184) Antibody10A9 10A9V2-30
DVVMTQSPLSLPVTLGQPASISCRSSQTIIHSNGNTYLEWFQQRPGQSPRR
LIYKVSNRFCGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQGSHPY
TFGQGTKVEIK (SEQ ID NO: 1186) 10A9V2-28
DIVMTQSPLSLPVTGPGEPAISCRSSQTIIHSNGNTYLEWYLQKPGQSP
QLLIYKVSNRFCGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1187) 10A9V4-1
DIVMTQSPDSLAVSLGERATINCRSSQTIIHSNGNTYLEWYQQKPGQP
PKWYKVSNRFCGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1188) 10A9V3-11
EIVLTQSPATLSLSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQAP
RLLIYKVSNRFCGIPARFSGSGSGTDFTLTISLQSEDFAVYYCFQGS
VPYTFGQGTKVEIK (SEQ ID NO: 1189) 10A9V3-15
EIVMTQSPATLSVSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQ
APRLLIYKVSNRFCGIPARFSGSGSGTEFTLTISLQSEDFAVYYCFQG
SHVPYTFGQGTKVEIK (SEQ ID NO: 1190) 10A9V1-33
DIQMTQSPSSLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPG
KAPKWYKVSNRFCGVPDRFSGSGSGTDFTLTISLQPEDFATYYCF
QGSHVPYTFGQGTKVEIK (SEQ ID NO: 1191) 10A9V3-20
EIVLTQSPGTLSPGERATLSCRSSQTIIHSNGNTYLEWYQQKPGQAP
RLLIYKVSNRFCGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCFQGS
VPYTFGQGTKVEIK (SEQ ID NO: 1192) 10A9V1-9
DIQLTQSPSFLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGKA
PKWYKVSNRFCGVPDRFSGSGSGTEFTLTISLQPEDFATYYCFQGS
HVPYTFGQGTKVEIK (SEQ ID NO: 1193) 10A9V1-5
DIQMTQSPSTLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGKA
PKWYKVSNRFCGVPDRFSGSGSGTEFTLTISLQPEDFATYYCFQGS
HVPYTFGQGTKVEIK (SEQ ID NO: 1194) 10A9V1-39
DIQMTQSPSSLSASVGDRVTITCRSSQTIIHSNGNTYLEWYQQKPGKA
PKWYKVSNRFCGVPDRFSGSGSGTDFTLTISLQPEDFATYYCFQGS
HVPYTFGQGTKVEIK (SEQ ID NO: 1195) Antibody11A8 11A8V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLNSGNQKKYLTWYQQKPG
QPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCQ
NDYGFPLTFGQGTKVEIK (SEQ ID NO: 1197) 11A8V2-30
DVVMTQSPLSLPVTLGQPASISCKSSQSLNSGNQKKYLTWFQQRPG

QSPRRLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVIYYCQ
NDYGFPLTFGQGGTKVEIK (SEQ ID NO: 1198) 11A8V2-28
DIVMTQSPSLPLVTPGEPASISCKSSQSLLNSGNQKKYLTWYLQKPG
QSPQLLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVIYYC
QNDYGFPLTFGQGGTKVEIK (SEQ ID NO: 1199) 11A8V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLNSGNQKKYLTWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTFITSSSLQPEDIATYYCQN
DYGFPLTFGQGGTKVEIK (SEQ ID NO: 1200) 11A8V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLLNSGNQKKYLTWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQN
DYGFPLTFGQGGTKVEIK (SEQ ID NO: 1201) 11A8V3-15
EIVMTQSPATLSVSPGERATLSCKSSQSLLNSGNQKKYLTWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQN
DYGFPLTFGQGGTKVEIK (SEQ ID NO: 1202) 11A8V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLNSGNQKKYLTWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISSELEPEDFAVYYCQ
NDYGFPLTFGQGGTKVEIK (SEQ ID NO: 1203) 11A8V3-20
EIVLTQSPGTLSPGERATLSCKSSQSLLNSGNQKKYLTWYQQKPG
QAPRLLIYWASTRESGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQN
DYGFPLTFGQGGTKVEIK (SEQ ID NO: 1204) 11A8V1-9
DIQLTQSPSFLSASVGDRVTITCKSSQSLLNSGNQKKYLTWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISSELEPEDFAVYYCQ
NDYGFPLTFGQGGTKVEIK (SEQ ID NO: 1205) 11A8V1-39
DIQMTQSPSSLSASVGDRVTITCKSSQSLLNSGNQKKYLTWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISSELEPEDFAVYYCQ
NDYGFPLTFGQGGTKVEIK (SEQ ID NO: 1206) Antibody12D9 12D9V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSGNQKNFLAWYQQKPG
QPPKLLIYWASTRESGVDPDRFSGSGSGTDFTLTISSELEPEDFAVYYCQ
QYYSYPFTFGQGGTKVEIK (SEQ ID NO: 1208) 12D9V2-28
DIVMTQSPSLPLVTPGEPASISCKSSQSLLYSGNQKNFLAWYLQKPGQ
SPQLLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVIYYCQ
YYSYPFTFGQGGTKVEIK (SEQ ID NO: 1209) 12D9V2-30
DVVMTQSPSLPLVTLGQPASISCKSSQSLLYSGNQKNFLAWFQQRPG
QSPRRLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVIYYCQ
QYYSYPFTFGQGGTKVEIK (SEQ ID NO: 1210) 12D 9V1-9
DIQLTQSPSFLSASVGDRVTITCKSSQSLLYSGNQKNFLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISSELEPEDFAVYYCQ
YYSYPFTFGQGGTKVEIK (SEQ ID NO: 1211) 12D9V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLYSGNQKNFLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISSELEPEDFAVYYCQ
QYYSYPFTFGQGGTKVEIK (SEQ ID NO: 1212) 12D9V3-15
EIVMTQSPATLSVSPGERATLSCKSSQSLLYSGNQKNFLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQ
YYSYPFTFGQGGTKVEIK (SEQ ID NO: 1213) 12D9V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSGNQKNFLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTFITSSSLQPEDIATYYCQ
YYSYPFTFGQGGTKVEIK (SEQ ID NO: 1214) 12D9V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLLYSGNQKNFLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISSELEPEDFAVYYCQ
YYSYPFTFGQGGTKVEIK (SEQ ID NO: 1215) 12D9V1-39

DIQMTQSPSTLSASVGDRTITCKSSQSLLYSSDQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQ
QYYSYPFTFGQGGTKVEIK (SEQ ID NO: 1216) 12D9V3-20
EIVLTQSPGTLSPGERATLSCCKSSQSLLYSGNQKNFLAWYQQKPG
QAPRLLIYWASTRESGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQ
YYSYPFTFGQGGTKVEIK (SEQ ID NO: 1217) Antibody12F9 12F9V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSSDQKNYLAWYQQKPG
QPPKLLIYWASTRESGVDPDRFSGSGSGTDFTLTISLQAEDVAVYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1120) 12F9V2-28
DIVMTQSPLSLPVTLPGEPAISCKSSQSLLYSSDQKNYLAWYLQKPG
QSPQLLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYC
QQYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1121) 12F9V2-30
DVVMTQSPLSLPVTLGQPASISCKSSQSLLYSSDQKNYLAWFQQRPG
QSPRRLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYC
QQYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1122) 12F9V3-15
EIVMTQSPATLSVSPGERATLSCCKSSQSLLYSSDQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTEFTLTISLQSEDFAVYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1125) 12F9V1-5
DIQMTQSPSTLSASVGDRTITCKSSQSLLYSSDQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1123) 12F9V1-9
DIQLTQSPSFLSASVGDRTITCKSSQSLLYSSDQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1124) 12F9V1-33
DIQMTQSPSSLSASVGDRTITCKSSQSLLYSSDQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTFTISLQPEDIATYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1128) 12F9V3-11
EIVLTQSPATLSLSPGERATLSCCKSSQSLLYSSDQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1127) 12F9V1-39
DIQMTQSPSSLSASVGDRTITCKSSQSLLYSSDQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1126) 12F9V3-20
EIVLTQSPGTLSPGERATLSCCKSSQSLLYSSDQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1129) Antibody10C1 10C1V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSVFYSSNQKNYLAWYQQKPG
QPPKLLIYWASTRESGVDPDRFSGSGSGTDFTLTISLQAEDVAVYYCH
QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1423) 10C1V2-30
DVVMTQSPLSLPVTLGQPASISCKSSQSVFYSSNQKNYLAWFQQRPG
QSPRRLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCH
QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1424) 10C1V2-28
DIVMTQSPLSLPVTLPGEPAISCKSSQSVFYSSNQKNYLAWYLQKPGQ
SPQLLIYWASTRESGVDPDRFSGSGSGTDFTLKISRVEAEDVGVYYCH
QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1425) 10C1V1-5
DIQMTQSPSTLSASVGDRTITCKSSQSVFYSSNQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCHQ
YLSSLTFGQGGTKVEIK (SEQ ID NO: 1426) 10C1V3-15
EIVMTQSPATLSVSPGERATLSCCKSSQSVFYSSNQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTEFTLTISLQSEDFAVYYCHQ

YLSSLTFGQGGTKVEIK (SEQ ID NO: 1427) 10C1V1-9
DIQLTQSPSFLSASVGDRVTITCKSSQSVFYSSNQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTEFTLTISLQPEDFATYYCH
QYLSSLTFGQGGTKVEIK (SEQ ID NO: 1428) 10C1V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSVFYSSNQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISLQPEDFAVYYCHQ
YLSSLTFGQGGTKVEIK (SEQ ID NO: 1429) 10C1V1-39
DIQMTQSPSSLSASVGDRVTITCKSSQSVFYSSNQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISLQPEDFATYYCHQ
YLSSLTFGQGGTKVEIK (SEQ ID NO: 1430) 10C1V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSVFYSSNQKNYLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTFTISSLQPEDATYYCHQ
YLSSLTFGQGGTKVEIK (SEQ ID NO: 1431) 10C1V3-20
EIVLTQSPGTLSPGERATLCKSSQSVFYSSNQKNYLAWYQQKPG
QAPRLLIYWASTRESGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCHQ
YLSSLTFGQGGTKVEIK (SEQ ID NO: 1432) Antibody 7E9 7E9V4-1
DIVMTQSPDSLAVSLGERATINCKSSQSLLYSSNQKNCLAWYQQKPG
QPPKLLIYWASTRESGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1434) 7E9V2-28
DIVMTQSPSLSPVTPGEPASISCKSSQSLLYSSNQKNCLAWYLQKPG
QSPQLLIYWASTRESGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYC
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1435) 7E9V2-30
DVVMTQSPSLSPVTLGQPASISCKSSQSLLYSSNQKNCLAWFQQRPG
QSPRRLIYWASTRESGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYC
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1436) 7E9V1-9
DIQLTQSPSFLSASVGDRVTITCKSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1437) 7E9V3-15
EIVMTQSPATLSVSPGERATLSCKSSQSLLYSSNQKNCLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTEFTLTISLQSEDFAVYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1438) 7E9V1-5
DIQMTQSPSTLSASVGDRVTITCKSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTEFTLTISLQPDDEFATYYCQ
QYYSYPLTFGQGGTKVEIK (SEQ ID NO: 1439) 7E9V1-33
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTFTISSLQPEDATYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1440) 7E9V1-39
DIQMTQSPSSLSASVGDRVTITCKSSQSLLYSSNQKNCLAWYQQKPG
KAPKLLIYWASTRESGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1441) 7E9V3-11
EIVLTQSPATLSLSPGERATLSCKSSQSLLYSSNQKNCLAWYQQKPG
QAPRLLIYWASTRESGIPARFSGSGSGTDFTLTISLQPEDFAVYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1442) 7E9V3-20
EIVLTQSPGTLSPGERATLSCKSSQSLLYSSNQKNCLAWYQQKPG
QAPRLLIYWASTRESGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCQQ
YYSYPLTFGQGGTKVEIK (SEQ ID NO: 1443) Antibody 8C3 8C3V2-30
DVVMTQSPSLSPVTLGQPASISCRSSQSLVHSNGNTYLHWFQQRPGQ
SPRRLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQ
STHVPPTFGQGGTKVEIK (SEQ ID NO: 1445) 8C3V2-28
DIVMTQSPSLSPVTPGEPASISCRSSQSLVHSNGNTYLHWYLLQKPGQSP

QLLIYKVSNNRFSGVPDRFSGSGSGTDFTLKISRVEAEDVGVYYCSQSTHV
PPTFGQGGTKVEIK (SEQ ID NO: 1446) 8C3V4-1
DIVMTQSPDSLAVSLGERATINCRSSQSLVHSNGNTYLHWYQQKPGQ
PPKWYKVSNNRFSGVPDRFSGSGSGTDFTLTISLQAEDVAVYYCSQS
THVPPTFGQGGTKVEIK (SEQ ID NO: 1447) 8C3V3-11
EIVLTQSPATLSLSPGERATLSCRSSQSLVHSNGNTYLHWYQQKPGQ
APRLLIYKVSNNRFSGIPARFSGSGSGTDFTLTISLEPEDFAVYYCSQST
HVPPTFGQGGTKVEIK (SEQ ID NO: 1448) 8C3V1-33
DIQMTQSPSSLSASVGDRVTITCRSSQSLVHSNGNTYLHWYQQKPGK
APKLLIYKVSNNRFSGVPSRFSGSGSGTDFTFTISSLQPEDIATYYCSQS
THVPPTFGQGGTKVEIK (SEQ ID NO: 1449) 8C3V1-5
DIQMTQSPSTLSASVGDRVTITCRSSQSLVHSNGNTYLHWYQQKPGK
APKLLIYKVSNNRFSGVPSRFSGSGSGTEFTLTISSLQPDDFATYYCSQST
HVPPTFGQGGTKVEIK (SEQ ID NO: 1450) 8C3V1-39
DIQMTQSPSSLSASVGDRVTITCRSSQSLVHSNGNTYLHWYQQKPGK
APKWYKVSNNRFSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCSQS
THVPPTFGQGGTKVEIK (SEQ ID NO: 1451) 8C3V1-9
DIQLTQSPSFLSASVGDRVTITCRSSQSLVHSNGNTYLHWYQQKPGK
APKLLIYKVSNNRFSGVPSRFSGSGSGTEFTLTISSLQPEDFATYYCSQS
THVPPTFGQGGTKVEIK (SEQ ID NO: 1452) 8C3V3-15
EIVMTQSPATLSVSPGERATLSCRSSQSLVHSNGNTYLHWYQQKPGQ
APRLLIYKVSNNRFSGIPARFSGSGSGTEFTLTISLQSEDFAVYYCSQST
HVPPTFGQGGTKVEIK (SEQ ID NO: 1453) 8C3V3-20
EIVLTQSPGTLSLSPGERATLSCRSSQSLVHSNGNTYLHWYQQKPGQ
APRLLIYKVSNNRFSGIPDRFSGSGSGTDFTLTISRLEPEDFAVYYCSQST
HVPPTFGQGGTKVEIK (SEQ ID NO: 1454)

TABLE-US-00029 TABLE D8 Humanized heavy chain variable region sequences

Antibody variant Humanized sequences Antibody 4D11 4D11V4-59
QVQLQESGPGLVKPSSETLSLTCTVSGFTLSSYAMSWIRQPPGKGLEW
VASISRGGSTYYPPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
TRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1220) 4D11V3-23
EVQLLES GGGLVQPGGSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPD SVKGRFTISRDN SKNTLYLQMNSLRAEDTAVY
YCTRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1221) 4D11V3-7
EVQLVES GGGLVQPGGSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPD SVKGRFTISRDN AKNSLYLQMNSLRAEDTAVY
YCTRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1222) 4D11V3-48
EVQLVES GGGLVQPGGSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPD SVKGRFTISRDN AKNSLYLQMNSLRAEDTAVY
YCTRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1222) 4D11V3-30
QVQLVES GGGVVQPG RSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPD SVKGRFTISRDN SKNTLYLQMNSLRAEDTAVY
YCTRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1223) 4D11V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGFTLSSYAMSWVRQAPGQGLE
WVASISRGGSTYYPQKFQGRVTITADESTSTAYMELSSLRSED TAVYY
CTRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1224) 4D11V1-46
QVQLVQSGAEVKKPGASVKV SCKASGFTLSSYAMSWVRQAPGQGLE
WVASISRGGSTYYPQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVY
YCTRGYGYRTPFANWGQGTLVTVSS (SEQ ID NO: 1225) 4D11V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTLSSYAMSWVRQMPGKGLE

WVASISRGGSTYYPPSFQGGVTVSSADKSISTAYLQWSSLKASDTAMY
CTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1226) 4D11V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFTLSSYAMSWIRQPPGKGLEW
VASISRGGSTYYPPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
TRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1227) 4D11V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGFTLSSYAMSWIRQPPGKGLEW
VASISRGGSTYYPPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
TRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1228) Antibody 7C5 7C5V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFTLSSYAMSWIRQPPGKGLEW
VASISRGGSTYYPPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
TRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1220) 7C5V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVY
YCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1221) 7C5V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1222) 7C5V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1222) 7C5V3-30
QVQLVESGGGVVQPGRLSLRLSCAASGFTLSSYAMSWVRQAPGKGLE
WVASISRGGSTYYPDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVY
YCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1223) 7C5V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGFTLSSYAMSWVRQAPGQGLE
WVASISRGGSTYYPQKFQGRVTITADESTSTAYMELSSLRSED TAVY
YCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1224) 7C5V1-46
QVQLVQSGAEVKKPGASVKV SCKASGFTLSSYAMSWVRQAPGQGL
EWVASISRGGSTYYPQKFQGRVTMTRDTSTSTVYMELSSLRSED TAV
YYCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1225) 7C5V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTLSSYAMSWVRQMPGKGLE
WVASISRGGSTYYPPSFQGGVTVSSADKSISTAYLQWSSLKASDTAMY
YCTRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1226) 7C5V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFTLSSYAMSWIRQPPGKGLEW
VASISRGGSTYYPPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
TRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1227) 7C5V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGFTLSSYAMSWIRQPPGKGLEW
VASISRGGSTYYPPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
TRGYGYRTPFANWGQGTTLVTVSS (SEQ ID NO: 1228) Antibody 6G12 6G12V1-46
QVQLVQSGAEVKKPGASVKV SCKASGYTFTEYTMHWVRQAPGQGL
EWIGGINPNNGGTSYSQKFQGRVTMTRDTSTSTVYMELSSLRSED TA
VYYCARGGSHYYAMDYWGQGTTLVTVSS (SEQ ID NO: 1230) 6G12V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYTFTEYTMHWVRQMPGKGLE
WIGGINPNNGGTSYSPSFQGGVTVSSADKSISTAYLQWSSLKASDTAMY
YCARGGSHYYAMDYWGQGTTLVTVSS (SEQ ID NO: 1231) 6G12V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGYTFTEYTM EIWVRQAPGQGL
EWIGGINPNNGGTSYSQKFQGRVTITADESTSTAYMELSSLRSED TAV
YYCARGGSHYYAMDYWGQGTTLVTVSS (SEQ ID NO: 1232) 6G12V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYTFTEYTMHWVRQAPGKGLE
WIGGINPNNGGTSYSDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGTTLVTVSS (SEQ ID NO: 1233) 6G12V3-30

QVQLVSGGGVQPGGSLRLSCAASGYTFTEYTMHWVRQAPGKGL
EWIGGINPNNGGTSYSDSVKGRFTISRDN SKNTLYLQMNSLRAEDTA
VYYCARGGSHYYAMDYWGQGT LVT VSS (SEQ ID NO: 1234) 6G12V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYTFTEYTMHWVRQAPGKGLE
WIGGINPNNGGTSYSDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGT LVT VSS (SEQ ID NO: 1235) 6G12V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYTFTEYTMHWVRQAPGKGLE
WIGGINPNNGGTSYSDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGT LVT VSS (SEQ ID NO: 1235) 6G12V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYTFTEYTMHWIRQPPGKGLEW
IGGINPNNGGTSYSPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYY
CARGGSHYYAMDYWGQGT LVT VSS (SEQ ID NO: 1236) 6G12V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYTFTEYTMHWVRQAPGKGLEWI
GGINPNNGGTSYSAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCA
RGGSHYYAMDYWGQGT LVT VSS (SEQ ID NO: 1237) 6G12V4-39
QLQLQESGPGLVKPSETLSLTCTVSGYTFTEYTMHWIRQPPGKGLEWIGG
INPNNGGTSYSPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCARGG
SHYYAMDYWGQGT LVT VSS (SEQ ID NO: 1238) Antibody 8E10 8E10V1-46
QVQLVQSGAEVKKPGASVKV SCKASGYTFTDYEMHWVRQAPGQGLEWI
GVIDPETGGTAYNQKFQGRVTMTRDTSTSTVYME LSSLRSED TAVYYCTS
PDYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1240) 8E10V5-51
EVQLVQSGAEVKKPGESLKISCKGSGY TFTDYEME IWVRQM PGKGLEWIG
VIDPETGGTAYNPSFQGQVTISADKSISTAYLQWSSLKASDTAMY YCTSPD
YYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1241) 8E10V3-30
QVQLVESGGGVVQPG RSLRLSCAASGYTFTDYEME IWVRQAPG KGLEWIG
VIDPETGGTAYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCTSP
DYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1242) 8E10V3-23
EVQLLES GGGLVQPGGSLRLSCAASGYTFTDYEME IWVRQAPG KGLEWIG
VIDPETGGTAYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCTSP
DYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1243) 8E10V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGYTFTDYEME IWVRQAPG QGLEWI
GVIDPETGGTAYNQKFQGRVTITADESTSTAYME LSSLRSED TAVYYCTSP
DYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1244) 8E10V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYTFTDYENIHWVRQAPGKGLEWIG
VIDPETGGTAYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCTSP
DYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1245) 8E10V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYTFTDYENIHWVRQAPGKGLEWIG
VIDPETGGTAYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCTSP
DYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1245) 8E10V4-59
QVQLQESGPGLVKPSETLSLTCTVSGY TFTDYENIHWIRQPPG KGLEWIGVI
DPETGGTAYNPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYYCTSPDYY
GSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1246) 8E10V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYTFTDYENIHWVRQAPGKGLEWIG
VIDPETGGTAYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCTSP
DYYGSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1247) 8E10V4-39
QLQLQESGPGLVKPSETLSLTCTVSGY TFTDYENIHWIRQPPG KGLEWIGVI
DPETGGTAYNPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYYCTSPDYY
GSSYPLYYAMDYWGQGT LVT VSS (SEQ ID NO: 1248) Antibody 7E5 7E5V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYAAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYY

CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1250) 7E5V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1251) 7E5V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1252) 7E5V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1251) 7E5V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1253) 7E5V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGFTFSDAWMGWVRQAPGQGLEWV
AEIRDKVKNHATYYAQKFQGRVTITADESTSTAYMELSSLRSED TAVYYC
RLGVFDYWGQGTTLTVSS (SEQ ID NO: 1254) 7E5V1-46
QVQLVQSGAEVKKPGASVKVSCKASGFTFSDAWMGWVRQAPGQGLEW
VAEIRDKVKNHATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVY
YCRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1255) 7E5V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTFSDAWMGWVRQMPPGKGLEWV
AEIRDKVKNHATYYAPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYCY
RLGVFDYWGQGTTLTVSS (SEQ ID NO: 1256) 7E5V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMGWIRQPPGKGLEWVAE
IRDKVKNHATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCRL
GVFDYWGQGTTLTVSS (SEQ ID NO: 1257) 7E5V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMGWIRQPPGKGLEWVAE
IRDKVKNHATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCRL
GVFDYWGQGTTLTVSS (SEQ ID NO: 1258) Antibody 7F8 7F8V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYAAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCV
RHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1260) 7F8V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1261) 7F8V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1262) 7F8V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1261) 7F8V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1263) 7F8V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGFSFNTYAMNWVRQAPGQGLEWI
ARIRSKSNNYATYYAQKFQGRVTITADESTSTAYMELSSLRSED TAVYYC
VRHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1264) 7F8V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFSFNTYAMNWVRQMPPGKGLEWIA
RIRSKSNNYATYYAPSFQGGQVTISADKSISTAYLQWSSLKASDTAMYCY
RHGDGNLWYIDVWGQGTTLTVSS (SEQ ID NO: 1265) 7F8V1-46
QVQLVQSGAEVKKPGASVKVSCKASGFSFNTYAMNWVRQAPGQGLEWI

ARRSKSNNYATYYAQKFQGRVTMTRDTSTVYMESSLRSEDNAVYYC
VRHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1266) 7F8V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFSFNTYAMNWIRQPPGKGLEWIARI
RSKSNNYATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVRH
GDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1267) 7F8V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGFSFNTYAMNWIRQPPGKGLEWIARI
RSKSNNYATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVRH
GDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1268) Antibody 8F8 8F8V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYTVSRYWMHWRQAPGQGLEWI
GRIDPNSGGTKYNQKFQGRVTMTRDTSTSTVYMESSLRSEDNAVYYCVL
TGTDFDYWGQGTLVTVSS (SEQ ID NO: 1270) 8F8V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYTVSRYWMHWRQAPGKGLEWI
GRIDPNSGGTKYND SVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCVL
TGTDFDYWGQGTLVTVSS (SEQ ID NO: 1271) 8F8V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYTVSRYWMHWRQAPGQGLEWI
GRIDPNSGGTKYNQKFQGRVTITADESTSTAYMESSLRSEDNAVYYCVLT
GTDFDYWGQGTLVTVSS (SEQ ID NO: 1272) 8F8V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYTVSRYWMHWRQMPGKGLEWI
GRIDPNSGGTKYNPSFQGQVTISADKSISTAYLQWSSLKASDTAMYCYCVLT
GTDFDYWGQGTLVTVSS (SEQ ID NO: 1273) 8F8V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYTVSRYWMHWRQAPGKGLEWI
GRIDPNSGGTKYND SVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCVL
TGTDFDYWGQGTLVTVSS (SEQ ID NO: 1274) 8F8V3-30
QVQLVESGGGVVQPGRSLRLSCAASGYTVSRYWMHWRQAPGKGLEWI
GRIDPNSGGTKYND SVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCVL
TGTDFDYWGQGTLVTVSS (SEQ ID NO: 1275) 8F8V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYTVSRYWMHWRQAPGKGLEWI
GRIDPNSGGTKYND SVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCVL
TGTDFDYWGQGTLVTVSS (SEQ ID NO: 1274) 8F8V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYTVSRYWMHWIRQPPGKGLEWIGR
IDPNSGGTKYNPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVLTGT
DFDYWGQGTLVTVSS (SEQ ID NO: 1276) 8F8V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYTVSRYWMHWRQAPGKGLEWI
GRIDPNSGGTKYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCVL
TGTDFDYWGQGTLVTVSS (SEQ ID NO: 1277) 8F8V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGYTVSRYWNIHWIRQPPGKGLEWIGR
IDPNSGGTKYNPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVLTGT
DFDYWGQGTLVTVSS (SEQ ID NO: 1278) Antibody 1H7 1H7V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYAAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1260) 1H7V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1262) 1H7V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1261) 1H7V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1261) 1H7V3-30

QVQLVSGGGVQPGSLRSGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1263) 1H7V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFSFNTYAMNWVRQMPGKGLEWIA
RIRSKSNNYATYYAPSFQGGQVTISADKSISTAYLQWSSLKASDTAMY YCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1265) 1H7V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGFSFNTYAMNWVRQAPGQGLEWI
ARIRSKSNNYATYYAQKFQGRVTITADESTSTAYMELSSLRSED TAVYYC
VRHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1264) 1H7V1-46
QVQLVQSGAEVKKPGASVKV SCKASGFSFNTYAMNWVRQAPGQGLEWI
ARIRSKSNNYATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYC
VRHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1266) 1H7V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFSFNTYAMNWIRQPPGKGLEW IARI
RSKSNNYATYYAPSLKSRVTISVDT SKNQFSLKLSSVTAADTAVYYCVRH
GDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1267) 1H7V4-30-4
QVQLQESGPGLVKPSQTL SLTCTVSGFSFNTYAMNWIRQPPGKGLEW IARI
RSKSNNYATYYAPSLKSRVTISVDT SKNQFSLKLSSVTAADTAVYYCVRH
GDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1268) Antibody 2H8 2H8V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYAAPVKGRFTISRDD SKNTLYLQMNSLKTEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1260) 2H8V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1261) 2H8V3-23
EVQLLES GGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1262) 2H8V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1261) 2H8V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFSFNTYAMNWVRQAPGKGLEWIA
RIRSKSNNYATYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1263) 2H8V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFSFNTYAMNWVRQMPGKGLEWIA
RIRSKSNNYATYYAPSFQGGQVTISADKSISTAYLQWSSLKASDTAMY YCV
RHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1265) 2H8V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGFSFNTYAMNWVRQAPGQGLEWI
ARIRSKSNNYATYYAQKFQGRVTITADESTSTAYMELSSLRSED TAVYYC
VRHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1264) 2H8V1-46
QVQLVQSGAEVKKPGASVKV SCKASGFSFNTYAMNWVRQAPGQGLEWI
ARIRSKSNNYATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYC
VRHGDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1266) 2H8V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFSFNTYAMNWIRQPPGKGLEW IARI
RSKSNNYATYYAPSLKSRVTISVDT SKNQFSLKLSSVTAADTAVYYCVRH
GDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1267) 2H8V4-30-4
QVQLQESGPGLVKPSQTL SLTCTVSGFSFNTYAMNWIRQPPGKGLEW IARI
RSKSNNYATYYAPSLKSRVTISVDT SKNQFSLKLSSVTAADTAVYYCVRH
GDGNLWYIDVWGQGTLVTVSS (SEQ ID NO: 1268) Antibody 3A2 3A2V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYPFSNFWITWVRQMPGKGLEWIGD
IYPGSDNSNYNPSFQGGQVTISADKSISTAYLQWSSLKASDTAMY YCAREAY

YTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1282) 3A2V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYPFSSNFWITWVRQAPGQGLEWIG
DIYPGSDNSNYNQKFQGRVTITADESTSTAYMELSSLRSED TAVYYCAREA
YYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1283) 3A2V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYPFSSNFWITWVRQAPGQGLEWIG
DIYPGSDNSNYNQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYCARE
AYYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1284) 3A2V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYPFSSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNDSVKGRFTISRDN AKNSLYLQMNSLRAED TAVYYCAREA
YYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1285) 3A2V3-30
QVQLVESGGGVVQPGRSLRLSCAASGYPFSSNFWITWVRQAPGKGLEWIG
DIYPGSDNSNYNDSVKGRFTISRDN SKNTLYLQMNSLRAED TAVYYCARE
AYYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1286) 3A2V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYPFSSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNDSVKGRFTISRDN AKNSLYLQMNSLRAED TAVYYCAREA
YYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1285) 3A2V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYPFSSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNDSVKGRFTISRDN SKNTLYLQMNSLRAED TAVYYCAREA
YYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1287) 3A2V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYPFSSNFWITWIRQPPGKGLEWIGDIY
PGSDNSNYNPSLKS RVTISVDTSKNQFSLKLSSVTAAD TAVYYCAREAYY
TNPGFAYWGQGTTLTVSS (SEQ ID NO: 1288) IGHV3-15
EVQLVESGGGLVKPGGSLRLSCAASGYPFSSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNAPVKGRFTISRDD SKNTLYLQMNSLKTED TAVYYCAREA
YYTNPGFAYWGQGTTLTVSS (SEQ ID NO: 1289) 3A2V4-39
QLQLQESGPGLVKPSETLSLTCTVSGYPFSSNFWITWIRQPPGKGLEWIGDIY
PGSDNSNYNPSLKS RVTISVDTSKNQFSLKLSSVTAAD TAVYYCAREAYY
TNPGFAYWGQGTTLTVSS (SEQ ID NO: 1290) Antibody 3A7 3A7V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYAAPVKGRFTISRDD SKNTLYLQMNSLKTED TAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1250) 3A7V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDN AKNSLYLQMNSLRAED TAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1251) 3A7V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDN SKNTLYLQMNSLRAED TAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1252) 3A7V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDN AKNSLYLQMNSLRAED TAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1251) 3A7V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFTFSDAWMGWVRQAPGKGLEWV
AEIRDKVKNHATYYADSVKGRFTISRDN SKNTLYLQMNSLRAED TAVYY
CRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1253) 3A7V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGFTFSDAWMGWVRQAPGQGLEWV
AEIRDKVKNHATYYAQKFQGRVTITADESTSTAYMELSSLRSED TAVYYC
RLGVFDYWGQGTTLTVSS (SEQ ID NO: 1254) 3A7V1-46
QVQLVQSGAEVKKPGASVKVSCKASGFTFSDAWMGWVRQAPGQGLEW
VAEIRDKVKNHATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVY
YCRLGVFDYWGQGTTLTVSS (SEQ ID NO: 1255) 3A7V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTFSDAWMGWVRQMPPGKGLEWV

AEIRDKNKHYATYQWSSFKASDTAMYYC
RLGVFDYWGQGTLVTVSS (SEQ ID NO: 1256) 3A7V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMGWIRQPPGKGLEWVAE
IRDKVKNHATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCRL
GVFDYWGQGTLVTVSS (SEQ ID NO: 1257) 3A7V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMGWIRQPPGKGLEWVAE
IRDKVKNHATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCRL
GVFDYWGQGTLVTVSS (SEQ ID NO: 1258) Antibody 3B10 3B10V3-15
EVQLVESGGGLVKPGGSLRLSCAASGLTSNTYTQTWVRQAPGKGLEWES
VIRSKSNFSTLYAAPVKGRFTISRDD SKNTLYLQMNSLKTEDTAVYYCV
RHKSNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1292) 3B10V3-30
QVQLVESGGGVVQPGRSLRLSCAASGLTSNTYTQTWVRQAPGKGLEWES
VIRSKSNFSTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCV
RHKSNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1293) 3B10V3-23
EVQLLES GGGLVQPGGSLRLSCAASGLTSNTYTQTWVRQAPGKGLEWES
VIRSKSNFSTLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCV
RHKSNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1294) 3B10V1-46
QVQLVQSGAEVKKPGASVKV SCKASGLTSNTYTQTWVRQAPGQGLEWE
SVIRSKSNFSTLYAQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYC
VRHKSNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1295) 3B10V3-48
EVQLVESGGGLVQPGGSLRLSCAASGLTSNTYTQTWVRQAPGKGLEWES
VIRSKSNFSTLYADSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCV
RHKSNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1296) 3B10V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGLTSNTYTQTWVRQAPGQGLEWES
VIRSKSNFSTLYAQKFQGRVTITADESTSTAYMELSSLRSED TAVYYCVR
HKS NRYPGVYWGQGTLVTVSS (SEQ ID NO: 1297) 3B10V3-7
EVQLVESGGGLVQPGGSLRLSCAASGLTSNTYTQTWVRQAPGKGLEWES
VIRSKSNFSTLYADSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCV
RHKSNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1296) 3B10V5-51
EVQLVQSGAEVKKPGESLKISCKGSGGLTSNTYTQTWVRQMPGKGLEWES
VIRSKSNFSTLYAPSFQGQVTISADKSISTAYLQWSSLKASDTAMYYCVR
HKS NRYPGVYWGQGTLVTVSS (SEQ ID NO: 1298) 3B10V4-59
QVQLQESGPGLVKPSETLSLTCTVSGLT SNTYTQTWIRQPPGKGLEWESVI
RSKSNFSTLYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVRHK
SNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1299) 3B10V4-39
QLQLQESGPGLVKPSETLSLTCTVSGLT SNTYTQTWIRQPPGKGLEWESVI
RSKSNFSTLYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVRHK
SNRYPGVYWGQGTLVTVSS (SEQ ID NO: 1300) Antibody 4F11 4F11V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYPFSNFWITWVRQMPGKGLEWIGD
IYPGSDNSNYNPSFQGQVTISADKSISTAYLQWSSLKASDTAMYYCAREAY
YTNP GFAYWGQGTLVTVSS (SEQ ID NO: 1282) 4F11V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGYPF SNFWITWVRQAPGQGLEWIG
DIYPGSDNSNYNQKFQGRVTITADESTSTAYMELSSLRSED TAVYYCAREA
YYTNP GFAYWGQGTLVTVSS (SEQ ID NO: 1283) 4F 11V1-46
QVQLVQSGAEVKKPGASVKV SCKASGYPF SNFWITWVRQAPGQGLEWIG
DIYPGSDNSNYNQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVYYCARE
AYYTNP GFAYWGQGTLVTVSS (SEQ ID NO: 1284) 4F11V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYPF SNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCAREA
YYTNP GFAYWGQGTLVTVSS (SEQ ID NO: 1285) 4F11V3-30

QVQLVESGGGVVQPGGSLRLSCAASGYPFSNFWITWVRQAPGKGLEWIG
DIYPGSDNSNYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARE
AYYTNPGFAYWGQGT LTVSS (SEQ ID NO: 1286) 4F11V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYPFSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYYCAREA
YYTNPGFAYWGQGT LTVSS (SEQ ID NO: 1285) 4F11V3-23
EVQLLES GGGLVQPGGSLRLSCAASGYPFSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAREA
YYTNPGFAYWGQGT LTVSS (SEQ ID NO: 1287) 4F11V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYPFSNFWITWIRQPPGKGLEWIGDIY
PGSDNSNYNPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYYCAREAYY
TNPGFAYWGQGT LTVSS (SEQ ID NO: 1288) 4F11V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYPFSNFWITWVRQAPGKGLEWIGD
IYPGSDNSNYNAPVKGRFTISRDD SKNTLYLQMNSLKTEDTAVYYCAREA
YYTNPGFAYWGQGT LTVSS (SEQ ID NO: 1289) 4F11V4-39
QLQLQESGPGLVKPSETLSLTCTVSGYPFSNFWITWIRQPPGKGLEWIGDIY
PGSDNSNYNPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYYCAREAYY
TNPGFAYWGQGT LTVSS (SEQ ID NO: 1290) Antibody 6H6 6H6V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLEWW
VAEIRNKVNNHATYYAPVKGRFTISRDD SKNTLYLQMNSLKTEDTAVYY
CCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1302) 6H6V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLEWW
VAEIRNKVNNHATYYDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYY
CCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1303) 6H6V3-23
EVQLLES GGGLVQPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLEWW
VAEIRNKVNNHATYYDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1304) 6H6V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLEWW
VAEIRNKVNNHATYYDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAVYY
CCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1303) 6H6V3-30
QVQLVESGGGVVQPGSLRLSCAASGFTFSDAWMDWVRQAPGKGLEWW
VAEIRNKVNNHATYYDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1305) 6H6V1-46
QVQLVQSGAEVKKPGASVKV SCKASGFTFSDAWMDWVRQAPGQGLEW
WVAEIRNKVNNHATYYQKFQGRVTMTRDTSTSTVYMELSSLRSED TAVY
YCCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1306) 6H6V 1-69
QVQLVQSGAEVKKPGSSVKV SCKASGFTFSDAWMDWVRQAPGQGLEW
WVAEIRNKVNNHATYYQKFQGRVTITADESTSTAYMELSSLRSED TAVYY
CCTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1307) 6H6V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTFSDAWMDWVRQMPPGKGLEWW
VAEIRNKVNNHATYYPSFQGQVTISADKSISTAYLQWSSLKASDTAMY C
CTSLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1308) 6H6V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMDWIRQPPGKGLEWWV
AEIRNKVNNHATYYPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYYCCT
SLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1309) 6H6V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMDWIRQPPGKGLEWWV
AEIRNKVNNHATYYPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYYCCT
SLYDGYYLRFAWGQGT LTVSS (SEQ ID NO: 1310) Antibody 7A9 7A9V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFTFNTYS MNWVRQAPGKGLEWVA
HIKTKZNNFATFYAAPVKGRFTISRDD SKNTLYLQMNSLKTEDTAVYYCV

ZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1312) 7A9V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFTFNTYSMNWVRQAPGKGLEWVA
HIKTKZNNFATFYADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCV
ZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1313) 7A9V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFTFNTYSMNWVRQAPGKGLEWVA
HIKTKZNNFATFYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCV
ZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1314) 7A9V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFTFNTYSMNWVRQAPGKGLEWVA
HIKTKZNNFATFYADSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCV
ZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1313) 7A9V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFTFNTYSMNWVRQAPGKGLEWV
AHIKTKZNNFATFYADSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYC
VZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1315) 7A9V1-46
QVQLVQSGAEVKKPGASVKVSCKASGFTFNTYSMNWVRQAPGQGLEWV
AHIKTKZNNFATFYAQKFQGRVTMTRDTSTSTVYMELSSLRSEDVAVYYC
VZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1316) 7A9V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGFTFNTYSMNWVRQAPGQGLEWV
AHIKTKZNNFATFYAQKFQGRVTITADESTSTAYMELSSLRSEDVAVYYCV
ZHZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1317) 7A9V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTFNTYSMNWVRQMPGKGLEWVA
HIKTKZNNFATFYAPSFQGQVTISADKSISTAYLQWSSLKASDTAMYVCVZ
HZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1318) 7A9V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFTFNTYSMNWIRQPPGKGLEWVAHI
KTKZNNFATFYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVZHZ
SNYPFAYWGQGTTLTVSS (SEQ ID NO: 1319) 7A9V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGFTFNTYSMNWIRQPPGKGLEWVAH
IKTKZNNFATFYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYCVZH
ZSNYPFAYWGQGTTLTVSS (SEQ ID NO: 1320) Antibody 7B3 7B3V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYTFTTYWIHWVRQAPGQGLEWIG
RNDPNSGGSNYNQKFQGRVTMTRDTSTSTVYMELSSLRSEDVAVYYCVR
TNWDGDFWGQGTTLTVSS (SEQ ID NO: 1322) 7B3V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYTFTTYWIHWVRQMPGKGLEWIG
RNDPNSGGSNYNPSFQGQVTISADKSISTAYLQWSSLKASDTAMYVCVRT
NWDGDFWGQGTTLTVSS (SEQ ID NO: 1323) 7B3V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYTFTTYWIHWVRQAPGQGLEWIG
RNDPNSGGSNYNQKFQGRVTITADESTSTAYMELSSLRSEDVAVYYCVRT
NWDGDFWGQGTTLTVSS (SEQ ID NO: 1324) 7B3V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYTFTTYWIHWVRQAPGKGLEWIG
RNDPNSGGSNYNDSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCVR
TNWDGDFWGQGTTLTVSS (SEQ ID NO: 1325) 7B3V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYTFTTYWIHWVRQAPGKGLEWIG
RNDPNSGGSNYNDSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCVR
TNWDGDFWGQGTTLTVSS (SEQ ID NO: 1326) 7B3V3 -30
QVQLVESGGGVVQPGRSLRLSCAASGYTFTTYWIHWVRQAPGKGLEWIG
RNDPNSGGSNYNDSVKGRFTISRDNANKNTLYLQMNSLRAEDTAVYYCVR
TNWDGDFWGQGTTLTVSS (SEQ ID NO: 1327) 7B3V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYTFTTYWIHWVRQAPGKGLEWIG
RNDPNSGGSNYNDSVKGRFTISRDNANKNSLYLQMNSLRAEDTAVYYCVR
TNWDGDFWGQGTTLTVSS (SEQ ID NO: 1326) 7B3V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYTFTTYWIHWIRQPPGKGLEWIGRN

DPNSGGSNYNSPNSRVTSVDTSKNQFSLKLSSVTAADTAVYYCVRTNW
DGDFWGQGTLVTVSS (SEQ ID NO: 1328) 7B3V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYTFTTYWIHWVRQAPGKGLE
WIGRNDPNSGGSNYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
YYCVRTNWDGDFWGQGTLVTVSS (SEQ ID NO: 1329) 7B3V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGYTFTTYWIHWIRQPPGKGLEWI
GRNDPNSGGSNYNPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYC
VRTNWDGDFWGQGTLVTVSS (SEQ ID NO: 1330) Antibody 8A1 8A1V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYAFSNIWMSWVRQMPGKGLE
WIGQIYPGDGDTKYNPSFQGQVTISADKSISTAYLQWSSLKASDTAM
YYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1332) 8A1V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYAFSNIWMSWVRQAPGQG
LEWIGQIYPGDGDTKYNQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1333) 8A1V3-23
EVQLLES GGGLVQPGGSLRLSCAASGYAFSNIWMSWVRQAPGKGLE
WIGQIYPGDGDTKYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1334) 8A1V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSNIWMSWVRQAPGQGL
EWIGQIYPGDGDTKYNQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1335) 8A1V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYAFSNIWMSWVRQAPGKGLE
WIGQIYPGDGDTKYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1336) 8A1V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYAFSNIWMSWVRQAPGKGLE
WIGQIYPGDGDTKYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1336) 8A1V3-30
QVQLVESGGGVVQPGRLRLSCAASGYAFSNIWMSWVRQAPGKGL
EWIGQIYPGDGDTKYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTA
VYYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1337) 8A1V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYAFSNIWMSWIRQPPGKGLE
WIGQIYPGDGDTKYNPSLKSRTISVDTSKNQFSLKLSSVTAADTAVY
YCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1338) 8A1V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYAFSNIWMSWVRQAPGKGLE
WIGQIYPGDGDTKYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
YYCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1339) 8A1V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGYAFSNIWMSWIRQPPGKGLE
WIGQIYPGDGDTKYNPSLKSRTISVDTSKNQFSLKLSSVTAADTAVY
YCSREKGADYYGSTYSAWFSYWQGTLVTVSS (SEQ ID NO: 1340) Antibody 9F5
9F5V5-51 EVQLVQSGAEVKKPGESLKISCKGSGYAFSSSWMNWVRQMPGKGLE
WIGRIYPGDGDTNYNPSFQGQVTISADKSISTAYLQWSSLKASDTAM
YYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1342) 9F5V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDTNYNQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1343) 9F5V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDTNYNQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1344) 9F5V3 -23
EVQLLES GGGLVQPGGSLRLSCAASGYAFSSSWMNWVRQAPGKGLE
WIGRIYPGDGDTNYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1345) 9F5V3-7

EVQLVESGGGLVQPGGSLRLSQAASGYAFSSSWMNWVRQAPGKGLE
WIGRIYPGDGDTNYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1346) 9F5V3-48
EVQLVESGGGLVQPGGSLRLSQAASGYAFSSSWMNWVRQAPGKGLE
WIGRIYPGDGDTNYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1346) 9F5V3-30
QVQLVESGGGVVQPGRLSLRLSQAASGYAFSSSWMNWVRQAPGKGLE
WIGRIYPGDGDTNYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1347) 9F5V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYAFSSSWMNWIRQPPGKGLE
WIGRIYPGDGDTNYNPSLKSRTVISVDTSKNQFSLKLSSVTAADTAVY
YCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1348) 9F5V3-15
EVQLVESGGGLVQPGGSLRLSQAASGYAFSSSWMNWVRQAPGKGLE
WIGRIYPGDGDTNYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
YYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1349) 9F5V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGYAFSSSWMNWIRQPPGKGLE
WIGRIYPGDGDTNYNPSLKSRTVISVDTSKNQFSLKLSSVTAADTAVY
YCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1350) 9F5-H1
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAQKFQGRVTMTRDTSTSTVYMELSSLRSED
AVYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1665) 9F5-H2
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDTNYAQKFQGRVTMTADTSTSTVYMELSSLRSED
VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1666) 9F5-H3
QVQLVQSGAEVKKPGASLKISCKASGYAFSSSWMNWVRQAPGQGLE
WIGRIYPGDGDTNYAQKFQGRATLTADTSTSTAYMELSSLRSED
TAVYCARLLRNQPGESYAMDYWGQGLVTVSS (SEQ ID NO: 1667) Antibody 9G1
9G1V5-51 EVQLVQSGAEVKKPGESLKISCKGSGYIFTTYWIHWVRQMPGKGLE
WIGRIDPNNGDTNYNPSFQGQVTISADKSISTAYLQWSSLKASDTAM
YYCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1352) 9G1V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYIFTTYWIHWVRQAPGQGLE
WIGRIDPNNGDTNYNQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1353) 9G1V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYIFTTYWIHWVRQAPGQGLE
WIGRIDPNNGDTNYNQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1354) 9G1V3-23
EVQLLES GGGLVQPGGSLRLSQAASGYIFTTYWIHWVRQAPGKGLEW
IGRIDPNNGDTNYNDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVY
YCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1355) 9G1V3-30
QVQLVESGGGVVQPGRLSLRLSQAASGYIFTTYWIHWVRQAPGKGLE
WIGRIDPNNGDTNYNDSVKGRFTISRDN SKNTLYLQMNSLRAED
TAVYYCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1356) 9G1V3-7
EVQLVESGGGLVQPGGSLRLSQAASGYIFTTYWIHWVRQAPGKGLE
WIGRIDPNNGDTNYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1357) 9G1V3-48
EVQLVESGGGLVQPGGSLRLSQAASGYIFTTYWIHWVRQAPGKGLE
WIGRIDPNNGDTNYNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YCVMTGTDFDYWGQGTLVTVSS (SEQ ID NO: 1357) 9G1V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYIFTTYWIHWIRQPPGKGLEW
GRIDPNNGDTNYNPSLKSRTVISVDTSKNQFSLKLSSVTAADTAVYY

CVMTGTDFTDYGQGTFLVTVSS (SEQ ID NO: 1358) 9G1V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYIFTTYWIHWVRQAPGKGLE
WIGRIDPNNGDTNYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
YYCVMTGTDFDYWGQGTFLVTVSS (SEQ ID NO: 1359) 9G1V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGYIFTTYWIHWIRQPPGKGLEWI
GRIDPNNGDTNYNPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYY
CVMTGTDFDYWGQGTFLVTVSS (SEQ ID NO: 1360) Antibody 9G3 9G3V3 -15
EVQLVESGGGLVKPGGSLRLSCAASGFNFNTYAMKWVRQAPGKGLE
WIARIRSNSNDYATNYSAPVKGRFTISRDDSKNTLYLQMNSLKTEDTA
VYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1456) 9G3V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFNFNTYAMKWVRQAPGKGLE
WIARIRSNSNDYATNYSDSVKGRFTISRDN SKNTLYLQMNSLRAEDT
AVYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1457) 9G3V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFNFNTYAMKWVRQAPGKGLE
WIARIRSNSNDYATNYSDSVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1458) 9G3V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFNFNTYAMKWVRQAPGKGLE
WIARIRSNSNDYATNYSDSVKGRFTISRDN AKNSLYLQMNSLRAEDT
AVYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1459) 9G3V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFNFNTYAMKWVRQAPGKGLE
WIARIRSNSNDYATNYSDSVKGRFTISRDN AKNSLYLQMNSLRAEDT
AVYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1460) 9G3V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGFNFNTYAMKWVRQAPGQGL
EWIARIRSNSNDYATNYSQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1461) 9G3V1-46
QVQLVQSGAEVKKPGASVKV SCKASGFNFNTYAMKWVRQAPGQGL
EWIARIRSNSNDYATNYSQKFQGRVTMTRDTSTSTVYME
LSSLRSED
TAVYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1462) 9G3V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFNNTYAMKWVRQM
PGKGL
EWIARIRSNSNDYATNYSQKFQGRVTMTRDTSTSTVYME
LSSLRSED
TAVYYCVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1463) 9G3V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFNNTYAMKWIRQPPGKGLEWI
ARIRSNSNDYATNYSQKFQGRVTISVDTSKNQFSLKLSSVTAADTAVYY
CVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1464) 9G3V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGFNNTYAMKWIRQPPGKGLEWI
ARIRSNSNDYATNYSQKFQGRVTISVDTSKNQFSLKLSSVTAADTAVYY
CVGHKINNYPFPAHWGQGTFLVTVSS (SEQ ID NO: 1465) Antibody 10A9 10A9V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYPFSNFWITWVRQM
PGKGLE
WIGDIYPGSDNRNFNPSFQGGV TISADKSISTAYLQWSSLKASDT
TAMYYCAREAYYTNP GFAYWGQGTFLVTVSS (SEQ ID NO: 1362) 10A9V1-69
QVQLVQSGAEVKKPGSSVKV SCKASGYPF SNFWITWVRQAPGQGLE
WIGDIYPGSDNRNFNQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCAREAYYTNP GFAYWGQGTFLVTVSS (SEQ ID NO: 1363) 10A9V1-46
QVQLVQSGAEVKKPGASVKV SCKASGYPF SNFWITWVRQAPGQGLE
WIGDIYPGSDNRNFNQKFQGRVTMTRDTSTSTVYME
LSSLRSED
TAVYYCAREAYYTNP GFAYWGQGTFLVTVSS (SEQ ID NO: 1364) 10A9V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYPF SNFWITWVRQAPGKGLE
WIGDIYPGSDNRNFNDSVKGRFTISRDN AKNSLYLQMNSLRAEDTAV
YYCAREAYYTNP GFAYWGQGTFLVTVSS (SEQ ID NO: 1365) 10A9V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYPF SNFWITWVRQAPGKGLE

WIGDIYPGSDNRNFNDNAKNSLYLQMNSLRAEDTAV
YYCAREAYYTNPGFAYWGQGTTLVTVSS (SEQ ID NO: 1365) 10A9V3-30
QVQLVESGGGVVQPGRSLRLSCAASGYPFNSFWITWVRQAPGKGLE
WIGDIYPGSDNRNFNDNAKNSLYLQMNSLRAEDTAV
YYCAREAYYTNPGFAYWGQGTTLVTVSS (SEQ ID NO: 1366) 10A9V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYPFNSFWITWVRQAPGKGLE
WIGDIYPGSDNRNFNDNAKNSLYLQMNSLRAEDTAV
YYCAREAYYTNPGFAYWGQGTTLVTVSS (SEQ ID NO: 1367) 10A9V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYPFNSFWITWIRQPPGKGLEWI
GDIYPGSDNRNFNPVSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
AREAYYTNPGFAYWGQGTTLVTVSS (SEQ ID NO: 1368) 10A9V3-15
EVQLVESGGGLVQPGGSLRLSCAASGYPFNSFWITWVRQAPGKGLE
WIGDIYPGSDNRNFNPVSLKSRVTISRDSDSKNTLYLQMNSLKTEDTAV
YYCAREAYYTNPGFAYWGQGTTLVTVSS (SEQ ID NO: 1369) 10A9V4-39
QLQLQESGPGLVKPSETLSLTCTVSGYPFNSFWITWIRQPPGKGLEWI
GDIYPGSDNRNFNPVSLKSRVTISVDTSKNQFSLKLSSVTAADTAVYYC
AREAYYTNPGFAYWGQGTTLVTVSS (SEQ ID NO: 1370) Antibody11A8 11A8V3-15
EVQLVESGGGLVQPGGSLRLSCAASGFNFNTYAMNWVRQAPGKGLE
WVARIRSKSNNYATYYAAPVKGRFTISRDDSKNTLYLQMNSLKTEDT
AVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1372) 11A8V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFNFNTYAMNWVRQAPGKGLE
WVARIRSKSNNYATYYADSVKGRFTISRDNNAKNSLYLQMNSLRAED
TAVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1373) 11A8V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFNFNTYAMNWVRQAPGKGLE
WVARIRSKSNNYATYYADSVKGRFTISRDNNAKNSLYLQMNSLRAEDT
AVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1374) 11A8V3-30
QVQLVESGGGVVQPGRSLRLSCAASGFNFNTYAMNWVRQAPGKGL
EWVARIRSKSNNYATYYADSVKGRFTISRDNNAKNSLYLQMNSLRAED
TAVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1375) 11A8V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFNFNTYAMNWVRQAPGKGLE
WVARIRSKSNNYATYYADSVKGRFTISRDNNAKNSLYLQMNSLRAED
TAVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1373) 11A8V1-69
QVQLVQSGAEVKKPGSSVKVSCASGFNFNTYAMNWVRQAPGQGL
EWVARIRSKSNNYATYYAQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1376) 11A8V1-46
QVQLVQSGAEVKKPGASVKVSCASGFNFNTYAMNWVRQAPGQGL
EWVARIRSKSNNYATYYAQKFQGRVTMTRDTSTSTVYMELSSLRSE
DTAVYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1377) 11A8V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFNNTYAMNWVRQMPPGKGLE
WVARIRSKSNNYATYYAPSFQGQVTISADKSISTAYLQWSSLKASDT
AMYYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1378) 11A8V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFNNTYAMNWVRQPPGKGLEW
VARIRSKSNNYATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAV
YYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1379) 11A8V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFNNTYAMNWVRQPPGKGLEW
VARIRSKSNNYATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAV
YYCVRHYSNYGWGFAYWGQGTTLVTVSS (SEQ ID NO: 1380) Antibody12D9
12D9V1-46 QVQLVQSGAEVKKPGASVKVSCASGYTFSDYYIHWVRQAPGQGLE
WIGYIYPNNGDNGYNQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCARRGYYGGSYDYWGQGTTLVTVSS (SEQ ID NO: 1382) 12D9V5-51

EVQLVQSGAEVKKPGSSSVKVSCKASGYTFSDYYIHWVRQMPGKGLE
WIGYIYPNNGDNGYNPSFQGGQVTISADKSISTAYLQWSSLKASDTAM
YYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1383) 12D9V1-69
QVQLVQSGAEVKKPGSSSVKVSCKASGYTFSDYYIHWVRQAPGQGLE
WIGYIYPNNGDNGYNQKFQGRVTITADESTSTAYMELSSLRSED
TAV YYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1384) 12D9V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYTFSDYYIHWVRQAPGKGLE
WIGYIYPNNGDNGYND SVKGRFTISRDN AKNSLYLQMNSLRAEDTA
VYYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1385) 12D9V3-30
QVQLVESGGGVVQPG RSLRLSCAASGYTFSDYYIHWVRQAPGKGLE
WIGYIYPNNGDNGYND SVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1386) 12D9V3-23
EVQLLES GGGLVQPGGSLRLSCAASGYTFSDYYIHWVRQAPGKGLE
WIGYIYPNNGDNGYND SVKGRFTISRDN SKNTLYLQMNSLRAEDTAV
YYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1387) 12D9V3-7
EVQLVES GGGLVQPGGSLRLSCAASGYTFSDYYIHWVRQAPGKGLE
WIGYIYPNNGDNGYND SVKGRFTISRDN AKNSLYLQMNSLRAEDTA
VYYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1385) 12D9V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYTFSDYYIHWIRQPPGKGLEWI
GYIYPNNGDNGYNPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYY
CARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1388) 12D9V3-15
EVQLVES GGGLVKPGGSLRLSCAASGYTFSDYYIHWVRQAPGKGLE
WIGYIYPNNGDNGYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
YYCARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1389) 12D9V4-30-4
QVQLQESGPGLVKPSQTL SLTCTVSGYTFSDYYTHWIRQPPGKGLEWI
GYIYPNNGDNGYNPSLKS RVTISVDTSKNQFSLKLSSVTAADTAVYY
CARRGYYGGSYDYWGQGTLVTVSS (SEQ ID NO: 1390) Antibody 12F9 12F9V3-15
EVQLVES GGGLVKPGGSLRLSCAASGFRFNTYAMTWVRQAPGKGLE
WEGVIRRKSSNFATLYAAPVKGRFTISRDDSKNTLYLQMNSLKTEDT
AVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1392) 12F9V3-23
EVQLLES GGGLVQPGGSLRLSCAASGFRFNTYAMTWVRQAPGKGLE
WEGVIRRKSSNFATLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDT
AVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1393) 12F9V3-48
EVQLVES GGGLVQPGGSLRLSCAASGFRFNTYAMTWVRQAPGKGLE
WEGVIRRKSSNFATLYADSVKGRFTISRDN AKNSLYLQMNSLRAEDT
AVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1394) 12F9V3-30
QVQLVES GGGVVQPG RSLRLSCAASGFRFNTYAMTWVRQAPGKGLE
WEGVIRRKSSNFATLYADSVKGRFTISRDN SKNTLYLQMNSLRAEDT
AVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1395) 12F9V3-7
EVQLVES GGGLVQPGGSLRLSCAASGFRFNTYAMTWVRQAPGKGLE
WEGVIRRKSSNFATLYADSVKGRFTISRDN AKNSLYLQMNSLRAEDT
AVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1394) 12F9V1-69
QVQLVQSGAEVKKPGSSSVKVSCKASGFRFNTYAMTWVRQAPGQGL
EWEGVIRRKSSNFATLYAQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1396) 12F9V1-46
QVQLVQSGAEVKKPGASVKVSCKASGFRFNTYAMTWVRQAPGQGL
EWEGVIRRKSSNFATLYAQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCVRHKS NKYPFVYWGQGTLVTVSS (SEQ ID NO: 1397) 12F9V5-51
EVQLVQSGAEVKKPGESLKISCKSGFRFNTYAMTWVRQMPGKGLE
WEGVIRRKSSNFATLYAPSFQGGQVTISADKSISTAYLQWSSLKASDTA

MYYCVRHKSNKYPFVYWGQGTTLTVSS (SEQ ID NO: 1398) 12F9V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFRFNTYAMTWIRQPPGKGLEW
EGVIRRKSSNFATLYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAV
YYCVRHKSNNKYPFVYWGQGTTLTVSS (SEQ ID NO: 1399) 12F9V4-39
QLQLQESGPGLVKPSETLSLTCTVSGFRFNTYAMTWIRQPPGKGLEW
EGVIRRKSSNFATLYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAV
YYCVRHKSNNKYPFVYWGQGTTLTVSS (SEQ ID NO: 1400) Antibodyl0C1 10C1V3-15
EVQLVESGGGLVKPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLE
WVAEIRNKINNHATYYAAPVKGRFTISRDDSKNTLYLQMNSLKTEDT
AVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1467) 10C1V3-7
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLE
WVAEIRNKINNHATYYADSVKGRFTISRDNKNSLYLQMNSLRAEDT
AVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1468) 10C1V3-23
EVQLLESGGGLVQPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLE
WVAEIRNKINNHATYYADSVKGRFTISRDNKNTLYLQMNSLRAEDT
AVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1469) 10C1V3-30
QVQLVESGGGVVQPGSLRLSCAASGFTFSDAWMDWVRQAPGKGL
EWVAEIRNKINNHATYYADSVKGRFTISRDNKNTLYLQMNSLRAED
TAVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1470) 10C1V3-48
EVQLVESGGGLVQPGGSLRLSCAASGFTFSDAWMDWVRQAPGKGLE
WVAEIRNKINNHATYYADSVKGRFTISRDNKNSLYLQMNSLRAEDT
AVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1471) 10C1V1-69
QVQLVQSGAEVKKPGSSVKVCKASGFTFSDAWMDWVRQAPGQGL
EWVAEIRNKINNHATYYAQQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1472) 10C1V1-46
QVQLVQSGAEVKKPGASVKVCKASGFTFSDAWMDWVRQAPGQGL
EWVAEIRNKINNHATYYAQQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1473) 10C1V5-51
EVQLVQSGAEVKKPGESLKISCKGSGFTFSDAWMDWVRQMPPGKGLE
WVAEIRNKINNHATYYAPSFQGGQVTISADKSISTAYLQWSSLKASDTA
MYYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1474) 10C1V4-59
QVQLQESGPGLVKPSETLSLTCTVSGFTFSDAWMDWIRQPPGKGLEW
VAEIRNKINNHATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAV
YYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1475) 10C1V4-30-4
QVQLQESGPGLVKPSQTLSTCTVSGFTFSDAWMDWIRQPPGKGLEW
VAEIRNKINNHATYYAPSLKSRVTISVDTSKNQFSLKLSSVTAADTAV
YYCTSLYDGSYLRFAFAYWGQGTTLTVSS (SEQ ID NO: 1476) Antibody 7E9 7E9V1-
46 QVQLVQSGAEVKKPGASVKVCKASGYTFTEYTMHWRQAPGQGL
EWIGGINPNNGGTSYKQKFQGRVTMTRDTSTSTVYMELSSLRSED
TAVYYCARGGSHYYAMDYWGQGTTLTVSS (SEQ ID NO: 1478) 7E9V1-69
QVQLVQSGAEVKKPGSSVKVCKASGYTFTEYTMHWRQAPGQGL
EWIGGINPNNGGTSYKQKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARGGSHYYAMDYWGQGTTLTVSS (SEQ ID NO: 1479) 7E9V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYTFTEYTMHWRQMPPGKGLE
WIGGINPNNGGTSYKPSFQGGQVTISADKSISTAYLQWSSLKASDTA
MYCARGGSHYYAMDYWGQGTTLTVSS (SEQ ID NO: 1480) 7E9V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYTFTEYTMHWRQAPGKGLE
WIGGINPNNGGTSYKDSVKGRFTISRDNKNTLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGTTLTVSS (SEQ ID NO: 1481) 7E9V3-30
QVQLVESGGGVVQPGSLRLSCAASGYTFTEYTMHWRQAPGKGLE

WIGGINPNNGGTSYKDSVKGRFTISRDNASKNTLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGTLVTVSS (SEQ ID NO: 1482) 7E9V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYTFTEYTMEIWVRQAPGKGLE
WIGGINPNNGGTSYKDSVKGRFTISRDNASKNTLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGTLVTVSS (SEQ ID NO: 1483) 7E9V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYTFTEYTMEIWVRQAPGKGLE
WIGGINPNNGGTSYKDSVKGRFTISRDNASKNTLYLQMNSLRAEDTAV
YYCARGGSHYYAMDYWGQGTLVTVSS (SEQ ID NO: 1484) 7E9V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYTFTEYTMEIWIRQPPGKGLEWI
GGINPNNGGTSYKPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYC
ARGGSHYYAMDYWGQGTLVTVSS (SEQ ID NO: 1485) 7E9V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYTFTEYTMEIWVRQAPGKGLE
WIGGINPNNGGTSYKAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
YYCARGGSHYYAMDYWGQGTLVTVSS (SEQ ID NO: 1486) 7E9V4-39
QLQLQESGPGLVKPSETLSLTCTVSGYTFTEYTMEIWIRQPPGKGLEWI
GGINPNNGGTSYKPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYYC
ARGGSHYYAMDYWGQGTLVTVSS (SEQ ID NO: 1487) Antibody 8C3 8C3V1-46
QVQLVQSGAEVKKPGASVKVSCKASGYSTGYMEIWVRQAPGQGL
EWIGRVNPNGGTSYNQKFQGRVTMTRDTSTSTVYMELSSLRSEDTA
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1489) 8C3V5-51
EVQLVQSGAEVKKPGESLKISCKGSGYSTGYMEIWVRQMPGKGLE
WIGRVNPNGGTSYNPSFQGGVTISADKSISTAYLQWSSLKASDTAM
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1490) 8C3V3-23
EVQLLESGGGLVQPGGSLRLSCAASGYSTGYMEIWVRQAPGKGLE
WIGRVNPNGGTSYNDSVKGRFTISRDNASKNTLYLQMNSLRAEDTA
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1491) 8C3V1-69
QVQLVQSGAEVKKPGSSVKVSCKASGYSTGYMHWRQAPGQGL
EWIGRVNPNGGTSYNQKFQGRVTITADESTSTAYMELSSLRSEDTA
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1492) 8C3V3-30
QVQLVESGGGVVQPGRSLRLSCAASGYSTGYMHWRQAPGKGL
EWIGRVNPNGGTSYNDSVKGRFTISRDNASKNTLYLQMNSLRAEDTA
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1493) 8C3V3-48
EVQLVESGGGLVQPGGSLRLSCAASGYSTGYMEIWVRQAPGKGLE
WIGRVNPNGGTSYNDSVKGRFTISRDNASKNTLYLQMNSLRAEDTA
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1494) 8C3V3-7
EVQLVESGGGLVQPGGSLRLSCAASGYSTGYMEIWVRQAPGKGLE
WIGRVNPNGGTSYNDSVKGRFTISRDNASKNTLYLQMNSLRAEDTA
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1495) 8C3V4-59
QVQLQESGPGLVKPSETLSLTCTVSGYSTGYMEIWIRQPPGKGLEW
IGRVNPNGGTSYNPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYY
CVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1496) 8C3V3-15
EVQLVESGGGLVKPGGSLRLSCAASGYSTGYMEIWVRQAPGKGLE
WIGRVNPNGGTSYNAPVKGRFTISRDDSKNTLYLQMNSLKTEDTAV
VYYCVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1497) 8C3V4-39
QLQLQESGPGLVKPSETLSLTCTVSGYSTGYMEIWIRQPPGKGLEW
IGRVNPNGGTSYNPSLKSRTISVDTSKNQFSLKLSSVTAADTAVYY
CVLTGGYFDYWGQGTLVTVSS (SEQ ID NO: 1498)

[0332] In some embodiments, anti-TREM2 antibodies of the present disclosure comprise a light chain variable region of any one of the antibodies listed in TABLES D1-D6, or selected from 1A7, 3A2, 3B10, 6G12, 6H6, 7A9, 7B3, 8A1, 8E10, 8F11, 8F8, 9F5, 9F5v2, 9G1, 9G3, 10A9, 1, 11A8,

12E2, 12F9, 12G6, 2C7, 2F5, 3C1, 4D7, 4D11, 6C11, 6G12, 7A3, 7C5, 7E9, 7F6, 7G1, 7H1, 8C3, 8F10, 12A1, 1E9, 2C5, 3C5, 4C12, 4F2, 5A2, 6B3, 7D1, 7D9, 11D8, 8A12, 10E7, 10B11, 10D2, 7D5, 2A7, 3G12, 6H9, 8G9, 9B4, 10A1, 11A8, 12F3, 2F8, 10E3, 1H7, 2F6, 2H8, 3A7, 7E5, 7E5v2, 7F8, 11H5, 7C5, 4F11, 12D9, 1B4v1, 1B4V2, 6H2, 7B11v1, 7B11v2, 18D8, 18E4v1, 18E4v2, 29F6v1, 29F6v2, 40D5v1, 40D5v2, 43B9, 44A8v1, 44A8v2, 44B4v1, and 44B4v2; and/or a heavy chain variable region of any one of the antibodies listed in TABLES D1-D6, or selected from 1A7, 3A2, 3B10, 6G12, 6H6, 7A9, 7B3, 8A1, 8E10, 8F11, 8F8, 9F5, 9G1, 9G3, 10A9, 10C1, 11A8, 12E2, 12F9, 12G6, 2C7, 2F5, 3C1, 4D7, 4D11, 6C11, 6G12, 7A3, 7C5, 7E9, 7F6, 7G1, 7H1, 8C3, 8F10, 12A1, 1E9, 2C5, 3C5, 4C12, 4F2, 5A2, 6B3, 7D1, 7D9, 11D8, 8A12, 10E7, 10B11, 10D2, 7D5, 2A7, 3G12, 6H9, 8G9, 9B4, 10A1, 11A8, 12F3, 2F8, 10E3, 1H7, 2F6, 2H8, 3A7, 7E5, 7F8, 11H5, 7C5, 4F11, 12D9, 1B4v1, 1B4V2, 6H2, 7B11v1, 7B11v2, 18D8, 18E4v1, 18E4v2, 29F6v1, 29F6v2, 40D5v1, 40D5v2, 43B9, 44A8v1, 44A8v2, 44B4v1, and 44B4v2.

[0333] In some embodiments, the anti-TREM2 antibody is an anti-TREM2 monoclonal antibody selected from 1A7, 3A2, 3B10, 6G12, 6H6, 7A9, 7B3, 8A1, 8E10, 8F11, 8F8, 9F5, 9G1, 9G3, 10A9, 10C1, 11A8, 12E2, 12F9, 12G6, 2C7, 2F5, 3C1, 4D7, 4D11, 6C11, 6G12, 7A3, 7C5, 7E9, 7F6, 7G1, 7H1, 8C3, 8F10, 12A1, 1E9, 2C5, 3C5, 4C12, 4F2, 5A2, 6B3, 7D1, 7D9, 11D8, 8A12, 10E7, 10B11, 10D2, 7D5, 2A7, 3G12, 6H9, 8G9, 9B4, 10A1, 11A8, 12F3, 2F8, 10E3, 1H7, 2F6, 2H8, 3A7, 7E5, 7F8, 11H5, 7C5, 4F11, 12D9, 1B4v1, 1B4V2, 6H2, 7B11v1, 7B11v2, 18D8, 18E4v1, 18E4v2, 29F6v1, 29F6v2, 40D5v1, 40D5v2, 43B9, 44A8v1, 44A8v2, 44B4v1, and 44B4v2, and humanized variants thereof.

[0334] In some embodiments, each of the light chain variable regions disclosed in listed in TABLES D1-D6, or selected from 1A7, 3A2, 3B10, 6G12, 6H6, 7A9, 7B3, 8A1, 8E10, 8F11, 8F8, 9F5, 9F5v2, 9G1, 9G3, 10A9, 10C1, 11A8, 12E2, 12F9, 12G6, 2C7, 2F5, 3C1, 4D7, 4D11, 6C11, 6G12, 7A3, 7C5, 7E9, 7F6, 7G1, 7H1, 8C3, 8F10, 12A1, 1E9, 2C5, 3C5, 4C12, 4F2, 5A2, 6B3, 7D1, 7D9, 11D8, 8A12, 10E7, 10B11, 10D2, 7D5, 2A7, 3G12, 6H9, 8G9, 9B4, 10A1, 11A8, 12F3, 2F8, 10E3, 1H7, 2F6, 2H8, 3A7, 7E5, 7E5v2, 7F8, 11H5, 7C5, 4F11, 12D9, 1B4v1, 1B4V2, 6H2, 7B11v1, 7B 11v2, 18D8, 18E4v1, 18E4v2, 29F6v1, 29F6v2, 40D5v1, 40D5v2, 43B9, 44A8v1, 44A8v2, 44B4v1, and 44B4v2; and/or each of the heavy chain variable region of any one of the antibodies listed in TABLES D1-D6, or selected from 1A7, 3A2, 3B10, 6G12, 6H6, 7A9, 7B3, 8A1, 8E10, 8F11, 8F8, 9F5, 9G1, 9G3, 10A9, 10C1, 11A8, 12E2, 12F9, 12G6, 2C7, 2F5, 3C1, 4D7, 4D11, 6C11, 6G12, 7A3, 7C5, 7E9, 7F6, 7G1, 7H1, 8C3, 8F10, 12A1, 1E9, 2C5, 3C5, 4C12, 4F2, 5A2, 6B3, 7D1, 7D9, 11D8, 8A12, 10E7, 10B11, 10D2, 7D5, 2A7, 3G12, 6H9, 8G9, 9B4, 10A1, 11A8, 12F3, 2F8, 10E3, 1H7, 2F6, 2H8, 3A7, 7E5, 7F8, 11H5, 7C5, 4F11, 12D9, 1B4v1, 1B4V2, 6H2, 7B11v1, 7B11v2, 18D8, 18E4v1, 18E4v2, 29F6v1, 29F6v2, 40D5v1, 40D5v2, 43B9, 44A8v1, 44A8v2, 44B4v1, and 44B4v2 may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

E. PCT Patent Application Publication No. WO2019/028292A1

[0335] In some embodiments, the TREM2 agonist is an antibody, or antigen binding fragment thereof, as described in PCT Patent Application Publication No. WO2019/028292A1 (“the '292 application”), which is incorporated by reference herein, in its entirety.

[0336] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3 (also referred to as HVR-L1, HVR-L2, and HVR-L3, respectively), and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 (also referred to as HVR-H1, HVR-H2, and HVR-H3, respectively) disclosed in the '573 application specification. In some embodiments, the TREM2 binding agent

comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '573 application specification.

[0337] In some embodiments, anti-TREM2 antibodies of the present disclosure bind both human and cynomolgus monkey TREM2 with an affinity that is at least about 1-fold higher than an anti-TREM2 antibody selected from anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763 (e.g., antibody AL2p-h50); an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1810 (e.g., antibody AL2p-h77); and an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1826 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1827 (e.g., antibody AL2). In some embodiments, anti-TREM2 antibodies of the present disclosure bind to primary human immune cells with an affinity that is at least about 10 times higher than that of an anti-TREM2 antibody selected from an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1810; and an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1826 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1827. In some embodiments, anti-TREM2 antibodies of the present disclosure cluster and activate TREM2 signaling in an amount that is at least about 1-fold greater than that of an anti-TREM2 antibody selected from an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1810; and an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1826 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1827. In some embodiments, anti-TREM2 antibodies of the present disclosure increase immune cell survival in vitro that to an extent that is greater than an anti-TREM2 antibody selected from an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1810; and an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1826 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1827. In some embodiments, anti-TREM2 antibodies of the present disclosure may also have improved in vivo half-lives. In some embodiments, anti-TREM2 antibodies of the present disclosure may also decrease plasma levels of soluble TREM2 in vivo. In some embodiments, anti-TREM2 antibodies of the present disclosure may also decrease soluble TREM2. In some embodiments, the soluble TREM2 is decreased about any of 10, 20, 30, 40, 50 or 60%.

[0338] In some embodiments, the antibody binds to a TREM2 protein, wherein the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises: an HVR-H1 comprising the sequence according to Formula I: YAFX.sub.1X.sub.2X.sub.3WMN, wherein X.sub.1 is S or W, X.sub.2 is S, L, or R. and X.sub.3 is S, D, H, Q, or E (SEQ ID NO: 1828); an HVR-H2 comprising the sequence according to Formula II: RIYPGX.sub.1GX.sub.2TNYAX.sub.3KX.sub.4X.sub.5G, wherein X.sub.1 is D, G, E, Q, or V,

X.sub.2 is D or Q, X.sub.3 is Q, R, H, W, Y, or G, X.sub.4 is F, R, or W, and X.sub.5 is Q, R, K, or H (SEQ ID NO: 1829); and an HVR-H3 comprising the sequence according to Formula III: ARLLRNX.sub.1PGX.sub.2SYAX.sub.3DY, wherein X.sub.1 is Q or K, X.sub.2 is E, S, or A, and X.sub.3 is M or H (SEQ ID NO: 1830), and wherein the antibody is not an antibody comprising a heavy chain variable region comprising an HVR-H1 comprising the sequence of YAFSSSWMN (SEQ ID NO:1831), an HVR-H2 comprising the sequence of RIYPGDGDTNYAQKFQG (SEQ ID NO: 1832), and an HVR-H3 comprising the sequence of ARLLRNQPGESYAMDY (SEQ ID NO: 1833). In some embodiments, the TREM2 agonist is an antibody that binds to a TREM2 protein, wherein the antibody comprises a heavy chain variable region and a light chain variable region, wherein the light chain variable region comprises: an HVR-L1 comprising the sequence according to Formula IV: RX.sub.1SX.sub.2SLX.sub.3HSNX.sub.4YTYLH, wherein X.sub.1 is S or T, X.sub.2 is Q, R, or S, X.sub.3 is V or I, and X.sub.4 is G, R, W, Q, or A (SEQ ID NO: 1834); an HVR-L2 comprising the sequence according to Formula V: KVSNRX.sub.1S, wherein X is F, R, V, or K (SEQ ID NO:1835); and an HVR-L3 comprising the sequence according to Formula V: SQSTRVPYT (SEQ ID NO: 1836), and wherein the antibody is not an antibody comprising a light chain variable region comprising an HVR-L1 comprising the sequence of RSSQSLVHSNGYTYLH (SEQ ID NO: 1837), an HVR-L2 comprising the sequence of KVSNRFS (SEQ ID NO: 1838), and an HVR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises: an HVR-H1 comprising the sequence according to Formula I: YAFX.sub.1X.sub.2X.sub.3WMN, wherein X.sub.1 is S or W, X.sub.2 is S, L, or R, and X.sub.3 is S, D, H, Q, or E (SEQ ID NO:1828); an HVR-H2 comprising the sequence according to Formula II: RIYPGX.sub.1GX.sub.2TNYAX.sub.3KX.sub.4X.sub.5G, wherein X.sub.1 is D, G, E, Q, or V, X.sub.2 is D or Q, X.sub.3 is Q, R, H, W, Y, or G, X.sub.4 is F, R, or W, and X.sub.5 is Q, R, K, or H (SEQ ID NO:1829); and an HVR-H3 comprising the sequence according to Formula III: ARLLRNX.sub.1PGX.sub.2SYAX.sub.3DY, wherein X.sub.1 is Q or K, X.sub.2 is E, S, or A, and X.sub.3 is M or H (SEQ ID NO:1830), and the light chain variable region comprises: an HVR-L1 comprising the sequence according to Formula IV: RX.sub.1SX.sub.2SLX.sub.3HSNX.sub.4YTYLH, wherein X.sub.1 is S or T, X.sub.2 is Q, R, or S, X.sub.3 is V or I, and X.sub.4 is G, R, W, Q, or A (SEQ ID NO:1834); an HVR-L2 comprising the sequence according to Formula V: KVSNRX.sub.1S, wherein X.sub.1 is F, R, V, or K (SEQ ID NO: 1835); and an HVR-L3 comprising the sequence: SQSTRVPYT (SEQ ID NO: 1836), and wherein the antibody is not an antibody comprising a heavy chain variable region comprising an HVR-H1 comprising the sequence of YAFSSSWMN (SEQ ID NO:1831), an HVR-H2 comprising the sequence of RIYPGDGDTNYAQKFQG (SEQ ID NO: 1832), and an HVR-H3 comprising the sequence of ARLLRNQPGESYAMDY (SEQ ID NO:1833), and comprising a light chain variable region comprising an HVR-L1 comprising the sequence of RSSQSLVHSNGYTYLH (SEQ ID NO: 1837), an HVR-L2 comprising the sequence of KVSNRFS (SEQ ID NO: 1838), and an HVR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836).

[0339] In some embodiments, the antibody binds to a TREM2 protein, wherein the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises: an HVR-H1 comprising a sequence selected from the group consisting of SEQ ID NOS:1839 and 1843; an HVR-H2 comprising a sequence selected from the group consisting of SEQ ID NOS:1840, 1842, 1844, and 1848; and an HVR-H3 comprising a sequence selected from the group consisting of SEQ ID NOS: 1833 and 1845; and/or the light chain variable region comprises: an HVR-L1 comprising a sequence selected from the group consisting of 1837, 1846, 1849, and 1851; an HVR-L2 comprising a sequence selected from the group consisting of SEQ ID NOS: 1838, 1841, and 1847; and an HVR-L3 comprising the sequence of SEQ ID NO: 1836. In some embodiments, the antibody comprises a heavy chain variable region

and a light chain variable region, wherein the heavy chain variable region comprises: an HVR-H1 comprising the sequence of SEQ ID NO: 1839; an HVR-H2 comprising a sequence selected from the group consisting of SEQ ID NOS: 1840, 1842, and 1848; and an HVR-H3 comprising the sequence of SEQ ID NO: 1833; and/or the light chain variable region comprises: an HVR-L1 comprising a sequence selected from the group consisting of 1837, 1849, and 1851; an HVR-L2 comprising a sequence selected from the group consisting of SEQ ID NOS: 1838 and 1841; and an HVR-L3 comprising the sequence of SEQ ID NO: 1836.

[0340] In some embodiments, the antibody binds to a TREM2 protein, wherein the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises the HVR-H1, HVR-H2, and HVR-H3 of antibody AL2p-2, AL2p-3, AL2p-4, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLES E1-E3). In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the light chain variable region comprises the HVR-L1, HVR-L2, and HVR-L3 of antibody AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLES E4-E6). In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises the HVR-H1, HVR-H2, and HVR-H3 of antibody AL2p-2, AL2p-3, AL2p-4, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLES E1-E3); and the light chain variable region comprises the HVR-L1, HVR-L2, and HVR-L3 of antibody AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLES E4-E6). In some embodiments, the antibody comprises a heavy chain variable region comprising an HVR-H1, HVR-H2, and HVR-H3 and a light chain variable region comprising an HVR-L1, HVR-L2, and HVR-L3, wherein the antibody comprises the HVR-H1, HVR-H2, HVR-H3, HVR-L1, HVR-L2, and HVR-L3 of antibody AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLES E1-E3 and TABLES E4-E6).

[0341] In some embodiments, the heavy chain variable region comprises one, two, three or four frame work regions selected from VH FR1, VH FR2, VH FR3, and VH FR4, wherein: the VH FR1 comprises a sequence selected from the group consisting of SEQ ID NOS: 1716-1718, the VH FR2 comprises a sequence selected from the group consisting of SEQ ID NOS: 1719 and 1720, the VH FR3 comprises a sequence selected from the group consisting of SEQ ID NOS: 1721 and 1722, and the VH FR4 comprises the sequence of SEQ ID NO: 1723; and/or the light chain variable region comprises one, two, three or four frame work regions selected from VL FR1, VL FR2, VL FR3, and VL FR4, wherein: the VL FR1 comprises a sequence selected from the group consisting of SEQ ID NOS: 1724-1727, the VL FR2 comprises a sequence selected from the group consisting of SEQ ID NOS: 1728 and 1729, the VL FR3 comprises a sequence selected from the group consisting of SEQ ID NOS: 1730 and 1731, and the VL FR4 comprises a sequence selected from the group consisting of SEQ ID NOS: 1732 and 1733. In some embodiments, the antibody comprises a heavy chain variable region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1734-1777 and 1798; and/or a light chain variable region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1799-1820 and 1825. In some embodiments, the antibody comprises the heavy chain variable region of antibody AL2p-h50, AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLE E15); and/or the antibody comprises the light chain variable region of antibody AL2p-h50, AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLE E17). In some embodiments: (a) the HVR-H1 comprises the amino acid sequence YAFSSQWMN (SEQ ID NO: 1839), the HVR-H2 comprises the amino acid sequence RIYPGGGDTNYARKFQG (SEQ ID NO: 1840), the HVR-H3 comprises the amino acid sequence ARLLRNQPGESYAMDY (SEQ ID NO: 1833), the HVR-L1 comprises the amino acid sequence RSSQSLVHSNGYTYLH (SEQ ID NO: 1837), the HVR-L2 comprises the amino acid sequence KVSNNRRS (SEQ ID NO: 1841), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836); (b) the HVR-H1 comprises the amino acid sequence YAFSSQWMN (SEQ ID NO: 1839), the HVR-H2 comprises the amino acid sequence RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842), the HVR-H3 comprises the amino acid sequence ARLLRNQPGESYAMDY (SEQ ID NO: 1833), the HVR-L1 comprises the amino acid sequence RSSQSLVHSNGYTYLH (SEQ ID NO: 1837), the HVR-L2 comprises the amino acid sequence KVSNNRFS (SEQ ID NO: 1838), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836); (c) the HVR-H1 comprises the amino acid sequence YAFSSDWMN (SEQ ID NO: 1843), the HVR-H2 comprises the amino acid sequence RIYPGEGDTNYARKFHG (SEQ ID NO: 1844) the HVR-H3 comprises the amino acid sequence ARLLRNKPGESYAMDY (SEQ ID NO: 1845) the HVR-L1 comprises the amino acid sequence RTSQSLVHSNAYTYLH (SEQ ID NO: 1846), the HVR-L2 comprises the amino acid sequence KVSNNRVS (SEQ ID NO: 1847), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836); (d) the HVR-H1 comprises the amino acid sequence YAFSSQWMN (SEQ ID NO: 1839), the HVR-H2 comprises the amino acid sequence RIYPGEGDTNYARKFQG (SEQ ID NO: 1848), the HVR-H3 comprises the amino acid sequence

[illegible]

KVSNRFS (SEQ ID NO: 1838), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the HVR-H1 comprises the amino acid sequence YAFSSQWMN (SEQ ID NO: 1839), the HVR-H2 comprises the amino acid sequence RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842), the HVR-H3 comprises the amino acid sequence ARLLRNQPGESYAMDY (SEQ ID NO: 1833), the HVR-L1 comprises the amino acid sequence RSSQSLVHSNRYTYLH (SEQ ID NO: 1851), the HVR-L2 comprises the amino acid sequence KVSNNRFS (SEQ ID NO: 1838), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the HVR-H1 comprises the amino acid sequence YAFSSQWMN (SEQ ID NO: 1839), the HVR-H2 comprises the amino acid sequence RIYPGGGDTNYARKFQG (SEQ ID NO: 1840), the HVR-H3 comprises the amino acid sequence ARLLRNQPGESYAMDY (SEQ ID NO: 1833), the HVR-L1 comprises the amino acid sequence RSSQSLVHSNRYTYLH (SEQ ID NO: 1851), the HVR-L2 comprises the amino acid sequence KVSNNRFS (SEQ ID NO: 1841), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the HVR-H1 comprises the amino acid sequence YAFSSQWMN (SEQ ID NO: 1839), the HVR-H2 comprises the amino acid sequence RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842), the HVR-H3 comprises the amino acid sequence ARLLRNQPGESYAMDY (SEQ ID NO: 1833), the HVR-L1 comprises the amino acid sequence RSSQSLVHSNRYTYLH (SEQ ID NO: 1851), the HVR-L2 comprises the amino acid sequence KVSNNRFS (SEQ ID NO: 1838), and the HVR-L3 comprises the amino acid sequence SQSTRVPYT (SEQ ID NO: 1836).

[0342] In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises Kabat CDRs; and/or the light chain variable region comprises Kabat CDRs. In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO: 1901), a CDR-H2 comprising the sequence of RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO: 1902). In some embodiments, the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNGYTYLH (SEQ ID NO: 1837), a CDR-L2 comprising the sequence of KVSNNRFS (SEQ ID NO: 1838); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO: 1901), a CDR-H2 comprising the sequence of RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO: 1902); and the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNGYTYLH (SEQ ID NO: 1837), a CDR-L2 comprising the sequence of KVSNNRFS (SEQ ID NO: 1838); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836).

[0343] In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises Kabat CDRs; and/or the light chain variable region comprises Kabat CDRs. In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SDWMN (SEQ ID NO: 1903), a CDR-H2 comprising the sequence of RIYPGEGDTNYARKFHG (SEQ ID NO: 1844); and a CDR-H3 comprising the sequence of LLRNKPGESYAMDY (SEQ ID NO: 1904). In some embodiments, the light chain variable region comprises a CDR-L1 comprising the sequence of RTSQSLVHSNAYTYLH (SEQ ID NO: 1846), a CDR-L2 comprising the sequence of KVSNNRVS (SEQ ID NO: 1847); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SDWMN (SEQ ID NO: 1903), a CDR-H2 comprising the sequence of RIYPGEGDTNYARKFHG (SEQ ID NO: 1844); and a CDR-H3 comprising the sequence of LLRNKPGESYAMDY (SEQ ID NO: 1904); and the light chain variable region comprises a CDR-L1 comprising the sequence of RTSQSLVHSNAYTYLH (SEQ ID NO: 1846), a CDR-L2

comprising the sequence of KVSNRVS (SEQ ID NO: 1847); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836).

[0344] In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises Kabat CDRs; and/or the light chain variable region comprises Kabat CDRs. In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO:1901), a CDR-H2 comprising the sequence of RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO:1902). In some embodiments, the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNRYTYLH (SEQ ID NO:1851), a CDR-L2 comprising the sequence of KVSNRFS (SEQ ID NO:1838); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO: 1901), a CDR-H2 comprising the sequence of RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842); and a Kabat CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO: 1902); and the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNRYTYLH (SEQ ID NO:1851), a CDR-L2 comprising the sequence of KVSNRFS (SEQ ID NO: 1838); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836).

[0345] In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises Kabat CDRs; and/or the light chain variable region comprises Kabat CDRs. In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO:1901), a CDR-H2 comprising the sequence of RIYPGGGDTNYARKKFQG (SEQ ID NO: 1840); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO:1902). In some embodiments, the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNRYTYLH (SEQ ID NO:1851), a CDR-L2 comprising the sequence of KVSNRRS (SEQ ID NO:1841); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO: 1901), a CDR-H2 comprising the sequence of RIYPGGGDTNYARKKFQG (SEQ ID NO: 1840); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO: 1902); and the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNRYTYLH (SEQ ID NO:1851), a CDR-L2 comprising the sequence of KVSNRRS (SEQ ID NO:1841); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836).

[0346] In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region comprises Kabat CDRs; and/or the light chain variable region comprises Kabat CDRs. In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO:1901), a CDR-H2 comprising the sequence of RIYPGEGDTNYARKKFQG (SEQ ID NO: 1848); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO:1902). In some embodiments, the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNQYTYLH (SEQ ID NO:1849), a CDR-L2 comprising the sequence of KVSNRRS (SEQ ID NO:1841); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836). In some embodiments, the heavy chain variable region comprises a CDR-H1 comprising the sequence of SQWMN (SEQ ID NO: 1901), a CDR-H2 comprising the sequence of RIYPGEGDTNYARKKFQG (SEQ ID NO:1848); and a CDR-H3 comprising the sequence of LLRNQPGESYAMDY (SEQ ID NO: 1902); and the light chain variable region comprises a CDR-L1 comprising the sequence of RSSQSLVHSNQYTYLH (SEQ ID NO:1849), a CDR-L2 comprising the sequence of KVSNRRS (SEQ ID NO:1841); and a CDR-L3 comprising the sequence of SQSTRVPYT (SEQ ID NO: 1836).

[0347] In some embodiments, the antibody comprises a heavy chain variable region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1734-1778 and 1798; and/or a light chain variable region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1799-1820 and 1825. In some embodiments, the antibody comprises the heavy chain variable region of antibody AL2p-h50, AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLE E15); and/or the antibody comprises the light chain variable region of antibody AL2p-h50, AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, or AL2p-62 (as shown in TABLE E17). In some embodiments: (a) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1760, and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1804; (b) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO:1766; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1811; (c) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO:1771; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1815; (d) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1777; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1817; (e) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1778; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1818; (f) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1766; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1819; or (g) the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1760; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1820. In some embodiments, the antibody comprises an Fc region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1853-1863. In some embodiments, the antibody comprises an Fe region comprising the amino acid sequence of SEQ ID NO:1853. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1854. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1855. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1856. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1857. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1858. In some embodiments, the antibody comprises an Fe region comprising the amino acid sequence of SEQ ID NO: 1859. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1860. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1861. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1862. In some embodiments, the antibody comprises an Fc region comprising the amino acid sequence of SEQ ID NO: 1863. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1905-

1920; and/or a light chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1921-1925. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1905 and 1906; and a light chain comprising the amino acid sequence of SEQ ID NO:1921. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1907 and 1908; and a light chain comprising the amino acid sequence of SEQ ID NO:1921. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1909 and 1910; and a light chain comprising the amino acid sequence of SEQ ID NO: 1922. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1911 and 1912; and a light chain comprising the amino acid sequence of SEQ ID NO:1922. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1913 and 1914; and a light chain comprising the amino acid sequence of SEQ ID NO: 1923. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1915 and 1916; and a light chain comprising the amino acid sequence of SEQ ID NO: 1925. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1917 and 1918; and a light chain comprising the amino acid sequence of SEQ ID NO:1925. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1919 and 1920; and a light chain comprising the amino acid sequence of SEQ ID NO: 1924.

[0348] In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1760, and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1804. In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1766; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1811. In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO:1771; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1815. In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1777; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1817. In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1778; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1718. In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1766; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1819. In some embodiments, the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1760; and/or the light chain variable region comprises the amino acid sequence of SEQ ID NO:1820.

[0349] In some embodiments, the antibody comprises a heavy chain variable region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1734, 1763 and 1779-1797; and/or a light chain variable region comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1799, 1811, and 1821-1824. In some embodiments, the antibody comprises the heavy chain variable region of antibody AL2p-h19, AL2p-h21, AL2p-h22, AL2p-h23, AL2p-h24, AL2p-h25, AL2p-h26, AL2p-h27, AL2p-h28, AL2p-h29, AL2p-h30, AL2p-h31, AL2p-h32, AL2p-h33, AL2p-h34, AL2p-h35, AL2p-h36, AL2p-h42, AL2p-h43, AL2p-h44, AL2p-h47, AL2p-h59, AL2p-h76, or AL2p-h90 (as shown in TABLE E15); and/or the antibody comprises the light chain variable region of antibody AL2p-h19, AL2p-h21, AL2p-h22, AL2p-h23, AL2p-h24, AL2p-h25, AL2p-h26, AL2p-h27, AL2p-h28, AL2p-h29, AL2p-h30, AL2p-h31, AL2p-h32, AL2p-h33, AL2p-h34, AL2p-h35, AL2p-h36, AL2p-h42, AL2p-h43, AL2p-h44, AL2p-h47, AL2p-h59, AL2p-h76, or AL2p-h90 (as shown in TABLE E17).

[0350] In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1905-1920; and/or a light chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1921-1925. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1905 and 1906; and a light chain comprising the amino acid sequence of SEQ ID NO:1921. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1907 and 1908; and a light chain comprising the amino acid sequence of SEQ ID NO:1921. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1909 and 1910; and a light chain comprising the amino acid sequence of SEQ ID NO:1922. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1911 and 1912; and a light chain comprising the amino acid sequence of SEQ ID NO: 1922. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1913 and 1914; and a light chain comprising the amino acid sequence of SEQ ID NO: 1923. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1915 and 1916; and a light chain comprising the amino acid sequence of SEQ ID NO:1925. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS: 1917 and 1918; and a light chain comprising the amino acid sequence of SEQ ID NO: 1925. In some embodiments, the antibody comprises a heavy chain comprising an amino acid sequence selected from the group consisting of SEQ ID NOS:1919 and 1920; and a light chain comprising the amino acid sequence of SEQ ID NO: 1924.

[0351] In some embodiments that may be combined with any of the preceding embodiments, the antibody is a bispecific antibody recognizing a first antigen and a second antigen, wherein the first antigen is human TREM2 or a naturally occurring variant thereof, and the second antigen is: (a) an antigen facilitating transport across the blood-brain-barrier; (b) an antigen facilitating transport across the blood-brain-barrier selected from the group consisting of transferrin receptor (TR), insulin receptor (HIR), insulin-like growth factor receptor (IGFR), low-density lipoprotein receptor related proteins 1 and 2 (LPR-1 and 2), diphtheria toxin receptor, CRM197, a llama single domain antibody, TMEM 30(A), a protein transduction domain, TAT, Syn-B, penetratin, a poly-arginine peptide, an angiopeptide, and ANG1005; (c) a disease-causing agent selected from the group consisting of disease-causing peptides or proteins or, disease-causing nucleic acids, wherein the disease-causing nucleic acids are antisense GGCCC (G2C4) repeat-expansion RNA, the disease-causing proteins are selected from the group consisting of amyloid beta, oligomeric amyloid beta, amyloid beta plaques, amyloid precursor protein or fragments thereof, Tau, TAPP, alpha-synuclein, TDP-43, FUS protein, C9orf72 (chromosome 9 open reading frame 72), c9RAN protein, prion protein, PrPSc, huntingtin, calcitonin, superoxide dismutase, ataxin, ataxin 1, ataxin 2, ataxin 3, ataxin 7, ataxin 8, ataxin 10, Lewy body, atrial natriuretic factor, islet amyloid polypeptide, insulin, apolipoprotein A1, serum amyloid A, medin, prolactin, transthyretin, lysozyme, beta 2 microglobulin, gelsolin, keratoepithelin, cystatin, immunoglobulin light chain AL, S-IBM protein, Repeat-associated non-ATG (RAN) translation products, DiPeptide repeat (DPR) peptides, glycine-alanine (GA) repeat peptides, glycine-proline (GP) repeat peptides, glycine-arginine (GR) repeat peptides, proline-alanine (PA) repeat peptides, ubiquitin, and proline-arginine (PR) repeat peptides; (d) ligands and/or proteins expressed on immune cells, wherein the ligands and/or proteins selected from the group consisting of CD40, OX40, ICOS, CD28, CD137/4-1BB, CD27, GITR, PD-L1, CTLA-4, PD-L2, PD-1, B7-H3, B7-H4, HVEM, BTLA, KIR, GAL9, TIM3, A2AR, LAG-3, and phosphatidylserine; and (e) a protein, lipid, polysaccharide, or glycolipid expressed on one or more tumor cells. In some embodiments, the antibody binds specifically to both human TREM2 and

cynomolgus monkey TREM2. In some embodiments, the antibody has a dissociation constant (K_{sub}.D) for human TREM2 and/or cynomolgus monkey TREM2 that is at least 1-fold lower than an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; or at least 1-fold lower than an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO:1810. In some embodiments, the antibody has a dissociation constant (K_{sub}.D) for human TREM2 that ranges from about 9 μ M to about 100 μ M, or less than 100 μ M, wherein the K_{sub}.D is determined at a temperature of approximately 25° C. In some embodiments, the antibody has a dissociation constant (K_{sub}.D) for cynomolgus monkey TREM2 that ranges from about 50 nM to about 100 μ M, or less than 100 μ M, wherein the K_{sub}.D is determined at a temperature of approximately 25° C. In some embodiments, the antibody binds to primary human immune cells with an affinity that is at least 10 times higher than that of an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; or at least 10 times higher than an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO:1810. In some embodiments, the antibody clusters and activates TREM2 signaling in an amount that is at least 1-fold greater than that of an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; or at least 1-fold greater than an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO:1810. In some embodiments, the antibody increases immune cell survival in vitro that to an extent that is greater than an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:1734 and a light chain variable region comprising the amino acid sequence of SEQ ID NO: 1763; or that is greater than an anti-TREM2 antibody comprising a heavy chain variable region comprising the amino acid sequence of SEQ ID NO:1798 and a light chain variable region comprising the amino acid sequence of SEQ ID NO:1810. In some embodiments, the antibody has an in vivo half-life that is lower than a human control IgG1 antibody. In some embodiments, the antibody decreases plasma levels of soluble TREM2 in vivo by an amount that is at least 25% greater than that of a human control IgG1 antibody. In some embodiments, the antibody decreases plasma levels of soluble TREM2 in vivo by blocking cleavage, by inhibiting one or more metalloproteases, and/or by inducing internalization. In some embodiments, soluble TREM2 is decreased by about any of 10, 20, 30, 40, or 50%. In some embodiments, the antibody competes with one or more antibodies selected from the group consisting of AL2p-h50, AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, AL2p-62, AL2p-h19, AL2p-h21, AL2p-h22, AL2p-h23, AL2p-h24, AL2p-h25, AL2p-h26, AL2p-h27, AL2p-h28, AL2p-h29, AL2p-h30, AL2p-h31, AL2p-h32, AL2p-h33, AL2p-h34, AL2p-h35, AL2p-h36, AL2p-h42, AL2p-h43, AL2p-h44, AL2p-h47, AL2p-1159, AL2p-h76, AL2p-h90, and any combination thereof for binding to TREM2. In some embodiments, the antibody binds essentially the same TREM2 epitope as an antibody selected from the group consisting of: AL2p-h50, AL2p-2, AL2p-3, AL2p-4, AL2p-5, AL2p-6, AL2p-7, AL2p-8, AL2p-9, AL2p-10, AL2p-11, AL2p-12, AL2p-13, AL2p-14, AL2p-15,

AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, AL2p-61, AL2p-62, AL2p-h19, AL2p-h21, AL2p-h22, AL2p-h23, AL2p-h24, AL2p-h25, AL2p-h26, AL2p-h27, AL2p-h28, AL2p-h29, AL2p-h30, AL2p-h31, AL2p-h32, AL2p-h33, AL2p-h34, AL2p-h35, AL2p-h36, AL2p-h42, AL2p-h43, AL2p-h44, AL2p-h47, AL2p-h59, AL2p-h76, and AL2p-h90. In some embodiments, the antibody binds to one or more amino acids within amino acid residues 149-157 of SEQ ID NO: 1. In some embodiments, the antibody binds to one or more amino acid residues selected from the group consisting of E151, D152, and E156 of SEQ ID NO: 1.

[0352] In some embodiments, the antibody is an antibody disclosed in Tables 2A, 2B, 2C, 3A, 3B, 3C, 4A-4D, 5A-5D, 6A, 6B, 7A or 7B of PCT Patent Application Publication No.

WO2019/028292A1, reproduced below as TABLES E1-E18.

TABLE-US-00030 TABLE E1 Heavy chain HVR H1 sequences of anti-TREM2 antibodies Ab HVR H1 AL2p-h50, AL2p-2, AL2p-3, YAFSSSWMN AL2p-4, AL2p-5, AL2p-6, (SEQ ID AL2p-33, AL2p-h77, and NO: 1831) AL2p-36 AL2p-29, AL2p-30, AL2p-31, YAFSSQWMN AL2p-37, AL2p-58, AL2p-60, (SEQ ID AL2p-61, and AL2p-62 NO: 1839) AL2p-10, AL2p-11, AL2p-45, YAFSSDWMN AL2p-46, AL2p-47, AL2p-48, (SEQ ID and AL2p-49 NO: 1843) AL2p-7 and AL2p-8 YAFSLSWMN (SEQ ID NO: 1864) AL2p-9 YAFSRSWMN (SEQ ID NO: 1865) AL2p-12, AL2p-13, AL2p-14, YAFSSHWMN AL2p-15, AL2p-16, AL2p-17, (SEQ ID AL2p-18, AL2p-19, AL2p-20, NO: 1866) AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26 AL2p-27, AL2p-28, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, and AL2p-59 AL2p-32 YAFSSEWMN (SEQ ID NO: 1867) AL2P-35 YAFWSSWMN (SEQ ID NO: 1868) Formula I

YAFX.sub.1X.sub.2X.sub.3WMN X.sub.1 is S or W X.sub.2 is S, L, or R X.sub.3 is S, D, H, Q, or E (SEQ ID NO: 1828)

TABLE-US-00031 TABLE E2 Heavy chain HVR H2 sequences of anti-TREM2 antibodies Ab HVR H2 AL2p-h50, AL2p-5, RIYPGDGDTNYAQKFQG AL2p-6, AL2p-9, (SEQ ID NO: 1832) AL2p-10, AL2p-14, AL2p-15, AL2p-29, AL2p-32, AL2p-33, AL2p-h77, and AL2p-35 AL2p-31 and AL2p-60 RIYPGGGDTNYARKFQG (SEQ ID NO: 1840) AL2p-37 and AL2p-58 RIYPGGGDTNYAGKFQG (SEQ ID NO: 1842) AL2p47, AL2p-48, RIYPGEGDTNYARKFHG AL2p-49 (SEQ ID NO: 1844) AL2p-45, AL2p46, RIYPGEGDTNYARKFQG and AL2p-61 (SEQ ID NO: 1848) AL2p-62 RIYPGEGDTNYAGKFQG (SEQ ID NO: 1850) AL2p-2 and AL2p-24 RIYPGGGDTNYAQKFQG (SEQ ID NO: 1869) AL2p-3 RIYPGEGDTNYAQKFQG (SEQ ID NO: 1870) AL2p-4 and AL2p-27 RIYPGQGDNTNYAQKFQG (SEQ ID NO: 1871) AL2p-7 and AL2p-16 RIYPGDGDTNYAQKFRG (SEQ ID NO: 1872) AL2p-8, AL2p-11, RIYPGDGDTNYARKFQG AL2p-19, AL2p-20, (SEQ ID NO: 1873) and AL2p-36 AL2p-12 RIYPGDGDTNYAHKFQG (SEQ ID NO: 1874) AL2p-13 RIYPGDGDTNYAQKFKG (SEQ ID NO: 1875) AL2p-17 RIYPGDGDTNYAQKRQG (SEQ ID NO: 1876) AL2p-18 RIYPGDGDTNYAQKWQG (SEQ ID NO: 1877) AL2p-21 and AL2p-30 RIYPGDGDTNYAWKFQG (SEQ ID NO: 1878) AL2p-22 RIYPGDGDTNYAYKFQG (SEQ ID NO: 1879) AL2p-23 RIYPGDGQDTNYAQKRQG (SEQ ID NO: 1880) AL2p-25, AL2p-38, RIYPGGGDTNYAQKFRG AL2p-39, and AL2p-40 (SEQ ID NO: 1881) AL2p-26 RIYPGGGDTNYAQKRQG (SEQ ID NO: 1882) AL2p-28 RIYPGVGDTNYAQKFQG (SEQ ID NO: 1883) AL2p-41 and AL2p-42 RIYPGEGDTNYAQKFRG (SEQ ID NO: 1884) AL2p-43 and AL2p44

RIYPGGDTNYARKFRG (SEQ ID NO: 1885) AL2p-50, AL2p-51, RIYPGEGDTNYAQKFHG AL2p-52, AL2p-53, (SEQ ID NO: 1886) AL2p-54, AL2p-55, AL2p-56, and AL2p-57 AL2p-59 RIYPGEGQTNYAQKRQG (SEQ ID NO: 1887) Formula II RIYPGX1GX2TNYAX.sub.3KX.sub.4 X.sub.5G X.sub.1 is D, G, E, Q, or V X.sub.2 is D or Q X.sub.1 is Q, R, H, W, Y, or G X.sub.4 is F, R, or W X.sub.5 is Q, R, K, or H (SEQ ID NO: 1829)

TABLE-US-00032 TABLE E3 Heavy chain HVR H3 sequences of anti-TREM2 antibodies Ab HVR 113 AL2p-h50, AL2p-2, AL2p-3, ARLLRNQPGESYAMDY AL2p-4, AL2p-5, AL2p-6, (SEQ ID NO: AL2p-7, AL2p-10, AL2p-11, 1833) AL2p-12, AL2p-13, AL2p-14. AL2p-15,, AL2p-17, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30. AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-37, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-58, AL2p-59. AL2p-60, AL2p-61, and AL2p-62 AL2p-45, AL2p-46, AL2p-47, ARLLRNKPGESYAMDY AL2p-48, AL2p-49, AL2p-54, (SEQ ID NO: AL2p-55, AL2p-56, and 1845) AL2p-57 AL2p-8 and AL2p-18 ARLLRNQPGSSYAMDY (SEQ ID NO: 1888) AL2p-9, AL2p-16, AL2p-36, ARLLRNQPGASYAMDY AL2p-38, AL2p-39, AL2p-40, (SEQ ID NO: AL2p-41, AL2p-42, AL2p-43, 1889) and AL2p-44 AL2p-35 ARLLRNQPGESYAHDY (SEQ ID NO: 1890) Formula III ARLLRNX.sub.1PGX.sub.2 SYAX.sub.3DY X.sub.1 is Q or K X.sub.2 is E, S, or A X.sub.3 is M or H (SEQ ID NO: 1830)

TABLE-US-00033 TABLE E4 Light chain HVR LI sequences of anti-TREM2 antibodies Ab HVR L1 AL2p-h50, AL2p-2, AL2p-3, RSSQSLVHSNGYTYLH AL2p-4, AL2p-10, AL2p-12, (SEQ ID NO: AL2p-31, AL2p-32, AL2p-h77, 1837) AL2p-35, AL2p-36. and AL2p-37 AL2p-45, AL2p-47, AL2p-50. RTSQSLVHSNAYTYLH AL2p-52, AL2p-55, and (SEQ ID NO: AL2p-56 1846) AL2p-61 and AL2p-62 RSSQSLVHSNQYTYLH (SEQ ID NO: 1849 AL2p-5, AL2p-58, and RSSQSLVHSNRYTYLH AL2p-60 (SEQ ID NO: 1851) AL2p-6 RSSQSLVHSNWTYTYLH (SEQ ID NO: 1891) AL2p-7, AL2p-8, AL2p-13, RSSQSLIHSNGYTYLH and AL2p-26 (SEQ ID NO: 1892) AL2p-9, AL2p-16, AL2p-18. RTSQSLVHSNGYTYLH AL2p-20, AL2p-23, AL2p-25, (SEQ ID NO: AL2p-28, and AL2p-33 1893) AL2p-11, AL2p-14, AL2p-17, RSSRSLVHSNGYTYLH AL2p-19, AL2p-22, AL2p-24, (SEQ ID NO: AL2p-27. and AL2p-29 1894) AL2p-15, AL2p-21, and RSSSSLVHSNGYTYLH AL2p-30 (SEQ ID NO: 1895) AL2p-38 and AL2p-43 RSSRSLVHSNRYTYLH (SEQ ID NO: 1896) AL2p-39 and AL2p-41 RSSRSLVHSNQYTYLH (SEQ ID NO: 1897) AL2p-40, AL2p-42, and RTSRSLVHSNRYTYLH AL2p-44 (SEQ ID NO: 1898) AL2p-46, AL2p-48, AL2p-49, RTSQSLVHSNQYTYLH AL2p-51, AL2p-53, AL2p-54, (SEQ ID NO: AL2p-57, and AL2p-59 1899) Formula IV RX.sub.1SX.sub.2SLX.sub.3HSNX.sub.4 YTYLH X.sub.1 is S or T X.sub.2 is Q, R, or S X.sub.3 is V or I X.sub.4 is G, R, W, Q or A (SEQ ID NO: 1834)

TABLE-US-00034 TABLE E5 Light chain HVR L2 sequences of anti-TREM2 antibodies Ab HVR L2 AL2p-h50, AL2p-2, AL2p-3, KVSNRFS AL2p-4, AL2p-5, AL2p-6, (SEQ ID AL2p-14, AL2p-24, AL2p-29, NO: 1838) AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-58, and AL2p-62 AL2p-7, AL2p 8, AL2p 10, KVSNNRRS AL2p 12, AL2p-13, AL2p 22, (SEQ ID AL2p-26, AL2p-31, AL2p-32, NO: 1841) AL2p38, AL2p39. AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-60, and AL2p-61 AL2p-9, AL2p-11, AL2p-16, KVSNNRVS AL2p-17, AL2p-18, AL2p-19, (SEQ ID AL2p-20, AL2p-23, AL2p-25, NO: 1847) AL2p-27, AL2p-28, AL2p-33, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, and AL2p-59

AL2p-15, AL2p-21, and KVSNRKS AL2p-30 (SEQ ID NO: 1900) Formula V
KVSNRX.sub.1S X.sub.1 is F, R, V, or K (SEQ ID NO: 1835)

TABLE-US-00035 TABLE E6 Light chain HVR L3 sequences of anti-TR FM2
antibodies Ab HVR L3 AL2p-h50, AL2p-2, AL2p-3, SQSTRVPYT AL2p-4, AL2p-
5, AL2p-6, (SEQ ID AL2p-7, AL2p-8, AL2p-9, NO: 1836) AL2p-10, AL2p-11,
AL2p-12, AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19,
AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27,
AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35,
AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43,
AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51,
AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59,
AL2p-60, AL2p-61, and AL2p-62

TABLE-US-00036 TABLE E7 Heavy chain framework I sequences of anti-
TREM2 antibodies Ab VH FR1 AL2p-h50, AL2p-2, AL2p-3, QVQLVQSGAE AL2p-
4, AL2p-5, AL2p-6, VKKPGSSVKV AL2p-7, AL2p-8, AL2p-9, SCKASG AL2p-
10, AL2p-11, AL2p-12, (SEQ ID AL2p-13, AL2p-14, AL2p-15, NO: 1716) AL2p-
16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-
24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-
32, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-
45, AL2p-46, AL2p-47, AL2p-48, AL2p-50, AL2p-51, AL2p-54, AL2p-59, AL2p-
60, and AL2p-61 AL2p-33, AL2p-49, AL2p-52, EVQLVQSGAE AL2p-53, AL2p-
55, AL2p-56, VKKPGSSVKV and AL2p-57 SCKASG (SEQ ID NO: 1717) AL2p-
h77, AL2p-35, AL2p-36, QVQLVQSGAE AL2p-37, AL2p-58, and AL2p-62
VKKPGASVKV SCKASG (SEQ ID NO: 1718)

TABLE-US-00037 TABLE E8 Heavy chain framework 2 sequences of anti-
TREM2 antibodies Ab VH FR2 AL2p-h50, AL2p-2, AL2p-3, AL2p-4,
WVRQAPGQGL AL2p-5, AL2p-6, AL2p-7, AL2p-8, EWMG AL2p-9, AL2p-10, AL2p-
11, AL2p-12, (SEQ ID AL2p-13, AL2p-14, AL2p-15, AL2p-16, NO: 1719) AL2p-17,
AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25,
AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33,
AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45,
AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53,
AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-59, AL2p-60, and AL2p-61 AL2p-
h77, AL2p-35, AL2p-36, WVRQAPGQRL AL2p-37, AL2p-58, and AL2p-62 EWIG
(SEQ ID NO: 1720)

TABLE-US-00038 TABLE E9 Heavy chain framework 3 sequences of anti-
TREM2 antibodies Ab VH FR3 AL2p-h50, AL2p-2, AL2p-3, AL2p-4, RVTITADEST
AL2p-5, AL2p-6, AL2p-7, AL2p-8, STAYMELSSL AL2p-9, AL2p-10, AL2p-11,
AL2p-12, RSED TAVYYC AL2p-13, AL2p-14, AL2p-15, AL2p-16, (SEQ ID AL2p-
17, AL2p-18, AL2p-19, AL2p-20, NO: 1721) AL2p-21, AL2p-22, AL2p-23, AL2p-
24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-
32, AL2p-33, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-
44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-
52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-59, AL2p-60, and
AL2p-61 AL2p-h77, AL2p-35, AL2p-36, AL2p-37- RVTITADTSA AL2p-58, and AL2p-
62 STAYMELSSL RSED TAVYYC (SEQ ID NO: 1722)

TABLE-US-00039 TABLE E10 Heavy chain framework 4 sequences of anti-
TREM2 antibodies Ab VH FR4 AL2p-h50, AL2p-2, AL2p-3, AL2p-4, WGQGTLV
AL2p-5, AL2p-6, AL2p-7, AL2p-8, TVSS AL2p-9, AL2p-10, AL2p-11, AL2p-12,
(SEQ ID AL2p-13, AL2p-14, AL2p-15, AL2p-16, NO: AL2p-17, AL2p-18, AL2p-

19, AL2p-20, 1723) AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-h77, AL2p-35, AL2p-36, AL2p-37, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60. AL2p-61, and AL2p-62

TABLE-US-00040 TABLE E11 Light chain framework 1 sequences of anti-TREM2 antibodies Ab VL FR1 AL2p-h50, AL2p-2, AL2p-3, AL2p-4, DVVMTQTPLS AL2p-5, AL2p-6, AL2p-11, AL2p-17, LSVTPGQPAS AL2p-19, AL2p-45, AL2p-46, AL2p-47, ISC AL2p-48, AL2p-49, AL2p-50, AL2p-51, (SEQ ID AL2p-52, AL2p-53, AL2p-54, AL2p-55, NO: 1724) AL2p-56, and AL2p-57 AL2p-7, AL2p-8, AL2p-9, AL2p-10, GVVMTQTPLS AL2p-12, AL2p-13, AL2p-14, AL2p-15, LSVTPGQPAS AL2p-16, AL2p-18, AL2p-20, AL2p-21, ISC AL2p-22, AL2p-23, AL2p-24, AL2p-25, (SEQ ID AL2p-26, AL2p-27, AL2p-28, AL2p-29, NO: 1725) AL2p-30, AL2p-31, AL2p-32, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-59, AL2p-60, and AL2p-61 AL2p-33 GVVMAQTPLS LSVTPGQPAS ISC (SEQ ID NO: 1726 AL2p-h77, AL2p-35, AL2p-36, AL2p-37, DVVMTQSPDS AL2p-58, and AL2p-62 LAVSLGERAT INC (SEQ ID NO: 1727)

TABLE-US-00041 TABLE E12 Light chain framework 2 sequences of anti-TREM2 antibodies Ab VL FR2 AL2p-h50, AL2p-2, AL2p-3, AL2p-4, WYLQKPGQSP AL2p-5, AL2p-6, AL2p-7, AL2p-8, QLLIY AL2p-9, AL2p-10, AL2p-11, AL2p-12, (SEQ ID AL2p-13, AL2p-14, AL2p-15, AL2p-16, NO: 1728) AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-59, AL2p-60 and AL2p-61 AL2p-h77, AL2p-35, AL2p-36, AL2p-37, WYQQKPGQSP AL2p-58, and AL2p-62 KLLIY (SEQ ID NO: 1729)

TABLE-US-00042 TABLE E13 Light chain framework 3 sequences of anti-TREM2 antibodies Ab VL FR3 AL2p-h50, AL2p-2, AL2p-3, AL2p-4, GVPDRFSG AL2p-5, AL2p-6, AL2p-7, AL2p-8, SGSGTDFT AL2p-9, AL2p-10, AL2p-11. AL2p-12, LKISRVEA AL2p-13, AL2p-14, AL2p-15, AL2p-16, EDVGVYYC AL2p-17, AL2p-18, AL2p-19, AL2p-20, (SEQ ID AL2p-21, AL2p-22, AL2p-23, AL2p-24, NO: 1730) AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51. AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, and AL2p-61 AL2p-h77, AL2p-35, AL2p-36, AL2p-37, GVPDRFSGS and AL2-67 GSGTDFTLT ISSLQAEDV AVYYC (SEQ ID NO: 1731)

TABLE-US-00043 TABLE E14 Light chain framework 4 sequences of anti-TREM2 antibodies Ab VL FR4 AL2p-h50, AL2p-2, AL2p-3, AL2p-4, FGQGTKLEIK AL2p-5. AL2p-6. AL2p-7, AL2p-8, (SEQ ID AL2p-9, AL2p-10, AL2p-11, AL2p-12, NO: 1732) AL2p-13, AL2p-14, AL2p-15, AL2p-16, AL2p-17, AL2p-18, AL2p-19, AL2p-20, AL2p-21, AL2p-22, AL2p-23, AL2p-24, AL2p-25, AL2p-26, AL2p-27, AL2p-28, AL2p-29, AL2p-30, AL2p-31, AL2p-32, AL2p-33, AL2p-38, AL2p-39, AL2p-40, AL2p-41, AL2p-42, AL2p-43, AL2p-44, AL2p-45, AL2p-46, AL2p-47, AL2p-48, AL2p-49, AL2p-50, AL2p-51, AL2p-52, AL2p-53, AL2p-54, AL2p-55, AL2p-56, AL2p-57, AL2p-58, AL2p-59, AL2p-60, and AL2p-61 AL2p-h77. AL2p-35, AL2p-36, FGGGTKVEIK AL2p-37, and AL2p-62 (SEQ ID NO: 1733

TABLE-US-00044 TABLE E15 Heavy chain variable region sequences of anti-TREM2 antibodies Ab HCVR AL2p-h50. AL2p-5, QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL and AL2p-6 EWMGRIYPGDGDTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1734) AL2p-2 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGLE WMGRIYPGGGDTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1735) AL2p-3 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL EWMGRIYPGEGDTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1736) AL2p-4 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL EWMGRIYPGQGDNTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1737) AL2p-7 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSLSWMNWVRQAPGQGL EWMGRIYPGDGDTNYARKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1738) AL2p-8 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSLSWMNWVRQAPGQGL EWMGRIYPGDGDTNYARKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGSSYAMDYWGQGTLVTVSS (SEQ ID NO: 1739) AL2p-9 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSRSWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGASYAMDYWGQGTLVTVSS (SEQ ID NO: 1740) AL2p-10 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1741) AL2p-11 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGL EWMGRIYPGDGDTNYARKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1742) AL2p-12 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYAHKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1743) AL2p-13 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKFKGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1744) AL2p-14 and AL2p-15 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1745) AL2p-I6 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKFRGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGASYAMDYWGQGTLVTVSS (SEQ ID NO: 1746) AL2p-17 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKRQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1747) AL2p-18 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYAQKWQGRVTITADESTSTAYMELSSLRSEDT AVYYCARLLRNQPGSSYAMDYWGQGTLVTVSS (SEQ ID NO: 1748) AL2p-19 and AL2p-20 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL EWMGRIYPGDGDTNYARKFQGRVTITADESTSTAYMELSSLRSEDTA VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1749) AL2p-21

QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAWKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1750) AL2p-22
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAYKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1751) AL2p-23
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAQRQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1752) AL2p-24
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGGGDTNYAQRKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1753) AL2p-25
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGGGDTNYAQRFRGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1754) AL2p-26
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGGGDTNYAQRQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1755) AL2p-27
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGQGD
TNYAQRKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1756) AL2p-28
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGVGD
TNYAQRKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1757) AL2p-29
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSQWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAQRKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1758) AL2p-30
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSQWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAWKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1759) AL2p-31,
AL2p-60, QVQLVQSGAEVKKPGSSVKVSCASGYAFSSQWMNWVRQAPGQGL and
AL2p-h31 EWMGRIYPGGGDTNYARKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1760) AL2p-32
QVQLVQSGAEVKKPGSSVKVSCASGYAFSSSEWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAQRKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1761) AL2p-33
EVQLVQSGAEVKKPGSSVKVSCASGYAFSSSWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAQRKFQGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1762) AL2p-h77,
AL2p-h26, QVQLVQSGAEVKKPGASVKVSCASGYAFSSSWMNWVRQAPGQRL and
AL2p-h90 EWIGRIYPGDGDTNYAQRKFQGRVTITADTSASTAYMELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGTLLVTVSS (SEQ ID NO: 1763) AL2p-35
QVQLVQSGAEVKKPGASVKVSCASGYAFWSSWMNWVRQAPGQR
LEWIGRIYPGDGDTNYAQRKFQGRVTITADTSASTAYMELSSLRSED
TAVYYCARLLRNQPGESYAH
DYWGQGTLLVTVSS (SEQ ID NO: 1764) AL2p-36
QVQLVQSGAEVKKPGASVKVSCASGYAFSSSWMNWVRQAPGQRL
EWIGRIYPGDGDTNYARKFQGRVTITADTSASTAYMELSSLRSED
TAVYYCARLLRNQPGASYAMDYWGQGTLLVTVSS (SEQ ID NO: 1765) AL2p-37 and
AL2p-58 QVQLVQSGAEVKKPGASVKVSCASGYAFSSQWMNWVRQAPGQRL
EWIGRIYPGGGDTNYAGKFQGRVTITADTSASTAYMELSSLRSED

VYYCARLLRNQPGASYAMDYWGQGTTLVTVSS (SEQ ID NO: 1766) AL2p-38,
AL2p-39, QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL and
AL2p-40 EWMGRIYPGGGDTNYAQKFRGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNQPGASYAMDYWGQGTTLVTVSS (SEQ ID NO: 1767) AL2p-41 and
AL2p-42 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGEGDTNYAQKFRGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNQPGASYAMDYWGQGTTLVTVSS (SEQ ID NO: 1768) AL2p-43 and
AL2p-44 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGGGDTNYARKFRGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNQPGASYAMDYWGQGTTLVTVSS (SEQ ID NO: 1769) AL2p-45 and
AL2p-46 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGL
EWMGRIYPGEGDTNYARKFQGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNKPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1770) AL2p-47 and
AL2p-48 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGL
EWMGRIYPGEGDTNYARKFHGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNKPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1771) AL2p-49
EVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGL
EWMGRIYPGEGDTNYARKFHGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNKPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1772) AL2p-50 and
AL2p-51 QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGEGDTNYAQKFHGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1773) AL2p-52 and
AL2p-53 EVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGEGDTNYAQKFHGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1774) AL2p-54
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGEGDTNYAQKFHGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNKPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1775) AL2p-55,
AL2p-56, EVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL and
AL2p-57 EWMGRIYPGEGDTNYAQKFHGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNKPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1776) AL2p-61
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSQWMNWVRQAPGQGL
EWMGRIYPGEGDTNYARKFQGRVTITADESTSTAYMELSSLRSEDTA
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1777) AL2p-62
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSQWMNWVRQAPGQRL
EWIGRIYPGEGDTNYAGKFQGRVTITADTSASTAYMELSSLRSEDTA
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1778) AL2p-h19
and AL2p- QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL h35
EWMGRIYPGDGDTNYAQKFQGRATITADTSTSTAYMELSSLRSEDTA
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1779) AL2p-h21
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWMGRIYPGDGDTNYAQKFQGRVTMTTRDTSTSTVYMELSSLRSED
AVYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1780) AL2p-h22
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDTNYAQKFQGRVTMTADTSTSTVYMELSSLRSED
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1781) AL2p-h23
QVQLVQSGAEVKKPGASLKISCKASGYAFSSSWMNWVRQAPGQGLE
WIGRIYPGDGDTNYAQKFQGRATLTADTSTSTAYMELSSLRSED
VYYCARLLRNQPGESYAMDYWGQGTTLVTVSS (SEQ ID NO: 1782) AL2p-h24
QVQLVQSGAEVKKPGASLKISCKASGYAFSSSWMNWVRQAPGQGLE

WIGRIYPGDGDGTNYNQKFQGRATLTADTSTSTAYMELSSLRSEDATV
YFCARLLRNQPGESYAMDYWGQGALVTVSS (SEQ ID NO: 1783) AL2p-h25
QVQLVQSGAEVKKPGASLKISCKASGYAFSSSWMNWVRQAPGQGLE
WIGRIYPGDGDGTNYNGEFRVRATLTADTSTSTAYMELSSLRSEDATV
YYCARLLRNQPGESYAMDYWGQGALVTVSS (SEQ ID NO: 1784) AL2p-h27
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDGTNYNGEFRVRATLTADTSTSTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1785) AL2p-h28
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDGTNYAQKFQGRATLTADTSTSTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1786) AL2p-h29
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDGTNYAQKFQGRATMTADTSTSTAYMELSSLRSEDATV
VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1787) AL2p-h30
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWMGRIYPGDGDGTNYAQKFQGRVTMTADTSTSTAYMELSSLRSEDATV
AVYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1788) AL2p-h32
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDGTNYNGEFRVRATLTADTSTTTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1789) AL2p-h33
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDGTNYAQKFQGRATLTADTSTTTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1790) AL2p-h34
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWIGRIYPGDGDGTNYAQKFQGRATITADTSTSTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1791) AL2p-b36
EVQLLESGLVQPGGSLRLSCAASGYAFSSSWMNWVRQAPGKGLE
WIGRIYPGDGDGTNYAQKFQGRATISADTSKNTAYLQMNSLRAEDATV
VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1792) AL2p-h42
and AL2p- QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQRL h59
EWMGRIYPGDGDGTNYAQKFQGRVTITRDTASTAYMELSSLRSEDATV
VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1793) AL2p-h43
QVQLVQSGAEVKKPGASLKVSCKASGYAFSSSWMNWVRQAPGQRL
EWIGRIYPGDGDGTNYNGEFRVRATLTADTSASTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1794) AL2p-h44
QVQLVQSGAEVKKPGASLKVSCKASGYAFSSSWMNWVRQAPGQRL
EWIGRIYPGDGDGTNYAQKFQGRATLTADTSASTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1795) AL2p-h47
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQGL
EWMGRIYPGDGDGTNYNGEFRVRVTMTRDTSTSTVYMELSSLRSEDATV
AVYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1796) AL2p-h76
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSSWMNWVRQAPGQRL
EWIGRIYPGDGDGTNYAQKFQGRATITADTSASTAYMELSSLRSEDATV
VYFCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1797) AL2p-59
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSHWMNWVRQAPGQGL
EWMGRIYPGEGQTNYAQKRQGRVTITADESTSTAYMELSSLRSEDATV
VYYCARLLRNQPGESYAMDYWGQGTLVTVSS (SEQ ID NO: 1798)
TABLE-US-00045 TABLE E16 Heavy chain sequences of anti-TREM2 antibodies Ab
HC AL2p-58 huIgG1
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSQWMNWVRQAPGQRLWEI

GRIYPGGGDTNYAGKFKQGRVTITADTSASTAYMELSSLRSED TAVYYCAR
LLRNQPGESYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAL
GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSL
GTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQP
REPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT
TPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALFINHYTQKSLSL SPGK
(SEQ ID NO: 1905) AL2p-58 huIgG1
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSQWMNWVRQAPGQRLEWI
GRIYPGGGDTNYAGKFKQGRVTITADTSASTAYMELSSLRSED TAVYYCAR
LLRNQPGESYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAL
GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSL
GTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSHEDPEVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQP
REPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT
TPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALFINHYTQKSLSL SPG (SEQ
ID NO: 1906) AL2p-58 huIgG1 PSEG
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSQWMNWVRQAPGQRLEWI
GRIYPGGGDTNYAGKFKQGRVTITADTSASTAYMELSSLRSED TAVYYCAR
LLRNQPGESYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAL
GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSL
GTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPASIEKTISKAKGQP
REPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT TPPVLD
SDGSFFLYSKLTVDKSRWQQGNVFSCSVMHGALFINHYTQKSLSL SPGK (SEQ ID
NO: 1907) AL2p-58 huIgG1 PSEG
QVQLVQSGAEVKKPGASVKVSCKASGYAFSSQWMNWVRQAPGQRLEWI
GRIYPGGGDTNYAGKFKQGRVTITADTSASTAYMELSSLRSED TAVYYCAR
LLRNQPGESYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAL
GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSSSVVTVPSSSL
GTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPASIEKTISKAKGQP
REPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT TPPVLD
SDGSFFLYSKLTVDKSRWQQGNVFSCSVMHGALHNHYTQKSLSL SPG (SEQ ID
NO: 1908) AL2p-47 huIgG1
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGLEW
MGRIYPGEGDTNYARKFHGRVTITADESTSTAYMELSSLRSED TAVYYCA
RLLRNKPGESYAMDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSS
LGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ
PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT
TPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSL SPGK (SEQ
ID NO: 1909) AL2p-47 hulG1
QVQLVQSGAEVKKPGSSVKVSCKASGYAFSSDWMNWVRQAPGQGLEW

MGRIYPGEGDTNYARKFHGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNKPGESYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALHNHYTQKSLSL
SPG (SEQ ID NO: 1910) AL2p-47 hulG1 PSEG
QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSDWMNWVRQAPGQGLEW
MGRIYPGEGDTNYARKFHGRVTITADESTSTAY
MELSSLRSED
TAVYYCARLLRNKPGESYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNF1KPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPASIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHGALFINHYTQKSLSL
SPGK (SEQ ID NO: 1911) AL2p-47 hulG1 PSEG
QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSDWMNWVRQAPGQGLEW
MGRIYPGEGDTNYARKFHGRVTITADESTSTAY
MELSSLRSED
TAVYYCARLLRNKPGESYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNFIKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPASIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHGALFINHYTQKSLSL
SPG (SEQ ID NO: 1912) AL2p-61 hulG1
QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSQWMNWVRQAPGQGLEW
MGRIYPGEGDTNYARKFQGRVTITADESTSTAY
MELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK
PKDTLMISRTPEV
CVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNS TYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTL
PPSRDELTKNQVSLTCLVKGFYPSDIAVEWES
NGQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALHNHYTQKSLSL
SPGK (SEQ ID NO: 1913) AL2p-61 gulG1
QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSQWMNWVRQAPGQGLEW
MGRIYPGEGDTNYARKFQGRVTITADESTSTAY
MELSSLRSED
TAVYYCARLLRNQPGESYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNFIKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALFINHYTQKSLSL
SPG (SEQ ID NO: 1914) AL2p-40 hulG1
QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSHWMNWVRQAPGQGLEW

MGRIYPPGGDTNYAQKFRGRVTITADESTSTAYMELSSLRSED
TAVYYCARLLRNQPGASYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALFINHYTQKSLSL
SPGK (SEQ ID NO: 1915) AL2p-40 huIgG1

QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSHWMNWVRQAPGQGLEW
MGRIYPPGGDTNYAQKFRGRVTITADEST
STAYMELSSLRSED
TAVYYCARLLRNQPGASYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNFIKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALFINHYTQKSLSL
SPG (SEQ ID NO: 1916) AL2p-44 huIgG1

QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSHWMNWVRQAPGQGLEW
MGRIYPPGGDTNYARKFRGRVTITADEST
STAYMELSSLRSED
TAVYYCARLLRNQPGASYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
ALGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTK
PREEQYNSTYRVVSVLTVLHQDWLNGKEYK
CKVSNKALPAPIEKTISKAKGQPREPQVYTL
PPSRDELTKNQVSLTCLVKGFYPSDIAVEW
ESNGQPENNYKTTPPVLDSDGSFFLYSKLT
VDKSRWQQGNVFSCSVMHEALFINHYTQK
SLSLSPGK (SEQ ID NO: 1917) AL2p-44 huIgG1

QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSHWMNWVRQAPGQGLEW
MGRIYPPGGDTNYARKFRGRVTITADEST
STAYMELSSLRSED
TAVYYCARLLRNQPGASYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALFINHYTQKSLSL
SPG (SEQ ID NO: 1918) AL2p-41 huIgG1

QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSHWMNWVRQAPGQGLEW
MGRIYPPGEGDTNYAQKFRGRVTITADEST
STAYMELSSLRSED
TAVYYCARLLRNQPGASYAMDYWGQGT
LTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVT
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSV
VTVPSSSLGTQTYICNVNHKPSNTKVDKK
VEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCV
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPP
SRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTPPVLDSDGSFFLYSKLTVDK
SRWQQGNVFSCSVMHEALFINHYTQKSLSL
SPGK (SEQ ID NO: 1919) AL2p-41 huIgG1

QVQLVQSGAEVKKPGSSVKV
SCKASGYAFSSHWMNWVRQAPGQGLEW

MGRIPYEGDNTNYAQKFRGTVSTAYMELSSLRSEDITAVYYCA
RLLRNQPGASYAMDYWGQGTLLTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSS
LGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ
PREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT
TPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALFINHYTQKSLSL SPG (SEQ
ID NO: 1920)

TABLE-US-00046 TABLE E17 Light chain variable region sequences of anti-
TREM2 antibodies Ab LCVR AL2p-h50, AL2p-2, AL2p-3, AL2p-
DVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWYL 4, AL2p-h42, AL2p-
h43, AL2p-h44, QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA and AL2p-
h47 EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1799) AL2p-5
DVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNRYTYLHWYL
QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1800) AL2p-6
DVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNWYTYLHWYL
QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1801) AL2p-7, AL2p-8,
AL2p-13, and GVVMTQTPLSLSVTPGQPASISCRSSQSLIHSNGYTYLHWYL AL2p-26
QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1802) AL2p-9, AL2p-16,
AL2p-18, AL2p- GVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWYL 20,
AL2p-23, AL2p-25, and AL2p-
QKPGQSPQLLIYKVSNRVSGVPDRFSGSGSGTDFTLKISRVEA 28
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1803) AL2p-10, AL2p-12,
AL2p-31, and GVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWYL AL2p-32
QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1804) AL2p-11, AL2p-17,
and AL2p-19 DVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNGYTYLHWYL
QKPGQSPQLLIYKVSNRVSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1805) AL2p-14, AL2p-24,
and AL2p-29 GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNGYTYLHWYL
QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAE
DVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1806) AL2p-15, AL2p-21,
and AL2p-30 GVVMTQTPLSLSVTPGQPASISCRSSSSLVHSNGYTYLHWYL
QKPGQSPQLLIYKVSNRKSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1807) AL2p-22
GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNGYTYLHWYL
QKPGQSPQLLIYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1808) AL2p-27
GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNGYTYLHWYL
QKPGQSPQLLIYKVSNRVSGVPDRFSGSGSGTDFTLKISRVEA
EDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1809) AL2p-33
GVVMAQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWY
LQKPGQSPQLLIYKVSNRVSGVPDRFSGSGSGTDFTLKISRVE
AEDVG VYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1810) AL2p-h77, AL2p-
35, AL2p-36, DVVMTQSPDSLAVSLGERATINCRSSQSLVHSNGYTYLHWY AL2p-37,
and AL2p-h76 QKPGQSPKLLIYKVSNRFSGVPDRFSGSGSGTDFTLTISLQ

AEDVAVYVYCSQSTRVPYTFGGGTKVEIK (SEQ ID NO: 1811) AL2p-38 and AL2p-43 GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNRYTYLHWYL QKPGQSPQLLIYKVSNNRRSGVPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1812) AL2p-39 and AL2p-41 GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNQYTYLHWYL QKPGQSPQLLIYKVSNNRRSGVPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1813) AL2p-40, AL2p-42, and AL2p-44 GVVMTQTPLSLSVTPGQPASISCRSRSRSLVHSNRYTYLHWYL QKPGQSPQLLIYKVSNNRRSGVPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1814) AL2p-45 AL2p-47 AL2p-50 DVVMTQTPLSLSVTPGQPASISCRTSQSLVHSNAYTYLHWY AL2p-52, AL2p-55, and AL2p-56 LQKPGQSPQLLIYKVSNNRVSGVPDRFSGSGSGTDFTLKISRVE AEDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1815) AL2p-46, AL2p-48, AL2p-49, DVVMTQTPLSLSVTPGQPASISCRTSQSLVHSNQYTYLHWY AL2p-51, AL2p-53, AL2p-54, and LQKPGQSPQLLIYKVSNNRVSGVPDRFSGSGSGTDFTLKISRVE AL2p-57 AEDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1816) AL2p-61 GVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNQYTYLHWYL QKPGQSPQLLIYKVSNNRRSGVPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1817) AL2p-62 DVVMTQSPDSLAVSLGERATINCRSSQSLVHSNQYTYLHWY QQKPGQSPKLLIYKVSNNRFSGVDPDRFSGSGSGTDFTLTISLQ AEDVAVYYCSQSTRVPYTFGGGTKVEIK (SEQ ID NO: 1818) AL2p-58 DVVMTQSPDSLAVSLGERATINCRSSQSLVHSNRYTYLHWY QQKPGQSPKLLIYKVSNNRFSGVDPDRFSGSGSGTDFTLKISRVE AEDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1819) AL2p-60 GVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNRYTYLHWYL QKPGQSPQLLIYKVSNNRRSGVPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1820) AL2p-h19 DIVMTQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWYL QKPGQSPQLLIYKVSNNRFSGVDPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1821) AL2p-h21, AL2p-h22, AL2p-h23, DVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNGYTYLHWYL AL2p-h24, AL2p-h25, AL2p-h26, QKPGQSPQLLIYKVSNNRFSGVDPDRFSGSGSGTDFTLKISRVEA AL2p-h27, AL2p-h28, AL2p-h29 EDLGVYFCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1822) AL2p-h30, AL2p-h31, AL2p-h32, AL2p-h33, AL2p-h34, AL2p-h35, AL2p-h36 DIVMTQSPLSLPVTGPGEASISCRSSQSLVHSNGYTYLHWYL AL2p-h59 QKPGQSPQLLIYKVSNNRFSGVDPDRFSGSGSGTDFTLKISRVEA EDVGVYYCSQSTRVPYTFGGGTKVEIK (SEQ ID NO: 1823) AL2p-h90 DVQMTQSPSSLSASVGDRTITCRSSQSLVHSNGYTYLHWY QQKPGKSPKLLIYKVSNNRFSGVPSRFSGSGSGTDFTLTISLQP EDFATYYCSQSTRVPYTFGGGTKVEIK (SEQ ID NO: 1824) AL2p-59 GVVMTQTPLSLSVTPGQPASISCRTSQSLVHSNQYTYLHWY LQKPGQSPQLLIYKVSNNRVSGVPDRFSGSGSGTDFTLKISRVE AEDVGVYYCSQSTRVPYTFGQGTKLEIK (SEQ ID NO: 1825)

TABLE-US-00047 TABLE E18 Light chain sequences of anti-TREM2 antibodies Ab LC AL2p-58 DVVMTQSPDSLAVSLGERATINCRSSQSLVHSNRYTYLH hulgG1, WYQQKPGQSPKLLIYKVSNNRFSGVDPDRFSGSGSGTDFTL and KISRVEAEDVGVYYCSQSTRVPYTFGQGTKLEIKRTVAA AL2p-58 PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD hulgG1

NALQSGNSQESVTEQDSKDYESTLSSTLTLSKADYEKHK PSEG
 VYACEVTHQGLSSPVTKSFNREGC (SEQ ID NO: 1921) AL2p-47
 DVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNAYTYLH huIgG1,
 WYLQKPGQSPQLLIYKVSNRVSGVPDRFSGSGSGTDFTL and
 KISRVEAEDVGVYYCSQSTRVPYTFGQGGTKLEIKRTVAA AL2p-47
 PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD huIgG1
 NALQSGNSQESVTEQDSKDYESTLSSTLTLSKADYEKHK PSEG
 VYACEVTHQGLSSPVTKSFNREGC (SEQ ID NO: 1922) AL2p-61
 GVVMTQTPLSLSVTPGQPASISCRSSQSLVHSNQYTYLH huIgG1
 WYLQKPGQSPQLLIYKVSNRSGVPDRFSGSGSGTDFTL
 KISRVEAEDVGVYYCSQSTRVPYTFGQGGTKLEIKRTVAA
 PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD
 NALQSGNSQESVTEQDSKDYESTLSSTLTLSKADYEKHK
 VYACEVTHQGLSSPVTKSFNREGC (SEQ ID NO: 1923) AL2p-41
 GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNQYTYLH huIgG1
 WYLQKPGQSPQLLIYKVSNRSGVPDRFSGSGSGTDFTL
 KISRVEAEDVGVYYCSQSTRVPYTFGQGGTKLEIKRTVAA
 PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD
 NALQSGNSQESVTEQDSKDYESTLSSTLTLSKADYEKHK
 VYACEVTHQGLSSPVTKSFNREGC (SEQ ID NO: 1924) AL2p-40
 GVVMTQTPLSLSVTPGQPASISCRSSRSLVHSNRYTYLH huIgG1,
 WYLQKPGQSPQLLIYKVSNRSGVPDRFSGSGSGTDFTL and
 KISRVEAEDVGVYYCSQSTRVPYTFGQGGTKLEIKRTVAA AL2p-44
 PSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVD huIgG1
 NALQSGNSQESVTEQDSKDYESTLSSTLTLSKADYEKHK
 VYACEVTHQGLSSPVTKSFNREGC (SEQ ID NO: 1925)

[0353] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in TABLES E1-E18 as well as specific combinations thereof and other embodiments of the anti-TREM2 antibody described in the '573 application and herein may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

F. PCT Patent Application Publication No. WO2018/015573A1

[0354] In some embodiments, the TREM2 agonist is an antibody, or antigen binding fragment thereof, that prevents the cleavage of TREM2 as described in PCT Patent Application Publication No. WO2018/015573A1 ("the '573 application"), which is incorporated by reference herein, in its entirety.

[0355] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3, and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 disclosed in the '573 application specification. In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '573 application specification.

[0356] In some embodiments, the antibody is a binding molecule that inhibits (preferably prevents) TREM2 cleavage. More specifically, in the context of the present invention cleavage (i.e. shedding) of the TREM2 ectodomain is inhibited by the binding molecule of the present invention. In some embodiments, the antibody is a binding molecule that inhibits (preferably prevents) TREM2

cleavage and activates TREM2 activity. In some embodiments, the herein provided binding molecule has a binding site within the ectodomain of TREM2, preferably the stalk region of the TREM2 ectodomain.

[0357] In some embodiments, the antibody is: [0358] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1955 and the light chain variable region comprises the sequence of SEQ ID NO: 1965; and wherein the antibody inhibits TREM2 cleavage; [0359] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1955, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1965; and wherein the antibody inhibits TREM2 cleavage; [0360] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1975; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1985; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1995; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2005; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2015; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2025; and wherein the antibody inhibits TREM2 cleavage; or [0361] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1975; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1985; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1995; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 2005; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO: 2015; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 2025; and wherein the antibody inhibits TREM2 cleavage.

[0362] In some embodiments, the antibody is antibody clone 14D3, which is: [0363] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1946 and the light chain variable region comprises the sequence of SEQ ID NO: 1956; and wherein the antibody inhibits TREM2 cleavage; [0364] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85% identity to SEQ ID NO: 1946, and the light chain variable region comprises a sequence having at least 85% identity to SEQ ID NO: 1956; and wherein the antibody inhibits TREM2 cleavage; [0365] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1966; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1976; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1986; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1996; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2006; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2016; and wherein the antibody inhibits TREM2 cleavage; or [0366] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1966; the CDR2 of the heavy chain variable region comprises an

amino acid sequence having at least 70% identity to SEQ ID NO: 1976; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1986; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1996; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60% identity to SEQ ID NO:2006; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO:2016; and wherein the antibody inhibits TREM2 cleavage.

[0367] In some embodiments, the antibody is antibody clone 14D8, which is: [0368] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO:1947 and the light chain variable region comprises the sequence of SEQ ID NO: 1957; and wherein the antibody inhibits TREM2 cleavage; [0369] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85% identity to SEQ ID NO:1947, and the light chain variable region comprises a sequence having at least 85% identity to SEQ ID NO: 1957; and wherein the antibody inhibits TREM2 cleavage; [0370] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO:1967; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1977; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1987; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1997; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2007; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2017; and wherein the antibody inhibits TREM2 cleavage; or [0371] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1967; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1977; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1987; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO: 1997; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60% identity to SEQ ID NO:2007; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70% identity to SEQ ID NO:2017; and wherein the antibody inhibits TREM2 cleavage.

[0372] In some embodiments, the antibody is antibody clone 7A12, which is: [0373] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1948 and the light chain variable region comprises the sequence of SEQ ID NO: 1958; and wherein the antibody inhibits TREM2 cleavage; [0374] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1948, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1958; and wherein the antibody inhibits TREM2 cleavage; [0375] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1968; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1978; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1988; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1998; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2008; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2018; and wherein the antibody inhibits TREM2 cleavage; or [0376] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1968; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%,

preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:1978; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1988; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1998; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO:2008; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO:2018; and wherein the antibody inhibits TREM2 cleavage.

[0377] In some embodiments, the antibody is antibody clone 8A11, which is: [0378] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1949 and the light chain variable region comprises the sequence of SEQ ID NO: 1959; and wherein the antibody inhibits TREM2 cleavage; [0379] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1949, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1959; and wherein the antibody inhibits TREM2 cleavage; [0380] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1969; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1979; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1989; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 1999; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2009; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2019; and wherein the antibody inhibits TREM2 cleavage; or [0381] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1969; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:1979; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1989; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1999; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO:2009; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO:2019; and wherein the antibody inhibits TREM2 cleavage.

[0382] In some embodiments, the antibody is antibody clone 21A3, which is: [0383] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1950 and the light chain variable region comprises the sequence of SEQ ID NO: 1960; and wherein the antibody inhibits TREM2 cleavage; [0384] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO:

1950, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1960; and wherein the antibody inhibits TREM2 cleavage; [0385] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1970; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1980; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1990; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2000; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2010; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2020; and wherein the antibody inhibits TREM2 cleavage; or [0386] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1970; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1980; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1990; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 2000; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO: 2010; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 2020; and wherein the antibody inhibits TREM2 cleavage.

[0387] In some embodiments, the antibody is antibody clone 10C3, which is: [0388] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1951 and the light chain variable region comprises the sequence of SEQ ID NO: 1961; and wherein the antibody inhibits TREM2 cleavage; [0389] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1951, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1961; and wherein the antibody inhibits TREM2 cleavage; [0390] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1971; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1981; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1991; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2001; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2011; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2021; and wherein the antibody inhibits TREM2 cleavage; or [0391] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1971; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1981; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID

NO:1991; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:2001; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO:2011; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO:2021; and wherein the antibody inhibits TREM2 cleavage.

[0392] In some embodiments, the antibody is antibody clone 18F9, which is: [0393] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1952 and the light chain variable region comprises the sequence of SEQ ID NO: 1962; and wherein the antibody inhibits TREM2 cleavage; [0394] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferred at least 99% identity to SEQ ID NO: 1952, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1962; and wherein the antibody inhibits TREM2 cleavage; [0395] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1972; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1982; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1992; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2002; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2012; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2022; and wherein the antibody inhibits TREM2 cleavage; or [0396] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1972; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:1982; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1992; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:2002; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO:2012; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO:2022; and wherein the antibody inhibits TREM2 cleavage.

[0397] In some embodiments, the antibody is antibody clone 15C5, which is: [0398] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1953 and the light chain variable region comprises the sequence of SEQ ID NO: 1963; and wherein the antibody inhibits TREM2 cleavage; [0399] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1953, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1963; and wherein the antibody inhibits TREM2 cleavage; [0400] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino

acid sequence of SEQ ID NO: 1973; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1983; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1993; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2003; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2013; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2023; and wherein the antibody inhibits TREM2 cleavage; or [0401] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1973; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:1983; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1993; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:2003; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID NO:2013; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO:2023; and wherein the antibody inhibits TREM2 cleavage.

[0402] In some embodiments, the antibody is antibody clone 1G6, which is: [0403] (1) an antibody, wherein the heavy chain variable region comprises the sequence of SEQ ID NO: 1954 and the light chain variable region comprises the sequence of SEQ ID NO: 1964; and wherein the antibody inhibits TREM2 cleavage; [0404] (2) an antibody, wherein the heavy chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1954, and the light chain variable region comprises a sequence having at least 85%, preferably at least 90%, more preferably at least 95%, even more preferably at least 98%, and most preferably at least 99% identity to SEQ ID NO: 1964; and wherein the antibody inhibits TREM2 cleavage; [0405] (3) an antibody, wherein the CDR1 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1974; the CDR2 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1984; the CDR3 of the heavy chain variable region comprises the amino acid sequence of SEQ ID NO: 1994; the CDR1 of the light chain variable region comprises the amino acid sequence of SEQ ID NO: 2004; the CDR2 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2014; and the CDR3 of the light chain variable region comprises the amino acid sequence of SEQ ID NO:2024; and wherein the antibody inhibits TREM2 cleavage; or [0406] (4) an antibody, wherein the CDR1 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO: 1974; the CDR2 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:1984; the CDR3 of the heavy chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO: 1994; the CDR1 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, even more preferably at least 85%, and most preferably at least 90% identity to SEQ ID NO:2004; the CDR2 of the light chain variable region comprises an amino acid sequence having at least 60%, preferably 100% identity to SEQ ID

NO:2014; and the CDR3 of the light chain variable region comprises an amino acid sequence having at least 70%, preferably at least 75%, more preferably at least 80%, and most preferably at least 85% identity to SEQ ID NO:2024; and wherein the antibody inhibits TREM2 cleavage. [0407] In some embodiments, the antibody is an antibody disclosed in FIG. 9 of PCT Patent Application Publication No. WO2018/015573A1, reproduced below as TABLES F1-F4.

TABLE-US-00048 TABLE F1 Clone name Variable region of the heavy chain 14D3 EVKLLLEFGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGRAPEWLGLIRNKTKGYTTEYNRSVKGRFTI

SRDNTQNMLYLQMNSLRPEDTATYYCARIGVNNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1946) 14D8 EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGKAPEWLGLIRNKANGYTTVYNPSVKGRFTI

SRDNTQNMLYLQMNTLRGEDTATYYCARIGINNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1947) 7A12 EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGKAPEWLGLIRNKANGYTTQYNPSVKGRFTI

SRDNTQNMLYLQMNTLRGEDTATYYCARIGINNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1948) 8A11 EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGKAPEWLGLIRNKTKGYTTEYNTSVKGRFTI

SRDNTQNMLYLQMNSLRPEDTATYYCARIGVNNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1949) 21A3 EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGKAPEWLGLIRNKANGYTTQYNPSVKGRFTI

SRDNTQNMLYLQMNTLRGEDTATYYCARIGINNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1950) 10C3 EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGETPEWLGLIRNKTKGYTTEYNPSVKGRFTI

SRDNTQNMLYLQMNSLRPEDTATYYCARIGTNNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1951) 18F9 EVKLLLESGGGLVQPGGSMRLSCVVSFGFTFTDFYMNW

IRQAAGKAPEWLGLIRNKVNGYRTEYNPSVKGRFTI

SRDNIQNMLYLQMNTLRAEDTATYYCARIGINNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1952) 15C5 EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW

IRQPAGKAPEWLGLIRNKAYGYTTEYNPSVKGRFTI

SRDNTQDMLYLQMNTLRAEDTATYYCARIGINYGGGS LDYWGQGVMVTVSS (SEQ ID NO: 1953) 1G6 EVKLLLESGGGLVQPGGSLRLSCVASGFTFTDFYMNW

IRQPAGKAPEWLGLIRNKANGFTTEYNPSVKGRFTI

SRDNTQHMLYLQMNTLRAEDTATYYCARIGINNGGS LDYWGQGVMVTVSS (SEQ ID NO: 1954) Consensus EVKLLLESGGGLVQPGGSMRLSCAASGFTFTDFYMNW sequence

IRQPAGKAPEWLGLIRNKANGYTTTEYNPSVKGRFTI

SRDNTQNMLYLQMNTLREDTATYYCARIGINNGGSL DYWGQGVMVTVSS (SEQ ID NO: 1955)

TABLE-US-00049 TABLE F2 Clone name Variable region of the light chain 14D3 DILIIQSPASLTVSAGARVTMSCKSSQSLLYSENNQ

DYLAWYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG

SGTDFTLTISVQAEDLADYYCEQTYSYPYTFGAGT KLELK (SEQ ID NO: 1956) 14D8 DILINQSPASLTVSTGEKVTMSCRSSQSLLYSEKNQ

DYLAWYQQKPGQFPKLLIYGASYRHTGVPDRFTGSG

SGTDFTLTISVQAEDLADYYCEQTYSYPYTFGAGT KLELK (SEQ ID NO: 1957) 7A12 DILINQSPASLTVSAGEKVTMSCKSSQSLLYSEKNQ

DYLAWYQQKPGQSPKLLMYGASYRHTGVPDRFTGSG

SGTDFTLTISVQAEDLADYYCEQTYSYPYTFGAGT KLELK (SEQ ID NO: 1958) 8A11 DILIIQSPASLTVSAGARVTMSCKSSQSLLYSENNQ

DYLAWYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG

SGTDFTLTISVQAEDLADYYCEQTYSYPYTFGAGT KLELK (SEQ ID NO: 1959)

21A3 DILINQSPASLTVSAGEKVTMSCKSSQSLLYSEKNQ
DYLAWEYQQKPGQSPKLLMYGASYRHTGVPDRFTGSG
SGTDFTLTISVQAEDLADYYCEQTYSSYPYTFGAGT KLELK (SEQ ID NO: 1960)
10C3 DILIIQSPASLTVSAGARVTMSCKSSQSLLYSENNQ
DYLAWEYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG
SGTDFTLTISVQAEDLADYYCEQTYSSYPYTFGAGT KLELK (SEQ ID NO: 1961)
18F9 DILINQSPASLTVSAGEKVTMSCKSSQSLLYSENNQ
DYLAWEYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG
SGTDFTLTISVQAEDLADYYCEQTYSSYPYTFGAGT KLELK (SEQ ID NO: 1962)
15C5 DILINQSPASLTVSAGEKVTMSCKSSQSLLYSENNQ
DYLAWEYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG
SGTDFTLTISVQAEDLAHYCEQTYSSYPYTFGAGT KLELK (SEQ ID NO: 1963)
1G6 DILINQSPASLTVSTGEKVTMSCKSSQSLLYSENNQ
DYLAWEYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG
SGTDFTLTINIVQAEDLADYYCEQTYSSYPYTFGAGT KLELK (SEQ ID NO: 1964)
Consensus DILINQSPASLTVSAGEKVTMSCKSSQSLLYSENNQ sequence
DYLAWEYQQKPGQFPKLLIYGASNRHTGVPDRFTGSG
SGTDFTLTISVQAEDLADYYCEQTYSSYPYTFGAGT KLELK (SEQ ID NO: 1965)
TABLE-US-00050 TABLE F3 Complementarity determining regions in Clone the
variable region of the heavy chain name CDR1 CDR2 CDR3 14D3 GFTFTDFY
IRNKTKGYTT ARIGVNNGGSLDYWG (SEQ ID (SEQ ID (SEQ ID NO: 1986)
NO: 1966) NO: 1976) 14D8 GFTFTDFY IRNKANGYTT ARIGINNGGSLDYWG (SEQ
ID (SEQ ID (SEQ ID NO: 1987) NO: 1967) NO: 1977) 7A12 GFTFTDFY
IRNKANGYTT ARIGINNGGSLDYWG (SEQ ID (SEQ ID (SEQ ID NO: 1988)
NO: 1968) NO: 1978) 8A11 GFTFTDFY IRNKTKGYTT ARIGVNNGGSLDYWG (SEQ
ID (SEQ ID (SEQ ID NO: 1989) NO: 1969) NO: 1979) 21A3 GFTFTDFY
IRNKANGYTT ARIGINNGGSLDYWG (SEQ ID (SEQ ID (SEQ ID NO: 1990)
NO: 1970) NO: 1980) 10C3 GFTFTDFY IRNKTKGYTT ARIGTNNGGSLDYWG (SEQ
ID (SEQ ID (SEQ ID NO: 1991) NO: 1971) NO: 1981) 18F9 GFTFTDFY
IRNKVNGYRT ARIGINNGGSLDYWG (SEQ ID (SEQ ID (SEQ ID NO: 1992)
NO: 1972) NO: 1982) 15C5 GFTFTDFY IRNKAYGYTT ARIGINYGGSLDYWG (SEQ
ID (SEQ ID (SEQ ID NO: 1993) NO: 1973) NO: 1983) 1G6 GFTFTDFY
IRNKANGFTT ARIGINNGGSLDYWG (SEQ ID (SEQ ID (SEQ ID NO: 1994)
NO: 1974) NO: 1984) Consensus GFTFTDFY IRNKANGYTT ARIGINNGGSLDYWG
(SEQ ID (SEQ ID (SEQ ID NO: 1995) NO: 1975) NO: 1985)
TABLE-US-00051 TABLE F4 Complementarity determining regions in Clone the
variable region of the light chain name CDR1 CDR2 CDR3 14D3 QSLLYSENNQDY
GAS EQTYSSYPYT (SEQ ID NO: 1996) (SEQ ID (SEQ ID NO: 2006) NO:
2016) 14D8 QSLLYSEKNQDY GAS EQTYSSYPYT (SEQ ID NO: 1997) (SEQ ID
(SEQ ID NO: 2007) NO: 2017) 7A12 QSLLYSEKNQDY GAS EQTYSSYPYT (SEQ
ID NO: 1998) (SEQ ID (SEQ ID NO: 2008) NO: 2018) 8A11 QSLLYSENNQDY
GAS EQTYSSYPYT (SEQ ID NO: 1999) (SEQ ID (SEQ ID NO: 2009) NO:
2019) 21A3 QSLLYSEKNQDY GAS EQTYSSYPYT (SEQ ID NO: 2000) (SEQ ID
(SEQ ID NO: 2010) NO: 2020) 10C3 QSLLYSENNQDY GAS EQTYSSYPYT (SEQ
ID NO: 2001) (SEQ ID (SEQ ID NO: 2011) NO: 2021) 18F9 QSLLYSENNQDY
GAS EQTYSSYPYT (SEQ ID NO: 2002) (SEQ ID (SEQ ID NO: 2012) NO:
2022) 15C5 QSLLYSENNQDY GAS EQTYSSYPYT (SEQ ID NO: 2003) (SEQ ID
(SEQ ID NO: 2013) NO: 2023) 1G6 QSLLYSENNQDY GAS EQTYSSYPYT (SEQ
ID NO: 2004) (SEQ ID (SEQ ID NO: 2014) NO: 2024) Consensus
QSLLYSENNQDY GAS EQTYSSYPYT (SEQ ID NO: 2005) (SEQ ID (SEQ ID

NO: 2015) NO: 2025)

[0408] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in in the above tables as well as specific combinations thereof and other embodiments of the anti-TREM2 antibody described in the '573 application and herein may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

G. PCT Patent Application Publication No. WO2019/055841A1

[0409] In some embodiments, the TREM2 agonist is an antibody or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2019/055841A1 (“the '841 application”), which is incorporated by reference herein, in its entirety.

[0410] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3, and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 disclosed in the '841 application specification. In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '841 application specification.

[0411] In some embodiments, the antibody comprises one or more (e.g., one, two, three, four, five, or all six) CDRs selected from the group consisting of: [0412] (a) a heavy chain CDR1 sequence having at least 90% sequence identity to the amino acid sequence of any one of SEQ ID NOS:2049, 2077, 2080, 2086, 2092, 2098, 2103, 2109, 2115, 2122, 2126, 2347, and 2355 or having up to two amino acid substitutions relative to the amino acid sequence of any one of SEQ ID NOS:2049, 2077, 2080, 2086, 2092, 2098, 2103, 2109, 2115, 2122, 2126, 2347, and 2355; [0413] (b) a heavy chain CDR2 sequence having at least 90% sequence identity to the amino acid sequence of any one of SEQ ID NOS:2050, 2078, 2081, 2087, 2093, 2099, 2104, 2110, 2116, 2120, 2123, 2127, 2348, and 2356 or having up to two amino acid substitutions relative to the amino acid sequence of any one of SEQ ID NOS:2050, 2078, 2081, 2087, 2093, 2099, 2104, 2110, 2116, 2120, 2123, 2127, 2348, and 2356; [0414] (c) a heavy chain CDR3 sequence having at least 90% sequence identity to the amino acid sequence of any one of SEQ ID NOS:2051, 2082, 2088, 2094, 2100, 2105, 2111, 2117, 2124, 2128, 2349, and 2357 or having up to two amino acid substitutions relative to the amino acid sequence of any one of SEQ ID NOS:2051, 2082, 2088, 2094, 2100, 2105, 2111, 2117, 2124, 2128, 2349, and 2357; [0415] (d) a light chain CDR1 sequence having at least 90% sequence identity to the amino acid sequence of any one of SEQ ID NOS:2052, 2083, 2089, 2095, 2101, 2106, 2112, 2118, 2129, and 2351 or having up to two amino acid substitutions relative to the amino acid sequence of any one of SEQ ID NOS:2052, 2083, 2089, 2095, 2101, 2106, 2112, 2118, 2129, and 2351; [0416] (e) a light chain CDR2 sequence having at least 90% sequence identity to the amino acid sequence of any one of SEQ ID NOS:2053, 2079, 2084, 2090, 2096, 2107, 2113, 2352, and 2359 or having up to two amino acid substitutions relative to the amino acid sequence of any one of SEQ ID NOS:2053, 2079, 2084, 2090, 2096, 2107, 2113, 2352, and 2359; and [0417] (f) a light chain CDR3 sequence having at least 90% sequence identity to the amino acid sequence of any one of SEQ ID NOS:2054, 2085, 2091, 2097, 2102, 2108, 2114, 2119, 2121, 2125, 2130, and 2353 or having up to two amino acid substitutions relative to the amino acid sequence of any one of SEQ ID NOS:2054, 2085, 2091, 2097, 2102, 2108, 2114, 2119, 2121, 2125, 2130, and 2353. [0418] In some embodiments, the antibody comprises: [0419] (a) a heavy chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2049, a heavy chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2050, a heavy chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2051, a light chain CDR1 sequence

[illegible]

[0427] (j) a heavy chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2115, a heavy chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2120, a heavy chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2117, a light chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2118, a light chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2079, and a light chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2121; or [0428] (k) a heavy chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2123, a heavy chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2132, a heavy chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2133, a light chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2102, a light chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2079, and a light chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2125; or [0429] (l) a heavy chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2126, a heavy chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2127, a heavy chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2128, a light chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2129, a light chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2079, and a light chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2130; or [0430] (m) a heavy chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2347, a heavy chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2348, a heavy chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2349, a light chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2351, a light chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2352, and a light chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2353; or [0431] (n) a heavy chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2355, a heavy chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2356, a heavy chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2357, a light chain CDR1 sequence comprising the amino acid sequence of SEQ ID NO:2089, a light chain CDR2 sequence comprising the amino acid sequence of SEQ ID NO:2359, and a light chain CDR3 sequence comprising the amino acid sequence of SEQ ID NO:2091.

[0432] In some embodiments, the antibody or antigen-binding portion thereof comprises: [0433]

(a) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2047; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2048; or [0434] (b) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2055; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2066; or [0435] (c) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2056; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2067; or [0436] (d) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2057; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2068; or [0437] (e) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2058; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2069; or [0438] (f) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2059; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2070; or [0439] (g) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2060; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence

identity to SEQ ID NO:2071; or [0440] (h) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2061; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2072; or [0441] (i) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2062; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2073; or [0442] (j) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2063; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2074; or [0443] (k) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2064; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2075; or [0444] (l) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2065; and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2076; or [0445] (m) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2346, and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2350; or [0446] (n) a heavy chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2354, and a light chain variable region comprising an amino acid sequence that has at least 90% sequence identity to SEQ ID NO:2358.

[0447] In some embodiments, the antibody is an antibody disclosed in Table 15 of PCT Patent Application Publication No. WO2019/05584A1, reproduced as TABLE G1 below. In some embodiments, the antibody comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3, and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 disclosed in TABLE G1.

TABLE-US-00052 TABLE G1 Description Sequence

muIgG1	3'	primer
GGACAGGGATCCAGAGTTCC	(SEQ ID NO: 2042)	muIgG2
AGCTGGGAAGGTGTGCACAC	(SEQ ID NO: 2043)	muIgG3
CAGGGGCCAGTGGATAGAC	(SEQ ID NO: 2044)	muCkappa.1
GACATTGATGTCTTTGGGGT	(SEQ ID NO: 2045)	primer muCkappa.2
TTCAGTGGCATCAATCTTCC	(SEQ ID NO: 2046)	primer R59.F6
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQSPGRGLE		VH amino acid sequence
WIGRSDPTTGGTNYNEKFKTKATLTVDKPSSTAYMQLSSLTSDDSAV		
YYCVRTSGTGDYWGQGTSLTVSSAKTTAPSVYPLAPVCGGTTGSSVT	(SEQ ID NO: 2047)	R59.F6
DVVMQTQPLSLPVSLGDQASISCRSSQSLVHNNGNTFLHWYLQKPGQ		acid sequence
SPKWYkVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYFCSQTT		
HVPPTFGGGTKLEIKRADAAPTVSIFPPSSEQLTSGGASVVC	(SEQ ID NO: 2048)	
R59.F6 CDR-H1	GYTFTSY (SEQ ID NO: 2049)	amino acid sequence R59.F6
CDR-H2	IGRSDPTTGGTNYNE (SEQ ID NO: 2050)	amino acid sequence R59.F6
and RS.F10	VRTSGTGDY (SEQ ID NO: 2051)	CDR-H3 amino acid sequence
R59.F6 and RS.F10	RSSQSLVHNNGNTFLH (SEQ ID NO: 2052)	CDR-L1 amino acid sequence
R59.F6 CDR-L2	VSNRFS (SEQ ID NO: 2053)	amino acid sequence
R59.F6 and RS.F10	SQTTHVPPT (SEQ ID NO: 2054)	CDR-L3 amino acid sequence
21D11	VH amino	
QVQLQQSGAELARPGASVKLSCKASGYTFTSYWIQWVKQRPGQGLE		acid sequence
WIGTIYPGDGDARYTQKFKGKATLTADKSSSTTYMQLNSLASEDSAV		
YYCARNGITTAGYYAMDYWGQGTSTVTVSS	(SEQ ID NO: 2055)	21D4.D1
VH amino	QVQLQQSGADLLRPGVSVKISCKGSGYTFTDHAMHWVKQSHAESLE	acid sequence
WIGVISTYSGDTGYNQKFKGKATMTVDKSSSTAYLELARLTSEDSAIY		

YCAREGYDDAMDYWGQGTSTVTSS (SEQ ID NO: 2056) 26D2 VH amino acid
EVQLQQSGPELVKPGASVKMSCKASGYTFTSYVMHWVKQKPGQGLE sequence
WIGYINPYTDGTTYNEKFKGKATLTSDKSSSTAYMDLSSLTSEDSAVY
YCARGEVRRYALDYWGQGTSTVTSS (SEQ ID NO: 2057) 26E2.A3 VH amino
QVHLQQSGSELRSFGSSVKLSCKDFDSEVFPISYMSWIRQKPGHGFEE acid sequence;
IGDILPSIGGRIYGVKFEDRATLDADTVSNTAYLELNSLTSEDSAIYYC 24B4.A1 VH
amino ARKDYGSLAYWGQGTSLTVSA (SEQ ID NO: 2058) acid sequence 3D3.A1
VH amino EVQLQQSGPELVKPGASVKISCKTSGYTLSEYTMHWVIQSHGKSLEWI acid
sequence GGVIPNSGGTSYNQKFRDKASLTVDKSSSTAYLELRSLTSEDSAVYYC
ARGDDSYRRGYALDYWGQGTSTVTSS (SEQ ID NO: 2059) 40H3.A4 VH amino
EVQLQQSGAEVVKPGASVKLSCTASGFNIKDTYMHVVKQRPEQGLE acid sequence
WIGRIDPANGNTKYDPKFQGKATITADTSSNTAYLQLSSLTSEDTAVY
YCATLFAYWGQGTSLTVSA (SEQ ID NO: 2060) 42E8.H1 VH amino
DVQLQESGPGVLKPSQSLSLTCTVTGYSITSDYAWNWRQFPGNKLE acid sequence
WMGYINYSGRTIYNPSLKSRIITRDTSKNHFFLQLISVTTEDTATYYC
ARWNGNYGFAYWGQGTSLTVSA (SEQ ID NO: 2061) 49H1 LB1 VH amino
DVQLQESGPGVLKPSQSLSLTCTVTGYSITSDYAWNWRQFPGNRLE acid sequence
WMGYISFSGSTSYNPSLKSRIITRDTSKNQFFLQLNSVTTEDTATYYC
ARWNGNYGFAYWGQGTSLTVSA (SEQ ID NO: 2062) 54C2.A1 VH amino
QVHLQQSGSELRSFGSSVKLSCKDFDSEVFPISYMSWVRQKPGHGFEE acid sequence
WIGDILPSIGRRIYGVKFEDKATLDADTVSNTAYLELNSLTSEDSAIYY
CTRKDYGSLAYWGQGTSLTVSA (SEQ ID NO: 2063) 57D7.A1 VH amino
QVQLKESGPGVLAPSQSLSLTCTVSGFSLRSYSVYWVRQPPGKGLEWL acid sequence
GMIWGGGNTDYNALSKSLRSISKDNSKSKVFLKMNSLQTDDSAMYY
CVQYGGMDYWGQGTSTVTSS (SEQ ID NO: 2064) RS9.F6 VH amino
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQSPGRGLE acid sequence;
WIGRSDPTTGGTNYNEKFKTKATLTVDKPSSTAYMQLSSLTSDDSAV RS.F10 VH amino
YYCVRTSGTGDYWGQGTSLTVSS (SEQ ID NO: 2065) acid sequence 2ID11 VL
amino DIQMTQSPASLSVSVGETVTITCRASENIYSNLAWYQQKQGRSPQLLV acid
sequence YAATNLADGVPSRFSGSGSGTQYSLKINSLQSEDFGYYCYCHFWGTP
YTFGGGTTKLEIK (SEQ ID NO: 2066) 21D4.D1 VL amino
DVVMTQTPLTSLVLTIGQPASFSCKSSQSLDSDGKTYLNWLLRRPGQS acid sequence
PKRLIYVVSRLDSGVPDRFTGSGSGTDFTLKISRVEAEDLGVYYCWQG
THFPYTFGGGTTKLEIK (SEQ ID NO: 2067) 26D2 VL amino acid
DIQMTQSSSSFSVSLGDRVITITCKASEDIYNRLAWYQQKPGNAPRLIS sequence
GATSLETGVPSRFSGSGSGKDYTLSTSLQTEDVATYYCQQYWSTPWT FGGGTTKLEIK
(SEQ ID NO: 2068) 26E2.A3 VL amino
DVVMTQTPLSLPVSLGDQASISCRSSQSLVHINGNTYLLQWFLQKPGQS acid sequence;
PKLLIYKVSNRFSGVPDRFSGSGSGTAFTLKISRVEAEDLGVYFCSQST 24B4.A1 VL
amino HVPYTFGGGTTKLEIK (SEQ ID NO: 2069) acid sequence 3D3.A1 VL amino
DIVMSQSPSSLAVSVGEKVTMSCKSSQSLLYSSNQKSYLAWYQQKPG acid sequence
QSPKLLIYWASTRESGVPDRFRGSGSGTDFTLTISVKAEDLAVYYCQ
QYFSYPPTFGGGTTKLEIK (SEQ ID NO: 2070) 40H3.A4 VL amino
DIVMTQAAFSNPVTLGTSASISCRSSKSLHNSGITYLYWYLQKPGQSP acid sequence
QLLIYQMSNLASGVPDRFSSSGSGIDFTLRINRVEAEDVGVYYCAQNL
ELPTFGSGTTKLEIK (SEQ ID NO: 2071) 42E8.H1 VL amino
DVVMTQNPLSLPVSLGDQASISCRSSQSLVHINGNTYLLHWYLQKPGQS acid sequence
PKWYKVSNRFSGVPDRFSGSGSGTDFTLKISRVEAEDLGVYFCSQTT
HALFTFGSGTTKLEIK (SEQ ID NO: 2072) 49H1 LB 1 VL amino
DVVMTQTPLSLPVSLGDQASISCRSSQSLVHINGNTYLLHWYLQKPGQS acid sequence

PKLLIYKVS NRFS GVPDR FSGSGSGTDFTL KISRVEAEDLG VYFCSQST
HVTFTFGSGTKLEIK (SEQ ID NO: 2073) 54C2.A1 VL amino
DVVMTQTPLSLPVSLGDQASISCRSSQSLVHINGNTYLQWYLQKPGQS acid
PKLLIYKVS NRFS GVPDR FSGSGSGTDFTL KISRVEAEDLG VYFCSQST
HLPYTFGGGTKLEIK (SEQ ID NO: 2074) 57D7.A1 VL amino
DVLMTQTPLSLPVSLGDQASISCRSSQSIVHSNGNTYLEWYLQKPGQS acid sequence
PKWYKVS NRFS GVPDR FSGSGSGTDFTL KISRVEAEDLG VYYCFQGS
HVPYTFGGGTKLEIK (SEQ ID NO: 2075) RS9.F6 VL amino
DVVMTQTPLSLPVSLGDQASISCRSSQSLVHNNGNTFLHWYLQKPGQ acid sequence;
SPKWYKVS NRFS GVPDR FSGSGSGTDFTL KISRVEAEDLG VYFCSQST RS.F10 VL amino
THVPPTFGGGTKLEIK (SEQ ID NO: 2076) acid sequence RS9.F6 and RS.F10
GYTFTSYWMH (SEQ ID NO: 2077) CDR-H1 R59.F6 and RS.F10
RSDPTTGGTNYNEKFKT (SEQ ID NO: 2078) CDR-H2 RS9.F6. RS.F10, KVS NRFS
(SEQ ID NO: 2079) 26E2.A3, 24B4.A1, 42E8.H1, 49H11.B1, 54C2.A1, and
57D7.A1 CDR-L2 2ID11 CDR-H1 GYTFTSYWIQ (SEQ ID NO: 2080) 2ID11 CDR-
H2 TIYPGDGDARYTQKFKG (SEQ ID NO: 2081) 2ID11 CDR-H3
ARNGITTAGYYAMDY (SEQ ID NO: 2082) 2ID11 CDR-L1 RASENIYSNLA
(SEQ ID NO: 2083) 2ID11 CDR-L2 AATNLAD (SEQ ID NO: 2084) 2ID11
CDR-L3 QHFWGTPYT (SEQ ID NO: 2085) 21D4.D1 CDR-H1 GYTFTDHAMH
(SEQ ID NO: 2086) 21D4.D1 CDR-H2 VISTYSGDTGYNQKFKG (SEQ ID NO:
2087) 21D4.D1 CDR-H3 AREGHYDDAMDY (SEQ ID NO: 2088) 21D4.D1 and
51D4 KSSQSLLDSDGKTYLN (SEQ ID NO: 2089) CDR-L1 21D4.D1 CDR-L2
VVSKLDS (SEQ ID NO: 2090) 21D4.D1 and 51D4 WQGTFFIPYT (SEQ ID
NO: 2091) CDR-L3 26D2 CDR-H1 GYTFTSYVMH (SEQ ID NO: 2092) 26D2
CDR-H2 YINPYTDGTYNEKFKG (SEQ ID NO: 2093) 26D2 CDR-H3
ARGEVRRYALDY (SEQ ID NO: 2094) 26D2 CDR-L1 KASEDIYNRLA (SEQ
ID NO: 2095) 26D2 CDR-L2 GATSLET (SEQ ID NO: 2096) 26D2 CDR-L3
QQYWSTPWT (SEQ ID NO: 2097) 26E2.A3 and DSEVFPISYMS (SEQ ID
NO: 2098) 24B4.A1 CDR-H1 26E2.A3 and DILPSIGGRIYGVKF (SEQ ID NO:
2099) 24B4.A1 CDR-H2 26E2.A3 and ARKDYGSLAY (SEQ ID NO: 2100)
24B4.A1 CDR-H3 26E2.A3, 24B4.A1, RSSQSLVHINGNTYLQ (SEQ ID NO: 2101)
and 54C2.A1 CDR-L1 26E2.A3 and SQSTHVPYT (SEQ ID NO: 2102) 24B4.A1
CDR-L3 3D3.A1 CDR-H1 GYTLSEYTMH (SEQ ID NO: 2103) 3D3.A1 CDR-H2
GVIPNSGGTSYNQKFRD (SEQ ID NO: 2104) 3D3.A1 CDR-H3
ARGDDSYRRGYALDY (SEQ ID NO: 2105) 3D3.A1 CDR-L1
KSSQSLLYSSNQKSYLA (SEQ ID NO: 2106) 3D3.A1 CDR-L2 WASTRES (SEQ
ID NO: 2107) 3D3.A1 CDR-L3 QQYFSYPPT (SEQ ID NO: 2108) 40H3.A4 CDR-
H1 GFNIKDTYMH (SEQ ID NO: 2109) 40H3.A4 CDR-H2
RIDPANGNTKYDPKFQG (SEQ ID NO: 2110) 40H3.A4 CDR-H3 ATLFAY (SEQ
ID NO: 2111) 40H3.A4 CDR-L1 RSSKSLLHSNGITYLY (SEQ ID NO: 2112)
40H3.A4 CDR-L2 QMSNLAS (SEQ ID NO: 2113) 40H3.A4 CDR-L3 AQNLELPT
(SEQ ID NO: 2114) 42E8.H1 and 49H11.B1 GYSITSDYAWN (SEQ ID NO:
2115) CDR-H1 42E8.H1 CDR-H2 YINYSGRITIYNPSLKS (SEQ ID NO: 2116)
42E8.H1 and 49H11.B1 ARWNGNYGFAY (SEQ ID NO: 2117) CDR-H3 42E8.H1
and 49H11.B1 RSSQSLVHINGNTYLH (SEQ ID NO: 2118) CDR-L1 42E8.H1 CDR-
L3 SQTTHALFT (SEQ ID NO: 2119) 49H11.B1 CDR-H2 YISFSGSTSYNPSLKS
(SEQ ID NO: 2120) 49H11.B1 CDR-L3 SQSTHVTFT (SEQ ID NO: 2121)
54C2.A1 CDR-H1 DSEVFPIAYMS (SEQ ID NO: 2122) 54C2.A1 CDR-H2
DILPSIGRRIYGVKFED (SEQ ID NO: 2123) 54C2.A1 CDR-H3 KDYGSLAY
(SEQ ID NO: 2124) 54C2.A1 CDR-L3 SQSTHLPYT (SEQ ID NO: 2125)

57D7.A1 CDR-H1 GFSLSRYSVY (SEQ ID NO: 2126) 57D7.A1 CDR-H2
MIWGGGNTDYN SALKS (SEQ ID NO: 2127) 57D7.A1 CDR-H3 YGGMDY (SEQ
ID NO: 2128) 57D7.A1 CDR-L1 RSSQSIVHSNGNTYLE (SEQ ID NO: 2129)
57D7.A1 CDR-L3 FQGSHVPYT (SEQ ID NO: 2130) RS9.F6-Fd
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQSPGRGLE
WIGRSDPTTGGTNYNEKFKTKATLTVDKPSSTAYMQLSSLTSDDSAV
YYCVRTSGTGDYWGQGTSLTVSSASTKGPSVFPLAPSSKSTSGGTAAL
GCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTH (SEQ ID NO: 2131) RS9.F6-Fd
fused to QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQSPGRGLE Fc
with LALAPG, WIGRSDPTTGGTNYNEKFKTKATLTVDKPSSTAYMQLSSLTSDDSAV
TfR binding, and YYCVRTSGTGDYWGQGTSLTVSSASTKGPSVFPLAPSSKSTSGGTAAL
knob mutations GCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPEAAGGP
SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH
NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAP
IEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGFYPSDIAVL
WESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQQGFVFSCSV
MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2132) RS9.F6-Fd fused to
QVQLQQPGAELVKPGASVKLSCKASGYTFTSYWMHWVKQSPGRGLE Fc with
LALAPG and WIGRSDPTTGGTNYNEKFKTKATLTVDKPSSTAYMQLSSLTSDDSAV hole
mutations YYCVRTSGTGDYWGQGTSLTVSSASTKGPSVFPLAPSSKSTSGGTAAL
GCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPEAAGGP
SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVH
NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALGAP
IEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGFYPSDIAVE
WESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSV
MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2133) 3D3.A1-Fd
EVQLQQSGPELVKPGASVKISCKTSGYTLSEYTMHWVIQSHGKSLEWI
GGVIPNSGGTSYNQKFRDKASLTVDKSSSTAYLELRSLTSEDSAVYYC
ARGDDSYRRGYALDYWGQGTSTVTVSSASTKGPSVFPLAPSSKSTSGG
TAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSV
TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTH (SEQ ID NO: 2134) 3D3.A1-
Fd fused to EVQLQQSGPELVKPGASVKISCKTSGYTLSEYTMHWVIQSHGKSLEWI Fc
with LALAPG, GGVIPNSGGTSYNQKFRDKASLTVDKSSSTAYLELRSLTSEDSAVYYC
TfR binding, and ARGDDSYRRGYALDYWGQGTSTVTVSSASTKGPSVFPLAPSSKSTSGG
knob mutations TAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSV
TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPEA
AGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNK
ALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGFYP
SDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQQGF
VFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2135) 3D3.A1-Fd fused to
EVQLQQSGPELVKPGASVKISCKTSGYTLSEYTMHWVIQSHGKSLEWI Fc with
LALAPG and GGVIPNSGGTSYNQKFRDKASLTVDKSSSTAYLELRSLTSEDSAVYYC
hole mutations ARGDDSYRRGYALDYWGQGTSTVTVSSASTKGPSVFPLAPSSKSTSGG
TAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSV
TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPEA
AGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG

VEVHNAKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGFYPS
ALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGFYPS
DIAVEWESNGQPENNYKTTTPVLDSDGSFFLVSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2136) Human TREM2
MEPLRLLILLFVTELSGAHNTTVFQGVAGQSLQVSCPYSMKH protein
WGRRKAWCRQLGEKGPCQRVSTHNLWLLSFLRRWNGSTAIT
DDTLGGTLTITLRNLQPHDAGLYQCQSLHGSEADTLRKVLVEVL
ADPLDHRDAGDLWFPGESESFEDAHVEHSISRSLLEGEIPFPPTSI
LLLACIFLIKILAAASALWAAAWHGQKPGTHPPSELDCGHDPGY QLQTLPLGRDT
(SEQ ID NO: 1) Human transferrin
MMDQARSAFSNLFGGPELSYTRFSLARQVDGDNHVMKLAVIDEEE receptor protein 1
NADNNTKANVTKPKRCSGSICYGTIAVIVFFLIGFMIGYLG YCKGVEP (TFR1)
KTECERLAGTESPVREEPGEDFPAARRLYWDDLKRKLSEKLDSTDFTG
TIKLLNENSYPREAGSQKDENLALYVENQFREFKLSKVWRDQHFVK
IQVKDSAQNSVIIVDKNGRLVYLVENPGGYVAYS KAATVTGKLVHAN
FGTKKDFEDLYTPVNGSIVIVRAGKITFAEKVANAESLNAIGVLIYMD
QTKFPIVNAELSFFGHAHLGTGDPYTPGFPSFNHTQFPSPRSSGLPNIPV
QTISRAAA EKLFGNMEGDCPSDWKTDSTCRMVTS ESKNVKLTVSNVL
KEIKILNIFGVIKGFVEPDHYVVVGAQRDAWGPGA AKSGVGTALLK
LAQMFSDMVLKDG FQPSRSIIFASWSAGDFG SVGATEWLEGY LSSLHL
KAFTYINLDKAVLGTSNFKVSASPLLYTLIEKTMQNVKHPVTGQFLYQ
DSNWASKVEKLTLDNAAFPFLAYSGIPAVSFCFCEDTDYPYL GTTMD
TYKELIERIPELNKVARAAAEVAGQFVIKLTHDVELNLDYERYNSQLL
SFVRDLNQYRADIKEMGLSLQWLYSARGDFFRATSRLTTDFGNAEKT
DRFVMKKLNDRVMRVEYHFLSPYVSPKES PFRHVFWGSGSHTLPALL
ENLKL RKQNNGAFNETLFRNQLALATWTIQGAANALSGDVWDIDNE F (SEQ ID
NO: 2137) Wild-type human Fc
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY sequence positions
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV 231-447 EU index
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF numbering
YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2138) Human IgG1 hinge
EPKSCDKTHTCPPCP (SEQ ID NO: 2139) sequence Clone CH3C.35.20
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2140) Clone CH3C.35.21
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2141) Clone CH3C.35.22
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSNYKTTTPVLDSDGSFFLYSKLTVTKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2142) Clone CH3C.35.23
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSNYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2143) Clone CH3C.35.24
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSNYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2144) Clone CH3C.35.21.17
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2145) Clone CH3C.35.20.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2146) Clone CH3C.35.20.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWASYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2147) Clone CH3C.35.20.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWVSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2148) Clone CH3C.35.20.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVSKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2149) Clone CH3C.35.20.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWASYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2150) Clone CH3C.35.20.6
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWVSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2151) Clone CH3C.35.21.a.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2152) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.21.a.2
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESYGTEWASWKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2153) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.21.a.3
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESYGTEWWSYKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2154) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.21.a.4
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESYGTEWSSYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2155) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.21.a.5
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESFGTEWASWKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2156) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.21.a.6
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESFGTEWWSYKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2157) Clone CH3C.35.23.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWSNYKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2158) Clone CH3C.35.23.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2159) Clone CH3C.35.23.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2160) Clone CH3C.35.23.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2161) Clone CH3C.35.23.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWANYKTTTPPVLDSDGSFFLYSKLTVTKKEEWQQ
GFVFSKSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 2162) Clone CH3C.35.23.6
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWVNYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2163) Clone CH3C.35.24.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESFGTEWSNYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2164) Clone CH3C.35.24.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWANYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2165) Clone CH3C.35.24.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWVNYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2166) Clone CH3C.35.24.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSNYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2167) Clone CH3C.35.24.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESFGTEWANYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2168) Clone CH3C.35.24.6
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESFGTEWVNYKTTTPVLDSDGSFFLSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2169) Clone
CH3C.35.21.17.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQG
FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2170) Clone CH3C.35.21.17.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWASYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2171) Clone
CH3C.35.21.17.3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWVSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2172) Clone
CH3C.35.21.17.4 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVSKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2173) Clone CH3C.35.21.17.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESFGTEWASYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2174) Clone
CH3C.35.21.17.6 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESFGTEWVSYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2175) Clones
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.N390
and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV CH3C.35.N163
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSNYKTTTPVLDSDGSFFLYSKLTVTKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2176) Clone CH3C.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGLVWVGKTTTPVLDSDGSFFLYSKLTVAKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2177) Clone CH3C.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTWVSHYKTTTPVLDSDGSFFLYSKLTVSKSEWQQ
GYVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2178) Clone CH3C.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSQYKTTTPVLDSDGSFFLYSKLTV EKSDWQQ
GHVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2179) Clone CH3C.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESVGPWALYKTTTPVLDSDGSFFLYSKLTVLKSEWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2180) Clone CH3C.17
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTWWSKYKTTTPVLDSDGSFFLYSKLTVSKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2181) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWAVYKTTTPVLDSDGSFFLYSKLTVPKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2182) Clone CH3C.21
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGLVWVG YKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2183) Clone CH3C.25
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESMGHVWVG YKTTTPPVLDSDGSFFLYSKLTVDKSTWQ
QGWFVSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2184) Clone CH3C.34
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGLVWVFSKTTTPPVLDSDGSFFLYSKLTPVKSTWQQG
WVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2185) Clone CH3C.35
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKSEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2186) Clone CH3C.44
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2187) Clone CH3C.51
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVG YKTTTPPVLDSDGSFFLYSKLTVSKSEWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2188) Clone CH3C.3.1-3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVATKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2189) Clone CH3C.3.1-9
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGPVWVHTKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2190) Clone CH3C.3.2-5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVDQKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2191) Clone CH3C.3.2-19
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVNQKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2192) Clone CH3C.3.2-1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVNFKTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2193) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY variant
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESLGHVWAVYKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2194) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY variant
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESLGHVWAVYKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2195) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY variant
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVYWESLGHVWAVYKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2196) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY variant
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWAVYQTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2197) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY variant
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWAVYFTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2198) Clone CH3C.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY variant
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWAVYHTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2199) Clone CH3C.35.13
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESLGHVWAVYKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2200) Clone CH3C.35.14
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWAVYQTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2201) Clone CH3C.35.15
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESLGHVWAVYQTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2202) Clone CH3C.35.16
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWVESLGHVWVNQKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2203) Clone CH3C.35.17
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVNQQTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2204) Clone CH3C.35.18
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWVESLGHVWVNQQTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2205) Clone CH3C.35.19
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWVESYGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2206) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.K165Q
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSSYQTTTPPVLDSDGSFFLYSKLTVTKSEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2207) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
CH3C.35.N163.K165Q
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSNYQTTTPPVLDSDGSFFLYSKLTVTKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2208) Clone CH3C.35.21.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKSEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2209) Clone CH3C.35.21.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTKSEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2210) Clone CH3C.35.21.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2211) Clone CH3C.35.21.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTGEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2212) Clone CH3C.35.21.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY

VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQG
FVFSCWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2213) Clone CH3C.35.21.6
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTKEEWQQG
FVFSCWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2214) Clone CH3C.35.21.7
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQG
FVFTCWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2215) Clone CH3C.35.21.8
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQG
FVFTCGVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2216) Clone CH3C.35.21.9
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQG
FVFECWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2217) Clone CH3C.35.21.10
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQG
FVFKCWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2218) Clone CH3C.35.21.11
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTPEEWQQG
FVFKCWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2219) Clone CH3C.35.21.12
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQ
GFVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2220) Clone CH3C.35.21.13
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTGEEWQQ
GFVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2221) Clone CH3C.35.21.14
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREEWQQ
GFVFTCWVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2222) Clone
CH3C.35.21.15 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY

VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTGEEWQQ
GFVFTCWMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2223) Clone
CH3C.35.21.16 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWEWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTREETWQQ
GFVFTCGVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2224) Clone
CH3C.35.21.18 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVLWESYGTEWSSYRTTPPVLDSDGSFFLYSKLTVTKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2225) Clone CH3B.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPRFDYVTTLPSPRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYGFHDLSLSPGK (SEQ ID NO: 2226) Clone CH3B.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPRFDMVTTLPSPRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYGFHDLSLSPGK (SEQ ID NO: 2227) Clone CH3B.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPRFEYVTTLPSPRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYGFHDLSLSPGK (SEQ ID NO: 2228) Clone CH3B.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPRFEMVTTLPSPRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYGFHDLSLSPGK (SEQ ID NO: 2229) Clone CH3B.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPRFELVTTLPSPRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYGFHDLSLSPGK (SEQ ID NO: 2230) Clone CH2A2.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVEFIWYV
DGVDVRYEWQLPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVS
NKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFY
PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQG
NVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2231) Clone CH2A2.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVGFVWY
VDGVPVSWEWYWPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2232) Clone CH2A2.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQFDWY

VDGVMVRREYVREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2233) Clone CH2A2.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVSFEWY
VDGVPVRWEWQWPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2234) Clone CH2A2.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVAFTWY
VDGVPVRWEWQNPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2235) Clone CH2C.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDPQTPPWEVKFNWY
VDGVEVHNAKTKPREEEYYTYRVSLSLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2236) Clone CH2C.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDPPSPWEVKFNWY
VDGVEVHNAKTKPREEEYYSNYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2237) Clone CH2C.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDPQTPPWEVKFNWY
VDGVEVHNAKTKPREEEYYSNYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2238) Clone CH2C.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVFRGPPWEVKFNWY
VDGVEVHNAKTKPREEEYYHDYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2239) Clone CH2C.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDPQTPVWEVKFNWY
VDGVEVHNAKTKPREEEYYSNYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2240) Clone CH2D.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSVPPRMVKFNWY
VDGVEVHNAKTKSLTSQHNSTVRVSVLTVLHQDWLNGKEYKCKVS
NKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFY
PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQG
NVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2241) Clone CH2D.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSVPPWMVKFNW
YVDGVEVHNAKTKSLTSQHNSTVRVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2242) Clone CH2D.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDMWEYVKFNW

YVDGVEVHNAKTKPWIAQPNSTWRVVSVLTVLHQDWLNGKEYKCK
KVS NKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2243) Clone CH2D.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDDWTVWKFNW
YVDGVEVHNAKTKPWIAQPNSTWRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2244) Clone CH2D.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDDWVWKFNW
YVDGVEVHNAKTKPWKLQLNSTWRVVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2245) Clone CH2E3.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPWVWFYW
YVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCS
VVNIALWWSIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2246) Clone CH2E3.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPVVGFRWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCRVS
NSALTWKIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2247) Clone CH2E3.3
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPVVGFRWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCRVS
NSALSWRIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2248) Clone CH2E3.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPIVGFRWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCRVS
NSALRWRIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2249) Clone CH2E3.5
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPAVGFEWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCQV
FNWALDWVIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRW
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2250) Fc sequence
with hole APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
mutations VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2251) Fc sequence with
hole APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY and LALA
mutations VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQ
GNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2252) Fc sequence with
hole APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY and YTE

mutations VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQ
GNVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2253) Fc sequence with
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY hole, LALA,
and YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQ
GNVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2254) Fc sequence with
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY knob mutation
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQ
QGNVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2255) Fc sequence with
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY knob and LALA
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQ
QGNVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2256) Fc sequence with
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY knob and YTE
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQ
QGNVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2257) Fc sequence with
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY knob, LALA,
and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE mutations
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQ
QGNVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2258) Clone CH3C.35.21
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob
mutation VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVWWESYGTEWSSYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2259) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVWWESYGTEWSSYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2260)
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY Clone CH3C.35.21
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV with knob and
YTE SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVWWESYGTEWSSYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2261) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE
mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVWWESYGTEWSSYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2262) Clone CH3C.35.21
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole

mutations VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVWWESYGTEWSSYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2263) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVWWESYGTEWSSYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2264) Clone CH3C.35.21
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVWWESYGTEWSSYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2265) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVWWESYGTEWSSYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2266) Clone CH3C.35.20.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob
mutation VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2267) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2268) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2269) Clone CH3C.35.20.1
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2270) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2271) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2272) Clone CH3C.35.20.1
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY hole mutations

VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSSYKTTTPVLDS DGSFFLVSKLTVTKEEWQQG
FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2273) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSSYKTTTPVLDS DGSFFLVSKLTVTKEEWQQG
FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2274) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVEWESFGTEWSSYKTTTPVLDS DGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2275) Clone CH3C.35.20.1
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSSYKTTTPVLDS DGSFFLVSKLTVTKEEWQQG
FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2276) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSSYKTTTPVLDS DGSFFLVSKLTVTKEEWQQG
FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2277) Clone CH3C.35.20.1
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVEWESFGTEWSSYKTTTPVLDS DGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2278) Clone CH3C.35.23.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob
mutation VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWANYKTTTPVLDS DGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2279) Clone
CH3C.35.23.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and LALA
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWANYKTTTPVLDS DGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2280) Clone
CH3C.35.23.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWANYKTTTPVLDS DGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2281) Clone
CH3C.35.23.2 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob and YTE
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG

FYPDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2282) Clone
CH3C.35.23.2 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
and YTE mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2283) Clone
CH3C.35.23.2 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, LALAPG,
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE
mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2284) Clone
CH3C.35.23.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with hole mutations
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2285) Clone CH3C.35.23.2
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2286) Clone CH3C.35.23.2
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2287) Clone
CH3C.35.23.2 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with hole and YTE
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2288) Clone CH3C.35.23.2
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2289) Clone CH3C.35.23.2
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2290) Clone
CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob mutation
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG

FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2291) Clone
CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and LALA
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2292) Clone
CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2293) Clone
CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob and YTE
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2294) Clone
CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
and YTE mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2295) Clone
CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, LALAPG,
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE
mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLYSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2296) Clone
CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with hole mutations
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLVSKLTVTKEEWQQ
GFVFSCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2297) Clone CH3C.35.23.3
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLVSKLTVTKEEWQQ
GFVFSCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2298) Clone CH3C.35.23.3
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPDIAVEWESYGTEWVNYKTTTPVLDSGDSFFLVSKLTVTKEEWQ
QGFFVFCSSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2299) Clone
CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with hole and YTE

VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2300) Clone CH3C.35.23.3
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2301) Clone CH3C.35.23.3
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2302) Clone
CH3C.35.23.4 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob mutation
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2303) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2304) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2305) Clone CH3C.35.23.4
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2306) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2307) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2308) Clone CH3C.35.23.4
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole
mutations VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLVSKLTVSKEEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2309) Clone CH3C.35.23.4

APEAAGGPSVFLFPPKPKDTLMSRTPEVTCVVVDVSHEDPEVKFNWY with hole and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2310) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2311) Clone CH3C.35.23.4
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2312) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole.
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2313) Clone CH3C.35.23.4
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2314) Clone
CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutation
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2315) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALA mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2316) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2317) Clone
CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
YTE mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2318) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALA, and YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF
mutations FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQ

QGFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2319) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG, and YTE
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2320) Clone
CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with hole VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2321) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with hole and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALA mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2322) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with hole and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2323) Clone
CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with hole and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
YTE mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2324) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with hole, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALA,
and YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations
YPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2325) Clone
CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with hole, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG, and YTE
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG mutations
FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2326) Clone CH3C.35.23
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob
mutation VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2327) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFVFSVMSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2328) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations

SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2329) Clone CH3C.35.23
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2330) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2331) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ
QGFFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2332) Clone CH3C.35.23
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole
mutations VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2333) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
LALA VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2334) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2335) Clone CH3C.35.23
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole and
YTE VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2336) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV YTE
mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2337) Clone CH3C.35.23
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2338) CH3C motif
YxTEWSS (SEQ ID NO: 2339) CH3C motif TxxExxxxF (SEQ ID NO: 2340)
mulgGI 3' VH PCR GGACAGGGATCCAGAGTTCC (SEQ ID NO: 2341) primer

mu3/4G3 3' VH PCR AGCTGGGAAGGTGTGCACAC (SEQ ID NO: 2342) primer
mu3/4G3 3' VH PCR CAGGGGCCAGTGGATAGAC (SEQ ID NO: 2343) primer
muCkappa.1 3' VL GACATTGATGTCTTTGGGGT (SEQ ID NO: 2344) PCR primer
muCkappa.2 3' VL TTCACTGCCATCAATCTTCC (SEQ ID NO: 2345) PCR primer
7B10.A2 VH amino
EVQLQQSGPELVKPGASVKMSCKASGYTFTDYNMHVVKQSHGKSLE acid sequence
WIGYINPNNGGTTYNQKFKGKATLTVNKSSTAYMELRSLTSEDSAV
YYCATYNNHYFDSWGQGTTTLTVSS (SEQ ID NO: 2346) 7B10.A2 CDR-H1
GYTFTDYNMH (SEQ ID NO: 2347) 7B10.A2 CDR-H2 YINPNNGGTTYNQKFKG
(SEQ ID NO: 2348) 7B10.A2 CDR-H3 ATYNNHYFDS (SEQ ID NO: 2349)
7B10.A2 VL amino
DIQMTQTTSSLSASLGDRVTISCSASQGISNYLNWYQQKPDGTVKLLI acid sequence
YYTSNLHSGVPSRFSGSGSGTDYSLTISNLEPEDIATYYCQQYSNLPYT FGGGGTKLEIK
(SEQ ID NO: 2350) 7B10.A2 CDR-L1 SASQGISNYLN (SEQ ID NO: 2351)
7B10.A2 CDR-L2 YTSNLHS (SEQ ID NO: 2352) 7B10.A2 CDR-L3 QQYSNLPYT
(SEQ ID NO: 2353) 51D4 VH amino acid
QVHLQQSGPEVVRPGVSVKISCKGSGYTFTDYGMHWVKQSHAKSLE sequence
WIGVISTYNGNTSYNQKYKGKATVTVDKPSSTAYMELVRLTSEDSAI
YYCARDFGYVPFDYWQGTTTLTVSS (SEQ ID NO: 2354) 51D4 CDR-H1
GYTFTDYGMH (SEQ ID NO: 2355) 51D4 CDR-H2 VISTYNGNTSYNQKYKG
(SEQ ID NO: 2356) 51D4 CDR-H3 ARDFGYVPFDY (SEQ ID NO: 2357) 51D4
VL amino acid DVVMTQTPLTSLVITIGQPASISCKSSQSLLDSDGKTYLNWLLQRP
sequence GQSPKRLIYLVSYLDSGVPDRFTGSGSGTDFTLKISRVEADDLGV
YYCWQGTTHFPYTFGGGGTKLEIK (SEQ ID NO: 2358) 51D4 CDR-L2 LVSYLDS
(SEQ ID NO: 2359) CDR-H1 consensus GX2X3X4X5X6X7X8X9X10X11, wherein
X2 is Y or F; X3 is T, N, or S; X4 is F, sequence L, or I; X5 is
T, S, or K; X6 is D, S, or E; X7 is D or absent; X8 is H, Y, or
T; X9 is A, N, G, V, W, T, or Y; X10 is M, I, or W; and X11 is
H, Q, or N (SEQ ID NO: 2360) CDR-H1 consensus
GYX3X4X5X6X7X8X9X10X11, wherein X3 is T or S; X4 is F, L, or I;
X5 is sequence T or S; X6 is D, S, or E; X7 is D or absent; X8 is H
or Y; X9 is A, N, G, V, W, T, or A; X10 is M, I, or W; and X11
is H, Q, or N (SEQ ID NO: 2361) CDR-H1 consensus
GX2X3X4X5X6X8X9X10X11, wherein X is Y or F; X3 is T or N; X4
is F, L, sequence or I; X is T, S, or K; X6 is D, S, or E; X8 is H,
Y, or T; X is A, N, G, V, W, T, Y, or A; X11 is M or I; and X
is H or Q (SEQ ID NO: 2362) CDR-H1 consensus GYTX4X5X6X8X9X10X11,
wherein X4 is F or L; X is T or S; X6 is D, S, or sequence E; X8
is H, Y; X9 is A, N, G, V, W, T; X10 is M or I; and X11 is H
or Q (SEQ ID NO: 2363) CDR-H2 consensus
X1X2X3X4X5X6X7X8X9X10YX12X13X14X15X16X17, wherein X1 is D, V, Y, R,
sequence G, or T; X2 is I, S, or V; X3 is L, S, N, D, I, or Y; X4
is P, T, or absent; X5 is S, Y, N, T, A, G, or F; X6 is I, S, N,
T, or D; X7 is G or D; X8 is G, D, N, R, or S; X9 is R, T, or
A; X10 is I, G, S, K, T, N, or R; X12 is G, N, D, or T; X13 is
V, Q, E, or P; X14 is K or S; X15 is F, Y or L; X16 is K, R,
Q, or is absent; and X17 is G, T, D, S, or is absent (SEQ ID NO:
2364) CDR-H2 consensus X1X2X3X4X5X6X7X8X9X10YX12X13X14X15X16X17,
wherein X1 is V, Y, R, G, sequence or T; X2 is I, S, or V; X3 is S,
N, D, I, or Y; X4 is P, T, or absent; X5 is Y, N, T, A, G, or F;

X6 is S, N, T, or D; X7 is G or D; X8 is G, D, N, R, or S; X9 is T, or A; X10 is I, G, S, K, T, N, or R; X12 is N, D, or T; X13 is Q, E, or P; X14 is K or S; X15 is F, Y or L; X16 is K, R, or Q; and Xi is G, T, D, or S (SEQ ID NO: 2365) CDR-H2 consensus X1X2X3X4X5X6X7X8X9X10YX12X13KX15X16X17, wherein Xi is V, Y, R, G, sequence or T; X2 is I, S, or V; X3 is S, N, D, I, or Y; X4 is P or T; X5 is Y, N, T, A, or G; X6 is S, N, T, or D; X7 is G or D; X8 is G, D, or N; X9 is T, or A; X10 is G, S, K, T, N, or R; X12 is N, D, or T; X13 is Q, E, or P; X15 is F or Y; X16 is K, R, or Q; and X17 is G, T, or D (SEQ ID NO: 2366) CDR-H3 consensus ARX3X4X5X6X7X8X9X10YAX13DY, wherein X3 is G or N; X4 is D or G; sequence X5 is D or I; X6 is S or T; X7 is Y or T; X8 is R or A; X9 is R or G; X10 is G or Y; and X13 is L or M (SEQ ID NO: 2367) CDR-L1 consensus X1SSX4SLX7X8X9X10X11X12X13X14X15LX17, wherein X1 is R or K; sequence X4 is Q or K; X7 is V or L; X8 is H, D, or Y; X9 is I, N, or S; X10 is S or absent; X11 is D or N; X12 is G or Q; X13 is N, I, or K; X14 is T or S; X15 is Y or F; and X17 is Q, H, Y, N, or A (SEQ ID NO: 2368) CDR-L1 consensus X1ASX4X5IX7X8X9LX11, wherein X1 is R, K, or S; X4 is E or Q; X5 is N, sequence D, or G; X7 is Y or S; X8 is S or N; X9 is N, R, or Y; and X11 is A or N (SEQ ID NO: 2369) CDR-L2 consensus X1X2SX4X5X6S, wherein Xi is K, Q, Y, V, or L; X2 is V, M, or T; X4 is N, sequence K, or Y; X5 is R or L; and X6 is F, A, H, or D (SEQ ID NO: 2370) CDR-L3 consensus X1X2X3X4X5X6X7X8T, wherein Xi is S, W, or Q; X2 is Q or H; X3 is S, T, sequence G, Y, or F; X4 is T, F, W, S; X5 is H, S, G, or N; X6 is V, A, F, Y, T, or L; X7 is P, T, or L; and X8 is Y, F, P, or W (SEQ ID NO: 2371) CDR-L3 consensus QX2X3X4X5X6PX8T, wherein X2 is Q or H; X3 is Y or F; X4 is F, W, or S; sequence X5 is S, G, or N; X6 is Y, T, or L; and X8 is P, Y, or W (SEQ ID NO: 2372) Human TREM2 SGAHNTTVFQGVAGQSLQVSCP YDSMKHWGRRKAWCRQLGEK extracellular domain GPCQRV VSTHNLWLLSFLRRWNGSTAITDDTLGGTTLTITLRNLQP (ECD) amino acid HDAGLYQCQSLHGSEADTLRKVLVEVLADPLDHRDAGDLWFPG sequence (without ESESFEDAHVEHSISRSLLEGEIPFPPTS (SEQ ID NO: 2373) signal peptide and His tag) Human TREM2 DLWFPG ESES (SEQ ID NO: 2374) peptide Human TREM2 DLWFPG ESE (SEQ ID NO: 2375) peptide 9-mer amino acid sequence Human TREM2 DLWFP (SEQ ID NO: 2376) peptide sequence (residues 140-144 of full-length TREM2) Clone CH3C.18.3.4-1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY (CH3C.3.4-1) VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVEWESWGFVWSTYKTTTPVLDS DGSFFLYSKLTPVPSNWQQ GFVVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2377) Clone CH3C.18.3.4-APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY 19 (CH3C.3.4-19) VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVEWESWGHVWSTYKTTTPVLDS DGSFFLYSKLTPVPSNWQQ GYVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2378) Clone CH3C.18.3.2-3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY (CH3C.3.2-3) VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVEQKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2379) Clone CH3C.18.3.2-
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY 14 (CH3C.3.2-14)
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVGKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2380) Clone CH3C.18.3.2-
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY 24 (CH3C.3.2-24)
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVHTKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2381) Clone CH3C.18.3.4-
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY 26 (CH3C.3.4-26)
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESWGTWGTYKTTTPPVLDSDGSFFLYSKLTPVKS NWQQ
GYVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2382) Clone CH3C.18.3.2-
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY 17 (CH3C.3.2-17)
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESLGHVWVGKTTTPPVLDSDGSFFLYSKLTPVKSTWQQ
GWVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2383) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWSSYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2384) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVSKSEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2385) Clone
CH3C.35.23.1.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2386) Clone CH3C.35.S413
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSSYKTTTPPVLDSDGSFFLYSKLTVSKSEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2387) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.3.1
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2388) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.N390.1
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESYGTEWSNYKTTTPVLDSDGSFFLYSKLTVSKSEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2389) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.6.1
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESFGTEWVNYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2390) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVWWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2391) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with knob,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVWWESYGTEWSSYKTTTPVLDSDGSFFLYSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2392) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVWWESYGTEWSSYKTTTPVLDSDGSFFLVSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2393) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
YTE mutations SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVWWESYGTEWSSYKTTTPVLDSDGSFFLVSKLTVTKEEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2394) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob mutation
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2395) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and
LALA SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVEWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2396) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and
LALAPG SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVEWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2397) Clone
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and
YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVEWESFGTEWSSYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ
GFVVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2398) Clone
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,

LALA, and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG YTE mutations FYPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLYSKLTVSKEEWQQ GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2399) Clone APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob, LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG YTE mutations FYPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLYSKLTVSKEEWQQ GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2400) Clone APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLVSKLTVSKEEWQQG FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2401) Clone APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and LALA SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLVSKLTVSKEEWQQG FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2402) Clone APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and LALAPG SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG mutations FYPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLVSKLTVSKEEWQQ GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2403) Clone APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLVSKLTVSKEEWQQG FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2404) Clone APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole, LALA, and YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLVSKLTVSKEEWQQG FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2405) Clone APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole, LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG YTE mutations FYPSDIAVEWESFGTEWSSYKTTTPVLDSGDGSFFLVSKLTVSKEEWQQ GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2406) Clone APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob mutation SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG FYPSDIAVEWESYGTEWANYKTTTPVLDSGDGSFFLYSKLTVSKSEWQ QGFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2407) Clone APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and LALA SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations FYPSDIAVEWESYGTEWANYKTTTPVLDSGDGSFFLYSKLTVSKSEWQ QGFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2408) Clone APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and

LALAPG SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVSKSEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2409) Clone
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and
YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVSKSEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2410) Clone
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,
LALA, and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG YTE
mutations FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVSKSEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2411) Clone
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,
LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
YTE mutations FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLYSKLTVSKSEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2412) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2413) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and
LALA SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2414) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and
LALAPG SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG mutations
FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2415) Clone
APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and
YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2416) Clone
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole, LALA,
and YTE SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations
YPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQQ
GFVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2417) Clone
APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole,
LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
YTE mutations FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQ
QG FVFSCSV MHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2418) Clone
CH3C.35.23.1.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutation

SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2419) Clone
CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALA mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2420) Clone
CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2421) Clone
CH3C.35.23.1.1 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
YTE mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2422) Clone
CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
and YTE mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2423) Clone
CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with knob, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
LALAPG, and YTE
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLYSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2424) Clone
CH3C.35.23.1.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY
with hole VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLVSKLTVSKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2425) Clone CH3C.35.23.1.1
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALA mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLVSKLTVSKEEWQQG
FVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2426) Clone CH3C.35.23.1.1
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG mutations
SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
FYPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLVSKLTVSKEEWQQ
GFVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2427) Clone
CH3C.35.23.1.1 APELLGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY
with hole and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
YTE mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLVSKLTVSKEEWQQG

FVFSCSVMSHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2428) Clone CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and YTE mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVEWESFGTEWSNYKTTTPVLDSGDSFFLVSKLTVSKEEWQQG

FVFSCSVMSHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2429) Clone CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLYITREPEVTCVVVDVSHEDPEVKFNWY with hole, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG, and YTE SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG mutations FYPDSIAVEWESFGTEWSNYKTTTPVLDSGDSFFLVSKLTVSKEEWQQG

GFVFSCSVMSHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2430) Clone CH3C.35.20.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY M428L and N434S VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVEWESFGTEWSSYKTTTPVLDSGDSFFLYSKLTVTKEEWQQG

FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2431) Clone CH3C.35.20.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF YPSDIAVEWESFGTEWSSYKTTTPVLDSGDSFFLYSKLTVTKEEWQQG

GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2432) Clone CH3C.35.20.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF N434S mutations FYPDSIAVEWESFGTEWSSYKTTTPVLDSGDSFFLYSKLTVTKEEWQQG

GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2433) Clone CH3C.35.20.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF N434S mutations FYPDSIAVEWESFGTEWSSYKTTTPVLDSGDSFFLYSKLTVTKEEWQQG

GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2434) Clone CH3C.35.20.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVEWESFGTEWSSYKTTTPVLDSGDSFFLVSKLTVTKEEWQQG

FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2435) Clone CH3C.35.20.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N434S SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESFGTEWSSYKTTTPVLDSGDSFFLVSKLTVTKEEWQQG

FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2436) Clone CH3C.35.20.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG N434S mutations FYPDSIAVEWESFGTEWSSYKTTTPVLDSGDSFFLVSKLTVTKEEWQQG

GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2437) Clone APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1 with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N434S SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF mutations

YPSDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLYSKLTVSKEEWQQG
FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2438) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and
M428L and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
N434S mutations FYPDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLYSKLTVSKEEWQQG
GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2439) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,
LALA, and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
M428L and N434S
FYPDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLYSKLTVSKEEWQQ mutations
GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2440) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,
LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
M428L and N434S
FYPDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLYSKLTVSKEEWQQ mutations
GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2441) APELLGGP
SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY Clone
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV CH3C.35.20.1.1
with SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF hole and
M428L and YPSDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLVSKLTVSKEEWQQG
N434S mutations FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2442) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole, LALA,
and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF M428L and
N434S YPSDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLVSKLTVSKEEWQQG mutations
FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2443) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.20.1.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole,
LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
M428L and N434S
FYPDIAVEWESFGTEWSSYKTTTPVLDSDFSFFLVSKLTVSKEEWQQ mutations
GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2444) Clone CH3C.35.21
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations
SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVWWESYGTEWSSYKTTTPVLDSDFSFFLYSKLTVTKEEWQQ
GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2445) Clone CH3C.35.21
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and
M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
N4345 mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
FYPDIAVWWESYGTEWSSYKTTTPVLDSDFSFFLYSKLTVTKEEWQ
QG FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2446) Clone CH3C.35.21
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob,
LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
M428L and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
N434S mutations FYPDIAVWWESYGTEWSSYKTTTPVLDSDFSFFLYSKLTVTKEEWQ
QG FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2447) Clone CH3C.35.21

APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPSDIAVWWESYGTEWSSYKTTTPVLDSGDSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2448) Clone CH3C.35.21 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVWWESYGTEWSSYKTTTPVLDSGDSFFLVSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2449) Clone CH3C.35.21 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole. LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVWWESYGTEWSSYKTTTPVLDSGDSFFLVSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2450) Clone CH3C.35.21 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF N434S mutations FYPSDIAVWWESYGTEWSSYKTTTPVLDSGDSFFLVSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2451) Clone CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVLWESYGTEWASYKTTTPVLDSGDSFFLYSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2452) Clone CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N434S SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations FYPSDIAVLWESYGTEWASYKTTTPVLDSGDSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2453) Clone CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALA, and M428L SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG and N4345 mutations FYPSDIAVLWESYGTEWASYKTTTPVLDSGDSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2454) Clone CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG M428L and N434S FYPSDIAVLWESYGTEWASYKTTTPVLDSGDSFFLYSKLTVTKEEWQ mutations QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2455) Clone CH3C.35.21.17.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVLWESYGTEWASYKTTTPVLDSGDSFFLVSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2456) Clone CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY

with hole, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALA, and M428L SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF and N4345 mutations YPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2457) Clone CH3C.35.21.17.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG M428L and N434S FYPSDIAVLWESYGTEWASYKTTPPVLDSDGSFFLVSKLTVTKEEWQ mutations QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2458) Clone CH3C.35.23 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2459) Clone CH3C.35.23 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N434S mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2460) Clone CH3C.35.23 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2461) Clone CH3C.35.23 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2462) Clone CH3C.35.23 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2463) Clone CH3C.35.23 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole. LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2464) Clone CH3C.35.23 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG N434S mutations FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2465) Clone CH3C.35.23.1.1 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVEWESFGTEWSNYKTTPPVLDSDGSFFLYSKLTVSKEEWQQG FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2466) Clone CH3C.35.23.1.1

APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations FYPDSIAVEWESFGTEWSNYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2467) Clone CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPDSIAVEWESFGTEWSNYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2468) Clone CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG M428L and N434S FYPDSIAVEWESFGTEWSNYKTTTPVLDSDGSFFLYSKLTVSKEEWQQ mutations GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2469) APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY Clone CH3C.35.23.1.1 VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV with hole and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF M428L and N434S YPSDIAVEWESFGTEWSNYKTTTPVLDSDGSFFLVSKLTVSKEEWQQG mutations FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2470) Clone CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF N434S mutations YPSDIAVEWESFGTEWSNYKTTTPVLDSDGSFFLVSKLTVSKEEWQQG FVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2471) Clone CH3C.35.23.1.1 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG M428L and N434S FYPDSIAVEWESFGTEWSNYKTTTPVLDSDGSFFLVSKLTVSKEEWQQ mutations GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2472) Clone CH3C.35.23.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF YPSDIAVEWESYGTEWANYKTTTPVLDSDGSFFLYSKLTVTKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2473) Clone CH3C.35.23.2 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG FYPDSIAVEWESYGTEWANYKTTTPVLDSDGSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2474) Clone CH3C.35.23.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPDSIAVEWESYGTEWANYKTTTPVLDSDGSFFLYSKLTVTKEEWQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2475) Clone CH3C.35.23.2 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPDSIAVEWESYGTEWANYKTTTPVLDSDGSFFLYSKLTVTKEEWQ

QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2476) Clone CH3C.35.23.2
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and
M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
N4345 mutations SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
YPSDIAVEWESYGTEWANYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2477) Clone CH3C.35.23.2
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole,
LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
M428L and N434S
SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations
YPSDIAVEWESYGTEWANYKTTPPVLDSDGSFFLVSKLTVTKEEWQQ
GFFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2478) Clone CH3C.35.23.2
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole,
LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and
M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG
N434S mutations FYPDSIAVEWESYGTEWANYKTTPPVLDSDGSFFLVSKLTVTKEEWQ
QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2479) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and
N4345 SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF mutations
YPSDIAVEWESYGTEWANYKTTPPVLDSDGSFFLYSKLTVSKSEWQQ
GFFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2480) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob and
M428L and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
N434S mutations FYPDSIAVEWESYGTEWANYKTTPPVLDSDGSFFLYSKLTVSKSEWQ
QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2481) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,
LALA, and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
M428L and N434 S
FYPDSIAVEWESYGTEWANYKTTPPVLDSDGSFFLYSKLTVSKSEWQ mutations
QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2482) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV knob,
LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG
M428L and N434S FYPDSIAVEWESYGTEWANYKTTPPVLDSDGSFFLYSKLTVSKSEWQ
mutations QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2483) Clone
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole and
M428L and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF
N434S mutations YPSDIAVEWESYGTEWANYKTTPPVLDSDGSFFLVSKLTVSKSEWQQ
GFFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2484) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole, LALA,
and SNKALPAIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF M428L and
N434S YPSDIAVEWESYGTEWANYKTTPPVLDSDGSFFLVSKLTVSKSEWQQ mutations
GFFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2485) Clone
APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY CH3C.35.23.2.1
with VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV hole,

LALAPG, and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGM428L and N434S FYPSDIAVEWESYGTEWANYKTTTPPVLDSDGSFFLVSKLTVSKSEWQ mutations QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2486) Clone CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2487) Clone CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2488) Clone CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF N434S mutations FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2489) Clone CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF N434S mutations FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLYSKLTVTKEEWQQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2490) Clone CH3C.35.23.3 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2491) Clone CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole. LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2492) Clone CH3C.35.23.3 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF N434S mutations FYPSDIAVEWESYGTEWVNYKTTTPPVLDSDGSFFLVSKLTVTKEEWQQ QGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2493) Clone CH3C.35.23.4 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2494) Clone CH3C.35.23.4 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKGF FYPSDIAVEWESYGTEWSNYKTTTPPVLDSDGSFFLYSKLTVSKEEWQQ GGFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2495) Clone CH3C.35.23.4 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob,

LALA, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2496) Clone CH3C.35.23.4 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with knob, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG N434S mutations FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLYSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2497) Clone CH3C.35.23.4 APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole and M428L VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and N4345 mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2498) Clone CH3C.35.23.4 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2499) Clone CH3C.35.23.4 APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY with hole, LALAPG, VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV and M428L and SNKALGAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKG N434S mutations FYPSDIAVEWESYGTEWSNYKTTPPVLDSDGSFFLVSKLTVSKEEWQQ GFVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2500) Fc sequence with hole APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY and M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQ GNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2501) Fc sequence with APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY hole, LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N4345 SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLSCAVKGF mutations YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQ GNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2502) Fc sequence with APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY knob and M428L and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV N434S mutations SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2503) Fc sequence with APEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWY knob, LALA, and VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV M428L and N434S SNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLWCLVKG mutations FYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQ QGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2504) SS2_NHis_TREM2 MPALLSLVSLLSVLLMGCVAETGGSGHHHHHSGTHNTTVFQGVAG QSLQVSCPYDSMIKHWGRRKAWCRQLGEKGPCQRVVSTHNLWLLSFL RRWNGSTAITDDTLGGTLTITLRNLQPHDAGLYQCQSLHGSEADTLR KVLVEVLADPLDHRDAGDLWFPGESESFEDAHVEHSISRSLLEGEIPFP PTSAS (SEQ ID NO: 2505)

[0448] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in the above tables as well as specific combinations thereof and other embodiments of the anti-TREM2 antibody described in the '841 application and herein may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

H. PCT Patent Application Publication No. WO2019/118513A1

[0449] In some embodiments, the TREM2 agonist is an antibody or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2019/118513A1 (“the '513 application”), which is incorporated by reference herein, in its entirety.

[0450] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3, and a heavy chain variable domain comprising a CDRH1, CDRH2, and CDRH3 disclosed in the '513 application specification. In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '513 application specification.

[0451] In some embodiments, the antibody comprises a CDR-H1 comprising the sequence set forth in SEQ ID NO:2514, a CDR-H2 comprising the sequence set forth in SEQ ID NO:2515, a CDR-H3 comprising the sequence set forth in SEQ ID NO: 11, a CDR-L1 comprising the sequence set forth in SEQ ID NO:2517, a CDR-L2 comprising the sequence set forth in SEQ ID NO:2518, and a CDR-L3 comprising the sequence set forth in SEQ ID NO:2519.

[0452] In some embodiments, the antibody is afucosylated and comprises the VH sequence shown in SEQ ID NO:2506; the VL sequence shown in SEQ ID NO:2507; and an active human IgG1 Fc region.

[0453] In some embodiments, the antibody comprises all 3 heavy chain CDRs of the sequence shown in SEQ ID NO:2512 and all 3 light chain CDRs of the sequence shown in SEQ ID NO:2513.

[0454] In some embodiments, the antibody comprises an A to T substitution at position 97 of the sequence shown in SEQ ID NO:2512; and a K to R substitution at position 98 of the sequence shown in SEQ ID NO:2512.

[0455] In some embodiments, the antibody comprises the VH sequence shown in SEQ ID NO:2506, 2508, or 2510.

[0456] In some embodiments, the antibody comprises the VH sequence shown in SEQ ID NO:2506, 2508, or 2510 and the VL sequence shown in SEQ ID NO:2507, 2509, or 2511. In some embodiments, the antibody comprises the VH sequence shown in SEQ ID NO:2506.

[0457] In some embodiments, the antibody comprises the VH sequence shown in SEQ ID NO:2506 and the VL sequence shown in SEQ ID NO:2507.

[0458] In some embodiments, the antibody is the 37012 antibody (see TABLE H1)

[0459] In some embodiments, the antibody is an antibody having a VL, VH, full heavy chain sequence or full light chain sequence disclosed in Table 1A or a CDR sequence as disclosed in Table 1B of PCT Patent Application Publication No. WO2019/118513A1, which are reproduced below as TABLES H1 and H2 respectively.

TABLE-US-00053 TABLE H1 Name Sequence 37012 VH
EVQLLESGGGLVQPGGSLRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VSSLTNSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CTREWAGSGYFDYWGGQTLTVSS (SEQ ID NO: 2506) 37012 VL
DIQMTQSPSSLSASVGDRVTITCKASQNVGNLAWYQQKPKAPKLLIY

YTSNRFTGVPSRFSGSGTDFTLTISLQPEDFATYYCQRIYNPWTFG QGTKLEIK
(SEQ ID NO: 2507) 37013VH
EVQLLES GGGGLVQP GGSRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VSSLTNSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CTREWAGSGYFDYWGGQGLTVTVSS (SEQ ID NO: 2508) 37013VL
DIQMTQSPSSLSASVGDRVTMTCKASQNVGNLAWYQQKPGKAPKLL
LYYTSNRFTGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQRIYNPWT FGQGTKLELK
(SEQ ID NO: 2509) 37014VH
EVQLLES GGGGLVQP GGSRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VASLTNSGGSTYYADSVKGRFTLSRDNSKNTLYLQMNSLRAEDTAVYY
CTREWAGSGYFDYWGGQGLTVTVSS (SEQ ID NO: 2510) 37014VL
DIQMTQSPSSLSASVGDRVTITCKASQNVGNLAWYQQKPGKAPKLLIY
YTSNRFTGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQRIYNPWTFG QGTKLEIK
(SEQ ID NO: 2511) 37017VH
EVQLLES GGGGLVQP GGSRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VSSLTNSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CAKEWAGSGYFDYWGGQGLTVTVSS (SEQ ID NO: 2512) 37017VL
DIQMTQSPSSLSASVGDRVTITCKASQNVGNLAWYQQKPGKAPKLLIY
YTSNRFTGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQRIYNPWTFG QGTKLEIK
(SEQ ID NO: 2513) Full 37012_H
EVQLLES GGGGLVQP GGSRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VSSLTNSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CTREWAGSGYFDYWGGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEAL
HNHYTQKSLSLSPGK (SEQ ID NO: 2529) Full 37012_L
DIQMTQSPSSLSASVGDRVTITCKASQNVGNLAWYQQKPGKAPKLLIY
YTSNRFTGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQRIYNPWTFG
QGTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQW
KVDNALQSGNSQESVTEQDSKDSYSTLSSTLTLSKADYEKHKVYACEV
THQGLSSPVTKSFNRGEC (SEQ ID NO: 2530) Full 37013_H
EVQLLES GGGGLVQP GGSRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VSSLTNSGGSTYYADSVKGRFTISRDN SKNTLYLQMNSLRAEDTAVYY
CTREWAGSGYFDYWGGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEAL
HNHYTQKSLSLSPGK (SEQ ID NO: 2531) Full 37013_L
DIQMTQSPSSLSASVGDRVTMTCKASQNVGNLAWYQQKPGKAPKLL
LYYTSNRFTGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQRIYNPWT
FGQGTKLELKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKV
QWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLTLSKADYEKHKVYAC
EVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 2532) Full 37017_H

EVQLLESGGGLVQPGGSLRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VASLTNSGGSTYYADSVKGRFTLSRDNSKNTLYLQMNSLRAEDTAVYY
CTREWAGSGYFDYWGGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEAL
HNHYTQKSLSLSPGK (SEQ ID NO: 2533) Full 37017_L
DIQMTQSPSSLSASVGDRVTITCKASQNVGNLAWYQQKPGKAPKLLIY
YTSNRFTGVPSRFSGSGGTDFTLTISSLQPEDFATYYCQRIYNPWTFG
QGKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQW
KVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEV
THQGLSSPVTKSFNRGEC (SEQ ID NO: 2534) Full 37017_H
EVQLLESGGGLVQPGGSLRLSCAASGFTFSNYYMAWVRQAPGKGLEW
VSSLTNSGGSTYYADSVKGRFTISRDNKNTLYLQMNSLRAEDTAVYY
CAKEWAGSGYFDYWGGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAA
LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS
SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPS
VFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESN
GQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEAL
HNHYTQKSLSLSPGK (SEQ ID NO: 2535) Full 37017_L
DIQMTQSPSSLSASVGDRVTITCKASQNVGNLAWYQQKPGKAPKLLIY
YTSNRFTGVPSRFSGSGGTDFTLTISSLQPEDFATYYCQRIYNPWTFG
QGKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQW
KVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEV
THQGLSSPVTKSFNRGEC (SEQ ID NO: 2536)

TABLE-US-00054 TABLE H2 CDR's of humanized antibodies CDR Sequence CDR-H1
FSNYYMA (SEQ ID NO: 2514) CDR-H2 SLTNSGGSTY (SEQ ID NO: 2515)
CDR-H3 EWAGSGY (SEQ ID NO: 2516) CDR-L1 NVGNL (SEQ ID NO:
2517) CDR-L2 YTSNRFT (SEQ ID NO: 2518) CDR-L3 RIYNPW (SEQ ID NO:
2519)

[0460] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in in the above tables as well as specific combinations thereof and other embodiments of the anti-TREM2 antibody described in the '513 application and herein may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

I. PCT Patent Application Publication No. WO2020/055975A1

[0461] In some embodiments, the TREM2 agonist is an antibody or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2020/055975A1 (“the '975 application”), which is incorporated by reference herein, in its entirety.

[0462] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain comprising a CDRL1, CDRL2, and CDRL3, and a heavy chain

variable domain comprising a CDRH1, CDRH2, and CDRH3 disclosed in the '975 application specification. In some embodiments, the TREM2 binding agent comprises an antibody that comprises a light chain variable domain and a heavy chain variable domain disclosed in the '975 application specification.

[0463] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 derived from SEQ ID NO:2539, an L2 derived from SEQ ID NO:2539, an L3 derived from of SEQ ID NO:2539, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 derived from SEQ ID NO:2540, an H2 derived from SEQ ID NO:2540, an H3 derived from SEQ ID NO:2540, or any combination thereof.

[0464] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 of SEQ ID NO:2541, an L2 comprising the amino acid sequence IVS, an L3 of SEQ ID NO:2542, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 comprising SEQ ID NO:2543, an H2 comprising SEQ ID NO:2544, an H3 comprising SEQ ID NO:2545, or any combination thereof.

[0465] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 derived from SEQ ID NO:2546, an L2 derived from SEQ ID NO:2546, an L3 derived from of SEQ ID NO:2546, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 derived from SEQ ID NO:2547, an H2 derived from SEQ ID NO:2547, an H3 derived from of SEQ ID NO:2547, or any combination thereof.

[0466] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 of SEQ ID NO:2548, an L2 comprising the amino acid sequence KVS, an L3 of SEQ ID NO:2549, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 comprising SEQ ID NO:2550, an H2 comprising SEQ ID NO:2551, an H3 comprising SEQ ID NO:2552, or any combination thereof.

[0467] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 derived from SEQ ID NO:2553, an L2 derived from SEQ ID NO:2553, an L3 derived from of SEQ ID NO:2553, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 derived from SEQ ID NO:2554, an H2 derived from SEQ ID NO:2554, an H3 derived from of SEQ ID NO:2554, or any combination thereof.

[0468] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 of SEQ ID NO:2555, an L2 comprising the amino acid sequence KVS, an L3 of SEQ ID NO:2556, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 comprising SEQ ID NO:2557, an H2 comprising SEQ ID NO:2558, an H3 comprising SEQ ID NO:2559, or any combination thereof.

[0469] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 derived from SEQ ID NO:2560, an L2 derived from SEQ ID NO:2560, an L3 derived from of SEQ ID NO:2560, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 derived from SEQ ID NO:2561, an H2 derived from SEQ ID NO:2561, an H3 derived from of SEQ ID NO:2561, or any combination thereof.

[0470] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 of SEQ ID NO:2562, an L2 comprising the amino acid sequence KVS, an L3 of SEQ ID NO:2563, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 comprising SEQ ID NO:2564, an H2 comprising SEQ ID NO:2565, an H3 comprising SEQ ID NO:2566, or any combination thereof.

[0471] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 derived from SEQ ID NO:2567, an L2 derived from SEQ ID NO:2567, an L3 derived from of SEQ ID NO:2567, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 derived from SEQ ID NO:2568, an H2 derived from SEQ ID NO:2568, an H3 derived from of SEQ ID NO:2568, or any combination thereof. Compositions comprising the antibody, including but not limited to pharmaceutical compositions, are contemplated herein. In

certain embodiments the antibody is a humanized antibody.

[0472] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 of SEQ ID NO:2569, an L2 comprising the amino acid sequence KVS, an L3 of SEQ ID NO:2570, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 comprising SEQ ID NO:2571, an H2 comprising SEQ ID NO:2572, an H3 comprising SEQ ID NO:2573, or any combination thereof.

[0473] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 derived from SEQ ID NO:2574, an L2 derived from SEQ ID NO:2574, an L3 derived from of SEQ ID NO:2574, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 derived from SEQ ID NO:2575, an H2 derived from SEQ ID NO:2575, an H3 derived from of SEQ ID NO:2575, or any combination thereof.

[0474] In some embodiments, the antibody comprises (a) a light chain variable region comprising an L1 of SEQ ID NO:2576, an L2 comprising the amino acid sequence WAS, an L3 of SEQ ID NO:2577, or any combination thereof; and/or (b) a heavy chain variable region comprising an H1 comprising SEQ ID NO:2578, an H2 comprising SEQ ID NO:2579, an H3 comprising SEQ ID NO:2580, or any combination thereof.

[0475] In some embodiments, the antibody is HJ23.4, HJ23.7, HJ23.8, HJ23.9, HJ23.10, or HJ23.13. In some embodiments, the antibody is a humanized antibody derived from HJ23.4, HJ23.7, HJ23.8, HJ23.9, HJ23.10, or HJ23.13. The accession number for the hybridoma that produced antibodies HJ23.4, HJ23.7, HJ23.8, HJ23.9, HJ23.10, and HJ23.13, and their respective light chain variable and heavy chain variable regions are noted in TABLE I1:

TABLE-US-00055 TABLE I1 Antibody (ATCC # Light chain Heavy chain of the hybridoma)
variable region variable region HJ23.4 (PTA-125168) SEQ ID NO: 2539 SEQ ID NO: 2540 HJ23.7 (PTA-125169) SEQ ID NO: 2546 SEQ ID NO: 2547 HJ23.8 (PTA-125170) SEQ ID NO: 2552 SEQ ID NO: 2553 HJ23.9 (PTA-125171) SEQ ID NO: 2561 SEQ ID NO: 2562 HJ23.10 (PTA-125172) SEQ ID NO: 2567 SEQ ID NO: 2568 HJ23.13 (PTA-125173) SEQ ID NO: 2574 SEQ ID NO: 2575

[0476] In some embodiments, the antibody is an antibody disclosed in Tables A and B or the summary table appended to Example 2 of PCT Patent Application Publication No.

WO2020/055975A1, reproduced below as TABLES I2-4.

TABLE-US-00056 TABLE I2 Light Chain HVR Heavy Chain HVR Antibody L1 L2 L3 H1 H2 H3
1 SEQ ID NO: 2541 2 SEQ ID NO: 2541 IVS 3 SEQ ID NO: 2541 IVS SEQ ID NO: 2542 4 IVS 5
IVS SEQ ID NO: 2542 6 SEQ ID NO: 2542 7 SEQ ID NO: 2541 SEQ ID NO: 2542 8 SEQ ID
NO: 2543 9 SEQ ID NO: 2543 SEQ ID NO: 2544 10 SEQ ID NO: 2543 SEQ ID NO: 2544 SEQ
ID NO: 2545 11 SEQ ID NO: 2544 12 SEQ ID NO: 2544 SEQ ID NO: 2545 13 SEQ ID NO: 2545
14 SEQ ID NO: 2543 SEQ ID NO: 2545 15 SEQ ID NO: 2541 SEQ ID NO: 2543 16 SEQ ID NO:
2541 SEQ ID NO: 2543 SEQ ID NO: 2544 17 SEQ ID NO: 2541 SEQ ID NO: 2543 SEQ ID NO:
2544 SEQ ID NO: 2545 18 SEQ ID NO: 2541 SEQ ID NO: 2544 19 SEQ ID NO: 2541 SEQ ID
NO: 2544 SEQ ID NO: 2545 20 SEQ ID NO: 2541 SEQ ID NO: 2545 21 SEQ ID NO: 2541 SEQ
ID NO: 2543 SEQ ID NO: 2545 22 SEQ ID NO: 2541 IVS SEQ ID NO: 2543 23 SEQ ID NO:
2541 IVS SEQ ID NO: 2543 SEQ ID NO: 2544 24 SEQ ID NO: 2541 IVS SEQ ID NO: 2543 SEQ
ID NO: 2544 SEQ ID NO: 2545 25 SEQ ID NO: 2541 IVS SEQ ID NO: 2544 26 SEQ ID NO:
2541 IVS SEQ ID NO: 2544 SEQ ID NO: 2545 27 SEQ ID NO: 2541 IVS SEQ ID NO: 2545 28
SEQ ID NO: 2541 IVS SEQ ID NO: 2543 SEQ ID NO: 2545 29 SEQ ID NO: 2541 IVS SEQ ID
NO: 2542 SEQ ID NO: 2543 30 SEQ ID NO: 2541 IVS SEQ ID NO: 2542 SEQ ID NO: 2543 SEQ
ID NO: 2544 31 SEQ ID NO: 2541 IVS SEQ ID NO: 2542 SEQ ID NO: 2543 SEQ ID NO: 2544
SEQ ID NO: 2545 32 SEQ ID NO: 2541 IVS SEQ ID NO: 2542 SEQ ID NO: 2544 33 SEQ ID
NO: 2541 IVS SEQ ID NO: 2542 SEQ ID NO: 2544 SEQ ID NO: 2545 34 SEQ ID NO: 2541 IVS
SEQ ID NO: 2542 SEQ ID NO: 2543 SEQ ID NO: 2545 35 SEQ ID NO: 2541 IVS SEQ ID NO:
2542 SEQ ID NO: 2545 36 IVS SEQ ID NO: 2543 37 IVS SEQ ID NO: 2543 SEQ ID NO: 2544

38 IVS SEQ ID NO: 2543 SEQ ID NO: 2544 SEQ ID NO: 2545 39 IVS SEQ ID NO: 2544 40 IVS
SEQ ID NO: 2544 SEQ ID NO: 2545 41 IVS SEQ ID NO: 2545 42 IVS SEQ ID NO: 2543 SEQ
ID NO: 2545 43 IVS SEQ ID NO: 2542 SEQ ID NO: 2543 44 IVS SEQ ID NO: 2542 SEQ ID NO:
2543 SEQ ID NO: 2544 45 IVS SEQ ID NO: 2542 SEQ ID NO: 2543 SEQ ID NO: 2544 SEQ ID
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SEQ ID NO: 2543 SEQ ID NO: 2545 50 SEQ ID NO: 2542 SEQ ID NO: 2543 51 SEQ ID NO:
2542 SEQ ID NO: 2543 SEQ ID NO: 2544 52 SEQ ID NO: 2542 SEQ ID NO: 2543 SEQ ID NO:
2544 SEQ ID NO: 2545 53 SEQ ID NO: 2542 SEQ ID NO: 2544 54 SEQ ID NO: 2542 SEQ ID
NO: 2544 SEQ ID NO: 2545 55 SEQ ID NO: 2542 SEQ ID NO: 2545 56 SEQ ID NO: 2542 SEQ
ID NO: 2543 SEQ ID NO: 2545 57 SEQ ID NO: 2541 SEQ ID NO: 2542 SEQ ID NO: 2543 58
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WAS SEQ ID NO: 2577 321 SEQ ID NO: 2577 322 SEQ ID NO: 2576 SEQ ID NO: 2577 323
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NO: 2578 338 SEQ ID NO: 2576 WAS SEQ ID NO: 2578 SEQ ID NO: 2579 339 SEQ ID NO:
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NO: 2576 WAS SEQ ID NO: 2580 343 SEQ ID NO: 2576 WAS SEQ ID NO: 2578 SEQ ID NO:
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WAS SEQ ID NO: 2577 SEQ ID NO: 2578 SEQ ID NO: 2579 346 SEQ ID NO: 2576 WAS SEQ
ID NO: 2577 SEQ ID NO: 2578 SEQ ID NO: 2579 SEQ ID NO: 2580 347 SEQ ID NO: 2576
WAS SEQ ID NO: 2577 SEQ ID NO: 2579 348 SEQ ID NO: 2576 WAS SEQ ID NO: 2577 SEQ
ID NO: 2579 SEQ ID NO: 2580 349 SEQ ID NO: 2576 WAS SEQ ID NO: 2577 SEQ ID NO:
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TABLE-US-00057 TABLE I3 Light Chain HVR Heavy Chain HVR Antibody L1 L2 L3 H1 H2 H3
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4 (I/V)KS 5 (I/V)KS SEQ ID NO: 2583 6 SEQ ID NO: 2583 7 SEQ ID NO: 2582 SEQ ID NO:
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2573 245 SEQ ID NO: 2573 246 SEQ ID NO: 2571 SEQ ID NO: 2573 246 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2571 247 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2571 SEQ ID NO: 2572 248 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2571 SEQ ID NO: 2572 SEQ ID NO: 2573 249 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2572 250 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2572 SEQ ID NO: 2573 251 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2573 252 SEQ ID NO: 2582 SEQ ID NO: 2583 SEQ ID NO: 2571 SEQ ID NO: 2573 TABLE-US-00058 TABLE 14 Comments Sequence HJ23.4 light chain variable DVVMTQTPLSLPVSLGDQAFISCRSSQNLVHSNGNTYLHWYLQK region PGQSPKLLIYIVSNRFSGVDPDRFSGSGSGTDFTLEISRVEAEDLGV YFCSQSTHVPLTFGAGTKLELK (SEQ ID NO: 2539) HJ23.4 heavy chain variable EVQLQQSGPDLVKPGASVKMSCKASGYTFTDYNIIHWVKQSHGK region TLEWIGYINPNTGGTYYNQKFKGKATMTV NKSSSTAYMELRSLT SEDSAVYYCVATRWDGVNWAQGTLTVSA (SEQ ID NO: 2540) HJ23.4 L1 QNLVHSNGNTY (SEQ ID NO: 2541) HJ23.4 L2 IVS HJ23.4 L3 SQSTHVPLT (SEQ ID NO: 2542) HJ23.4 H1 GYTFTDYN (SEQ ID NO: 2543) HJ23.4 H2 INPNTGGT (SEQ ID NO: 2544) HJ23.4 H3 VATRWDGVN (SEQ ID NO: 2545) HJ23.7 light chain variable DVVMTQTPLSLPVSLGDQASISCRSSQSLVHSNGNTYLHWYLQK region PGQSPKWKVSNRFSGVDPDRFSGSGSGTDFTLKISRVEAEDLGV YFCSQSTHVPLTFGAGTKLELK (SEQ ID NO: 2546) HJ23.7 heavy chain variable EVQLQQSGAELVKPGASVKLSCTSSGFNIKGYIHWVKQRTEQG region LEWIGRIDPEDGETKNAPKFQGKATFGTDTFSNTAYLRSLTSE DTGVYYCVRTESTRGAYWGPGLTVSA (SEQ ID NO: 2547) HJ23.7 L1 QSLVHSNGNTY (SEQ ID NO: 2548) HJ23.7 L2 KVS HJ23.7 L3 SQSTHVPLT (SEQ ID NO: 2549) HJ23.7 H1 GFNIKGYI (SEQ ID NO: 2550) HJ23.7 H2 IDPEDGET (SEQ ID NO: 2551) HJ23.7 H3 VRTETRGAY (SEQ ID NO: 2552) HJ23.8 light chain variable DVVMTQTPLSLPVSLGDQASISCRSSQSLVHSNGDNTYLHWYLQK region RGQSPKWKVSNRFSGVDPDRFSGSGSGTDFTLKISRVEAEDLGV YFCSQTTHVPLTFGAGTKLELK (SEQ ID NO: 2553) HJ23.8 heavy chain variable QVPLQQPGAELVKPGASVKLSCKASAYTFTRYWMHWVKQRPRGR region GLEWIGRIDPNSGGTNYNEKFKSKATFTVDKPSSTSYMQLSSLTS EDSAVYFCVFTGTLFDYWGQGTTTLTVSS (SEQ ID NO: 2554) HJ23.8 L1 QSLVHSNGDNTY (SEQ ID NO: 2555) HJ23.8 L2 KVS HJ23.8 L3 SQTTHVPLT (SEQ ID NO: 2556) HJ23.8 H1 AYTFTRYW (SEQ ID NO: 2557) HJ23.8 H2 IDPNSGGT (SEQ ID NO: 2558) HJ23.8 H3 VFTGTLFDY (SEQ ID NO: 2559) HJ23.9 light chain variable DVVMTQTPLSLPVSLGDQASISCKSSQSLVHSNGNTYLHWYLQK region PGQSPKWKVSNRFSGVDPDRFSGSGSGTDFTLKISRVEAEDLGV YFCSQSTHVPPTFGGGTKLEIK (SEQ ID NO: 2560) HJ23.9 heavy chain variable EVQLQHSGPVLVKPGASVKMSCKSSGYTFTDYILNWVKQSHGK region SPEWIGVINPNTGSTSYNQKFKGKATLTVDKSSSTAYMDLNSLTS EDSAVYYCATHYYGSIYKQAWFAYWGQGTLVT (SEQ ID NO: 2561) HJ23.9 L1 QSLVHSNGNTY (SEQ ID NO: 2562) HJ23.9 L2 KVS HJ23.9 L3 SQSTHVPPT (SEQ ID NO: 2563) HJ23.9 H1 GYTFTDYY (SEQ ID NO: 2564) HJ23.9 H2 INPNTGST (SEQ ID NO: 2565) HJ23.9 H3 ATHYYGSIYKQAWFAY (SEQ ID NO: 2566) HJ23.10 light chain variable DVVMTQTPLSLPVSLGDQASISCKSSQSLVHSNGNTYLHWYLQK region PGQSPKWKVSNRFSGVDPDRFSGSGSGTDFTLKISRVEAEDLGI YFCSQSTHVPPTFGGGTKLEIK (SEQ ID NO: 2567) HJ23.10 heavy chain

EVQLQHSGLPKGVKQVGLKASQYGFYTFDYYMNWVKQSHG variable region
KSPIEWIGVINPNTGSTSYNQKFKGKATLTVDKSSSTAYMDLNSLT
SEDSAVYYCATHYYGSIYKQAWFAYWGQGTLLTV (SEQ ID NO: 2568) HJ23.10 L1
QSLVHSNGNTY (SEQ ID NO: 2569) HJ23.10 L2 KVS HJ23.10 L3 SQSTHVPPT
(SEQ ID NO: 2570) HJ23.10 H1 GYTFTDYY (SEQ ID NO: 2571) HJ23.10 H2
INPNTGST (SEQ ID NO: 2572) HJ23.10 H3 ATHYYGSIYKQAWFAY (SEQ ID
NO: 2573) HJ23.13 light chain variable
DIVMSQSPSSLAVSVGEKVTMSCKSSQSLLYSSNLKNYLAWFQQ region
KPGQSPKLLIYWASIRESGVPDRFTGSGSGTDFTLTINSVKAEDLA
VYYCQQYYTFPLTFGAGTKLELK (SEQ ID NO: 2574) HJ23.13 heavy chain
EVQLVETGGGLVQPKGSLKLSAASGFSFNINAMHWVRQAPGT variable region
GLKWVARIRSGSNDFATYYADSVKDRFTISRDDSHSMLYLQMN
NLKTEDTAIYFCVREYVNYFVHWGQGTLLTVSA (SEQ ID NO: 2575) HJ23.13 L1
QSLLYSSNLKNY (SEQ ID NO: 2576) HJ23.13 L2 WAS HJ23.13 L3 QQYYTFPLT
(SEQ ID NO: 2577) HJ23.13 H1 GFSFNINA (SEQ ID NO: 2578) HJ23.13 H2
IRSGSNDFAT (SEQ ID NO: 2579) HJ23.13 H3 VREYVNYFVH (SEQ ID NO:
2580) DHRDAGDLWFPGES (SEQ ID NO: 2581) Consensus L1 QX1LVHSNGX2TY,
where X1 is S, T, N, or Q and X2 is D or N (SEQ ID NO: 2582)
Consensus L2 X1VS, where X1 is I or K Consensus L3 SQX1THVPX2T,
where X1 is S, T, N, or Q and X2 is P or L (SEQ ID NO: 2583)
[0477] In some embodiments, each of the light chain variable regions and each of the heavy chain
variable regions disclosed above, including those in Table 17B (e.g., antibody 1-378) and Table
17C (antibody 1-252) may be attached to the light chain constant regions (TABLE EN1) and heavy
chain constant regions (TABLE EN2) to form complete antibody light and heavy chains,
respectively, as further discussed below. Further, each of the generated heavy and light chain
sequences may be combined to form a complete antibody structure. It should be understood that the
heavy chain and light chain variable regions provided herein can also be attached to other constant
domains having different sequences than the exemplary sequences listed herein.

J. PCT Patent Application Publication No. WO2020/079580A1

[0478] In some embodiments, the TREM2 agonist is an antibody, or an antigen-binding fragment
thereof, as described in PCT Patent Application Publication No. WO2020/079580A1 (“the ‘580
application”), which is incorporated by reference herein, in its entirety.

[0479] In some embodiments, the TREM2 binding agent comprises an antibody that comprises a
light chain variable domain comprising a CDRL1, CDRL2, and CDRL3, and a heavy chain
variable domain comprising a CDRH1, CDRH2, and CDRH3 disclosed in the ‘580 application
specification. In some embodiments, the TREM2 binding agent comprises an antibody that
comprises a light chain variable domain and a heavy chain variable domain disclosed in the ‘580
application specification.

[0480] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0481]
a) a heavy chain variable region CDR1 comprising SEQ ID NO:2623 or SEQ ID NO:2626 or SEQ
ID NO:2627 or SEQ ID NO:2629; a heavy chain variable region CDR2 comprising SEQ ID
NO:2624 or SEQ ID NO:2628, or SEQ ID NO:2630; a heavy chain variable region CDR3
comprising SEQ ID NO:2625 or SEQ ID NO:2631; a light chain variable region CDR1 comprising
SEQ ID NO:2636 or SEQ ID NO:2639 or SEQ ID NO:2642; a light chain variable region CDR2
comprising SEQ ID NO:2637 or SEQ ID NO:2640; and a light chain variable region CDR3
comprising SEQ ID NO:2638 or SEQ ID NO:2641; [0482] b) a heavy chain variable region CDR1
comprising SEQ ID NO:2586 or SEQ ID NO:2589 or SEQ ID NO:2590 or SEQ ID NO:2592; a
heavy chain variable region CDR2 comprising SEQ ID NO:2587 or SEQ ID NO:2591 or SEQ ID
NO:2593; a heavy chain variable region CDR3 comprising SEQ ID NO:2588 or SEQ ID NO:2594;
a light chain variable region CDR1 comprising SEQ ID NO:2599 or SEQ ID NO:2602 or SEQ ID

NO:2605; a light chain variable region CDR2 comprising SEQ ID NO:2600 or SEQ ID NO:2603; and a light chain variable region CDR3 comprising SEQ ID NO:2601 or SEQ ID NO:2604; [0483] c) a heavy chain variable region CDR1 comprising SEQ ID NO:2586 or SEQ ID NO:2589 or SEQ ID NO:2590 or SEQ ID NO:2592; a heavy chain variable region CDR2 comprising SEQ ID NO:2587 or SEQ ID NO:2591 or SEQ ID NO:2593; a heavy chain variable region CDR3 comprising SEQ ID NO:2588 or SEQ ID NO:2594; a light chain variable region CDR1 comprising SEQ ID NO:2599 or SEQ ID NO:2602 or SEQ ID NO:2605; a light chain variable region CDR2 comprising SEQ ID NO:2600 or SEQ ID NO:2603; and a light chain variable region CDR3 comprising SEQ ID NO:2660 or SEQ ID NO:2661; or [0484] d) a heavy chain variable region CDR1 comprising SEQ ID NO:2666 or SEQ ID NO:2669 or SEQ ID NO:2670 or SEQ ID NO:2672; a heavy chain variable region CDR2 comprising SEQ ID NO:2667 or SEQ ID NO:2671 or SEQ ID NO:2673; a heavy chain variable region CDR3 comprising SEQ ID NO:2668 or SEQ ID NO:2674; a light chain variable region CDR1 comprising SEQ ID NO:2679 or SEQ ID NO:2682 or SEQ ID NO:2685; a light chain variable region CDR2 comprising SEQ ID NO:2680 or SEQ ID NO:2683; and a light chain variable region CDR3 comprising SEQ ID NO:2681 or SEQ ID NO:2684.

[0485] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0486] a) a VH polypeptide sequence having at least 95% sequence identity to SEQ ID NO:2595 or to SEQ ID NO:2632, and a VL polypeptide sequence having at least 95% sequence identity to SEQ ID NO:2606 or to SEQ ID NO:2643; or [0487] b) a VH polypeptide sequence having at least 95% sequence identity to SEQ ID NO:2595 or to SEQ ID NO:2675, and a VL polypeptide sequence having at least 95% sequence identity to SEQ ID NO:2662 or to SEQ ID NO:2686.

[0488] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0489] a) a heavy chain variable region CDR1 comprising SEQ ID NO:2589; a heavy chain variable region CDR2 comprising SEQ ID NO:2587; a heavy chain variable region CDR3 comprising SEQ ID NO:2588; a light chain variable region CDR1 comprising SEQ ID NO:2599; a light chain variable region CDR2 comprising SEQ ID NO:2600; and a light chain variable region CDR3 comprising SEQ ID NO:2601; b) a heavy chain variable region CDR1 comprising SEQ ID NO:2626; a heavy chain variable region CDR2 comprising SEQ ID NO:2624; a heavy chain variable region CDR3 comprising SEQ ID NO:2625; a light chain variable region CDR1 comprising SEQ ID NO:2636; a light chain variable region CDR2 comprising, e.g., consisting of SEQ ID NO:2637; and a light chain variable region CDR3 comprising SEQ ID NO:2638; c) a heavy chain variable region CDR1 comprising SEQ ID NO:2589; a heavy chain variable region CDR2 comprising SEQ ID NO:2587; a heavy chain variable region CDR3 comprising SEQ ID NO:2588; a light chain variable region CDR1 comprising SEQ ID NO:2599; a light chain variable region CDR2 comprising SEQ ID NO:2600; and a light chain variable region CDR3 comprising SEQ ID NO:2660; or d) a heavy chain variable region CDR1 comprising SEQ ID NO:2669; a heavy chain variable region CDR2 comprising SEQ ID NO:2667; a heavy chain variable region CDR3 comprising SEQ ID NO:2668; a light chain variable region CDR1 comprising SEQ ID NO:2679; a light chain variable region CDR2 comprising SEQ ID NO:2680; and a light chain variable region CDR3 comprising SEQ ID NO:2681.

[0490] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0491] a) a heavy chain variable region CDR1 of SEQ ID NO:2590; a heavy chain variable region CDR2 comprising SEQ ID NO:2591; a heavy chain variable region CDR3 comprising SEQ ID NO:2588; a light chain variable region CDR1 comprising SEQ ID NO:2602; a light chain variable region CDR2 comprising SEQ ID NO:2603; and a light chain variable region CDR3 comprising SEQ ID NO:2604; [0492] b) a heavy chain variable region CDR1 comprising SEQ ID NO:2627; a heavy chain variable region CDR2 comprising SEQ ID NO:2628; a heavy chain variable region CDR3 comprising SEQ ID NO:2625; a light chain variable region CDR1 comprising SEQ ID NO:2639; a light chain variable region CDR2 comprising SEQ ID NO:2640; and a light chain variable region

CDR3 comprising SEQ ID NO:2641; [0493] c) a heavy chain variable region CDR1 comprising SEQ ID NO:2590; a heavy chain variable region CDR2 comprising SEQ ID NO:2591; a heavy chain variable region CDR3 comprising SEQ ID NO:2588; a light chain variable region CDR1 comprising SEQ ID NO:2602; a light chain variable region CDR2 comprising SEQ ID NO:2603; and a light chain variable region CDR3 comprising SEQ ID NO:2661; or [0494] d) a heavy chain variable region CDR1 comprising SEQ ID NO:2670; a heavy chain variable region CDR2 comprising SEQ ID NO:2671; a heavy chain variable region CDR3 comprising SEQ ID NO:2668; a light chain variable region CDR1 comprising SEQ ID NO:2682; a light chain variable region CDR2 comprising SEQ ID NO:2683; and a light chain variable region CDR3 comprising SEQ ID NO:2684.

[0495] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0496] a) a heavy chain variable region CDR1 comprising SEQ ID NO:2592; a heavy chain variable region CDR2 comprising SEQ ID NO:2593; a heavy chain variable region CDR3 comprising SEQ ID NO:2594; a light chain variable region CDR1 comprising SEQ ID NO:2605; a light chain variable region CDR2 comprising SEQ ID NO:2603; and a light chain variable region CDR3 comprising SEQ ID NO:2601; [0497] b) a heavy chain variable region CDR1 comprising SEQ ID NO:2629; a heavy chain variable region CDR2 comprising SEQ ID NO:2630; a heavy chain variable region CDR3 comprising SEQ ID NO:2631; a light chain variable region CDR1 comprising SEQ ID NO:2642; a light chain variable region CDR2 comprising SEQ ID NO:2640; and a light chain variable region CDR3 comprising SEQ ID NO:2638; [0498] c) a heavy chain variable region CDR1 comprising SEQ ID NO:2592; a heavy chain variable region CDR2 comprising SEQ ID NO:2593; a heavy chain variable region CDR3 comprising SEQ ID NO:2594; a light chain variable region CDR1 comprising SEQ ID NO:2605; a light chain variable region CDR2 comprising SEQ ID NO:2603; and a light chain variable region CDR3 comprising SEQ ID NO:2660; or [0499] d) a heavy chain variable region CDR1 comprising SEQ ID NO:2672; a heavy chain variable region CDR2 comprising SEQ ID NO:2673; a heavy chain variable region CDR3 comprising SEQ ID NO:2674; a light chain variable region CDR1 comprising SEQ ID NO:2685; a light chain variable region CDR2 comprising SEQ ID NO:2683; and a light chain variable region CDR3 comprising SEQ ID NO:2681.

[0500] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0501] a) a heavy chain variable region CDR1 comprising SEQ ID NO:2586; a heavy chain variable region CDR2 comprising SEQ ID NO:2587; a heavy chain variable region CDR3 comprising SEQ ID NO:2588; a light chain variable region CDR1 comprising SEQ ID NO:2599; a light chain variable region CDR2 comprising SEQ ID NO:2600; and a light chain variable region CDR3 comprising SEQ ID NO:2601; [0502] b) a heavy chain variable region CDR1 comprising SEQ ID NO:2623; a heavy chain variable region CDR2 comprising SEQ ID NO:2624; a heavy chain variable region CDR3 comprising SEQ ID NO:2625; a light chain variable region CDR1 comprising SEQ ID NO:2636; a light chain variable region CDR2 comprising SEQ ID NO:2637; and a light chain variable region CDR3 comprising SEQ ID NO:2638; [0503] c) a heavy chain variable region CDR1 comprising SEQ ID NO:2586; a heavy chain variable region CDR2 comprising SEQ ID NO:2587; a heavy chain variable region CDR3 comprising SEQ ID NO:2588; a light chain variable region CDR1 comprising SEQ ID NO:2599; a light chain variable region CDR2 comprising SEQ ID NO:2600; and a light chain variable region CDR3 comprising SEQ ID NO:2660; or [0504] d) a heavy chain variable region CDR1 comprising SEQ ID NO:2666; a heavy chain variable region CDR2 comprising SEQ ID NO:2667; a heavy chain variable region CDR3 comprising SEQ ID NO:2668; a light chain variable region CDR1 comprising SEQ ID NO:2679; a light chain variable region CDR2 comprising SEQ ID NO:2680; and a light chain variable region CDR3 comprising SEQ ID NO:2681.

[0505] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0506] a) a VH comprising SEQ ID NO:2595 and a VL comprising SEQ ID NO:2606; or [0507] b) a VH

comprising SEQ ID NO:2632 and a VL comprising SEQ ID NO:2643; or [0508] c) a VH comprising a sequence having at least 95% homology to SEQ ID NO:2595 and a VL comprising a sequence having at least 95% homology to SEQ ID NO:2606; or [0509] d) a VH comprising a sequence having at least 95% homology to SEQ ID NO:2632 and a VL comprising a sequence having at least 95% homology to SEQ ID NO:2643; or [0510] e) a VH comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2595 and a VL comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2606; or [0511] f) a VH comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2632 and a VL comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2643. g) a VH comprising SEQ ID NO:2595 and a VL comprising SEQ ID NO:2662; or [0512] h) a VH comprising SEQ ID NO:2675 and a VL comprising SEQ ID NO:2686; or [0513] i) a VH comprising a sequence having at least 95% homology to SEQ ID NO:2595 and a VL comprising a sequence having at least 95% homology to SEQ ID NO:2662; or [0514] j) a VH comprising a sequence having at least 95% homology to SEQ ID NO:2675 and a VL comprising a sequence having at least 95% homology to SEQ ID NO:2686; or [0515] k) a VH comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2595 and a VL comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2662; or [0516] l) a VH comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2675 and a VL comprising, e.g. consisting of, a sequence that differs by at least 1, 2, 3, 4, 5, or 6 amino acids from SEQ ID NO:2686.

[0517] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0518] a) a heavy chain amino acid sequence comprising SEQ ID NO:2597, SEQ ID NO:2611, SEQ ID NO:2615, SEQ ID NO:2617, SEQ ID NO:2619, or SEQ ID NO:2621, and a light chain amino acid sequence comprising SEQ ID NO:2608; b) a heavy chain amino acid sequence comprising SEQ ID NO:2634, SEQ ID NO:2648, SEQ ID NO:2652, SEQ ID NO:2654, SEQ ID NO:2656, or SEQ ID NO:2658, and a light chain amino acid sequence comprising SEQ ID NO:2645; c) a heavy chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2597, SEQ ID NO:2611, SEQ ID NO:2615, SEQ ID NO:2617, SEQ ID NO:2619, or SEQ ID NO:2621, and a light chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2608; [0519] d) a heavy chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2634, SEQ ID NO:2648, SEQ ID NO:2652, SEQ ID NO:2654, SEQ ID NO:2656, or SEQ ID NO:2658, and a light chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2645; [0520] e) a heavy chain amino acid sequence comprising SEQ ID NO:2597, and a light chain amino acid sequence comprising SEQ ID NO:2664; [0521] f) a heavy chain amino acid sequence comprising SEQ ID NO:2677, and a light chain amino acid sequence comprising SEQ ID NO:2688; [0522] g) a heavy chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2597, and a light chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2664; or [0523] h) a heavy chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2677, and a light chain amino acid sequence having at least 95% sequence identity to SEQ ID NO:2688.

[0524] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0525] a) a heavy chain sequence comprising SEQ ID NO:2597 and a light chain sequence comprising SEQ ID NO:2608; [0526] b) a heavy chain sequence comprising SEQ ID NO:2611 and a light chain sequence comprising SEQ ID NO:2608; [0527] c) a heavy chain sequence comprising SEQ ID NO:2615 and a light chain sequence comprising SEQ ID NO:2608; [0528] d) a heavy chain sequence comprising SEQ ID NO:2617 and a light chain sequence comprising SEQ ID NO:2608; [0529] e) a heavy chain sequence comprising SEQ ID NO:2619 and a light chain sequence comprising SEQ ID NO:2608; [0530] f) a heavy chain sequence comprising SEQ ID NO:2621 and a light chain sequence comprising SEQ ID NO:2608; [0531] g) a heavy chain sequence comprising

SEQ ID NO:2634 and a light chain sequence comprising SEQ ID NO:2645; [0532] h) a heavy chain sequence comprising SEQ ID NO:2648 and light chain sequence comprising SEQ ID NO:2645; [0533] i) a heavy chain sequence comprising SEQ ID NO:2652 and light chain sequence comprising SEQ ID NO:2645; [0534] j) a heavy chain sequence comprising SEQ ID NO:2654 and light chain sequence comprising SEQ ID NO:2645; [0535] k) a heavy chain sequence comprising SEQ ID NO:2656 and light chain sequence comprising SEQ ID NO:2645; [0536] l) a heavy chain sequence comprising SEQ ID NO:2658 and light chain sequence comprising SEQ ID NO:2645; [0537] m) a heavy chain sequence comprising SEQ ID NO:2597 and light chain sequence comprising SEQ ID NO:2664; or [0538] n) a heavy chain sequence comprising SEQ ID NO:2677 and light chain sequence comprising SEQ ID NO:2688.

[0539] In some embodiments, the antibody is an antibody disclosed in Table 1 of PCT Patent Application Publication No. WO2020/079580A1, reproduced below as TABLE J1.

TABLE-US-00059 TABLE J1 Sequences of Exemplary Monoclonal Antibodies That Bind Human TREM2 Description Sequence MOR44698A HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY (SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat) HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Kabat) HCDR3 IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO: 2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3 IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO: 2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3 ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH

QVQLVQSGAEVKKPGASVKVSCASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLRSEDTAVYYCARIPSYT
YAFDYWGQGLTVTVSS (SEQ ID NO: 2595) DNA VH
CAGGTGCAATTGGTGCAGAGCGGTGCGGAAGTGAAAAAACCGGGTGCCAG
CGTGAAAGTTAGCTGCAAAGCGTCCGGATATACCTTCACTGGTTACCATAT
GTCTTGGGTGCGCCAGGCCCCGGGCCAGGGCCTCGAGTGGATGGGCGTTAT
CAACCCGGTTTCTGGCAACACGGTTTACGCGCAGAAATTTTCAGGGCCGGGT
GACCATGACCCGTGATACCAGCATTAGCACCGCGTATATGGAAGTGAAGCC
GTCTGCGTAGCGAAGATACGGCCGTGTATTATTGCGCGCGTATCCCGTCTT
ACACTTACGCTTTTCGATTACTGGGGCCAAGGCACCCTGGTGACTGTTAGCT CA
(SEQ ID NO: 2596) Heavy Chain

QVQLVQSGAEVKKPGASVKVSCASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLRSEDTAVYYCARIPSYT
YAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2597) DNA Heavy
Chain
CAGGTGCAATTGGTGCAGAGCGGTGCGGAAGTGAAAAAACCGGGTGCCAG
CGTGAAAGTTAGCTGCAAAGCGTCCGGATATACCTTCACTGGTTACCATAT
GTCTTGGGTGCGCCAGGCCCCGGGCCAGGGCCTCGAGTGGATGGGCGTTAT
CAACCCGGTTTCTGGCAACACGGTTTACGCGCAGAAATTTTCAGGGCCGGGT
GACCATGACCCGTGATACCAGCATTAGCACCGCGTATATGGAAGTGAAGCC
GTCTGCGTAGCGAAGATACGGCCGTGTATTATTGCGCGCGTATCCCGTCTT
ACACTTACGCTTTTCGATTACTGGGGCCAAGGCACCCTGGTGACTGTTAGCT
CAGCCTCCACCAAGGGTCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGA

GCACCTCTGGCTGGCGACAGCGGCTGGCTGCTCAAGGACTACTTCC
CCGAACCGGTGACGGTGTCTGTGGAAGCTCAGGCGCCCTGACCAGCGGCGTG
CACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGC
GTGGTGACCGTGCCCTCCAGCAGCTTGGGACCCAGACCTACATCTGCAAC
GTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAA
ATCTTGTGACAAAACCTCACACATGCCCACCGTGCCCAGCACCTGAAGCAGC
GGGGGGACCGTCAGTCTTCTCTTCCCCCCTAAAACCCAAGGACACCCTCAT
GATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACG
AAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCAT
AATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGGGT
GGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGT
ACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACC
ATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCC
CCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGG
TCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGG
CAGCCGGAGAACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGG
CTCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCA
GGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTA
CACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA (SEQ ID NO: 2598) LCDR1
RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Combined) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Combined)
LCDR1 RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Kabat) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Kabat) LCDR1
SQDISNY (SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603)
(Chothia) LCDR3 YRHMPST (SEQ ID NO: 2604) (Chothia) LCDR1 QDISNY (SEQ
ID NO: 2605) (IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3
FQYRHMPST (SEQ ID NO: 2601) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFGQGTKVEIK (SEQ ID
NO: 2606) DNA VL
GATATCCAGATGACCCAGAGCCCGAGCAGCCTGAGCGCCAGCGTGGGCGA
TCGCGTGACCATTACCTGCAGAGCCAGCCAGGACATTTCTAACTACCTGGC
TTGGTACCAGCAGAAACCGGGCAAAGCGCCGAACTATTAATCTACCGTG
CTTCTTCTCTGCAAAGCGGCGTGCCGAGCCGCTTTAGCGGCAGCGGATCCG
GCACCGATTTACCCCTGACCATTAGCTCTCTGCAACCGGAAGACTTTGCGA
CCTATTATTGCTTCCAGTACCGTCATATGCCGTCTCAGACCTTTGGCCAGGG
CACGAAAGTTGAAATTA (SEQ ID NO: 2607) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFGQGTKVEIK
RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
ESVTEQDSKDSSTLSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C (SEQ
ID NO: 2608) DNA Light
GATATCCAGATGACCCAGAGCCCGAGCAGCCTGAGCGCCAGCGTGGGCGA Chain
TCGCGTGACCATTACCTGCAGAGCCAGCCAGGACATTTCTAACTACCTGGC
TTGGTACCAGCAGAAACCGGGCAAAGCGCCGAACTATTAATCTACCGTG
CTTCTTCTCTGCAAAGCGGCGTGCCGAGCCGCTTTAGCGGCAGCGGATCCG
GCACCGATTTACCCCTGACCATTAGCTCTCTGCAACCGGAAGACTTTGCGA
CCTATTATTGCTTCCAGTACCGTCATATGCCGTCTCAGACCTTTGGCCAGGG
CACGAAAGTTGAAATTAACGTACGGTGGCCGCTCCCAGCGTGTTTCATCTT
CCCCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCC

GTGCTGAACAACTTACCTACAGCCTGAGCAGCACCCCTGACCCTGAGCAAGGCC
AACGCCCTGCAGAGCGGCAACAGCCAGGAAAGCGTCACCGAGCAGGACA
GCAAGGACTCCACCTACAGCCTGAGCAGCACCCCTGACCCTGAGCAAGGCC
GACTACGAGAAGCACAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCT
GTCCAGCCCCGTGACCAAGAGCTTCAACCGGGGCGAGTGT (SEQ ID NO: 2609)
MOR44698B HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2
VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY
(SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat)
HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Kabat) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO:
2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO:
2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3
ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH
QVQLVQSGAEVKKPGASVKVSCASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLESDTAVYYCARIPSYT
YAFDYWGQGLTVTVSS (SEQ ID NO: 2595) DNA VH
CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCACAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAATAACCGTGTACGCCCAGAAGTTCCAGGGTCGCG
TGACCATGACCCGGGATACTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATAACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTACCCGTGTCCT CG
(SEQ ID NO: 2610) Heavy Chain
QVQLVQSGAEVKKPGASVKVSCASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLESDTAVYYCARIPSYT
YAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2611) DNA Heavy
CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC Chain
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCACAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAATAACCGTGTACGCCCAGAAGTTCCAGGGTCGCG
TGACCATGACCCGGGATACTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATAACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTACCCGTGTCCT
CGGCCTCCACTAAGGGCCCAAGTGTGTTTCCCCTGGCCCCCAGCAGCAAGT
CTACTTCCGGCGGAACTGCTGCCCTGGGTTGCCTGGTGAAGGACTACTTCC
CCGAGCCCGTGACAGTGTCTGGAAGTCTGGGGCTCTGACTTCCGGCGTGC
ACACCTTCCCCGCCGTGCTGCAGAGCAGCGGCCTGTACAGCCTGAGCAGCG
TGGTGACAGTGCCCTCCAGCTCTCTGGGAACCCAGACCTATATCTGCAACG
TGAACCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTGGAGCCCAA
GAGCTGCGACAAGACCCACACCTGCCCCCCTGCCAGCTCCAGAACTGCT
GGGAGGGCCTTCCGTGTTCCCTGTTCCCCCACAAGCCCAAGGACACCCTGAT
GATCAGCAGGACCCCCGAGGTGACCTGCGTGGTGGTGGACGTGTCCCACG

AGGACCCAGAGCTTCAACTGGGTACGTGGACGGCTGGAGGTGCAC
AACGCCAAGACCAAGCCCAGAGAGGAGCAGTACAACAGCACCTACAGGGT
GGTGTCCGTGCTGACCGTGCTGCACCAGGACTGGCTGAACGGCAAAGAAT
ACAAGTGCAAAGTCTCCAACAAGGCCCTGCCAGCCCCAATCGAAAAGACA
ATCAGCAAGGCCAAGGGCCAGCCACGGGAGCCCCAGGTGTACACCCTGCC
CCCCAGCCGGGAGGAGATGACCAAGAACCAGGTGTCCCTGACCTGTCTGG
TGAAGGGGCTTCTACCCCAGCGATATCGCCGTGGAGTGGGAGAGCAACGGC
CAGCCCGAGAACAACACTACAAGACCACCCCCCAGTGCTGGACAGCGACGG
CAGCTTCTTCCTGTACAGCAAGCTGACCGTGGACAAGTCCAGGTGGCAGCA
GGGCAACGTGTTTCAGCTGCAGCGTGATGCACGAGGCCCTGCACAACCACT
ACACCCAGAAGTCCCTGAGCCTGAGCCCCGGCAAG (SEQ ID NO: 2612) LCDR1
RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Combined) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Combined)
LCDR1 RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Kabat) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Kabat) LCDR1
SQDISNY (SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603)
(Chothia) LCDR3 YRHMPST (SEQ ID NO: 2604) (Chothia) LCDR1 QDISNY (SEQ
ID NO: 2605) (IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3
FQYRHMPST (SEQ ID NO: 2601) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFGQGTKVEIK (SEQ ID
NO: 2606) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC
AGAGTCACCATCACTTGCCGGGCCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCTGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAG (SEQ ID NO: 2613) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFGQGTKVEIK
RTVAAPS FIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
ESVTEQDSKDSSTLSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C (SEQ
ID NO: 2608) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCTGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTT
CCCCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGCTGTGCC
TGCTGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGAC
AACGCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACA
GCAAGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCC
GACTACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCT
GTCCAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2614)
MOR44698C HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2
VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY
(SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat)

HCDR2 VINPVSGNTVYAQKFQ (SEQ ID NO: 2587) (Kabat) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO:
2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO:
2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3
ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLRSEDVAVYYCARIPSYT
YAFDYWGQGLTVTVSS (SEQ ID NO: 2595) DNA VH
CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAATACCGTGTACGCCCAGAAGTTCCAGGGTCGCG
TGACCATGACCCGGGATACCTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTACCCGTGTCCT CG
(SEQ ID NO: 2610) Heavy Chain
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLRSEDVAVYYCARIPSYT
YAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALAAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2615) DNA Heavy
Chain
CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAATACCGTGTACGCCCAGAAGTTCCAGGGTCGCG
TGACCATGACCCGGGATACCTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTACCCGTGTCCT
CGGCCTCCACTAAGGGCCCGTCAGTGTTCCCCCTTGCGCCATCCTCGAAGT
CAACCTCCGGAGGAAGTGCCTGCTGGGTTGCCTCGTGAAAGACTATTTCC
CGGAACCCGTCCTGTCTCCTGGAAGTCAAGGAGCGCTCACCAGCGGAGTGC
ATACCTTTCTGCGGTGCTGCAGTCCAGCGGCCTGTACTCCCTGAGCTCCGT
CGTGACCGTCCCCTCGTCGTCCCTGGGAACCCAAACCTACATTTGCAACGT
CAATCACAAGCCAAGCAACACTAAGGTGGACAAGAGAGTGGAGCCCAAGT
CCTGCGATAAGACCCACACCTGTCCTCCCTGTCCGGCACCTGAACTGCTTG
GTGGACCTTCCGTGTTCCCTGTTCCCGCCCAAGCCAAAAGACACCCTGATGA
TCTCCCGCACTCCGGAAGTCACTTGCCTGGTTCGTGGCCGTGTCCACGAGG
ACCCCGAGGTCAAGTTTAATTGGTACGTGGACGGAGTGAAGTGCACAAC
GCCAAGACCAAGCCGCGGGAAGAACAGTACAACCTACCGCGTGGT
GTCCGTCTGACTGTGCTCCACCAGGACTGGCTGAACGGAAAGGAGTACA
AGTGCAAAGTGTCCAACAAGGCACTGGCTGCCCCTATCGAAAAGACTATCT
CCAAGGCCAAGGGCCAACCTAGGGAGCCCCAGGTGTACACGTTGCCTCCTT
CCCGCGAAGAAATGACTAAGAACCAGGTGTCGCTGACCTGTCTCGTGAAA
GGGTTCTACCCCTCTGACATCGCCGTGGAATGGGAGTCAAACGGACAGCCT
GAGAACAACATAAGACCACACCACCTGTCCTGGACTCCGACGGCTCCTTC

TTCTGTACTGAACTTGACCTGCGCGGTGGCAAACAGGGCAAC
GTGTTCTCTTGCTCCGTGATGCACGAAGCCCTGCACAACCACTACACCCAA
AAGTCGCTCAGCCTCTCCCCCGGAAAG (SEQ ID NO: 2616) LCDR1
RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Combined) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Combined)
LCDR1 RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Kabat) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Kabat) LCDR1
SQDISNY (SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603)
(Chothia) LCDR3 YRHMPST (SEQ ID NO: 2604) (Chothia) LCDR1 QDISNY (SEQ
ID NO: 2605) (IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3
FQYRHMPST (SEQ ID NO: 2601) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFGQGTKVEIK (SEQ ID
NO: 2606) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC
AGAGTCACCATCACTTGCCGGGCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCTGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAG (SEQ ID NO: 2613) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFGQGTKVEIK
RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
ESVTEQDSKDSSTLSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C (SEQ
ID NO: 2608) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCTGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTATCTT
CCCCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGTTGTGCC
TGCTGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGAC
AACGCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACA
GCAAGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCC
GACTACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCT
GTCCAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2614)
MOR44698D HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2
VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY
(SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat)
HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Kabat) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO:
2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO:
2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3
ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLESDTAVYYCARIPSYT

YAFDYWGQGLTVTVSS (SEQ ID NO: 2595) DNA VH
CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCACAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAATAACCGTGTACGCCCAGAAGTTCAGGGTCGCG
TGACCATGACCCGGGATACTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATAACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGGCCAGGGGACTCTTGTACCCGTGTCCT CG

(SEQ ID NO: 2610) Heavy Chain

QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLESDTAVYYCARIPSYT
YAFDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ

QGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2617) DNA Heavy Chain

CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCACAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAATAACCGTGTACGCCCAGAAGTTCAGGGTCGCG
TGACCATGACCCGGGATACTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATAACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGGCCAGGGGACTCTTGTACCCGTGTCCT
CGGCCTCCACTAAGGGCCCCGTCAGTGTTCCCCCTTGCGCCATCCTCGAAGT
CAACCTCCGGAGGAACTGCCGCACTGGGTTGCCTCGTGAAAGACTATTTCC
CGGAACCCGTCCTGTCTCCTGGAAGTCAAGGAGCGCTCACCAGCGGAGTGC
ATACCTTTTCTGCGGTGCTGCAGTCCAGCGGCCTGTACTCCCTGAGCTCCGT
CGTGACCGTCCCCTCGTCGTCCCTGGGAACCCAAACCTACATTTGCAACGT
CAATCACAAGCCAAGCAACACTAAGGTGGACAAGAGAGTGGAGCCCAAGT
CCTGCGATAAGACCCACACCTGTCTCCTGTCCGGCACCTGAACTGCTTG
GTGGACCTTCCGTGTTCTGTTCCTGTTCCCGCCCAAGCCAAAAGACACCCTGATGA
TCTCCCGCACTCCGGAAGTCACTTGCCTGGTTCGTGGACGTGTCCCACGAGG
ACCCCGAGGTCAAGTTTAATTGGTACGTGGACGGAGTGGAAAGTGCACAAC
GCCAAGACCAAGCCGCGGGAAGAACAGTACAACCTCCACCTACCGCGTGGT
GTCCGTCTGACTGTGCTCCACCAGGACTGGCTGAACGGAAAGGAGTACA
AGTGCAAAGTGTCCAACAAGGCACTGCCAGCCCCCTATCGAAAAGACTATC
TCCAAGGCCAAGGGCCAACCTAGGGAGCCCCAGGTGTACACGTTGCCTCCT
TCCCGCGAAGAAATGACTAAGAACCAGGTGTCGCTGACCTGTCTCGTGAAA
GGGTTCTACCCCTCTGACATCGCCGTGGAATGGGAGTCAAACGGACAGCCT
GAGAACAACCTATAAGACCACACCACCTGTCTTGGACTCCGACGGCTCCTTC
TTCCTGTACTCAAAGTTGACCGTGGACAAGTCGCGGTGGCAACAGGGCAAC
GTGTTCTCTTGCTCCGTGCTGCACGAAGCCCTGCACAGCCACTACACCCAA
AAGTCGCTCAGCCTCTCCCCCGGAAAG (SEQ ID NO: 2618) LCDR1

RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Combined) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Combined)
LCDR1 RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Kabat) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Kabat) LCDR1
SQDISNY (SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603)

(Chothia) LCDR3 YRHMPST (SEQ ID NO: 2604) (Chothia) LCDR1 QDISNY (SEQ ID NO: 2605) (IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCFQYRHMPSTFGQGTKVEIK (SEQ ID NO: 2606) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC
AGAGTCACCATCACTTGCCGGGCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCTGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAG (SEQ ID NO: 2613) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCFQYRHMPSTFGQGTKVEIK
RTVAAPS VFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
ESVTEQDSKDSSTLSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C (SEQ ID NO: 2608) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCTGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTT
CCCCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGTTGTGCC
TGCTGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGAC
AACGCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACA
GCAAGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCC
GACTACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCT
GTCCAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2614)
MOR44698E HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2
VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY
(SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat)
HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Kabat) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO:
2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO:
2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3
ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH
QVQLVQSGAEVKKPGASVKVSKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTSDTSISTAYMELSRLESDTAVYYCARIPSYT
YAFDYWGQGTLVTVSS (SEQ ID NO: 2595) DNA VH
CAAGTGCAACTCGTGCAAGTCAAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCAT
GTCGTGGGTGAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCGGGTAAATACCGTGTACGCCAGAAAGTTCCAGGGTCCGG
TGACCATGACCCGGGATACTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATAACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTACCGTGTCTCT CG

(SEQ ID NO: 2610) Heavy Chain

QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLRSEDTAVYYCARIPSYT
YAFDYWGQGTLLTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALAAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2619) DNA Heavy
CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC Chain
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACACTTTCCTGGATACCACAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTGGGTAATACCGTGTACGCCCAGAAGTTCAGGGTTCGCG
TGACCATGACCCGGGATACCTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTACCGTGTCTCT
CGGCCTCCACTAAGGGCCCGTCAGTGTTCCCCCTTGCGCCATCCTCGAAGT
CAACCTCCGGAGGAAGTGGCGCACTGGGTTGCCTCGTGAAAGACTATTTCC
CGGAACCCGTCCTGTCTCCTGGAAGTCAAGGAGCGCTCACCAGCGGAGTGC
ATACCTTTCCTGCGGTGCTGCAGTCCAGCGGCCTGTACTCCCTGAGCTCCGT
CGTGACCGTCCCCTCGTCGTCCCTGGGAACCCAAACCTACATTTGCAACGT
CAATCACAAGCCAAGCAACACTAAGGTGGACAAGAGAGTGGAGCCCAAGT
CCTGCGATAAGACCCACACCTGTCCTCCCTGTCCGGCACCTGAACTGCTTG
GTGGACCTTCCGTGTTCCCTGTTCCCGCCCAAGCCAAAAGACACCCTGATGA
TCTCCCGCACTCCGGAAGTCACTTGCCTGGTGGTGGCCGTGTCCACGAGG
ACCCCGAGGTCAAGTTTAATTGGTACGTGGACGGAGTGGAAAGTGCACAAC
GCCAAGACCAAGCCGCGGGAAGAACAGTACAACCTACCGCGTGGT
GTCCGTCTGACTGTGCTCCACCAGGACTGGCTGAACGGAAAGGAGTACA
AGTGCAAAGTGTCCAACAAGGCACTGGCTGCCCCTATCGAAAAGACTATCT
CCAAGGCCAAGGGCCAACCTAGGGAGCCCCAGGTGTACACGTTGCCTCCTT
CCCGCGAAGAAATGACTAAGAACCAGGTGTCGCTGACCTGTCTCGTGAAA
GGGTTCTACCCCTCTGACATCGCCGTGGAATGGGAGTCAAACGGACAGCCT
GAGAACAACCTATAAGACCACACCACCTGTCCTGGACTCCGACGGCTCCTTC
TTCCTGTACTCAAAGTTGACCGTGGACAAGTCGCGGTGGCAACAGGGCAAC
GTGTTCTCTTGCTCCGTGCTGCACGAAGCCCTGCACAGCCACTACACCCAA
AAGTCGCTCAGCCTCTCCCCCGGAAAG (SEQ ID NO: 2620) LCDR1

RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Combined) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Combined)
LCDR1 RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Kabat) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Kabat) LCDR1
SQDISNY (SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603)
(Chothia) LCDR3 YRHMPST (SEQ ID NO: 2604) (Chothia) LCDR1 QDISNY (SEQ
ID NO: 2605) (IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3
FQYRHMPST (SEQ ID NO: 2601) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCFQYRHMPSTFGQGQTKVEIK (SEQ ID
NO: 2606) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC
AGAGTCACCATCACTTGCCGGGCCTCACAGGATATTTCCAACCTACCTGGCC

TGGTATCAGTACGAGGAGCGGAGCGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAG (SEQ ID NO: 2613) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCFQYRHMPSTFGQGTKVEIK
RTVAAPSVFIFPP SDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
ESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C (SEQ
ID NO: 2608) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGATATTTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTT
CCCCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCC
TGCTGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGAC
AACGCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACA
GCAAGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCC
GACTACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCT
GTCCAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2614)
MOR44698F HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2
VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY
(SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat)
HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Kabat) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO:
2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO:
2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3
ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLESDTAVYYCARIPSYT
YAFDYWGQGTLVTVSS (SEQ ID NO: 2595) DNA VH
CAAGTGCAACTCGTGCAAGTCAAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC
GGTCAAGGTGTCCTGCAAGGCCAGCGGATACTTTCACTGGATAACCAT
GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
TCAACCCGGTGTCTGGGTAAATACCGTGTACGCCCAGAAGTTCAGGGTCTCGG
TGACCATGACCCGGGATACCTCCATTAGCACCGCGTACATGGAGCTCAGCC
GGTTGAGATCCGAGGATACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
ACACTTACGCCTTCGACTATTGGGGGCCAGGGGACTCTTGTACCCGTGTCCT CG
(SEQ ID NO: 2610) Heavy Chain
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSRLESDTAVYYCARIPSYT
YAFDYWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLYITREPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSL

TCLVKGFYPSDIAVEWESNEQENPKYNTTTPVLDSDFLYSKLTVDKSRWQ
 QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2621) DNA Heavy
 CAAGTGCAACTCGTGCAGTCAGGAGCCGAAGTCAAGAAGCCTGGAGCCTC Chain
 GGTCAAGGTGTCCTGCAAGGCCAGCGGATACACTTTCACTGGATACCACAT
 GTCGTGGGTCAGACAGGCTCCTGGCCAAGGGCTGGAGTGGATGGGCGTCA
 TCAACCCGGTGTCTGGGTAATACCGTGTACGCCCAAGAAGTTCCAGGGTTCGCG
 TGACCATGACCCGGGATACCTCCATTAGCACCGCGTACATGGAGCTCAGCC
 GGTTGAGATCCGAGGATACCGCCGTGTACTACTGTGCGCGGATCCCGTCCT
 AACTTACGCCTTCGACTATTGGGGCCAGGGGACTCTTGTCACCGTGTCTCT
 CGGCCTCCACTAAGGGCCCCGTCAGTGTTCCCCCTTGCGCCATCCTCGAAGT
 CAACCTCCGGAGGAACTGCCGCACTGGGTTGCCTCGTGAAAGACTATTTCC
 CGGAACCCGTCCTGTCTCCTGGAACCTCAGGAGCGCTCACCAGCGGAGTGC
 ATACCTTTCTCTGCGGTGCTGCAGTCCAGCGGCCTGTACTCCCTGAGCTCCGT
 CGTGACCGTCCCCTCGTCGTCCCTGGGAACCCAAACCTACATTTGCAACGT
 CAATCACAAGCCAAGCAACACTAAGGTGGACAAGAGAGTGGAGCCCAAGT
 CCTGCGATAAGACCCACACCTGTCCTCCCTGTCCGGCACCTGAACTGCTTG
 GTGGACCTTCCGTGTTCTCTGTTCCCGCCCCAAGCCAAAAGACACCCTGTATA
 TCACTCGCGAACCAGGAAGTCACTTGCGTGGTTCGTGGACGTGTCCACGAGG
 ACCCCGAGGTCAAGTTTAATTGGTACGTGGACGGAGTGGAAAGTGACAAC
 GCCAAGACCAAGCCGCGGGAAGAAGAGTACAACCTCCACCTACCGCGTGGT
 GTCCGTCTCTGACTGTGCTCCACCAGGACTGGCTGAACGGAAAGGAGTACA
 AGTGCAAAGTGTCCAACAAGGCACTGCCAGCCCCTATCGAAAAGACTATC
 TCCAAGGCCAAGGGCCAACCTAGGGAGCCCCAGGTGTACACGTTGCCTCCT
 TCCCGCGAAGAAATGACTAAGAACCAGGTGTGCTGACCTGTCTCGTGAAA
 GGGTTCTACCCCTCTGACATCGCCGTGGAATGGGAGTCAAACGGACAGCCT
 GAGAACAACTATAAGACCACACCACCTGTCTTGACTCCGACGGCTCCTTC
 TTCCTGTACTCAAAGTTGACCGTGGACAAGTCGCGGTGGCAACAGGGCAAC
 GTGTTCTCTTGCTCCGTGATGCACGAAGCCCTGCACAACCACTACACCCAA
 AAGTCGCTCAGCCTCTCCCCCGGAAAG (SEQ ID NO: 2622) LCDR1
 RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
 NO: 2600) (Combined) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Combined)
 LCDR1 RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID
 NO: 2600) (Kabat) LCDR3 FQYRHMPST (SEQ ID NO: 2601) (Kabat) LCDR1
 SQDISNY (SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603)
 (Chothia) LCDR3 YRHMPST (SEQ ID NO: 2604) (Chothia) LCDR1 QDISNY (SEQ
 ID NO: 2605) (IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3
 FQYRHMPST (SEQ ID NO: 2601) (IMGT) VL
 DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
 QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFTGQGTKVEIK (SEQ ID
 NO: 2606) DNA VL
 GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC
 AGAGTCACCATCACTTGCCGGGCCTCACAGGATATTTCCAACCTGGCC
 TGGTATCAGCAGAAGCCTGGAAAGGCCCCGAAGCTGCTGATCTACCGGGC
 GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
 GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTGCTAC
 CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
 CACCAAAGTCGAGATCAAG (SEQ ID NO: 2613) Light Chain
 DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
 QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCFQYRHMPSTFTGQGTKVEIK
 RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ

ESVTEQDSKSTLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGE C (SEQ ID NO: 2608) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGATATTCCAACCTACCTGGCC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACCGGGC
GTCCTCCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTCGCTAC
CTACTACTGCTTCCAGTACCGGCACATGCCCTCACAAACCTTCGGACAGGG
CACCAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTT
CCCCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGCTGTGCC
TGCTGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGAC
AACGCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACA
GCAAGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCC
GACTACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCT
GTCCAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2614)
MOR44746A HCDR1 GDSVSSSSAAWN (SEQ ID NO: 2623) (Combined) HCDR2
HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Combined) HCDR3
GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Combined) HCDR1 SSSAAWN (SEQ ID NO: 2626) (Kabat) HCDR2 HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Kabat) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Kabat) HCDR1 GDSVSSSSA (SEQ ID NO: 2627) (Chothia) HCDR2 GYRSKWY (SEQ ID NO: 2628) (Chothia) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Chothia) HCDR1 GDSVSSSSAA (SEQ ID NO: 2629) (IMGT) HCDR2 IGYRSKWYN (SEQ ID NO: 2630) (IMGT) HCDR3 ARGMYGSVPYKEGYFDI (SEQ ID NO: 2631) (IMGT)
VH QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGHI
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFDIWGQGLTVTVSS (SEQ ID NO: 2632) DNA VH
CAGGTGCAATTGCAGCAGAGCGGTCCGGGCTGGTGAAACCGAGCCAGAC
CCTGAGCCTGACCTGCGCGATTTCCGGAGATAGCGTGAGCTCTTCTTCTGCT
GCTTGGAACCTGGATTCGTCAGAGCCCGAGCCGTGGCCTCGAGTGGCTGGGC
CATATCGGTTACCGTAGCAAATGGTACAACGAATATGCCGTGAGCGTGAA
AAGCCGCATTACCATTAACCCGGATACTTCGAAAAACCAAGTTTAGCCTGCA
ACTGAACAGCGTGACCCCGGAAGATACGGCCGTGTATTATTGCGCGCGTGG
TATGTACGGTTCTGTTCCCTACAAAGAAGGTTACTACTTCGATATTTGGGGC
CAAGGCACCCTGGTGACTGTTAGCTCA (SEQ ID NO: 2633) Heavy Chain
QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGHI
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFDIWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISR
TPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMT
KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTV
DKSRWQQGNVVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2634) DNA
Heavy CAGGTGCAATTGCAGCAGAGCGGTCCGGGCTGGTGAAACCGAGCCAGAC
Chain CCTGAGCCTGACCTGCGCGATTTCCGGAGATAGCGTGAGCTCTTCTTCTGCT
GCTTGGAACCTGGATTCGTCAGAGCCCGAGCCGTGGCCTCGAGTGGCTGGGC
CATATCGGTTACCGTAGCAAATGGTACAACGAATATGCCGTGAGCGTGAA
AAGCCGCATTACCATTAACCCGGATACTTCGAAAAACCAAGTTTAGCCTGCA
ACTGAACAGCGTGACCCCGGAAGATACGGCCGTGTATTATTGCGCGCGTGG

TATGTAGGTTCTTCACTACAAAGGTTACTTTCGATATTGGGGC
CAAGGCACCCTGGTGACTGTTAGCTCAGCCTCCACCAAGGGTCCATCGGTC
TTCCCCCTGGCACCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCCCTG
GGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCGTGGAAC
TCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAGTCC
TCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTG
GGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAA
GGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACATGCC
CACCGTGCCCAGCACCTGAAGCAGCGGGGGGACCGTCAGTCTTCCCTCTTCC
CCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACAT
GCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGG
TACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGA
GCAGTACAACAGCACGTACCGGGTGGTCAGCGTCCTCACCGTCCTGCACCA
GGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCC
TCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGA
GAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAA
CCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGC
CGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACCTACAAGACCACG
CCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCCTCTACAGCAAGCTCACCG
TGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATG
CATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCG GGTA
(SEQ ID NO: 2635) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Combined)
LCDR2 AASNLS (SEQ ID NO: 2637) (Combined) LCDR3 QQYTDESMT (SEQ
ID NO: 2638) (Combined) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Kabat)
LCDR2 AASNLS (SEQ ID NO: 2637) (Kabat) LCDR3 QQYTDESMT (SEQ ID
NO: 2638) (Kabat) LCDR1 SQGISSD (SEQ ID NO: 2639) (Chothia) LCDR2 AAS
(SEQ ID NO: 2640) (Chothia) LCDR3 YTDESM (SEQ ID NO: 2641) (Chothia)
LCDR1 QGISSD (SEQ ID NO: 2642) (IMGT) LCDR2 AAS (SEQ ID NO: 2640)
(IMGT) LCDR3 QQYTDESMT (SEQ ID NO: 2638) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNHWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYTDESMTFGQGTKVEIK (SEQ ID
NO: 2643) DNA VL
GATATCCAGATGACCCAGAGCCCGAGCAGCCTGAGCGCCAGCGTGGGCGA
TCGCGTGACCATTACCTGCAGAGCCAGCCAGGGTATTTCTTCTGACCTGAA
CTGGTACCAGCAGAAACCGGGCAAAGCGCCGAAACTATTAATCTACGCTG
CTTCTAACCTGCAAAGCGGCGTGCCGAGCCGCTTTAGCGGCAGCGGATCCG
GCACCGATTTACCCCTGACCATTAGCTCTCTGCAACCGGAAGACTTTGCGA
CCTATTATTGCCAGCAGTACACTGACGAATCTATGACCTTTGGCCAGGGCA
CGAAAGTTGAAATTA (SEQ ID NO: 2644) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNHWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYTDESMTFGQGTKVEIKR
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQE
SVTEQDSKDSSTLSLTLSKADYEEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID
NO: 2645) DNA Light
GATATCCAGATGACCCAGAGCCCGAGCAGCCTGAGCGCCAGCGTGGGCGA Chain
TCGCGTGACCATTACCTGCAGAGCCAGCCAGGGTATTTCTTCTGACCTGAA
CTGGTACCAGCAGAAACCGGGCAAAGCGCCGAAACTATTAATCTACGCTG
CTTCTAACCTGCAAAGCGGCGTGCCGAGCCGCTTTAGCGGCAGCGGATCCG
GCACCGATTTACCCCTGACCATTAGCTCTCTGCAACCGGAAGACTTTGCGA
CCTATTATTGCCAGCAGTACACTGACGAATCTATGACCTTTGGCCAGGGCA

CGAAAGTTGAAATTAACGTACCTTCCAGCTGCCAGCTTCCTCCTCC
CCCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTG GTGTGCCTG
CTGAACA ACTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAA
CGCCCTGCAGAGCGGCAACAGCCAGGAAAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTTGACCCTGAGCAAGGCCGAC
TACGAGAAGCACAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACCGGGGCGAGTGT (SEQ ID NO: 2646)
MOR44746B HCDR1 GDSVSSSSAAWN (SEQ ID NO: 2623) (Combined) HCDR2
HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Combined) HCDR3
GMYGSPYPYKEGYFYDI (SEQ ID NO: 2625) (Combined) HCDR1 SSSAAWN (SEQ
ID NO: 2626) (Kabat) HCDR2 HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624)
(Kabat) HCDR3 GMYGSVPYKEGYFYDI (SEQ ID NO: 2625) (Kabat) HCDR1
GDSVSSSSA (SEQ ID NO: 2627) (Chothia) HCDR2 YRSKWY (SEQ ID NO:
2628) (Chothia) HCDR3 GMYGSVPYKEGYFYDI (SEQ ID NO: 2625) (Chothia) HCDR1
GDSVSSSSAA (SEQ ID NO: 2629) (IMGT) HCDR2 IGYRSKWYN (SEQ ID
NO: 2630) (IMGT) HCDR3 ARGMYGSVPYKEGYFYDI (SEQ ID NO: 2631) (IMGT)
VH QVQLQQSGPGLVKPSQTLSTLCAISGDSVSSSSAAWNWIRQSPSRGLEWLGH
I
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFYDIWGQGTLVTVSS (SEQ ID NO: 2632) DNA VH
CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCGTCGTCGTCAGC
CGCCTGGA ACTGGATTAGACAGTCCCCTTCCCGAGGGCTGGAGTGGCTGGG
CCACATCGGATAACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCTCG (SEQ ID NO: 2647) Heavy Chain
QVQLQQSGPGLVKPSQTLSTLCAISGDSVSSSSAAWNWIRQSPSRGLEWLGH
I
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFYDIWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICN
VNHKPSNTKVDRVEPKSCDKTHSTCPGPCAPELLGGPSVFLFPPKPKDTLMISR
TPEVTCTVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMT
KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLT
V
DKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2648) DNA
Heavy CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC Chain
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCGTCGTCGTCAGC
CGCCTGGA ACTGGATTAGACAGTCCCCTTCCCGAGGGCTGGAGTGGCTGGG
CCACATCGGATAACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCTCGGCCTCCACTAAGGGCCCAAGTG
TGTTTCCCCTGGCCCCCAGCAGCAAGTCTACTTCCGGCGGAACTGCTGCCC
TGGGTTGCCTGGTGAAGGACTACTTCCCCGAGCCCGTGACAGTGTCTTGA
ACTCTGGGGCTCTGACTTCCGGCGTGCACACCTTCCCCGCCGTGCTGCAGA
GCAGCGGCCTGTACAGCCTGAGCAGCGTGGTGACAGTGCCCTCCAGCTCTC
TGGAACCCAGACCTATATCTGCAACGTGAACCAACAAGCCCAGCAACACC
AAGGTGGACAAGAGAGTGGAGCCCAAGAGCTGCGACAAGACCCACACCTG

CCCCCGCCCTCCAGAACTTGCGGGAGGCTTCCGTGTT
CCCCCCCCAAGCCCAAGGACACCCTGATGATCAGCAGGACCCCCGAGGTGA
CCTGCGTGGTGGTGGACGTGTCCCACGAGGACCCAGAGGTGAAGTTCAACT
GGTACGTGGACGGCGTGGAGGTGCACAACGCCAAGACCAAGCCCAGAGAG
GAGCAGTACAACAGCACCTACAGGGTGGTGTCCGTGCTGACCGTGCTGCA
CCAGGACTGGCTGAACGGCAAAGAATACAAGTGCAAAGTCTCCAACAAGG
CCCTGCCAGCCCCAATCGAAAAGACAATCAGCAAGGCCAAGGGCCAGCCA
CGGGAGCCCCAGGTGTACACCCTGCCCCCCAGCCGGGAGGAGATGACCAA
GAACCAGGTGTCCCTGACCTGTCTGGTGAAGGGCTTCTACCCCAGCGATAT
CGCCGTGGAGTGGGAGAGCAACGGCCAGCCCCGAGAACAACACTACAAGACCA
CCCCCCCAGTGCTGGACAGCGACGGCAGCTTCTTCCTGTACAGCAAGCTGA
CCGTGGACAAGTCCAGGTGGCAGCAGGGCAACGTGTTTCAGCTGCAGCGTG
ATGCACGAGGCCCTGCACAACCACTACACCCAGAAGTCCCTGAGCCTGAG
CCCCGGCAAG (SEQ ID NO: 2649) LCDR1 RASQGISSDLN (SEQ ID NO:
2636) (Combined) LCDR2 AASNLS (SEQ ID NO: 2637) (Combined) LCDR3
QQYTDESMT (SEQ ID NO: 2638) (Combined) LCDR1 RASQGISSDLN (SEQ ID
NO: 2636) (Kabat) LCDR2 AASNLS (SEQ ID NO: 2637) (Kabat) LCDR3
QQYTDESMT (SEQ ID NO: 2638) (Kabat) LCDR1 SQGISSD (SEQ ID NO:
2639) (Chothia) LCDR2 AAS (SEQ ID NO: 2640) (Chothia) LCDR3 YTDISM (SEQ
ID NO: 2641) (Chothia) LCDR1 QGISSD (SEQ ID NO: 2642) (IMGT) LCDR2 AAS
(SEQ ID NO: 2640) (IMGT) LCDR3 QQYTDESMT (SEQ ID NO: 2638) (IMGT)
VL DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNWFYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYTDESMTFGQGTKVEIK (SEQ ID
NO: 2643) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC
AGAGTCACCATCACTTGCCGGGCCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAG (SEQ ID NO: 2650) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNWFYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYTDESMTFGQGTKVEIKR
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQE
SVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID
NO: 2645) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTTCCC
CCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCCTGC
TGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAAC
GCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCCGAC
TACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2651)
MOR44746C HCDR1 GDSVSSSSAAWN (SEQ ID NO: 2623) (Combined) HCDR2

HIGYRSKYNEYAVSVKS (SEQ ID NO: 2624) (Combined) HCDR3
GMYGSVPYKEGYFFDI (SEQ ID NO: 2625) (Combined) HCDR1 SSSAAWN (SEQ
ID NO: 2626) (Kabat) HCDR2 HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624)
(Kabat) HCDR3 GMYGSVPYKEGYFFDI (SEQ ID NO: 2625) (Kabat) HCDR1
GDSVSSSSA (SEQ ID NO: 2627) (Chothia) HCDR2 GYRSKWY (SEQ ID NO:
2628) (Chothia) HCDR3 GMYGSVPYKEGYFFDI (SEQ ID NO: 2625) (Chothia) HCDR1
GDSVSSSSAA (SEQ ID NO: 2629) (IMGT) HCDR2 IGYRSKWYN (SEQ ID
NO: 2630) (IMGT) HCDR3 ARGMYGSVPYKEGYFFDI (SEQ ID NO: 2631) (IMGT)
VH QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGH
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFFDIWGQGLTVTVSS (SEQ ID NO: 2632) DNA VH
CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCTCGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCCTCG (SEQ ID NO: 2647) Heavy Chain
QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGH
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFFDIWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISR
TPEVTCVAVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREPQVYTLPPSREEMT
KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTV
DKSRWQQGNVVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2652) DNA
Heavy CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC Chain
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCTCGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCCTCGGCCTCCACTAAGGGCCCGTCAG
TGTTCCCCCTTGCGCCATCCTCGAAGTCAACCTCCGGAGGAACTGCCGCAC
TGGGTTGCCTCGTGAAAGACTATTTCCCGGAACCCGTCAGTGTCTCCTGGA
ACTCAGGAGCGCTCACCAGCGGAGTGCATACCTTTCTGCGGTGCTGCAGT
CCAGCGGCCTGTACTCCCTGAGCTCCGTCGTGACCGTCCCCTCGTCGTCCCT
GGGAACCCAAACCTACATTTGCAACGTCAATCACAAGCCAAGCAACACTA
AGGTGGACAAGAGAGTGGAGCCCAAGTCCTGCGATAAGACCCACACCTGT
CCTCCCTGTCCGGCACCTGAACTGCTTGGTGGACCTTCCGTGTTCTGTTCC
CGCCCAAGCCAAAAGACACCCTGATGATCTCCCGCACTCCGGAAGTCACTT
GCGTGGTCGTGGCCGTGTCCACGAGGACCCCGAGGTCAAGTTTAATTGGT
ACGTGGACGGAGTGGAAAGTGCACAACGCCAAGACCAAGCCGCGGGAAGA
ACAGTACAACCTCCACCTACCGCGTGGTGTCCGTCTGACTGTGCTCCACCA
GGACTGGCTGAACGGAAGGAGTACAAGTGCAAAGTGTCCAACAAGGCAC
TGGCTGCCCCTATCGAAAAGACTATCTCCAAGGCCAAGGGCCAACCTAGG
GAGCCCCAGGTGTACACGTTGCCTCCTTCCCGCGAAGAAATGACTAAGAAC

CAGGTGTGCTGCTGACCTGTCTCGTGTCTGCTGACATCGCC
GTGGAATGGGAGTCAAACGGACAGCCTGAGAACAACCTATAAGACCACACC
ACCTGTCCTGGACTCCGACGGCTCCTTCTTCTCTGTACTCAAAGTTGACCGTG
GACAAGTCGCGGTGGCAACAGGGCAACGTGTTCTCTTGCTCCGTGATGCAC
GAAGCCCTGCACAACCACTACACCCAAAAGTCGCTCAGCCTCTCCCCCGGA AAG
(SEQ ID NO: 2653) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Combined)
LCDR2 AASNLS (SEQ ID NO: 2637) (Combined) LCDR3 QQYTDESMT (SEQ
ID NO: 2638) (Combined) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Kabat)
LCDR2 AASNLS (SEQ ID NO: 2637) (Kabat) LCDR3 QQYTDESMT (SEQ ID
NO: 2638) (Kabat) LCDR1 SQGISSD (SEQ ID NO: 2639) (Chothia) LCDR2 AAS
(SEQ ID NO: 2640) (Chothia) LCDR3 YTDESM (SEQ ID NO: 2641) (Chothia)
LCDR1 (SEQ ID NO: 2642) (IMGT) LCDR2 AAS (SEQ ID NO: 2640) (IMGT)
LCDR3 QQYTDESMT (SEQ ID NO: 2638) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYTDESMTFGQGTKVEIK (SEQ ID
NO: 2643) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC
AGAGTCACCATCACTTGCCGGGCCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAG (SEQ ID NO: 2650) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYTDESMTFGQGTKVEIKR
TVAAPSVMFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQE
SVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC (SEQ ID
NO: 2645) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTTTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTTCCC
CCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCCTGC
TGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAAC
GCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCCGAC
TACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2651)
MOR44746D HCDR1 GDSVSSSSAAWN (SEQ ID NO: 2623) (Combined) HCDR2
HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Combined) HCDR3
GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Combined) HCDR1 SSSAAWN (SEQ
ID NO: 2626) (Kabat) HCDR2 HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624)
(Kabat) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Kabat) HCDR1
GDSVSSSSA (SEQ ID NO: 2627) (Chothia) HCDR2 GYRSKWY (SEQ ID NO:
2628) (Chothia) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Chothia) HCDR1
GDSVSSSSAA (SEQ ID NO: 2629) (IMGT) HCDR2 IGYRSKWYN (SEQ ID
NO: 2630) (IMGT) HCDR3 ARGMYGSVPYKEGYFDI (SEQ ID NO: 2631) (IMGT)

VH QVQLQQSGPGLVSPQSLTSLTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGHI
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFYFDIWGQGTLTVSS (SEQ ID NO: 2632) DNA VH
CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCCTCG (SEQ ID NO: 2647) Heavy Chain
QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGHI
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFYFDIWGQGTLTVSSASTKGPSVFPLAPSSKSTSGGTAAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVKDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISR
TPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMT
KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTV
DKSRWQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2654) DNA
Heavy CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC Chain
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCCTCGGCCTCCACTAAGGGCCCGTCAG
TGTTCCCCCTTGCGCCATCCTCGAAGTCAACCTCCGGAGGAACTGCCGCAC
TGGGTTGCCTCGTGAAAGACTATTTCCCGGAACCCGTCACTGTCTCCTGGA
ACTCAGGAGCGCTCACCAGCGGAGTGCATACCTTTCCTGCGGTGCTGCAGT
CCAGCGGCCTGTACTCCCTGAGCTCCGTGTCGTGACCGTCCCCTCGTCGTCCCT
GGGAACCCAAACCTACATTTGCAACGTCAATCACAAGCCAAGCAACACTA
AGGTGGACAAGAGAGTGGAGCCCAAGTCCTGCGATAAGACCCACACCTGT
CCTCCCTGTCCGGCACCTGAACTGCTTGGTGGACCTTCCGTGTTCTGTTC
CGCCCAAGCCAAAAGACACCCTGATGATCTCCCGCACTCCGGAAGTCACTT
GCGTGGTCGTGGACGTGTCCCACGAGGACCCCGAGGTCAAGTTTAATTGGT
ACGTGGACGGAGTGGAAAGTGCACAACGCCAAGACCAAGCCGCGGGAAGA
ACAGTACAACCTCCACCTACCGCGTGGTGTCCGTCTGACTGTGCTCCACCA
GGACTGGCTGAACGGAAAGGAGTACAAGTGCAAAGTGTCCAACAAGGCAC
TGCCAGCCCCTATCGAAAAGACTATCTCCAAGGCCAAGGGCCAACCTAGG
GAGCCCCAGGTGTACACGTTGCCTCCTTCCCGCGAAGAAATGACTAAGAAC
CAGGTGTCGCTGACCTGTCTCGTGAAAGGGTTCTACCCCTCTGACATCGCC
GTGGAATGGGAGTCAAACGGACAGCCTGAGAACAACCTATAAGACCACACC
ACCTGTCCTGGACTCCGACGGCTCCTTCTTCTGTACTCAAAGTTGACCGTG
GACAAGTCGCGGTGGCAACAGGGCAACGTGTTCTCTTGCTCCGTGCTGCAC
GAAGCCCTGCACAGCCACTACACCCAAAAGTCGCTCAGCCTCTCCCCCGGA AAG
(SEQ ID NO: 2655) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Combined)
LCDR2 AASNLQS (SEQ ID NO: 2637) (Combined) LCDR3 QQYTDESMT (SEQ
ID NO: 2638) (Combined) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Kabat)

LCDR2 AASNQS (SEQ ID NO: 2637) (Kabat) LCDR3 QQYTDESMT (SEQ ID NO: 2638) (Kabat) LCDR1 SQGISSD (SEQ ID NO: 2639) (Chothia) LCDR2 AAS (SEQ ID NO: 2640) (Chothia) LCDR3 YTDESM (SEQ ID NO: 2641) (Chothia) LCDR1 QGISSD (SEQ ID NO: 2642) (IMGT) LCDR2 AAS (SEQ ID NO: 2640) (IMGT) LCDR3 QQYTDESMT (SEQ ID NO: 2638) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNWWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYTDESMTFGQGTKVEIK (SEQ ID NO: 2643) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC
AGAGTCACCATCACTTGCCGGGCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAG (SEQ ID NO: 2650) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNWWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYTDESMTFGQGTKVEIKR
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQE
SVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO: 2645) DNA Light Chain
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTTCCC
CCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCCTGC
TGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAAC
GCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCCGAC
TACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2651)
MOR44746E HCDR1 GDSVSSSSAAWN (SEQ ID NO: 2623) (Combined) HCDR2
HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Combined) HCDR3
GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Combined) HCDR1 SSSAAWN (SEQ ID NO: 2626) (Kabat) HCDR2 HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Kabat) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Kabat) HCDR1
GDSVSSSSA (SEQ ID NO: 2627) (Chothia) HCDR2 GYRSKWY (SEQ ID NO: 2628) (Chothia) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Chothia) HCDR1
GDSVSSSSAA (SEQ ID NO: 2629) (IMGT) HCDR2 IGYRSKWYN (SEQ ID NO: 2630) (IMGT) HCDR3 ARGMYGSVPYKEGYFDI (SEQ ID NO: 2631) (IMGT)
VH QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGHI
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFDIWGQGLTVTVSS (SEQ ID NO: 2632) DNA VH
CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCTCGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC

AGTTGACCTGACCTCCGCGGAGGATACCCGCCGTGTACTACTGTGCGCGGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCCTCG (SEQ ID NO: 2647) Heavy Chain
QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGHI
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFYFDIWGQGTLTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISR
TPEVTCVAVVAVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKALAAPIEKTISKAKGQPREPQVYTLPPSREEMT
KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWQQGNVFCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 2656) DNA
Heavy CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC Chain
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCTCGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCCTCGGCCTCCACTAAGGGCCCGTCAG
TGTTCCCCCTTGCGCCATCCTCGAAGTCAACCTCCGGAGGAACTGCCGCAC
TGGGTTGCCTCGTGAAAGACTATTTCCCGGAACCCGTCCTGTCTCCTGGA
ACTCAGGAGCGCTCACCAGCGGAGTGCATACCTTTCCTGCGGTGCTGCAGT
CCAGCGGCCTGTACTCCCTGAGCTCCGTCGTGACCGTCCCCTCGTCGTCCCT
GGGAACCCAAACCTACATTTGCAACGTCAATCACAAGCCAAGCAACACTA
AGGTGGACAAGAGAGTGGAGCCCAAGTCCTGCGATAAGACCCACACCTGT
CCTCCCTGTCCGGCACCTGAACTGCTTGGTGGACCTTCCGTGTTCTGTTC
CGCCCAAGCCAAAAGACACCCTGATGATCTCCCGCACTCCGGAAGTCACTT
GCGTGGTCGTGGCCGTGTCCACGAGGACCCCGAGGTCAAGTTTAATTGGT
ACGTGGACGGAGTGGAAAGTGCACAACGCCAAGACCAAGCCGCGGGAAGA
ACAGTACAACCTCCACCTACCGCGTGGTGTCCGTCTGACTGTGCTCCACCA
GGACTGGCTGAACGGAAGGAGTACAAGTGCAAAGTGTCCAACAAGGCAC
TGGCTGCCCCCTATCGAAAAGACTATCTCCAAGGCCAAGGGCCAACCTAGG
GAGCCCCAGGTGTACACGTTGCCTCCTTCCCGCGAAGAAATGACTAAGAAC
CAGGTGTCGCTGACCTGTCTCGTGAAAGGGTTCTACCCCTCTGACATCGCC
GTGGAATGGGAGTCAAACGGACAGCCTGAGAACAACCTATAAGACCACACC
ACCTGTCCTGGACTCCGACGGCTCCTTCTTCTGTACTCAAAGTTGACCGTG
GACAAGTCGCGGTGGCAACAGGGCAACGTGTTCTCTTGCTCCGTGCTGCAC
GAAGCCCTGCACAGCCACTACACCCAAAAGTCGCTCAGCCTCTCCCCCGGA AAG
(SEQ ID NO: 2657) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Combined)
LCDR2 AASNLS (SEQ ID NO: 2637) (Combined) LCDR3 QQYTDESMT (SEQ
ID NO: 2638) (Combined) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Kabat)
LCDR2 AASNLS (SEQ ID NO: 2637) (Kabat) LCDR3 QQYTDESMT (SEQ ID
NO: 2638) (Kabat) LCDR1 SQGISSD (SEQ ID NO: 2639) (Chothia) LCDR2 AAS
(SEQ ID NO: 2640) (Chothia) LCDR3 YTDESM (SEQ ID NO: 2641) (Chothia)
LCDR1 QGISSD (SEQ ID NO: 2642) (IMGT) LCDR2 AAS (SEQ ID NO: 2640)
(IMGT) LCDR3 QQYTDESMT (SEQ ID NO: 2638) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLN WYQQKPKAPKLLIYAASNL (SEQ
ID NO: 2643) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC

AGATCATCAGCACTTCCGGGAGGGAATTTCCTCCGCAAGCTGCTGATCTACGCCGC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAG (SEQ ID NO: 2650) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQGISDLNWFYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQYTDSEMTFGQGTKVEIKR
TVAAPSVEFIPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQE
SVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID
NO: 2645) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTTCCC
CCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCCTGC
TGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAAC
GCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCCGAC
TACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2651)
MOR44746F HCDR1 GDSVSSSSAAWN (SEQ ID NO: 2623) (Combined) HCDR2
HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624) (Combined) HCDR3
GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Combined) HCDR1 SSSAAWN (SEQ
ID NO: 2626) (Kabat) HCDR2 HIGYRSKWYNEYAVSVKS (SEQ ID NO: 2624)
(Kabat) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Kabat) HCDR1
GDSVSSSSA (SEQ ID NO: 2627) (Chothia) HCDR2 GYRSKWY (SEQ ID NO:
2628) (Chothia) HCDR3 GMYGSVPYKEGYFDI (SEQ ID NO: 2625) (Chothia) HCDR1
GDSVSSSSAA (SEQ ID NO: 2629) (IMGT) HCDR2 IGYRSKWYN (SEQ ID
NO: 2630) (IMGT) HCDR3 ARGMYGSVPYKEGYFDI (SEQ ID NO: 2631) (IMGT)
VH QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGH
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFDIWGQGLTVTVSS (SEQ ID NO: 2632) DNA VH
CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATAACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCTCTCG (SEQ ID NO: 2647) Heavy Chain
QVQLQQSGPGLVKPSQTLSTCAISGDSVSSSSAAWNWIRQSPSRGLEWLGH
GYRSKWYNEYAVSVKSRITINPDTSKNQFSLQLNSVTPEDTAVYYCARGMYG
SVPYKEGYFDIWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK
DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICN
VNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLYITR
EPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVL

TVLHQDWLKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFFLYSKLTV
DKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2658) DNA
Heavy CAAGTGCAACTCCAGCAGTCAGGACCGGGGTTGGTCAAGCCTTCGCAGAC Chain
CCTGTCCCTCACTTGCGCCATTAGCGGAGATTCGGTGTCTGTCGTCGTCAGC
CGCCTGGAAGTGGATTAGACAGTCCCCTTCCCGAGGGGCTGGAGTGGCTGGG
CCACATCGGATACCGCAGCAAGTGGTACAACGAATACGCCGTCAGCGTGA
AGTCACGCATCACCATCAACCCGGATACTAGCAAGAACCAGTTCAGCCTCC
AGTTGAACTCCGTGACCCCGGAGGATACCGCCGTGTACTACTGTGCGCGGG
GCATGTACGGATCCGTGCCGTACAAGGAGGGATACTACTTCGACATTTGGG
GCCAGGGGACTCTTGTCACCGTGTCTCGGCCTCCACTAAGGGCCCGTCAG
TGTTCCCCCTTGCGCCATCCTCGAAGTCAACCTCCGGAGGAACTGCCGCAC
TGGGTTGCCTCGTGAAAGACTATTTCCCGGAACCCGTCAGTGTCTCCTGGA
ACTCAGGAGCGCTCACCAGCGGAGTGCATACCTTTCCTGCGGTGCTGCAGT
CCAGCGGCCTGTACTCCCTGAGCTCCGTCTGTGACCGTCCCCTCGTCGTCCCT
GGGAACCCAAACCTACATTTGCAACGTCAATCACAAGCCAAGCAACACTA
AGGTGGACAAGAGAGTGGAGCCCAAGTCCTGCGATAAGACCCACACCTGT
CCTCCCTGTCCGGCACCTGAACTGCTTGGTGGACCTTCCGTGTTCTGTTC
CGCCCAAGCCAAAAGACACCCTGTATATCACTCGCGAACCGGAAGTCACTT
GCGTGGTCGTGGACGTGTCCACGAGGACCCCGAGGTCAAGTTTAATTGGT
ACGTGGACGGAGTGGAAGTGCACAACGCCAAGACCAAGCCGCGGGAAGA
ACAGTACAACCTCCACCTACCGCGTGGTGTCCGTCTGACTGTGCTCCACCA
GGACTGGCTGAACGGAAGGAGTACAAGTGCAAAGTGTCCAACAAGGCAC
TGCCAGCCCCTATCGAAAAGACTATCTCCAAGGCCAAGGGCCAACCTAGG
GAGCCCCAGGTGTACACGTTGCCTCCTTCCCGCGAAGAAATGACTAAGAAC
CAGGTGTCTGCTGACCTGTCTCGTGAAAGGGTTCTACCCCTCTGACATCGCC
GTGGAATGGGAGTCAAACGGACAGCCTGAGAACAACTATAAGACCACACC
ACCTGTCCTGGACTCCGACGGCTCCTTCTTCTGTACTCAAAGTTGACCGTG
GACAAGTCGCGGTGGCAACAGGGCAACGTGTTCTCTTGCTCCGTGATGCAC
GAAGCCCTGCACAACCACTACACCCAAAAGTCGCTCAGCCTCTCCCCCGGA AAG
(SEQ ID NO: 2659) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Combined)
LCDR2 AASNLS (SEQ ID NO: 2637) (Combined) LCDR3 QQYTDESMT (SEQ
ID NO: 2638) (Combined) LCDR1 RASQGISSDLN (SEQ ID NO: 2636) (Kabat)
LCDR2 AASNLS (SEQ ID NO: 2637) (Kabat) LCDR3 QQYTDESMT (SEQ ID
NO: 2638) (Kabat) LCDR1 SQGISSD (SEQ ID NO: 2639) (Chothia) LCDR2 AAS
(SEQ ID NO: 2640) (Chothia) LCDR3 YTDESM (SEQ ID NO: 2641) (Chothia)
LCDR1 QGISSD (SEQ ID NO: 2642) (IMGT) LCDR2 AAS (SEQ ID NO: 2640)
(IMGT) LCDR3 QQYTDESMT (SEQ ID NO: 2638) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNHWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYTDESMTFGQGTKVEIK (SEQ ID
NO: 2643) DNA VL
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC
AGAGTCACCATCACTTGCCGGGCCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACGCCGC
GTCCAACCTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAG (SEQ ID NO: 2650) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQGISSDLNHWYQQKPGKAPKLLIYAASNL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYTDESMTFGQGTKVEIKR

TVAAPSPVFIPPSPDEQLKSTASVCLLNFFYPREAKVQWKVDNALQSGNSQE
SVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID
NO: 2645) DNA Light
GACATTCAGATGACCCAGTCCCCGTCGTCCCTGTCCGCATCCGTGGGCGAC Chain
AGAGTCACCATCACTTGCCGGGCTCACAGGGAATTCCTCCGACCTGAAC
TGGTATCAGCAGAAGCCTGGAAAGGCCCGAAGCTGCTGATCTACGCCGC
GTCCAACTTGCAATCGGGAGTGCCAAGCCGCTTTTCTGGTTCCGGGAGCGG
GACTGACTTCACCCTGACTATTAGCAGCCTGCAGCCCGAAGATTCGCTAC
CTACTACTGCCAACAGTACACAGATGAATCCATGACCTTCGGACAGGGCAC
CAAAGTCGAGATCAAGCGTACGGTGGCCGCTCCCAGCGTGTTTCATCTTCCC
CCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCCTGC
TGAACAACTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAAC
GCCCTGCAGAGCGGCAACAGCCAGGAGAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCCGAC
TACGAGAAGCATAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACAGGGGCGAGTGC (SEQ ID NO: 2651)
MOR042596 HCDR1 GYTFTGYHMS (SEQ ID NO: 2586) (Combined) HCDR2
VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Combined) HCDR3 IPSYTYAFDY
(SEQ ID NO: 2588) (Combined) HCDR1 GYHMS (SEQ ID NO: 2589) (Kabat)
HCDR2 VINPVSGNTVYAQKFQG (SEQ ID NO: 2587) (Kabat) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Kabat) HCDR1 GYTFTGY (SEQ ID NO:
2590) (Chothia) HCDR2 NPVSGN (SEQ ID NO: 2591) (Chothia) HCDR3
IPSYTYAFDY (SEQ ID NO: 2588) (Chothia) HCDR1 GYTFTGYH (SEQ ID NO:
2592) (IMGT) HCDR2 INPVSGNT (SEQ ID NO: 2593) (IMGT) HCDR3
ARIPSYTYAFDY (SEQ ID NO: 2594) (IMGT) VH
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSLRSEDVAVYYCARIPSYT
YAFDYWGQGTTLVTVSS (SEQ ID NO: 2595) DNA VH
CAGGTGCAATTGGTGCAGAGCGGTGCGGAAGTGAAAAAACCGGGTGCCAG
CGTGAAAGTTAGCTGCAAAGCGTCCGGATATACCTTCACTGGTTACCATAT
GTCTTGGGTGCGCCAGGCCCCGGGCCAGGGCCTCGAGTGGATGGGCGTTAT
CAACCCGGTTTCTGGCAACACGGTTTACGCGCAGAAATTCAGGGCCGGGT
GACCATGACCCGTGATACCAGCATTAGCACCGCGTATATGGAAGTGAAGCC
GTCTGCGTAGCGAAGATACGGCCGTGTATTATTGCGCGCGTATCCCGTCTT
ACACTTACGCTTTCGATTACTGGGGCCAAGGCACCCTGGTGACTGTTAGCT CA
(SEQ ID NO: 2596) Heavy Chain
QVQLVQSGAEVKKPGASVKVSCKASGYTFTGYHMSWVRQAPGQGLEWMGV
INPVSGNTVYAQKFQGRVTMTRDTSISTAYMELSLRSEDVAVYYCARIPSYT
YAFDYWGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV
TVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2597) DNA Heavy
CAGGTGCAATTGGTGCAGAGCGGTGCGGAAGTGAAAAAACCGGGTGCCAG Chain
CGTGAAAGTTAGCTGCAAAGCGTCCGGATATACCTTCACTGGTTACCATAT
GTCTTGGGTGCGCCAGGCCCCGGGCCAGGGCCTCGAGTGGATGGGCGTTAT
CAACCCGGTTTCTGGCAACACGGTTTACGCGCAGAAATTCAGGGCCGGGT
GACCATGACCCGTGATACCAGCATTAGCACCGCGTATATGGAAGTGAAGCC

GTCTGATCGATAGCAATACGCTGATATTGCGCGGTATCCCGTCTT
ACACTTACGCTTTTCGATTACTGGGGCCAAGGCACCCTGGTGACTGTTAGCT
CAGCCTCCACCAAGGGTCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGA
GCACCTCTGGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCC
CCGAACCGGTGACGGTGTCTGTGGAACCTCAGGCGCCCTGACCAGCGGCGTG
CACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGC
GTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAAC
GTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAA
ATCTTGTGACAAAACCTCACACATGCCACCCGTGCCCAGCACCTGAAGCAGC
GGGGGGACCGTCAGTCTTCTCTTCCCCCCTAAAACCCAAGGACACCCTCAT
GATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACG
AAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCAT
AATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGGGT
GGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGT
ACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACC
ATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCC
CCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGG
TCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGG
CAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGG
CTCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCA
GGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTA
CACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA (SEQ ID NO: 2598) LCDR1
RASQDISNYLA (SEQ ID NO: 2599) (Combined) LCDR2 RASSLQS (SEQ ID
NO: 2600) (Combined) LCDR3 QQHGHSPPT (SEQ ID NO: 2660) (Combined) LCDR1
RASQDISNYLA (SEQ ID NO: 2599) (Kabat) LCDR2 RASSLQS (SEQ ID NO:
2600) (Kabat) LCDR3 QQHGHSPPT (SEQ ID NO: 2660) (Kabat) LCDR1 SQDISNY
(SEQ ID NO: 2602) (Chothia) LCDR2 RAS (SEQ ID NO: 2603) (Chothia) LCDR3
HGHSPPT (SEQ ID NO: 2661) (Chothia) LCDR1 QDISNY (SEQ ID NO: 2605)
(IMGT) LCDR2 RAS (SEQ ID NO: 2603) (IMGT) LCDR3 QQHGHSPPT (SEQ ID
NO: 2660) (IMGT) VL
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQHGHSPPTTFGQGTKVEIK (SEQ ID
NO: 2662) DNA VL
GATATCCAGATGACCCAGAGCCCGAGCAGCCTGAGCGCCAGCGTGGGCGA
TCGCGTGACCATTACCTGCAGAGCCAGCCAGGACATTTCTAACTACCTGGC
TTGGTACCAGCAGAAACCGGGCAAAGCGCCGAAACTATTAATCTACCGTG
CTTCTTCTCTGCAAAGCGGCGTGCCGAGCCGCTTTAGCGGCAGCGGATCCG
GCACCGATTTACCCCTGACCATTAGCTCTCTGCAACCGGAAGACTTTGCGA
CCTATTATTGCCAGCAGCATGGTCATTCTCCGACTACCTTTGGCCAGGGCA
CGAAAGTTGAAATTAAA (SEQ ID NO: 2663) Light Chain
DIQMTQSPSSLSASVGDRVTITCRASQDISNYLAWYQQKPGKAPKLLIYRASSL
QSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQHGHSPPTTFGQGTKVEIKRT
VAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQES
VTEQDSKDSSTLSLTLSKADYEEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID
NO: 2664) DNA Light
GATATCCAGATGACCCAGAGCCCGAGCAGCCTGAGCGCCAGCGTGGGCGA Chain
TCGCGTGACCATTACCTGCAGAGCCAGCCAGGACATTTCTAACTACCTGGC
TTGGTACCAGCAGAAACCGGGCAAAGCGCCGAAACTATTAATCTACCGTG
CTTCTTCTCTGCAAAGCGGCGTGCCGAGCCGCTTTAGCGGCAGCGGATCCG
GCACCGATTTACCCCTGACCATTAGCTCTCTGCAACCGGAAGACTTTGCGA

CCTATTATTCAGCATACCTTTCGACCTACCTTTGGCCAGGGCA
CGAAAGTTGAAATTAAACGTACGGTGGCCGCTCCCAGCGTGTTTCATCTTCC
CCCCAGCGACGAGCAGCTGAAGAGCGGCACCGCCAGCGTGGTGTGCCTG
CTGAACAACCTTCTACCCCCGGGAGGCCAAGGTGCAGTGGAAGGTGGACAA
CGCCCTGCAGAGCGGCAACAGCCAGGAAAGCGTCACCGAGCAGGACAGCA
AGGACTCCACCTACAGCCTGAGCAGCACCTGACCCTGAGCAAGGCCGAC
TACGAGAAGCACAAAGGTGTACGCCTGCGAGGTGACCCACCAGGGCCTGTC
CAGCCCCGTGACCAAGAGCTTCAACCGGGGCGAGTGT (SEQ ID NO: 2665)
MOR041877 HCDR1 GFSLSTSGVGV (SEQ ID NO: 2666) (Combined) HCDR2
LIFSDHDKIYSTSLKT (SEQ ID NO: 2667) (Combined) HCDR3 TLIDRSVYFDY
(SEQ ID NO: 2668) (Combined) HCDR1 TSGVGV (SEQ ID NO: 2669) (Kabat)
HCDR2 LIFSDHDKIYSTSLKT (SEQ ID NO: 2667) (Kabat) HCDR3 TLIDRSVYFDY
(SEQ ID NO: 2668) (Kabat) HCDR1 GFSLSTSGV (SEQ ID NO: 2670) (Chothia)
HCDR2 FSDHD (SEQ ID NO: 2671) (Chothia) HCDR3 TLIDRSVYFDY (SEQ ID
NO: 2668) (Chothia) HCDR1 GFSLSTSGV (SEQ ID NO: 2672) (IMGT) HCDR2
IFSDHDK (SEQ ID NO: 2673) (IMGT) HCDR3 ARTLIDRSVYFDY (SEQ ID
NO: 2674) (IMGT) VH QVQLKESGPALVKPTQTLTLTCTF SGF
SLSTSGVGVSWIRQPPGKALEWLALIF
SDHDKIYSTSLKTRLTISKDTSKNQVVLMTNMDPVDATYYCARTLIDRSVY
FDYWGQGLTVTVSS (SEQ ID NO: 2675) DNA VH
CAGGTGCAATTGAAAGAAAGCGGTCCGGCGCTGGTGAACCGACCCAGAC
CCTGACCCTGACGTGCACCTTTTCCGGATTTCAGCCTGTCTACTTCCGGTGT
GGTGTGAGCTGGATTCGCCAGCCGCCGGGCAAAGCGCTCGAGTGGCTGGC
GCTGATCTTCTCTGACCATGACAAGATCTATAGCACCAGCCTGAAAACCCG
TCTGACCATTAGCAAAGATACTTCGAAAAACCAGGTGGTGTGACCATGAC
CAACATGGACCCGGTGGATACCGCGACCTATTATTGCGCGCGTACTCTGAT
CGACCGTTCTGTTTACTTCGATTACTGGGGCCAAGGCACCCTGGTGACTGTT
AGCTCA (SEQ ID NO: 2676) Heavy Chain QVQLKESGPALVKPTQ TLTLTCTF
SGF SLSTSGVGVSWIRQPPGKALEWLALIF
SDHDKIYSTSLKTRLTISKDTSKNQVVLMTNMDPVDATYYCARTLIDRSVY
FDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
SWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTK
VDKRVEPKSCDKTHTCPPCPAPEAAGGPSVFLFPPKPKDTLMISRTPEVTCVVV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN
GKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCL
VKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQG
NVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2677) DNA Heavy
CAGGTGCAATTGAAAGAAAGCGGTCCGGCGCTGGTGAACCGACCCAGAC Chain
CCTGACCCTGACGTGCACCTTTTCCGGATTTCAGCCTGTCTACTTCCGGTGT
GGTGTGAGCTGGATTCGCCAGCCGCCGGGCAAAGCGCTCGAGTGGCTGGC
GCTGATCTTCTCTGACCATGACAAGATCTATAGCACCAGCCTGAAAACCCG
TCTGACCATTAGCAAAGATACTTCGAAAAACCAGGTGGTGTGACCATGAC
CAACATGGACCCGGTGGATACCGCGACCTATTATTGCGCGCGTACTCTGAT
CGACCGTTCTGTTTACTTCGATTACTGGGGCCAAGGCACCCTGGTGACTGTT
AGCTCAGCCTCCACCAAGGGTCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
AAGAGCACCTCTGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTA
CTTCCCCGAACCGGTGACGGTGTCTGGAACCTCAGGCGCCCTGACCAGCGG
CGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGC
AGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGC
AACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCC

CAAATCTTGTGACGAACTCACTACCATGCCACCGTGCCCGACCTGGAAGC
AGCGGGGGGACCGTCAGTCTTCTCTTCCCCC AAAACCCAAGGACACCCT
CATGATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCA
CGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGC
ATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGG
GTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGA
GTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAA
CCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTG
CCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCT
GGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATG
GGCAGCCGGAGAACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGAC
GGCTCCTTCTTCTCTACAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAG
CAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCAC
TACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA (SEQ ID NO: 2678) LCDR1
SGSSSNIGHHYVS (SEQ ID NO: 2679) (Combined) LCDR2 DNTNRPS (SEQ ID
NO: 2680) (Combined) LCDR3 ATWDGLMNSIV (SEQ ID NO: 2681) (Combined)
LCDR1 SGSSSNIGHHYVS (SEQ ID NO: 2679) (Kabat) LCDR2 DNTNRPS (SEQ
ID NO: 2680) (Kabat) LCDR3 ATWDGLMNSIV (SEQ ID NO: 2681) (Kabat) LCDR1
SSSNIGHHY (SEQ ID NO: 2682) (Chothia) LCDR2 DNT (SEQ ID NO: 2683)
(Chothia) LCDR3 WDGLMNSI (SEQ ID NO: 2684) (Chothia) LCDR1 SSNIGHHY
(SEQ ID NO: 2685) (IMGT) LCDR2 DNT (SEQ ID NO: 2683) (IMGT) LCDR3
ATWDGLMNSIV (SEQ ID NO: 2681) (IMGT) VL
DIVLTQPPSVSGAPGQRVTISCSGSSSNIGHHYVSWYQQLPGTAPKLLIYDNTN
RPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCATWDGLMNSIVFGGGTKL TVL
(SEQ ID NO: 2686) DNA VL
GATATCGTGCTGACCCAGCCGCGGAGCGTGAGCGGTGCACCGGGGCCAGCG
CGTGACCATTAGCTGTAGCGGCAGCAGCAGCAACATTGGTCATCATTACGT
GTCTTGGTACCAGCAGCTGCCGGGCACGGCGCCGAAACTGCTGATCTACGA
CAACACTAACCGCCCCGAGCGGCGTGCCGGATCGCTTTAGCGGATCCAAAA
GCGGCACCAGCGCCAGCCTGGCGATTACCGGCCTGCAAGCAGAAGACGAA
GCGGATTATTACTGCGCTACTTGGGACGGTCTGATGAACTCTATCGTGTTTG
GCGGCGGCACGAAGTTAACCGTCCTA (SEQ ID NO: 2687) Light Chain
DIVLTQPPSVSGAPGQRVTISCSGSSSNIGHHYVSWYQQLPGTAPKLLIYDNTN
RPSGVPDRFSGSKSGTSASLAITGLQAEDEADYYCATWDGLMNSIVFGGGTKL
TVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVK
AGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPT ECS
(SEQ ID NO: 2688) DNA Light Chain
GATATCGTGCTGACCCAGCCGCGGAGCGTGAGCGGTGCACCGGGGCCAGCG Chain
CGTGACCATTAGCTGTAGCGGCAGCAGCAGCAACATTGGTCATCATTACGT
GTCTTGGTACCAGCAGCTGCCGGGCACGGCGCCGAAACTGCTGATCTACGA
CAACACTAACCGCCCCGAGCGGCGTGCCGGATCGCTTTAGCGGATCCAAAA
GCGGCACCAGCGCCAGCCTGGCGATTACCGGCCTGCAAGCAGAAGACGAA
GCGGATTATTACTGCGCTACTTGGGACGGTCTGATGAACTCTATCGTGTTTG
GCGGCGGCACGAAGTTAACCGTCCTAGGTCAGCCCAAGGCTGCCCCCTCG
GTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAGCCAACAAGGCCACA
CTGGTGTGTCTCATAAGTGA CTTCTACCCGGGAGCCGTGACAGTGGCCTGG
AAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGGAGACCACCACACCCTC
CAAACAAAGCAACAACAAGTACGCGGCCAGCAGCTATCTGAGCCTGACGC
CTGAGCAGTGGAAGTCCCACAGAAGCTACAGCTGCCAGGTCACGCATGAA
GGGAGCACCGTGAGAAAGACAGTGGCCCCTACAGAATGTTCA (SEQ ID NO: 2689)

[0540] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed above, including those in TABLE J1 above, may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

K. KR Patent Application Publication No. KR20200048069A

[0541] In some embodiments, the TREM2 agonist is an antibody, or an antigen-binding fragment thereof, as described in KR Patent Application Publication No. KR20200048069A, which is incorporated by reference herein, in its entirety.

[0542] In some embodiments, the TREM2 antibody comprises the CDR L1, CDR L2 and CDR L3 in the light chain variable region of the antibody produced by hybridoma cells with accession number KCTC 13471BP or hybridoma cells with accession number KTC 13470BP.

[0543] In some embodiments, the TREM2 antibody comprises the CDR H1, CDR H2 and CDR H3 in the heavy chain variable region of the antibody produced by hybridoma cells with accession number KCTC 13471BP or hybridoma cells with accession number KTC 13470BP.

[0544] In some embodiments, the TREM2 antibody comprises the CDR L1, CDR L2 and CDR L3 in the light chain variable region and the CDR H1, CDR H2 and CDR H3 in the heavy chain variable region of the antibody produced by hybridoma cells with accession number KCTC 13471BP or hybridoma cells with accession number KTC 13470BP.

[0545] In some embodiments, the TREM2 antibody comprises the light chain variable region and the heavy chain variable region of the antibody produced by hybridoma cells with accession number KCTC 13471BP or hybridoma cells with accession number KTC 13470BP.

[0546] In some embodiments, the TREM2 agonist is an antibody produced by hybridoma cells with accession number KCTC 13471BP or hybridoma cells with accession number KTC 13470BP.

[0547] In some embodiments, the light chain variable regions and the heavy chain variable regions describe above for the antibody produced by hybridoma cells with accession number KCTC 13471BP or hybridoma cells with accession number KTC 13470BP may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

L. PCT Patent Application Publication No. WO2020/172450A1

[0548] In some embodiments, the TREM2 agonist is an antibody, or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2020/172450A1 (“the ‘450 application”), which is incorporated by reference herein, in its entirety.

[0549] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0550] (a) a CDR-H1 sequence comprising the sequence of GFSIEDFYIH (SEQ ID NO:2717); [0551] (b) a CDR-H2 sequence comprising the sequence of W-I-D-P-E- β .sub.6-G- β .sub.8-S-K-Y-A-P-K-F-Q-G (SEQ ID NO:2735), wherein β .sub.6 is N or Q and β .sub.8 is D or E; [0552] (c) a CDR-H3 sequence comprising the sequence of HADHGNYGSTMDY (SEQ ID NO:2719); [0553] (d) CDR-L1 sequence comprising the sequence of HASQHINVWLS (SEQ ID NO:2720); [0554] (e) a CDR-L2 sequence comprising the sequence of KASNLHT (SEQ ID NO:2721); and [0555] (f) a CDR-L3 sequence comprising the sequence of QQGQTYPR (SEQ ID NO: 2722).

[0556] In some embodiments, the CDR-H2 sequence is selected from SEQ ID NOS:2718, 2727, 2729, and 2731.

[0557] In some embodiments, the antibody or antigen-binding fragment comprises: [0558] (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2717, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2718, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2719, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2720, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2721, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2722; or [0559] (b) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2717, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2727, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2719, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2720, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2721, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2722; or [0560] (c) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2717, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2729, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2719, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2720, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2721, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2722; or [0561] (d) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2717, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2731, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2719, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2720, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2721, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2722.

[0562] In some embodiments, the antibody or antigen-binding fragment comprises a V H sequence that has at least 85% sequence identity to any one of SEQ ID NOS:2715, 2723, 2725, 2726, 2728, 2730, 2732, 2733, and 2734. In some embodiments, the V H sequence has at least 90% sequence identity to SEQ ID NO:2715. In some embodiments, the V.sub.H sequence has at least 95% sequence identity to SEQ ID NO:2715. In some embodiments, the VH sequence comprises SEQ ID NO:2715. In some embodiments, the V.sub.H sequence has at least 90% sequence identity to SEQ ID NO:2730. In some embodiments, the V.sub.H sequence has at least 95% sequence identity to SEQ ID NO:2730. In some embodiments, the V.sub.H sequence comprises SEQ ID NO:2730. In some embodiments, the V.sub.H sequence has at least 90% sequence identity to SEQ ID NO:2733. In some embodiments, the V.sub.H sequence has at least 95% sequence identity to SEQ ID NO:2733. In some embodiments, the V.sub.H sequence comprises SEQ ID NO:2733.

[0563] In some embodiments, the antibody or antigen-binding fragment comprises a V.sub.L sequence that has at least 85% sequence identity to SEQ ID NO:2716 or SEQ ID NO:2724. In some embodiments, the V.sub.L sequence has at least 90% sequence identity to SEQ ID NO:2716. In some embodiments, the V.sub.L sequence has at least 95% sequence identity to SEQ ID NO:2716. In some embodiments, the V.sub.L sequence comprises SEQ ID NO:2716. In some embodiments, the V.sub.L sequence has at least 90% sequence identity to SEQ ID NO:2724. In some embodiments, the V.sub.L sequence has at least 95% sequence identity to SEQ ID NO:2724. In some embodiments, the V.sub.L sequence comprises SEQ ID NO:2724.

[0564] In some embodiments, the antibody or antigen-binding fragment comprises: [0565] (a) a VH sequence comprising SEQ ID NO:2715 and a V.sub.L sequence comprising SEQ ID NO:2716; or [0566] (b) a VH sequence comprising SEQ ID NO:2723 and a V.sub.L sequence comprising SEQ ID NO:2724; or [0567] (c) a VH sequence comprising SEQ ID NO:2725 and a V.sub.L sequence comprising SEQ ID NO:2724; or [0568] (d) a VH sequence comprising SEQ ID NO:2726 and a V.sub.L sequence comprising SEQ ID NO:2724; or [0569] (e) a VH sequence comprising SEQ ID NO:2728 and a V.sub.L sequence comprising SEQ ID NO:2724; or [0570] (f) a VH sequence comprising SEQ ID NO:2730 and a V.sub.L sequence comprising SEQ ID NO:2724; or [0571] (g) a VH sequence comprising SEQ ID NO:2732 and a V.sub.L sequence comprising SEQ ID NO:2724; or [0572] (h) a VH sequence comprising SEQ ID NO:2733 and a VL sequence comprising SEQ ID NO:2724; or [0573] (i) a VH sequence comprising SEQ ID NO:2734 and a VL sequence comprising SEQ ID NO:2724.

[0574] In some embodiments, an antibody or antigen-binding fragment thereof that specifically binds to TREM2 comprises: [0575] (a) a CDR-H1 sequence comprising the sequence of G-F-T-F-T- α .sub.6-F-Y-M-S (SEQ ID NO:2736), wherein α .sub.6 is D or N; [0576] (b) a CDR-H2 sequence comprising the sequence of V-I-R-N- β .sub.5- β .sub.6-N- β .sub.8-Y-T- β .sub.11- β .sub.12-Y-N-P-S-V-K-G (SEQ ID NO:2737), wherein β .sub.5 is K or R; β .sub.6 is A or P; β .sub.8 is G or A; β .sub.11 is A or T; and β .sub.12 is G or D; [0577] (c) a CDR-H3 sequence comprising the sequence of γ .sub.1-R-L- γ .sub.4-Y-G-F-D-Y (SEQ ID NO:2738), wherein γ .sub.1 is A or T; and γ 4 is T or S; [0578] (d) a CDR-L1 sequence comprising the sequence of Q-S-S-K-S-L-L-H-S- δ to-G-K-T-Y-L-N (SEQ ID NO:2739), wherein δ to is N or T; a CDR-L2 sequence comprising the sequence of WMSTRAS (SEQ ID NO: 2696); and [0579] (e) a CDR-L3 sequence comprising the sequence of Q-Q-F-L-E- ϕ .sub.6-P-F-T (SEQ ID NO:2740), wherein ϕ .sub.6 is Y or F.

[0580] In some embodiments, the CDR-H1 sequence is selected from any one of SEQ ID NOS:2692 and 2700. In some embodiments, the CDR-H2 sequence is selected from any one of SEQ ID NOS:2693, 2701, and 2713. In some embodiments, the CDR-H3 sequence is selected from any one of SEQ ID NOS:2694, 2702, and 2705. In some embodiments, the CDR-L1 sequence is selected from any one of SEQ ID NOS:2695 and 2711. In some embodiments, the CDR-L3 sequence is selected from any one of SEQ ID NOS:2697 and 2706.

[0581] In some embodiments, the antibody or antigen-binding fragment comprises: [0582] (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2693, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2705, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2695, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2706; or [0583] (b) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2693, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2705, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2711, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2706; or [0584] (c) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2713, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2705, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2695, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2706; or [0585] (d) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2713, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2705, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2711, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2706; or [0586] (e) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2693, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2694, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2695, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2697; or [0587] (f) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2700, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2701, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2702, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2695, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2697; or [0588] (g) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2713, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2705, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2695, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2696, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2697.

[0589] In some embodiments, the antibody or antigen-binding fragment comprises a VH sequence that has at least 85% sequence identity to any one of SEQ ID NOS:2690, 2698, 2703, 2708, 2709, 2712, 2714, and 2752. In some embodiments, the VH sequence has at least 90% sequence identity to SEQ ID NO:2703. In some embodiments, the VH sequence has at least 95% sequence identity to SEQ ID NO:2703. In some embodiments, the VH sequence comprises SEQ ID NO:2703. In some embodiments, the VH sequence has at least 90% sequence identity to SEQ ID NO:2712. In some embodiments, the VH sequence has at least 95% sequence identity to SEQ ID NO:2712. In some embodiments, the VH sequence comprises SEQ ID NO:2712. In some embodiments, the VH sequence has at least 90% sequence identity to SEQ ID NO:79. In some embodiments, the VH sequence has at least 95% sequence identity to SEQ ID NO:79. In some embodiments, the VH sequence comprises SEQ ID NO:79.

[0590] In some embodiments, the antibody or antigen-binding fragment comprises a VL sequence that has at least 85% sequence identity to any one of SEQ ID NOS:2691, 2699, 2704, 2708, 2710, and 2741. In some embodiments, the VL sequence has at least 90% sequence identity to SEQ ID NO:2704. In some embodiments, the VL sequence has at least 95% sequence identity to SEQ ID NO:2704. In some embodiments, the VL sequence comprises SEQ ID NO:2704. In some embodiments, the VL sequence has at least 90% sequence identity to SEQ ID NO:2710. In some embodiments, the VL sequence has at least 95% sequence identity to SEQ ID NO:2710. In some embodiments, the VL sequence comprises SEQ ID NO:2710. In some embodiments, the VL sequence has at least 90% sequence identity to SEQ ID NO:2741. In some embodiments, the VL sequence has at least 95% sequence identity to SEQ ID NO:2741. In some embodiments, the VL sequence comprises SEQ ID NO:2741.

[0591] In some embodiments, the antibody or antigen-binding fragment comprises: [0592] (a) a VH sequence comprising SEQ ID NO:2703 and a VL sequence comprising SEQ ID NO:2704; or [0593] (b) a VH sequence comprising SEQ ID NO:2707 and a VL sequence comprising SEQ ID NO:2708; or [0594] (c) a VH sequence comprising SEQ ID NO:2709 and a VL sequence comprising SEQ ID NO:2708; or [0595] (d) a VH sequence comprising SEQ ID NO:2707 and a VL sequence comprising SEQ ID NO:2710; or [0596] (e) a VH sequence comprising SEQ ID NO:79 and a VL sequence comprising SEQ ID NO:2710; or [0597] (f) a VH sequence comprising SEQ ID NO:2712 and a VL sequence comprising SEQ ID NO:2708; or [0598] (g) a VH sequence comprising SEQ ID NO:2714 and a VL sequence comprising SEQ ID NO:2708; or [0599] (h) a VH sequence comprising SEQ ID NO:2712 and a VL sequence comprising SEQ ID NO:2710; or [0600] (i) a VH sequence comprising SEQ ID NO:2714 and a VL sequence comprising SEQ ID NO:2710; or [0601] (j) a VH sequence comprising SEQ ID NO:2690 and a VL sequence comprising SEQ ID NO:2691; or [0602] (k) a VH sequence comprising SEQ ID NO:2698 and a VL sequence comprising SEQ ID NO:2699; or [0603] (1) a VH sequence comprising SEQ ID NO:2712 and a VL sequence comprising SEQ ID NO:2741.

[0604] In some embodiments, an antibody or antigen-binding fragment thereof that specifically binds to TREM2 comprises: [0605] (a) a CDR-H1 sequence comprising the amino acid sequence of any one of SEQ ID NOS:2692, 2700, and 2717; [0606] (b) a CDR-H2 sequence comprising the amino acid sequence of any one of SEQ ID NOS:2693, 2701, 2713, 2718, 2727, 2729, and 2731; [0607] (c) a CDR-H3 sequence comprising the amino acid sequence of any one of SEQ ID NOS:2694, 2702, 2705, and 2719; [0608] (d) a CDR-L1 sequence comprising the amino acid sequence of any one of SEQ ID NOS:2695, 2711, and 2720; [0609] (e) a CDR-L2 sequence comprising the amino acid sequence of any one of SEQ ID NOS:2696 and 2721; and [0610] (f) a CDR-L3 sequence comprising the amino acid sequence of any one of SEQ ID NOS:2697, 2706, and 2722.

[0611] In some embodiments, the antibody or antigen-binding fragment comprises: [0612] (a) a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2692, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2693, a CDR-H3 comprising the amino acid sequence of SEQ

has at least 85% sequence identity to SEQ ID NO:2691; or [0625] (b) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2698 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2699; or [0626] (c) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2703 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2704; or [0627] (d) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2707 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2708; or a VH sequence that has at least 85% sequence identity to SEQ ID NO:2709 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2708; or [0628] (f) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2707 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2710; or [0629] (g) a VH sequence that has at least 85% sequence identity to SEQ ID NO:79 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2710; or [0630] (h) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2712 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2708; or [0631] (i) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2714 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2708; or [0632] (j) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2712 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2710; or [0633] (k) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2714 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2710; or [0634] (l) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2715 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2716; or [0635] (m) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2723 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0636] (n) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2725 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0637] (o) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2726 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0638] (p) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2728 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0639] (q) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2730 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0640] (r) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2732 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0641] (s) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2733 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0642] (t) a VH sequence that has at least 85% sequence identity to SEQ ID NO: 2734 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2724; or [0643] (u) a VH sequence that has at least 85% sequence identity to SEQ ID NO:2712 and a VL sequence that has at least 85% sequence identity to SEQ ID NO:2741.

[0644] In some embodiments, an antibody or antigen-binding fragment thereof that specifically binds to TREM2 recognizes an epitope that is the same or substantially the same as the epitope recognized by antibody clone selected from the group consisting of: CL0020306, Clone CL0020188, Clone CL0020188-1, Clone CL0020188-2, Clone CL0020188-3, Clone CL0020188-4, Clone CL0020188-5, Clone CL0020188-6, Clone CL0020188-7, Clone CL0020188-8, Clone CL0020307, Clone CL0020123, Clone CL0020123-1, Clone CL0020123-2, Clone CL0020123-3, Clone CL0020123-4, Clone CL0020123-5, Clone CL0020123-6, Clone CL0020123-7, and Clone CL0020123-8.

[0645] In some embodiments, the antibody or antigen-binding fragment recognizes an epitope that is the same or substantially the same as the epitope recognized by an antibody clone selected from the group consisting of: Clone CL0020123, Clone CL0020123-1, Clone CL0020123-2, Clone CL0020123-3, Clone CL0020123-4, Clone CL0020123-5, Clone CL0020123-6, Clone CL0020123-7, and Clone CL0020123-8. In particular embodiments, the antibody or antigen-binding fragment recognizes one or more of the following epitopes in SEQ ID NO: 1: (i) amino

acid residues 55-63 (GEKGPCQRV (SEQ ID NO:2743)), (ii) amino acids 96-107 (TLRNLQPHDAGL (SEQ ID NO:2744)), and (iii) amino acid residues 126-129 (VEVL (SEQ ID NO:2745)). In another aspect, the disclosure features an isolated antibody or antigen-binding fragment thereof that specifically binds to a human TREM2, wherein the antibody or antigen-binding fragment thereof recognizes an epitope comprising or consisting of one or more of the following epitopes in SEQ ID NO: 1: (i) amino acid residues 55-63 (GEKGPCQRV (SEQ ID NO:2743)), (ii) amino acids 96-107 (TLRNLQPHDAGL (SEQ ID NO:2744)), and (iii) amino acid residues 126-129 (VEVL (SEQ ID NO:2745)). In some embodiments, the antibody or antigen-binding fragment recognizes an epitope that is the same or substantially the same as the epitope recognized by an antibody clone selected from the group consisting of: Clone CL0020188, Clone CL0020188-1, Clone CL0020188-2, Clone CL0020188-3, Clone CL0020188-4, Clone CL0020188-5, Clone CL0020188-6, Clone CL0020188-7, Clone CL0020188-8, Clone CL0020307, and Clone CL0020306. In particular embodiments, the antibody or antigen-binding fragment recognizes amino acid residues 143-149 (FPGESES (SEQ ID NO:2742)) in SEQ ID NO: 1. In another aspect, the disclosure features an isolated antibody or antigen-binding fragment thereof that specifically binds to a human TREM2, wherein the antibody or antigen-binding fragment thereof recognizes an epitope comprising or consisting of amino acid residues 143-149 (FPGESES (SEQ ID NO:2742)) in SEQ ID NO:1.

[0646] In some embodiments, an antibody or antigen-binding fragment as disclosed herein decreases levels of soluble TREM2 protein (sTREM2). In some embodiments, an antibody or antigen-binding fragment as disclosed herein binds soluble TREM2 protein (sTREM2) in healthy human CSF or cynomolgus CSF with better potency compared to a reference antibody. In some embodiments, the reference antibody is represented by a combination of sequences selected from the group consisting of: SEQ ID NOS:2746 and 2747; SEQ ID NOS:2748 and 2749; and SEQ ID NOS:2750 and 2751.

[0647] In some embodiments, the antibody is an antibody having a VL, VH, full heavy chain sequence, full light chain sequence, a CDR sequence, or a full sequence disclosed in the "Informal Sequence Listing" Table IX of PCT Patent Application Publication No. WO 2020/172450 A1, which are reproduced below as TABLE L1.

TABLE-US-00060	TABLE L1	Description	Sequence	Human	TREM2	Protein
			MEPLRLILLFVTELSGAHNTTVFQGQVAGQSLQVS			
			CPYDSMKHWGRRKAWCRQLGEKGPCQRRVSTH			
			NLWLLSFLRRWNGSTAITDDTLGGTLTITLRNLQP			
			HDAGLYQCQSLHGSEADTLRKVLVEVLADPLDH			
			RDAGDLWFPGESSEFEDAHVEHSISRSLLEGEIPFP			
			PTSILLLLACIFLIKILAASALWAAAWHGQKPGTH			
			PPSELDCGHDPGYQLQTLPLGLRDT			
		(SEQ ID NO: 1) CL0020306	VH EVKLLDSGGGLVQAGGSLRLSCAGSGFTFTDFYM			
			SWIRQPPGKAPEWLGVIRNKANGYTAGYNPSVK			
		GRFTISRDNNTQNILYLQMNTL				
		RAEDTAIYYCARLSYGFDYWGQGVMTVSS (SEQ ID NO: 2690)	CL0020306			VL
			DIVMTQGALPNPVPSPGESASITCQSSKSLHSNGK			
			TYLNWYLQRPGQSPQLLIYWMSTRASGVSDRFSG			
		SGSGTDFTLKISSVEAEDVG				
		VYYCQQFLEFPFTFGSGTKLEIK (SEQ ID NO :2691)	CL0020306			CDR-H1; CDR-
		H1 for GFTFTDFYMS (SEQ ID NO: 2692)	CL0020188			and variants
			CL0020188-1, CL0020188-2, CL0020188-3, CL0020188-4, CL0020188-5, CL0020188-6, CL0020188-7, and CL0020188-8			
			CL0020306			CDR-H2; CDR-H2 for
			VIRNKANGYTAGYNPSVKG (SEQ ID NO: 2693)			CL0020188 and variants
			CL0020188-1, CL0020188-2, CL0020188-3, and CL0020188-4			
			CL0020306			CDR-H3
		ARLSYGFDY (SEQ ID NO: 2694)	CL0020306			CDR-L1; CL0020307
			CDR-QSSKSLHSNGKTYLN (SEQ ID NO: 2695)			L1; CL0020307-1
			CDR-L1; CDR-L1 for CL0020188			and variants
			CL0020188-1, CL0020188-2, CL0020188-5, and			

CL0020188-6 CDR-L2; CL0020306 CDR-L2; WMSTRAS (SEQ ID NO: 2696) CL0020307-1 CDR-L2; CDR-L2 for CL0020188 and variants CL0020188-1, CL0020188-2, CL0020188-3, CL0020188-4, CL0020188-5, CL0020188-6, CL0020188-7, and CL0020188-8 CL0020306 CDR-L3; CL0020307 CDR- QQFLEFPFT (SEQ ID NO: 2697) L3; CL0020307-1 CDR-L3 CL0020307 V.sub.H EVKLLES GGGGLVQPGGSLRLSCAASGFTFTNFYM SWIRQPPGRAPEWLGVIRNRPNGYTTDYNPSVKG RFTISRDNNTQNILYLQMSTLRADDTAFYYCTRLTY GFDYWGQGVMVTVSS (SEQ ID NO: 2698) CL0020307 VL DIVMTQGALPNPVPSGESASITCQSSKSLHNSNGK TYLNWYLQRPGQSPQLLIYWMSTRASGVSDRFSG SGSGTDFTLKIS SVEAEVVG VYYCQQFLEFPFTFGSGTKLEIK (SEQ ID NO: 2699) CL0020307 CDR-H1 GFTFTNFYMS (SEQ ID NO: 2700) CL0020307 CDR-H2 VIRNRPNGYTTDYNPSVKG (SEQ ID NO: 2701) CL0020307 CDR-H3 TRLTYGFDY (SEQ ID NO: 2702) CL0020188 VH EVKLDSGGGLVQAGGSLRLSCAGSGFTFTDFYM SWIRQPPGKAPEWLGVIRNKANGYTAGYNPSVK GRFTISRDNNTQNILYLQMNTLRAEDTAIYYCARLT YGFDYWGQGVMVTVSS (SEQ ID NO: 2703) CL0020188 VL DIVMTQGALPNPVPSGESASITCQSSKSLHNSNGK TYLNWYLQR PGQSPQLLIYWMSTRASGVSDRFSGSGSGTDFTLK ISSVEAEDVG VYYCQQFLEYPFTFGSGTKLEIK (SEQ ID NO: 2704) CDR-H3 for CL0020188 and variants ARLTYGFDY (SEQ ID NO: 2705) CL0020188-1, CL0020188-2, CL0020188-3, CL0020188-4, CL0020188-5, CL0020188-6, CL0020188-7, and CL0020188-8 CDR-L3 for CL0020188 and variants QQFLEYPFT (SEQ ID NO: 2706) CL0020188-1, CL0020188-2, CL0020188-3, CL0020188-4, CL0020188-5, CL0020188-6, CL0020188-7, and CL0020188-8 CL0020188-1 VH; CL0020188-3 VH EVQLVESGGGLVQPGGSLRLSCAASGFTFTDFYM SWVRQAPGKGLEWVSVIRNKANGYTAGYNPSVK GRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARL TYGFDYWGQGTLVTVSS (SEQ ID NO: 2707) CL0020188-1 VL; CL0020188-2 VL; DIVMTQTPLSLPVTPGEPASISCQSSKSLHNSNGKT CL0020188-5 VL; CL0020188-6 VL YLNWYLQKPGQSPQLLIYWMSTRASGVDPDRFSGS GSGTDFTLKISRVEAEDVG VYYCQQFLEYPFTFG QGTKVEIK (SEQ ID NO: 2708) CL0020188-2 VH EVQLVESGGGLVQPGGSLRLSCAGSGFTFTDFYM SWVRQAPGKGLEWVSVIRNKANGYTAGYNPSVK GRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARL TYGFDYWGQGTLVTVSS (SEQ ID NO: 2709) CL0020188-3 VL; CL0020188-4 VL; DIVMTQTPLSLPVTPGEPASISCQSSKSLHSTGKT CL0020188-7 VL; CL0020188-8 VL YLNWYLQKPGQSPQLLIYWMSTRASGVDPDRFSGS GSGTDFTLKISRVEAEDVG VYYCQQFLEYPFTFG QGTKVEIK (SEQ ID NO: 2710) CDR-L1 for variants CL0020188-3, QSSKSLHSTGKTYLN (SEQ ID NO: 2711) CL0020188-4, CL0020188-7, and CL0020188-8 CL0020188-5 VH; CL0020188-7 VH EVQLVESGGGLVQPGGSLRLSCAASGFTFTDFYM SWVRQAPGKGLEWVSVIRNKANAYTAGYNPSVK GRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARL TYGFDYWGQGTLVTVSS (SEQ ID NO: 2712) CDR-H2 for variants CL0020188-5, VIRNKANAYTAGYNPSVKG (SEQ ID NO: 2713) CL0020188-6, CL0020188-7, and CL0020188-8 CL0020188-6 VH; CL0020188-8 VH EVQLVESGGGLVQPGGSLRLSCAGSGFTFTDFYM SWVRQAPGKGPEWLSVIRNKANAYTAGYNPSVK GRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARL TYGFDYWGQGTLVTVSS (SEQ ID NO: 2714) CL0020123 VH EVQLQQSGAELVRSGASVKLSCTASGFSIEDFYIH

WVKQRPEQGLEWIDPENGDSKYAPKFQGKA
TMTADTSSNTAYLHLSSLTSED TAVYYCHADHGN YGSTMDYWGQGTSVTVSS (SEQ
ID NO: 2715) CL0020123 VL DIQMNQSPSSLSASLGDTVITICHASQHINWLSW
YQQKPGDHPKLLIYKASNLHTGVPSRFSGSGSGT
GFTLTISLQPEDATYYCQQGQTYPRTFGGGTKLEIK (SEQ ID NO: 2716) CDR-
H1 for CL0020123 and variants GFSIEDFYIH (SEQ ID NO: 2717) CL0020123-
1, CL0020123-2, CL0020123-3, CL0020123-4, CL0020123-5, CL0020123-6,
CL0020123-7, and CL0020123-8 CDR-H2 for CL0020123 and variants
WIDPENGDSKYAPKFQG (SEQ ID NO: 2718) CL0020123-1 and CL0020123-2
CDR-H3 for CL0020123 and variants HADHGN YGSTMDY (SEQ ID NO: 2719)
CL0020123-1, CL0020123-2, CL0020123-3, CL0020123-4, CL0020123-5, CL0020123-
6, CL0020123-7, and CL0020123-8 CDR-L1 for CL0020123 and variants
HASQHINWLS (SEQ ID NO: 2720) CL0020123-1, CL0020123-2, CL0020123-3,
CL0020123-4, CL0020123-5, CL0020123-6, CL0020123-7, and CL0020123-8 CDR-L2
for CL0020123 and variants KASNLHT (SEQ ID NO: 2721) CL0020123-1,
CL0020123-2, CL0020123-3, CL0020123-4, CL0020123-5, CL0020123-6, CL0020123-7,
and CL0020123-8 CDR-L3 for CL0020123 and variants QQGQTYPRT (SEQ ID
NO: 2722) CL0020123-1, CL0020123-2, CL0020123-3, CL0020123-4, CL0020123-5,
CL0020123-6, CL0020123-7, and CL0020123-8 CL0020123-1 VH
QVQLVQSGAEVKKPGASVKVSC KASGFSIEDFYI
HWVRQAPGQGLEWMGWIDPENGDSKYAPKFQG
RATITADTSTSTAYMELSSLRSED TAVYYCHADH GNYGSTMDYWGQGTLVTVSS (SEQ
ID NO: 2723) CL0020123-1 VL; CL0020123-2 VL;
DIQMTQSPSSLSASVGDRVTITICHASQHINWLS CL0020123-3 VL; CL0020123-4 VL
WYQQKPGKAPKLLIYKASNLHTGVPSRFSGSGSG CL0020123-5 VL; CL0020123-6
VL; TDFLTISLQPEDFATYYCQQGQTYPRTFGQGTK CL0020123-7 VL; CL0020123-
8 VL VEIK (SEQ ID NO: 2724) CL0020123-2 VH
QVQLVQSGAEVKKPGASVKVSC KASGFSIEDFYI
HWVRQAPGQGLEWMGWIDPENGDSKYAPKFQG
RVTITADTSTSTAYMELSSLRSED TAVYYCHADH GNYGSTMDYWGQGTLVTVSS
(SEQ ID NO: 2725) CL0020123-3 VH
QVQLVQSGAEVKKPGASVKVSC KASGFSIEDFYI
HWVRQAPGQGLEWMGWIDPEQGDSKYAPKFQG
RATITADTSTSTAYMELSSLRSED TAVYYCHADH GNYGSTMDYWGQGTLVTVSS (SEQ
ID NO: 2726) CDR-H2 for variants CL0020123-3 and WIDPEQGDSKYAPKFQG
(SEQ ID NO: 2727) CL0020123-6 CL0020123-4 VH
QVQLVQSGAEVKKPGASVKVSC KASGFSIEDFYI
HWVRQAPGQGLEWMGWIDPENGE SKYAPKFQG
RATITADTSTSTAYMELSSLRSED TAVYYCHADH GNYGSTMDYWGQGTLVTVSS (SEQ
ID NO: 2728) CDR-H2 for variants CL0020123-4 and WIDPENGE SKYAPKFQG
(SEQ ID NO: 2729) CL0020123-7 CL0020123-5 VH
QVQLVQSGAEVKKPGASVKVSC KASGFSIEDFYI
HWVRQAPGQGLEWMGWIDPEQGESKYAPKFQG
RATITADTSTSTAYMELSSLRSED TAVYYCHADH GNYGSTMDYWGQGTLVTVSS (SEQ
ID NO: 2730) CDR-H2 for variants CL0020123-5 and WIDPEQGESKYAPKFQG
(SEQ ID NO: 2731) CL0020123-8 CL0020123-6 VH
QVQLVQSGAEVKKPGASVKVSC KASGFSIEDFYI
HWVRQAPGQGLEWMGWIDPEQGDSKYAPKFQG
RVTITADTSTSTAYMELSSLRSED TAVYYCHADH GNYGSTMDYWGQGTLVTVSS
(SEQ ID NO: 2732) CL0020123-7 VH

QVQLVQSGAEVKKPGASVKVSCKASGFSIEDFYI
 HWVRQAPGQGLEWMGWIDPENGESESKYAPKFQG
 RVTITADTSTSTAYMELSSLRSEDTAVYYCHADH GNYGSTMDYWGQGTLVTVSS
 (SEQ ID NO: 2733) CL0020123-8 VH
 QVQLVQSGAEVKKPGASVKVSCKASGFSIEDFYI
 HWVRQAPGQGLEWMGWIDPEQGESKYAPKFQG
 RVTITADTSTSTAYMELSSLRSEDTAVYYCHADH GNYGSTMDYWGQGTLVTVSS
 (SEQ ID NO: 2734) CDR-H2 consensus sequence W-I-D-P-E- β 6-G- β 8-S-K-Y-A-P-K-F-
 Q-G, wherein (β 6 is N or Q and (β 8 is D or E (SEQ ID NO: 2735)
 CDR-H1 consensus sequence G-F-T-F-T- α 6-F-Y-M-S, wherein α 6 is D or N
 (SEQ ID NO: 2736) CDR-H2 consensus sequence V-I-R-N- β 5- β 6-N- β 8-Y-T- β 11- β 12-Y-
 N-P-S-V-K-G, wherein β 5 is K or R; β 6 is A or P; β 8 is G or A; β 11
 is A or T; and β 12 is G or D (SEQ ID NO: 2737) CDR-H3 consensus
 sequence γ 1-R-L- γ 4-Y-G-F-D-Y, wherein γ 1 is A or T; and γ 4 is T or S
 (SEQ ID NO: 2738) CDR-L1 consensus sequence Q-S-S-K-S-L-L-H-S- δ 10-G-K-T-Y-L-
 N, wherein δ 10 is N or T (SEQ ID NO: 2739) CDR-L3 consensus sequence Q-
 Q-F-L-E-f6-P-F-T, wherein f6 is Y or F (SEQ ID NO: 2740) CL0020307-1 VL
 DIVMTQSPDSLAVSLGERATINCQSSKSLLSNGK
 TYLNWYQQKPGQPPKLLIYWMSTRASGVPRFS
 GSGSGTDFTLTISSLQAEDVAVYYCQQFLEFPFTF GQGTKVEIK (SEQ ID NO: 2741)
 TREM2 epitope FPGESSES (SEQ ID NO: 2742) TREM2 epitope GEKGPCQRV
 (SEQ ID NO: 2743) TREM2 epitope TLRNLQPHDAGL (SEQ ID NO: 2744)
 TREM2 epitope VEVL (SEQ ID NO: 2745) Reference antibody #1 VL
 DIQMTQSPSSVSASVGDRVTITCRASQGISNWLA
 WYQQKPGKAPKLLIYAASSLQVGVPLRFSGSGSG
 TDFTLTISSLQPEDFATYYCQQADSFPNFGQGTLKLEIK (SEQ ID NO: 2746)
 Reference antibody #1 VH EVQLVQSGAEVKKPGESLKISCKGSGHSFTNYWI
 AWVRQMPGKGLEWMGIIYPGDS DTRYSPSFQGGQ
 VTISADKSISTAYLQWSSLKASDTAVYFCARQRTF YYDSSGYFDYWGQGTLVTVSS
 (SEQ ID NO: 2747) Reference antibody #2 VL
 DIQMTQSPSSVSASVGDRVTITCRASQGISSWLAW
 YQQKPGKAPKLLIYAASSLQNGVPSRFSGSGSGT
 DFTLTISLQPEDFATYFCQQADSFPRTFGQGTKLEIK (SEQ ID NO: 2748)
 Reference antibody #2 VH EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIA
 WVRQMPGKGLEWMGIIYPGDS DTRYSPSFQGGQ
 TISADKSISTAYLQWSSLKASDTAMYFCARQRTF YYDSSDYFDYWGQGTLVTVSS
 (SEQ ID NO: 2749) Reference antibody #3 VL
 DVVMTQSPDSLAVSLGERATINCRSSQSLVHSNR
 YTYLHWYQQKPGQSPKWKVSNRFSGVPRFS
 GSGSGTDFTLKISRVEAEDVGVYYCSQSTRVPYTF GQGTKLEIK (SEQ ID NO: 2750)
 Reference antibody #3 VH QVQLVQSGAEVKKPGASVKVSCKASGYAFSSQW
 MNWVRQAPGQRLEWIGRIYPGGGDTNYAGKFQG
 RVTITADTSASTAYMELSSLRSEDTAVYYCARLL RNQPGESYAMDYWGQGTLVTVSS
 (SEQ ID NO: 2751) CL0020188-4 VH
 EVQLVESGGGLVQPGLSLRLSCAGSGFTFTDFYM
 SWVRQAPGKGPWLSVIRNKANGYTAGYNPSVK
 GRFTISRDN SKNTLYLQMNSLRAEDTAVYYCARL TYGFDYWGQGTLVTVSS (SEQ
 ID NO: 2752)

[0648] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in TABLE L1 as well as specific combinations thereof and other

embodiments of the anti-TREM2 antibody described in the '450 application and herein may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

M. PCT Patent Application Publication No. WO2021/101823A1

[0649] In some embodiments, the TREM2 agonist is an antibody, or an antigen-binding fragment thereof, as described in PCT Patent Application Publication No. WO2021101823A1 (“the '823 application”), which is incorporated by reference herein, in its entirety.

[0650] In some embodiments, the antibody or antigen-binding fragment thereof comprises: [0651]

(a) a CDR-H1 sequence comprising the sequence of GFSFNTYWIG (SEQ ID NO:2753); [0652]

(b) a CDR-H2 sequence comprising the sequence of IIYPGDQDIRYSPSFQG (SEQ ID NO:2754;

[0653] (c) a CDR-H3 sequence comprising the sequence of ARYGRYIYGGYHGMDV (SEQ

ID NO:2755; [0654] (d) CDR-L1 sequence comprising the sequence of RASQAIRDDL (SEQ ID

NO:2756); [0655] (e) a CDR-L2 sequence comprising the sequence of YAASSLQS (SEQ ID

NO:2757); and [0656] (f) a CDR-L3 sequence comprising the sequence of LQNYNYPHT (SEQ ID NO:2758).

[0657] In some embodiments, the antibody or antigen-binding fragment comprises a CDR-H1 comprising the amino acid sequence of SEQ ID NO:2753, a CDR-H2 comprising the amino acid sequence of SEQ ID NO:2754, a CDR-H3 comprising the amino acid sequence of SEQ ID NO:2755, a CDR-L1 comprising the amino acid sequence of SEQ ID NO:2756, a CDR-L2 comprising the amino acid sequence of SEQ ID NO:2757, and a CDR-L3 comprising the amino acid sequence of SEQ ID NO:2758.

[0658] In some embodiments, the antibody or antigen-binding fragment comprises a VH sequence that has at least 85% sequence identity to SEQ ID NO:2759. In some embodiments, the V.sub.H sequence has at least 90% sequence identity to SEQ ID NO:2759. In some embodiments, the V.sub.H sequence has at least 95% sequence identity to SEQ ID NO:2759. In some embodiments, the V.sub.H sequence comprises SEQ ID NO:2759.

[0659] In some embodiments, the antibody or antigen-binding fragment comprises a V.sub.L sequence that has at least 85% sequence identity to SEQ ID NO:2760. In some embodiments, the V.sub.L sequence has at least 90% sequence identity to SEQ ID NO:2760. In some embodiments, the V.sub.L sequence has at least 95% sequence identity to SEQ ID NO:2760. In some embodiments, the V.sub.L sequence comprises SEQ ID NO:2760.

[0660] In some embodiments, the antibody or antigen-binding fragment comprises a VH sequence comprising SEQ ID NO:2759 and a V.sub.L sequence comprising SEQ ID NO:2760.

[0661] In some embodiments, an antibody or antigen-binding fragment thereof that specifically binds to TREM2 recognizes an epitope that is the same or substantially the same as the epitope recognized by Antibody 1 of the '823 application.

[0662] In some embodiments, the antibody or antigen-binding fragment recognizes an epitope present on the extracellular domain of a human TREM2. In particular embodiments, the antibody or antigen-binding fragment recognizes an epitope present on the extracellular domain of a human TREM2 in SEQ ID NO:2763. In some embodiments, the antibody or antigen-binding fragment recognizes an epitope present on the extracellular domain of a mouse TREM2. In particular embodiments, the antibody or antigen-binding fragment recognizes an epitope present on the extracellular domain of a mouse TREM2 in SEQ ID NO:2764. In some embodiments, the antibody or antigen-binding fragment recognizes an epitope present on the extracellular domain of a rat TREM2. In particular embodiments, the antibody or antigen-binding fragment recognizes an epitope present on the extracellular domain of a rat TREM2 in SEQ ID NO:2765. In some

QDAGVYQCQSLQGAEVLTQKLVLEPLEHEHAGDFWVP
EESGSFEDPPVERSSSRSPSEGEPSFPPASGGGGQHHHHHHH (SEQ ID NO: 2765)
Rabbit TREM2 ECD-His NTTVLQGVAGQSLRVSCYDALRHWGRRKAWCRQLAEEGPC
QRVVSTHGVWLLAFLRKQNGSTVITDDTLAGTVTITLRNLQA
GDAGLYQCQSLRGREAEVLQKVVEVLEDPDLDQDAGDLWV
PEESESFEQAQVEHSTSRSQSGGGGGQHHHHHHH (SEQ ID NO: 2766) Cynomolgus
monkey HNTTVFQGVGQSLQVSCPYSMKHWGRRKAWCRQLGEKGP TREM2 ECD-His
CQRVVSTHNLWLLSFLRRRNGSTAITDDTLGGTLTITLRNLQPH
DAGFYQCQSLHGSEADTLRKVLVEVLADPLDHRDAGDLWVP
GESESFEDAHVEHSISRPSQGSHLPSCLSKEGGGGQHHHHHHH (SEQ ID NO: 2767)

[0664] In some embodiments, each of the light chain variable regions and each of the heavy chain variable regions disclosed in TABLE M1 as well as specific combinations thereof and other embodiments of the anti-TREM2 antibody described in the '823 application and herein may be attached to the light chain constant regions (TABLE EN1) and heavy chain constant regions (TABLE EN2) to form complete antibody light and heavy chains, respectively, as further discussed below. Further, each of the generated heavy and light chain sequences may be combined to form a complete antibody structure. It should be understood that the heavy chain and light chain variable regions provided herein can also be attached to other constant domains having different sequences than the exemplary sequences listed herein.

Exemplary Anti-TREM2 Antibodies

[0665] In some embodiments, the TREM2 agonist antigen binding protein comprises a CDRL1 or a variant thereof having one, two, three or four amino acid substitutions; a CDRL2, or a variant thereof having one, two, three or four amino acid substitutions; a CDRL3, or a variant thereof having one, two, three or four amino acid substitutions; a CDRH1, or a variant thereof having one, two, three or four amino acid substitutions; a CDRH2, or a variant thereof having one, two, three or four amino acid substitutions; and a CDRH3, or a variant thereof having one, two, three or four amino acid substitutions, where the amino acid sequences of the CDRL1, CDRL2, CDRL3, CDRH1, CDRH2, and CDRH3 are provided in TABLES EX1 and EX2 below, along with exemplary light chain and variable regions.

TABLE-US-00062 TABLE EX1 Exemplary Anti-TREM2 Antibody Light Chain Variable Regions VL Amino Acid Sequence CDRL1 CDRL2 CDRL3
EIVMTQSPATLSVSPGERATLSCR RASQSVSSNLA GASTRAT LQDNNFPPT
ASQSVSSNLAWFQQKPGQAPRL (SEQ ID NO: 10) (SEQ ID NO: 23) (SEQ ID NO: 372) IYGASTRATGIPARFSGSGSGTEF LTISLQPEDFAVYYCLQDNNFP
PTFGQGTVKDIK (SEQ ID NO: 330)

TABLE-US-00063 TABLE EX2 Exemplary Anti-TREM2 Antibody Heavy Chain Variable Regions VH Amino Acid Sequence CDRH1 CDRH2 CDRH3
EVQLVQSGAEVKKPGESLKIS SWIG IIYPGDADARYSPSF RRQGIFGDALDF
CKGSGYSFTSYWIGWVRQMP (SEQ ID NO: 81) QG (SEQ ID NO: 373)
(SEQ ID NO: 374) GKGLEWMGIIYPGDADARYS PSFQGQVTISADKSISTAYLQ
WSSLKASDTAMYFCARRRQ GIFGDALDFWQGQGLTVTVSS (SEQ ID NO: 331)

[0666] As noted above, anti-TREM2 antibodies may comprise one or more of the CDRs presented in TABLE EX1 (light chain CDRs; i.e. CDRLs) and TABLE EX2 (heavy chain CDRs, i.e. CDRHs).

[0667] In some embodiments, an anti-TREM2 antibody comprises a light chain comprising a CDRL1 having an amino acid sequence according to SEQ ID NO:10, a CDRL2 having an amino acid sequence according to SEQ ID NO:23, a CDRL3 having an amino acid sequence according to SEQ ID NO:372, or any CDRL1, CDRL2, or CDRL3 amino acid sequence that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids to

any of SEQ ID NOS: 10, 23, and 372. Such substitutions, deletions, and insertions would retain significant anti-TREM2 binding activity. In these and other embodiments, an anti-TREM2 antibody comprises a CDRH1 having an amino acid sequence according to SEQ ID NO:81, a CDRH2 having an amino acid sequence according to SEQ ID NO:373, a CDRH3 having an amino acid sequence according to SEQ ID NO:374, or any CDRH1, CDRH2, or CDRH3 having an amino acid sequence that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids to any of SEQ ID NOS:81, 373, and 374. Such substitutions, deletions, and insertions would retain significant anti-TREM2 binding activity.

[0668] In some embodiments, an anti-TREM2 antibody comprises a light chain variable region comprising a CDRL1 having an amino acid sequence according to SEQ ID NO: 10; a CDRL2 having an amino acid sequence according to SEQ ID NO:23; and a CDRL3 having an amino acid sequence according to SEQ ID NO:372, and a heavy chain variable region comprising a CDRH1 having an amino acid sequence according to SEQ ID NO:81; a CDRH2 having an amino acid sequence according to SEQ ID NO:373; and a CDRH3 having an amino acid sequence according to SEQ ID NO:374.

[0669] In some embodiments, an anti-TREM2 antibody comprises a light chain variable region having an amino acid sequence according to SEQ ID NO:330, or any amino acid sequence that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids to SEQ ID NO:330. Such substitutions, deletions, and insertions would retain significant anti-TREM2 binding activity. In some embodiments, an anti-TREM2 antibody comprises a heavy chain variable region having an amino acid sequence according to SEQ ID NO:331, or any amino acid sequence that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids to SEQ ID NO:331. Such substitutions, deletions, and insertions would retain significant anti-TREM2 binding activity.

[0670] In specific embodiments, an anti-TREM2 antibody comprises a light chain variable region having an amino acid sequence according to SEQ ID NO:330, and a heavy chain variable region having an amino acid sequence according to SEQ ID NO:331.

[0671] In some embodiments, an anti-TREM2 antibody comprises a heavy chain amino acid sequence, and/or a light chain amino acid sequence selected from TABLE EX3. TABLE EX3 shows exemplary anti-TREM2 antibody heavy and light chains for exemplary antibodies “Ab-1”, “Ab-2”, and “Ab-3”.

TABLE-US-00064 TABLE EX3 Exemplary Anti-TREM2 Antibody Heavy and Light Chains

Chain	Description	Amino Acid Sequence
Light Ab-1		EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWFQQKPGQAPRLLI
Heavy Ab-1		YGASTRATGIPARFSGSGSGTEFTLTISLQPEDFAVYYCLQDNNFPPT
Light Ab-1		FGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAK
Heavy Ab-1		VQWKVDNALQSGNSQESVTEQDSKDYESTLSSTLTLSKADYEKHKV
Light Ab-1		YACEVTHQGLSSPVTKSFNRGEC
Heavy Ab-1	(SEQ ID NO: 2777)	Ab-1 w/ MDMRVPAAQLLGLLLLWLRGARCEIVMTQSPATLSVSPGERATLSCR leader
Light Ab-1		ASQSVSSNLAWFQQKPGQAPRLLI
Heavy Ab-1		YGASTRATGIPARFSGSGSGTEFTLTISLQPEDFAVYYCLQDNNFPPT
Light Ab-1		FGQGTKVDIKRTVAAPSVFIFPPS
Heavy Ab-1		DEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQ
Light Ab-1		DSKDYESTLSSTLTLSKADYEKHKV
Heavy Ab-1		YACEVTHQGLSSPVTKSFNRGEC
Light Ab-2	(SEQ ID NO: 339)	Ab-2 EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWFQQKPGQAPRLLI
Heavy Ab-2		YGASTRATGIPARFSGSGSGTEFTLTISLQPEDFAVYYCLQDNNFPPT
Light Ab-2		FGQGTKVDIKRADAAPTIVSIFPPSSEQLTSGGASVVCFLNNFYPKDIN
Heavy Ab-2		VKWKIDGSEKQNGVLNSWTDQDSKDYESTLSSTLTLSKADYEKHKV

YTCEATHKSTSPINVEK (SEQ ID NO: 2780) Ab-3
EIVMTQSPATLSVSPGERATLSCRASQSVSSNLAWFQQKPGQAPRLLI
YGASTRATGIPARFSGSGSGTEFTLTISSLQPEDFAVYYCLQDNNFPPT
FGQGTKVDIKRADAAPTVSIFPPSSEQLTSGGASVVCFLNNFYPKDIN
VKWKIDGSERQNGVLNSWTDQDSKDSTYSMSSTLTTLTKDEYERHNS
YTCEATHKSTSTSPIVKSFNREK (SEQ ID NO: 2780) Heavy Ab-1
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLE Chain
WMGIIYPGDADARYSPSFQGQVTISADKSISTAYLQWSSLKASDTAM
YFCARRRQGIFGDALDFWGQGTLLTVSSASTKGPSVFPLAPSSKSTS
GGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSL
SVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNW
YVDGVEVHNAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCK
VSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRW
QQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 2774) Ab-1 w/
MDMRVPAQLLGLLLLWLRGARCEVQLVQSGAEVKKPGESLKISCK leader
GSGYSFTSYWIGWVRQMPGKGLEWMGITYPGDADARYSPSFQGQVT sequence
ISADKSISTAYLQWSSLKASDTAMYFCARRRQGIFGDALDFWGQGTLL
TVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWN
SGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPS
NTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISR
TPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTY
RCVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREP
QVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYK
TTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK
(SEQ ID NO: 340) Ab-2
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLE
WMGIIYPGDADARYSPSFQGQVTISADKSISTAYLQWSSLKASDTAM
YFCARRRQGIFGDALDFWGQGTLLTVSSAKTTPPSVYPLAPGSAAQT
NSMVTLGCLVKGYFPEPVTVTWNSGSLSSGVHTFPAVLQSDLYTLSS
SVTVPSSPRPSETVTCNVAHPASSTKVDKKIVPRDCGCKPCICTVPEV
SSVFIFPPKPKDVLITITLTPKVTCTVVVDISKDDPEVQFSWFVDDVEVH
TAQTQPREEQFNSTFRSVSELPIMHQDWLNGKEFKCRVNSAAFPAPI
EKTISKTKGRPKAPQVYTIPPPKEQMAKDKVSLTCMITDFFPEDITVE
WQWNGQPAENYKNTQPIMNNGSYFVYSKLVNQKSNWEAGNTFT
CSVLHEGLHNHHTTEKSLSHSPGK (SEQ ID NO: 2781) Ab-3
EVQLVQSGAEVKKPGESLKISCKGSGYSFTSYWIGWVRQMPGKGLE
WMGIIYPGDADARYSPSFQGQVTISADKSISTAYLQWSSLKASDTAM
YFCARRRQGIFGDALDFWGQGTLLTVSSAKTTPPSVYPLAPGSAAQT
NSMVTLGCLVKGYFPEPVTVTWNSGSLSSGVHTFPAVLQSDLYTLSS
SVTVPSSTWPSETVTCNVAHPASSTKVDKKIVPRDCGCKPCICTVPEV
SSVFIFPPKPKDVLITITLTPKVTCTVVVDISKDDPEVQFSWFVDDVEVH
TAQTQPREEQFGSTFRSVSELPIMHQDWLNGKEFKCRVNSAAFPAPI
EKTISKTKGRPKAPQVYTIPPPKEQMAKDKVSLTCMITDFFPEDITVE
WQWNGQPAENYKNTQPIMDTDGSYFVYSKLVNQKSNWEAGNTFT
CSVLHEGLHNHHTTEKSLSHSPGK (SEQ ID NO: 2779)

[0672] In some embodiments, an anti-TREM2 antibody comprises a light chain having an amino acid sequence according to any one of SEQ ID NOS:2777, 339, and 2780, or any amino acid sequence that contains one or more, e.g., one, two, three, four or more amino acid substitutions

(e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids to any one of SEQ ID NOS:2777, 339, and 2780. Such substitutions, deletions, and insertions would retain significant anti-TREM2 binding activity. In these and other embodiments, an anti-TREM2 antibody comprises a heavy chain having an amino acid sequence according to any one of SEQ ID NOS:2774, 340, 2779, and 2781, or any amino acid sequence that contains one or more, e.g., one, two, three, four or more amino acid substitutions (e.g., conservative amino acid substitutions), deletions or insertions of no more than five, four, three, two, or one amino acids to any one of SEQ ID NOS:2774, 340, 2779, and 2781. Such substitutions, deletions, and insertions would retain significant anti-TREM2 binding activity.

[0673] In some embodiments, an anti-TREM2 antibody comprises a light chain having an amino acid sequence according to SEQ ID NO:2777, and a heavy chain variable region having an amino acid sequence according to SEQ ID NO:2774. In some embodiments, an anti-TREM2 antibody comprises a light chain having an amino acid sequence according to SEQ ID NO:339, and a heavy chain variable region having an amino acid sequence according to SEQ ID NO:340. In some embodiments, an anti-TREM2 antibody comprises a light chain having an amino acid sequence according to SEQ ID NO:2780, and a heavy chain variable region having an amino acid sequence according to SEQ ID NO:2781. In some embodiments, an anti-TREM2 antibody comprises a light chain having an amino acid sequence according to SEQ ID NO:2780, and a heavy chain variable region having an amino acid sequence according to SEQ ID NO:2779.

Antibody Constant Domains and Engineered Constant Regions

[0674] In some embodiments, any of the antigen binding agents, can have a constant domain on the light chain and/or the heavy chain of any origin. The term “constant region” as used herein refers to all domains of an antibody other than the variable region. The constant domain can be that of rodent, primate or other mammals. In some embodiments, the constant domain is of human origin. Accordingly, in some embodiments, any of the antigen binding agents described herein can have a human constant region, some of which are described above.

[0675] In some embodiments, a human constant region is, for example, a human light chain constant region or a human constant heavy chain region.

[0676] The term “light chain” or “immunoglobulin light chain” refers to a polypeptide comprising, from amino terminus to carboxyl terminus, a single immunoglobulin light chain variable region (VL) and a single immunoglobulin light chain constant domain (CL). The immunoglobulin light chain constant domain (CL) can be a human kappa (κ) or human lambda (λ) constant domain.

[0677] The term “heavy chain” or “immunoglobulin heavy chain” refers to a polypeptide comprising, from amino terminus to carboxyl terminus, a single immunoglobulin heavy chain variable region (VH), an immunoglobulin heavy chain constant domain 1 (CH1), an immunoglobulin hinge region, an immunoglobulin heavy chain constant domain 2 (CH2), an immunoglobulin heavy chain constant domain 3 (CH3), and optionally an immunoglobulin heavy chain constant domain 4 (CH4). Heavy chains are classified as mu (μ), delta (Δ), gamma (γ), alpha (α), and epsilon (ϵ), and define the antibody's isotype as IgM, IgD, IgG, IgA, and IgE, respectively. The IgG-class and IgA-class antibodies are further divided into subclasses, namely, IgG1, IgG2, IgG3, and IgG4, and IgA1 and IgA2, respectively. The heavy chains in IgG, IgA, and IgD antibodies have three domains (CH1, CH2, and CH3), whereas the heavy chains in IgM and IgE antibodies have four domains (CH1, CH2, CH3, and CH4). The immunoglobulin heavy chain constant domains can be from any immunoglobulin isotype, including subtypes. The antibody chains are linked together via inter-polypeptide disulfide bonds between the CL domain and the CH1 domain (i.e. between the light and heavy chain) and between the hinge regions of the antibody heavy chains.

[0678] In some embodiments, the human light chain constant region comprises a human kappa or human lambda constant region. In some embodiments, the antigen binding agents based on any light chain variable region or CDRs of a light chain variable region described herein includes a

human light chain constant region, such as a kappa or lambda constant region sequences, which are found in all five antibody isotypes. Examples of human immunoglobulin light chain constant region sequences are shown in the following TABLE EN1.

Constant Regions	Designation	CL Domain	Amino Acid Sequence	Human Immunoglobulin Light Chain
GQPKANPTVT	LFPPSSEELQANKATLVCLISDFYPGAVTVAWKAD			v1
GSPVKAGVET	TKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVT			
HEGSTVEKTV	APTECS (SEQ ID NO: 191)			Human lambda v2
GQPKAAPSV	TLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADS			
SPVKAGVET	TTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTH			
EGSTVEKTV	APTECS (SEQ ID NO: 192)			Human lambda v3
QPKAAPSV	TLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSS			
PVKAGVET	TTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHE			
GSTVEKTV	APTECS (SEQ ID NO: 193)			Human lambda v4
GQPKAAPSV	TLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADS			
SPVKAGVET	TTTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTH			
EGSTVEKTV	APTECS (SEQ ID NO: 194)			Human lambda v5
GQPKAAPSV	TLFPPSSEELQANKATLVCLVSDFYPGAVTVAWKAD			
GSPVKVGV	ETTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCRVT			
HEGSTVEKTV	APAECS (SEQ ID NO: 195)			Human kappa v1
TVAAPSV	FIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA			
LQSGNSQES	VTEQDSKDYSLSTLTLSKADYEKHKVYACEVTH			
QGLSSPV	TKSFNRGEC (SEQ ID NO: 196)			Human kappa v2
RTVAAPSV	FIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDN			
ALQSGNSQES	VTEQDSKDYSLSTLTLSKADYEKHKVYACEVT			
HQGLSSPV	TKSFNRGEC (SEQ ID NO: 197)			

[0679] In some embodiments, a human constant region comprises at least one or all of the following: a human CH1, human Hinge, human CH2, and CH3 domain. In some embodiments, the heavy chain constant region comprises an Fc region, where the Fc portion is a human IgG.sub.1, IgG.sub.2, IgG.sub.3, IgG.sub.4 or IgM isotype. The term “Fc region” refers to the C-terminal region of an immunoglobulin heavy chain which may be generated by papain digestion of an intact antibody. The Fc region of an immunoglobulin generally comprises two constant domains, a CH2 domain and a CH3 domain, and optionally comprises a CH4 domain. In certain embodiments, the Fc region is an Fc region from an IgG1, IgG2, IgG3, or IgG4 immunoglobulin. In some embodiments, the Fc region comprises CH2 and CH3 domains from a human IgG1 or human IgG2 immunoglobulin. The Fc region may retain effector function, such as C1q binding, complement-dependent cytotoxicity (CDC), Fc receptor binding, antibody-dependent cell-mediated cytotoxicity (ADCC), and phagocytosis. In other embodiments, the Fc region may be modified to reduce or eliminate effector function as described in further detail below.

[0680] In some embodiments, the antigen binding agents based on any heavy chain variable region or CDRs of a heavy chain variable region described herein includes a human heavy chain constant region., for example a human constant region comprising at least one or all of a human CH1, human Hinge, human CH2, and CH3 domain. In some embodiments, the antigen binding agents based on any heavy chain variable region or CDRs of a heavy chain variable region described herein includes an Fc region, where the Fc region is a human IgG.sub.1, IgG.sub.2, IgG.sub.3, IgG.sub.4 or IgM isotype. Examples of human IgG1, IgG2, and IgG4 heavy chain constant region sequences are shown below in TABLE EN2.

Constant Regions	Ig isotype	Heavy Chain Constant Region	Amino Acid Sequence
Human IgG1z	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT	TVSWNSGALT	

SGVHTFPFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHNKPSNTKV
DKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
PVLDSGDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKS LSLSPGK (SEQ
ID NO: 198) Human IgG1za

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHNKPSNTKV
DKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
VLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSL SLSPGK (SEQ
ID NO: 199) Human IgG1f

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHNKPSNTKV
DKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
PVLDSGDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKS LSLSPGK (SEQ
ID NO: 200) Human IgG1fa

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHNKPSNTKV
DKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
VLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSL SLSPGK (SEQ
ID NO: 201) Human IgG1z

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT aglycosylated
SGVHTFPFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHNKPSNTKV v1
DKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYGSTYRVV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
PVLDSGDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKS LSLSPGK (SEQ
ID NO: 202) Human IgG1z

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT aglycosylated v2
SGVHTFPFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHNKPSNTKV
DKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP PVLD
SDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKS LSLSPGK (SEQ ID
NO: 203) Human IgG2

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVNHNKPSNTK
VDKTVKCCVECPKPPVAGPSVFLFPPKPKDTLMISRTPEVTC

VVVDVSHEDVQFNWYVDGVEVHNAKTKPREEQFNSTFRVVSVL
TVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLP
PSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPML
DSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSL SPGK (SESQ ID
NO: 204) Human IgG4
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKV
DKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV
TCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPCEEQYGSTYRCV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTP
PVLDSFGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKS LSLSPGK (SESQ
ID NO: 205)

[0681] In some embodiments, the heavy chain constant region, particularly the Fc region, is an engineered heavy chain constant region. In some embodiments, the antigen binding proteins, e.g. monoclonal antibodies, comprise one or more amino acid substitutions in the Fc region to enhance effector function, including ADCC activity, CDC activity, ADCP activity, and/or the clearance or half-life of the antigen binding protein. Exemplary amino acid substitutions (according to EU numbering scheme) that can enhance effector function include, but are not limited to, E233L, L234I, L234Y, L235S, G236A, S239D, F243L, F243V, P247I, D280H, K290S, K290E, K290N, K290Y, R292P, E294L, Y296W, S298A, S298D, S298V, S298G, S298T, T299A, Y300L, V305I, Q31 I_M, K326A, K326E, K326W, A330S, A330L, A330M, A330F, I332E, D333A, E333S, E333A, K334A, K334V, A339D, A339Q, P396L, or combinations of any of the foregoing.

[0682] In some embodiments, the TREM2 antigen binding proteins (e.g. monoclonal antibodies) comprise one or more amino acid substitutions in a heavy chain constant region to reduce effector function. Exemplary amino acid substitutions (according to EU numbering scheme) that can reduce effector function include, but are not limited to, C220S, C226S, C229S, E233P, L234A, L234V, V234A, L234F, L235A, L235E, G237A, P238S, S267E, H268Q, N297A, N297G, N297Q, V309L, E318A, L328F, A330S, A331S, P331S or combinations of any of the foregoing.

[0683] In some embodiments, the TREM2 agonist antigen binding proteins comprise one or more amino acid substitutions that affect the level or type of glycosylation of the binding proteins. Glycosylation can contribute to the effector function of antibodies, particularly IgG1 antibodies. Glycosylation of polypeptides is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tri-peptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tri-peptide sequences in a polypeptide creates a potential glycosylation site. O-linked glycosylation refers to the attachment of one of the sugars N-acetylgalactosamine, galactose, or xylose, to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used.

[0684] In some embodiments, glycosylation of the TREM2 agonist antigen binding proteins described herein is increased by adding one or more glycosylation sites, e.g., to the Fc region of the binding protein. Addition of glycosylation sites to the antigen binding protein can be conveniently accomplished by altering the amino acid sequence such that it contains one or more of the above-described tri-peptide sequences (for N-linked glycosylation sites). The alteration may also be made by the addition of, or substitution by, one or more serine or threonine residues to the starting sequence (for O-linked glycosylation sites). For ease, the antigen binding protein amino acid sequence may be altered through changes at the DNA level, particularly by mutating the DNA encoding the target polypeptide at preselected bases such that codons are generated that will translate into the desired amino acids.

[0685] The invention also encompasses production of TREM2 antigen binding protein molecules with altered carbohydrate structure resulting in altered effector activity, including antigen binding proteins with absent or reduced fucosylation that exhibit improved ADCC activity. Various methods are known in the art to reduce or eliminate fucosylation. For example, ADCC effector activity is mediated by binding of the antibody molecule to the FcγRIII receptor, which has been shown to be dependent on the carbohydrate structure of the N-linked glycosylation at the N297 residue of the CH2 domain. Non-fucosylated antibodies bind this receptor with increased affinity and trigger FcγRIII-mediated effector functions more efficiently than native, fucosylated antibodies. For example, recombinant production of non-fucosylated antibody in CHO cells in which the alpha-1,6-fucosyl transferase enzyme has been knocked out results in antibody with 100-fold increased ADCC activity (see Yamane-Ohnuki et al., *Biotechnol Bioeng.* 87(5):614-22, 2004). Similar effects can be accomplished through decreasing the activity of alpha-1,6-fucosyl transferase enzyme or other enzymes in the fucosylation pathway, e.g., through siRNA or antisense RNA treatment, engineering cell lines to knockout the enzyme(s), or culturing with selective glycosylation inhibitors (see Rothman et al., *Mol Immunol.* 26(12):1113-23, 1989). Some host cell strains, e.g. Lec13 or rat hybridoma YB2/0 cell line naturally produce antibodies with lower fucosylation levels (see Shields et al., *J Biol Chem.* 277(30):26733-40, 2002 and Shinkawa et al., *J Biol Chem.* 278(5):3466-73, 2003). An increase in the level of bisected carbohydrate, e.g. through recombinantly producing antibody in cells that overexpress GnTIII enzyme, has also been determined to increase ADCC activity (see Umana et al., *Nat Biotechnol.* 17(2):176-80, 1999).

[0686] In other embodiments, glycosylation of the TREM2 agonist antigen binding proteins described herein is decreased or eliminated by removing one or more glycosylation sites, e.g., from the Fc region of the binding protein. In some embodiments, the TREM2 agonist antigen binding protein is an aglycosylated human monoclonal antibody, e.g. an aglycosylated human IgG1 monoclonal antibody. Amino acid substitutions that eliminate or alter N-linked glycosylation sites can reduce or eliminate N-linked glycosylation of the antigen binding protein. In certain embodiments, the TREM2 agonist antigen binding proteins described herein comprise a mutation at position N297 (according to EU numbering scheme), such as N297Q, N297A, or N297G. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise an Fc region from a human IgG1 antibody with a mutation at position N297. In one particular embodiment, the TREM2 agonist antigen binding proteins of the invention comprise an Fc region from a human IgG1 antibody with a N297G mutation. For instance, in some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain constant region comprising the sequence of SEQ ID NO:202.

[0687] To improve the stability of molecules comprising a N297 mutation, the Fc region of the TREM2 agonist antigen binding proteins may be further engineered. For instance, in some embodiments, one or more amino acids in the Fc region are substituted with cysteine to promote disulfide bond formation in the dimeric state. Residues corresponding to V259, A287, R292, V302, L306, V323, or 1332 (according to EU numbering scheme) of an IgG1 Fc region may thus be substituted with cysteine. Preferably, specific pairs of residues are substituted with cysteine such that they preferentially form a disulfide bond with each other, thus limiting or preventing disulfide bond scrambling. Preferred pairs include, but are not limited to, A287C and L306C, V259C and L306C, R292C and V302C, and V323C and 1332C. In certain embodiments, the TREM2 agonist antigen binding proteins described herein comprise an Fc region from a human IgG1 antibody with mutations R292C and V302C. In such embodiments, the Fc region may also comprise a N297 mutation, such as a N297G mutation. In some embodiments, the TREM2 agonist antigen binding proteins of the invention comprise a heavy chain constant region comprising the sequence of SEQ ID NO:203.

[0688] Modifications to the hinge region and/or CH1 domain of the heavy chain and/or the constant region of the light chain of the TREM2 agonist antigen binding proteins (e.g. monoclonal

antibodies) of the invention can be made to reduce or eliminate disulfide heterogeneity. Structural heterogeneity of IgG2 antibodies has been observed where the disulfide bonds in the hinge and CH1 regions of IgG2 antibodies can be shuffled to create different structural disulfide isoforms (IgG2A, IgG2B, and IgG2A-B), which can have different levels of activity. See, e.g., Dillon et al., J. Biol. Chem., Vol. 283: 16206-16215; Martinez et al., Biochemistry, Vol. 47: 7496-7508, 2008; and White et al., Cancer Cell, Vol. 27: 138-148, 2015. Amino acid substitutions can be made in the hinge region, CH1 domain, and/or light chain constant region to promote the formation of a single disulfide isoform or lock the antigen binding protein (e.g. monoclonal antibody) into a particular disulfide isoform (e.g. IgG2A or IgG2B). Such mutations are described in WO 2009/036209 and White et al., Cancer Cell, Vol. 27: 138-148, 2015, both of which are hereby incorporated by reference in its entirety, and include C131S, C219S, and C220S (according to EU numbering scheme) mutations in the heavy chain and a C214S (according to EU numbering scheme) mutation in the light chain. In certain embodiments, the TREM2 agonist antigen binding proteins of the invention are human IgG2 anti-TREM2 agonist antibodies. In some such embodiments, the TREM2 agonist antibodies comprise a C131S mutation (according to the EU numbering scheme) in their heavy chains. In other embodiments, the TREM2 agonist antibodies comprise a C214S mutation (according to the EU numbering scheme) in their light chains and a C220S mutation (according to the EU numbering scheme) in their heavy chains. In still other embodiments, the TREM2 agonist antibodies comprise a C214S mutation (according to the EU numbering scheme) in their light chains and a C219S mutation (according to the EU numbering scheme) in their heavy chains.

[0689] In other embodiments, the TREM2 agonist antigen binding proteins of the invention are anti-TREM2 agonist antibodies comprising a CH1 region and hinge region from a human IgG2 antibody and an Fc region from a human IgG1 antibody. The unique arrangement of the disulfide bonds in the hinge region of IgG2 antibodies has been reported to impart enhanced stimulatory activity for certain anticancer antibodies (White et al., Cancer Cell, Vol. 27: 138-148, 2015). This enhanced activity could be transferred to IgG1-type antibodies by exchanging the CH1 and hinge regions of the IgG1 antibody for those in the IgG2 antibody (White et al., 2015). The IgG2 hinge region includes the amino acid sequence ERKCCVECPCP (SEQ ID NO:206). The amino acid sequence of the CH1 and hinge regions from a human IgG2 antibody may comprise the following amino acid sequence:

TABLE-US-00067 (SEQ ID NO: 207) ASTKGPSVFP LAPCSRSTSE STAALGCLVK
DYFPEPVTVS WNSGALTSGV HTFPAVLQSS GLYSLSSVVT VPSSNFGTQT
YTCNVDPHKPS NTKVDKTVR KCCVECPCP.

[0690] In some embodiments, the antigen binding agents based on any heavy chain variable region or CThus, in some embodiments, the anti-TREM2 agonist antibodies comprise the sequence of SEQ ID NO:207 in combination with an Fc region from a human IgG1 antibody. In such embodiments, the anti-TREM2 antibodies can comprise one or more of the mutations described above to lock the anti-TREM2 antibodies into a particular disulfide isoform. For instance, in one embodiment, the anti-TREM2 antibody comprises a CH1 region and hinge region from a human IgG2 antibody and an Fc region from a human IgG1 antibody and comprises a C131S mutation (according to the EU numbering scheme) in its heavy chain. In another embodiment, the anti-TREM2 antibody comprises a CH1 region and hinge region from a human IgG2 antibody and an Fc region from a human IgG1 antibody and comprises a C214S mutation (according to the EU numbering scheme) in its light chain and a C220S mutation (according to the EU numbering scheme) in its heavy chain. In yet another embodiment, the anti-TREM2 antibody comprises a CH1 region and hinge region from a human IgG2 antibody and an Fc region from a human IgG1 antibody and comprises a C214S mutation (according to the EU numbering scheme) in its light chain and a C219S mutation (according to the EU numbering scheme) in its heavy chain.

[0691] In embodiments in which the anti-TREM2 antibodies comprise a CH1 region and hinge

region from a human IgG2 antibody and an Fc region from a human IgG1 antibody, the anti-TREM2 antibodies may comprise any of the mutations in the Fc region described above to modulate the glycosylation of the antibodies. For instance, the human IgG1 Fc region of such anti-TREM2 antibodies may comprise a mutation at amino acid position N297 (according to the EU numbering scheme) in its heavy chain. In one particular embodiment, the N297 mutation is a N297G mutation. In certain embodiments, the Fc region may further comprise R292C and V302C mutations (according to the EU numbering scheme) in its heavy chain.

[0692] In certain embodiments, the anti-TREM2 antibodies of the invention comprise a CH1 region and hinge region from a human IgG2 antibody and an Fc region from a human IgG1 antibody, wherein the Fc region comprises the amino acid sequence of:

TABLE-US-00068 (SEQ ID NO: 281)

APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD
GVEVHNAKTKPREEQYGSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVE
WESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHE
ALHNHYTQKSLSLSPGK.

[0693] In other embodiments, the anti-TREM2 antibodies of the invention comprise a CH1 region and hinge region from a human IgG2 antibody and an Fc region from a human IgG1 antibody, wherein the Fc region comprises the amino acid sequence of:

TABLE-US-00069 (SEQ ID NO: 282)

APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD
GVEVHNAKTKPCEEQYGSTYRCVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVE
WESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHE
ALHNHYTQKSLSLSPGK.

[0694] Modifications of the TREM2 agonist antigen binding proteins of the invention to increase serum half-life also may be desirable, for example, by incorporation of or addition of a salvage receptor binding epitope (e.g., by mutation of the appropriate region or by incorporating the epitope into a peptide tag that is then fused to the antigen binding protein at either end or in the middle, e.g., by DNA or peptide synthesis; see, e.g., WO96/32478) or adding molecules such as PEG or other water soluble polymers, including polysaccharide polymers. The salvage receptor binding epitope preferably constitutes a region wherein any one or more amino acid residues from one or two loops of an Fc region are transferred to an analogous position in the antigen binding protein. Even more preferably, three or more residues from one or two loops of the Fc region are transferred. Still more preferred, the epitope is taken from the CH2 domain of the Fc region (e.g., an IgG Fc region) and transferred to the CH1, CH3, or VH region, or more than one such region, of the antigen binding protein. Alternatively, the epitope is taken from the CH2 domain of the Fc region and transferred to the CL region or VL region, or both, of the antigen binding protein. See International applications WO 97/34631 and WO 96/32478 for a description of Fc variants and their interaction with the salvage receptor.

Antibody Fragments

[0695] In some embodiments, the antigen binding agent can be a fragment of the antibody of the present disclosure, including portions of a full length antibody, and includes the antigen binding or variable region. Exemplary antibody fragments include Fab, Fab', F(ab').sub.2 and Fv fragments. In some embodiments, proteolytic digestion with papain produces two identical antigen binding fragments, the Fab' fragment, each with a single antigen binding site. In some embodiments, proteolytic digestion with pepsin yields an F(ab').sub.2 fragment that has two antigen binding fragments which are capable of cross-linking antigen, and a residual pFc' fragment. In some embodiments, antibody fragments are produced directly in recombinant host-cells, for example host cells that have a polynucleotide encoding an antigen binding agent described herein. For

example, Fab, Fv and scFv antibody fragments can all be expressed in and secreted from *E. coli*, thus allowing the straightforward production of large amounts of these fragments. Anti-TREM2 antibody fragments can also be isolated from the antibody phage libraries as discussed above. Alternatively, Fab'-SH fragments can be directly recovered from *E. coli* and chemically coupled to form F(ab')₂ fragments (Carter et al., Bio/Technology 10:163-167 (1992)). According to another approach, F(ab')₂ fragments can be isolated directly from recombinant host-cell culture. Production of Fab and F(ab')₂ antibody fragments with increased in vivo half-lives are described in U.S. Pat. No. 5,869,046. In other embodiments, the antibody of choice is a single chain Fv fragment (scFv). See WO 93/16185; U.S. Pat. Nos. 5,571,894, 5,587,458. Accordingly, other types of fragments can include diabodies, linear antibodies, single-chain antibodies, and multispecific antibodies formed from antibody fragments. In some embodiments, the antibody fragments are functional in that they retain the desired antigen binding properties, e.g., specific binding to TREM2, activation of TREM2 activities, and the like as described herein.

Bispecific Antibodies

[0696] In some embodiments, the TREM2 binding protein is a bispecific antibody that binds to a TREM2 protein of the present disclosure and a second antigen. In some embodiments, bispecific antibodies of the present disclosure bind to one or more amino acid residues of human TREM2 (SEQ ID NO: 1), or amino acid residues on a TREM2 protein corresponding to amino acid residues of SEQ ID NO: 1. In some embodiments, any of the TREM2 binding proteins described herein can be used to prepare the bispecific antibody.

[0697] In some embodiments, bispecific antibodies of the present disclosure recognize a first antigen and a second antigen. In some embodiments, the first antigen is human TREM2 or a naturally occurring variant thereof. In some embodiments, the second antigen is DAP12, or other proteins or ligand that interact with TREM2. In some embodiments, the second antigen is (a) an antigen facilitating transport across the blood-brain-barrier; (b) an antigen facilitating transport across the blood-brain-barrier, for example transferrin receptor (TR), insulin receptor (HIR), insulin-like growth factor receptor (IGFR), low-density lipoprotein receptor related proteins 1 and 2 (LPR-1 and 2), diphtheria toxin receptor, CRM 197, a llama single domain antibody, TMEM 30(A), a protein transduction domain, TAT, Syn-B, penetratin, a poly-arginine peptide, an angiopoietin peptide, and ANG1005; (c) a disease-causing protein selected from amyloid beta, oligomeric amyloid beta, amyloid beta plaques, amyloid precursor protein or fragments thereof, Tau, IAPP, alpha-synuclein, TDP-43, FUS protein, C9orf72 (chromosome 9 open reading frame 72), c9RAN protein, prion protein, PrP^{Sc}, huntingtin, calcitonin, superoxide dismutase, ataxin, ataxin 1, ataxin 2, ataxin 3, ataxin 7, ataxin 8, ataxin 10, Lewy body, atrial natriuretic factor, islet amyloid polypeptide, insulin, apolipoprotein A1, serum amyloid A, medin, prolactin, transthyretin, lysozyme, beta 2 microglobulin, gelsolin, keratoepithelin, cystatin, immunoglobulin light chain AL, S-IBM protein, Repeat-associated non-ATG (RAN) translation products, DiPeptide repeat (DPR) peptides, glycine-alanine (GA) repeat peptides, glycine-proline (GP) repeat peptides, glycine-arginine (GR) repeat peptides, proline-alanine (PA) repeat peptides, ubiquitin, and proline-arginine (PR) repeat peptides; and (d) ligands and/or proteins expressed on immune cells, wherein the ligands and/or proteins selected from CD40, OX40, ICOS, CD28, CD137/4-1BB, CD27, GITR, PD-L1, CTLA-4, PD-L2, PD-1, B7-H3, B7-H4, HVEM, BTLA, KIR, GAL9, TIM3, A2AR, LAG-3, and phosphatidylserine; and (e) a protein, lipid, polysaccharide, or glycolipid expressed on one or more tumor cells and any combination thereof.

[0698] Methods for making bispecific antibodies are known in the art. Traditional production of full-length bispecific antibodies is based on the coexpression of two immunoglobulin heavy-chain/light chain pairs, where the two chains have different specificities. Millstein et al., Nature, 305:537-539 (1983). Because of the random assortment of immunoglobulin heavy and light chains, these hybridomas (quadromas) produce a potential mixture of 10 different antibody molecules, of which only one has the correct bispecific structure. Purification of the correct molecule, which is

usually done by affinity chromatography steps, is rather cumbersome, and the product yields are low. Similar procedures are disclosed in WO 93/08829 and in Traunecker et al., EMBO J., 10:3655-3659 (1991).

[0699] In some embodiments, antibody variable domains with the desired binding specificities (antibody-antigen combining sites) are fused to immunoglobulin constant domain sequences. The fusion preferably is with an immunoglobulin heavy chain constant domain, comprising at least part of the hinge, CH2, and CH3 regions. It is preferred to have the first heavy-chain constant region (CH1) containing the site necessary for light chain binding, present in at least one of the fusions. DNAs encoding the immunoglobulin heavy chain fusions and, if desired, the immunoglobulin light chain, are inserted into separate expression vectors, and are co-transfected into a suitable host organism. This provides for great flexibility in adjusting the mutual proportions of the three polypeptide fragments in embodiments when unequal ratios of the three polypeptide chains used in the construction provide the optimum yields. It is, however, possible to insert the coding sequences for two or all three polypeptide chains in one expression vector when the expression of at least two polypeptide chains in equal ratios results in high yields or when the ratios are of no particular significance.

[0700] In some embodiments, the bispecific antibodies are composed of a hybrid immunoglobulin heavy chain with a first binding specificity in one arm, and a hybrid immunoglobulin heavy chain-light chain pair (providing a second binding specificity) in the other arm. It was found that this asymmetric structure facilitates the separation of the desired bispecific compound from unwanted immunoglobulin chain combinations, as the presence of an immunoglobulin light chain in only half of the bispecific molecules provides for an easy way of separation. This approach is disclosed in WO 94/04690. For further details of generating bispecific antibodies, see, for example, Suresh et al., *Methods in Enzymology* 121: 210 (1986); and Garber, *Nature Reviews Drug Discovery* 13, 799-801 (2014).

[0701] In some embodiments, the bispecific antibody can be prepared as described in WO 96/27011 or U.S. Pat. No. 5,731,168. In these embodiments, the interface between a pair of antibody molecules can be engineered to maximize the percentage of heterodimers which are recovered from recombinant-cell culture. The preferred interface comprises at least a part of the CH3 region of an antibody constant domain. In this method, one or more small amino acid side chains from the interface of the first antibody molecule are replaced with larger side chains (e.g., tyrosine or tryptophan). Compensatory "cavities" of identical or similar size to the large side chains(s) are created on the interface of the second antibody molecule by replacing large amino acid side chains with smaller ones (e.g., alanine or threonine). This provides a mechanism for increasing the yield of the heterodimer over other unwanted end-products such as homodimers.

[0702] In some embodiments, bispecific antibody can be prepared. Techniques for generating bispecific antibodies from antibody fragments have been described in for example, Brennan et al., *Science*, 1985, 229:81, which describe proteolytic cleavage of intact antibodies to generate F(ab')₂ fragments, which are then reduced in the presence of the dithiol complexing agent sodium arsenite to stabilize vicinal dithiols and prevent intermolecular disulfide formation. The Fab' fragments generated are then converted to thionitrobenzoate (TNB) derivatives. One of the Fab'-TNB derivatives is then reconverted to the Fab'-TNB derivative to form the bispecific antibody. The bispecific antibodies produced can be used as agents for the selective immobilization of enzyme.

[0703] Various techniques for making and isolating bivalent antibody fragments directly from recombinant-cell culture have also been described. For example, bivalent heterodimers have been produced using leucine zippers. Kostelny et al., *Immunol.*, 1992, 148(5):1547-1553. The "diabody" technology described by Hollinger et al., *Proc. Nat'l Acad. Sci. USA*, 1993, 90: 6444-6448, provides an alternative mechanism for making bispecific/bivalent antibody fragments. The fragments comprise a heavy-chain variable domain (VH) connected to a light-chain variable domain (VL) by a linker which is too short to allow pairing between the two domains on the same

chain. Accordingly, the VH and VL domains of one fragment are forced to pair with the complementary VL and VH domains of another fragment, thereby forming two antigen-binding sites. Another strategy for making bispecific/bivalent antibody fragments by the use of single-chain Fv (sFv) dimers (see, e.g., Gruber et al., *Immunol*, 152:5368 (1994)).

Single Chain Antibodies

[0704] In some embodiments, the TREM2 binding protein is a single chain antibody, e.g., single chain Fv (sFv or scFv) antibodies, in which a variable heavy and a variable light chain are joined together (directly or through a peptide linker) to form a continuous polypeptide. A single-chain Fv” or “sFv” antibody fragments comprise the VH and VL domains of an antibody, where these domains are present in a single polypeptide chain. Generally, the Fv polypeptide further comprises a polypeptide linker between the VH and VL domains which enables the sFv to form the desired structure for antigen binding. For a review of sFv, see Pluckthun in *The Pharmacology of Monoclonal Antibodies*, Vol. 113, pp. 269-315, Rosenberg and Moore, eds., Springer-Verlag, New York (1994). Any of the TREM2 binding agents described herein can be used to prepare a single chain antibody.

[0705] In some embodiments, single chain antibody can be prepared by phage display methods, where the antigen binding domain is expressed as a single polypeptide and screened for specific binding activity. Alternatively, the single chain antibody can be prepared by cloning the heavy and light chains from a cell, typically a hybridoma cell line expressing a desired antibody. Generally, a linker peptide, typically from 10 to 25 amino acids in length is used to link the heavy and light chains. The linker can be glycine, serine, and/or threonine rich to impart flexibility and solubility to the single chain antibody. Specific methods for generating single chain antibodies are described in, for example, Loffler et al., 2000, *Blood* 95(6):2098-103; Worn and Pluckthun, 2001, *J Mol Biol.* 305, 989-1010; Pluckthun, In *The Pharmacology of Monoclonal Antibodies*, Vol. 113, pp. 269-315, Rosenberg and Moore, eds., Springer-Verlag, New York (1994); U.S. Pat. Nos. 5,840,301; 5,844,093; and 5,892,020; all of which are incorporated herein by reference.

Multivalent Antibodies

[0706] In some embodiments, the anti-TREM2 antibody is a multivalent antibody, which may be internalized (and/or catabolized) faster than a bivalent antibody by a cell expressing an antigen to which the antibodies bind. In some embodiments, the anti-TREM2 antibodies of the present disclosure or antibody fragments thereof can be multivalent antibodies (which are other than of the IgM class) with three or more antigen binding sites (e.g., tetravalent antibodies), which can be readily produced by recombinant expression of nucleic acid encoding the polypeptide chains of the antibody. The multivalent antibody can comprise a dimerization domain and three or more antigen binding sites. A preferred dimerization domain comprises an Fc region or a hinge region. In this scenario, the antibody will comprise an Fc region and three or more antigen binding sites amino-terminal to the Fc region. The preferred multivalent antibody herein contains three to about eight, but preferably four, antigen binding sites. The multivalent antibody contains at least one polypeptide chain (and preferably two polypeptide chains), wherein the polypeptide chain or chains comprise two or more variable domains. For instance, the polypeptide chain or chains may comprise VD1-(X1)_n-VD2-(X2)_n-Fc, wherein VD1 is a first variable domain, VD2 is a second variable domain, Fc is one polypeptide chain of an Fc region, X1 and X2 represent an amino acid or polypeptide, and n is 0 or 1. Similarly, the polypeptide chain or chains may comprise VH-CH1-flexible linker- VH-CH1-Fc region chain; or VH-CH1-VH-CH1-FC region chain. The multivalent antibody herein preferably further comprises at least two (and preferably four) light chain variable domain polypeptides. The multivalent antibody herein may, for instance, comprise from about two to about eight light chain variable domain polypeptides. The light chain variable domain polypeptides contemplated here comprise a light chain variable domain and, optionally, further comprise a CL domain.

[0707] Multivalent antibodies may recognize the TREM2 antigen as well as without limitation

additional antigens A beta peptide, antigen or an alpha synuclein protein antigen or, Tau protein antigen or, TDP-43 protein antigen or, prion protein antigen or, huntingtin protein antigen, or RAN, translation Products antigen, including the DiPeptide Repeats, (DPRs peptides) composed of glycine-alanine (GA), glycine-proline (GP), glycine-arginine (GR), proline-alanine (PA), or proline-arginine (PR), Insulin receptor, insulin like growth factor receptor. Transferrin receptor or any other antigen that facilitate antibody transfer across the blood brain barrier.

Polynucleotides Encoding TREM12 Antibodies

[0708] In another aspect, the present disclosure provides polynucleotides encoding the antibodies or antigen binding regions of the described herein. In particular, the polynucleotides are isolated polynucleotides. The polynucleotides may be operatively linked to one or more heterologous control sequences that control gene expression to create a recombinant polynucleotide capable of expressing the polypeptide of interest. Expression constructs containing a heterologous polynucleotide encoding the relevant polypeptide or protein can be introduced into appropriate host cells to express the corresponding polypeptide.

[0709] As will be appreciated by those in the art, due to the degeneracy of the genetic code, where the same amino acids are encoded by alternative or synonymous codons, an extremely large number of nucleic acids can be made, all of which encode the CDRs, variable regions, and heavy and light chains or other components of the antigen binding proteins described herein. Thus, having identified a particular amino acid sequence, those skilled in the art could make any number of different nucleic acids, by simply modifying the sequence of one or more codons in a way which does not change the amino acid sequence of the encoded protein. In this regard, the present disclosure includes each and every possible variation of polynucleotides that encode the polypeptides disclosed herein.

[0710] An “isolated nucleic acid,” which is used interchangeably herein with “isolated polynucleotide,” is a nucleic acid that has been separated from adjacent genetic sequences present in the genome of the organism from which the nucleic acid was isolated, in the case of nucleic acids isolated from naturally-occurring sources. In the case of nucleic acids synthesized enzymatically from a template or chemically, such as PCR products, cDNA molecules, or oligonucleotides for example, it is understood that the nucleic acids resulting from such processes are isolated nucleic acids. An isolated nucleic acid molecule refers to a nucleic acid molecule in the form of a separate fragment or as a component of a larger nucleic acid construct. In one preferred embodiment, the nucleic acids are substantially free from contaminating endogenous material.

[0711] In some embodiments, the polynucleotide encodes a CDR L1, CDR L2 and CDR L3 of a light chain variable region described herein. In some embodiments, the polynucleotide encodes a CDR H1, CDR H2 and CDR H3 of a heavy chain variable region described herein.

[0712] In some embodiments, the polynucleotide encodes a CDR L1, CDR L2 and CDR L3 of a light chain variable region and a CDR H1, CDR H2 and CDR H3 of a heavy chain variable region described herein.

[0713] In some embodiments, the polynucleotide encodes a light chain variable region VL having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence identity to the amino acid sequence of a variable light chain disclosed herein.

[0714] In some embodiments, the polynucleotide encodes a heavy chain variable region VH having at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence identity to the amino acid sequence of a variable heavy chain disclosed herein.

[0715] In some embodiments, the polynucleotides herein may be manipulated in a variety of ways to provide for expression of the encoded polypeptide. In some embodiments, the polynucleotide is operably linked to control sequences, including among others, transcription promoters, leader sequences, transcription enhancers, ribosome binding or entry sites, termination sequences, and polyadenylation sequences for expression of the polynucleotide and/or corresponding polypeptide. Manipulation of the isolated polynucleotide prior to its insertion into a vector may be desirable or

necessary depending on the expression vector. The techniques for modifying polynucleotides and nucleic acid sequences utilizing recombinant DNA methods are well known in the art. Guidance is provided in Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 3rd Ed., Cold Spring Harbor Laboratory Press (2001); and *Current Protocols in Molecular Biology*, Ausubel, F. ed., Greene Pub. Associates (1998), updates to 2013.

[0716] In some embodiments, variants of the antigen binding proteins, including the variants described herein, can be prepared by site-specific mutagenesis of nucleotides in the DNA encoding the polypeptide, using cassette or PCR mutagenesis or other techniques well known in the art, to produce DNA encoding the variant, and thereafter expressing the recombinant DNA in cell culture as outlined herein. However, antigen binding proteins comprising variant CDRs having up to about 100-150 residues may be prepared by in vitro synthesis using established techniques. The variants typically exhibit the same qualitative biological activity as the naturally occurring analogue, e.g., binding to antigen. Such variants include, for example, deletions and/or insertions and/or substitutions of residues within the amino acid sequences of the antigen binding proteins. Any combination of deletion, insertion, and substitution is made to arrive at the final construct, provided that the final construct possesses the desired characteristics. The amino acid changes also may alter post-translational processes of the antigen binding protein, such as changing the number or position of glycosylation sites. In some embodiments, antigen binding protein variants are prepared with the intent to modify those amino acid residues which are directly involved in epitope binding. In other embodiments, modification of residues which are not directly involved in epitope binding or residues not involved in epitope binding in any way, is desirable, for purposes discussed herein. Mutagenesis within any of the CDR regions, framework regions, and/or constant regions is contemplated. Covariance analysis techniques can be employed by the skilled artisan to design useful modifications in the amino acid sequence of the antigen binding protein. See, e.g., Choulrier, et al., *Proteins* 41:475-484, 2000; Demarest et al., *J. Mol. Biol.*, 2004, 335:41-48; Hugo et al., *Protein Engineering*, 2003, 16(5):381-86; Aurora et al., US Patent Publication No. 2008/0318207 A1; Glaser et al., US Patent Publication No. 2009/0048122 A1; Urech et al., WO 2008/110348 A1; Borrás et al., WO 2009/000099 A2. Such modifications determined by covariance analysis can improve potency, pharmacokinetic, pharmacodynamic, and/or manufacturability characteristics of an antigen binding protein.

[0717] In another aspect, the present invention also provides vectors comprising one or more nucleic acids or polynucleotides encoding one or more components of the antigen binding proteins describe herein (e.g. variable regions, light chains, and heavy chains). As used herein, the term “vector” refers to any molecule or entity (e.g., nucleic acid, plasmid, bacteriophage or virus) used to transfer protein coding information into a host cell. Examples of vectors include, but are not limited to, plasmids, viral vectors, non-episomal mammalian vectors and expression vectors, for example, recombinant expression vectors. The term “expression vector” or “expression construct” as used herein refers to a recombinant DNA molecule containing a desired coding sequence and appropriate nucleic acid control sequences necessary for the expression of the operably linked coding sequence in a particular host cell. An expression vector can include, but is not limited to, sequences that affect or control transcription, translation, and, if introns are present, affect RNA splicing of a coding region operably linked thereto. Nucleic acid sequences necessary for expression in prokaryotes include a promoter, optionally an operator sequence, a ribosome binding site and possibly other sequences. Eukaryotic cells are known to utilize promoters, enhancers, and termination and polyadenylation signals. A secretory signal peptide sequence can also, optionally, be encoded by the expression vector, operably linked to the coding sequence of interest, so that the expressed polypeptide can be secreted by the recombinant host cell, for more facile isolation of the polypeptide of interest from the cell, if desired.

[0718] The recombinant expression vector may be any vector (e.g., a plasmid or virus), which can be conveniently subjected to recombinant DNA procedures and can bring about the expression of

the polynucleotide sequence. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vectors may be linear or closed circular plasmids. Exemplary expression vectors include, among others, vectors based on T7 or T7lac promoters (pACY: Novagen; pET); vectors based on Baculovirus promoters (e.g., pBAC); vectors based on Ef1- α and HTLV promoters (e.g., pFUSE2; Invitrogen, CA, USA); vectors based on CMV enhancer and human ferritin light chain gene promoters (e.g., pFUSE: Invitrogen, CA, USA); vectors based on CMV promoters (e.g., pFLAG: Sigma, USA); and vectors based on dihydrofolate reductase promoters (e.g., pEASE: Amgen, USA). Various vectors can be used for transient or stable expression of the polypeptides of interest.

Host Cells

[0719] In another aspect, the polynucleotide encoding the antigen binding proteins described herein (e.g. variable regions, light chains, and heavy chains) is operatively linked to one or more control sequences for expression of the polypeptide in the host cell. Accordingly, in a further aspect, the present disclosure provides a host cell comprising one or more expression vectors encoding the components of the TREM2 agonist antigen binding proteins described herein.

[0720] Exemplary host cells include prokaryote, yeast, or higher eukaryote cells. Prokaryotic host cells include eubacteria, such as Gram-negative or Gram-positive organisms, for example, Enterobacteriaceae such as *Escherichia*, e.g., *E. coli*, *Enterobacter*, *Erwinia*, *Klebsiella*, *Proteus*, *Salmonella*, e.g., *Salmonella typhimurium*, *Serratia*, e.g., *Serratia marcescans*, and *Shigella*, as well as *Bacillus*, such as *B. subtilis* and *B. licheniformis*, *Pseudomonas*, and *Streptomyces*. Eukaryotic microbes such as filamentous fungi or yeast are suitable cloning or expression hosts for recombinant polypeptides. *Saccharomyces cerevisiae*, or common baker's yeast, is the most commonly used among lower eukaryotic host microorganisms. However, a number of other genera, species, and strains are commonly available and useful herein, such as *Pichia*, e.g. *P. pastoris*, *Schizosaccharomyces pombe*; *Kluyveromyces*, *Yarrowia*; *Candida*; *Trichoderma reesia*; *Neurospora crassa*; *Schwanniomyces*, such as *Schwanniomyces occidentalis*; and filamentous fungi, such as, e.g., *Neurospora*, *Penicillium*, *Tolypocladium*, and *Aspergillus* hosts such as *A. nidulans* and *A. niger*.

[0721] Host cells for the expression of glycosylated antigen binding proteins can be derived from multicellular organisms. Examples of invertebrate cells include plant and insect cells. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruitfly), and *Bombyx mori* have been identified. A variety of viral strains for transfection of such cells are publicly available, e.g., the L-1 variant of *Autographa californica* NPV and the Bm-5 strain of *Bombyx mori* NPV.

[0722] Vertebrate host cells are also suitable hosts, and recombinant production of antigen binding proteins from such cells has become routine procedure. Mammalian cell lines available as hosts for expression are well known in the art and include, but are not limited to, immortalized cell lines available from the American Type Culture Collection (ATCC), including but not limited to Chinese hamster ovary (CHO) cells, including CHOK1 cells (ATCC CCL61), DXB-11, DG-44, and Chinese hamster ovary cells/-DHFR (CHO, Urlaub et al., Proc. Natl. Acad. Sci. USA, 1980, 77: 4216); monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651); human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture, (Graham et al., J. Gen Virol. 36: 59, 1977); baby hamster kidney cells (BHK, ATCC CCL 10); mouse sertoli cells (TM4, Mather, Biol. Reprod., 1980, 23:243-251); monkey kidney cells (CV1 ATCC CCL 70); African green monkey kidney cells (VERO-76, ATCC CRL-1587); human cervical carcinoma cells (HELA, ATCC CCL 2); canine kidney cells (MDCK, ATCC CCL 34); buffalo rat liver cells (BRL 3A, ATCC CRL 1442); human lung cells (W138, ATCC CCL 75); human hepatoma cells (Hep G2, HB8065); mouse mammary tumor (MMT 060562, ATCC CCL51); TRI cells (Mather et al., Annals N.Y Acad. Sci., 1982, 383:44-68); MRC 5 cells or FS4 cells; mammalian myeloma cells, and a

number of other cell lines. In certain embodiments, cell lines may be selected through determining which cell lines have high expression levels and constitutively produce antigen binding proteins with human TREM2 binding properties. In another embodiment, a cell line from the B cell lineage that does not make its own antibody but has a capacity to make and secrete a heterologous antibody can be selected. CHO cells are preferred host cells in some embodiments for expressing the TREM2 agonist antigen binding proteins of the invention.

[0723] In various embodiments, introduction and transformation of a host cell with a polynucleotide of the present disclosure, such as an expression vector for expressing an antigen binding protein, is accomplished by methods that including transfection, infection, calcium phosphate co-precipitation, electroporation, microinjection, lipofection, DEAE-dextran mediated transfection, or other known techniques. In some embodiments, the method selected can be guided by the type of host cell used. Suitable methods are described in, for example, Sambrook et al., 2001.

Expression and Isolation

[0724] In some embodiments, the host cell comprising a polynucleotide encoding one or more components of the antigen binding proteins described herein (e.g. variable regions, light chains, and heavy chains) is used to express the antigen binding protein of interest. In some embodiments, a method for expressing the antigen binding protein comprises culturing the host cell in suitable media and conditions appropriate for expression of the protein of interest.

[0725] The type of media and culture conditions selected is based on the type of host cell. In some embodiments, exemplary media for mammalian host cells include, by way of example and not limitation, Ham's F10 (Sigma), Minimal Essential Medium (MEM, Sigma), RPMI-1640 (Sigma), and Dulbecco's Modified Eagle's Medium (DMEM, Sigma. In some embodiments, the media can be supplemented as necessary with hormones and/or other growth factors (such as insulin, transferrin, or epidermal growth factor), salts (such as sodium chloride, calcium, magnesium, and phosphate), buffers (such as HEPES), nucleotides (such as adenosine and thymidine), antibiotics (such as Gentamycin™ drug), trace elements (defined as inorganic compounds usually present at final concentrations in the micromolar range), and glucose or an equivalent energy source. In some embodiments, culture conditions, such as temperature, pH, % CO₂, and the like, can use conditions available and known to the skilled artisan.

[0726] In some embodiments, the expressed antigen binding protein is isolate and/or purified from the host cell. In some embodiments in which the expressed protein is present in the media, the media containing the expressed protein is subject to isolation procedures. In some embodiments in which the antigen binding protein is produced intracellularly, the cells are subject to disruption, and as a first step, the particulate debris, either host cells or lysed fragments, is removed, for example, by centrifugation or ultrafiltration. Subsequently, the antigen binding protein can be isolated and further purified by various known techniques. Such isolation techniques include affinity chromatography with Protein-A Sepharose, size-exclusion chromatography, ion-exchange chromatography, high performance liquid chromatography, differential solubility, and the like (see, e.g., Fisher, Laboratory Techniques, In Biochemistry And Molecular Biology, Work and Burdon, eds., Elsevier (1980); Antibodies: A Laboratory Manual, Greenfield, E.A., ed., Cold Spring Harbor Laboratory Press, New York (2012); Coligan, et al., supra, sections 2.7.1-2.7.12 and sections 2.9.1-2.9.3; Barnes, et al., Purification of Immunoglobulin G (IgG), in Methods Mol. Biol., Vol. 10, pages 79-104, Humana Press (1992)).

[0727] In some embodiments, the isolated antibody can be further purified as measurable by: (1) weight of protein as determined using the Lowry method; (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning-cup sequencer; or (3) to homogeneity by SDS-PAGE under reducing or non-reducing conditions using Coomassie blue or, preferably, silver stain. The purified antibody can be 85% or greater, 90% or greater, 95% or greater, or at least 99% by weight as determined by the foregoing methods.

Antibody Formulations

[0728] In certain embodiments, the invention provides a composition (e.g. a pharmaceutical composition) comprising one or a plurality of the TREM2 activating antibodies and TREM2 agonist antibodies and antigen binding proteins disclosed herein together with pharmaceutically acceptable diluents, carriers, excipients, solubilizers, emulsifiers, preservatives, and/or adjuvants. Pharmaceutical compositions of the invention include, but are not limited to, liquid, frozen, and lyophilized compositions. "Pharmaceutically-acceptable" refers to molecules, compounds, and compositions that are non-toxic to human recipients at the dosages and concentrations employed and/or do not produce allergic or adverse reactions when administered to humans. In some embodiments, the pharmaceutical composition may contain formulation materials for modifying, maintaining or preserving, for example, the pH, osmolarity, viscosity, clarity, color, isotonicity, odor, sterility, stability, rate of dissolution or release, adsorption or penetration of the composition. In such embodiments, suitable formulation materials include, but are not limited to, amino acids (such as glycine, glutamine, asparagine, arginine or lysine); antimicrobials; antioxidants (such as ascorbic acid, sodium sulfite or sodium hydrogen-sulfite); buffers (such as borate, bicarbonate, Tris-HCl, citrates, phosphates or other organic acids); bulking agents (such as mannitol or glycine); chelating agents (such as ethylenediamine tetraacetic acid (EDTA)); complexing agents (such as caffeine, polyvinylpyrrolidone, beta-cyclodextrin or hydroxypropyl-beta-cyclodextrin); fillers; monosaccharides; disaccharides; and other carbohydrates (such as glucose, mannose or dextrans); proteins (such as serum albumin, gelatin or immunoglobulins); coloring, flavoring and diluting agents; emulsifying agents; hydrophilic polymers (such as polyvinylpyrrolidone); low molecular weight polypeptides; salt-forming counterions (such as sodium); preservatives (such as benzalkonium chloride, benzoic acid, salicylic acid, thimerosal, phenethyl alcohol, methylparaben, propylparaben, chlorhexidine, sorbic acid or hydrogen peroxide); solvents (such as glycerin, propylene glycol or polyethylene glycol); sugar alcohols (such as mannitol or sorbitol); suspending agents; surfactants or wetting agents (such as pluronics, PEG, sorbitan esters, polysorbates such as polysorbate 20, polysorbate 80, triton, tromethamine, lecithin, cholesterol, tyloxapal); stability enhancing agents (such as sucrose or sorbitol); tonicity enhancing agents (such as alkali metal halides, preferably sodium or potassium chloride, mannitol sorbitol); delivery vehicles; diluents; excipients and/or pharmaceutical adjuvants. Methods and suitable materials for formulating molecules for therapeutic use are known in the pharmaceutical arts, and are described, for example, in Remington's Pharmaceutical Sciences, 18th Ed., (A.R. Genrmo, ed.), 1990, Mack Publishing Company.

[0729] In some embodiments, the pharmaceutical composition of the invention comprises a standard pharmaceutical carrier, such as a sterile phosphate buffered saline solution, bacteriostatic water, and the like. A variety of aqueous carriers may be used, e.g., water, buffered water, 0.4% saline, 0.3% glycine and the like, and may include other proteins for enhanced stability, such as albumin, lipoprotein, globulin, etc., subjected to mild chemical modifications or the like.

[0730] Exemplary concentrations of the antigen binding proteins in the formulation may range from about 0.1 mg/ml to about 200 mg/ml or from about 0.1 mg/mL to about 50 mg/mL, or from about 0.5 mg/mL to about 25 mg/mL, or alternatively from about 2 mg/mL to about 10 mg/mL. An aqueous formulation of the antigen binding protein may be prepared in a pH-buffered solution, for example, at pH ranging from about 4.5 to about 6.5, or from about 4.8 to about 5.5, or alternatively about 5.0. Examples of buffers that are suitable for a pH within this range include acetate (e.g. sodium acetate), succinate (such as sodium succinate), gluconate, histidine, citrate and other organic acid buffers. The buffer concentration can be from about 1 mM to about 200 mM, or from about 10 mM to about 60 mM, depending, for example, on the buffer and the desired isotonicity of the formulation.

[0731] A tonicity agent, which may also stabilize the antigen binding protein, may be included in the formulation. Exemplary tonicity agents include polyols, such as mannitol, sucrose or trehalose.

Preferably the aqueous formulation is isotonic, although hypertonic or hypotonic solutions may be suitable. Exemplary concentrations of the polyol in the formulation may range from about 1% to about 150% w/v.

[0732] A surfactant may also be added to the antigen binding protein formulation to reduce aggregation of the formulated antigen binding protein and/or minimize the formation of particulates in the formulation and/or reduce adsorption. Exemplary surfactants include nonionic surfactants such as polysorbates (e.g., polysorbate 20 or polysorbate 80) or poloxamers (e.g., poloxamer 188). Exemplary concentrations of surfactant may range from about 0.001% to about 0.5%, or from about 0.005% to about 0.2%, or alternatively from about 0.004% to about 0.01% w/v.

[0733] In one embodiment, the formulation contains the above-identified agents (i.e. antigen binding protein, buffer, polyol and surfactant) and is essentially free of one or more preservatives, such as benzyl alcohol, phenol, m-cresol, chlorobutanol and benzethonium chloride. In another embodiment, a preservative may be included in the formulation, e.g., at concentrations ranging from about 0.1% to about 2%, or alternatively from about 0.5% to about 1%. One or more other pharmaceutically acceptable carriers, excipients or stabilizers such as those described in REMINGTON'S PHARMACEUTICAL SCIENCES, 18th Edition, (A.R. Genrmo, ed.), 1990, Mack Publishing Company, may be included in the formulation provided that they do not adversely affect the desired characteristics of the formulation.

[0734] Therapeutic formulations of the antigen binding protein are prepared for storage by mixing the antigen binding protein having the desired degree of purity with optional physiologically acceptable carriers, excipients or stabilizers (Remington's Pharmaceutical Sciences, 18th Ed., (A.R. Genrmo, ed.), 1990, Mack Publishing Company), in the form of lyophilized formulations or aqueous solutions. Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include buffers (e.g. phosphate, citrate, and other organic acids); antioxidants (e.g. ascorbic acid and methionine); preservatives (such as octadecyldimethylbenzyl ammonium chloride, hexamethonium chloride, benzalkonium chloride, benzethonium chloride, phenol, butyl or benzyl alcohol, alkyl parabens such as methyl or propyl paraben, catechol; resorcinol, cyclohexanol, 3-pentanol, and m-cresol); low molecular weight (e.g. less than about 10 residues) polypeptides; proteins (such as serum albumin, gelatin, or immunoglobulins); hydrophilic polymers (e.g. polyvinylpyrrolidone); amino acids (e.g. glycine, glutamine, asparagine, histidine, arginine, or lysine); monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, maltose, or dextrans; chelating agents such as EDTA; sugars such as sucrose, mannitol, trehalose or sorbitol; salt-forming counter-ions such as sodium; metal complexes (e.g., Zn-protein complexes); and/or non-ionic surfactants, such as polysorbates (e.g. polysorbate 20 or polysorbate 80) or poloxamers (e.g. poloxamer 188); or polyethylene glycol (PEG).

[0735] In one embodiment, a suitable formulation of the claimed invention contains an isotonic buffer such as a phosphate, acetate, or TRIS buffer in combination with a tonicity agent, such as a polyol, sorbitol, sucrose or sodium chloride, which tonicifies and stabilizes. One example of such a tonicity agent is 5% sorbitol or sucrose. In addition, the formulation could optionally include a surfactant at 0.01% to 0.02% wt/vol, for example, to prevent aggregation or improve stability. The pH of the formulation may range from 4.5 to 6.5 or 4.5 to 5.5. Other exemplary descriptions of pharmaceutical formulations for antigen binding proteins may be found in US Patent Publication No. 2003/0113316 and U.S. Pat. No. 6,171,586, each of which is hereby incorporated by reference in its entirety.

[0736] Suspensions and crystal forms of antigen binding proteins are also contemplated. Methods to make suspensions and crystal forms are known to one of skill in the art.

[0737] The formulations to be used for in vivo administration must be sterile. The compositions of the invention may be sterilized by conventional, well-known sterilization techniques. For example, sterilization is readily accomplished by filtration through sterile filtration membranes. The resulting

solutions may be packaged for use or filtered under aseptic conditions and lyophilized, the lyophilized preparation being combined with a sterile solution prior to administration.

[0738] The process of freeze-drying is often employed to stabilize polypeptides for long-term storage, particularly when the polypeptide is relatively unstable in liquid compositions. A lyophilization cycle is usually composed of three steps: freezing, primary drying, and secondary drying (see Williams and Polli, *Journal of Parenteral Science and Technology*, 1984, 38(2):48-59). In the freezing step, the solution is cooled until it is adequately frozen. Bulk water in the solution forms ice at this stage. The ice sublimates in the primary drying stage, which is conducted by reducing chamber pressure below the vapor pressure of the ice, using a vacuum. Finally, sorbed or bound water is removed at the secondary drying stage under reduced chamber pressure and an elevated shelf temperature. The process produces a material known as a lyophilized cake. Thereafter the cake can be reconstituted prior to use.

[0739] The standard reconstitution practice for lyophilized material is to add back a volume of pure water (typically equivalent to the volume removed during lyophilization), although dilute solutions of antibacterial agents are sometimes used in the production of pharmaceuticals for parenteral administration (see Chen, *Drug Development and Industrial Pharmacy*, Volume 18: 1311-1354, 1992).

[0740] Excipients have been noted in some cases to act as stabilizers for freeze-dried products (see Carpenter et al., Volume 74: 225-239, 1991). For example, known excipients include polyols (including mannitol, sorbitol and glycerol); sugars (including glucose and sucrose); and amino acids (including alanine, glycine and glutamic acid).

[0741] In addition, polyols and sugars are also often used to protect polypeptides from freezing and drying-induced damage and to enhance the stability during storage in the dried state. In general, sugars, in particular disaccharides, are effective in both the freeze-drying process and during storage. Other classes of molecules, including mono- and di-saccharides and polymers such as PVP, have also been reported as stabilizers of lyophilized products.

[0742] For injection, the pharmaceutical formulation and/or medicament may be a powder suitable for reconstitution with an appropriate solution as described above. Examples of these include, but are not limited to, freeze dried, rotary dried or spray dried powders, amorphous powders, granules, precipitates, or particulates. For injection, the formulations may optionally contain stabilizers, pH modifiers, surfactants, bioavailability modifiers and combinations of these.

[0743] Sustained-release preparations may be prepared. Suitable examples of sustained-release preparations include semipermeable matrices of solid hydrophobic polymers containing the antigen binding protein, which matrices are in the form of shaped articles, e.g., films, or microcapsule. Examples of sustained-release matrices include polyesters, hydrogels (for example, poly(2-hydroxyethyl-methacrylate), or poly(vinylalcohol)), polylactides (U.S. Pat. No. 3,773,919), copolymers of L-glutamic acid and γ ethyl-L-glutamate, non-degradable ethylene-vinyl acetate, degradable lactic acid-glycolic acid copolymers such as the Lupron Depot™ (injectable microspheres composed of lactic acid-glycolic acid copolymer and leuprolide acetate), and poly-D-(-)-3-hydroxybutyric acid. While polymers such as ethylene-vinyl acetate and lactic acid-glycolic acid enable release of molecules for over 100 days, certain hydrogels release proteins for shorter time periods. When encapsulated polypeptides remain in the body for a long time, they may denature or aggregate as a result of exposure to moisture at 37° C., resulting in a loss of biological activity and possible changes in immunogenicity. Rational strategies can be devised for stabilization depending on the mechanism involved. For example, if the aggregation mechanism is discovered to be intermolecular S-S bond formation through thio-disulfide interchange, stabilization may be achieved by modifying sulfhydryl residues, lyophilizing from acidic solutions, controlling moisture content, using appropriate additives, and developing specific polymer matrix compositions.

[0744] The formulations of the invention may be designed to be short-acting, fast-releasing, long-

acting, or sustained-releasing. Thus, the pharmaceutical formulations may also be formulated for controlled release or for slow release.

[0745] Specific dosages may be adjusted depending on the disease, disorder, or condition to be treated, the age, body weight, general health conditions, sex, and diet of the subject, dose intervals, administration routes, excretion rate, and combinations of drugs.

[0746] The TREM2 agonist antigen binding proteins of the invention can be administered by any suitable means, including parenteral, subcutaneous, intraperitoneal, intrapulmonary, intrathecal, intracerebral, intracerebroventricular, and intranasal, and, if desired for local treatment, intralesional administration. Parenteral administration includes intravenous, intraarterial, intraperitoneal, intramuscular, intradermal or subcutaneous administration. In addition, the antigen binding protein is suitably administered by pulse infusion, particularly with declining doses of the antigen binding protein. Preferably, the dosing is given by injections, most preferably intravenous or subcutaneous injections, depending in part on whether the administration is brief or chronic. Other administration methods are contemplated, including topical, particularly transdermal, transmucosal, rectal, oral or local administration e.g. through a catheter placed close to the desired site. In certain embodiments, the TREM2 agonist antigen binding protein of the invention is administered intravenously or subcutaneously in a physiological solution at a dose ranging between 0.01 mg/kg to 100 mg/kg at a frequency ranging from daily to weekly to monthly (e.g. every day, every other day, every third day, or 2, 3, 4, 5, or 6 times per week), preferably a dose ranging from 0.1 to 45 mg/kg, 0.1 to 15 mg/kg or 0.1 to 10 mg/kg at a frequency of once per week, once every two weeks, or once a month.

[0747] In some embodiments, the anti-TREM2 antibody is administered to a human patient via an IV infusion. In some embodiments, an IV infusion of anti-TREM2 antibody is up to about 5 hours, up to about 4 hours, up to about 3 hours, up to about 2 hours, or up to about 60 minutes. In some embodiments, an IV infusion of anti-TREM2 antibody is from about 5 minutes to about 5 hours, from about 5 minutes to about 4 hours, from about 5 minutes to about 3 hours, from about 5 minutes to about 2 hours, or from about 5 minutes to about 60 minutes. In some embodiments, an IV infusion of anti-TREM2 antibody is about 5 minutes, about 10 minutes, about 15 minutes, about 20 minutes, about 25 minutes, about 30 minutes, about 35 minutes, about 40 minutes, about 45 minutes, about 50 minutes, about 55 minutes, about 60 minutes, about 70 minutes, about 80 minutes, or about 90 minutes.

[0748] In some embodiments, anti-TREM2 antibody is administered to a human patient at a dose of up to about 200 mg/kg. In some embodiments, anti-TREM2 antibody is administered to a human patient at a dose of up to about 150 mg/kg. In some embodiments, anti-TREM2 antibody is administered to a human patient at a dose of up to about 100 mg/kg. In some embodiments, anti-TREM2 antibody is administered to a human patient at a dose of from about 1 mg/kg to about 100 mg/kg, from about 1 mg/kg to about 90 mg/kg, from about 1 mg/kg to about 80 mg/kg, from about 1 mg/kg to about 70 mg/kg, or from about 1 mg/kg to about 60 mg/kg. In some embodiments, anti-TREM2 antibody is administered to a human patient at a dose of about 1 mg/kg, about 2 mg/kg, about 3 mg/kg, about 5 mg/kg, about 10 mg/kg, about 15 mg/kg, about 20 mg/kg, about 25 mg/kg, about 30 mg/kg, about 35 mg/kg, about 40 mg/kg, about 45 mg/kg, about 50 mg/kg, about 55 mg/kg, or about 60 mg/kg.

[0749] In some embodiments, anti-TREM2 antibody is administered to a human patient once daily. In some embodiments, anti-TREM2 antibody is administered to a human patient 1, 2, 3 or 4 times weekly. In some embodiments, anti-TREM2 antibody is administered to a human patient 1, 2, 3 or 4 times monthly. In some embodiments, anti-TREM2 antibody is administered to a human patient once every 1, 2, 3, or 4 weeks. In some embodiments, anti-TREM2 antibody is administered to a human patient once every 2 days, 3 days, 4 days, 5 days, 6 days, 7 days, 8 days, 9 days, 10 days, or 14 days. In some embodiments, anti-TREM2 antibody is administered to a human patient once weekly.

[0750] In some embodiments, the present invention provides a liquid formulation, comprising anti-TREM2 antibody at a concentration of up to about 300 mg/mL. In some embodiments, the present invention provides a liquid formulation, comprising anti-TREM2 antibody at a concentration of up to about 250 mg/mL. In some embodiments, the present invention provides a liquid formulation, comprising anti-TREM2 antibody at a concentration of up to about 200 mg/mL. In some embodiments, the present invention provides a liquid formulation, comprising anti-TREM2 antibody at a concentration of up to about 150 mg/mL. In some embodiments, the present invention provides a liquid formulation, comprising anti-TREM2 antibody at a concentration of about 300 mg/mL, about 250 mg/mL, about 200 mg/mL, about 180 mg/mL, about 170 mg/mL, about 160 mg/mL, about 150 mg/mL, about 140 mg/mL, about 130 mg/mL, about 120 mg/mL, about 110 mg/mL, or about 100 mg/mL. In some embodiments, the present invention provides a liquid formulation, comprising anti-TREM2 antibody at a concentration of about 140 mg/mL. In some embodiments, a method of the invention comprises administering to a human patient a liquid formulation as described herein. In some embodiments, a method of the invention comprises administering to a human patient a liquid formulation, comprising anti-TREM2 antibody at a concentration of about 140 mg/mL.

[0751] The TREM2 agonist antigen binding proteins described herein (e.g. anti-TREM2 agonist monoclonal antibodies and binding fragments thereof) are useful for preventing, treating, or ameliorating a condition associated with TREM2 deficiency or loss of biological function of TREM2 in a patient in need thereof. As used herein, the term “treating” or “treatment” is an intervention performed with the intention of preventing the development or altering the pathology of a disorder. Accordingly, “treatment” refers to both therapeutic treatment and prophylactic or preventative measures. Patients in need of treatment include those already diagnosed with or suffering from the disorder or condition as well as those in which the disorder or condition is to be prevented, such as patients who are at risk of developing the disorder or condition based on, for example, genetic markers. “Treatment” includes any indicia of success in the amelioration of an injury, pathology or condition, including any objective or subjective parameter such as abatement, remission, diminishing of symptoms, or making the injury, pathology or condition more tolerable to the patient, slowing in the rate of degeneration or decline, making the final point of degeneration less debilitating, or improving a patient's physical or mental well-being. The treatment or amelioration of symptoms can be based on objective or subjective parameters, including the results of a physical examination, self-reporting by a patient, cognitive tests, motor function tests, neuropsychiatric exams, and/or a psychiatric evaluation.

III. Small Molecule TREM2 Agonists

[0752] In some embodiments, the agonist of TREM2 is a small molecule agonist of TREM2.

[0753] In some embodiments, the agonist of TREM2 is a lipid ligand of TREM2. In some embodiments, the lipid ligand of TREM2 is selected from 1-palmitoyl-2-(5'-oxo-valeroyl)-sn-glycero-3-phosphocholine (POVPC), 2-Arachidonoylglycerol (2-AG), 7-ketocholesterol (7-KC), 24(S)hydroxycholesterol (24OHC), 25(S)hydroxycholesterol (25OHC), 27-hydroxycholesterol (27OHC), Acyl Carnitine (AC), alkylacylglycerophosphocholine (PAF), a-galactosylceramide (KRN7000), Bis(monoacylglycero)phosphate (BMP), Cardiolipin (CL), Ceramide, Ceramide-1-phosphate (CIP), Cholesteryl ester (CE), Cholesterol phosphate (CP), Diacylglycerol 34: 1 (DG 34: 1), Diacylglycerol 38:4 (DG 38:4), Diacylglycerol pyrophosphate (DGPP), Dihydrosphingomyelin (DhCer), Dihydrosphingomyelin (DhSM), Ether phosphatidylcholine (PCe), Free cholesterol (FC), Galactosylceramide (GalCer), Galactosylsphingosine (GalSo), Ganglioside GM1, Ganglioside GM3, Glucosylsphingosine (GlcSo), Hank's Balanced Salt Solution (HBSS), Kdo2-Lipid A (KLA), Lactosylceramide (LacCer), lysoalkylacylglycerophosphocholine (LPAF), Lysophosphatidic acid (LPA), Lysophosphatidylcholine (LPC), Lysophosphatidylethanolamine (LPE), Lysophosphatidylglycerol (LPG), Lysophosphatidylinositol (LPI), Lysosphingomyelin (LSM), Lysophosphatidylserine (LPS), N-Acyl-phosphatidylethanolamine (NAPE), N-Acyl-Serine (NSer),

Oxidized phosphatidylcholine (oxPC), Palmitic-acid-9-hydroxy-stearic-acid (PAHSA), Phosphatidylethanolamine (PE), Phosphatidylethanol (PEtOH), Phosphatidic acid (PA), Phosphatidylcholine (PC), Phosphatidylglycerol (PG), Phosphatidylinositol (PI), Phosphatidylserine (PS), Sphinganine, Sphinganine-1-phosphate (SalP), Sphingomyelin (SM), Sphingosine, Sphingosine-1-phosphate (SolP), or Sulfatide, or a salt thereof.

[0754] In some embodiments, the agonist of TREM2 is a lipopolysaccharide.

[0755] In some embodiments, the agonist of TREM2 is a small molecule disclosed in PCT Application Publication WO2019/079529, which is incorporated by reference herein in its entirety. In some embodiments, the agonist of TREM2 is Tyrphostin AG 538, AC1NS458, IN1040, Butein, Okanin, AGL 2263, GB19, GB16, GB20, GB17, GB18, GB21, GB22, GB27, GB44, GB42, GB2, 4,4'-Dihydroxychalcone, or 3,4-Dihydroxybenzophenone, or a derivative or salt of any of the aforementioned.

[0756] In some embodiments, the agonist of TREM2 is a small molecule identified by a method disclosed in PCT Application Publication WO2019/079529. In some embodiments, the small molecule agonist of TREM2 is identified by applying the small molecule compound to a host cell expressing TREM2 and tyrosine kinase binding protein (TYROBP), wherein the host cell has a synthetic sequence comprising an NFAT-response element and a nucleotide sequence encoding a reporter, and measuring a signal emitted by the reporter.

[0757] In some embodiments, the agonist of TREM2 is a small molecule disclosed in PCT Application Publication WO2021/226135 or WO2021/226629.

[0758] In some embodiments, the agonist of TREM2 is a TREM2 agonist compound comprising a bicyclic core. In some embodiments, the bicyclic core is a 10-membered heteroaryl core. In some embodiments, the TREM2 agonist compound comprises a 10-membered heteroaryl core, comprising 1-4 nitrogen atoms as part of the core ring structure. In some embodiments, the TREM2 agonist compound comprises a 10-membered heteroaryl core, comprising 3 or 4 substituent groups.

IV. Other TREM2 Agonists

[0759] In some embodiments, the agonist of TREM2 is heat shock protein 60 (HSP60).

[0760] In some embodiments, the agonist of TREM2 is apolipoprotein E (ApoE).

V. Pharmaceutically Acceptable Compositions

[0761] In certain embodiments, a TREM2 activating antibody or small molecule disclosed herein is formulated as a composition for administration to a patient in need of such composition.

[0762] The term “pharmaceutically acceptable carrier, adjuvant, or vehicle” refers to a non-toxic carrier, adjuvant, or vehicle that does not destroy the pharmacological activity of the compound with which it is formulated. Pharmaceutically acceptable carriers, adjuvants or vehicles that may be used in the compositions of this invention include, but are not limited to, ion exchangers, alumina, aluminum stearate, lecithin, serum proteins, such as human serum albumin, buffer substances such as phosphates, glycine, sorbic acid, potassium sorbate, partial glyceride mixtures of saturated vegetable fatty acids, water, salts or electrolytes, such as protamine sulfate, disodium hydrogen phosphate, potassium hydrogen phosphate, sodium chloride, zinc salts, colloidal silica, magnesium trisilicate, polyvinyl pyrrolidone, cellulose-based substances, polyethylene glycol, sodium carboxymethylcellulose, polyacrylates, waxes, polyethylene-polyoxypropylene-block polymers, polyethylene glycol and wool fat.

[0763] A “pharmaceutically acceptable derivative” means any non-toxic salt, ester, salt of an ester or other derivative of a compound of this invention that, upon administration to a recipient, is capable of providing, either directly or indirectly, a compound of this invention or an inhibitorily active metabolite or residue thereof.

[0764] Compositions of the present invention may be administered orally, parenterally, by inhalation spray, topically, rectally, nasally, buccally, vaginally or via an implanted reservoir. The term “parenteral” as used herein includes subcutaneous, intravenous, intramuscular, intra-articular, intra-synovial, intrasternal, intrathecal, intrahepatic, intralesional and intracranial injection or

infusion techniques. In some embodiments, the compositions are administered orally, intraperitoneally or intravenously. Sterile injectable forms of the compositions of this invention may be aqueous or oleaginous suspension. These suspensions may be formulated according to techniques known in the art using suitable dispersing or wetting agents and suspending agents. The sterile injectable preparation may also be a sterile injectable solution or suspension in a non-toxic parenterally acceptable diluent or solvent, for example as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that may be employed are water, Ringer's solution and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium.

[0765] For this purpose, any bland fixed oil may be employed including synthetic mono- or diglycerides. Fatty acids, such as oleic acid and its glyceride derivatives are useful in the preparation of injectables, as are natural pharmaceutically-acceptable oils, such as olive oil or castor oil, especially in their polyoxyethylated versions. These oil solutions or suspensions may also contain a long-chain alcohol diluent or dispersant, such as carboxymethyl cellulose or similar dispersing agents that are commonly used in the formulation of pharmaceutically acceptable dosage forms including emulsions and suspensions. Other commonly used surfactants, such as Tweens, Spans and other emulsifying agents or bioavailability enhancers which are commonly used in the manufacture of pharmaceutically acceptable solid, liquid, or other dosage forms may also be used for the purposes of formulation.

[0766] Pharmaceutically acceptable compositions of this invention may be orally administered in any orally acceptable dosage form including, but not limited to, capsules, tablets, aqueous suspensions or solutions. In the case of tablets for oral use, carriers commonly used include lactose and corn starch. Lubricating agents, such as magnesium stearate, are also typically added. For oral administration in a capsule form, useful diluents include lactose and dried cornstarch. When aqueous suspensions are required for oral use, the active ingredient is combined with emulsifying and suspending agents. If desired, certain sweetening, flavoring or coloring agents may also be added.

[0767] Alternatively, pharmaceutically acceptable compositions of this invention may be administered in the form of suppositories for rectal administration. These can be prepared by mixing the agent with a suitable non-irritating excipient that is solid at room temperature but liquid at rectal temperature and therefore will melt in the rectum to release the drug. Such materials include cocoa butter, beeswax and polyethylene glycols.

[0768] Pharmaceutically acceptable compositions of this invention may also be administered topically, especially when the target of treatment includes areas or organs readily accessible by topical application, including diseases of the eye, the skin, or the lower intestinal tract. Suitable topical formulations are readily prepared for each of these areas or organs.

[0769] Topical application for the lower intestinal tract can be effected in a rectal suppository formulation (see above) or in a suitable enema formulation. Topically-transdermal patches may also be used.

[0770] For topical applications, provided pharmaceutically acceptable compositions may be formulated in a suitable ointment containing the active component suspended or dissolved in one or more carriers. Carriers for topical administration of compounds of this invention include, but are not limited to, mineral oil, liquid petrolatum, white petrolatum, propylene glycol, polyoxyethylene, polyoxypropylene compound, emulsifying wax and water. Alternatively, provided pharmaceutically acceptable compositions can be formulated in a suitable lotion or cream containing the active components suspended or dissolved in one or more pharmaceutically acceptable carriers. Suitable carriers include, but are not limited to, mineral oil, sorbitan monostearate, polysorbate 60, cetyl esters wax, cetaryl alcohol, 2-octyldodecanol, benzyl alcohol and water.

[0771] For ophthalmic use, provided pharmaceutically acceptable compositions may be formulated as micronized suspensions in isotonic, pH adjusted sterile saline, or as solutions in isotonic, pH

adjusted sterile saline, either with or without a preservative such as benzylalkonium chloride. Alternatively, for ophthalmic uses, the pharmaceutically acceptable compositions may be formulated in an ointment such as petrolatum.

[0772] Pharmaceutically acceptable compositions of this invention may also be administered by nasal aerosol or inhalation. Such compositions are prepared according to techniques well-known in the art of pharmaceutical formulation and may be prepared as solutions in saline, employing benzyl alcohol or other suitable preservatives, absorption promoters to enhance bioavailability, fluorocarbons, and/or other conventional solubilizing or dispersing agents.

[0773] In some embodiments, pharmaceutically acceptable compositions of this invention are formulated for oral administration. Such formulations may be administered with or without food. In some embodiments, pharmaceutically acceptable compositions of this invention are administered without food. In other embodiments, pharmaceutically acceptable compositions of this invention are administered with food.

[0774] In other embodiments, pharmaceutically acceptable compositions of this invention are formulated for intravenous (IV) administration.

[0775] The amount of compounds of the present invention that may be combined with the carrier materials to produce a composition in a single dosage form will vary depending upon the host treated, the particular mode of administration. Preferably, provided compositions should be formulated so that a dosage of between 0.01-100 mg/kg body weight/day of the inhibitor can be administered to a patient receiving these compositions.

[0776] It should also be understood that a specific dosage and treatment regimen for any particular patient will depend upon a variety of factors, including the activity of the specific compound employed, the age, body weight, general health, sex, diet, time of administration, rate of excretion, drug combination, and the judgment of the treating physician and the severity of the particular disease being treated. The amount of a compound of the present invention in the composition will also depend upon the particular compound in the composition.

[0777] All features of each of the aspects of the disclosure apply to all other aspects mutatis mutandis. Each of the references referred to herein, including but not limited to patents, patent applications and journal articles, is incorporated by reference herein as though fully set forth in its entirety.

[0778] In order that the disclosure described herein may be more fully understood, the following examples are set forth. It should be understood that these examples are for illustrative purposes only and are not to be construed as limiting this disclosure in any manner.

EXAMPLES

General Procedures

Microglial Differentiation Protocol

[0779] Pluripotent stem cell (PSC) differentiation is induced with mTeSR Custom medium (STEMCELL Technologies®) containing 80 ng/mL BMP4. At day 4, cells are induced with 25 ng/mL basic fibroblast growth factor, 100 ng/mL stem cell factor (SCF), and 80 ng/mL vascular endothelial growth factor in StemPro-34 SFM (with 2 mM GlutaMAX, Life Technologies). Two days later, the medium is supplemented with 50 ng/mL SCF, 50 ng/mL IL-3, 5 ng/mL thrombopoietin, 50 ng/mL macrophage CSF (M-CSF) and 50 ng/mL Flt31, and from day 14 with 50 ng/mL M-CSF, 50 ng/mL Flt31, and 25 ng/mL GM-CSF. Between days 25 and 50, CD14⁺ or CD14⁺CX3CR1⁺ progenitors are isolated and plated onto tissue culture-treated dishes or Thermanox plastic coverslips (all from Thermo Fisher Scientific) in Microglial Medium (RPMI-1640 [Life Technologies] with 2 mM GlutaMAX-I, 10 ng/mL GM-CSF, and 100 ng/mL IL-34). Medium is replenished every 3-4 days for at least 2 weeks.

Generation of Induced Pluripotent Stem Cells (iPSCs) from Primary Healthy,

Adrenomyeloneuropathy (AMN) and Cerebral Childhood ALD (cALD) Fibroblasts

[0780] Fibroblast cell cultures from healthy individuals (AGO1439, male 3 days old),

Adrenomyeloneuropathy (AMN) (GM 07530, male 26 years old) and cALD (GM04934, male 7 years old with VLCFA abnormality and clinical X-ALD disease) patients are obtained. A control human IPSC ATCC-DYRO100 cell line is also obtained. Fibroblasts are cultured in DMEM with 10% FBS, 2 mM L-glutamine and 1% penicillin/streptomycin at 37° C. with 5% CO₂. [0781] Fibroblast cells seeded at 0.2×10⁶ cells/well of a 6-well plate in fibroblast medium (DMEM+10% FBS) are transduced with six lentiviral vectors designed to deliver human OCT4, SOX2, c-MYC, KLF4, Nanog and Lin28 cDNA sequences. On the next day, fresh fibroblast media is added to the cells 24 hours after transduction. At 48 hours after transduction the media is changed to half E8 medium and half fibroblast medium. When the cells reach about 60% confluence they are transferred to 10 cm Matrigel-coated plates (one well of a 6-well plate into one 10 cm dish) in E8 medium (StemCell Technologies) and media is replaced daily. Between day 15 and day 30 in culture, individual hiPS clones are manually picked using a Leica stereomicroscope. Each hiPS clone is expanded and characterized by immunofluorescence for the expression of Oct4 and Tra-1-60. IPSCs are cultured on a Matrigel (BD-Biosciences) coated plate in IPSC medium (mTeSR media from Stemcell technologies, Vancouver, Canada) and media is changed daily until cells are ready for passage.

Microglia Characterization Assays and Methods

[0782] Microglia are analyzed using flow cytometry, immunohistochemistry and cell quantification procedures such as those disclosed by Masuda et al. (Masuda et al., *Nature*. 2019; 566: 388-394.)

Monocyte Derived Macrophage Isolation Protocols

[0783] Monocyte derived macrophages (MDMs) are derived from PBMCs collected from patients with adrenomyeloneuropathy (AMN) or cALD and isolated by magnetic bead separation and differentiation into macrophages as reported in Jin, et al. *J Vis Exp*. 2016; (112): 54244. CD14⁺ monocytes are collected for use in the Examples described below.

Example 1. The Effects of TREM2 Agonists on Microglia after VLCFA Challenge Conceptual Basis

[0784] Saturated very long chain fatty acids (VLCFAs; ≥C22:0) accumulate in the microglia of patients suffering from X-linked adrenoleukodystrophy (X-ALD, OMIM 300100), a severe hereditary neurodegenerative disease, due to peroxisomal impairment. Previous studies analysed the development of X-ALD in humans and gene knockout animal models. However, the toxic effect of VLCFA leading to severe symptoms with progressive and multifocal demyelination, adrenal insufficiency and inflammation still remains unclear. To understand the toxic effects of VLCFA in the brain, neural cells are exposed to VLCFA and the effects are analyzed.

Oligodendrocytes and astrocytes challenged with docosanoic-(C22:0), tetracosanoic- (C24:0) and hexacosanoic acids (C26:0) die within 24 h. VLCFA-induced depolarization of mitochondria in situ and increased intracellular Ca²⁺ level in all three brain cell types provides indications about the mechanism of toxicity of VLCFA. VLCFAs affect to the largest degree the myelin-producing oligodendrocytes. In isolated mitochondria, VLCFAs exert a detrimental effect by affecting the inner mitochondrial membrane and promoting the permeability transition. Without intending to be limited to any particular theory, it is reasonable to conclude that there is potent toxic activity of VLCFA due to dramatic cell physiological effects with mitochondrial dysfunction and Ca²⁺ deregulation. This provides evidence for mitochondrial-based cell death mechanisms in neurodegenerative diseases with peroxisomal defects and subsequent VLCFA accumulation. Treatment with a TREM2 agonist rescues at least some of the deleterious effects of VLCFA accumulation in microglia, by preventing apoptosis and activating phagocytic mechanisms to clear myelin debris to lower inflammation in the vicinity of axonopathy.

Experiments

[0785] Healthy patient-derived microglia are plated in 96 well microtiter plates. Each well also contains either a TREM2 agonist (eg. a TREM2 antibody agonist or TREM2 small molecule agonist) or a control compound (eg. an isotype control IgG as a control antibody or DMSO as a

control small molecule). An exemplary method uses an immobilized TREM2 antibody agonist in the test wells at 10 pg/well and an isotype control plated in the control wells at 10 pg/well. Another exemplary method uses a solubilized TREM2 antibody agonist or small molecule agonist in the test wells. Cells are maintained in CSF1-containing culture medium (50 ng/mL) for 2 days prior to adding VLCFA (e.g., C26:0, C24:0, C22:0 added at 10-20 uM concentration to culture medium; Hein et al., Human Molecular Genetics. 2008; 17: 1750-176.). After 24 hrs of VLCFA challenge, wells are analyzed microscopically by immunohistochemical staining, e.g., by staining for Iba1, P2YR12 or other myeloid cell surface markers, staining for caspases, which indicate cell apoptosis, and for the number of viable cells as well as for cell morphology. Other microtiter wells can be treated with lysis buffer to collect mRNA for qPCR analysis for markers of myeloid phenotype, such as homeostatic or activated cell states (Keren-Shaul et al., Cell. 2017; 169: 1276-1290; Decskowska et al., Cell. 2018; 173:1073-1081.). Other microtiter wells can be analyzed for total cell death by measuring lactate dehydrogenase levels, a measure of cell lysis, from the culture supernatant (Hein et al., Human Molecular Genetics. 2008; 17: 1750-176.). This experiment can be modified to measure the kinetics of phenotypic transition following the pilot described above by analyzing changes over time and at various doses of VGL101 to understand the dose dependence of rescue. In addition, the Incucyte method of monitoring cell cultures can be used to monitor for morphological changes in real time.

[0786] The above experiments are also repeated using monocytes derived macrophages, in place of the microglia.

Example 2. The Effects of TREM2 Agonists on Microglia with an ABCD1 Dysfunction Conceptual Basis

[0787] Microglia derived from patients suffering from cALD, such as by differentiating patient iPSC or patient monocytes, have transcriptional and biochemical signatures distinct from healthy donor-derived microglia, e.g., potentially enriched for disease associated microglia (DAM). Treatment of cALD patient-derived microglia with VLCFA in vitro will cause toxic and pro-inflammatory effects due to accumulation of fatty acids from dysfunctional peroxisome-mediated metabolism. cALD microglia may also autonomously accumulate VLCFAs without an extracellular challenge. Rescue of the cytotoxicity and inflammatory state by treating with TREM2 agonists during VLCFA challenge can be characterized through transcriptional and biochemical analyses.

Experiments

[0788] iPSC or monocyte-derived microglia, derived from patients suffering from an ABCD1 dysfunction, such as cALD or AMN, are plated in 96 well microtiter plates. Each well also contains either a TREM2 agonist (eg. a TREM2 antibody agonist or TREM2 small molecule agonist) or a control compound (eg. an isotype control IgG as a control antibody or DMSO as a control small molecule). An exemplary method uses an immobilized TREM2 antibody agonist in the test wells at 10 pg/well and an isotype control plated in the control wells at 10 pg/well. Another exemplary method uses a solubilized TREM2 antibody agonist or small molecule agonist in the test wells. Cells are maintained in a CSF1-containing culture medium (50 ng/mL) for 2 days prior to adding VLCFA (e.g., C26:0, C24:0, C22:0 added at 10-20 uM concentration to culture medium; Hein et al., Human Molecular Genetics. 2008; 17: 1750-176.). Vehicle is added to select wells, in place of the VLCFA. After 24 hrs of challenge, wells are analyzed microscopically by immunohistochemical staining, e.g., by staining for Iba1, PYR12 or other myeloid cell surface markers, staining for caspases, which indicate cell apoptosis, and for the number of viable cells as well as for cell morphology. Other microtiter wells can be treated with lysis buffer to collect mRNA for qPCR analysis for markers of myeloid phenotype, such as homeostatic or activated cell states (Keren-Shaul et al., Cell. 2017; 169: 1276-1290; Decskowska et al., Cell. 2018; 173:1073-1081.). Other microtiter wells can be analyzed for total cell death by measuring lactate dehydrogenase levels, a measure of cell lysis, from the culture supernatant (Hein et al., Human Molecular Genetics. 2008; 17: 1750-176.). This experiment can be modified to measure the kinetics of

phenotypic transition following the pilot described above by analyzing changes over time and at various doses of VLCFA or VGL101 to understand the dose dependence of rescue. In addition, the Incucyte method of monitoring cell cultures can be used to monitor for morphological changes in real time.

[0789] The above experiments are also repeated using monocytes derived macrophages, in place of the iPSC or microglia. The above experiments may also be repeated, wherein the VLCFA challenge is replaced with alternative additions known to induce increased accumulation of VLCFAs in cALD microglia. For example, myelin debris or lipids can be added to the wells in place of the VLCFA.

Example 3. Neurofilament Light Chain as a Biomarker for Tracking x-ALD Treatment Efficacy

[0790] Monitoring of serum from patients with x-ALD for levels of neurofilament light chain (NfL) in order to select patients for treatment, and to monitor the efficacy of treatment will be done as follows. Serum is collected from patients at various time points as required for the use. Serum is stored in sample aliquots at -80° C. When ready for analysis, samples are thawed on ice.

Measurement of NfL is determined using an assay run on a Simoa® HD-1 instrument (QUANTERIX) using a 2-step Assay Neat 2.0 protocol; 100 μ l of sample or calibrator (diluent: Tris-buffered saline [TBS], 0.1% Tween 20, 1% milk powder, 400 μ g/ml Heteroblock [Omega Biologicals, Bozeman, MT]), 25 μ l conjugated beads (diluent: TBS, 0.1% Tween 20, 1% milk powder, 300 μ g/ml Heteroblock), and 20 μ l of mAB2:1 (0.1 μ g/ml; diluent: TBS, 0.1% Tween 20, 1% milk powder, 300 μ g/ml Heteroblock) are incubated for 47 cadences (1 cadence=45 seconds). After washing, 100 μ l of streptavidin-conjugated b-galactosidase (150 μ M; Quanterix) is added, followed by a 7-cadence incubation and a wash. Prior to reading, 25 μ l resorufin b-D-galactopyranoside (QUANTERIX) is added. Calibrators (neat) and samples (serum: 1:4 dilution) are measured in duplicates. Bovine lyophilized NfL is obtained from UmanDiagnostics. Calibrators ranged from 0 to 2,000 μ g/ml for serum and from 0 to 10,000 μ g/ml for CSF measurements. Batch prepared calibrators are stored at -80° C. Final NfL levels measured by the above method are used to help both select patients to treat with an agonist of TREM2 and guide response to treatment with an agonist of TREM2.

Claims

1. A method of treating a disease or disorder caused by and/or associated with ATP-binding cassette transporter 1 (ABCD1) dysfunction in a human patient, the method comprising administering to the patient an effective amount of an agonist of triggering receptor expressed on myeloid cells 2 (TREM2).
2. The method of claim 1, wherein the disease or disorder is selected from X-linked adrenoleukodystrophy (x-ALD), Globoid cell leukodystrophy, Metachromatic leukodystrophy (MLD), Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL), Alexander disease, fragile X-associated tremor ataxia syndrome (FXTAS), adult-onset autosomal dominant leukodystrophy (ADLD), and X-linked Charcot-Marie-Tooth disease (CMTX).
3. The method of claim 1, wherein the disease or disorder is selected from Nasu-Hakola disease, Alzheimer's disease, frontotemporal dementia, multiple sclerosis, Guillain-Barre syndrome, amyotrophic lateral sclerosis (ALS), or Parkinson's disease; wherein the patient exhibits ABCD1 dysfunction, and/or has a mutation in a gene affecting the function of ABCD1.
4. The method of claim 1, wherein the disease or disorder is x-ALD.
5. The method of claim 1, wherein the disease or disorder is Addison disease wherein the patient has been found to have a mutation in one or more ABCD1 genes affecting ABCD1 function.
6. The method of claim 1, wherein the administration of the agonist of TREM2 increases microglia function in the patient.

7. The method of claim 1, wherein the agonist of TREM2 activates TREM2/DAP12 signaling in myeloid cells.

8. The method of claim 1, wherein the agonist of TREM2 activates, induces, promotes, stimulates, or otherwise increases one or more TREM2 activities selected from: (a) TREM2 binding to DAP12; DAP12 binding to TREM2; TREM2 phosphorylation; DAP12 phosphorylation; (b) PI3K activation; (c) increased levels of soluble TREM2 (sTREM2); (d) increased levels of soluble CSF1R (sCSF1R); (e) increased expression of one or more anti-inflammatory mediators selected from the group consisting of IL-12p70, IL-4, IL-6, and IL-1 β ; (f) reduced expression of one or more pro-inflammatory mediators selected from the group consisting of IFN- α 4, IFN-b, IL-6, IL-12 p70, IL-12 p40, IL-1 β , TNF, TNF- α , IL-1 β , IL-8, CRP, TGF-beta members of the chemokine protein families, IL-20 family members, IL-33, LIF, IFN-gamma, OSM, CNTF, TGF-beta, GM-CSF, IL-11, IL-12, IL-17, IL-18, and CRP; (g) increased expression of one or more chemokines selected from the group consisting of CCL2, CCL4, CXCL10, CCL3 and CST7; (h) reduced expression of TNF- α , IL-6, or both; extracellular signal-regulated kinase (ERK) phosphorylation; increased expression of C-C chemokine receptor 7 (CCR7); (i) induction of microglial cell chemotaxis toward CCL19 and CCL21 expressing cells; (j) an increase, normalization, or both of the ability of bone marrow-derived dendritic cells to induce antigen-specific T-cell proliferation; (k) induction of osteoclast production, increased rate of osteoclastogenesis, or both; increasing the survival and/or function of one or more of dendritic cells, macrophages, microglial cells, M1 macrophages and/or microglial cells, activated M1 macrophages and/or microglial cells, M2 macrophages and/or microglial cells, monocytes, osteoclasts, Langerhans cells of skin, and Kupffer cells; (l) induction of one or more types of clearance selected from the group consisting of apoptotic neuron clearance, nerve tissue debris clearance, non-nerve tissue debris clearance, bacteria or other foreign body clearance, disease-causing protein clearance, disease-causing peptide clearance, and disease-causing nucleic acid clearance; (m) induction of phagocytosis of one or more of apoptotic neurons, nerve tissue debris, non-nerve tissue debris, bacteria, other foreign bodies, disease-causing proteins, disease-causing peptides, or disease-causing nucleic acids; normalization of disrupted TREM2/DAP12-dependent gene expression; (n) recruitment of Syk, ZAP70, or both to the TREM2/DAP12 complex; Syk phosphorylation; increased expression of CD83 and/or CD86 on dendritic cells, macrophages, monocytes, and/or microglia; (o) reduced secretion of one or more inflammatory cytokines selected from the group consisting of TNF- α , IL-1 β , IL-6, MCP-1, IFN- α 4, IFN-b, IL-1 β , IL-8, CRP, TGF-beta members of the chemokine protein families, IL-20 family members, IL-33, LIF, IFN-gamma, OSM, CNTF, TGF-beta, GM-CSF, IL-11, IL-12, IL-17, IL-18, and CRP; (p) reduced expression of one or more inflammatory receptors; increasing phagocytosis by macrophages, dendritic cells, monocytes, and/or microglia under conditions of reduced levels of MCSF; (q) decreasing phagocytosis by macrophages, dendritic cells, monocytes, and/or microglia in the presence of normal levels of MCSF; increasing activity of one or more TREM2-dependent genes; (r) increased levels of one or more of CSF1, CSF2 and IL-34; or (s) any combination thereof.

9. The method of claim 1, wherein the agonist of TREM2 is an antigen binding protein or an antibody, or an antigen-binding fragment thereof.

10. The method of claim 9, wherein the agonist of TREM2 is a monoclonal antibody, a humanized antibody, or a human antibody.

11-12. (canceled)

13. The method of claim 1, wherein the agonist of TREM2 is an antibody or antigen-binding fragment thereof that specifically binds to the polypeptide of SEQ ID NO: 1.

14. The method of claim 13, wherein the antibody or antigen-binding fragment thereof binds specifically to a polypeptide of amino acid residues 19-174 of SEQ ID NO: 1 or a polypeptide of amino acid residues 19-140 of SEQ ID NO: 1.

15. (canceled)

16. The method of claim 1, wherein the agonist of TREM2 is an antibody or antigen-binding fragment thereof comprising a light chain variable region having a CDRL1, CDRL2, and CDRL3 selected from Table EX1 and A10, and a heavy chain variable region having a CDRH1, CDRH2, and CDRH3 selected from Table EX2 and A11.

17. (canceled)

18. The method of claim 16, wherein the TREM2 agonist is an antibody or antigen binding fragment thereof comprising: (a) a CDRL1, CDRL2, and CDRL3 comprising the sequence of SEQ ID NOs: 8, 22, and 35, respectively, and a CDRH1, CDRH2, and CDRH3 comprising the sequence of SEQ ID NOs: 77, 368, and 98, respectively; (b) a CDRL1, CDRL2, and CDRL3 comprising the sequence of SEQ ID NOs: 16, 369, and 370, respectively, and a CDRH1, CDRH2, and CDRH3 comprising the sequence of SEQ ID NOs: 85, 371, and 107, respectively; (c) a CDRL1, CDRL2, and CDRL3 comprising the sequence of SEQ ID NOs: 10, 23, and 372, respectively, and a CDRH1, CDRH2, and CDRH3 comprising the sequence of SEQ ID NOs: 81, 373, and 374, respectively; or (d) a CDRL1, CDRL2, and CDRL3 comprising the sequence of SEQ ID NOs: 17, 29, and 44, respectively, and a CDRH1, CDRH2, and CDRH3 comprising the sequence of SEQ ID NOs: 86, 94, and 375, respectively.

19. The method of claim 1, wherein the agonist of TREM2 is an antibody or antigen-binding fragment thereof comprising a light chain variable region selected from Table EX1 or A14, and a heavy chain variable region selected from Table EX2 and A15.

20. (canceled)

21. The method of claim 19, wherein the antibody or antigen-binding fragment thereof comprises: (a) a light chain variable region comprising the amino acid sequence of SEQ ID NO: 326 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 327; (b) a light chain variable region comprising the amino acid sequence of SEQ ID NO: 328 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 329; (c) a light chain variable region comprising the amino acid sequence of SEQ ID NO: 330 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 331; or (d) a light chain variable region comprising the amino acid sequence of SEQ ID NO: 332 and a heavy chain variable region comprising the amino acid sequence of SEQ ID NO: 333.

22. The method of claim 1, wherein the agonist of TREM2 is a small molecule agonist of TREM2.

23. The method of claim 22, wherein the agonist of TREM2 is a lipid ligand of TREM2.

24. The method of claim 23, wherein the agonist of TREM2 is selected from 1-palmitoyl-2-(5'-oxo-valeroyl)-sn-glycero-3-phosphocholine (POVPC), 2-Arachidonoylglycerol (2-AG), 7-ketocholesterol (7-KC), 24(S)hydroxycholesterol (240HC), 25(S)hydroxycholesterol (250HC), 27-hydroxycholesterol (270HC), Acyl Carnitine (AC), alkylacylglycerophosphocholine (PAF), α -galactosylceramide (KRN7000), Bis(monoacylglycerol)phosphate (BMP), Cardiolipin (CL), Ceramide, Ceramide-1-phosphate (CIP), Cholesteryl ester (CE), Cholesterol phosphate (CP), Diacylglycerol 34: 1 (DG 34: 1), Diacylglycerol 38:4 (DG 38:4), Diacylglycerol pyrophosphate (DGPP), Dihydrosphingomyelin (DhSM), Ether phosphatidylcholine (PCe), Free cholesterol (FC), Galactosylceramide (GalCer), Galactosylsphingosine (GalSo), Ganglioside GM1, Ganglioside GM3, Glucosylsphingosine (GlcSo), Hank's Balanced Salt Solution (HBSS), Kdo2-Lipid A (KLA), Lactosylceramide (LacCer), lysoalkylacylglycerophosphocholine (LPAF), Lysophosphatidic acid (LPA), Lysophosphatidylcholine (LPC), Lysophosphatidylethanolamine (LPE), Lysophosphatidylglycerol (LPG), Lysophosphatidylinositol (LPI), Lysosphingomyelin (LSM), Lysophosphatidylserine (LPS), N-Acyl-phosphatidylethanolamine (NAPE), N-Acyl-Serine (NSer), Oxidized phosphatidylcholine (oxPC), Palmitic-acid-9-hydroxy-stearic-acid (PAHSA), Phosphatidylethanolamine (PE), Phosphatidylethanol (PEtOH), Phosphatidic acid (PA), Phosphatidylcholine (PC), Phosphatidylglycerol (PG), Phosphatidylinositol (PI), Phosphatidylserine (PS), Sphinganine, Sphinganine-1-phosphate (SalP), Sphingomyelin (SM), Sphingosine, Sphingosine-1-phosphate

(SolP), or Sulfatide, or a salt thereof.

25. (canceled)

26. The method of claim 22, wherein the agonist of TREM2 is selected from Tyrphostin AG 538, AC1NS458, IN1040, Butein, Okanin, AGL 2263, GB19, GB16, GB20, GB17, GB18, GB21, GB22, GB27, GB44, GB42, GB2, 4,4'-Dihydroxychalcone, or 3,4-Dihydroxybenzophenone, or a salt thereof.

27-28. (canceled)
